Complex Analysis and Riemann Surfaces: A Graduate Path to Algebraic Geometry

Gunhee Cho

with contributions by
Bae Dongsong, Junhyuk Boo, Byungjoo Jeon, Yonghyun Ji, Sumin Kim, Namho Kim, Minseung Kwak, Hojae Jung, and Hyunmin Yoon

Department of Mathematics, Texas State University

January 10, 2026

Contents

I Perspective from contributors **10**

II Foundations of Complex Analysis **19**

1 Baby Complex Analysis **19**

1.1 Holomorphic Functions, Cauchy–Riemann Equations, and Local Conformality 19
1.2 Cauchy Integral Theorem and the Holographic Property 23
1.3 Complex Analytic Functions and Power Series Representations 29
1.4 Cauchy Estimates, Liouville's Theorem, and Applications 35
1.5 Identity Theorem and the Holomorphic Logarithm 40
1.6 Multivalued Analytic Functions and Riemann Surfaces 44
1.7 Meromorphic Functions, Singularities, and the Residue 47
1.8 Laurent Series, Weierstrass M-test, and Cauchy Theory on Annuli 53
1.9 Laurent Coefficients, Singularity Classification, and the Residue Formula 58
1.10 Poles, Residues, Holomorphic Simple Connectivity, and the Residue Theorem 62
1.11 Meromorphic Functions, Singularities at ∞, and Rationality 67
1.12 Concrete Examples with the Residue Theorem 70
1.13 Applications of the Residue Theorem: Multi–Contour Integrals and Systematic Residue Extraction 74
1.14 Applications of the Residue Theorem to Real Integrals 79
1.15 Advanced Contour Integration: Jordan's Lemma, Oscillatory Integrals, and Branch-Cut Methods 84
1.16 Zeros, Poles, and the Argument Principle 88
1.17 Rouché's Theorem and Zero Counting by Boundary Data 91

2 Advanced Topics in Complex Analysis **93**
2.1 Maximum and Minimum Modulus Principles and Consequences 93
2.2 Schwarz Lemma, Schwarz–Pick Lemma, and the Poincaré Metric on the Disk 95
2.3 Montel's Theorem and Normal Families 101
2.4 Hurwitz's Theorem and the Open Mapping Theorem 103
2.5 Picard Theorems and Essential Singularities 105
2.6 Infinite Products and Weierstrass Factorization 108
2.7 Entire Functions: Order and Type 111
2.8 Harmonic Functions and the Poisson Kernel 114
2.9 Subharmonic Functions and Riesz Decomposition 122
2.10 Analytic Continuation and Natural Boundaries 126
2.11 Gamma and Beta Functions via Complex Integration 131
2.12 Conformal Mapping Theory and the Riemann Mapping Theorem 134
2.13 Elliptic Functions and Doubly Periodic Functions 139
2.14 Appendix: Standard Contour Integrals and Techniques 145

3 Branched Double Covers and Gluing via Branch Cuts **151**
3.1 General two-sheeted picture and local models 151
3.2 The case $y^2 = x$: branch values $\{0, \infty\}$ and genus 0 152
3.3 The case $y^2 = x(x-1)(x-2)$: branch values $\{0, 1, 2, \infty\}$ and genus 1 . . 153

III Differential Forms and the Generalized Stokes Theorem **154**

4 Differential Forms, Exterior Calculus, and Applications **154**
4.1 Differential Forms and Exterior Derivatives 154
4.2 Line Integrals and Conservative Vector Fields 161
4.3 Complex Reformulation of Line Integrals 167
4.4 Stokes' Theorem, Cauchy–Green, and Surface Area as a 2-Form 172
4.5 Gaussian Curvature and the Gauss–Bonnet Theorem 180

5 Hodge–Weyl Theorem on a Compact Riemann Surface: Weak Existence, Uniqueness, Smooth Regularity, and Explicit Examples **184**
5.1 Standing assumptions, norms, and local coordinate dictionary 184
5.2 Weak formulation and the mean-zero Hilbert space 184
5.3 Poincaré inequality, coercivity, and Lax–Milgram 185
5.4 Local smoothing: Weyl's lemma (weakly harmonic $\Rightarrow$ smooth) 186
5.5 From weak to smooth solutions on (M, ω) 186
5.6 Hodge–Weyl theorem (functions) 187
5.7 Concrete model examples with full calculations 187
5.8 Optional: mean-value property from the weak formulation 189
5.9 Exercises 189

IV Sheaves and Cohomology on Complex Manifolds **190**

6 Holomorphic line bundles, divisors, and the Picard group 190
6.1 Holomorphic line bundles: cocycles, gauge changes, and explicit models . 190
6.2 Divisors and the associated line bundle 194
6.3 Picard group and classification 196
6.4 Exercises 197
6.5 Divisors, meromorphic sections, and the bundle $\mathcal{O}(D)$ 198
6.6 Picard group and the divisor correspondence: explicit computations . . . 199
6.7 Global sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$: chart computation and divisors 200
6.8 Picard group operations and degree 201

7 Sheaves: Definitions, Morphisms, and Locality 202
7.1 Presheaves and restriction maps 202
7.2 Sheaves: locality and gluing 204
7.3 Morphisms of sheaves and exact sequences 205
7.4 Stalks, germs, and local criteria 206
7.5 Examples and non-examples 206
7.6 Exercises 207

8 Čech Cohomology of a Sheaf 207
8.1 Open covers, cochains, and the coboundary operator 207
8.2 Refinements and the direct limit construction 209
8.3 Degree zero and degree one: gluing and patching 210
8.4 Examples and basic computations 210
8.5 Exercises 213

9 From Čech Cohomology to Sheaf Cohomology 213
9.1 Acyclic sheaves and Leray covers 213
9.2 Sheaf cohomology via injective resolutions 214
9.3 Recovering Čech cohomology from derived functors 215
9.4 Fine, soft, and flabby sheaves 215
9.5 Exercises 216

10 The Exponential Sequence and First Applications 216
10.1 Holomorphic logarithms on simply connected domains 216
10.2 The exponential short exact sequence of sheaves 217
10.3 Connecting homomorphisms and cohomological meaning 218
10.4 Examples: line bundles and transition functions 218
10.5 Exercises 218

11 Short Exact Sequences of Sheaves and Long Exact Sequences 219
11.1 Long exact sequences in sheaf cohomology 219
11.2 Functoriality and naturality 220
11.3 Examples and applications 220
11.4 Exercises 221

12 Fine Sheaves, Mayer–Vietoris, and the de Rham Theorem 221
12.1 Local exactness and the Poincaré lemma 221
12.2 Partitions of unity and fine sheaves 223
12.3 The Mayer–Vietoris sequence for complexes 225

12.4 de Rham cohomology via sheaves . . . 227
12.5 Computations: spheres and explicit generators . . . 227
12.6 Dolbeault sheaves and the $\bar{\partial}$-resolution . . . 228
12.7 Exercises . . . 229

13 Canonical Bundles, Chern Classes, and Curvature 229
13.1 Local expressions in complex coordinates . . . 229
13.2 Gaussian curvature and the Ricci form . . . 231
13.3 The exponential sequence and the degree/Chern class map . . . 233
13.4 Canonical divisors . . . 235
13.5 Metrics and concrete examples . . . 236
13.6 Exercises . . . 237

14 Divisors, the Degree Map, and the Fundamental Theorem of Algebra 237
14.1 Divisors on compact Riemann surfaces . . . 237
14.2 The degree map and its properties . . . 239
14.3 Divisors, divisor classes, and the Picard group . . . 240
14.4 Application: the Fundamental Theorem of Algebra . . . 241
14.5 Examples and explicit computations . . . 242
14.6 Exercises . . . 242

V Duality 243

15 Duality on Compact Riemann Surfaces 243
15.1 Perfect pairings and Poincaré duality . . . 243
15.2 Hodge theory on a compact surface . . . 245
15.3 Serre duality on a compact Riemann surface . . . 246
15.4 Concrete examples with explicit calculations . . . 249
15.5 A compact dictionary for later use . . . 250

VI Riemann–Roch Theorem 250

16 The Riemann–Roch Theorem on Compact Riemann Surfaces 250
16.1 Notation from earlier chapters . . . 250
16.2 Degree of principal divisors . . . 250
16.3 The spaces $L(D)$ and $I(D)$ and their sheaf interpretations . . . 251
16.4 Serre duality input: $i(D) = \ell(K - D)$ . . . 251
16.5 Euler characteristic and degree additivity . . . 252
16.6 Riemann–Roch: statement and proof . . . 253
16.7 Degree of the canonical divisor . . . 253
16.8 Explicit examples and computations . . . 253
16.9 Asymptotic form and a practical vanishing criterion . . . 254
16.10Geometric meaning and how to use Riemann–Roch . . . 254
16.11Exercises . . . 254

17 A Complete Proof of Riemann–Roch **255**
17.1 Fine resolutions and Dolbeault models 255
17.2 A long exact sequence relating $\mathbb{C}$, $\mathcal{O}$, and Ω 255
17.3 Serre duality as a perfect pairing 256
17.4 A key exact sequence for adding a point 256
17.5 Base case $D = 0$ 257
17.6 Proof of Riemann–Roch by induction on degree 257
17.7 Corollaries and asymptotics 257
17.8 Summary of the mechanism 258

18 Intersection Theory and Riemann–Roch for Surfaces **258**
18.1 Intersection pairing: local definition and global properties 258
18.2 Chern classes and the cohomological intersection product 259
18.3 Adjunction and the canonical class 260
18.4 Riemann–Roch for surfaces: statement and proof mechanisms 261
18.5 Worked examples: explicit intersection computations and RR checks . . . 261
18.6 Using RR + Serre duality: vanishing tests and effectivity heuristics . . . 264
18.7 Exercises 264

VII Jacobian of Compact Riemann Surfaces **266**

19 Jacobian, Picard Variety, and Abel–Jacobi Theory **266**
19.1 From holomorphic differentials to a period lattice 266
19.2 Abel–Jacobi map on points and on divisors 267
19.3 Picard group and the exponential sequence 268
19.4 Identifying $\mathrm{Pic}^0(X)$ with $J(X)$ 269
19.5 Divisors, line bundles, and complete linear systems 270
19.6 Abel's theorem: fibers of α_d and linear equivalence 271
19.7 Jacobi inversion: surjectivity of α_g 272
19.8 Riemann–Roch from geometry of α_d 272
19.9 Abel's theorem in degree zero: principal divisors 272
19.10 Worked examples: genus 0 and genus 1 273

VIII Algebraic Curves and Algebraic Geometry **273**

20 Compact Riemann Surfaces as Algebraic Curves **273**
20.1 Very ample divisors on curves 274
20.2 Two meromorphic generators and a plane model 274
20.3 Examples 275
20.4 Exercises 275

21 Algebraic Intersection Multiplicity of Plane Curves **276**
21.1 Local algebraic definition via length 276
21.2 Basic properties (with proofs) 277
21.3 Multiplicity of a curve at a point and the tangent cone bound 277
21.4 Computation techniques (with concrete worked demonstrations) 278
21.5 Examples 279

21.6 Global relation: Bézout's theorem 279
21.7 Compatibility with surface intersection theory 280
21.8 Exercises 281

22 Jacobi Theta Functions from the Viewpoint of Riemann Surfaces and Jacobians 281
22.1 From a Riemann surface to its Jacobian: period matrices and lattices . . 281
22.2 The Riemann theta function: convergence and quasi-periodicity 283
22.3 Theta as a section of a line bundle on $\mathrm{Jac}(X)$ 284
22.4 Genus 1: Jacobi theta functions and explicit computations 285
22.5 From theta to Weierstrass: σ, ζ, and $\wp$ via logarithmic derivatives 285
22.6 Exercises (with guidance) 287
22.7 Why theta is the "right" object: modular transformations and geometry 288

23 Galois Group and Riemann Surface 288
23.1 Function fields and pullback 288
23.1.1 Local branches, norm, and trace (explicit formulas) 288
23.1.2 Why the extension degree equals $\deg \pi$ 290
23.2 Galois coverings and Galois extensions 290
23.2.1 Deck transformations (unramified and branched) 290
23.2.2 Main theorem: deck transformations = field automorphisms . . . 291
23.3 Examples 292
23.3.1 Example A: a hyperelliptic double cover 292
23.3.2 Example B: cyclic n-covers (Kummer) with full Galois group . . . 292
23.4 Uniformization and the Galois picture 293
23.4.1 Normalizer formula 293
23.5 Exercises 294

24 Connection to Number Theory 294

Introduction

This document is written in the spirit of a *lecture note*: concise, computation–first, and organized to help a reader move from hands-on calculations to conceptual structure. They grew out of the *Enjoying Math* YouTube channel and, in particular, from several cohort-style study readings led by **Bu Junhyuk** and others. In those groups, participants met regularly (often online) to work through the *details*—derivations, estimates, normal forms, local models—and then stepped back collectively to distill the abstractions. The present document systematizes that *compute–then–abstract* pathway.

Study groups, YouTube, and collaborative computation

The notes should be read as a snapshot of an ongoing learning community rather than as a solitary monograph. Many examples and proof-checks were first developed on the board or tablet during live sessions, refined in discussion, and only later written up in a coherent form. The *Enjoying Math* channel serves as a parallel "visual layer": videos illustrate computations, branch-cut gluings, and geometric pictures that appear here in formulaic form.

To reflect this collaborative origin, we adopt the following convention: at the beginning of many major chapters, a short note of the form *"verified by . . . "* records the participants who checked the computations, examples, and local arguments in that part of the notes. This is not merely acknowledgment; it documents computational labor and emphasizes that what appears as a clean proof typically started life as several pages of shared rough work. Any remaining errors, misprints, or gaps are, of course, our responsibility.

Philosophy and pathway

The conceptual arc is Riemann's: begin with *complex analysis* (Cauchy theory, harmonicity, residues), use it to build and control *Riemann surfaces* (branch points, branch cuts, local models), then develop geometric structure via *differential forms* and *Hodge theory*, and finally cross to *algebraic geometry* (divisors, line bundles, sheaves, cohomology), culminating in the *Jacobian* and the *Riemann–Roch theorem*. At every stage, abstraction is anchored in explicit computations: Taylor/Laurent expansions, Stokes/Cauchy–Green identities, residue extraction, branch-cut gluings, curvature and area forms, Mayer–Vietoris patching, and concrete section-counting by Riemann–Roch.

A guiding principle is:

compute first on test cases, then abstract only after the calculation is in your hands.

Prerequisites and style

A working knowledge of multivariable calculus and linear algebra is assumed. Familiarity with basic topology and differential geometry (manifolds, forms) is helpful but not required at the outset; we introduce what we need by computation and local models. Proofs are included whenever a short, illuminating argument fits the flow; otherwise we give a clear reference frame and return later with a fuller treatment. The style is deliberately "lecture-note-like": terse statements, many examples, and a bias toward derivations you can reproduce on paper.

Organization of the notes (guided by the table of contents)

The document is organized so that each major tool is first built analytically, then reinterpreted geometrically, and finally reused in algebraic form.

- **Part I: Foundations of Complex Analysis (Chapters 1–3).** We develop the analytic backbone: holomorphicity and the Cauchy–Riemann equations, Cauchy's integral theorem and its "holographic" consequences, power series, Cauchy estimates and Liouville, the identity theorem and holomorphic logarithms, and a systematic treatment of Laurent series, residues, and residue calculus (including real integrals, Jordan's lemma, oscillatory integrals, and branch-cut methods). We end with zero/pole counting via the argument principle and Rouché.

  The transition to geometry begins in Chapter 3, where we construct Riemann surfaces *by hand* using two-sheeted branched covers and explicit gluing along branch cuts. The models $y^2 = x$ (genus 0) and $y^2 = x(x-1)(x-2)$ (genus 1) serve as the main "computational laboratories" for later chapters.

- **Part II: Differential Forms and Generalized Stokes (Chapters 4–5).** We recast vector calculus in the language of differential forms and Stokes' theorem, then connect it back to complex analysis via Cauchy–Green. We compute area forms, curvature, and Gauss–Bonnet in concrete examples. The part culminates in a detailed, functional-analytic proof package for Hodge–Weyl on compact Riemann surfaces: weak formulation, Poincaré inequality, Lax–Milgram, Weyl's lemma (regularity), and explicit mode computations (with full calculations and exercises).

- **Part III: Sheaves and Cohomology on Complex Manifolds (Chapters 6–14).** This is the algebraic–geometric core. We build holomorphic line bundles from cocycles and gauge changes, relate bundles to divisors, and develop the Picard group via explicit operations. We then introduce sheaves carefully (locality/gluing, stalks, exactness), compute Čech cohomology by hand (including degree 0 and 1 gluing problems), and explain how derived-functor sheaf cohomology recovers Čech cohomology under Leray/acyclic hypotheses. The exponential sequence is treated concretely, with connecting homomorphisms computed in examples. We then prove the de Rham theorem sheaf-theoretically, set up Dolbeault resolutions, and connect curvature/Chern classes/degree to canonical bundles and divisors. The section on divisors and degree culminates in an application to the Fundamental Theorem of Algebra, emphasizing "global topology forces algebraic constraints."

- **Part IV: Duality (Chapter 15).** We establish perfect pairings (Poincaré duality, Hodge-theoretic pairings) and then Serre duality on a compact Riemann surface, with concrete calculations that the reader can verify directly on basic examples. A compact dictionary summarizes the analytic and algebraic languages for later use.

- **Part V: Riemann–Roch (Chapters 16–18).** We present the statement, proof mechanisms, and practical consequences of Riemann–Roch, with explicit computations and vanishing heuristics. A full proof is given using the exponential sequence, Dolbeault/de Rham models, and an inductive "add one point" exact sequence. The section on surfaces extends the reader's intuition to intersection theory and adjunction, providing worked intersection computations and RR checks.

- **Part VI: Jacobians (Chapter 19).** We build the Jacobian from holomorphic differentials and a period lattice, define and exploit the Abel–Jacobi map (on points and divisors), identify $\mathrm{Pic}^0(X)$ with $J(X)$, and prove Abel's theorem in a computation-driven style. Jacobi inversion appears as a surjectivity statement whose meaning becomes transparent once one has computed enough examples and matched them to divisor theory.

- **Part VII: Algebraic Curves and Algebraic Geometry (Chapters 20–21).** We explain how compact Riemann surfaces become algebraic curves: very ample divisors, plane models from meromorphic generators, normalization, and genus/degree control. Intersection multiplicity is developed from local algebra (length) and then used for global results such as Bézout, with explicit computation techniques and examples.

- **Part VIII: Bridges to special functions, Galois theory, and arithmetic (Chapters 22–24).** Chapter 22 revisits the Jacobian through theta functions: Riemann theta as an entire series on $\mathbb{C}^g$ with controlled quasi-periodicity, its descent to a canonical line bundle section on $Jac(X)$, and in genus 1 the recovery of Jacobi theta functions and Weierstrass theory via logarithmic derivatives. Chapter 23 develops the Galois dictionary for Riemann surfaces: holomorphic maps $\leftrightarrow$ function field extensions, deck transformations $\leftrightarrow$ field automorphisms, with detailed examples (hyperelliptic double covers, cyclic covers) and a uniformization viewpoint. Chapter 24 then gestures toward number theory, using modular curves as a guiding example where analytic quotients, algebraic models over $\mathbb{Q}$, and arithmetic Galois actions meet in a single object.

How to read these notes

A recommended workflow is:

1. Skim the statements to see the landscape.

2. Work through the *examples and computations* before reading full proofs.

3. When a construction feels abstract (sheaves, cohomology, Jacobians), return to the test cases (two-sheeted covers, $\mathbb{CP}^1$, tori) and redo the local calculations.

There are (at least) three natural reading paths:

- **Analysis-first path:** Chapters 1–2 $\to$ residues and applications $\to$ Chapter 3 (branch cuts) $\to$ Chapter 22 (elliptic/theta).

- **Geometry-first path:** Chapters 4–5 $\to$ Gauss–Bonnet and Hodge–Weyl $\to$ Chapters 12–15 (de Rham, duality) $\to$ Riemann–Roch.

- **Algebraic-geometry-first path:** Chapters 6–11 (bundles, divisors, sheaves, exponential sequence) $\to$ Chapter 16 (Riemann–Roch) $\to$ Chapters 19–22 (Jacobian, theta).

The intended reward is the moment when calculus you can do by hand and geometry you can picture snap into the same idea: analytic identities become global invariants, and global invariants explain why certain computations *must* come out the way they do.

Part I
Perspective from contributors

Reflections on the Importance of Calculus Computation

These are individual reflections from participants in the "Riemann's Complex Analysis II" lecture note project on the importance of calculus computations. This is a preliminary draft; for later revised versions, please refer to the Discord channel.

Junhyuk Boo (Leader)

When I study abstract mathematics written as a long list of definitions and theorems, there are so many moments when I find myself asking: "Why on earth are we defining *this*?" Then, when I see an important proposition being proved using those very definitions, I find myself wondering: "Who are these people that they could even think of such things?" From within that emotional distance, I think I gradually drifted away from mathematics. I lived for a long time feeling that mathematics is a subject for geniuses, and that I am simply not someone who can "do math."

But through *Enjoying Math*, and especially through the calculus computations we have been doing (and continue to do), I have had a series of precious experiences: moments when I genuinely feel, "Ah, even someone like me can actually do mathematics." I believe this will remain true going forward as well. I deeply feel that calculus computation is not only a tool that demonstrates the validity of the mathematical theories that have been developed so far, but in fact is a purpose in itself.

I still vividly remember the moment in our study when we locally computed the potential of the winding form, applied the FTLI, and obtained 2π *independently* of the choice of closed curve. Up to that point, I had only thought of this as a situation where the fundamental theorem of calculus (FTC) fails because there is a hole at the origin. But through that first computation, I realized that locally we *can* apply FTC.

Of course, the computation

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

is a calculation I had done so often that it felt utterly familiar. However, when I actually plugged the endpoints of the curve into the local expression of arctan and added everything up, seeing that this was *really* the FTC that I thought I knew, the FTC suddenly felt extremely unfamiliar. For the first time while doing a computation, I experienced something like catharsis.

This is very different from simply remembering, in the context of Green's theorem, that "because the curl of the vector field is zero, the line integral along any boundary curve is the same." Instead, by experiencing this directly, the fact that locally the form is written as a gradient, and thus its curl is trivially zero, and that using FTLI locally forces the integrals to agree, became something I knew in a very intuitive and obvious way. Through this, I've been constantly feeling just how big a role the intuition coming from computation plays.

In the context of complex analysis, when we complexify the winding form, we get dz/z, and one can say that Riemann, in a sense, algebraically extended the worldview of calculus through the computation of integrals of this simple form. Even though I myself have been looking at the winding form for almost a year now, I still do not fully understand it. This makes me feel that it is incredibly important to examine seemingly elementary objects very thoroughly and meticulously.

The world of mathematics that I get to know through computation is truly fun, and it feels as if I am having a conversation with mathematicians of the past. The numerical values produced by computations do not change; through them we can observe which patterns they discovered and how they formulated theories. Looking at these values, I feel a kind of awe, in a sense different from what I wrote at the beginning: "Ah, so this is what it means for someone to do mathematics."

Even so, the reason I still think I can continue doing mathematics is that I have come to understand the criteria by which mathematical theories are built, and that the core of those criteria lies in calculus computation.

Byungjoo Jeon

I would like to share some reflections on the "importance of calculus computation." Since my thoughts may be somewhat vague, if someone were to ask me, "So why exactly is calculus computation important?" I am not sure this reflection would serve as a fully satisfactory answer to that question. But once I started writing, I could not take it back, so here is a short reflection anyway!

For me, Riemann's complex analysis in *Enjoying Math* seems to provide a lens through which we can reinterpret abstract algebra, topology, and differential geometry, all written in abstract language, as calculus performed in $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$, implemented via the FTC.

Practically speaking, all the computational tools we used were essentially just calculus and linear algebra that we learned in our first year of university.

When I look at the logically complete proofs of countless theorems, I find that, line by line, I can follow the argument, but I do not really gain intuition from it. When I ask myself whether I could ever discover such a proof on my own, it feels impossible; it is just a cold bundle of logic.

However, calculations in Euclidean space are precisely the source of the ideas that appear in such proofs; from carefully observing those computations, I have experienced that it is possible to make new discoveries.

At first I knew nothing, but as my learning accumulated, I slowly began to realize that all abstract objects ultimately come from concrete models and concrete objects. If I had to pick one mathematically concrete object, I would, without hesitation, say differential forms. Conceptually, but also in terms of their role in computations, they have extremely wide applicability. It feels as though differential forms have appeared in almost every area we have touched. If I had not known differential forms, I think I would have lost my way, trying to follow all the theories purely logically, without any computations.

One charm of computation is that as you deepen your understanding of mathematics, even the calculations you performed in the past start to look different, and the same object can be interpreted differently, giving you a new perspective. The belief—or perhaps the attempt—that everything in mathematics can be described in the language of calculus and linear algebra is, I think, both a gateway to understanding abstract objects that seem to live in the heavens, and at the same time a way of learning not to dismiss trivial and

elementary objects at our feet as "easy," but instead to see beauty and depth in them.

Short exact sequences, cohomology, homology, homotopy, group, ring, field, module, tensor product, Zariski topology, projective space, ideal, sheaf, and so on: these objects, which look abstract when you only see their definitions, can all be interpreted through the lens of elementary calculus and linear algebra, and their computations can be carried out concretely. That is what I now consider the main charm of complex analysis as I learned it in *Enjoying Math*.

I am still learning and still in the process of realizing new things, but each year, I look forward to gaining a new viewpoint and a deeper mathematical literacy.

Yonghyun Ji

A reflection on the "importance of calculus computation"

For a long time, I was doing what you might call "picky mathematics." Rather than doing computations in calculus, I preferred to set up algebraic structures and look for systems and patterns within them. I felt much more at home in the world where structures built from sets and operations— like groups, rings, and vector spaces—are defined and then studied.

So, concretely speaking, calculus computations were not really to my taste. The grad, curl, and div that I had learned as a first-year student were merely "Oh, so such things exist." I solved a few problems for exams, but I never seriously tried to think about why they had those particular forms, or how they are related to each other. Computation was just a procedure for obtaining results and felt a bit removed from the kind of "structures" I liked.

But after encountering differential forms, my thinking changed a lot. I saw that grad, curl, and div do not exist separately but appear naturally unified in a single language—and the thing that really allowed me to actually understand this language was none other than the calculus computations that I had done half-heartedly in the past.

When we carry this language of differential forms over to complex space, the story extends in yet another direction. Studying Riemann's complex analysis, I realized that, in a sense, I had been preparing to cross over into the "algebraic worldview" via calculus computations.

One of the biggest shocks was that my view of polynomials completely changed. Before studying Riemann's complex analysis, a polynomial was, to me, just an algebraic object formed by multiplying and adding variables and coefficients. Within that framework, I was mainly interested in structure—what properties the polynomial ring has, what ideals look like, and so on.

But when I learned that the most basic data of a polynomial—its coefficients and the order of its zeros—can be recovered from integrals, it was quite a shock. By integrating along a suitable curve, you see the coefficients pop out, and the order of poles or zeros becomes visible. I realized that the information I thought of as "algebraic" was actually hidden inside integrals.

At that moment, I began to feel that the dichotomy of "algebra vs. analysis" is, in actual mathematics, a somewhat artificial division. In complex analysis, computing integrals—for example, actually computing integrals along specific paths—is not just a matter of polishing calculation techniques. It plays the role of reading off algebraic information. Without doing the integrals, you simply do not see certain things. Through computation, structures that were invisible suddenly reappear in a new form.

So now I interpret the phrase "the importance of calculus computation" somewhat differently. In the past, I thought of computation as just a short rite of passage before entering the world of structures. But through complex analysis and differential forms, I came to feel that computation is what illuminates the very structures I want to see.

For me, calculus computations became a bridge connecting the algebraic mathematics I liked with complex analysis. Whereas I used to want to avoid computation and see only structure, now I accept that we can approach structure precisely through computation. That realization itself is something I value very highly.

Hyunmin Yoon

Before talking about the importance of calculus, I would like to organize how I used to think about complex numbers.

1. Thoughts on complex numbers. I first encountered complex numbers when I was told that the equation $x^2 + 1 = 0$ actually *has* solutions. At that time, I did not feel anything in particular; I just thought, "Oh, this is part of the curriculum." I only thought of complex conjugates and computations with them as something that appears in problems about the relation between roots and coefficients.

When I first saw Euler's formula, I was impressed for a moment: all these strange symbols e, π, i appearing together in a single equation! What on earth is its true nature? But then, when I saw it again in university, that feeling collapsed. I thought, "Oh, it's just polar coordinates." Even when complex numbers appeared together with the definition of e^x in analysis, it did not move me.

So after that, complex numbers were really out of sight and out of mind. Later, in an electrical engineering course on signal processing, I encountered complex numbers again. But hearing about them from the engineering side, I ended up thinking, "Complex numbers are just a kind of seasoning you bring in when needed." This made me distance complex numbers even further from my mind. And so, when I graduated from the math department, I felt as if I had graduated from complex numbers as well.

2. The importance of calculus. Looking back on when I studied mathematics as an undergraduate, I was obsessed with how quickly I could reach more advanced courses. As a result, each course was merely a stepping stone to the next, and in my last semester I took graduate-level functional analysis and Riemannian geometry. At that time I was intoxicated with the feeling of "I am finally studying such difficult mathematics!" I had no concept of reflecting on or "appreciating" mathematics.

Even when I decided to pursue a career in AI, I thought that "bringing mathematics into my career" meant having been exposed to as difficult mathematics as possible.

But as I began learning differential forms in *Enjoying Math*, I realized I had been studying mathematics the wrong way. Whenever I faced a difficult proposition, I used to chew over each word in the statement. However, studying differential forms was nothing like that. Once you study differential forms, you spend ages talking about what integration is.

Until then, integration to me was just "Riemann integral: first–second year integration," "Lebesgue integral: third–fourth year integration." Even when solving exercise problems, I often skipped the actual computation in integrals.

Differential forms grabbed me by the collar and asked: "Do you really know what an integral is?" I responded, "Of course I know what an integral is—why are you asking?" But the more I learned, the more I felt: "I actually have no clue what integration really is." I realized I did not truly understand even the one-variable integral $\int_0^1 x\,dx$. I came to feel sorry toward differential forms, realizing that even the exam-type problems I had breezed through had been supported by differential forms in the background.

Differential forms are also a good friend that encourages self-reflection. As we study mathematics, we often begin to think that studying means encountering many difficult concepts and hearing about them. It seems I am not the only one. But after studying like that, I found I could not extend my thinking by myself.

In those moments, differential forms keep poking me from the side: "Friend, what can you actually do?" At the same time, they remind me that the only thing I can truly do is calculus.

Ironically, I think the importance of calculus is precisely that "calculus is important in order to study calculus." There is an enormous amount to unpack even in a single integral problem, and likewise in a single differentiation problem. As I started to unpack them one by one, I was finally able to appreciate mathematics. The mathematics I learned afterward was built by piling up such observations.

Difficult things are still difficult, of course, but the difficulty is not in how alien or impressive something looks. Rather, it lies in how densely calculus has been condensed into it. I think of calculus as a lens through which we can appreciate mathematics. What I truly have to do is computation, and those computations can only be carried out through the lens (or tool) of calculus.

3. Then why should we look at complex numbers? I know differential forms only a little, and I know even less about complex numbers. If I ask myself why I should care about complex numbers, the first thing that comes to mind is: "I can't do this integral—complex numbers, please help." When I examined the winding form at the level of computations, I came to understand that this calculation really measures rotation. When complex numbers are introduced, angles become more clearly visible. Angles themselves are real numbers, so even though we bring in complex numbers, it still feels as if we are ultimately revisiting the same calculus we were looking at before.

So I do not feel that complex numbers "flip the entire setup"—rather, they serve as a window that allows us to carry calculus further. To be honest, I still cannot confidently explain "why we should look at complex numbers." But when I decide to study calculus and sit down with complex numbers at my side, I find that I can see integrals from a broader perspective.

Sumin Kim

Here are my reflections.

Perhaps this sounds overly highbrow, but I think we have to start with the question, "Why do we study mathematics at all?" In a way, it is obvious: we study in order to understand. People will have very different standards for what it means "to know" something, but I personally think that you should at least be able to explain it to someone else before you can say you know it.

However, as abstraction increases, explanations often start to self-reference: you end up recycling your previous explanations, and they risk turning into something like a

religious text. I do not think that is a good explanation. A good explanation, in my view, is one that is easy for the listener to understand.

For that reason, I think computation is important. No matter how difficult a concept is, it must ultimately have been created to compute something. If you show it through computation, at the very least you can persuade someone through the numerical outcome. (Of course, persuading someone why we suddenly perform a certain computation is also a very important problem in itself.)

Also, even if, in the distant future, theory becomes ever more abstract, the concrete results of computations will not change. So I believe that computations themselves can become a kind of lantern that illuminates hidden stories (perhaps future stories) behind the scenes.

It might be better if I share a concrete computation experience. Among the computations I have cherished while studying Riemann–complex analysis, I would like to share one particular (unknown) example.

At first, I could not understand at all why we needed to look at the Mayer–Vietoris sequence. When I asked myself, "Why can't I accept this?", my self-reflection led me to the conclusion: "The real issue is that I don't even understand why we should decompose S^2 into open sets U and V in the first place."

From there, I came to realize that the fundamental problem was my inability to accept that the integral over the sphere is not zero. Since polar coordinates felt inconvenient, I tried to compute potentials in (x, y)-coordinates, and through this I came to accept that the local potential difference is $2\,d\phi$. Then, after computing two different local potentials, I finally began to sympathize with the intention behind the partition of unity—namely, to somehow glue these together.

In this way, computation is the only tool that allows someone who does not yet know the theory to *relate* to it, and at the same time it is the tool that enables you to resonate with the very intention behind the theory. In that sense, I think computation is important to the act of *knowing* in mathematics.

Hojae Jung

I like discovering patterns in numbers and seeing geometric information. Perhaps because of that, as a middle and high school student I liked number theory, where you find patterns in integers, and Euclidean geometry, where you use proofs to derive information about lengths and angles.

Calculus, for me, was not like that. More precisely, differentiating and integrating functions seemed useful for computing areas under graphs or extrema, but I did not see much connection to Euclidean geometry or number theory.

However, studying with people in *Enjoying Math*, I came to see that geometry, number theory, and calculus are all connected, and that treating them as separate subjects can cause serious misunderstandings.

You can see this, for example, when computing the de Rham cohomology of S^1. It is a very simple yet astonishing idea that different spaces have different de Rham cohomology. It is obvious in one sense, but also quite striking.

The domain of a differential form or function is precisely the underlying space, so it is obvious that the behavior of differential forms and functions varies depending on the space. But de Rham cohomology is a story about calculus, and at the same time it contains information about the space itself. This is astonishing.

Moreover, this story is not limited to S^1. It naturally extends to n-dimensional spheres and tori. The most characteristic visual difference between the two-dimensional sphere S^2 and the two-dimensional torus is that every loop on S^2 is contractible, whereas on the torus there are two distinct types of loops that are not contractible. Even more surprising is that there are differential forms dx and dy associated to these two types of loops.

And de Rham cohomology is not something detached from geometric information and our geometric intuition. On the contrary, calculus, in a way that is deeply aligned with our geometric instincts, speaks about those instincts in a clear and profound language.

For that reason, calculus is important. When visual intuition is not available, calculus, in turn, provides us with a calculable way of obtaining geometric intuition.

When complex numbers enter calculus, even more intriguing phenomena appear. Complex numbers have many good properties. Unlike real two-dimensional vectors, they come with a natural multiplication. We introduced i in order to consider $\sqrt{-1}$, but we do not need to introduce any new number system for $\sqrt{i}$: complex numbers already contain it. Moreover, by Euler's formula, we get a natural polar form.

Thus complex numbers have an in-built computational convenience. It is very natural, then, to consider complex-valued functions and to attempt to differentiate and integrate them.

Complex functions have the very nice property that, once differentiable, they admit power series expansions. And in the process of integrating complex functions, we gain the perspective that line integrals of vector fields can be viewed as integrals of complex functions.

Then, just as we study patterns in numbers, we begin to consider the *order* of complex functions. The study of numerical patterns is closely related to the notion of order in complex analysis, and that order, in turn, tells us about geometric information (such as the winding number).

I believe we are all born with a bit of curiosity and ability to detect numerical patterns and geometric structure. Calculus gradually reveals those pieces of information in a way that is very closely aligned with our instincts.

And I want to emphasize: calculus is a computational method that is very familiar to us (even if not always easy). As a tool that clarifies our thoughts in a computable way, calculus is something we could study all our lives and still never exhaust.

Hyunsoo Yoo

I am not satisfied with this draft because I have not revised it, but I will post it as it is for now.

1. Do you like mathematics, or do you dislike it? Why? Perhaps I have spent my younger years asking and pondering this question. There were times when I would stubbornly cling to my pride because I could not understand a problem that seemed easy at first glance. There were times when I was satisfied to be recognized by others through good grades. There were times when I felt overwhelmed by the vastness of mathematics, which never seems to end no matter how much you study. There were times when I felt very small in front of the seemingly unreachable genius or greatness of others. Sharing moments of wonder and difficulty with peers was always a joy.

If I were to quantify it dichotomously, it seems that for me—and for many people I met— the suffering side of mathematics occupies a larger portion than the joy. Yet I

clearly do like mathematics.

2. Some emotions overwhelm and sometimes control us; others are so subtle that we barely realize we are experiencing them. For me, the joy of mathematics seems to belong to the latter. It is often said that for a mathematics major, the serious research that follows undergraduate and master's studies is still only a preparation for the beginning. We have to create our own path in a place where we cannot see even one step ahead. Even if you fully master one textbook, they say there are dozens or hundreds more like it.

No matter how much I studied, it felt as if nothing remained. What do I have left?

3. This article follows Riemann's work from complex analysis to understanding Riemann surfaces and their geometry, and then reaches the connection with algebraic geometry and the Riemann–Roch theorem. I had heard that it deals with very important and beautiful content in mathematics. But what remained for me was not its splendor.

What gained meaning were the countless famous statements I had heard over and over again, when I tried to reproduce them with my own hands, to compute them myself, and to get stuck. I slowly came to suspect that I actually did not know at all things like the winding form and the residue theorem, which I had taken for granted. I even realized that I did not truly understand the FTC I had learned as a freshman.

The more I saw how many things I did not know, the more I began to recognize that even the things I thought I knew were, in fact, unknown. What scares me more than not knowing is the realization that I had been *thinking* I knew something when I really did not.

4. Why should we understand Riemann's worldview? There is something here that cannot be experienced in standard complex analysis courses. It is hard to deny that standard complex analysis is really just a preparation for this deeper worldview.

Because of that, when we experience more difficult and complex theories, it becomes natural to suspect that there is a wealth of hidden information in things we previously thought of as simple and easy.

Of course, I am still in the middle of this process. Learning mathematics still feels like swimming alone in a deep ocean. Again, I feel that even the mathematics that freshmen learn is something I still do not understand at all.

Even so, the reason I must not stop has become clear. This is how the process is supposed to be; everyone finds it difficult. Those who begin and keep going will be the ones who get to enjoy the joy of mathematics.

5. Before I hit adolescence, I left the neighborhood where I had lived my whole life and transferred to a new school in a completely different area. Nothing there was familiar to me, and I had to build a new life on my own. I kept discovering that the world I had lived in was not all there was.

I still vaguely remember mustering the courage one day to talk to the kid next to me during lunchtime. I do not remember what we talked about, but I remember the feeling of courage and the expression on that kid's face who received it.

Studying mathematics feels like that to me. In the midst of many things I do not know, I hold on to one calculation that I have done with my own hands and carry it

with me. That emotion became the motivation that helped me overcome other emotional hardships, and it is the reason I enjoy mathematics and the reason I like mathematics.

Namho Kim

I used to be someone who could not do complex analysis. I disliked the word "complex" itself, and I disliked the word "analysis" even more. Complex numbers felt like symbols that had nothing to do with my life, and analysis seemed like a world reserved for math majors who walk around with their shoulders full of tension.

That I came to find complex analysis interesting was, surprisingly, thanks to computation. Computation was almost the only thing I could actually hold on to, and one of the few ways I could verify something with the sense of my own hands rather than relying on someone else's authority.

In fact, I had never heard terms like differential forms or Green's theorem. I did not know even the names of Stokes's theorem or the divergence theorem. It was the same for grad, curl, and div. I could not properly integrate $\tan x$. So the only thing I could really do was perform the given calculations, and with that one small tool called calculus, I trudged forward.

But strangely, through repeatedly doing such simple calculations, a door opened. Though I knew almost no mathematics, by holding on to calculus alone I managed to endure complex analysis, and along the way I directly saw moments when algebraic structure seeped out of the computations. Even though I did not know advanced tools like Mayer–Vietoris, by looking at the winding form through the lens of calculus and checking with my own hands why certain values appear, I physically felt that calculus could naturally lead into algebraic stories.

In hindsight, that was a kind of "naive intuition." Because I did not know abstract terminology, I think I was able to directly touch the structures hidden inside calculations.

Complex analysis showed me where it connects with algebra: why boundary integrals carry topological information, and so on. I learned this not through general theory but gradually through computation.

What remains in my hands even now is, in the end, calculus. With that one small tool, I am able to grab hold of propositions that once felt like rocks rolled in from Mars—strange, unfriendly claims with no visible origin— and tear them apart to see what is inside. That is my only consolation.

In the past, algebra felt like hell. Symbols appeared and disappeared; proofs ended somewhere far away from where they began, without any visible starting point; the more I read, the more only *why* remained. But when I revisited complex analysis holding only calculus in my hand, those hellish sentences started to look a little more human.

When you approach something through computation, you can grope for its meaning with your fingertips, even if you do not know the full background.

The winding form, for instance, was utterly unfamiliar at first, but as I followed it with a calculus mindset, I experienced the quiet realization of "Ah, this is how it connects."

I still do not know the stronghold of abstract algebra. I still do not fully understand its language. But thanks to that one familiar technique called calculus, I can now dissect propositions that once looked like extraterrestrial life. That experience, I think, is the reason I continue to hold on to complex analysis.

Part II
Foundations of Complex Analysis

Complex analysis—also called complex function theory—is often regarded as the pinnacle of mathematical elegance: simple in statement, profound in consequences, and remarkably beautiful. This part lays the analytic foundations we will need later for branched covers, Riemann surfaces, Hodge theory, and duality. We emphasize both intuition and rigor, so that the transition to algebraic curves and branch-cut gluing is seamless.

1. Baby Complex Analysis

1.1. Holomorphic Functions, Cauchy–Riemann Equations, and Local Conformality

Goal and Big Picture (What you should get from this subsection)

Goal. We explain why complex differentiability is much more rigid than real differentiability. Concretely, we will prove:

1. A C^1 function $f = u + iv$ is holomorphic $\iff$ it satisfies the Cauchy–Riemann (CR) equations.
2. Holomorphicity implies a *local rotation + scaling* geometry: the differential has the form $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$.
3. Complex line integrals $\int_\gamma f(z)\,dz$ are just coupled real line integrals; the first unit circle computations already foreshadow residues.

1. Domains and Complex Differentiability

Why we care. In real analysis, differentiability at a point controls only the best linear approximation along real directions. In complex analysis, differentiability along *one* complex direction forces compatibility along *all* real directions simultaneously. This rigidity will manifest as the CR equations.

Definition 1.1 (Domain)**.** An open set $U \subset \mathbb{C}$ is a *domain* if it is connected.

Definition 1.2 (Complex differentiability and holomorphicity)**.** Let $U \subset \mathbb{C}$ be a domain and $f : U \to \mathbb{C}$. We say that f is *complex differentiable* at $z_0 \in U$ if the limit

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists. If f is complex differentiable at every point of U, we say f is *holomorphic* on U, and write $f \in \mathrm{Hol}(U)$.

First sanity check (two-direction test). A useful beginner test: try the difference quotient limit along two paths (e.g. $z = z_0 + h$ and $z = z_0 + ih$). If the two limits disagree, $f'(z_0)$ cannot exist.

Exercise 1.3 (Two-path test)**.** Let $f(z) = \overline{z}$ and fix $z_0 \in \mathbb{C}$. Compute the difference quotient along $z = z_0 + h$ and $z = z_0 + ih$ ($h \in \mathbb{R}$, $h \to 0$) and show the two limits are different. Conclude that f is nowhere complex differentiable.

2. Real Jacobian viewpoint: what should Df look like?

Key idea. Write $z = x+iy$ and $f(z) = u(x,y)+iv(x,y)$. Then f is a map $F : \mathbb{R}^2 \to \mathbb{R}^2$, $F(x,y) = (u(x,y), v(x,y))$. If f is complex differentiable at z_0, then the best real linear approximation $DF(z_0)$ must be equal to *multiplication by a complex number.*

$$DF(x_0,y_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}_{(x_0,y_0)}.$$

Lemma 1.4 (Real-linear maps that are complex multiplications)**.** *A real-linear map $L : \mathbb{R}^2 \to \mathbb{R}^2$ is multiplication by a complex number $a+ib$ (i.e. $L(h) = (a+ib)\,h$ under $\mathbb{R}^2 \simeq \mathbb{C}$) if and only if its matrix has the form*

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Proof. If L is multiplication by $a+ib$, then

$$(a+ib)(h_1+ih_2) = (ah_1 - bh_2) + i(bh_1 + ah_2),$$

so in coordinates (h_1,h_2) the matrix is $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$. Conversely, a map with this matrix satisfies exactly the above identity, hence is multiplication by $a+ib$. □

Exercise 1.5 (Recognizing complex multiplication)**.** Show that a real matrix $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$ represents multiplication by some complex number if and only if $\alpha = \delta$ and $\gamma = -\beta$. Interpret this as: "columns are obtained by a 90° rotation and same scaling."

3. Wirtinger derivatives and the CR equations

Why Wirtinger derivatives. They package the CR equations into the single condition $\partial f/\partial \overline{z} = 0$, which becomes the gateway to $\overline{\partial}$-methods, differential forms, and sheaf cohomology later.

Definition 1.6 (Wirtinger derivatives)**.** Write $z = x+iy$ and $\overline{z} = x-iy$. Define

$$\frac{\partial}{\partial z} := \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big), \qquad \frac{\partial}{\partial \overline{z}} := \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big).$$

For $f = u+iv$ set

$$\frac{\partial f}{\partial z} = \frac{1}{2}(f_x - if_y), \qquad \frac{\partial f}{\partial \overline{z}} = \frac{1}{2}(f_x + if_y),$$

where $f_x = u_x + iv_x$ and $f_y = u_y + iv_y$.

Lemma 1.7 (CR equations in Wirtinger form)**.** *Let $f = u+iv \in C^1(U)$. Then*

$$\frac{\partial f}{\partial \overline{z}} = 0 \quad \Longleftrightarrow \quad u_x = v_y \ \text{ and } \ u_y = -v_x.$$

Proof. Compute:

$$\frac{\partial f}{\partial \overline{z}} = \frac{1}{2}(f_x + if_y) = \frac{1}{2}\big((u_x+iv_x) + i(u_y+iv_y)\big) = \frac{1}{2}\big((u_x - v_y) + i(v_x+u_y)\big).$$

This equals 0 iff both real and imaginary parts vanish: $u_x = v_y$ and $v_x = -u_y$. □

Concrete calculation (two examples).

Example 1.8 (A holomorphic and a non-holomorphic map)**.**

1. $f(z) = z^2$. Write $z = x + iy$:

$$f = (x+iy)^2 = (x^2 - y^2) + i(2xy), \quad u = x^2 - y^2,\ v = 2xy.$$

   Then $u_x = 2x,\ u_y = -2y,\ v_x = 2y,\ v_y = 2x$, so CR holds everywhere. Hence f is entire and $f'(z) = 2z$.

2. $f(z) = \overline{z} = x - iy$. Here $u = x,\ v = -y$ so $u_x = 1$ while $v_y = -1$, violating CR everywhere. Thus f is nowhere complex differentiable. Geometrically it is a reflection (orientation reversing), which cannot be holomorphic.

Exercise 1.9 (Practice with Wirtinger)**.**

1. Compute $\partial z/\partial z$, $\partial z/\partial \overline{z}$, $\partial \overline{z}/\partial z$, $\partial \overline{z}/\partial \overline{z}$.
2. Show that for $f(z) = z^n$ ($n \in \mathbb{Z}_{\geq 0}$), we have $\partial f/\partial \overline{z} = 0$.
3. Show that for $f(z) = |z|^2 = z\overline{z}$, we have $\partial f/\partial \overline{z} = z \neq 0$. Conclude $|z|^2$ is not holomorphic.

4. CR equations ⇒ complex differentiability (detailed proof)

Idea of the proof. Use real differentiability: a C^1 map admits a first-order approximation by its Jacobian. Then CR forces the Jacobian to be a complex multiplication matrix, hence the complex difference quotient converges.

Theorem 1.10 (CR equations imply complex differentiability)**.** *Let $U \subset \mathbb{C}$ be open and let $f = u + iv \in C^1(U)$. If the CR equations hold in a neighborhood of $z_0 = x_0 + iy_0$, then f is complex differentiable at z_0 and*

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

Proof. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$, $F(x, y) = (u(x, y), v(x, y))$. Since $f \in C^1$, F is Fréchet differentiable at (x_0, y_0), so for $h = (h_1, h_2) \to 0$,

$$F(z_0 + h) - F(z_0) = DF(z_0)\, h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \to 0.$$

Here

$$DF(z_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}_{(x_0, y_0)}.$$

By CR, $u_x = v_y$ and $u_y = -v_x$, hence

$$DF(z_0) = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}_{(x_0, y_0)} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad a := u_x(x_0, y_0),\ b := v_x(x_0, y_0).$$

By the previous lemma, this matrix is multiplication by $a + ib$ under $\mathbb{R}^2 \simeq \mathbb{C}$. Identifying $h = (h_1, h_2)$ with $h_1 + ih_2 \in \mathbb{C}$, we obtain

$$f(z_0 + h) - f(z_0) = (a + ib)\, h + \varepsilon(h), \qquad \text{where } \frac{|\varepsilon(h)|}{|h|} \to 0.$$

Divide by $h \neq 0$:

$$\frac{f(z_0+h)-f(z_0)}{h} = (a+ib) + \frac{\varepsilon(h)}{h} \;\longrightarrow\; a+ib.$$

Therefore $f'(z_0) = a+ib$. □

Exercise 1.11 (Direct derivative computation)**.** Let $f(z) = z^2$ and $z_0 \neq 0$. Compute $f'(z_0)$ by the difference quotient definition. Then compute $u_x(x_0,y_0) + iv_x(x_0,y_0)$ and verify they match.

5. Equivalent characterizations of holomorphicity

Theorem 1.12 (Equivalent characterizations)**.** *Let $U \subset \mathbb{C}$ be a domain and $f : U \to \mathbb{C}$. The following are equivalent:*

1. *f is holomorphic on U.*
2. *$f \in C^1(U)$ and $\dfrac{\partial f}{\partial \overline{z}} = 0$ on U.*
3. *Writing $f = u + iv$ with $u, v \in C^1(U)$, the CR equations hold on U:*

$$u_x = v_y, \qquad u_y = -v_x.$$

Proof. (2) ⇔ (3) is Lemma 1.7. (3) ⇒ (1) follows from Theorem 1.10 applied pointwise. (1) ⇒ (3) is standard: if f is complex differentiable, compare the difference quotient along the real and imaginary directions to obtain CR. (You may treat this as an exercise below.) □

Exercise 1.13 (Holomorphic ⇒ CR)**.** Assume f is complex differentiable at $z_0 = x_0 + iy_0$. Show that the limits along $z = z_0 + h$ and $z = z_0 + ih$ force $u_x(z_0) = v_y(z_0)$ and $u_y(z_0) = -v_x(z_0)$.

6. Local conformality: infinitesimal rotation and scaling

Geometric meaning. At a point where f is holomorphic and $f'(z_0) \neq 0$, the Jacobian is complex multiplication:

$$DF(z_0) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a+ib = f'(z_0).$$

Thus $DF(z_0)$ maps every infinitesimal vector by the same scaling factor $|f'(z_0)|$ and the same rotation angle $\arg f'(z_0)$. Hence angles are preserved: f is locally conformal.

Exercise 1.14 (Angle preservation from matrices)**.** Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ have matrix $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$ with $a^2 + b^2 \neq 0$. Show that for any nonzero vectors v, w, the angle between Lv and Lw equals the angle between v and w. (Hint: write $L = \lambda R$ where $\lambda = \sqrt{a^2+b^2}$ and R is orthogonal.)

7. Complex line integrals and first experiments

Why this definition. Once holomorphic functions are viewed as geometric maps, integration along curves becomes natural. This is also the entry point to Cauchy's theorem and residues later.

Definition 1.15 (Piecewise C^1 curve and complex line integral)**.** A continuous curve $\gamma : [a, b] \to U$ is *piecewise* C^1 if there is a finite partition of $[a, b]$ such that γ is C^1 on each subinterval. For such γ and $f : U \to \mathbb{C}$, define

$$\int_\gamma f(z)\, dz := \int_a^b f(\gamma(t))\, \gamma'(t)\, dt.$$

Remark 1.16 (Real and imaginary decomposition (as two coupled real integrals))**.** Write $\gamma(t) = x(t) + iy(t)$ and $f = u + iv$. Then $\gamma'(t) = x'(t) + iy'(t)$ and

$$f(\gamma(t))\gamma'(t) = (u + iv)(x' + iy') = (ux' - vy') + i(vx' + uy').$$

Hence

$$\int_\gamma f(z)\, dz = \int_a^b (ux' - vy')\, dt + i \int_a^b (vx' + uy')\, dt.$$

This shows complex line integrals are not mysterious: they are precisely two real line integrals.

Example 1.17 (Unit circle integral: the first residue glimpse)**.** Let $\gamma(t) = e^{it}$ for $0 \le t \le 2\pi$ and $f(z) = z^n$ with $n \in \mathbb{Z}$. Then $dz = ie^{it}\, dt$, so

$$\int_\gamma z^n\, dz = \int_0^{2\pi} (e^{it})^n\, ie^{it}\, dt = i \int_0^{2\pi} e^{i(n+1)t}\, dt = \begin{cases} 0, & n \neq -1, \\ 2\pi i, & n = -1. \end{cases}$$

Exercise 1.18 (Path dependence vs path independence)**.**

1. Compute $\displaystyle\int_\gamma z\, dz$ for $\gamma(t) = t(1 + i)$, $0 \le t \le 1$.
2. Compute the same integral along the broken path $0 \to 1 \to 1 + i$.
3. Explain why both answers agree (hint: find an antiderivative of z).

Exercise 1.19 (More circle integrals)**.** Let $\gamma(t) = Re^{it}$, $0 \le t \le 2\pi$ with $R > 0$. Compute $\int_\gamma \frac{1}{z}\, dz$ and $\int_\gamma z^n\, dz$ for $n \in \mathbb{Z}$.

1.2. Cauchy Integral Theorem and the Holographic Property

Goal and Philosophy: Why boundary data controls the interior

Goal. We prove the two central results:

1. **Cauchy Integral Theorem (CIT):** if f is holomorphic on and inside a simple closed curve γ, then $\int_\gamma f(z)\, dz = 0$.

2. **Cauchy Integral Formula (CIF):** for z_0 inside γ,

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0}\, dz.$$

Holographic principle. For holomorphic functions, boundary values determine interior values. This is the prototype of the local–global philosophy that later reappears via sheaves and cohomology.

0. Geometry/topology input: simple closed curves and orientation

Why this definition. We want a curve γ that bounds a region D so that we can apply Green's theorem (a planar form of Stokes). Positivity means "counterclockwise around the interior."

Definition 1.20 (Simple closed curve)**.** A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ is *simple closed* if $\gamma(a) = \gamma(b)$ and γ is injective on (a, b). It is *positively oriented* if it winds counterclockwise around its interior.

Exercise 1.21 (Orientation sanity check)**.** Let $\gamma(t) = e^{it}$, $0 \le t \le 2\pi$. Show it is positively oriented. Now consider $\widetilde{\gamma}(t) = e^{-it}$. Show $\int_{\widetilde{\gamma}} f(z)\, dz = -\int_\gamma f(z)\, dz$.

1. Cauchy Integral Theorem on planar domains (CIT)

What CIT says. If f is holomorphic on and inside the boundary, then the integral around the boundary is 0. This is the "first global rigidity" of holomorphicity.

Theorem 1.22 (Cauchy Integral Theorem (CIT))**.** *Let γ be a simple, positively oriented, piecewise C^1 closed curve in $\mathbb{C}$, and let D be the bounded domain enclosed by γ. If f is holomorphic on an open neighborhood of $\overline{D}$, then*

$$\int_\gamma f(z)\, dz = 0.$$

Proof strategy. Rewrite the complex integral as two real line integrals. Apply Green's theorem to convert them into area integrals. Holomorphicity (CR equations) kills the area integrands.

Proof. Write $f = u + iv$ and $dz = dx + i\, dy$. Then

$$\int_\gamma f(z)\, dz = \int_\gamma (u + iv)(dx + i\, dy) = \int_\gamma (u\, dx - v\, dy) + i \int_\gamma (v\, dx + u\, dy).$$

Green's theorem states: for $P, Q \in C^1$,

$$\int_{\partial D} P\, dx + Q\, dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\, dA.$$

Apply it to $P = u$, $Q = -v$ to get

$$\int_\gamma (u\, dx - v\, dy) = \iint_D \big((-v)_x - u_y\big)\, dA = \iint_D (-v_x - u_y)\, dA,$$

and to $P = v$, $Q = u$ to get

$$\int_\gamma (v\,dx + u\,dy) = \iint_D (u_x - v_y)\,dA.$$

Since f is holomorphic, the Cauchy–Riemann equations hold on D:

$$u_x = v_y, \qquad u_y = -v_x.$$

Hence $-v_x - u_y = 0$ and $u_x - v_y = 0$ pointwise, so both area integrals vanish. Therefore $\int_\gamma f(z)\,dz = 0$. □

Remark 1.23 (What really happened: Stokes + $\overline{\partial} f = 0$)**.** CIT is a 2D Stokes theorem in disguise: the contour integral is the integral of a 1-form over ∂D, which becomes an integral of d(that 1-form) over D. Holomorphicity forces that exterior derivative to vanish. Later we will rewrite this cleanly as a $\overline{\partial}$-statement on forms.

Concrete computation (the easiest check). If f has a global primitive on D (i.e. $f = g'$), then by the fundamental theorem of calculus along curves, $\int_\gamma f(z)\,dz = \int_\gamma g'(z)\,dz = 0$ for closed γ. CIT says this remains true far beyond "having a primitive".

Exercise 1.24 (Warm-up: a primitive case)**.** Let $f(z) = z^3$. Compute $\int_{|z|=2} z^3\,dz$ directly using the parameterization $z = 2e^{it}$. Explain why the answer must be 0 using primitives.

2. The basic kernel and why $1/z$ matters

Why isolate $1/z$. CIT says holomorphic integrands give 0. The first nontrivial phenomenon is when the integrand fails to be holomorphic at one point inside the curve. The kernel $1/(z - z_0)$ is the simplest such singularity.

Definition 1.25 (Disks and circles)**.** For $z_0 \in \mathbb{C}$ and $r > 0$, define

$$B(z_0, r) = \{z \in \mathbb{C} : \ |z - z_0| < r\}, \qquad \partial B(z_0, r) = \{z \in \mathbb{C} : \ |z - z_0| = r\}.$$

Lemma 1.26 (The fundamental circle integral)**.** *For every $r > 0$,*

$$\int_{\partial B(0,r)} \frac{1}{z}\,dz = 2\pi i.$$

Proof. Parameterize $z = re^{it}$, $0 \le t \le 2\pi$. Then $dz = ire^{it}\,dt$ and

$$\int_{\partial B(0,r)} \frac{1}{z}\,dz = \int_0^{2\pi} \frac{1}{re^{it}}(ire^{it})\,dt = i\int_0^{2\pi} dt = 2\pi i.$$

□

Remark 1.27 (Topology in one line: winding number preview)**.** The value $2\pi i$ is not an accident: it detects that the curve winds once around the origin. Later we will express $\int_\gamma \frac{1}{z}\,dz = 2\pi i \cdot \mathrm{wind}(\gamma, 0)$.

Exercise 1.28 (Translate the kernel)**.** Let $z_0 \in \mathbb{C}$ and $\gamma = \partial B(z_0, r)$ positively oriented. Show $\displaystyle\int_\gamma \frac{1}{z - z_0}\,dz = 2\pi i$ by substituting $w = z - z_0$.

3. Cauchy Integral Formula (CIF): boundary determines interior

What CIF does. It gives an explicit reconstruction of $f(z_0)$ from boundary values of f. This is the first precise form of "holography" in complex analysis.

Theorem 1.29 (Cauchy Integral Formula (CIF))**.** *Let $U \subset \mathbb{C}$ be open and let f be holomorphic on U. Let γ be a simple, positively oriented closed curve in U, and let z_0 lie inside γ. Assume f is holomorphic on a neighborhood of γ and its interior. Then*

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0}\, dz.$$

Key idea of the proof. The function $\dfrac{f(z) - f(z_0)}{z - z_0}$ *looks* singular at z_0 but is actually holomorphic there (removable singularity). Then CIT applies.

Proof. Fix z_0 and define

$$g(z) = \frac{f(z) - f(z_0)}{z - z_0} \qquad (z \neq z_0).$$

Since f is holomorphic, the limit $\lim_{z \to z_0} g(z) = f'(z_0)$ exists, so g extends holomorphically across z_0 (the singularity is removable). Hence g is holomorphic on the region bounded by γ, and by CIT,

$$\int_\gamma g(z)\, dz = 0.$$

Expanding,

$$0 = \int_\gamma \frac{f(z) - f(z_0)}{z - z_0}\, dz = \int_\gamma \frac{f(z)}{z - z_0}\, dz - f(z_0) \int_\gamma \frac{1}{z - z_0}\, dz.$$

By the translation $w = z - z_0$ (or Exercise above), we have $\displaystyle\int_\gamma \frac{1}{z - z_0}\, dz = 2\pi i$. Rearranging yields

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0}\, dz.$$

□

Remark 1.30 (Holographic viewpoint)**.** CIF says: the interior value $f(z_0)$ is a weighted average of boundary data with kernel $(z - z_0)^{-1}$. This is a concrete local–global mechanism: *sections on a domain are determined by boundary/overlap data*, foreshadowing sheaves and cohomology.

A fully worked example (CIF as a calculator).

Example 1.31 (Compute a contour integral using CIF)**.** Let $\gamma = \partial B(0, 2)$ oriented counterclockwise. Compute

$$\int_\gamma \frac{\cos z}{z}\, dz.$$

Solution. Here $f(z) = \cos z$ is entire. CIF with $z_0 = 0$ gives

$$\cos(0) = \frac{1}{2\pi i} \int_\gamma \frac{\cos z}{z - 0}\, dz \quad \Longrightarrow \quad \int_\gamma \frac{\cos z}{z}\, dz = 2\pi i \cos(0) = 2\pi i.$$

Exercise 1.32 (CIF as an integral-evaluation machine)**.** Let $\gamma = \partial B(1,3)$ oriented counterclockwise. Compute $\displaystyle\int_\gamma \frac{e^z}{z-1}\,dz$ and $\displaystyle\int_\gamma \frac{e^z}{(z-1)^2}\,dz$. (Use Theorem 1.29 and the derivative formula in the next subsection.)

4. Consequences: smoothness, derivative formulas, and power series

Why this matters. In real analysis, differentiability does not automatically imply analyticity. In complex analysis, holomorphicity forces infinite differentiability and power series expansions.

Corollary 1.33 (Cauchy estimates and C^∞-regularity)**.** If f is holomorphic on U, then $f \in C^\infty(U)$. Moreover, for any integer $n \geq 0$ and any simple closed curve γ enclosing z_0 inside U,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{n+1}}\,dz.$$

In particular, f admits a convergent power series expansion around every $z_0 \in U$.

Proof sketch (what to justify). Differentiate under the integral sign. The only analytic input is that f is bounded on γ and the kernels $(z-z_0)^{-(n+1)}$ behave uniformly on γ away from z_0.

Proof. Fix z_0 and choose γ enclosing z_0. From CIF,

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-z_0}\,dz.$$

Differentiate both sides with respect to z_0 (justified since the integrand is smooth in z_0 and uniformly bounded on γ away from z_0), obtaining

$$f'(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^2}\,dz.$$

Iterating yields the general formula with $n!$. The existence of the power series follows by expanding $\frac{1}{z-z_0}$ as a geometric series on a smaller disk and interchanging sum and integral (uniform convergence). □

Exercise 1.34 (Derivatives of e^z from the integral formula)**.** Use Corollary 1.33 with $f(z) = e^z$ to compute $f^{(n)}(z_0)$. Conclude $e^{(n)}(z_0) = e^{z_0}$ for all n.

5. Morera's theorem (one direction): integrals detect holomorphicity

Meaning. CIT says holomorphic $\Rightarrow$ all closed integrals vanish. Morera says (under continuity) the converse holds: vanishing integrals force holomorphicity. This is a bridge between integral conditions and differential conditions ($\overline{\partial} f = 0$).

Corollary 1.35 (Morera's theorem, one direction)**.** If f is continuous on U and $\int_\gamma f(z)\,dz = 0$ for every simple closed curve $\gamma \subset U$, then f is holomorphic on U.

Proof idea (roadmap). One shows that the vanishing of integrals implies f has a local primitive on disks, hence satisfies a Cauchy-type formula, hence is holomorphic. (A full proof is a good "mini-project" exercise once we develop more tools.)

Proof. Sketch: fix a disk $B(z_0, r) \subset U$. The hypothesis implies path-independence of $\int f\,dz$ on the disk, so define $F(z) = \int_{z_0}^{z} f(w)\,dw$ (well-defined). Then F is complex differentiable with $F'(z) = f(z)$, so f is holomorphic on the disk. Cover U by disks to conclude. □

Exercise 1.36 (Mini-project: fill in Morera carefully)**.** Provide a complete proof of Corollary 1.35 by:

1. proving path-independence on a disk from the vanishing of closed integrals,
2. defining a primitive F and proving $F'(z) = f(z)$ using the definition of complex derivative,
3. concluding holomorphicity.

6. Examples and first applications (with explicit computations)

Example 1.37 (Integrals of monomials around a circle)**.** Let $\gamma = \partial B(z_0, r)$ oriented counterclockwise. For $n \in \mathbb{Z}$,

$$\int_\gamma (z - z_0)^n\,dz = \begin{cases} 0, & n \neq -1, \\ 2\pi i, & n = -1. \end{cases}$$

Reason. If $n \geq 0$, the integrand is holomorphic, so CIT gives 0. If $n \leq -2$, write $(z - z_0)^n = \frac{d}{dz}\big(\frac{(z-z_0)^{n+1}}{n+1}\big)$, which has a primitive on $\mathbb{C} \setminus \{z_0\}$, and the circle avoids z_0 only when interpreted appropriately; alternatively, parameterize $z = z_0 + re^{it}$ and integrate directly. The special case $n = -1$ is Lemma 1.26.

Example 1.38 (Branch cut and topology: $z^{3/2}$)**.** Let $f(z) = z^{3/2}$ defined on $\mathbb{C} \setminus (-\infty, 0]$ using the principal branch of $\arg z$. Then f is holomorphic on this slit plane but cannot extend to a single-valued holomorphic function on all of $\mathbb{C}$: looping once around 0 changes $\arg z$ by 2π, hence $z^{3/2}$ picks up a factor $e^{3\pi i} = -1$. CIF holds on simply connected subdomains avoiding 0, illustrating that holomorphicity and domain topology both matter for global integral representations.

Exercises (core / optional / challenge)

Exercise 1.39 (Core: Green's theorem computation)**.** Use Green's theorem to show that if $f = u + iv$ satisfies the Cauchy–Riemann equations on a bounded domain D, then

$$\iint_D \Big(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\Big)\,dA = 0.$$

(Then explain why this identity is exactly what appears in the proof of CIT.)

Exercise 1.40 (Core: CIF on a square)**.** Let γ be the positively oriented square centered at 0 with side length 2. Compute $\displaystyle\int_\gamma \frac{\cos z}{z}\,dz$ using CIF and interpret the answer.

Exercise 1.41 (Optional: independence of curve for CIF)**.** Assume f is holomorphic on a neighborhood of two simple closed curves γ_1, γ_2 that both enclose z_0 and are homotopic in $U \setminus \{z_0\}$. Show

$$\int_{\gamma_1} \frac{f(z)}{z - z_0}\, dz = \int_{\gamma_2} \frac{f(z)}{z - z_0}\, dz.$$

(Hint: apply CIT to the boundary of the annular region between them.)

Exercise 1.42 (Challenge: winding number preview)**.** Let γ be a simple closed positively oriented curve and assume 0 lies inside. Show $\int_\gamma \frac{1}{z}\, dz = 2\pi i$ by mapping the interior conformally to a disk (or by a homotopy argument once you establish that the integral is invariant under deformation avoiding 0).

1.3. Complex Analytic Functions and Power Series Representations

Goal and Big Picture: Why power series are "rigid" local models

Goal. We make precise the rigidity principle:

1. A power series expansion near p determines *everything* about the function near p.
2. Complex-analytic functions are automatically C^∞, with derivatives obtained by termwise differentiation.
3. The coefficients are unique and satisfy $a_k = \dfrac{f^{(k)}(p)}{k!}$.
4. **Holomorphic $\Rightarrow$ analytic:** Cauchy's integral formula forces a convergent Taylor series.

Algebraic-geometry preview. A convergent power series is a first model of a *germ* and a *local coordinate*: near p, a holomorphic function is "just" a convergent polynomial in $(z - p)$.

1. Analyticity and radius of convergence

Why define analyticity separately? In real analysis, differentiability (even smoothness) does *not* imply analytic expandability. In complex analysis, holomorphicity will imply analyticity; but it is instructive to isolate the notion of "power series representation" first.

Definition 1.43 (Complex analytic function)**.** Let $U \subset \mathbb{C}$ be open. A function $f : U \to \mathbb{C}$ is *complex analytic at* $p \in U$ if there exists $r > 0$ such that $B(p, r) \subset U$ and

$$f(z) = \sum_{n=0}^{\infty} a_n (z - p)^n, \qquad z \in B(p, r),$$

where the series converges uniformly on every compact subset of $B(p, r)$. The number r is called a (local) *radius of convergence at* p.

Remark 1.44 (What "uniform on compacts" buys you)**.** Uniform convergence on compact sets is exactly the hypothesis that allows:

- termwise differentiation and integration,
- uniform control of tails,
- continuity/smoothness of the sum.

Thus once f is analytic at p, its behavior on $B(p,r)$ is encoded in the coefficient sequence $\{a_n\}$.

Remark 1.45 (Radius of convergence and singularities (preview))**.** Later we will relate the maximal radius of convergence of the Taylor series at p to the distance from p to the nearest singularity of f (often on $\widehat{\mathbb{C}}$). This connects local expansions to global geometry on Riemann surfaces.

Exercise 1.46 (Warm-up: geometric series and radius)**.** Show that for $|z-p| < R$,

$$\frac{1}{1-\frac{z-p}{R}} = \sum_{n=0}^{\infty} \frac{(z-p)^n}{R^n}.$$

What is the radius of convergence at p? Where is the singularity?

2. Analytic $\Rightarrow C^\infty$: termwise differentiation with full estimates

Key mechanism. On a slightly smaller disk, the power series behaves like a uniformly convergent sum of smooth functions, and the derivative series converges uniformly by the Weierstrass M-test. This justifies differentiating term-by-term.

Lemma 1.47 (Analytic $\Rightarrow C^\infty$ and termwise differentiation)**.** *Let $f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n$ on $B(p,r)$. Then $f \in C^\infty(B(p,r))$, and for every integer $k \geq 0$,*

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)\, a_n (z-p)^{n-k}, \qquad z \in B(p,r),$$

with uniform convergence on every compact subset of $B(p,r)$.

Proof. Fix $z_0 \in B(p,r)$. Choose $0 < \rho < r - |z_0 - p|$, so that $\overline{B(z_0,\rho)} \subset B(p,r)$. Set

$$R := \sup_{z \in \overline{B(z_0,\rho)}} |z-p| \leq |z_0 - p| + \rho < r.$$

Since the power series converges for all $|z-p| < r$, the series of nonnegative terms

$$\sum_{n=0}^{\infty} |a_n| R^n$$

converges. Hence there exists N and a constant $M > 0$ such that for all $n \geq N$,

$$|a_n| R^n \leq M 2^{-n}.$$

(For instance, take M to dominate the tail after some N; any geometric domination suffices.)

Now fix $k \geq 1$ and consider the k-th derivative term:

$$n(n-1)\cdots(n-k+1)\, a_n (z-p)^{n-k}.$$

For $z \in \overline{B(z_0,\rho)}$ we have $|z-p| \le R$, so

$$\left| n(n-1)\cdots(n-k+1)\, a_n (z-p)^{n-k} \right| \le n^k |a_n| R^{n-k}.$$

For $n \ge N$ we use $|a_n|R^n \le M 2^{-n}$ to get

$$n^k |a_n| R^{n-k} = n^k (|a_n| R^n)\, R^{-k} \le (MR^{-k})\, n^k 2^{-n}.$$

The numerical series $\sum_{n \ge N} n^k 2^{-n}$ converges, so by the Weierstrass M-test, the derivative series converges uniformly on $\overline{B(z_0,\rho)}$.

Uniform convergence of the derivative series implies we may differentiate termwise on $\overline{B(z_0,\rho)}$, hence on a neighborhood of z_0. Since z_0 was arbitrary, $f \in C^\infty(B(p,r))$ and the displayed formula holds. □

Remark 1.48 (What the estimates are really doing)**.** We shrink from $B(p,r)$ to a smaller closed disk where $|z-p| \le R < r$. On that closed disk, the tails of the power series are controlled by a geometric sequence, which is precisely what the M-test needs to justify exchanging $\sum$ and $\frac{d}{dz}$.

Exercise 1.49 (Do it once by hand: differentiate a power series)**.** Let $f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n$ converge for $|z-p| < r$. Fix $0 < R < r$ and show directly (using M-test) that

$$\sum_{n=1}^{\infty} n a_n (z-p)^{n-1}$$

converges uniformly on $|z-p| \le R$.

3. Uniqueness of coefficients: Taylor formula and rigidity

Why uniqueness matters. If a function has a power series expansion, the coefficients are not "choices"—they are forced. This is the first hard rigidity statement: the infinite jet at one point determines the function near that point.

Theorem 1.50 (Uniqueness of coefficients and Taylor formula)**.** *Let $f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n$ on $B(p,r)$. Then for every $k \ge 0$,*

$$a_k = \frac{f^{(k)}(p)}{k!}.$$

In particular, the analytic expansion of f at p is unique.

Proof. By Lemma 1.47, we may differentiate termwise:

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)\, a_n (z-p)^{n-k}.$$

Evaluate at $z = p$. Then $(p-p)^{n-k} = 0$ for all $n > k$, and only the $n = k$ term survives:

$$f^{(k)}(p) = k!\, a_k.$$

Hence $a_k = f^{(k)}(p)/k!$. □

Remark 1.51 (Rigidity rephrased: infinite jets determine germs)**.** For analytic functions, the infinite jet

$$\big(f(p), f'(p), f''(p), \dots\big)$$

determines the entire function on some disk around p. In real analysis, there exist C^∞ functions whose Taylor series at a point does *not* recover the function. This is one of the deep reasons complex-analytic methods are powerful in geometry and number theory.

Exercise 1.52 (Two analytic functions with same jet)**.** Assume f and g are analytic near p and satisfy $f^{(k)}(p) = g^{(k)}(p)$ for all $k \geq 0$. Show that their power series coefficients at p agree, hence $f = g$ on a neighborhood of p.

4. Holomorphic functions are analytic: Cauchy expansion and explicit coefficients

Why this is the main theorem here. So far, analyticity was a separate definition. Now we prove holomorphicity *forces* analyticity using Cauchy's integral formula. This is the central rigidity theorem: complex differentiability implies power series.

Proposition 1.53 (Holomorphic $\Rightarrow$ analytic (Cauchy expansion))**.** *Let $U \subset \mathbb{C}$ be open and let $f : U \to \mathbb{C}$ be holomorphic. Then for each $p \in U$ there exists $r > 0$ such that for all $z \in B(p, r)$,*

$$f(z) = \sum_{k=0}^{\infty} a_k (z-p)^k,$$

where for any $0 < \rho < r$,

$$a_k = \frac{1}{2\pi i} \int_{|\zeta - p| = \rho} \frac{f(\zeta)}{(\zeta - p)^{k+1}}\, d\zeta.$$

Equivalently, $a_k = \dfrac{f^{(k)}(p)}{k!}$.

Proof strategy. CIF gives an integral representation involving the kernel $\frac{1}{\zeta - z}$. When $|z - p| < |\zeta - p| = \rho$, expand $\frac{1}{\zeta - z}$ as a geometric series in $(z - p)$, then interchange sum and integral (uniform convergence on $|z - p| \leq r_0 < \rho$).

Proof. Translate so that $p = 0$ (otherwise replace z by $z - p$ and ζ by $\zeta - p$). Choose $r > 0$ such that $\overline{B(0, r)} \subset U$. Fix $0 < \rho < r$. For $|z| < \rho$, Cauchy's integral formula gives

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta| = \rho} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

For $|\zeta| = \rho$ and $|z| < \rho$,

$$\frac{1}{\zeta - z} = \frac{1}{\zeta} \cdot \frac{1}{1 - \frac{z}{\zeta}} = \sum_{k=0}^{\infty} \frac{z^k}{\zeta^{k+1}},$$

and the series converges uniformly for $|z| \le r_0 < \rho$ since $\left|\frac{z}{\zeta}\right| \le \frac{r_0}{\rho} < 1$. Hence we may substitute into the integral and exchange sum and integral:

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{|\zeta|=\rho} f(\zeta) \sum_{k=0}^{\infty} \frac{z^k}{\zeta^{k+1}}\, d\zeta \\ &= \sum_{k=0}^{\infty} \left(\frac{1}{2\pi i} \int_{|\zeta|=\rho} \frac{f(\zeta)}{\zeta^{k+1}}\, d\zeta \right) z^k. \end{aligned}$$

Define

$$a_k := \frac{1}{2\pi i} \int_{|\zeta|=\rho} \frac{f(\zeta)}{\zeta^{k+1}}\, d\zeta.$$

This yields a power series expansion valid for $|z| < \rho$. Since ρ can be chosen with $0 < \rho < r$, we obtain analyticity on $B(0, r)$.

Finally, by Cauchy's differentiation formula (Corollary 1.33 in the previous subsection),

$$f^{(k)}(0) = \frac{k!}{2\pi i} \int_{|\zeta|=\rho} \frac{f(\zeta)}{\zeta^{k+1}}\, d\zeta \quad \Longrightarrow \quad a_k = \frac{f^{(k)}(0)}{k!}.$$

□

Remark 1.54 (Holomorphic $\iff$ analytic near a point)**.** Combining Proposition 1.53 with Theorem 1.50, we get

holomorphic near p $\iff$ analytic near p $\iff$ admits a convergent Taylor series at p.

This equivalence has no real-variable analogue and is a cornerstone of complex geometry.

Exercise 1.55 (Uniform convergence justification)**.** Fix $0 < r_0 < \rho$. Show that for $|\zeta| = \rho$,

$$\sum_{k=0}^{\infty} \left| \frac{z^k}{\zeta^{k+1}} \right| \le \frac{1}{\rho} \sum_{k=0}^{\infty} \left(\frac{r_0}{\rho} \right)^k = \frac{1}{\rho} \cdot \frac{1}{1 - r_0/\rho}$$

for all $|z| \le r_0$. Use this to justify exchanging $\sum$ and $\int$ in the proof above.

5. Examples: radius of convergence as a detector of singularities

Example A: a pole determines the radius.

Example 1.56 (Geometric series and a boundary pole)**.** For $|z| < 1$,

$$\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n.$$

The radius of convergence at 0 is 1, and the nearest singularity is the pole at $z = 1$ (distance 1 from 0).

Example B: entire functions have infinite radius.

Example 1.57 (Exponential function)**.** For all $z \in \mathbb{C}$,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

The radius of convergence is ∞ since $\limsup_{n\to\infty} |1/n!|^{1/n} = 0$.

Example C: branch points and local expansions.

Example 1.58 (A local branch of $\log z$ around $z = 1$)**.** On $|z - 1| < 1$, define a holomorphic branch of $\log z$ and expand

$$\log z = \log\big(1 + (z-1)\big) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(z-1)^n}{n}.$$

The radius of convergence is 1, and the obstruction comes from the branch point at $z = 0$.

Exercise 1.59 (Compute a radius from coefficients)**.** Use the root test to show that the power series $\sum_{n=0}^{\infty} a_n (z-p)^n$ has radius r satisfying

$$\limsup_{n\to\infty} |a_n|^{1/n} = \frac{1}{r}.$$

Exercises (compute first, then abstract)

Type A (Computational).

Exercise 1.60. Let f be holomorphic on $B(0,1)$ and assume $|f(z)| \le M$ on $|z| < 1$. Using Cauchy's estimate, prove

$$\left|\frac{f^{(k)}(0)}{k!}\right| \le \frac{M}{\rho^k} \qquad (0 < \rho < 1).$$

Deduce that if $|f(z)| \le M$ on $|z| < 1$, then the Taylor coefficients satisfy $|a_k| \le M$.

Exercise 1.61. Compute the power series expansion on $|z| < 1$:

$$\frac{1}{1-z^2} = \sum_{n=0}^{\infty} z^{2n}.$$

Find its radius of convergence and relate it to the poles at $z = \pm 1$.

Exercise 1.62. Find the Taylor series of $\frac{1}{(1-z)^2}$ on $|z| < 1$ by differentiating the geometric series. Then use Theorem 1.50 to identify $f^{(n)}(0)$.

Type B (Conceptual / geometric).

Exercise 1.63 (Removable singularity and radius)**.** Define f on $U = B(0,2)$ by $f(z) = \frac{\sin z}{z}$ for $z \neq 0$ and $f(0) = 1$.

1. Show that f is holomorphic on U (removable singularity).
2. Compute the Taylor series at 0 and determine its radius of convergence.
3. Explain why the radius is governed by the nearest *non-removable* singularity of $\sin z/z$ on $\widehat{\mathbb{C}}$.

Exercise 1.64 (Germs and jets)**.** Assume f and g are holomorphic near p and satisfy $f^{(k)}(p) = g^{(k)}(p)$ for all $k \ge 0$. Show $f = g$ near p. Interpret this as: *a holomorphic germ is determined by its infinite jet.*

Exercise 1.65 (Real-variable contrast)**.** Give an example of a smooth function $h : \mathbb{R} \to \mathbb{R}$ whose Taylor series at 0 is identically 0 but $h \not\equiv 0$. Explain why this cannot happen for holomorphic functions (use the uniqueness of coefficients and the identity principle).

1.4. Cauchy Estimates, Liouville's Theorem, and Applications

Roadmap: from boundary control to global rigidity

Cauchy's integral formula is not just a representation theorem; it is a *quantitative control mechanism*. In this subsection we develop three layers of rigidity:

- **Local quantitative rigidity (Cauchy estimates).** Boundary size on a circle controls every derivative at the center.
- **Global qualitative rigidity (Liouville).** A bounded entire function has no room to vary.
- **Growth ⇒ algebra (polynomial growth ⇒ polynomial).** If an entire function grows like $|z|^n$, then all derivatives beyond order n vanish.

These results are prototypes for later complex-geometry statements: growth conditions at infinity (or at divisors) cut down holomorphic objects to finite-dimensional spaces.

1. Cauchy estimates

Geometric intuition. Fix a disk $\overline{B(p,r)} \subset U$. Once $|f|$ is controlled on the boundary circle, holomorphicity forces the interior Taylor coefficients to be controlled, hence controls every derivative at p. This is a precise form of the holographic principle.

Proposition 1.66 (Cauchy estimates: boundary controls derivatives)**.** *Let $U \subset \mathbb{C}$ be open, $p \in U$, and assume $\overline{B(p,r)} \subset U$. Let $f \in \mathrm{Hol}(U)$ and define*

$$M_r := \max_{|\zeta - p| = r} |f(\zeta)|.$$

Then for every integer $k \geq 0$,

$$|f^{(k)}(p)| \leq \frac{k!}{r^k} M_r.$$

Proof. By Cauchy's integral formula for derivatives,

$$f^{(k)}(p) = \frac{k!}{2\pi i} \int_{|\zeta - p| = r} \frac{f(\zeta)}{(\zeta - p)^{k+1}} \, d\zeta.$$

Take absolute values:

$$|f^{(k)}(p)| \leq \frac{k!}{2\pi} \int_{|\zeta - p| = r} \frac{|f(\zeta)|}{|\zeta - p|^{k+1}} \, |d\zeta| \leq \frac{k!}{2\pi} \int_{|\zeta - p| = r} \frac{M_r}{r^{k+1}} \, |d\zeta|.$$

Since the circle has length $2\pi r$, we obtain

$$|f^{(k)}(p)| \leq \frac{k!}{2\pi} \cdot \frac{M_r}{r^{k+1}} \cdot 2\pi r = \frac{k!}{r^k} M_r.$$

□

Remark 1.67 (A concrete reading of the inequality)**.** For $k = 1$ the estimate says:

$$|f'(p)| \leq \frac{1}{r} \max_{|\zeta - p| = r} |f(\zeta)|.$$

So if f is small on a circle, it cannot change rapidly at the center. Higher k bounds control curvature and higher-order "wiggles" of f at p.

Sharpness and extremizers. The estimate is sharp (as an inequality with the constant $\frac{k!}{r^k}$).

Lemma 1.68 (Sharpness of Cauchy estimates)**.** *Fix $k \geq 0$ and $r > 0$. On $B(p, r)$, the holomorphic function $f(\zeta) = (\zeta - p)^k$ satisfies*

$$|f^{(k)}(p)| = \frac{k!}{r^k} M_r.$$

Proof. We have $f^{(k)}(p) = k!$ and $M_r = \max_{|\zeta - p| = r} |\zeta - p|^k = r^k$. □

Worked example

Example 1.69 (Using Cauchy estimates for e^z)**.** Let $f(z) = e^z$ and $p = 0$. On $|z| = r$ we have $|e^z| = e^{\Re z} \leq e^{|z|} = e^r$. Hence $M_r \leq e^r$ and

$$|f^{(k)}(0)| \leq \frac{k!}{r^k} e^r.$$

Since $f^{(k)}(0) = 1$, we get the inequality

$$1 \leq \frac{k!}{r^k} e^r \qquad \text{for all } r > 0.$$

Optimizing in r gives a crude-but-useful bound on $k!$ (choose $r = k$ to get $k! \geq k^k e^{-k}$).

Exercise 1.70 (Cauchy estimate on a disk)**.** Let f be holomorphic on $B(0, 2)$ and assume $|f(z)| \leq 5$ on $|z| = 2$. Show that $|f^{(3)}(0)| \leq \frac{3!}{2^3} \cdot 5$.

Exercise 1.71 (Coefficient bounds)**.** If $f(z) = \sum_{n \geq 0} a_n (z - p)^n$ is holomorphic on $B(p, r)$, show that

$$|a_n| = \left| \frac{f^{(n)}(p)}{n!} \right| \leq \frac{M_r}{r^n}.$$

Interpret this as: boundary bounds control Taylor coefficients.

2. Liouville's theorem and immediate consequences

What changes from local to global? On $\mathbb{C}$, you may take circles of arbitrarily large radius. Cauchy estimates then force derivatives to vanish.

Theorem 1.72 (Liouville: bounded entire ⇒ constant)**.** *If $f : \mathbb{C} \to \mathbb{C}$ is entire and bounded, then f is constant.*

Proof. Assume $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Fix $p \in \mathbb{C}$ and apply Proposition 1.66 to the disk $\overline{B(p, R)}$ for any $R > 0$:

$$|f'(p)| \leq \frac{1!}{R} \max_{|\zeta - p| = R} |f(\zeta)| \leq \frac{M}{R}.$$

Letting $R \to \infty$ yields $f'(p) = 0$. Since p was arbitrary, $f' \equiv 0$ and f is constant. □

Remark 1.73 (Compact Riemann surfaces: a preview)**.** On a compact Riemann surface X, every holomorphic function is automatically bounded, so Liouville implies: *every global holomorphic function on X is constant.* This is the first glimpse of how compactness forces rigidity.

A standard corollary: bounded entire ⇒ maximum principle global form.

Corollary 1.74 (A nonconstant entire function is unbounded)**.** If f is entire and nonconstant, then $f(\mathbb{C})$ is unbounded.

Proof. Contrapositive of Theorem 1.72. □

Exercise 1.75 (Trigonometric functions are unbounded on $\mathbb{C}$)**.** Show that $\sin z$ and $\cos z$ are unbounded on $\mathbb{C}$. (Hint: evaluate at $z = iy$ and let $y \to \infty$.)

3. Polynomial growth ⇒ polynomial

Key idea. Polynomial growth beats the decay R^{-k} in Cauchy estimates when k is large. So sufficiently high derivatives must vanish.

Lemma 1.76 (Polynomial growth kills high derivatives (centered version))**.** *Let f be entire. Assume there exist constants $C > 0$ and integer $n \geq 0$ such that*

$$|f(z)| \leq C(1 + |z|)^n \quad \forall z \in \mathbb{C}.$$

Then for every integer $k > n$,

$$f^{(k)}(0) = 0.$$

Proof. Fix $k > n$. On $|z| = R$ we have $|f(z)| \leq C(1 + R)^n$. By Cauchy estimates (Proposition 1.66 with $p = 0$),

$$|f^{(k)}(0)| \leq \frac{k!}{R^k} \max_{|z|=R} |f(z)| \leq \frac{k!}{R^k} C(1 + R)^n.$$

Let $R \to \infty$. Since $(1 + R)^n / R^k \to 0$ for $k > n$, the right-hand side tends to 0, so $f^{(k)}(0) = 0$. □

Upgrade: vanishing at every center. The lemma at 0 implies the same at every point by translating.

Lemma 1.77 (Polynomial growth kills high derivatives (any center))**.** *Under the assumptions of Lemma 1.76, for every $p \in \mathbb{C}$ and every $k > n$,*

$$f^{(k)}(p) = 0.$$

Proof. Fix p and define $g(z) = f(z + p)$. Then g is entire and satisfies the same type of growth bound:

$$|g(z)| = |f(z + p)| \leq C(1 + |z + p|)^n \leq C'(1 + |z|)^n$$

for a constant C' (use $1 + |z + p| \leq 1 + |z| + |p| \leq (1 + |p|)(1 + |z|)$). Apply Lemma 1.76 to g to get $g^{(k)}(0) = 0$ for $k > n$. But $g^{(k)}(0) = f^{(k)}(p)$, so $f^{(k)}(p) = 0$. □

Proposition 1.78 (Entire function of polynomial growth is a polynomial)**.** *If f is entire and $|f(z)| \leq C(1 + |z|)^n$, then f is a polynomial of degree at most n.*

Proof. By Lemma 1.76, $f^{(k)}(0) = 0$ for all $k > n$. Hence the Taylor series at 0 truncates:

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} z^k,$$

so f is a polynomial of degree $\leq n$. □

Remark 1.79 (Bridge to Riemann–Roch language)**.** On a compact Riemann surface X, controlling pole orders (growth near finitely many points) produces finite-dimensional spaces of meromorphic functions. Proposition 1.78 is the simplest flat-model analogue of that principle.

Exercise 1.80 (Linear growth implies affine)**.** Let f be entire and satisfy $|f(z)| \leq A + B|z|$ for all $z \in \mathbb{C}$. Show that f is a polynomial of degree at most 1, i.e. $f(z) = \alpha z + \beta$.

4. Fundamental Theorem of Algebra via Liouville

Theorem 1.81 (Fundamental Theorem of Algebra)**.** *Every nonconstant complex polynomial $P(z)$ has at least one root in $\mathbb{C}$.*

Proof. Assume P is nonconstant and has no zeros. Then $f(z) := 1/P(z)$ is entire. Write $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ with $a_n \neq 0$. Factor:

$$P(z) = a_n z^n \Big(1 + \varepsilon(z)\Big), \qquad \varepsilon(z) := \frac{a_{n-1}}{a_n} \frac{1}{z} + \cdots + \frac{a_0}{a_n} \frac{1}{z^n}.$$

Then $\varepsilon(z) \to 0$ as $|z| \to \infty$. Choose $R > 0$ such that $|\varepsilon(z)| \leq \frac{1}{2}$ when $|z| \geq R$. For $|z| \geq R$,

$$|P(z)| = |a_n||z|^n |1 + \varepsilon(z)| \geq |a_n||z|^n \Big(1 - \tfrac{1}{2}\Big) = \frac{1}{2}|a_n||z|^n,$$

so

$$|f(z)| = \frac{1}{|P(z)|} \leq \frac{2}{|a_n|} |z|^{-n} \leq \frac{2}{|a_n|} R^{-n} \quad \text{for } |z| \geq R.$$

Thus f is bounded on $\mathbb{C} \setminus B(0, R)$, and since f is continuous on the compact disk $\overline{B(0, R)}$, it is bounded there as well. Hence f is bounded on all of $\mathbb{C}$. By Liouville (Theorem 1.72), f is constant, so P is constant, contradiction. Therefore P must have a root. □

Remark 1.82 (What to remember)**.** FTA is a global rigidity theorem disguised as algebra: "a polynomial cannot avoid 0 everywhere without forcing $1/P$ to be a bounded entire function."

Exercise 1.83 (Concrete estimate for a specific polynomial)**.** Let $P(z) = z^n + 1$. Show that for $|z| \geq 2$,

$$\left|\frac{1}{P(z)}\right| \leq \frac{2}{|z|^n}.$$

Conclude $1/P$ is bounded outside $B(0, 2)$ and complete the Liouville argument to deduce that P has a root.

5. Uniform limits, holomorphicity, and convergence of derivatives

Why this matters. Many existence/compactness arguments in complex analysis (e.g. normal families, Montel) pass to limits of holomorphic functions. Cauchy's formula makes holomorphicity stable under uniform convergence on compact subsets.

Proposition 1.84 (Uniform limit on compacts of holomorphic functions is holomorphic)**.** *Let $U \subset \mathbb{C}$ be open and $f_n \in \mathrm{Hol}(U)$. Assume $f_n \to f$ uniformly on every compact subset of U. Then $f \in \mathrm{Hol}(U)$.*

Proof. Fix $p \in U$ and choose $r > 0$ with $\overline{B(p,r)} \subset U$. For any z with $|z-p| < r$, Cauchy's integral formula gives

$$f_n(z) = \frac{1}{2\pi i} \int_{|\zeta - p| = r} \frac{f_n(\zeta)}{\zeta - z}\, d\zeta.$$

Since the circle $|\zeta - p| = r$ is compact and $f_n \to f$ uniformly on it, we may pass to the limit inside the integral:

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta - p| = r} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

The right-hand side is a holomorphic function of z on $B(p,r)$ (as an integral of a holomorphic kernel in z), so f is holomorphic on $B(p,r)$. Since p was arbitrary, $f \in \mathrm{Hol}(U)$. □

Theorem 1.85 (Convergence of derivatives under uniform convergence on compacts)**.** *Under the assumptions of Proposition 1.84, for every integer $k \geq 0$,*

$$f_n^{(k)} \to f^{(k)} \quad \textit{uniformly on every compact subset of } U.$$

Proof. Let $K \subset U$ be compact. Choose finitely many disks covering K; it suffices to prove uniform convergence on each. So fix one disk $\overline{B(p,r)} \subset U$ with $K \subset B(p, r_0)$ for some $r_0 < r$. For $z \in K$, Cauchy's formula for derivatives gives

$$f_n^{(k)}(z) = \frac{k!}{2\pi i} \int_{|\zeta - p| = r} \frac{f_n(\zeta)}{(\zeta - z)^{k+1}}\, d\zeta.$$

On the circle $|\zeta - p| = r$ and for $z \in K \subset B(p, r_0)$ we have $|\zeta - z| \geq r - r_0 =: \delta > 0$, so

$$\left| \frac{1}{(\zeta - z)^{k+1}} \right| \leq \delta^{-(k+1)}$$

uniformly in ζ and $z \in K$. Since $f_n \to f$ uniformly on the circle,

$$\sup_{|\zeta - p| = r} |f_n(\zeta) - f(\zeta)| \to 0.$$

Therefore,

$$\begin{aligned} \sup_{z \in K} |f_n^{(k)}(z) - f^{(k)}(z)| &\leq \frac{k!}{2\pi} \int_{|\zeta - p| = r} \sup_{z \in K} \frac{|f_n(\zeta) - f(\zeta)|}{|\zeta - z|^{k+1}}\, |d\zeta| \\ &\leq \frac{k!}{2\pi} \cdot (2\pi r) \cdot \delta^{-(k+1)} \cdot \sup_{|\zeta - p| = r} |f_n(\zeta) - f(\zeta)|. \end{aligned}$$

The right-hand side tends to 0, proving uniform convergence on K. □

Remark 1.86 (Normal families: preview)**.** Montel's theorem will upgrade "uniform boundedness on compact sets" into subsequential compactness. Proposition 1.84 and Theorem 1.85 are the analytic engine behind such compactness arguments.

Exercise 1.87 (Local boundedness from uniform convergence)**.** Let $f_n \to f$ uniformly on compact subsets of U. Prove: for each compact $K \subset U$ there exists M_K such that $|f_n(z)| \leq M_K$ for all $z \in K$ and all n.

Exercise 1.88 (A counterexample if uniform-on-compacts fails)**.** Give an example of holomorphic f_n on $U = \mathbb{C} \setminus \{0\}$ such that $f_n \to f$ pointwise on U, but f fails to be holomorphic. Identify exactly where uniform convergence on compacts breaks down. (Hint: try $f_n(z) = z^n$ on a punctured neighborhood and look at different regions.)

Exercises

Type A: Computational / estimate-based.

Exercise 1.89. Let f be holomorphic on $\overline{B(0,3)}$ and assume $\max_{|z|=3} |f(z)| \leq 10$. Show that $|f^{(5)}(0)| \leq \dfrac{5!}{3^5} \cdot 10$.

Exercise 1.90. Let f be entire and satisfy $|f(z)| \leq 7(1 + |z|)^4$. Prove that f is a polynomial of degree at most 4, and explain why you cannot conclude degree ≤ 3 from this data.

Type B: Conceptual / geometric.

Exercise 1.91. Explain why Liouville's theorem can be read as: "if a holomorphic function does not grow anywhere on $\mathbb{C}$, then it cannot vary anywhere." Compare this with the compact Riemann surface statement "holomorphic implies constant": what is the geometric reason boundedness becomes automatic on compact domains?

Exercise 1.92 (Where Liouville stops working)**.** Let f be entire and assume $|f(z)| \leq A + Be^{|z|}$ for all z. Explain why Cauchy estimates do not force f' to vanish. Suggest a growth condition that *would* still force f to be a polynomial (you may refer to Proposition 1.78 as guidance).

1.5. Identity Theorem and the Holomorphic Logarithm

Roadmap: two rigidity principles

This subsection contains two closely related "no-room-to-wiggle" phenomena.

- **Identity theorem / zero-set rigidity.** A holomorphic function cannot have a "thick" zero set inside a domain unless it is the zero function. Equivalently, knowing a holomorphic function on any subset with an interior accumulation point determines it everywhere.
- **Holomorphic logarithm / topology and branches.** The obstruction to defining $\log z$ globally on $\mathbb{C} \setminus \{0\}$ is *topological*: loops that wind around 0 force a $2\pi i$ ambiguity. On simply connected domains avoiding 0, one can choose a single-valued holomorphic branch.

1. Identity theorem and zero sets

Definition 1.93 (Zero set)**.** Let $U \subset \mathbb{C}$ be open and $f \in \mathrm{Hol}(U)$. The *zero set* of f is

$$Z(f) := \{z \in U : f(z) = 0\}.$$

Key local fact: zeros are isolated unless $f \equiv 0$.

Lemma 1.94 (Zeros are isolated unless everything is zero)**.** *Let $U \subset \mathbb{C}$ be open and $f \in \mathrm{Hol}(U)$. Then either $f \equiv 0$ on U, or else every zero $z_0 \in Z(f)$ is* isolated*: there exists $r > 0$ such that*

$$0 < |z - z_0| < r \quad \Longrightarrow \quad f(z) \neq 0.$$

Proof. Fix $z_0 \in U$. Since f is holomorphic, it admits a power series expansion on some disk $B(z_0, r_0) \subset U$:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \qquad (|z - z_0| < r_0).$$

If $a_n = 0$ for all n, then $f \equiv 0$ on $B(z_0, r_0)$.

Otherwise, let $m \geq 0$ be the smallest index such that $a_m \neq 0$. Factor out $(z - z_0)^m$:

$$f(z) = (z - z_0)^m \, g(z), \qquad g(z) := \sum_{n=0}^{\infty} a_{n+m} (z - z_0)^n.$$

Then g is holomorphic on $B(z_0, r_0)$ and satisfies $g(z_0) = a_m \neq 0$. By continuity, there exists $0 < r \leq r_0$ such that $g(z) \neq 0$ for all $|z - z_0| < r$. Therefore, in $B(z_0, r)$ the only zero of f is z_0 itself (with multiplicity m), so z_0 is isolated. □

A refinement: order of vanishing.

Definition 1.95 (Order of a zero)**.** Let $f \in \mathrm{Hol}(U)$ and $z_0 \in U$ with $f(z_0) = 0$ but $f \not\equiv 0$. The unique integer $m \geq 1$ such that

$$f(z) = (z - z_0)^m g(z), \qquad g \in \mathrm{Hol}(U),\ g(z_0) \neq 0,$$

is called the *order* (or *multiplicity*) of the zero of f at z_0, and we write

$$\mathrm{ord}_{z_0}(f) = m.$$

Theorem 1.96 (Identity theorem)**.** *Let $U \subset \mathbb{C}$ be open and connected, and let $f, g \in \mathrm{Hol}(U)$. Assume the coincidence set*

$$E := \{z \in U : f(z) = g(z)\}$$

has a limit point in U. Then $f \equiv g$ on U.

Equivalently: if $h \in \mathrm{Hol}(U)$ has zeros accumulating at a point of U, then $h \equiv 0$.

Proof. Let $h := f - g \in \mathrm{Hol}(U)$. Then $E = Z(h)$. If $h \not\equiv 0$, Lemma 1.94 implies every zero of h is isolated, so $Z(h)$ cannot have a limit point in U, contradicting the hypothesis. Therefore $h \equiv 0$, i.e. $f \equiv g$. □

Corollary 1.97 (Uniqueness of holomorphic extension)**.** Let $U \subset \mathbb{C}$ be connected and $f, g \in \mathrm{Hol}(U)$. If $f = g$ on a subset $A \subset U$ that has an accumulation point in U, then $f \equiv g$ on U.

Example 1.98 (Multiplicity at 0)**.** Let $f(z) = z^m e^z$ with $m \geq 1$. Then $\mathrm{ord}_0(f) = m$ because

$$f(z) = z^m \underbrace{e^z}_{g(z)}, \qquad g(0) = 1 \neq 0.$$

Equivalently, $f^{(k)}(0) = 0$ for $k = 0, 1, \ldots, m-1$ but $f^{(m)}(0) \neq 0$.

Exercise 1.99 (Order of vanishing via derivatives)**.** Suppose $f \in \mathrm{Hol}(U)$ and $z_0 \in U$ with $f(z_0) = 0$ and $f \not\equiv 0$. Show that $\mathrm{ord}_{z_0}(f) = m$ if and only if

$$f(z_0) = f'(z_0) = \cdots = f^{(m-1)}(z_0) = 0 \quad \text{and} \quad f^{(m)}(z_0) \neq 0.$$

2. The holomorphic logarithm and branches

The exponential map $\exp : \mathbb{C} \to \mathbb{C}$ satisfies $\exp(w + 2\pi i) = \exp(w)$, hence is not injective. Therefore it cannot have a global holomorphic inverse on $\mathbb{C}$. Instead, $\log z$ is naturally multi-valued.

Definition 1.100 (Multi-valued logarithm)**.** Write $z = re^{i\theta}$ with $r > 0$ and $\theta \in \mathbb{R}$. Define the multi-valued logarithm

$$\log z := \{\ln r + i(\theta + 2\pi k) \mid k \in \mathbb{Z}\}.$$

Theorem 1.101 (Existence of a holomorphic logarithm)**.** *Let $\Omega \subset \mathbb{C}$ be simply connected, and fix $z_* \in \Omega$. Then there exists a holomorphic function $L : \Omega \to \mathbb{C}$ such that*

$$e^{L(z)} = z \quad \textit{for all } z \in \Omega,$$

and $L(z_*)$ can be chosen to be any element of the multi-valued set $\log(z_*)$. Any two such branches differ by an additive constant $2\pi i k$, $k \in \mathbb{Z}$.

Proof. Since $\Omega \subset \mathbb{C}$, the function $1/z$ is holomorphic on Ω. Fix $z_* \in \Omega$ and define

$$L(z) := \int_{\gamma_{z_*,z}} \frac{1}{\zeta}\, d\zeta + c,$$

where $\gamma_{z_*,z}$ is any piecewise C^1 path in Ω from z_* to z.

Step 1: path-independence. If γ_1, γ_2 are two paths from z_* to z, then $\gamma_1 \cdot \overline{\gamma_2}$ is a closed loop in Ω. Because Ω is simply connected, this loop bounds a region in Ω. By the Cauchy integral theorem applied to $1/\zeta$ on that region,

$$\int_{\gamma_1} \frac{1}{\zeta}\, d\zeta - \int_{\gamma_2} \frac{1}{\zeta}\, d\zeta = \int_{\gamma_1 \cdot \overline{\gamma_2}} \frac{1}{\zeta}\, d\zeta = 0,$$

so the integral is path-independent and L is well-defined.

Step 2: holomorphy and derivative. For z near z_0, take a short path from z_0 to z inside Ω:

$$L(z) - L(z_0) = \int_{z_0}^{z} \frac{1}{\zeta}\, d\zeta.$$

Dividing by $z - z_0$ and letting $z \to z_0$ gives $L'(z_0) = 1/z_0$. Hence L is holomorphic and satisfies $L'(z) = 1/z$.

Step 3: exponentiating. Define $F(z) := e^{L(z)}/z$ on Ω. Then F is holomorphic and

$$\frac{F'(z)}{F(z)} = L'(z) - \frac{1}{z} = 0,$$

so F is constant on connected Ω. Choose the constant c so that $F(z_*) = 1$, i.e. $e^{L(z_*)} = z_*$. Then $F \equiv 1$ and $e^{L(z)} = z$ for all $z \in \Omega$.

Finally, if L_1, L_2 are two branches, then $e^{L_1 - L_2} = 1$, so $L_1 - L_2$ takes values in $2\pi i\mathbb{Z}$. By connectedness it must be constant, hence equals $2\pi i k$ for some $k \in \mathbb{Z}$. □

Definition 1.102 (Principal branch on the slit plane)**.** Let $\Omega_0 = \mathbb{C} \setminus (-\infty, 0]$. Define the *principal argument* $\operatorname{Arg} z \in (-\pi, \pi)$ and the *principal logarithm*

$$\operatorname{Log} z := \ln|z| + i \operatorname{Arg} z \qquad (z \in \Omega_0).$$

Proposition 1.103 (Holomorphy and derivative of the principal branch)**.** *The function* Log *is holomorphic on* Ω_0 *and satisfies*

$$(\operatorname{Log} z)' = \frac{1}{z}.$$

Proof. This is a special case of Theorem 1.101 since Ω_0 is simply connected and avoids 0. The normalization $\operatorname{Arg} z \in (-\pi, \pi)$ pins down the branch uniquely. □

Example 1.104 (No holomorphic logarithm on an annulus)**.** Let $\Omega = \{z \in \mathbb{C} : 1 < |z| < 2\}$. Assume for contradiction that $L \in \operatorname{Hol}(\Omega)$ satisfies $e^{L(z)} = z$. Differentiating gives $L'(z) = 1/z$ on Ω. Let $\gamma(t) = \frac{3}{2}e^{it}$, $0 \le t \le 2\pi$. Then γ is a closed loop in Ω and

$$\int_\gamma L'(z)\,dz = \int_\gamma \frac{1}{z}\,dz.$$

The left-hand side equals 0 because L is single-valued:

$$\int_\gamma L'(z)\,dz = L(\gamma(2\pi)) - L(\gamma(0)) = 0.$$

But $\int_\gamma \frac{1}{z}\,dz = 2\pi i$, a contradiction. Hence no holomorphic logarithm exists on Ω.

Exercise 1.105 (Rotated branch cuts)**.** Fix $\alpha \in \mathbb{R}$ and set

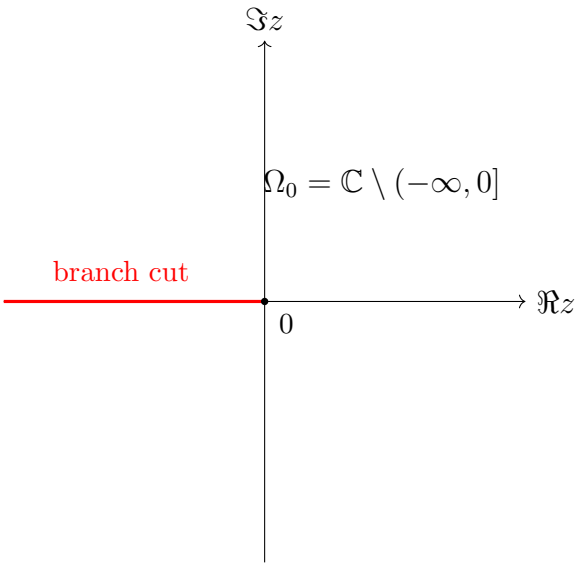
$$\Omega_\alpha := \mathbb{C} \setminus \{re^{i\alpha} : r \ge 0\}.$$

1. Show Ω_α is simply connected.
2. Construct a branch $\operatorname{Log}_\alpha$ on Ω_α and prove $(\operatorname{Log}_\alpha)'(z) = 1/z$.
3. Compare $\operatorname{Log}_\alpha$ and Log on $\Omega_\alpha \cap \Omega_0$.

Exercise 1.106 (Winding and $2\pi i$ jumps)**.** Let $\gamma : [0,1] \to \mathbb{C}$ be a closed piecewise C^1 curve. Show that

$$\frac{1}{2\pi i}\int_\gamma \frac{1}{z}\,dz \in \mathbb{Z}.$$

(Preview: this integer will later be identified with the winding number of γ about 0.)



Section takeaway. Identity theorem: zeros do not accumulate in the interior unless $f \equiv 0$. Holomorphic logarithm: log exists precisely when the domain has no winding around 0 (equivalently, the domain is simply connected inside $\mathbb{C}$).

1.6. Multivalued Analytic Functions and Riemann Surfaces

In this subsection we revisit *multivalued* analytic functions such as $\log z$ and z^α, and explain a standard resolution principle:

Multivalued on the z–plane $\quad\rightsquigarrow\quad$ *single-valued on a suitable Riemann surface.*

Rather than forcing the function to live on the flat z–plane, we build a new surface by cutting and gluing *sheets*. On that surface the function becomes an honest single-valued holomorphic function.

Cut the plane $\longrightarrow$ *glue sheets along the cut* $\longrightarrow$ *obtain a new surface.*

The multivalued logarithm, branches, and monodromy

Every $z \in \mathbb{C}^*$ admits a polar form $z = re^{i\theta}$ with $r > 0$ and $\theta \in \mathbb{R}$. Since $\exp(w + 2\pi i) = \exp(w)$, the inverse cannot be single-valued on $\mathbb{C}^*$. Indeed, the logarithm is naturally multivalued:

$$\log z := \left\{ \ln r + i(\theta + 2\pi k) \;\middle|\; k \in \mathbb{Z} \right\}.$$

If one analytically continues along a loop winding once about 0, then θ increases by 2π and the value shifts by $2\pi i$:

$$\log\big(re^{i(\theta+2\pi)}\big) = \log\big(re^{i\theta}\big) + 2\pi i.$$

This phenomenon—a value changing after continuation along a closed loop—is the first instance of *monodromy*.

Definition 1.107 (Holomorphic branch)**.** Let F be a multivalued analytic function on a domain $\Omega \subset \mathbb{C}$. A *holomorphic branch* of F on Ω is a holomorphic function $f \in \mathrm{Hol}(\Omega)$ such that $f(z) \in F(z)$ for every $z \in \Omega$. In particular, a holomorphic branch of $\log z$ on $\Omega \subset \mathbb{C}^*$ is a holomorphic function $L \in \mathrm{Hol}(\Omega)$ satisfying $e^{L(z)} = z$ for all $z \in \Omega$.

A practical criterion is: if $\Omega \subset \mathbb{C}^*$ is simply connected, then there exists a holomorphic branch of $\log z$ on Ω (equivalently, one can choose a continuous argument on Ω).

Definition 1.108 (Principal branch)**.** Let $\Omega_0 := \mathbb{C} \setminus (-\infty, 0]$. The *principal branch* of the logarithm is

$$\operatorname{Log} z := \ln|z| + i \operatorname{Arg} z, \qquad \operatorname{Arg} z \in (-\pi, \pi).$$

Proposition 1.109 (Holomorphy and derivative of Log)**.** *The principal branch* Log *is holomorphic on* Ω_0 *and satisfies*

$$(\operatorname{Log} z)' = \frac{1}{z} \qquad (z \in \Omega_0).$$

Proof. On Ω_0 we have $e^{\operatorname{Log} z} = z$. Differentiating gives

$$e^{\operatorname{Log} z} \cdot (\operatorname{Log} z)' = 1.$$

Since $e^{\operatorname{Log} z} = z \neq 0$ on Ω_0, we obtain $(\operatorname{Log} z)' = 1/z$. □

Example 1.110 (Walking around the origin)**.** Let $\gamma(t) = re^{it}$ for $0 \leq t \leq 2\pi$. Analytic continuation of $\log z$ along γ returns to the same point $z = r$ but changes the value by $2\pi i$. This is the obstruction to having a global single-valued logarithm on $\mathbb{C}^*$.

Complex powers z^α and (multi)valuedness

Fix $\alpha \in \mathbb{C}$. Using the multivalued logarithm, define the multivalued power

$$z^\alpha := e^{\alpha \log z}.$$

Concretely, if $z = re^{i\theta}$ then the set of values is

$$z^\alpha = \left\{ r^\alpha e^{\,i\alpha(\theta+2\pi k)} \;\middle|\; k \in \mathbb{Z} \right\}.$$

Theorem 1.111 (Single-, finite-, and infinite-valuedness of z^α)**.** *Let* $\alpha \in \mathbb{C}$.

(1) *If* $\alpha \in \mathbb{Z}$*, then* z^α *is single-valued on* $\mathbb{C}^*$ *(and extends holomorphically across* 0 *if* $\alpha \geq 0$*).*

(2) *If* $\alpha \in \mathbb{Q} \setminus \mathbb{Z}$ *and* $\alpha = \frac{m}{n}$ *with* $\gcd(m, n) = 1$ *and* $n \geq 2$*, then* z^α *has exactly* n *distinct values at each* $z \neq 0$*.*

(3) *If* $\alpha \notin \mathbb{Q}$*, then* z^α *is infinitely multivalued: different* $k \in \mathbb{Z}$ *give infinitely many distinct values.*

Proof. Write $z = re^{i\theta}$.
(1) If $\alpha = n \in \mathbb{Z}$, then $e^{in(\theta+2\pi k)} = e^{in\theta}$ for all $k \in \mathbb{Z}$, so the value is independent of k.
(2) If $\alpha = \frac{m}{n}$ with $\gcd(m, n) = 1$, then increasing k by n adds $2\pi i m$ to the exponent, hence does not change the value. Thus there are at most n values. They are distinct for $k = 0, 1, \ldots, n-1$ since $\gcd(m, n) = 1$ forces the angles $\frac{m}{n}(\theta + 2\pi k)$ to be distinct modulo 2π.
(3) If $\alpha \notin \mathbb{Q}$, then the numbers $\alpha(\theta + 2\pi k)$ are all distinct modulo 2π, so the arguments differ and hence the values are infinitely many. □

Example 1.112. 1. $\alpha = \frac{1}{2}$: $z^{1/2}$ has two values (the two square roots).

2. $\alpha = \frac{2}{3}$: $z^{2/3}$ has three values.

Geometric picture: the Riemann surface of $z^{1/2}$

The function $z^{1/2}$ cannot be single-valued on $\mathbb{C}^*$: going once around 0 changes the sign of the square root. We resolve this by passing to a two-sheeted surface built from two copies of a cut plane.

Cut-and-glue model. Cut the z–plane along the ray $[0, \infty)$ and take two copies of

$$\mathbb{C} \setminus [0, \infty).$$

On each sheet we have a continuous argument ϑ. Glue the sheets so that crossing the cut moves you to the other sheet. On the resulting surface the argument becomes globally continuous with range $\vartheta \in (0, 4\pi)$, and we may define

$$\sqrt{z} = \sqrt{r}\, e^{i\vartheta/2}.$$

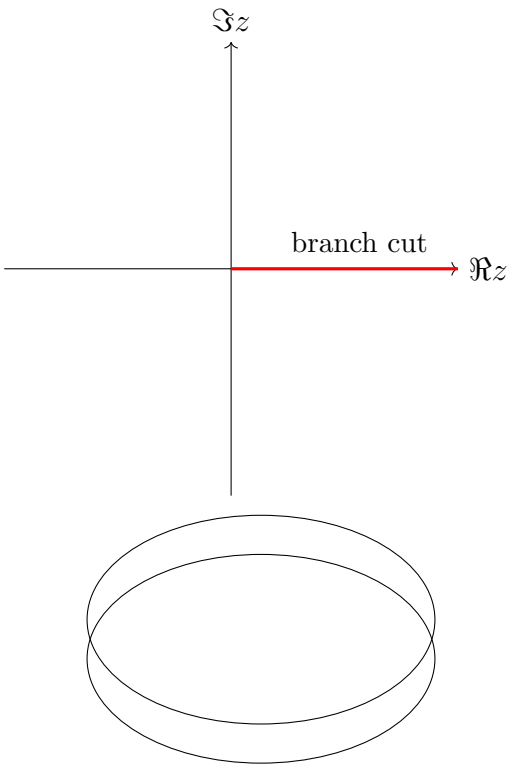
This is now single-valued and holomorphic on the glued surface.

Algebraic model. Equivalently, consider the affine curve

$$X := \{(z, w) \in \mathbb{C}^2 : w^2 = z\}.$$

The projection $\pi : X \to \mathbb{C}$, $\pi(z, w) = z$, is 2-to-1 away from $z = 0$. On X the function w is a single-valued holomorphic square root of z.

Proposition 1.113 (Two-sheeted square-root surface)**.** *On $X = \{(z, w) \in \mathbb{C}^2 : w^2 = z\}$, the function w is holomorphic and satisfies $w^2 = z$. The projection $\pi : X \to \mathbb{C}$ is a 2-to-1 holomorphic map away from $z = 0$, and it is branched at $z = 0$ (and also at $z = \infty$ after compactification to $\mathbb{CP}^1$).*



Two sheets glued along a cut

Three sheets: the Riemann surface of $z^{1/3}$

For the cube root we obtain a three-sheeted surface. Consider

$$\mathcal{C} := \{(z, w) \in \mathbb{C}^2 : w^3 = z\}.$$

Then $\pi(z, w) = z$ is 3-to-1 away from $z = 0$ (and, after compactification, away from ∞). Writing $z = re^{i\theta}$, the three values

$$w = r^{1/3} e^{i(\theta+2\pi k)/3}, \qquad k = 0, 1, 2,$$

are realized as three sheets. Cutting along a ray and gluing three copies cyclically produces the same surface as the algebraic model $\mathcal{C}$.

Exercise 1.114 (Three-sheeted gluing)**.** Construct the Riemann surface of $z^{1/3}$ by:

(1) taking three copies of $\mathbb{C} \setminus [0, \infty)$,

(2) gluing the edges of the cut cyclically (sheet $1 \to 2 \to 3 \to 1$),

(3) explaining why, after adding the points over ∞, the resulting surface is simply connected, and why the projection to the z–sphere is branched at 0 and ∞.

Exercise 1.115 (n sheets for $\alpha = \frac{m}{n}$)**.** Let $\alpha = \frac{m}{n} \in \mathbb{Q}$ with $\gcd(m, n) = 1$ and $n \geq 2$. Construct an n-sheeted surface by gluing n copies of a slit plane so that the argument runs through an interval of length $2\pi n$. Explain why the projection to $\mathbb{CP}^1$ is an n-sheeted branched covering with branch points at 0 and ∞.

Exercise 1.116 (Where can z^α be single-valued?)**.** Let $\alpha \in \mathbb{C}$. Determine the domains $\Omega \subset \mathbb{C}^*$ on which one can choose a single-valued holomorphic branch of z^α. Phrase your answer in terms of winding numbers of closed curves in Ω around 0.

1.7. Meromorphic Functions, Singularities, and the Residue

This subsection introduces *meromorphic* functions and explains how their behavior near isolated points is encoded by the Laurent expansion. Conceptually, this is about *classifying the local geometry of a function near a puncture*:

- removable singularities behave like ordinary points after filling in a hole,
- poles look like "finite order blow-ups",
- essential singularities exhibit chaotic behavior (Casorati–Weierstrass).

The residue will appear as the coefficient of the $(z - p)^{-1}$ term and will serve as a basic counting tool in residue theory.

Meromorphic functions and Laurent expansions

Definition 1.117 (Meromorphic function)**.** Let $\Omega \subset \mathbb{C}$ be open. A function $f : \Omega \to \mathbb{C}$ is *meromorphic on* Ω if there exists a discrete set $S \subset \Omega$ such that

(1) f is holomorphic on $\Omega \setminus S$, and

(2) for every $p \in S$, the point p is a *pole* of f.

Equivalently, f is holomorphic on $\Omega \setminus S$ and has only isolated poles (no essential singularities) in Ω.

Remark 1.118. Meromorphic functions are to holomorphic functions what rational functions are to polynomials: "holomorphic except for controlled poles".

Theorem 1.119 (Laurent expansion on a punctured disk)**.** *Let $p \in \Omega$, and suppose f is holomorphic on $0 < |z-p| < r$ for some $r > 0$. Then there exist unique coefficients $\{a_k\}_{k\in\mathbb{Z}}$ such that*

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z-p)^k,$$

and the series converges absolutely and uniformly on every annulus $\rho \le |z-p| \le R$ with $0 < \rho < R < r$.

Proof. Fix numbers $0 < \rho < R < r$. Choose two positively oriented circles

$$C_R : \ \zeta = p + Re^{i\theta}, \qquad C_\rho : \ \zeta = p + \rho e^{i\theta}, \qquad 0 \le \theta \le 2\pi.$$

Let $A_{\rho,R} := \{z : \rho < |z-p| < R\}$ be the annulus. Since f is holomorphic on $A_{\rho,R}$ and on a neighborhood of its closure, Cauchy's integral formula on the annulus (equivalently, applying Cauchy on the boundary $\partial A_{\rho,R} = C_R - C_\rho$) gives, for every $z \in A_{\rho,R}$,

$$f(z) = \frac{1}{2\pi i}\int_{C_R} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i}\int_{C_\rho} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

We expand each kernel as a geometric series.

(Outer circle contribution). On C_R we have $|\zeta - p| = R$ and for $z \in A_{\rho,R}$ we have $|z-p| < R$, hence

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta - p) - (z-p)} = \frac{1}{\zeta - p}\cdot \frac{1}{1 - \frac{z-p}{\zeta - p}} = \frac{1}{\zeta - p}\sum_{n=0}^{\infty}\Big(\frac{z-p}{\zeta - p}\Big)^n,$$

where the series converges absolutely and uniformly because $\left|\frac{z-p}{\zeta-p}\right| \le \frac{R'}{R} < 1$ on $\{|z-p| \le R'\}$ for any fixed $R' < R$.

Thus

$$\frac{1}{2\pi i}\int_{C_R} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\int_{C_R} \frac{f(\zeta)}{(\zeta - p)^{n+1}}\, d\zeta\right)(z-p)^n.$$

Define for $n \ge 0$

$$a_n := \frac{1}{2\pi i}\int_{C_R} \frac{f(\zeta)}{(\zeta - p)^{n+1}}\, d\zeta.$$

These do not depend on the chosen R as long as $\rho < R < r$, by deforming contours inside the holomorphic region (Cauchy's theorem).

(Inner circle contribution). On C_ρ we have $|\zeta - p| = \rho$ and for $z \in A_{\rho,R}$ we have $|z-p| > \rho$, hence

$$\frac{1}{\zeta - z} = -\frac{1}{z - \zeta} = -\frac{1}{(z-p) - (\zeta - p)} = -\frac{1}{z-p}\cdot \frac{1}{1 - \frac{\zeta - p}{z-p}} = -\frac{1}{z-p}\sum_{n=0}^{\infty}\Big(\frac{\zeta - p}{z-p}\Big)^n,$$

with absolute and uniform convergence on $\{|z-p| \geq \rho'\}$ for any $\rho' > \rho$.

Therefore

$$-\frac{1}{2\pi i}\int_{C_\rho} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i}\int_{C_\rho} f(\zeta)(\zeta-p)^n\, d\zeta \right) (z-p)^{-(n+1)}.$$

Define for $m \geq 1$ (set $m = n+1$)

$$a_{-m} := \frac{1}{2\pi i}\int_{C_\rho} f(\zeta)(\zeta-p)^{m-1}\, d\zeta.$$

Again a_{-m} is independent of ρ by contour deformation.

(Assembling). Combining the two expansions yields, for every $z \in A_{\rho,R}$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n + \sum_{m=1}^{\infty} a_{-m}(z-p)^{-m} = \sum_{k=-\infty}^{\infty} a_k (z-p)^k.$$

Uniform convergence on compact subannuli follows from uniform convergence of the geometric series and boundedness of f on each circle.

Uniqueness. If two Laurent series represent f on the annulus, subtracting them gives a Laurent series representing 0. Integrating term-by-term on a circle and using

$$\frac{1}{2\pi i}\int_{|\zeta-p|=\rho} (\zeta-p)^k\, d\zeta = \begin{cases} 1, & k=-1, \\ 0, & k \neq -1, \end{cases}$$

shows each coefficient must vanish. Hence the coefficients are unique. □

Definition 1.120 (Residue)**.** Under the assumptions of Theorem 1.119, the coefficient a_{-1} is called the *residue* of f at p and is denoted by

$$\operatorname{Res}(f,p) := a_{-1}.$$

Theorem 1.121 (Cauchy integral formula for residues)**.** *Let $p \in \Omega$ and let γ be a positively oriented simple closed C^1 curve contained in a punctured disk $0 < |z-p| < r$ and winding once around p. If f is holomorphic on and inside γ except possibly at p, then*

$$\frac{1}{2\pi i}\int_\gamma f(z)\, dz = \operatorname{Res}(f,p).$$

Proof. Since f is holomorphic on an annulus containing γ, Theorem 1.119 gives a Laurent expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z-p)^k$$

valid on that annulus, with uniform convergence on γ. Hence we may integrate term-by-term:

$$\int_\gamma f(z)\, dz = \sum_{k=-\infty}^{\infty} a_k \int_\gamma (z-p)^k\, dz.$$

For $k \neq -1$, the function $(z-p)^k$ has a holomorphic primitive on $\mathbb{C} \setminus \{p\}$ near γ:

$$(z-p)^k = \frac{d}{dz}\left(\frac{(z-p)^{k+1}}{k+1} \right),$$

so by the fundamental theorem of calculus for holomorphic functions, $\int_\gamma (z-p)^k\,dz = 0$ for all $k \neq -1$.

For $k = -1$ we parameterize γ as a loop winding once around p. By homotopy invariance, it suffices to compute on the circle $|z-p| = \rho$:

$$\int_{|z-p|=\rho} \frac{1}{z-p}\,dz = \int_0^{2\pi} \frac{1}{\rho e^{it}}\, i\rho e^{it}\,dt = i\int_0^{2\pi} dt = 2\pi i.$$

Therefore

$$\int_\gamma f(z)\,dz = a_{-1}\cdot 2\pi i,$$

and dividing by $2\pi i$ yields the claim. □

Classification of isolated singularities

Definition 1.122 (Isolated singularities)**.** Let f be holomorphic on $0 < |z-p| < r$.

- p is a *removable singularity* if f extends holomorphically to p.
- p is a *pole* if $|f(z)| \to \infty$ as $z \to p$.
- p is an *essential singularity* if it is neither removable nor a pole.

Proposition 1.123 (Laurent-series criteria)**.** *Let*

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z-p)^k$$

be the Laurent expansion on $0 < |z-p| < r$.

(1) *p is removable* $\iff$ *$a_k = 0$ for all $k < 0$.*

(2) *p is a pole of order $m \geq 1$* $\iff$ *$a_{-m} \neq 0$ and $a_k = 0$ for all $k < -m$.*

(3) *p is essential* $\iff$ *$a_k \neq 0$ for infinitely many $k < 0$.*

Proof. (1) If $a_k = 0$ for all $k < 0$, then $f(z) = \sum_{k\geq 0} a_k (z-p)^k$ on $0 < |z-p| < r$, and the series converges at $z = p$ as well, giving a holomorphic extension.

Conversely, if f extends holomorphically to p, then f is holomorphic on $|z-p| < r$ and has a Taylor series there, so all negative Laurent coefficients must be 0 by uniqueness.

(2) If $a_k = 0$ for all $k < -m$ and $a_{-m} \neq 0$, define

$$g(z) := (z-p)^m f(z) = \sum_{k=-m}^{\infty} a_k (z-p)^{k+m} = \sum_{j=0}^{\infty} a_{j-m}(z-p)^j.$$

Then g extends holomorphically to p with $g(p) = a_{-m} \neq 0$, hence is nonzero near p. Thus $f(z) = (z-p)^{-m} g(z)$ blows up like $|z-p|^{-m}$, so p is a pole of order m.

Conversely, if p is a pole of order m, then $(z-p)^m f(z)$ extends holomorphically and nonvanishingly to p, forcing the Laurent series of f to have precisely m negative powers.

(3) If infinitely many negative powers occur, then for every m the product $(z-p)^m f(z)$ still has a negative term in its Laurent series, hence cannot extend holomorphically to p. In particular f is not removable, and also not a pole of finite order. Thus p is essential.

Conversely, if p is essential, it is neither removable nor a pole, so the principal part cannot be finite and cannot be empty; hence there must be infinitely many negative terms. □

Remark 1.124 (Order notation)**.** If p is a pole of order m, write $\operatorname{ord}_p(f) = -m$. If f has a zero of order $m \geq 1$ at p, write $\operatorname{ord}_p(f) = m$.

Examples

Example 1.125 (Removable singularity)**.** Consider $f(z) = \dfrac{\sin z}{z}$ on $\mathbb{C}^*$. Since

$$\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} - \cdots,$$

we obtain

$$\frac{\sin z}{z} = 1 - \frac{z^2}{6} + \frac{z^4}{120} - \cdots,$$

which converges at $z = 0$. Defining $f(0) = 1$ fills in the hole, so $z = 0$ is removable.

Example 1.126 (Pole and residue)**.** Let $f(z) = \dfrac{1}{(z-p)^3}$. Then p is a pole of order 3 and the residue is 0. Indeed the Laurent series is exactly $(z-p)^{-3}$, with no $(z-p)^{-1}$ term.

Example 1.127 (Essential singularity)**.** The function $f(z) = e^{1/z}$ has an essential singularity at $z = 0$, since

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!}\, z^{-n}$$

has infinitely many negative powers.

Removable singularities and extension

Theorem 1.128 (Riemann removable singularity theorem)**.** *Let f be holomorphic on $0 < |z-p| < r$ and bounded near p. Then f extends uniquely to a holomorphic function on $|z-p| < r$.*

Proof. Assume $|f(z)| \leq M$ for all $0 < |z-p| < r_0$ for some $r_0 \leq r$. Fix a radius ρ with $0 < \rho < r_0$. By Theorem 1.119, f has a Laurent expansion on $0 < |z-p| < \rho$:

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z-p)^k, \qquad a_k = \frac{1}{2\pi i} \int_{|\zeta - p| = \rho} \frac{f(\zeta)}{(\zeta - p)^{k+1}}\, d\zeta.$$

We show $a_k = 0$ for all $k < 0$. Let $k = -m$ with $m \geq 1$. Then

$$a_{-m} = \frac{1}{2\pi i} \int_{|\zeta - p| = \rho} f(\zeta)\, (\zeta - p)^{m-1}\, d\zeta.$$

Estimate its absolute value:

$$|a_{-m}| \leq \frac{1}{2\pi} \int_{|\zeta - p| = \rho} |f(\zeta)|\, |\zeta - p|^{m-1}\, |d\zeta| \leq \frac{1}{2\pi} \int_{|\zeta - p| = \rho} M\, \rho^{m-1}\, |d\zeta| = M\rho^m.$$

Since this holds for every $0 < \rho < r_0$, letting $\rho \to 0$ forces $|a_{-m}| \leq M\rho^m \to 0$, hence $a_{-m} = 0$ for each $m \geq 1$. Therefore all negative coefficients vanish and

$$f(z) = \sum_{k=0}^{\infty} a_k (z-p)^k$$

on $0 < |z-p| < r_0$. The right-hand side converges at $z = p$, so defining $f(p) := a_0$ extends f holomorphically to $|z-p| < r_0$ (and hence to $|z-p| < r$ by analyticity on overlaps).

Uniqueness: if two holomorphic extensions existed, their difference would vanish on $0 < |z-p| < r_0$, hence everywhere by the Identity Theorem. □

Essential singularities and Casorati–Weierstrass

Theorem 1.129 (Casorati–Weierstrass)**.** *If f has an essential singularity at p, then for every $r > 0$ the image*

$$f\big(\{z : 0 < |z-p| < r\}\big)$$

is dense in $\mathbb{C}$.

Proof. Fix $r > 0$ and suppose the image is not dense. Then there exist $a \in \mathbb{C}$ and $\varepsilon > 0$ such that

$$f\big(\{z : 0 < |z-p| < r\}\big) \cap D(a, \varepsilon) = \emptyset,$$

equivalently,

$$|f(z) - a| \geq \varepsilon \qquad (0 < |z-p| < r).$$

Define

$$g(z) := \frac{1}{f(z) - a}.$$

Then g is holomorphic on $0 < |z-p| < r$ and satisfies $|g(z)| \leq 1/\varepsilon$ there, so g is bounded near p. By Theorem 1.128, g extends holomorphically to $|z-p| < r$; denote the extension again by g.

Now consider the value $g(p)$.

Case 1: $g(p) \neq 0$**.** Then $1/g$ is holomorphic near p, and hence

$$f(z) = a + \frac{1}{g(z)}$$

extends holomorphically to p. This means p is removable for f.

Case 2: $g(p) = 0$**.** Then g has a zero at p of some order $m \geq 1$, so we can write

$$g(z) = (z-p)^m h(z)$$

with h holomorphic and $h(p) \neq 0$. Hence $1/g$ has a pole of order m at p, and thus

$$f(z) = a + \frac{1}{g(z)}$$

has a pole at p.

In both cases, p is not an essential singularity of f, contradicting the hypothesis. Therefore the image must be dense in $\mathbb{C}$. □

Exercises

Exercise 1.130 (Principal part and residue)**.** Compute the principal part of

$$f(z) = \frac{z^2+1}{(z-2)^3(z+1)}$$

at $z = 2$ and determine $\mathrm{Res}(f, 2)$. Hint: write the partial fraction expansion near $z = 2$ and read off the $(z-2)^{-1}$ term.

Exercise 1.131 (Density near $z = 0$)**.** Using the Laurent expansion of $e^{1/z}$, show directly that for any $w \in \mathbb{C}$ and any $\varepsilon > 0$ there exists z with $0 < |z| < \delta$ (for sufficiently small δ) such that

$$|e^{1/z} - w| < \varepsilon.$$

Exercise 1.132 (Meromorphic functions on the Riemann sphere)**.** A meromorphic function on the Riemann sphere $\widehat{\mathbb{C}} \simeq \mathbb{CP}^1$ is the same thing as a rational function $P(z)/Q(z)$. Explain why:

(1) a meromorphic function on $\widehat{\mathbb{C}}$ has only finitely many poles (including possibly a pole at ∞),

(2) using partial fractions (and the behavior at ∞), one can write it as a ratio of polynomials.

This is a first step toward viewing meromorphic functions as morphisms of compact Riemann surfaces.

1.8. Laurent Series, Weierstrass M-test, and Cauchy Theory on Annuli

In this subsection we collect the analytic tools needed to work comfortably with Laurent series on annuli:

- the Weierstrass M-test for uniform convergence,
- uniform convergence of power series on disks and of principal parts on punctured disks,
- simultaneous convergence of both parts on an annulus,
- and the Cauchy theory (coefficient formulas and contour deformation) adapted to annular regions.

The guiding picture is that a Laurent series is the superposition of *(i)* an ordinary power series near a finite center p and *(ii)* a power series at ∞ in the variable $w = 1/(z-p)$, glued on a common annulus.

Weierstrass M-test and uniform convergence

Theorem 1.133 (Weierstrass M-test)**.** *Let $\{f_k\}$ be complex-valued functions on a set A. Suppose there exist numbers $M_k \geq 0$ such that*

$$|f_k(z)| \leq M_k \qquad \text{for all } z \in A,$$

and $\sum_{k=0}^{\infty} M_k < \infty$. Then $\sum_{k=0}^{\infty} f_k(z)$ converges absolutely and uniformly on A.

Remark 1.134 (Why we care)**.** On a set where the M-test applies (typically a compact subset), we may justify term-by-term limits, differentiation, and integration for series of holomorphic functions. This is the main engine behind Laurent theory on annuli.

Lemma 1.135 (Uniform convergence on smaller disks)**.** *Let $\sum_{k=0}^{\infty} a_k(z-p)^k$ be a power series that converges at some point $z_0 \neq p$. Set $r_0 := |z_0 - p|$. Then the series converges absolutely and uniformly on every closed disk $\overline{B(p,r)}$ with $0 < r < r_0$; hence it converges uniformly on every compact subset of $B(p, r_0)$.*

Proof. Since the series converges at z_0, the terms $a_k(z_0-p)^k$ tend to 0, hence are bounded: there exists $C > 0$ such that

$$|a_k(z_0 - p)^k| \leq C \qquad \text{for all } k \geq 0.$$

Fix r with $0 < r < r_0$, and take z with $|z - p| \leq r$. Then

$$|a_k(z-p)^k| = |a_k(z_0-p)^k| \left|\frac{z-p}{z_0-p}\right|^k \leq C\left(\frac{r}{r_0}\right)^k.$$

The geometric series $\sum_{k=0}^{\infty} C(r/r_0)^k$ converges. By the Weierstrass M-test (Theorem 1.133), $\sum_{k=0}^{\infty} a_k(z-p)^k$ converges absolutely and uniformly on $\overline{B(p,r)}$. Since every compact $K \Subset B(p, r_0)$ is contained in some $\overline{B(p,r)}$ with $r < r_0$, the convergence is uniform on K. □

Inner series (principal parts) as power series in $1/(z-p)$

A principal part

$$\sum_{k=1}^{\infty} a_{-k}(z-p)^{-k}$$

is a genuine power series in the variable $w := 1/(z-p)$:

$$\sum_{k=1}^{\infty} a_{-k}(z-p)^{-k} = \sum_{k=1}^{\infty} a_{-k}w^k.$$

Thus it behaves like an ordinary power series near $w = 0$, i.e. near $z = \infty$ relative to the center p.

Lemma 1.136 (Uniform convergence of principal parts outside a disk)**.** *Suppose $\sum_{k=1}^{\infty} a_{-k}(z-p)^{-k}$ converges at some point $z_1 \neq p$. Let $r_1 := |z_1 - p|$. Then for every $r > r_1$, the series converges absolutely and uniformly on the exterior region $\{z : |z-p| \geq r\}$ (and hence uniformly on every compact subset of $\{z : |z-p| > r_1\}$).*

Proof. Convergence at z_1 implies the terms $a_{-k}(z_1-p)^{-k}$ are bounded: there exists $C>0$ such that

$$|a_{-k}(z_1-p)^{-k}| \le C \qquad \text{for all } k \ge 1.$$

Fix $r > r_1$ and take z with $|z-p| \ge r$. Then

$$|a_{-k}(z-p)^{-k}| = |a_{-k}(z_1-p)^{-k}| \left|\frac{z_1-p}{z-p}\right|^k \le C\left(\frac{r_1}{r}\right)^k.$$

Again $\sum_{k=1}^{\infty} C(r_1/r)^k$ is geometric, so by the M-test the principal-part series converges absolutely and uniformly on $\{|z-p| \ge r\}$. As before, any compact subset of $\{|z-p| > r_1\}$ lies in some $\{|z-p| \ge r\}$ with $r > r_1$, giving uniform convergence on compacts. □

Simultaneous convergence on an annulus

A Laurent series about p is the formal sum of an "outer" (nonnegative) and an "inner" (negative) part:

$$\sum_{k=-\infty}^{\infty} a_k(z-p)^k = \underbrace{\sum_{k=0}^{\infty} a_k(z-p)^k}_{\text{outer part}} + \underbrace{\sum_{k=1}^{\infty} a_{-k}(z-p)^{-k}}_{\text{inner part}}.$$

Lemma 1.137 (Convergence on an annulus)**.** *Assume the outer series $\sum_{k=0}^{\infty} a_k(z-p)^k$ converges at some point z_0 with $r_0 := |z_0-p| > 0$, and the inner series $\sum_{k=1}^{\infty} a_{-k}(z-p)^{-k}$ converges at some point z_1 with $r_1 := |z_1-p| > 0$. Then both series converge absolutely and uniformly on every compact subset of the annulus*

$$A_{r_1,r_0}(p) := \{z \in \mathbb{C} : r_1 < |z-p| < r_0\}.$$

In particular, the full Laurent series converges absolutely and uniformly on every compact subset of $A_{r_1,r_0}(p)$.

Proof. By Lemma 1.135, the outer series converges uniformly on every compact subset of $\{|z-p| < r_0\}$. By Lemma 1.136, the inner series converges uniformly on every compact subset of $\{|z-p| > r_1\}$. Intersecting these regions yields uniform convergence of both parts on every compact subset of $A_{r_1,r_0}(p)$. Summing two uniformly convergent series gives uniform convergence of the full Laurent series on the same compact sets. □

Corollary 1.138 (Uniform convergence on closed subannuli)**.** Fix $\varepsilon > 0$ with $r_1 + \varepsilon < r_0 - \varepsilon$. Then on the closed subannulus

$$\{z : r_1 + \varepsilon \le |z-p| \le r_0 - \varepsilon\},$$

both the inner and outer series converge absolutely and uniformly, hence the Laurent series may be differentiated and integrated term-by-term.

Laurent coefficients: uniqueness and contour formulas

Proposition 1.139 (Coefficient formula and uniqueness)**.** *Let f be holomorphic on the annulus $A_{r_1,r_0}(p)$ and suppose*

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z-p)^k$$

converges absolutely and uniformly on each compact subset of the annulus. Then for every integer n and every circle $C_\rho := \{|\zeta - p| = \rho\}$ with $r_1 < \rho < r_0$,

$$a_n = \frac{1}{2\pi i} \int_{C_\rho} \frac{f(\zeta)}{(\zeta - p)^{n+1}}\, d\zeta.$$

Moreover, the right-hand side is independent of ρ, so the coefficients are uniquely determined by f. In particular, the Laurent expansion of f about p is unique.

Proof. Fix n and ρ. On C_ρ the Laurent series converges uniformly, hence we may divide term-by-term:

$$\frac{f(\zeta)}{(\zeta - p)^{n+1}} = \sum_{k=-\infty}^{\infty} a_k (\zeta - p)^{k-n-1}, \qquad \zeta \in C_\rho.$$

Uniform convergence on C_ρ justifies term-by-term integration:

$$\frac{1}{2\pi i} \int_{C_\rho} \frac{f(\zeta)}{(\zeta - p)^{n+1}}\, d\zeta = \sum_{k=-\infty}^{\infty} a_k \left(\frac{1}{2\pi i} \int_{C_\rho} (\zeta - p)^{k-n-1}\, d\zeta \right).$$

Parameterize C_ρ by $\zeta - p = \rho e^{it}$ $(0 \le t \le 2\pi)$. Then

$$\int_{C_\rho} (\zeta - p)^m\, d\zeta = \int_0^{2\pi} (\rho e^{it})^m\, i\rho e^{it}\, dt = i\rho^{m+1} \int_0^{2\pi} e^{i(m+1)t}\, dt = \begin{cases} 2\pi i, & m = -1, \\ 0, & m \neq -1. \end{cases}$$

Hence the only surviving term is the one with $k - n - 1 = -1$, i.e. $k = n$, and the sum equals a_n.

To see independence of ρ, fix $r_1 < \rho_1 < \rho_2 < r_0$ and consider

$$h(\zeta) := \frac{f(\zeta)}{(\zeta - p)^{n+1}},$$

which is holomorphic on $A_{r_1, r_0}(p)$. By contour deformation on an annulus (Theorem 1.140 below), we have

$$\int_{C_{\rho_2}} h(\zeta)\, d\zeta = \int_{C_{\rho_1}} h(\zeta)\, d\zeta,$$

so the coefficient formula does not depend on the chosen radius. □

Cauchy theory on annuli

Theorem 1.140 (Contour deformation on an annulus)**.** *Let f be holomorphic on the annulus $A_{r_1, r_0}(p)$. For any radii $r_1 < \rho_1 < \rho_2 < r_0$ we have*

$$\int_{C_{\rho_2}} f(\zeta)\, d\zeta = \int_{C_{\rho_1}} f(\zeta)\, d\zeta,$$

where each C_ρ is positively oriented.

Proof. Consider the closed annular region

$$\overline{A_{\rho_1, \rho_2}}(p) := \{\zeta : \rho_1 \le |\zeta - p| \le \rho_2\}.$$

Its boundary is $\partial\overline{A_{\rho_1,\rho_2}}(p) = C_{\rho_2} - C_{\rho_1}$, since the inner boundary is negatively oriented. By Cauchy's integral theorem applied to f on a neighborhood of this closed region,

$$0 = \int_{\partial\overline{A_{\rho_1,\rho_2}}(p)} f(\zeta)\,d\zeta = \int_{C_{\rho_2}} f(\zeta)\,d\zeta - \int_{C_{\rho_1}} f(\zeta)\,d\zeta,$$

which gives the claim. □

Theorem 1.141 (Cauchy integral formula on an annulus)**.** *Let f be holomorphic on $A_{r_1,r_0}(p)$ and choose radii $r_1 < \rho_1 < \rho_2 < r_0$. Then for every z with $\rho_1 < |z-p| < \rho_2$,*

$$f(z) = \frac{1}{2\pi i}\left(\int_{C_{\rho_2}} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \int_{C_{\rho_1}} \frac{f(\zeta)}{\zeta - z}\,d\zeta\right).$$

Equivalently,

$$\int_{C_{\rho_2}} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \int_{C_{\rho_1}} \frac{f(\zeta)}{\zeta - z}\,d\zeta = 2\pi i\, f(z).$$

Proof. Fix z with $\rho_1 < |z-p| < \rho_2$. Choose $\varepsilon > 0$ so small that $\overline{B(z,\varepsilon)} \subset \{\rho_1 < |\zeta - p| < \rho_2\}$, and set

$$R := \{\zeta : \rho_1 < |\zeta - p| < \rho_2\} \setminus \overline{B(z,\varepsilon)}.$$

Define $g(\zeta) := f(\zeta)/(\zeta - z)$. Then g is holomorphic on R. The boundary of R has three components:

$$\partial R = C_{\rho_2} - C_{\rho_1} - \partial B(z,\varepsilon),$$

where $\partial B(z,\varepsilon)$ is positively oriented as a circle, and appears with a minus sign because it is an inner boundary of R.

By Cauchy's integral theorem applied to g on R,

$$0 = \int_{\partial R} g(\zeta)\,d\zeta = \int_{C_{\rho_2}} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \int_{C_{\rho_1}} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \int_{\partial B(z,\varepsilon)} \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

Hence

$$\int_{C_{\rho_2}} \frac{f(\zeta)}{\zeta - z}\,d\zeta - \int_{C_{\rho_1}} \frac{f(\zeta)}{\zeta - z}\,d\zeta = \int_{\partial B(z,\varepsilon)} \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

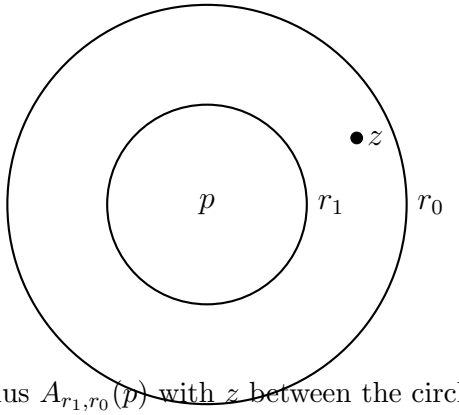
Since f is holomorphic on and inside $\partial B(z,\varepsilon)$, the usual Cauchy integral formula on a disk gives

$$\int_{\partial B(z,\varepsilon)} \frac{f(\zeta)}{\zeta - z}\,d\zeta = 2\pi i\, f(z).$$

Substituting yields the claimed identity, and dividing by $2\pi i$ gives the formula for $f(z)$. □

Remark 1.142 (Two boundary components)**.** On a disk there is only one boundary component, so Cauchy's formula uses one contour. On an annulus, the correct formula is "outer minus inner". This is exactly the analytic reflection of why Laurent series (with positive and negative powers) are the natural expansions on annular domains.

Visualization of the annulus



Annulus $A_{r_1,r_0}(p)$ with z between the circles

1.9. Laurent Coefficients, Singularity Classification, and the Residue Formula

In this subsection we connect the abstract Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-p)^n$$

to concrete analytic data near an isolated point p: how the negative-power terms classify singularities, how poles can be detected from growth, and how the coefficient a_{-1} governs contour integrals (the residue formula).

Laurent coefficients from contour integrals

Let f be holomorphic on an annulus

$$A_{r_1,r_2}(p) := \{z \in \mathbb{C} : r_1 < |z-p| < r_2\}.$$

Assume f admits a Laurent expansion on $A_{r_1,r_2}(p)$ that converges absolutely and uniformly on every compact subannulus:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-p)^n.$$

Proposition 1.143 (Laurent coefficient formula)**.** *For every integer n and every circle $C_\rho := \{|\zeta - p| = \rho\}$ with $r_1 < \rho < r_2$,*

$$a_n = \frac{1}{2\pi i} \int_{C_\rho} \frac{f(\zeta)}{(\zeta-p)^{n+1}}\, d\zeta.$$

The right-hand side is independent of ρ.

Proof. Fix n and $r_1 < \rho < r_2$. On the circle C_ρ the Laurent series converges uniformly, so we may divide term-by-term:

$$\frac{f(\zeta)}{(\zeta-p)^{n+1}} = \sum_{k=-\infty}^{\infty} a_k (\zeta-p)^{k-n-1}, \qquad \zeta \in C_\rho.$$

Uniform convergence on C_ρ implies uniform convergence of the divided series (because $|(\zeta-p)^{-(n+1)}| = \rho^{-(n+1)}$ is constant on C_ρ), hence we may integrate term-by-term:

$$\frac{1}{2\pi i}\int_{C_\rho}\frac{f(\zeta)}{(\zeta-p)^{n+1}}\,d\zeta = \sum_{k=-\infty}^{\infty} a_k\left(\frac{1}{2\pi i}\int_{C_\rho}(\zeta-p)^{k-n-1}\,d\zeta\right).$$

Write $\zeta - p = \rho e^{it}$, $0 \le t \le 2\pi$. Then $d\zeta = i\rho e^{it}\,dt$ and

$$\int_{C_\rho}(\zeta-p)^m\,d\zeta = \int_0^{2\pi}(\rho e^{it})^m\, i\rho e^{it}\,dt = i\rho^{m+1}\int_0^{2\pi} e^{i(m+1)t}\,dt = \begin{cases} 2\pi i, & m=-1,\\ 0, & m\neq -1.\end{cases}$$

Hence every term in the sum vanishes except when $k-n-1=-1$, i.e. $k=n$. Therefore the integral equals a_n.

To see that the value is independent of ρ, define

$$h(\zeta) := \frac{f(\zeta)}{(\zeta-p)^{n+1}},$$

which is holomorphic on the annulus $A_{r_1,r_2}(p)$. If $r_1 < \rho_1 < \rho_2 < r_2$, Cauchy's integral theorem on the closed annulus $\{\rho_1 \le |\zeta-p| \le \rho_2\}$ gives

$$0 = \int_{\partial(\rho_1\le|\zeta-p|\le\rho_2)} h(\zeta)\,d\zeta = \int_{C_{\rho_2}} h(\zeta)\,d\zeta - \int_{C_{\rho_1}} h(\zeta)\,d\zeta,$$

so $\int_{C_\rho} h(\zeta)\,d\zeta$ does not depend on ρ. □

Laurent expansion via splitting of the Cauchy kernel

Let f be holomorphic on $A_{r_1,r_2}(p)$ and choose radii

$$r_1 < \rho_1 < |z-p| < \rho_2 < r_2.$$

The annulus version of Cauchy's formula (derived by removing a small disk around z and applying Cauchy's theorem) gives

$$f(z) = \frac{1}{2\pi i}\int_{C_{\rho_2}}\frac{f(\zeta)}{\zeta-z}\,d\zeta - \frac{1}{2\pi i}\int_{C_{\rho_1}}\frac{f(\zeta)}{\zeta-z}\,d\zeta, \tag{1}$$

where both C_{ρ_j} are positively oriented circles centered at p.

On the outer circle C_{ρ_2} we have $|\zeta-p| = \rho_2 > |z-p|$, so

$$\frac{1}{\zeta-z} = \frac{1}{\zeta-p}\cdot\frac{1}{1-\frac{z-p}{\zeta-p}} = \sum_{k=0}^{\infty}\frac{(z-p)^k}{(\zeta-p)^{k+1}}.$$

On the inner circle C_{ρ_1} we have $|\zeta-p| = \rho_1 < |z-p|$, so

$$\frac{1}{\zeta-z} = -\frac{1}{z-p}\cdot\frac{1}{1-\frac{\zeta-p}{z-p}} = -\sum_{k=0}^{\infty}\frac{(\zeta-p)^k}{(z-p)^{k+1}}.$$

Each geometric series converges uniformly on the corresponding circle (because the ratio has modulus <1), so we may substitute into (1) and interchange integral and summation. The C_{ρ_2} integral produces the nonnegative powers $(z-p)^k$, while the C_{ρ_1} integral produces negative powers $(z-p)^{-k-1}$, recovering the Laurent expansion.

Classification of isolated singularities via Laurent coefficients

Assume f is holomorphic on $0 < |z-p| < r$ and admits a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-p)^n, \qquad 0 < |z-p| < r.$$

Theorem 1.144 (Classification of isolated singularities)**.**

- *p is removable if and only if $a_{-k} = 0$ for all $k \geq 1$.*
- *p is a pole of order m if and only if $a_{-m} \neq 0$ and $a_{-k} = 0$ for all $k > m$.*
- *p is essential if and only if infinitely many coefficients a_{-k} ($k \geq 1$) are nonzero.*

Proof. First assume $a_{-k} = 0$ for all $k \geq 1$. Then the series is a power series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n$$

convergent for $|z-p| < r$, so defining $f(p) := a_0$ extends f holomorphically across p. Hence p is removable.

Next assume there is an $m \geq 1$ with $a_{-m} \neq 0$ and $a_{-k} = 0$ for all $k > m$. Then in $0 < |z-p| < r$ we can factor

$$f(z) = (z-p)^{-m} g(z), \qquad g(z) := \sum_{j=0}^{\infty} a_{j-m} (z-p)^j.$$

The series defining g is a power series convergent for $|z-p| < r$, hence g extends holomorphically to p with $g(p) = a_{-m} \neq 0$. Therefore $|f(z)| = |(z-p)^{-m} g(z)| \to \infty$ as $z \to p$, so p is a pole, and the smallest negative exponent $-m$ shows the pole order is exactly m.

Finally, assume infinitely many negative coefficients occur. Then for every $m \geq 0$, the function $(z-p)^m f(z)$ still has infinitely many negative powers in its Laurent series, hence cannot extend holomorphically to p (a holomorphic function has a power series with no negative powers). In particular, f is not removable. Also, if p were a pole of order m, then $(z-p)^m f(z)$ would extend holomorphically and be nonzero at p, contradicting what we just observed. Hence p is neither removable nor a pole, so it is essential. □

Detecting poles via growth

Proposition 1.145 (Growth criterion for poles)**.** *Let f be holomorphic on $0 < |z-p| < r$. If $|f(z)| \to \infty$ as $z \to p$, then p is a pole of f.*

Proof. Choose $0 < r_0 < r$ so that f is holomorphic on $0 < |z-p| < r_0$. Define

$$g(z) := \frac{1}{f(z)} \qquad (0 < |z-p| < r_0).$$

Then g is holomorphic on the punctured disk and $g(z) \to 0$ as $z \to p$. In particular, g is bounded near p: there exists $M > 0$ and $0 < r_1 < r_0$ such that $|g(z)| \leq M$ for $0 < |z-p| < r_1$.

By Riemann's removable singularity theorem, g extends holomorphically to p. Denote the extension again by g, so $g(p) = 0$. Since g is holomorphic and not identically zero

on $B(p, r_1)$ (otherwise f would be identically infinite), the zero of g at p has finite order: there exist an integer $m \geq 1$ and a holomorphic function h on $B(p, r_1)$ with $h(p) \neq 0$ such that

$$g(z) = (z-p)^m h(z).$$

Therefore, on $0 < |z-p| < r_1$,

$$f(z) = \frac{1}{g(z)} = \frac{1}{(z-p)^m h(z)} = (z-p)^{-m} \cdot \frac{1}{h(z)}.$$

Since $h(p) \neq 0$, the function $1/h$ is holomorphic near p, so f has a pole of order m at p. □

Principal part and holomorphic part

If p is a pole of order m, then the Laurent series splits uniquely as

$$f(z) = \underbrace{\sum_{k=-m}^{-1} a_k (z-p)^k}_{\text{principal part}} + \underbrace{\sum_{k=0}^{\infty} a_k (z-p)^k}_{\text{holomorphic part}}.$$

The integer m is intrinsic.

Lemma 1.146 (Uniqueness of the pole order)**.** *Suppose the Laurent expansion of f at p satisfies*

$$a_{-m} \neq 0, \qquad a_{-k} = 0 \text{ for all } k > m.$$

Then m is the unique pole order of f at p.

Proof. Consider $(z-p)^m f(z)$. Multiplying the Laurent expansion by $(z-p)^m$ yields

$$(z-p)^m f(z) = a_{-m} + a_{-m+1}(z-p) + a_{-m+2}(z-p)^2 + \cdots,$$

which is a power series convergent near p and takes the nonzero value a_{-m} at p. Hence $(z-p)^m f(z)$ extends holomorphically and is nonvanishing at p.

If $N < m$, then $(z-p)^N f(z)$ still has a term $a_{-m}(z-p)^{N-m}$ with negative exponent $N - m < 0$, so it cannot extend holomorphically to p. If $N > m$, then

$$(z-p)^N f(z) = (z-p)^{N-m}\big((z-p)^m f(z)\big)$$

extends holomorphically but vanishes at p (since $N - m \geq 1$). Thus the only exponent producing a holomorphic *nonvanishing* extension at p is $N = m$, which is exactly the pole order. □

Residue and the residue integral formula

Definition 1.147 (Residue)**.** Let f be holomorphic on $0 < |z-p| < r$ with Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-p)^n.$$

The *residue* of f at p is the coefficient of $(z-p)^{-1}$:

$$\operatorname{Res}(f, p) := a_{-1}.$$

Theorem 1.148 (Residue integral formula)**.** *Let f be holomorphic on $0 < |z-p| < r$. Let γ be a positively oriented, simple closed curve contained in $\{0 < |z-p| < r\}$ that winds once around p and does not enclose any other singularities of f. Then*

$$\int_\gamma f(z)\,dz = 2\pi i\ \mathrm{Res}(f,p).$$

Proof. Step 1: reduction to a small circle. Choose $0 < \rho < r$ so small that the circle $C_\rho := \{|z-p| = \rho\}$ lies inside the region where f is holomorphic (away from p). Since f is holomorphic on the domain between γ and C_ρ (no singularities are crossed), Cauchy's integral theorem on that doubly connected region implies

$$\int_\gamma f(z)\,dz = \int_{C_\rho} f(z)\,dz.$$

Step 2: compute the integral on C_ρ using the Laurent series. On C_ρ the Laurent series converges uniformly, so we may integrate term-by-term:

$$\int_{C_\rho} f(z)\,dz = \sum_{n=-\infty}^{\infty} a_n \int_{C_\rho} (z-p)^n\,dz.$$

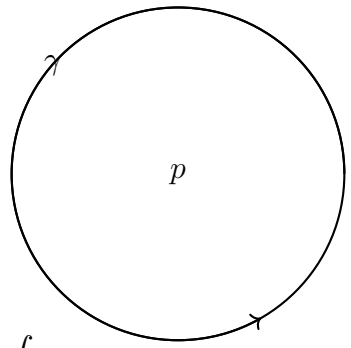
As in the coefficient computation, parameterize C_ρ by $z - p = \rho e^{it}$, $0 \le t \le 2\pi$. Then

$$\int_{C_\rho} (z-p)^n\,dz = \int_0^{2\pi} (\rho e^{it})^n\, i\rho e^{it}\,dt = i\rho^{n+1}\int_0^{2\pi} e^{i(n+1)t}\,dt = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Therefore only the $n = -1$ term survives:

$$\int_{C_\rho} f(z)\,dz = 2\pi i\, a_{-1} = 2\pi i\ \mathrm{Res}(f,p).$$

Combining with Step 1 yields the desired identity. □



$$\int_\gamma f(z)\,dz = 2\pi i\ \mathrm{Res}(f,p)$$

1.10. Poles, Residues, Holomorphic Simple Connectivity, and the Residue Theorem

We now globalize the local residue formula to contours enclosing many poles, and relate it to the notion of *holomorphic simple connectivity* (also called *Cauchy domains* in some texts): domains where integrals of holomorphic functions over closed curves always vanish.

Coefficient formulas at a pole

Let f have a pole of order $k \geq 1$ at p. Then f is holomorphic on $0 < |z-p| < r$ for some $r > 0$ and has a Laurent expansion

$$f(z) = \sum_{n=-k}^{\infty} a_n (z-p)^n, \qquad a_{-k} \neq 0.$$

Equivalently,

$$(z-p)^k f(z) = a_{-k} + a_{-k+1}(z-p) + a_{-k+2}(z-p)^2 + \cdots$$

is holomorphic near p.

Proposition 1.149 (Coefficient formulas at a pole)**.** *Let f have a pole of order k at p. Then*

$$a_{-k+j} = \frac{1}{j!} \left[\frac{d^j}{dz^j} \left((z-p)^k f(z) \right) \right]_{z=p} \qquad (j = 0, 1, 2, \dots).$$

In particular, the residue is obtained by taking $j = k-1$:

$$\operatorname{Res}(f, p) = a_{-1} = \frac{1}{(k-1)!} \left[\frac{d^{k-1}}{dz^{k-1}} \left((z-p)^k f(z) \right) \right]_{z=p}.$$

Proof. Set

$$g(z) := (z-p)^k f(z).$$

By assumption g is holomorphic in a neighborhood of p, so it admits a Taylor expansion

$$g(z) = \sum_{j=0}^{\infty} b_j (z-p)^j \quad \text{for } |z-p| < r$$

with coefficients

$$b_j = \frac{1}{j!} g^{(j)}(p) = \frac{1}{j!} \left[\frac{d^j}{dz^j} \left((z-p)^k f(z) \right) \right]_{z=p}.$$

But also

$$g(z) = (z-p)^k f(z) = (z-p)^k \sum_{n=-k}^{\infty} a_n (z-p)^n = \sum_{n=-k}^{\infty} a_n (z-p)^{n+k} = \sum_{j=0}^{\infty} a_{-k+j} (z-p)^j,$$

where we relabeled $j = n+k$. Comparing with the Taylor expansion of g shows $b_j = a_{-k+j}$ for every $j \geq 0$, giving the stated coefficient formula. Taking $j = k-1$ yields a_{-1}, i.e. the residue formula. □

Holomorphically simply connected domains

Definition 1.150 (Holomorphically simply connected)**.** A domain $\Omega \subset \mathbb{C}$ is *holomorphically simply connected* if it is connected and

$$\int_\gamma f(z)\, dz = 0 \quad \text{for every closed piecewise } C^1 \text{ curve } \gamma \subset \Omega \quad \text{and every } f \in \operatorname{Hol}(\Omega).$$

Equivalently, every holomorphic function on Ω admits a global antiderivative.

Proposition 1.151 (Equivalence with existence of primitives)**.** *For a domain $\Omega \subset \mathbb{C}$, the following are equivalent:*

(i) $\displaystyle\int_\gamma f(z)\,dz = 0$ *for every closed piecewise C^1 curve $\gamma \subset \Omega$ and every $f \in \mathrm{Hol}(\Omega)$.*

(ii) Every $f \in \mathrm{Hol}(\Omega)$ has an antiderivative $F \in \mathrm{Hol}(\Omega)$ with $F' = f$.

Proof. (i)⇒(ii). Fix $f \in \mathrm{Hol}(\Omega)$ and fix a base point $z_0 \in \Omega$. For any $z \in \Omega$, choose a piecewise C^1 curve $\gamma_{z_0,z}$ in Ω from z_0 to z, and define

$$F(z) := \int_{\gamma_{z_0,z}} f(\zeta)\,d\zeta.$$

We first show F is well-defined (independent of the chosen path). If γ_1, γ_2 are two curves from z_0 to z, then $\gamma_1 - \gamma_2$ (the curve obtained by traversing γ_1 then γ_2 backwards) is a closed curve in Ω, so by (i),

$$0 = \int_{\gamma_1-\gamma_2} f(\zeta)\,d\zeta = \int_{\gamma_1} f(\zeta)\,d\zeta - \int_{\gamma_2} f(\zeta)\,d\zeta.$$

Hence the two path integrals agree, and F is well-defined.

Next we show $F' = f$. Fix $z \in \Omega$ and choose $\varepsilon > 0$ so that the disk $B(z,\varepsilon) \subset \Omega$. For h with $|h| < \varepsilon$, let σ_h be the straight line segment from z to $z+h$. By additivity of path integrals,

$$F(z+h) - F(z) = \int_{\sigma_h} f(\zeta)\,d\zeta.$$

Parameterize $\sigma_h(t) = z + th$, $0 \le t \le 1$. Then $d\zeta = h\,dt$ and

$$\frac{F(z+h)-F(z)}{h} = \int_0^1 f(z+th)\,dt.$$

As $h \to 0$, continuity of f implies $f(z+th) \to f(z)$ uniformly in $t \in [0,1]$, so by dominated convergence,

$$\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = \int_0^1 f(z)\,dt = f(z).$$

Thus $F'(z) = f(z)$ for all $z \in \Omega$, and F is holomorphic with derivative f.

(ii)⇒(i). If $f = F'$ on Ω, then for any closed piecewise C^1 curve γ, the fundamental theorem of calculus for complex line integrals gives

$$\int_\gamma f(z)\,dz = \int_\gamma F'(z)\,dz = 0,$$

because the integral of an exact differential over a closed curve vanishes. □

Remark 1.152. If Ω is (topologically) simply connected, then Ω is holomorphically simply connected; the converse need not hold in more general complex manifolds, but in the plane this notion is often used as a convenient analytic substitute for simply connectedness in residue theory.

Contour decomposition: reducing a global integral to local circles

Let f be meromorphic on a domain Ω, and let Γ be a positively oriented, simple closed contour in Ω. Assume f is holomorphic on a neighborhood of Γ and meromorphic in the interior of Γ, with only finitely many poles $p_1, \ldots, p_n$ inside Γ.

Choose disjoint disks $B(p_j, \varepsilon_j) \subset \Omega$ so small that:

(a) f is holomorphic on each punctured disk $0 < |z - p_j| < \varepsilon_j$,

(b) the closed disks $\overline{B(p_j, \varepsilon_j)}$ are pairwise disjoint, and

(c) all these disks lie strictly inside Γ.

Let $C_j := \partial B(p_j, \varepsilon_j)$, oriented positively (counterclockwise).

Proposition 1.153 (Contour decomposition)**.** *With the notation above,*

$$\int_\Gamma f(z)\,dz = \sum_{j=1}^{n} \int_{C_j} f(z)\,dz.$$

Proof. Let

$$D := \mathrm{Int}(\Gamma) \setminus \bigcup_{j=1}^{n} B(p_j, \varepsilon_j).$$

Then D is a region whose boundary consists of the outer curve Γ and the inner circles C_j with the *opposite* orientation (because each C_j bounds a hole removed from the interior). More precisely,

$$\partial D = \Gamma - \sum_{j=1}^{n} C_j,$$

where "$-$" means reversing orientation.

By construction, f is holomorphic on an open neighborhood of $\overline{D}$. Therefore, by Cauchy's integral theorem,

$$0 = \int_{\partial D} f(z)\,dz = \int_\Gamma f(z)\,dz - \sum_{j=1}^{n} \int_{C_j} f(z)\,dz.$$

Rearranging gives the stated decomposition. □

Residue theorem

Theorem 1.154 (Residue theorem)**.** *Let f be meromorphic on an open set containing a positively oriented simple closed contour Γ and its interior. Suppose that the poles of f inside Γ are $p_1, \ldots, p_n$ (finitely many). Then*

$$\int_\Gamma f(z)\,dz = 2\pi i \sum_{j=1}^{n} \mathrm{Res}(f, p_j).$$

Proof. Choose small circles C_j around each pole as in Proposition 1.153. By contour decomposition,

$$\int_\Gamma f(z)\,dz = \sum_{j=1}^{n} \int_{C_j} f(z)\,dz.$$

For each j, the function f is holomorphic on a punctured disk about p_j, so the local residue formula (applied to C_j) yields

$$\int_{C_j} f(z)\,dz = 2\pi i\ \mathrm{Res}(f, p_j).$$

Summing over $j = 1, \ldots, n$ gives

$$\int_\Gamma f(z)\,dz = \sum_{j=1}^{n} 2\pi i\ \mathrm{Res}(f, p_j) = 2\pi i \sum_{j=1}^{n} \mathrm{Res}(f, p_j),$$

which is the desired identity. □

Winding number and the special case $f(z) = 1/(z-p)$

The integrand $\dfrac{1}{z-p}$ has residue 1 at p and no other singularities. Its contour integral measures how many times the curve winds around p.

Definition 1.155 (Winding number / index)**.** Let $\gamma : [a,b] \to \mathbb{C} \setminus \{p\}$ be a closed piecewise C^1 curve. The winding number of γ around p is

$$\mathrm{Ind}(\gamma, p) := \frac{1}{2\pi i} \int_\gamma \frac{1}{z-p}\,dz.$$

Proposition 1.156 (Integer-valuedness of the winding number)**.** *For every closed piecewise C^1 curve γ avoiding p,*

$$\mathrm{Ind}(\gamma, p) \in \mathbb{Z}.$$

Moreover, if γ is a positively oriented simple closed curve and p lies inside γ, then $\mathrm{Ind}(\gamma, p) = 1$*, while if p lies outside, then* $\mathrm{Ind}(\gamma, p) = 0$*.*

Proof. Step 1: lift the curve to an argument function. Let

$$\alpha(t) := \gamma(t) - p \in \mathbb{C}^*.$$

The map $\alpha : [a,b] \to \mathbb{C}^*$ is continuous and closed: $\alpha(a) = \alpha(b)$. Write $\alpha(t)$ in polar form $\alpha(t) = r(t)e^{i\theta(t)}$. Because $[a,b]$ is simply connected and $\alpha(t) \neq 0$ for all t, one can choose a continuous argument function $\theta : [a,b] \to \mathbb{R}$ such that

$$\alpha(t) = r(t)e^{i\theta(t)} \quad \text{for all } t,$$

with $r(t) = |\alpha(t)| > 0$. (Equivalently, we choose a continuous branch of Arg along the curve.)

Step 2: compute the integrand as a total derivative. Differentiate $\log \alpha(t) = \ln r(t) + i\theta(t)$:

$$\frac{d}{dt} \log(\alpha(t)) = \frac{\alpha'(t)}{\alpha(t)} = \frac{\gamma'(t)}{\gamma(t) - p}.$$

Hence

$$\int_\gamma \frac{1}{z-p}\,dz = \int_a^b \frac{\gamma'(t)}{\gamma(t)-p}\,dt = \int_a^b \frac{d}{dt} \log(\alpha(t))\,dt = \log(\alpha(b)) - \log(\alpha(a)).$$

Since $\alpha(a) = \alpha(b)$, the real parts $\ln r(b) - \ln r(a)$ cancel, leaving

$$\int_\gamma \frac{1}{z-p}\,dz = i\big(\theta(b) - \theta(a)\big).$$

Step 3: conclude integrality. Because $\alpha(a) = \alpha(b)$, we must have $r(a) = r(b)$ and $e^{i\theta(a)} = e^{i\theta(b)}$, so $\theta(b) - \theta(a) \in 2\pi\mathbb{Z}$. Therefore

$$\mathrm{Ind}(\gamma, p) = \frac{1}{2\pi i}\int_\gamma \frac{1}{z-p}\,dz = \frac{1}{2\pi i}\cdot i\big(\theta(b) - \theta(a)\big) = \frac{\theta(b) - \theta(a)}{2\pi} \in \mathbb{Z}.$$

The final statement about simple closed curves follows from the geometric meaning of $\theta(b) - \theta(a)$: it records the net change of the argument, which is 2π times the number of turns around p (one turn if p is inside, none if outside). □

Remark 1.157. The residue theorem admits an equivalent "winding number" form: if f is meromorphic and Γ is a closed contour avoiding the poles, then

$$\int_\Gamma f(z)\,dz = 2\pi i \sum_{p \in \mathrm{Poles}(f)} \mathrm{Ind}(\Gamma, p)\ \mathrm{Res}(f, p),$$

where only finitely many terms are nonzero when Γ is compact and f has isolated poles. This version is useful when Γ winds multiple times around some poles.

1.11. Meromorphic Functions, Singularities at ∞, and Rationality

Meromorphic functions on a domain

Let $\Omega \subset \mathbb{C}$ be a domain. A function $f : \Omega \to \mathbb{C}$ is *meromorphic* on Ω if there exists a *closed discrete* subset $S \subset \Omega$ such that

(i) f is holomorphic on $\Omega \setminus S$, and

(ii) for each $p \in S$, the point p is a pole of f.

We write $f \in M(\Omega)$.

Remark 1.158. The condition "closed discrete" means: for every compact $K \subset \Omega$, the intersection $S \cap K$ is finite. Equivalently, S has no accumulation point in Ω.

Proposition 1.159 (Changing values at poles does not change meromorphicity)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and $S \subset \Omega$ be closed and discrete. Suppose $f \in M(\Omega)$ has all its poles contained in S. Define a function $g : \Omega \to \widehat{\mathbb{C}}$ by*

$$g(z) = f(z) \quad (z \in \Omega \setminus S), \qquad \text{and assign arbitrary values } g(p) \in \widehat{\mathbb{C}} \text{ for } p \in S.$$

Then g is meromorphic on Ω and has poles (contained in) S.

Proof. Fix $p \in S$. Since S is discrete, there exists $r > 0$ such that $B_r(p) \cap S = \{p\}$. On the punctured disk $0 < |z - p| < r$ we have $g(z) = f(z)$, so g is holomorphic there.

Because f has a pole at p, there exists $m \geq 1$ and a holomorphic function h on $B_r(p)$ with $h(p) \neq 0$ such that

$$f(z) = \frac{h(z)}{(z-p)^m} \qquad (0 < |z-p| < r).$$

The same representation holds for g on $0 < |z-p| < r$ because $g = f$ there. Hence p is a pole of g (and its order agrees with that of f).

Since g is holomorphic on $\Omega \setminus S$ (it equals f there) and every point of S is an isolated pole, g is meromorphic on Ω with poles contained in S. □

Example 1.160. The function

$$g(z) = \frac{1}{\sin z}$$

is meromorphic on $\mathbb{C}$ with pole set $S = \{k\pi : k \in \mathbb{Z}\}$. Indeed, $\sin z$ is holomorphic and has simple zeros at each $k\pi$, so $1/\sin z$ has simple poles there and is holomorphic elsewhere.

Meromorphicity and singularities at infinity

Let $\Omega \subset \mathbb{C}$ be an unbounded domain, and suppose f is holomorphic on $\{z \in \Omega : |z| > R\}$ for some $R > 0$. To study the behavior "near ∞" we invert the coordinate:

$$w = \frac{1}{z}.$$

A neighborhood of ∞ in the z-plane corresponds to a punctured neighborhood of 0 in the w-plane. Accordingly, define the transformed function

$$G(w) := f\left(\frac{1}{w}\right),$$

defined for $0 < |w| < 1/R$ (and with w restricted so that $1/w \in \Omega$).

Definition 1.161 (Classification of the singularity at ∞)**.** We say f has:

(i) a *removable singularity at* ∞ if G extends holomorphically to $w = 0$;

(ii) a *pole at* ∞ if G has a pole at $w = 0$;

(iii) an *essential singularity at* ∞ if G has an essential singularity at $w = 0$.

Equivalently, the type of singularity of f at ∞ is the same as the type of singularity of $G(w) = f(1/w)$ at 0.

Remark 1.162 (Laurent expansions near ∞)**.** If G has a Laurent expansion near 0,

$$G(w) = \sum_{n=-m}^{\infty} a_n w^n \qquad (0 < |w| < \varepsilon),$$

then substituting $w = 1/z$ gives an expansion for f for $|z| > 1/\varepsilon$:

$$f(z) = G\left(\frac{1}{z}\right) = \sum_{n=-m}^{\infty} a_n z^{-n}.$$

Thus f is holomorphic at ∞ precisely when only nonnegative powers of w appear in G (equivalently, only nonpositive powers of z appear in the expansion of f), and f has a pole at ∞ precisely when only finitely many *positive* powers of z appear.

Entire functions and growth at infinity

Proposition 1.163 (Pole at ∞ forces a polynomial)**.** *If f is entire and has a pole at ∞, then f is a polynomial. Moreover, if the pole order of f at ∞ is m, then* $\deg f = m$.

Proof. Assume f is entire and has a pole of order $m \geq 1$ at ∞. Then $G(w) = f(1/w)$ has a pole of order m at $w = 0$, so there exists a Laurent expansion

$$G(w) = \sum_{n=-m}^{\infty} a_n w^n \qquad (0 < |w| < \varepsilon)$$

with $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$.

Substitute $w = 1/z$ (so $|z| > 1/\varepsilon$):

$$f(z) = G\left(\frac{1}{z}\right) = \sum_{n=-m}^{\infty} a_n z^{-n} = \sum_{j=0}^{m} a_{-j} z^j \; + \; \sum_{n=1}^{\infty} a_n z^{-n}.$$

We claim the negative-power tail $\sum_{n=1}^{\infty} a_n z^{-n}$ must vanish identically. Indeed, the function

$$H(z) := \sum_{n=1}^{\infty} a_n z^{-n}$$

is holomorphic on $\{|z| > 1/\varepsilon\}$, and it tends to 0 as $|z| \to \infty$ (because it is a convergent series in $1/z$). If some coefficient $a_N \neq 0$ with $N \geq 1$, then $z^N H(z)$ would have a nonzero limit a_N at ∞, contradicting that $z^N H(z) \to 0$ as $|z| \to \infty$ once higher terms are controlled. A more direct argument uses analytic continuation: since f is entire, the expression above describes f on the unbounded region $|z| > 1/\varepsilon$. If any $a_n \neq 0$ for $n \geq 1$, then f would have a singularity at $z = 0$ coming from the term $a_n z^{-n}$; but f is holomorphic at 0. Hence $a_n = 0$ for all $n \geq 1$.

Therefore

$$f(z) = \sum_{j=0}^{m} a_{-j} z^j,$$

a polynomial of degree exactly m because $a_{-m} \neq 0$. □

Corollary 1.164 (Nonpolynomial entire functions are essential at ∞)**.** If f is entire and not a polynomial, then f has an essential singularity at ∞.

Proof. Let f be entire and not a polynomial. By Proposition 1.163, f cannot have a pole at ∞.

If ∞ were a removable singularity, then $G(w) = f(1/w)$ would extend holomorphically to $w = 0$, so f would extend to a holomorphic function on the compact Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. A holomorphic function on a compact Riemann surface is constant, so f would be constant, contradicting that f is nonpolynomial (in particular, not constant). Hence ∞ is neither removable nor a pole, so it must be essential. □

Meromorphic functions on the Riemann sphere are rational

Let $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the Riemann sphere.

Theorem 1.165 (Rationality criterion)**.** *A function $f : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ is meromorphic on $\widehat{\mathbb{C}}$ if and only if f is a rational function.*

Proof. ($\Leftarrow$) Suppose f is rational: $f(z) = P(z)/Q(z)$ with polynomials P, Q and $Q \not\equiv 0$. Then f is holomorphic on $\mathbb{C} \setminus \{Q = 0\}$, and each zero of Q gives a pole of finite order. Since Q has finitely many zeros, f has finitely many poles in $\mathbb{C}$. At ∞, write $z = 1/w$ and consider $f(1/w) = P(1/w)/Q(1/w)$; multiplying numerator and denominator by a suitable power of w shows $f(1/w)$ has at worst a pole at $w = 0$. Hence f is meromorphic on $\widehat{\mathbb{C}}$.

($\Rightarrow$) Conversely, suppose f is meromorphic on $\widehat{\mathbb{C}}$. Then f has only isolated poles on $\widehat{\mathbb{C}}$. Because $\widehat{\mathbb{C}}$ is compact, any infinite set of poles would have an accumulation point, contradicting that poles are isolated. Hence f has only finitely many poles in $\widehat{\mathbb{C}}$.

List the finite poles in $\mathbb{C}$ as $p_1, \ldots, p_k$ with orders $m_1, \ldots, m_k$. Define

$$D(z) := \prod_{j=1}^{k} (z - p_j)^{m_j}, \qquad F(z) := D(z)\, f(z).$$

On $\mathbb{C} \setminus \{p_1, \ldots, p_k\}$, the product is holomorphic. Near each p_j, the factor $(z - p_j)^{m_j}$ cancels the pole of order m_j, so F extends holomorphically across p_j. Therefore F is entire on $\mathbb{C}$.

We now analyze F at ∞. Since f is meromorphic on $\widehat{\mathbb{C}}$, it is meromorphic at ∞; multiplying by the polynomial D cannot create an essential singularity at ∞. Hence F is entire and has at worst a pole at ∞. By Proposition 1.163, F must be a polynomial.

Therefore,

$$f(z) = \frac{F(z)}{D(z)} = \frac{F(z)}{\prod_{j=1}^{k} (z - p_j)^{m_j}},$$

which is a rational function. □

Exercises

Exercise 1.166 (Pole at ∞ and growth)**.** Let f be entire. Prove that f has a pole at ∞ if and only if there exist constants $C > 0$ and $m \in \mathbb{N}$ such that

$$|f(z)| \le C(1 + |z|)^m \qquad \text{for all } z \in \mathbb{C}.$$

Conclude that an entire function has a pole at ∞ if and only if it is a polynomial.

Exercise 1.167 (Laurent expansion at infinity)**.** Show that f is meromorphic at ∞ if and only if there exist $R > 0$ and integers $m \ge 0$ such that for $|z| > R$,

$$f(z) = \sum_{n=-m}^{\infty} a_n z^{-n}$$

with convergence as a Laurent series in $1/z$.

Exercise 1.168 (Degree and the value at ∞)**.** Let f be meromorphic on $\widehat{\mathbb{C}}$ and write $f(z) = P(z)/Q(z)$ in lowest terms. Prove that $f(\infty) = \infty$ if and only if $\deg P > \deg Q$, and that in this case the pole order of f at ∞ equals $\deg P - \deg Q$.

1.12. Concrete Examples with the Residue Theorem

This subsection develops explicit computational techniques for determining isolated singularities, classifying poles and essential singularities, and extracting residues using Laurent expansions, limit formulas, and structural algebraic manipulations.

Local order language: zeros and poles

Definition 1.169 (Zero of order n)**.** Let f be holomorphic in a neighborhood of z_0. We say that f has a *zero of order* n at z_0 if there exists a holomorphic function g near z_0 with $g(z_0) \neq 0$ such that

$$f(z) = (z - z_0)^n g(z).$$

Equivalently, $f(z_0) = f'(z_0) = \cdots = f^{(n-1)}(z_0) = 0$ and $f^{(n)}(z_0) \neq 0$.

Definition 1.170 (Pole of order n)**.** Let f be holomorphic on a punctured neighborhood $0 < |z - z_0| < r$. We say that f has a *pole of order* n at z_0 if the Laurent expansion has the form

$$f(z) = \sum_{k=-n}^{\infty} a_k (z - z_0)^k, \qquad a_{-n} \neq 0.$$

The coefficient a_{-1} is the residue $\operatorname{Res}(f, z_0)$. A pole of order 1 is called a *simple pole*.

Remark 1.171 (A quick order test)**.** If $f = \dfrac{u}{v}$ with u, v holomorphic near z_0 and $v \not\equiv 0$, then

$$\operatorname{ord}_{z_0}(f) = \operatorname{ord}_{z_0}(u) - \operatorname{ord}_{z_0}(v).$$

In particular, f has a pole at z_0 if $\operatorname{ord}_{z_0}(v) > \operatorname{ord}_{z_0}(u)$.

Three standard residue extraction tools

Proposition 1.172 (Simple poles: limit formula)**.** *If f has a simple pole at z_0, then*

$$\operatorname{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z).$$

Proof. Write the Laurent expansion at z_0:

$$f(z) = \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots .$$

Multiplying by $(z - z_0)$ gives

$$(z - z_0) f(z) = a_{-1} + a_0(z - z_0) + a_1(z - z_0)^2 + \cdots ,$$

which is holomorphic near z_0 and tends to a_{-1} as $z \to z_0$. Hence the limit equals $\operatorname{Res}(f, z_0) = a_{-1}$. □

Proposition 1.173 (Higher-order poles: derivative formula)**.** *If f has a pole of order k at z_0, then*

$$\operatorname{Res}(f, z_0) = \frac{1}{(k-1)!} \left[\frac{d^{k-1}}{dz^{k-1}} \Big((z - z_0)^k f(z) \Big) \right]_{z = z_0}.$$

Proof. Since f has a pole of order k at z_0, the product

$$h(z) := (z - z_0)^k f(z)$$

is holomorphic near z_0 and admits a Taylor expansion

$$h(z) = \sum_{m=0}^{\infty} b_m (z - z_0)^m.$$

Then

$$f(z) = \frac{h(z)}{(z-z_0)^k} = \sum_{m=0}^{\infty} b_m (z-z_0)^{m-k}.$$

The residue is the coefficient of $(z-z_0)^{-1}$, which corresponds to $m-k=-1$, i.e. $m=k-1$. Hence $\operatorname{Res}(f,z_0) = b_{k-1}$. On the other hand, $h^{(k-1)}(z_0) = (k-1)!\, b_{k-1}$, so

$$\operatorname{Res}(f,z_0) = \frac{h^{(k-1)}(z_0)}{(k-1)!} = \frac{1}{(k-1)!}\left[\frac{d^{k-1}}{dz^{k-1}}\Big((z-z_0)^k f(z)\Big)\right]_{z=z_0}.$$

□

Proposition 1.174 (Reciprocal of a simple zero)**.** *Let h be holomorphic near z_0 and suppose h has a simple zero at z_0. Then $1/h$ has a simple pole at z_0 and*

$$\operatorname{Res}\left(\frac{1}{h}, z_0\right) = \frac{1}{h'(z_0)}.$$

Proof. Since h has a simple zero at z_0, write $h(z) = (z-z_0)u(z)$ with u holomorphic and $u(z_0) \neq 0$. Then

$$\frac{1}{h(z)} = \frac{1}{z-z_0} \cdot \frac{1}{u(z)}.$$

Thus the residue is $1/u(z_0)$. Differentiating $h(z) = (z-z_0)u(z)$ gives $h'(z_0) = u(z_0)$, so the residue equals $1/h'(z_0)$. □

Worked examples

Example 1.175 (A rational function with a zero and poles)**.** Consider

$$f(z) = \frac{z+1}{z^3(z-1)}.$$

The numerator vanishes at $z=-1$ while the denominator does not, hence f has a simple zero at $z=-1$.

At $z=0$, the factor z^3 in the denominator forces a pole of order 3. Indeed,

$$g(z) := z^3 f(z) = \frac{z+1}{z-1}$$

is holomorphic near 0 with $g(0) = -1 \neq 0$.

To compute $\operatorname{Res}(f,0)$, expand near 0:

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{k=0}^{\infty} z^k, \qquad |z|<1.$$

Hence

$$g(z) = -(z+1)\sum_{k=0}^{\infty} z^k = -(1+2z+2z^2+2z^3+\cdots),$$

so

$$f(z) = \frac{g(z)}{z^3} = -\frac{1}{z^3} - \frac{2}{z^2} - \frac{2}{z} + \cdots, \qquad \operatorname{Res}(f,0) = -2.$$

At $z=1$ we have a simple pole, and

$$\operatorname{Res}(f,1) = \lim_{z\to 1}(z-1)\frac{z+1}{z^3(z-1)} = \frac{2}{1^3} = 2.$$

Example 1.176 (Residues at several simple poles)**.** Let

$$f(z) = \frac{z^2 + z^7}{(z-2)(z-3)(z-4)(z-5)}.$$

Each pole is simple. For instance,

$$\operatorname{Res}(f, 2) = \lim_{z \to 2} (z-2) f(z) = \frac{2^2 + 2^7}{(2-3)(2-4)(2-5)}.$$

Analogous formulas hold at $z = 3, 4, 5$.

Example 1.177 (Essential singularity)**.** The function

$$f(z) = \exp\left(\frac{1}{z}\right)$$

has an essential singularity at $z = 0$ since

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!}\, z^{-n} = 1 + \frac{1}{z} + \frac{1}{2! z^2} + \cdots .$$

Hence $\operatorname{Res}(f, 0) = 1$.

Example 1.178 (Combining zeros and poles)**.** Consider

$$f(z) = \frac{\sin z}{z^2}.$$

Since $\sin z = z - \frac{z^3}{3!} + \cdots$, we obtain

$$\frac{\sin z}{z^2} = \frac{1}{z} - \frac{z}{3!} + \cdots ,$$

so f has a simple pole at $z = 0$ and $\operatorname{Res}(f, 0) = 1$.

Example 1.179 (Higher-order pole)**.** Let

$$f(z) = \frac{1}{z^2} + \frac{7}{z} + 6z.$$

Then f has a pole of order 2 at $z = 0$ and $\operatorname{Res}(f, 0) = 7$.

Example 1.180 (Simple pole from algebraic manipulation)**.** Let $f(z) = \dfrac{z}{z+1}$. Then

$$f(z) = \frac{z + 1 - 1}{z + 1} = 1 - \frac{1}{z+1},$$

so f has a simple pole at $z = -1$ and $\operatorname{Res}(f, -1) = -1$.

Example 1.181 (Convergence region and residues)**.** Let

$$f(z) = \frac{1}{z(1-z)}.$$

On $0 < |z| < 1$ we have $\dfrac{1}{1-z} = \sum_{n=0}^{\infty} z^n$, hence

$$f(z) = \frac{1}{z} + 1 + z + z^2 + \cdots , \qquad \operatorname{Res}(f, 0) = 1.$$

Example 1.182 (Branch point: no residue)**.** The function $f(z) = \log z$ has a branch point at $z = 0$, so the singularity is not isolated (as a single-valued holomorphic function on a punctured disk). Therefore a residue is not defined.

Example 1.183 (Residue via reciprocal of a simple zero)**.** Since $\sin z$ has simple zeros at $z = n\pi$ and $\sin'(n\pi) = \cos(n\pi) = (-1)^n \neq 0$, each pole of $\csc z = 1/\sin z$ is simple and

$$\operatorname{Res}(\csc z, n\pi) = \frac{1}{\cos(n\pi)} = (-1)^n.$$

Example 1.184 (Residue of a rational function at $z = 2$)**.** Let

$$f(z) = \frac{2z + 7z^2}{(z-2)(z-3)(z-4)(z-5)}.$$

Then

$$\operatorname{Res}(f, 2) = \lim_{z\to 2}(z-2)f(z) = \frac{2 \cdot 2 + 7 \cdot 2^2}{(2-3)(2-4)(2-5)}.$$

Exercises

Exercise 1.185. Compute the residues of the following functions:

(a) $f(z) = \dfrac{z^2}{(z-1)^3}$ at $z = 1$.

(b) $f(z) = \dfrac{1}{z^2+4}$ at $z = 2i$.

(c) $f(z) = \dfrac{e^{1/z}}{z^2}$ at $z = 0$.

(d) $f(z) = \dfrac{\cos z}{z^4}$ at $z = 0$.

1.13. Applications of the Residue Theorem: Multi–Contour Integrals and Systematic Residue Extraction

This subsection develops explicit techniques for evaluating complex contour integrals using the Residue Theorem. We emphasize: (i) identifying which singularities lie inside a chosen contour, (ii) extracting residues efficiently (small–circle limits, Laurent expansions, derivative formulas), and (iii) two useful principles for quotients f/g when g has a simple zero, including the special case when $f(a) = g(a) = 0$.

Residue Theorem (global form)

Theorem 1.186 (Residue Theorem)**.** *Let f be analytic on a domain $\Omega \subset \mathbb{C}$ except for isolated singularities $p_1, \ldots, p_n$. Let C be a positively oriented, simple, closed, piecewise C^1 curve in Ω that avoids the singularities and encloses $p_1, \ldots, p_n$ (and no other singularities). Then*

$$\int_C f(z)\,dz = 2\pi i \sum_{j=1}^{n} \operatorname{Res}(f, p_j).$$

Proof. Choose pairwise disjoint small circles C_j around each p_j contained in the interior of C and contained in Ω, oriented positively. Set

$$U := \operatorname{Int}(C) \setminus \bigcup_{j=1}^{n} \operatorname{Int}(C_j).$$

Then U is a region whose boundary is the oriented union

$$\partial U = C - \sum_{j=1}^{n} C_j,$$

where each C_j appears with negative orientation relative to U (equivalently, C_j is traversed clockwise when viewed as part of ∂U). Since f is analytic on U (all singularities have been removed), Cauchy's integral theorem applied to f on U gives

$$0 = \int_{\partial U} f(z)\,dz = \int_{C} f(z)\,dz - \sum_{j=1}^{n} \int_{C_j} f(z)\,dz,$$

so

$$\int_{C} f(z)\,dz = \sum_{j=1}^{n} \int_{C_j} f(z)\,dz.$$

For each j, the local residue formula on a small circle implies

$$\int_{C_j} f(z)\,dz = 2\pi i \operatorname{Res}(f, p_j).$$

Summing over j yields the stated formula. □

Example 1: A rational function with poles at 0 and 1

Consider

$$f(z) = \frac{z^2}{z(z-1)} = \frac{z}{z-1}.$$

Let C be a positively oriented contour enclosing both 0 and 1.

At $z = 0$, the singularity is removable because $f(z) = z/(z-1)$ is analytic at 0: indeed,

$$\operatorname{Res}(f, 0) = 0.$$

At $z = 1$, the pole is simple, and

$$\operatorname{Res}(f, 1) = \lim_{z \to 1} (z-1)\frac{z}{z-1} = 1.$$

Therefore

$$\int_{C} f(z)\,dz = 2\pi i\,[0 + 1] = 2\pi i.$$

Example 2: A rational integrand (quick residues, or partial fractions)

Compute

$$\int_C \frac{5z-2}{z(z-1)}\,dz,$$

where C encloses 0 and 1.

Both poles are simple. At $z=0$,

$$\operatorname{Res}\left(\frac{5z-2}{z(z-1)},0\right)=\lim_{z\to 0}\frac{5z-2}{z-1}=2.$$

At $z=1$,

$$\operatorname{Res}\left(\frac{5z-2}{z(z-1)},1\right)=\lim_{z\to 1}\frac{5z-2}{z}=3.$$

Hence

$$\int_C \frac{5z-2}{z(z-1)}\,dz=2\pi i(2+3)=10\pi i.$$

Example 3: Laurent expansion at 0 and detecting the residue

Compute

$$\int_{|z|=1} z^{-2}\sin\left(\frac{1}{z}\right)\,dz.$$

On $0<|z|<\infty$ we have

$$\sin\left(\frac{1}{z}\right)=\frac{1}{z}-\frac{1}{3!z^3}+\frac{1}{5!z^5}-\cdots,$$

so

$$z^{-2}\sin\left(\frac{1}{z}\right)=\frac{1}{z^3}-\frac{1}{3!z^5}+\frac{1}{5!z^7}-\cdots.$$

There is no $(1/z)$ term, hence $\operatorname{Res}(f,0)=0$ and therefore

$$\int_{|z|=1} z^{-2}\sin(1/z)\,dz=0.$$

Example 4: A pole inside the contour and a careful expansion around $z=2$

Compute

$$\int_{|z-2|=1}\frac{dz}{z^2(z-2)^4}.$$

The contour encloses $z=2$ and excludes $z=0$. Thus only the pole at $z=2$ contributes.

Write

$$\frac{1}{z^2}=\frac{1}{(2+(z-2))^2}=\frac{1}{4}\,\frac{1}{\left(1+\frac{z-2}{2}\right)^2}.$$

For $|z-2|<2$ we may expand (binomial series)

$$\frac{1}{(1+w)^2}=\sum_{m=0}^{\infty}(-1)^m(m+1)w^m=1-2w+3w^2-4w^3+\cdots.$$

With $w = \frac{z-2}{2}$ this gives

$$\frac{1}{z^2} = \frac{1}{4}\sum_{m=0}^{\infty}(-1)^m(m+1)\left(\frac{z-2}{2}\right)^m.$$

Therefore

$$\frac{1}{z^2(z-2)^4} = \frac{1}{4}\sum_{m=0}^{\infty}(-1)^m(m+1)\left(\frac{1}{2}\right)^m(z-2)^{m-4}.$$

The residue at $z=2$ is the coefficient of $(z-2)^{-1}$, i.e. the term with $m-4=-1$, so $m=3$. Hence

$$\operatorname{Res}\left(\frac{1}{z^2(z-2)^4}, 2\right) = \frac{1}{4}(-1)^3(3+1)\left(\frac{1}{2}\right)^3 = -\frac{1}{8}.$$

By the residue theorem (here: single pole inside),

$$\int_{|z-2|=1}\frac{dz}{z^2(z-2)^4} = 2\pi i\left(-\frac{1}{8}\right) = -\frac{\pi i}{4}.$$

Example 5: Residues of $\csc z$ at finitely many enclosed poles

Consider $f(z) = \csc z = 1/\sin z$. The zeros of $\sin z$ are $k\pi$, and

$$\sin'(k\pi) = \cos(k\pi) = (-1)^k \neq 0,$$

so each pole of $\csc z$ is simple. If a contour C encloses $0, \pi, 2\pi$ and no other poles, then

$$\operatorname{Res}(\csc z, k\pi) = \frac{1}{\sin'(k\pi)} = \frac{1}{\cos(k\pi)} = (-1)^k,$$

hence

$$\int_C \frac{dz}{\sin z} = 2\pi i\left[1 - 1 + 1\right] = 2\pi i.$$

Quotients, simple zeros, and residues

The next two propositions are often the fastest way to compute residues of quotients f/g.

Proposition 1.187 (Residue at a simple zero of the denominator)**.** *Let f and g be analytic near a, and assume g has a simple zero at a:*

$$g(a) = 0, \qquad g'(a) \neq 0.$$

Then $\dfrac{f}{g}$ has a simple pole at a with

$$\operatorname{Res}\left(\frac{f}{g}, a\right) = \frac{f(a)}{g'(a)}.$$

Proof. Since $g(a) = 0$ and $g'(a) \neq 0$, the Taylor expansion of g at a begins with a nonzero linear term. Define

$$h(z) := \begin{cases} \dfrac{g(z)}{z-a}, & z \neq a, \\ g'(a), & z = a. \end{cases}$$

We show h is analytic near a and satisfies $h(a) = g'(a) \neq 0$. Indeed, write the Taylor series

$$g(z) = \sum_{m=1}^{\infty} \frac{g^{(m)}(a)}{m!}(z-a)^m,$$

(valid near a), then for $z \neq a$,

$$\frac{g(z)}{z-a} = \sum_{m=1}^{\infty} \frac{g^{(m)}(a)}{m!}(z-a)^{m-1} = \sum_{m=0}^{\infty} \frac{g^{(m+1)}(a)}{(m+1)!}(z-a)^m,$$

which extends holomorphically to $z = a$ with value $g'(a)$. Thus h is analytic and nonvanishing near a.

Now write $g(z) = (z-a)h(z)$. Then

$$\frac{f(z)}{g(z)} = \frac{1}{z-a} \cdot \frac{f(z)}{h(z)}.$$

Since f and h are analytic and $h(a) \neq 0$, the quotient f/h is analytic near a. Its value at a equals $f(a)/h(a) = f(a)/g'(a)$. Therefore the Laurent expansion of f/g at a has the form

$$\frac{f(z)}{g(z)} = \frac{f(a)}{g'(a)} \cdot \frac{1}{z-a} + \text{holomorphic terms},$$

so the residue is $f(a)/g'(a)$. □

Proposition 1.188 (Limit of a quotient when both numerator and denominator vanish)**.** *Let f and g be analytic near a with*

$$f(a) = g(a) = 0, \qquad g'(a) \neq 0.$$

Then f/g extends holomorphically across a and

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{f'(a)}{g'(a)}.$$

Proof. As above, since g has a simple zero at a, write $g(z) = (z-a)h(z)$ with h analytic and $h(a) = g'(a) \neq 0$. Because $f(a) = 0$, define

$$k(z) := \begin{cases} \dfrac{f(z)}{z-a}, & z \neq a, \\ f'(a), & z = a. \end{cases}$$

The same Taylor-series argument shows k is analytic near a and $k(a) = f'(a)$, and we have $f(z) = (z-a)k(z)$. Hence for $z \neq a$,

$$\frac{f(z)}{g(z)} = \frac{(z-a)k(z)}{(z-a)h(z)} = \frac{k(z)}{h(z)}.$$

The right-hand side is analytic near a (since $h(a) \neq 0$), so f/g extends holomorphically across a. Taking $z \to a$ gives

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{k(a)}{h(a)} = \frac{f'(a)}{g'(a)}.$$

□

1.14. Applications of the Residue Theorem to Real Integrals

This subsection develops systematic techniques for converting real integrals into complex contour integrals, locating poles, computing residues, and justifying the vanishing of large–arc contributions (Jordan-type estimates). We also treat the classical indentation argument for $\int_0^\infty \frac{\sin x}{x}\,dx$.

Simple poles and residues

Definition 1.189 (Simple pole and residue)**.** A point z_0 is a *simple pole* of a meromorphic function f if near z_0 one has

$$f(z) = \frac{a_{-1}}{z - z_0} + h(z),$$

where h is holomorphic near z_0. The coefficient a_{-1} is the residue:

$$\operatorname{Res}(f, z_0) := a_{-1}.$$

Lemma 1.190 (Residue at a simple pole via a limit)**.** *If f has a simple pole at z_0, then*

$$\operatorname{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z).$$

Proof. By definition, write $f(z) = \frac{a_{-1}}{z - z_0} + h(z)$ with h holomorphic near z_0. Then

$$(z - z_0) f(z) = a_{-1} + (z - z_0) h(z) \xrightarrow[z \to z_0]{} a_{-1}.$$

□

A standard large–arc estimate (Jordan-type lemma)

Lemma 1.191 (Upper–half–plane exponential decay)**.** *Fix $a > 0$. Let $R > 0$ and parametrize the upper semicircle*

$$\Gamma_R := \{z = Re^{it} : 0 \le t \le \pi\}.$$

Then for every $z \in \Gamma_R$,

$$|e^{iaz}| = e^{-a\,\Im z} = e^{-aR\sin t} \le 1.$$

Moreover, if H is meromorphic and satisfies $|H(z)| \le \frac{M}{R^{1+\delta}}$ on Γ_R for some constants $M, \delta > 0$ (uniformly in t), then

$$\left| \int_{\Gamma_R} e^{iaz} H(z)\,dz \right| \le \pi R \cdot \frac{M}{R^{1+\delta}} \longrightarrow 0 \qquad (R \to \infty).$$

Proof. For $z = Re^{it}$ we have $\Im z = R\sin t \ge 0$, hence

$$|e^{iaz}| = \exp(\Re(iaz)) = \exp(-a\,\Im z) = e^{-aR\sin t} \le 1.$$

Also $|dz| = R\,dt$ on Γ_R. Therefore,

$$\left| \int_{\Gamma_R} e^{iaz} H(z)\,dz \right| \le \int_0^\pi |e^{iaRe^{it}}|\,|H(Re^{it})|\,|dz| \le \int_0^\pi 1 \cdot \frac{M}{R^{1+\delta}} \cdot R\,dt = \frac{\pi M}{R^\delta} \xrightarrow[R \to \infty]{} 0.$$

□

Example 1: $\displaystyle\int_0^\infty \frac{\cos x}{1+x^4}\,dx$

Consider

$$I := \int_0^\infty \frac{\cos x}{1+x^4}\,dx.$$

Let

$$f(z) = \frac{e^{iz}}{1+z^4}.$$

Let C_R be the positively oriented contour consisting of the real segment $[-R, R]$ and the upper semicircle Γ_R.

Step 1: poles in the upper half–plane. The equation $1 + z^4 = 0$ is $z^4 = e^{i\pi}$, so the roots are

$$z_k = e^{i(\pi/4 + k\pi/2)} \qquad (k = 0, 1, 2, 3).$$

Those in the upper half–plane are

$$z_1 = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}, \qquad z_2 = e^{3i\pi/4} = \frac{-1+i}{\sqrt{2}}.$$

Each is a simple pole because $(1 + z^4)' = 4z^3$ does not vanish at a root.

Step 2: residues. By Lemma 1.190,

$$\operatorname{Res}(f, z_0) = \frac{e^{iz_0}}{(1+z^4)'|_{z=z_0}} = \frac{e^{iz_0}}{4z_0^3}.$$

Hence

$$\int_{C_R} f(z)\,dz = 2\pi i\Big(\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2)\Big).$$

Step 3: large arc vanishes. On Γ_R we have $|1 + z^4| \geq ||z|^4 - 1| = R^4 - 1$, so

$$|f(z)| = \left|\frac{e^{iz}}{1+z^4}\right| \leq \frac{|e^{iz}|}{R^4 - 1} \leq \frac{1}{R^4 - 1}.$$

Therefore by Lemma 1.191 (with $\delta = 3$),

$$\left|\int_{\Gamma_R} f(z)\,dz\right| \leq \pi R \cdot \frac{1}{R^4 - 1} \longrightarrow 0 \qquad (R \to \infty).$$

Step 4: pass to the limit and extract the real part. We have

$$\int_{-R}^{R} \frac{e^{ix}}{1+x^4}\,dx + \int_{\Gamma_R} f(z)\,dz = 2\pi i\Big(\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2)\Big).$$

Letting $R \to \infty$ and using $\int_{\Gamma_R} f \to 0$ gives

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{1+x^4}\,dx = 2\pi i\Big(\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2)\Big).$$

Taking real parts,

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^4}\,dx = \Re\left(2\pi i\Big(\operatorname{Res}(f,z_1)+\operatorname{Res}(f,z_2)\Big)\right).$$

Since the integrand $\frac{\cos x}{1+x^4}$ is even,

$$2I = \int_{-\infty}^{\infty} \frac{\cos x}{1+x^4}\,dx.$$

A direct computation (plugging z_1, z_2 and simplifying) yields the classical closed form

$$\int_{0}^{\infty} \frac{\cos x}{1+x^4}\,dx = \frac{\pi}{2\sqrt{2}}\,e^{-1/\sqrt{2}}\Big(\cos(1/\sqrt{2})+\sin(1/\sqrt{2})\Big).$$

Example 2: $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{1+x^2}\,dx$

Let

$$f(z) = \frac{e^{iz}}{1+z^2}.$$

The poles are at $z = \pm i$ and are simple. Only $z = i$ lies in the upper half–plane.

Step 1: residue at $z = i$. Since $(1+z^2)' = 2z$, by Lemma 1.190,

$$\operatorname{Res}(f,i) = \frac{e^{ii}}{2i} = \frac{e^{-1}}{2i}.$$

Step 2: apply the semicircle contour and let $R \to \infty$. Exactly as in Example 1, the arc contribution vanishes because $|f(z)| \le \frac{1}{|z|^2-1}$ on Γ_R (for R large) and $|e^{iz}| \le 1$ in the upper half–plane. Thus

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{1+x^2}\,dx = 2\pi i\,\operatorname{Res}(f,i) = 2\pi i \cdot \frac{e^{-1}}{2i} = \pi e^{-1}.$$

The right-hand side is real, so taking imaginary parts gives

$$\int_{-\infty}^{\infty} \frac{\sin x}{1+x^2}\,dx = 0.$$

Example 3: $\displaystyle\int_{0}^{\infty} \frac{\sin x}{x}\,dx$ **(indentation at the origin)**

Consider

$$f(z) = \frac{e^{iz}}{z}.$$

The point $z = 0$ is a simple pole with residue 1.

Contour. Fix $R > 1$ and $0 < \varepsilon < 1$. Let $C_{R,\varepsilon}$ be the positively oriented contour consisting of:

- the segment $[-R, -\varepsilon]$ on the real axis,
- the small semicircle γ_ε of radius ε in the upper half–plane joining $-\varepsilon$ to $+\varepsilon$ (clockwise or counterclockwise? see below),
- the segment $[\varepsilon, R]$ on the real axis,
- the large semicircle Γ_R in the upper half–plane joining R to $-R$.

We take γ_ε to be *counterclockwise* (positively oriented as a boundary of the upper half–disk), so the contour encloses the pole at 0.

Residue theorem on $C_{R,\varepsilon}$. Since f is analytic on and inside $C_{R,\varepsilon}$ except at 0 and 0 is enclosed,

$$\int_{C_{R,\varepsilon}} f(z)\,dz = 2\pi i \operatorname{Res}(f, 0) = 2\pi i.$$

Writing the contour integral as a sum gives

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x}\,dx + \int_{\gamma_\varepsilon} \frac{e^{iz}}{z}\,dz + \int_{\varepsilon}^{R} \frac{e^{ix}}{x}\,dx + \int_{\Gamma_R} \frac{e^{iz}}{z}\,dz = 2\pi i. \tag{2}$$

Large arc Γ_R vanishes. On Γ_R, $|e^{iz}| \le 1$ and $|z| = R$, hence $|e^{iz}/z| \le 1/R$ and

$$\left| \int_{\Gamma_R} \frac{e^{iz}}{z}\,dz \right| \le \pi R \cdot \frac{1}{R} = \pi.$$

To get vanishing, we use the sharper decay $|e^{iz}| = e^{-R\sin t}$:

$$\left| \int_{\Gamma_R} \frac{e^{iz}}{z}\,dz \right| \le \int_0^\pi \frac{e^{-R\sin t}}{R}\,R\,dt = \int_0^\pi e^{-R\sin t}\,dt \longrightarrow 0 \qquad (R \to \infty),$$

by dominated convergence (for $t \in (0, \pi)$, $\sin t > 0$ so $e^{-R\sin t} \to 0$, and the integrand is bounded by 1).

Small arc γ_ε. Parametrize γ_ε by $z = \varepsilon e^{it}$, $t \in [0, \pi]$. Then $dz = i\varepsilon e^{it} dt$ and

$$\int_{\gamma_\varepsilon} \frac{e^{iz}}{z}\,dz = \int_0^\pi \frac{e^{i\varepsilon e^{it}}}{\varepsilon e^{it}}\,i\varepsilon e^{it}\,dt = i\int_0^\pi e^{i\varepsilon e^{it}}\,dt \longrightarrow i\int_0^\pi 1\,dt = i\pi \qquad (\varepsilon \to 0),$$

again by dominated convergence (since $|e^{i\varepsilon e^{it}}| = e^{-\varepsilon \sin t} \le 1$).

Combine the real-axis pieces. Use the substitution $x \mapsto -x$ in the first integral:

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x}\,dx = \int_{\varepsilon}^{R} \frac{e^{-ix}}{x}\,dx.$$

Hence the sum of the two real-axis integrals is

$$\int_{\varepsilon}^{R} \frac{e^{-ix}}{x}\,dx + \int_{\varepsilon}^{R} \frac{e^{ix}}{x}\,dx = \int_{\varepsilon}^{R} \frac{e^{ix}+e^{-ix}}{x}\,dx = 2\int_{\varepsilon}^{R} \frac{\cos x}{x}\,dx.$$

But we want $\sin x/x$, so instead take the *difference* of the two real-axis pieces:

$$\int_{\varepsilon}^{R} \frac{e^{ix}}{x}\,dx - \int_{\varepsilon}^{R} \frac{e^{-ix}}{x}\,dx = 2i\int_{\varepsilon}^{R} \frac{\sin x}{x}\,dx.$$

From (2), move the $[-R,-\varepsilon]$ integral to the other side:

$$\int_{\varepsilon}^{R} \frac{e^{ix}}{x}\,dx - \int_{\varepsilon}^{R} \frac{e^{-ix}}{x}\,dx = 2\pi i - \int_{\gamma_\varepsilon} \frac{e^{iz}}{z}\,dz - \int_{\Gamma_R} \frac{e^{iz}}{z}\,dz.$$

Now let $R \to \infty$ (large arc $\to 0$) and then $\varepsilon \to 0$ (small arc $\to i\pi$), giving

$$2i\int_{0}^{\infty} \frac{\sin x}{x}\,dx = 2\pi i - i\pi = i\pi,$$

hence

$$\int_{0}^{\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

Example 4: A general template for $\cos(ax)$ and $\sin(ax)$ integrals

Let $a > 0$ and consider real integrals of the form

$$\int_{0}^{\infty} \frac{\cos(ax)}{1+x^n}\,dx, \qquad \int_{0}^{\infty} \frac{\sin(ax)}{1+x^n}\,dx, \qquad (n \in \mathbb{N},\ n \geq 2).$$

Set

$$f(z) = \frac{e^{iaz}}{1+z^n}.$$

The poles are the roots of $1 + z^n = 0$, i.e.

$$z_k = e^{i((2k+1)\pi/n)}, \qquad k = 0, 1, \ldots, n-1,$$

and the relevant ones are those with $\Im z_k > 0$. Each pole is simple since $(1+z^n)' = nz^{n-1} \neq 0$ at a root. For such a root z_0,

$$\operatorname{Res}\left(\frac{e^{iaz}}{1+z^n}, z_0\right) = \frac{e^{iaz_0}}{nz_0^{n-1}}.$$

On the upper semicircle Γ_R, the exponential factor satisfies $|e^{iaz}| \leq 1$ (Lemma 1.191), and the denominator grows like $|z|^n$, so the large arc contribution vanishes as $R \to \infty$. Thus the residue theorem reduces the real integral to a finite sum of residues, after taking real or imaginary parts and using symmetry when applicable.

Exercises

Exercise 1.192. Evaluate $\displaystyle\int_0^\infty \frac{\cos x}{x^2+4}\,dx$ using the residue theorem.

Exercise 1.193. Show that $\displaystyle\int_0^\infty \frac{\sin(ax)}{x}\,dx = \frac{\pi}{2}$ for all $a > 0$. (Hint: scale $x \mapsto ax$, or redo Example 3 with e^{iaz}/z.)

Exercise 1.194. Compute $\displaystyle\int_{-\infty}^\infty \frac{x\sin x}{1+x^4}\,dx$. (Hint: consider $\int_{-\infty}^\infty \frac{ze^{iz}}{1+z^4}\,dz$ and take imaginary parts.)

1.15. Advanced Contour Integration: Jordan's Lemma, Oscillatory Integrals, and Branch-Cut Methods

Conventions. Throughout, all contours are piecewise C^1 and positively oriented unless stated otherwise. For residues we write $\operatorname{Res}(f, p)$ (so no undefined `\Res` macro is needed). When using branch cuts, we specify the branch explicitly.

Jordan's Lemma

Lemma 1.195 (Jordan's lemma)**.** *Let $a > 0$ and let g be continuous on the closed upper half-disk*

$$D_R^+ := \{z \in \mathbb{C} : |z| \le R,\ \Im z \ge 0\},$$

holomorphic on its interior, and bounded by $|g(z)| \le M$ on D_R^+. Let $C_R := \{Re^{it} : 0 \le t \le \pi\}$ be the upper semicircle. Then

$$\left|\int_{C_R} e^{iaz} g(z)\,dz\right| \le \frac{\pi M}{a}.$$

Moreover, if in addition $g(z) = O(1/|z|)$ as $|z| \to \infty$ in $\Im z \ge 0$ (for instance $|g(z)| \le M/R$ on C_R), then

$$\left|\int_{C_R} e^{iaz} g(z)\,dz\right| \le \frac{\pi M}{aR} \xrightarrow[R\to\infty]{} 0.$$

Proof. Parametrize C_R by $z = Re^{it}$, $0 \le t \le \pi$, so $dz = iRe^{it}\,dt$ and

$$\left|\int_{C_R} e^{iaz} g(z)\,dz\right| \le \int_0^\pi \left|e^{iaRe^{it}}\right| |g(Re^{it})|\,R\,dt.$$

Now $e^{iaRe^{it}} = e^{iaR(\cos t + i\sin t)}$, hence

$$\left|e^{iaRe^{it}}\right| = e^{-aR\sin t}.$$

Using $|g| \le M$ on D_R^+ gives

$$\left|\int_{C_R} e^{iaz} g(z)\,dz\right| \le MR\int_0^\pi e^{-aR\sin t}\,dt.$$

For $t \in [0, \pi]$ we have the elementary bound

$$\sin t \geq \frac{2}{\pi}\, t \quad \text{for } 0 \leq t \leq \frac{\pi}{2}, \qquad \sin t \geq \frac{2}{\pi}\,(\pi - t) \quad \text{for } \frac{\pi}{2} \leq t \leq \pi.$$

Splitting at $\pi/2$ and changing variables,

$$\begin{aligned}\int_0^{\pi} e^{-aR\sin t}\, dt &\leq \int_0^{\pi/2} e^{-aR(2t/\pi)}\, dt + \int_{\pi/2}^{\pi} e^{-aR(2(\pi-t)/\pi)}\, dt \\ &= 2\int_0^{\pi/2} e^{-2aRt/\pi}\, dt = 2 \cdot \frac{\pi}{2aR}\left(1 - e^{-aR}\right) \leq \frac{\pi}{aR}.\end{aligned}$$

Therefore

$$\left| \int_{C_R} e^{iaz} g(z)\, dz \right| \leq MR \cdot \frac{\pi}{aR} = \frac{\pi M}{a}.$$

If additionally $|g(z)| \leq M/R$ on C_R, the same estimate yields

$$\left| \int_{C_R} e^{iaz} g(z)\, dz \right| \leq \frac{M}{R}\, R \int_0^{\pi} e^{-aR\sin t}\, dt \leq \frac{M}{R}\, R \cdot \frac{\pi}{aR} = \frac{\pi M}{aR} \to 0.$$

□

Dirichlet Integral: $\displaystyle\int_0^{\infty} \frac{\sin x}{x}\, dx$

We evaluate

$$\int_0^{\infty} \frac{\sin x}{x}\, dx.$$

Consider

$$f(z) = \frac{e^{iz}}{z},$$

which has a simple pole at $z = 0$ with residue $\operatorname{Res}(f, 0) = 1$.

Let $\Gamma_{R,\varepsilon}$ be the standard indented contour in the upper half-plane:

- the segment $[-R, -\varepsilon]$ on the real axis,
- the small semicircle $C_\varepsilon := \{\varepsilon e^{it} : 0 \leq t \leq \pi\}$ above 0,
- the segment $[\varepsilon, R]$ on the real axis,
- the large semicircle $C_R := \{Re^{it} : 0 \leq t \leq \pi\}$.

Step 1: Apply the residue theorem. The contour $\Gamma_{R,\varepsilon}$ encloses $z = 0$ once, so

$$\int_{\Gamma_{R,\varepsilon}} f(z)\, dz = 2\pi i \operatorname{Res}(f, 0) = 2\pi i.$$

Writing the contour integral as the sum of pieces gives

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x}\, dx + \int_{C_\varepsilon} \frac{e^{iz}}{z}\, dz + \int_{\varepsilon}^{R} \frac{e^{ix}}{x}\, dx + \int_{C_R} \frac{e^{iz}}{z}\, dz = 2\pi i. \tag{3}$$

Step 2: Large arc vanishes. On C_R, we have $|1/z| = 1/R$, so with $g(z) = 1/z$ Lemma 2.233 (second part) yields

$$\int_{C_R} \frac{e^{iz}}{z}\, dz \xrightarrow[R\to\infty]{} 0.$$

Step 3: Small arc contributes $i\pi$. Parametrize C_ε by $z = \varepsilon e^{it}$, $0 \le t \le \pi$. Then

$$\int_{C_\varepsilon} \frac{e^{iz}}{z}\, dz = \int_0^\pi \frac{e^{i\varepsilon e^{it}}}{\varepsilon e^{it}}\, (i\varepsilon e^{it})\, dt = i \int_0^\pi e^{i\varepsilon e^{it}}\, dt.$$

Since $e^{i\varepsilon e^{it}} \to 1$ uniformly in t as $\varepsilon \to 0$,

$$\int_{C_\varepsilon} \frac{e^{iz}}{z}\, dz \xrightarrow[\varepsilon\to 0]{} i \int_0^\pi 1\, dt = i\pi.$$

Step 4: Take limits and extract the sine integral. Let $R \to \infty$ in (3), then $\varepsilon \to 0$:

$$\text{p. v.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x}\, dx + i\pi = 2\pi i, \qquad \text{hence} \qquad \text{p. v.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x}\, dx = i\pi.$$

Taking imaginary parts,

$$\text{p. v.} \int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi.$$

Since $\sin x/x$ is even,

$$\int_0^\infty \frac{\sin x}{x}\, dx = \frac{1}{2}\ \text{p. v.} \int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \frac{\pi}{2}.$$

A warning example: $\displaystyle\int_0^\infty \frac{x}{1+x^2}\, dx$

The integral

$$\int_0^\infty \frac{x}{1+x^2}\, dx = \left[\frac{1}{2}\log(1+x^2)\right]_0^\infty = \infty$$

diverges (logarithmically). Nevertheless,

$$\text{p. v.} \int_{-\infty}^{\infty} \frac{x}{1+x^2}\, dx = 0$$

by oddness. Contour methods frequently compute *principal values* when poles lie on (or symmetry is used across) the real axis.

Branch-cut method: $\displaystyle\int_0^\infty \frac{x^{1/3}}{1+x^2}\, dx$

We now compute the genuinely branch-cut-driven integral

$$I := \int_0^\infty \frac{x^{1/3}}{1+x^2}\, dx.$$

Step 0: Choose a branch. Let $\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z$ with $0 < \operatorname{Arg} z < 2\pi$ (branch cut along the *positive* real axis), and define

$$z^{1/3} := \exp\left(\frac{1}{3}\operatorname{Log} z\right).$$

With this choice, for $x > 0$:

$$(x)^{1/3}_{\text{upper}} = x^{1/3} \quad (\operatorname{Arg} x = 0^+), \qquad (x)^{1/3}_{\text{lower}} = x^{1/3} e^{2\pi i/3} \quad (\operatorname{Arg} x = 2\pi^-).$$

Set

$$f(z) = \frac{z^{1/3}}{1+z^2}.$$

The poles of f are at $z = i$ and $z = -i$, both simple.

Step 1: Keyhole contour and reduction to $(1 - e^{2\pi i/3})I$. Let $\Gamma_{R,\varepsilon}$ be the keyhole contour around the positive real axis:

- upper ray: $z = x$ from ε to R (just above the cut),
- outer circle: $|z| = R$ from angle 0 to 2π avoiding the cut,
- lower ray: $z = x$ from R back to ε (just below the cut),
- inner circle: $|z| = \varepsilon$ closing the contour around the origin.

Because $f(z) = O(|z|^{-5/3})$ as $|z| \to \infty$ and $f(z) = O(|z|^{1/3})$ as $z \to 0$, the outer and inner circular contributions vanish as $R \to \infty$ and $\varepsilon \to 0$:

$$\int_{|z|=R} f(z)\,dz \to 0, \qquad \int_{|z|=\varepsilon} f(z)\,dz \to 0.$$

Along the upper ray we have $z^{1/3} = x^{1/3}$ and $dz = dx$, so the contribution tends to I. Along the lower ray, the contour runs from R to ε (reverse direction), and $z^{1/3} = x^{1/3} e^{2\pi i/3}$, hence the contribution tends to

$$\int_R^\varepsilon \frac{(x)^{1/3} e^{2\pi i/3}}{1+x^2}\,dx = -e^{2\pi i/3} \int_\varepsilon^R \frac{x^{1/3}}{1+x^2}\,dx \to -e^{2\pi i/3} I.$$

Therefore,

$$\lim_{R\to\infty,\ \varepsilon\to 0} \int_{\Gamma_{R,\varepsilon}} f(z)\,dz = (1 - e^{2\pi i/3})\, I. \tag{4}$$

Step 2: Compute residues at $\pm i$ on this branch. Since $1 + z^2 = (z-i)(z+i)$, for a simple pole at $z = z_0 \in \{i, -i\}$,

$$\operatorname{Res}(f, z_0) = \lim_{z\to z_0} (z - z_0) \frac{z^{1/3}}{(z-i)(z+i)} = \frac{z_0^{1/3}}{2z_0}.$$

On our branch $0 < \operatorname{Arg} z < 2\pi$, we have

$$i = e^{i\pi/2} \ \Rightarrow\ i^{1/3} = e^{i\pi/6}, \qquad -i = e^{i3\pi/2} \ \Rightarrow\ (-i)^{1/3} = e^{i\pi/2} = i.$$

Hence

$$\operatorname{Res}(f, i) = \frac{i^{1/3}}{2i} = \frac{e^{i\pi/6}}{2i}, \qquad \operatorname{Res}(f, -i) = \frac{(-i)^{1/3}}{-2i} = \frac{i}{-2i} = -\frac{1}{2}.$$

Therefore,

$$\sum \operatorname{Res}(f, \pm i) = \frac{e^{i\pi/6}}{2i} - \frac{1}{2}. \tag{5}$$

Step 3: Apply the residue theorem and solve for I. The keyhole contour encloses both poles $\pm i$, so

$$\int_{\Gamma_{R,\varepsilon}} f(z)\,dz = 2\pi i\left(\operatorname{Res}(f,i) + \operatorname{Res}(f,-i)\right).$$

Taking limits and using (4)–(5),

$$(1 - e^{2\pi i/3})\,I = 2\pi i\left(\frac{e^{i\pi/6}}{2i} - \frac{1}{2}\right).$$

Compute the right-hand side:

$$2\pi i \cdot \frac{e^{i\pi/6}}{2i} = \pi e^{i\pi/6}, \qquad 2\pi i \cdot \left(-\frac{1}{2}\right) = -\pi i,$$

so

$$(1 - e^{2\pi i/3})\,I = \pi e^{i\pi/6} - \pi i.$$

Now simplify using $e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $e^{i\pi/6} = \frac{\sqrt{3}}{2} + i\frac{1}{2}$. Then

$$1 - e^{2\pi i/3} = \frac{3}{2} - i\frac{\sqrt{3}}{2}, \qquad \pi e^{i\pi/6} - \pi i = \pi\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right).$$

Thus

$$I = \pi\,\frac{\frac{\sqrt{3}}{2} - i\frac{1}{2}}{\frac{3}{2} - i\frac{\sqrt{3}}{2}}.$$

Multiply numerator and denominator by the conjugate $\frac{3}{2} + i\frac{\sqrt{3}}{2}$:

$$I = \pi\,\frac{\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)\left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)}{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}.$$

The denominator equals $\frac{9}{4} + \frac{3}{4} = 3$. The numerator expands to

$$\left(\frac{\sqrt{3}}{2}\cdot\frac{3}{2} + \frac{\sqrt{3}}{2}\cdot i\frac{\sqrt{3}}{2}\right) + \left(-i\frac{1}{2}\cdot\frac{3}{2} - i\frac{1}{2}\cdot i\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{4} + i\frac{3}{4} - i\frac{3}{4} + \frac{\sqrt{3}}{4} = \sqrt{3}.$$

Hence

$$I = \pi\cdot\frac{\sqrt{3}}{3} = \frac{\pi}{\sqrt{3}}.$$

$$\int_0^\infty \frac{x^{1/3}}{1+x^2}\,dx = \frac{\pi}{\sqrt{3}}.$$

1.16. Zeros, Poles, and the Argument Principle

In this subsection we summarize the structure of zeros and poles of holomorphic and meromorphic functions. We discuss (i) the order of a zero, (ii) integral formulas for computing multiplicities, (iii) the argument principle, and (iv) geometric intuition behind local covering behavior.

Order of a Zero

Let $\Omega \subset \mathbb{C}$ be a domain and let $f \in \mathrm{Hol}(\Omega)$. Assume $f \not\equiv 0$ and $p \in \Omega$ satisfies $f(p) = 0$. Then there exists a unique integer $n \geq 1$ and a function $g \in \mathrm{Hol}(B_r(p))$ for some $r > 0$ such that

$$f(z) = (z-p)^n g(z), \qquad g(p) \neq 0.$$

The integer n is called the *order* (or *multiplicity*) of the zero at p, and we write

$$\mathrm{ord}_p(f) = n.$$

In particular, zeros of a nonzero holomorphic function are isolated; hence f has only finitely many zeros in any compact set.

Remark 1.196. If $g(p) \neq 0$ then g is nonvanishing on a sufficiently small neighborhood of p. Thus locally, all the vanishing of f near p is captured by the factor $(z-p)^n$.

Computing the Order via a Contour Integral

Let $r > 0$ be small enough so that f has no zeros on the circle

$$\partial B_r(p) = \{z : |z-p| = r\}.$$

Then

$$\frac{1}{2\pi i} \int_{\partial B_r(p)} \frac{f'(z)}{f(z)}\, dz = \mathrm{ord}_p(f). \tag{6}$$

Proof. Write $f(z) = (z-p)^n g(z)$ with $g \in \mathrm{Hol}(B_r(p))$ and $g(p) \neq 0$. Then g has no zeros on $\overline{B_r(p)}$ for r small, so g'/g is holomorphic on $B_r(p)$. Differentiate and divide:

$$\frac{f'(z)}{f(z)} = \frac{n}{z-p} + \frac{g'(z)}{g(z)}.$$

Integrate over $\partial B_r(p)$:

$$\frac{1}{2\pi i} \int_{\partial B_r(p)} \frac{f'}{f}\, dz = \frac{1}{2\pi i} \int_{\partial B_r(p)} \frac{n}{z-p}\, dz + \frac{1}{2\pi i} \int_{\partial B_r(p)} \frac{g'}{g}\, dz.$$

The second integral is 0 because g'/g is holomorphic on $B_r(p)$. The first integral equals n because $\int_{\partial B_r(p)} \frac{dz}{z-p} = 2\pi i$. □

Argument Principle (Holomorphic Case)

Let $f \in \mathrm{Hol}(\Omega)$ be nonzero and let γ be a positively oriented simple closed curve whose image lies in Ω. Assume f has no zeros on γ. Let $z_1, \ldots, z_k$ be the zeros of f in the interior of γ, and let $m_j = \mathrm{ord}_{z_j}(f)$. Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)}\, dz = \sum_{j=1}^{k} m_j. \tag{7}$$

Proof. The function f'/f is meromorphic, and at a zero z_j of order m_j we have $f(z) = (z - z_j)^{m_j} g(z)$ with $g(z_j) \neq 0$, hence

$$\frac{f'(z)}{f(z)} = \frac{m_j}{z - z_j} + \frac{g'(z)}{g(z)}.$$

So $\frac{f'}{f}$ has a simple pole at z_j with residue m_j. By the residue theorem,

$$\int_\gamma \frac{f'}{f}\, dz = 2\pi i \sum_{j=1}^{k} \operatorname{Res}\left(\frac{f'}{f}, z_j\right) = 2\pi i \sum_{j=1}^{k} m_j,$$

which gives (7). □

Remark 1.197 (Geometric meaning: change of argument)**.** If γ is parametrized by $z = \gamma(t)$ and $f(\gamma(t)) \neq 0$, then $f(\gamma(t))$ traces a closed curve in $\mathbb{C}^*$. The integral $\frac{1}{2\pi i}\int_\gamma f'/f$ equals the winding number of $f(\gamma)$ around 0, i.e. the total change of $\arg f(\gamma(t))$ divided by 2π.

Argument Principle (Meromorphic Case)

Let $f \in M(\Omega)$ be meromorphic and let γ be a positively oriented simple closed curve in Ω such that γ contains no zeros and no poles of f. Let $z_1, \ldots, z_k$ be the zeros of f inside γ with multiplicities m_j, and let $p_1, \ldots, p_\ell$ be the poles of f inside γ with orders μ_i. Then

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\, dz = \sum_{j=1}^{k} m_j - \sum_{i=1}^{\ell} \mu_i. \tag{8}$$

Proof. The logarithmic derivative f'/f is meromorphic. At a zero z_j of order m_j, it has a simple pole with residue m_j as above. At a pole p_i of order μ_i, write $f(z) = (z-p_i)^{-\mu_i} h(z)$ with $h(p_i) \neq 0$, so

$$\frac{f'(z)}{f(z)} = -\frac{\mu_i}{z - p_i} + \frac{h'(z)}{h(z)},$$

hence the residue is $-\mu_i$. Apply the residue theorem to $\frac{f'}{f}$ on γ to obtain (8). □

Local geometric intuition

If f is holomorphic and $\operatorname{ord}_p(f) = n$, then in a small punctured neighborhood of p, the map behaves like $z \mapsto (z - p)^n$ up to multiplication by a nonzero holomorphic factor. Geometrically, small circles around p are mapped to loops winding n times around 0. This is the local "n-to-1 covering" picture behind why $\frac{1}{2\pi i}\int f'/f$ counts multiplicities.

Exercises

Exercise 1.198. Let $f(z) = (z - 1)^3(z + 2)^2$.

1. Find $\operatorname{ord}_1(f)$ and $\operatorname{ord}_{-2}(f)$.
2. Compute
$$\frac{1}{2\pi i}\int_{|z-1|=r} \frac{f'(z)}{f(z)}\, dz$$
for sufficiently small $r > 0$, and interpret the result via (6).

Exercise 1.199. Let $f(z) = z^5 - 2z^3 + z$.

1. Factor f and list all zeros with multiplicities.
2. Use (7) to compute
$$\frac{1}{2\pi i}\int_{|z|=2} \frac{f'(z)}{f(z)}\,dz,$$
and check that it matches the total multiplicity of zeros inside $|z| < 2$.

Exercise 1.200. Let $f(z) = \dfrac{z^2+1}{z^2(z-3)}$. Compute
$$\frac{1}{2\pi i}\int_{|z|=2} \frac{f'(z)}{f(z)}\,dz.$$
Identify explicitly the zeros and poles of f inside $|z| < 2$, and verify (8).

Exercise 1.201. Let $p(z) = z^4 + 3z^2 + 1$. Use the argument principle on $|z| = R$ for R large to determine the number of zeros of p in the right half-plane $\{z : \Re z > 0\}$. (*Hint:* compare $p(z)$ to z^4 on $|z| = R$ and use the symmetry $p(\overline{z}) = \overline{p(z)}$.)

1.17. Rouché's Theorem and Zero Counting by Boundary Data

Motivation: zeros are stable under small boundary perturbations

The argument principle shows that the integer
$$N_\Gamma(f) := \frac{1}{2\pi i}\int_\Gamma \frac{f'(z)}{f(z)}\,dz$$
counts the number of zeros of a holomorphic function f inside Γ (with multiplicity), as long as f has no zeros on Γ. Rouché's theorem is a robust version of this idea: if two holomorphic functions are close on the boundary, then they must have the same number of zeros inside.

Rouché's Theorem

Theorem 1.202 (Rouché's Theorem)**.** *Let $\Omega \subset \mathbb{C}$ be a domain, and let $f, g \in \mathcal{H}(\Omega)$. Let Γ be a positively oriented simple closed contour with $\Gamma \subset \Omega$. Assume f and $f+g$ have no zeros on Γ. If*
$$|g(z)| < |f(z)| \qquad \text{for all } z \in \Gamma,$$
then f and $f+g$ have the same number of zeros (counted with multiplicity) in the interior of Γ.

Proof. For $t \in [0,1]$ define $h_t(z) = f(z) + t\,g(z)$. On Γ,
$$|h_t(z) - f(z)| = t|g(z)| \le |g(z)| < |f(z)|.$$
By the reverse triangle inequality,
$$|h_t(z)| \ge \big|\,|f(z)| - |h_t(z) - f(z)|\,\big| > 0,$$

so h_t has no zeros on Γ for all $t \in [0,1]$. Hence h_t'/h_t is holomorphic on a neighborhood of Γ and

$$N_\Gamma(h_t) = \frac{1}{2\pi i} \int_\Gamma \frac{h_t'(z)}{h_t(z)}\, dz$$

is integer-valued and continuous in t, thus constant in t. Evaluating at $t = 0$ and $t = 1$ gives $N_\Gamma(f) = N_\Gamma(f+g)$. □

Remark 1.203. The hypothesis $|g| < |f|$ is used only on the boundary. No comparison is needed in the interior; the conclusion is topological (stability of the winding number of $h_t(\Gamma)$ about the origin).

Example 1: counting zeros via a dominant term

Example 1.204 (All zeros of $z^4 + 4z + 1$ lie in $|z| < 2$)**.** Let $p(z) = z^4 + 4z + 1$. On the circle $|z| = 2$ we estimate

$$|z^4| = 16, \qquad |4z + 1| \le 8 + 1 = 9,$$

so $|4z + 1| < |z^4|$ on $|z| = 2$. By Rouché with $f(z) = z^4$ and $g(z) = 4z + 1$, the functions $p(z) = f(z) + g(z)$ and z^4 have the same number of zeros in $|z| < 2$. Since z^4 has four zeros (counted with multiplicity) at 0, p also has four zeros in $|z| < 2$. In particular, *all* zeros of p lie in $|z| < 2$.

Example 2: a small circle and an explicit root count

Example 1.205 (Zeros of $z^3 + z^2$ in $|z| < \frac{1}{2}$)**.** Let $f(z) = z^3$ and $g(z) = z^2$. On $|z| = \frac{1}{2}$,

$$|f(z)| = |z|^3 = \frac{1}{8}, \qquad |g(z)| = |z|^2 = \frac{1}{4}.$$

Here $|g| > |f|$, so Rouché *does not* apply in the direction (f, g).

Instead write $z^3 + z^2 = z^2(z + 1)$ and count directly: the zeros are $z = 0$ (multiplicity 2) and $z = -1$ (multiplicity 1). Inside $|z| < \frac{1}{2}$ only $z = 0$ lies inside, so there are exactly 2 zeros (counted with multiplicity) in $|z| < \frac{1}{2}$.

Remark 1.206. This example is pedagogically useful: it shows that one should not apply Rouché blindly. If the inequality fails, either choose a different contour or compare different pairs (e.g. compare z^2 vs z^3 on a different radius).

Example 3: using Rouché to show "exactly one zero"

Example 1.207 (Exactly one zero in a disk)**.** Let $q(z) = z^5 + 10z - 3$ and consider $|z| = 1$. On $|z| = 1$ we have

$$|10z - 3| \ge 10 - 3 = 7, \qquad |z^5| = 1.$$

Thus $|z^5| < |10z - 3|$ on $|z| = 1$. By Rouché, $q(z) = (10z - 3) + z^5$ has the same number of zeros in $|z| < 1$ as $10z - 3$, which has exactly one zero at $z = 3/10$. Hence q has exactly one zero in the unit disk.

Exercises

Exercise 1.208 (Rouché: choose the right dominant term)**.** Let $p(z) = z^6 + 3z^2 + 2$.

1. Show that on $|z| = 2$ one has $|3z^2 + 2| < |z^6|$.
2. Deduce that p has six zeros in $|z| < 2$.

Exercise 1.209 (Rouché on an annulus boundary)**.** Let $p(z) = z^4 + z^3 + 1$.

1. Show that p has no zeros on $|z| = 2$ and conclude that it has four zeros in $|z| < 2$.
2. Show that p has no zeros on $|z| = \frac{1}{2}$ and conclude that it has no zeros in $|z| < \frac{1}{2}$.
3. Conclude that all zeros lie in $\frac{1}{2} < |z| < 2$.

Exercise 1.210 (Counting zeros in the right half-plane (outline))**.** Let $p(z) = z^4+3z^2+1$. Use symmetry and the argument principle on a large semicircle in $\Re z \geq 0$ to determine how many zeros lie in the right half-plane. (You may assume there are no zeros on the imaginary axis.)

Exercise 1.211 (A "one zero" disk problem)**.** Show that $z^7 - 8z + 1$ has exactly one zero in $|z| < 1$. (Choose an appropriate dominant term on $|z| = 1$.)

2. Advanced Topics in Complex Analysis

2.1. Maximum and Minimum Modulus Principles and Consequences

Maximum modulus principle (local and global forms)

Theorem 2.1 (Maximum Modulus Principle)**.** *Let $\Omega \subset \mathbb{C}$ be a connected open set and let $f \in \mathcal{H}(\Omega)$. If $|f|$ attains a local maximum at some $z_0 \in \Omega$, then f is constant on Ω.*

Proof. Assume f is nonconstant. By the open mapping theorem, $f(\Omega)$ is open. In particular, $f(z_0)$ is an interior point of $f(\Omega)$, so there exists $\varepsilon > 0$ with $D_\varepsilon(f(z_0)) \subset f(\Omega)$. Thus there exists z_1 arbitrarily close to z_0 such that $f(z_1) \in D_\varepsilon(f(z_0))$ and $f(z_1) \neq f(z_0)$. Choosing such z_1 with $|f(z_1) - f(z_0)|$ small forces

$$|f(z_1)| > |f(z_0)|$$

for some nearby point z_1, contradicting that $|f|$ has a local maximum at z_0. Hence f must be constant. □

Corollary 2.2 (Boundary maximum on a bounded domain)**.** Let Ω be bounded and $f \in \mathcal{H}(\Omega) \cap C^0(\overline{\Omega})$. Then

$$\max_{\overline{\Omega}} |f| = \max_{\partial\Omega} |f|.$$

Proof. The continuous function $|f|$ attains its maximum on compact $\overline{\Omega}$. If the maximizer lies in the interior, f is constant by the maximum modulus principle, and the statement holds trivially. Otherwise the maximum lies on $\partial\Omega$. □

Minimum modulus principle

Theorem 2.3 (Minimum Modulus Principle)**.** *Let Ω be a connected open set and let $f \in \mathcal{H}(\Omega)$ be nonconstant. If f has no zeros in Ω, then $|f|$ has no local minimum in the interior of Ω.*

Proof. If f has no zeros, then $1/f \in \mathcal{H}(\Omega)$. A local minimum of $|f|$ at z_0 is equivalent to a local maximum of $|1/f|$ at z_0. By the maximum modulus principle, $1/f$ must be constant, hence f is constant, contradicting the hypothesis. □

Remark 2.4. If f has zeros, then $\min_{\overline{\Omega}} |f| = 0$ can occur (at zeros), so the zero-free hypothesis is essential.

Worked examples with explicit estimates

Example 2.5 (Maximum modulus for e^z on a disk)**.** On $\Omega = B_R(0)$ let $f(z) = e^z$. Then $|f(z)| = e^{\Re z}$. On $|z| \le R$, the inequality $\Re z \le |z| \le R$ yields

$$|e^z| \le e^R,$$

with equality at $z = R$ (and more generally at points with $\Re z = R$ on the boundary). Thus $\max_{|z|\le R} |e^z| = e^R$ and it occurs only on the boundary.

Example 2.6 (Minimum modulus for a zero-free function)**.** On $\Omega = B_1(0)$ let $f(z) = e^z$. Since $e^z \neq 0$ on $\mathbb{C}$, the minimum modulus principle implies that $|e^z|$ cannot have an interior local minimum. Indeed, $|e^z| = e^{\Re z}$ is minimized on the boundary where $\Re z$ is smallest, namely near $z = -1$.

Example 2.7 (A function with zeros: minimum equals 0)**.** Let $f(z) = z^2 + 1$ on $|z| < 2$. The zeros $z = \pm i$ lie inside the disk, so

$$\min_{|z|\le 2} |z^2 + 1| = 0,$$

and the minimum is attained exactly at $z = \pm i$.

Strong maximum principle (identity-theorem viewpoint)

Theorem 2.8 (Strong Maximum Principle for holomorphic functions)**.** *Let $f \in \mathcal{H}(\Omega)$ and suppose that $|f(z)|$ is constant on a nonempty open set $U \subset \Omega$. Then f is constant on Ω.*

Proof. Fix $z_0 \in U$ and set $M := |f(z_0)|$. Consider

$$g(z) := f(z)\,\overline{f(z_0)} - M^2.$$

The function g is holomorphic on Ω and satisfies $g(z_0) = 0$. On U we have $|f(z)| = M$, hence $f(z)\overline{f(z_0)} = M^2$ for all $z \in U$ (after rotating by a unimodular constant if desired; equivalently, note that $|f|$ constant forces f to have constant argument locally). Thus g vanishes on the open set U, so by the identity theorem $g \equiv 0$ on Ω. Hence f is constant on Ω. □

A standard application: uniqueness for the Dirichlet problem

Example 2.9 (Uniqueness on the disk (harmonic maximum principle))**.** Let u be harmonic on the unit disk and continuous on the closed disk. If u attains a maximum at an interior point, then u is constant. Indeed, on a simply connected neighborhood, take a harmonic conjugate v and set $F = u + iv$ holomorphic. Then e^F is holomorphic and

$$|e^F| = e^u.$$

A maximum of u yields a maximum of $|e^F|$, hence e^F is constant by the maximum modulus principle, so u is constant. This implies uniqueness of solutions to the Dirichlet problem on the disk.

Exercises

Exercise 2.10 (Liouville via maximum modulus on large disks)**.** Let f be entire and assume there exist $R, M > 0$ such that $|f(z)| \leq M$ for all $|z| \geq R$. Show that f is a polynomial. (Hint: apply the maximum modulus principle to f on $|z| \leq n$ and let $n \to \infty$.)

Exercise 2.11 (Minimum modulus: interior minimum forces a zero)**.** Let Ω be bounded, $f \in \mathcal{H}(\Omega) \cap C^0(\overline{\Omega})$, and assume $|f(z_0)| = \min_{\overline{\Omega}} |f|$ for some $z_0 \in \Omega$. Show that either $f(z_0) = 0$ or f is constant.

Exercise 2.12 (Coefficient bounds from boundary control)**.** Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be holomorphic in $|z| < 1$ and continuous on $\overline{\mathbb{D}}$. Assume $\max_{|z|\leq 1} |f(z)| \leq 1$.

1. Use Cauchy's integral formula to show $|a_n| \leq 1$ for all $n \geq 0$.
2. Show that if $|a_k| = 1$ for some $k \geq 1$, then $f(z) = e^{i\theta} z^k$ for some $\theta \in \mathbb{R}$.

Exercise 2.13 (A holomorphic function cannot have both a maximum and minimum inside)**.** Let $f \in \mathcal{H}(\Omega)$ on a connected domain Ω. Show that if $|f|$ attains both a local maximum and a local minimum at interior points, then f is constant.

Exercise 2.14 (Harmonic maximum principle from holomorphic maximum modulus)**.** Let u be harmonic on a bounded domain Ω and continuous on $\overline{\Omega}$. Assume Ω is simply connected.

1. Let v be a harmonic conjugate of u and set $F = u + iv$. Show that $|e^F| = e^u$.
2. Deduce the maximum principle for u: $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.

Exercise 2.15 (Strong maximum principle via identity theorem)**.** Let $f \in \mathcal{H}(\Omega)$ and assume there exists a sequence $z_n \to z_0 \in \Omega$ such that $|f(z_n)| = \sup_{\Omega} |f|$ for all n. Show that f is constant on Ω.

2.2. Schwarz Lemma, Schwarz–Pick Lemma, and the Poincaré Metric on the Disk

Goal. We develop the Schwarz lemma and its Möbius-invariant extension, the Schwarz–Pick lemma. We then introduce the Poincaré metric on $\mathbb{D}$ and explain how Schwarz–Pick is exactly the statement that holomorphic self-maps of the disk are non-expanding for the hyperbolic metric. As an application we show that disk automorphisms are precisely the hyperbolic isometries and record the explicit structure of $\mathrm{Aut}(\mathbb{D})$.

Preliminaries: the unit disk and its automorphisms

Definition 2.16 (Unit disk)**.**

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}.$$

Definition 2.17 (Möbius automorphisms of the disk)**.** For $a \in \mathbb{D}$ define

$$\varphi_a(z) := \frac{a - z}{1 - \overline{a}z}.$$

Then $\varphi_a : \mathbb{D} \to \mathbb{D}$ is biholomorphic, satisfies $\varphi_a(a) = 0$ and $\varphi_a(0) = a$, and $\varphi_a^{-1} = \varphi_a$.

Lemma 2.18 (Basic identities for φ_a)**.** *For $a \in \mathbb{D}$ and $z \in \mathbb{D}$,*

$$1 - |\varphi_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \overline{a}z|^2}, \tag{9}$$

$$\varphi_a'(z) = -\frac{1 - |a|^2}{(1 - \overline{a}z)^2}, \qquad |\varphi_a'(z)| = \frac{1 - |a|^2}{|1 - \overline{a}z|^2}. \tag{10}$$

Proof. First compute

$$|\varphi_a(z)|^2 = \frac{|a - z|^2}{|1 - \overline{a}z|^2}.$$

Then

$$1 - |\varphi_a(z)|^2 = \frac{|1 - \overline{a}z|^2 - |a - z|^2}{|1 - \overline{a}z|^2}.$$

Expand the numerator:

$$|1 - \overline{a}z|^2 = (1 - \overline{a}z)(1 - a\overline{z}) = 1 - a\overline{z} - \overline{a}z + |a|^2|z|^2,$$
$$|a - z|^2 = (a - z)(\overline{a} - \overline{z}) = |a|^2 - a\overline{z} - \overline{a}z + |z|^2.$$

Subtracting gives

$$|1 - \overline{a}z|^2 - |a - z|^2 = (1 - |a|^2)(1 - |z|^2),$$

which proves (9).

For the derivative,

$$\varphi_a(z) = \frac{a - z}{1 - \overline{a}z} \quad \Longrightarrow \quad \varphi_a'(z) = \frac{-(1 - \overline{a}z) - (a - z)(-\overline{a})}{(1 - \overline{a}z)^2} = -\frac{1 - |a|^2}{(1 - \overline{a}z)^2}.$$

Taking absolute values yields (10). □

1) Schwarz lemma

Theorem 2.19 (Schwarz lemma)**.** *Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic and assume $f(0) = 0$. Then*

$$|f(z)| \le |z| \quad (z \in \mathbb{D}), \qquad \text{and} \qquad |f'(0)| \le 1.$$

Moreover, if equality holds at some $z \neq 0$ (i.e. $|f(z)| = |z|$) or if $|f'(0)| = 1$, then there exists $\theta \in \mathbb{R}$ such that $f(z) = e^{i\theta}z$ for all z.

Proof. Define

$$g(z) := \begin{cases} \dfrac{f(z)}{z}, & z \neq 0, \\ f'(0), & z = 0. \end{cases}$$

Since $f(0) = 0$, $f(z) = z \cdot g(z)$ for $z \neq 0$. We show g is holomorphic on $\mathbb{D}$. Indeed, f has a power series $f(z) = a_1 z + a_2 z^2 + \cdots$ near 0, hence

$$\frac{f(z)}{z} = a_1 + a_2 z + a_3 z^2 + \cdots,$$

which is holomorphic near 0 and equals $a_1 = f'(0)$ at 0. Thus $g \in \mathcal{H}(\mathbb{D})$.

Now fix $0 < r < 1$. On the circle $|z| = r$ we have $|f(z)| < 1$, hence

$$|g(z)| = \left|\frac{f(z)}{z}\right| \leq \frac{|f(z)|}{r} < \frac{1}{r}.$$

This is not yet the desired bound. Instead we use the maximum modulus principle on g as follows.

For $|z| = r$,

$$|g(z)| = \left|\frac{f(z)}{z}\right| \leq \frac{|f(z)|}{|z|} \leq \frac{1}{r}.$$

Therefore $\max_{|z|=r} |g(z)| \leq 1/r$. Letting $r \uparrow 1$, we obtain $\sup_{|z|<1} |g(z)| \leq 1$. More explicitly: for any fixed z with $|z| < 1$, choose r with $|z| < r < 1$. Then by the maximum modulus principle applied to g on $|w| \leq r$,

$$|g(z)| \leq \max_{|w|=r} |g(w)| = \max_{|w|=r} \left|\frac{f(w)}{w}\right| \leq \frac{1}{r}.$$

Now let $r \downarrow 1$ to conclude $|g(z)| \leq 1$ for all $z \in \mathbb{D}$.

Hence for $z \neq 0$,

$$|f(z)| = |z|\,|g(z)| \leq |z|.$$

Also $|f'(0)| = |g(0)| \leq 1$.

For the rigidity: if $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then $|g(z_0)| = 1$. Since $|g| \leq 1$ on $\mathbb{D}$ and g attains its maximum modulus at an interior point, the maximum modulus principle forces g to be constant: $g(z) \equiv e^{i\theta}$ for some θ. Thus $f(z) = e^{i\theta} z$. Similarly, if $|f'(0)| = |g(0)| = 1$, then $|g|$ attains its maximum at 0 and again g is constant, giving the same conclusion. □

2) Schwarz–Pick lemma

Theorem 2.20 (Schwarz–Pick lemma)**.** *Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic. Then for all $z, w \in \mathbb{D}$,*

$$\left|\frac{f(z) - f(w)}{1 - \overline{f(w)}\, f(z)}\right| \leq \left|\frac{z - w}{1 - \overline{w}\, z}\right|. \tag{11}$$

Equivalently, for all $z \in \mathbb{D}$,

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}. \tag{12}$$

Moreover, equality in (11) *for some distinct $z \neq w$ (or equality in* (12) *at some point) holds iff $f \in \mathrm{Aut}(\mathbb{D})$.*

Proof. Fix $w \in \mathbb{D}$ and set $a := f(w) \in \mathbb{D}$. Consider the holomorphic map

$$F(z) := \big(\varphi_a \circ f \circ \varphi_w\big)(z).$$

We have $\varphi_w(0) = w$ and $\varphi_a(a) = 0$, so

$$F(0) = \varphi_a(f(\varphi_w(0))) = \varphi_a(f(w)) = \varphi_a(a) = 0.$$

Also $F : \mathbb{D} \to \mathbb{D}$ since each factor maps $\mathbb{D}$ to itself. By Schwarz lemma applied to F,

$$|F(z)| \le |z| \qquad (z \in \mathbb{D}).$$

Now substitute $z = \varphi_w(z_0)$ (i.e. evaluate at $z = \varphi_w(z_0)$ and rename):

$$|\varphi_a(f(z_0))| = |(\varphi_a \circ f)(z_0)| = |F(\varphi_w(z_0))| \le |\varphi_w(z_0)|.$$

Using the explicit formula for φ_a and φ_w gives exactly (11).

To derive the differential form, set $z = w + h$ and let $h \to 0$ in (11), or more cleanly differentiate the identity obtained from Schwarz lemma for F at 0. From Schwarz lemma we also have $|F'(0)| \le 1$. Compute $F'(0)$ by the chain rule:

$$F'(0) = \varphi_a'(f(w)) \cdot f'(w) \cdot \varphi_w'(0).$$

Using (10) and the fact $\varphi_w'(0) = -(1 - |w|^2)$, we get

$$|\varphi_a'(a)| = \frac{1 - |a|^2}{|1 - \overline{a}a|^2} = \frac{1 - |a|^2}{(1 - |a|^2)^2} = \frac{1}{1 - |a|^2}, \qquad |\varphi_w'(0)| = 1 - |w|^2.$$

Hence

$$|F'(0)| = \frac{1}{1 - |a|^2}\,|f'(w)|\,(1 - |w|^2) \le 1.$$

Recalling $a = f(w)$ yields (12).

For rigidity: if equality holds in (11) for some $z \neq w$, then equality holds in Schwarz lemma for F at some point, forcing F to be a rotation: $F(\zeta) = e^{i\theta}\zeta$. Then $f = \varphi_a^{-1} \circ F \circ \varphi_w^{-1}$ is a disk automorphism. Conversely, if $f \in \mathrm{Aut}(\mathbb{D})$, then composing with automorphisms preserves equality, so (11) and (12) hold with equality everywhere. □

3) The Poincaré (hyperbolic) metric and curvature

Definition 2.21 (Poincaré metric on $\mathbb{D}$)**.** The hyperbolic (Poincaré) metric on $\mathbb{D}$ is the conformal Riemannian metric

$$ds^2_{\mathbb{D}} = \lambda_{\mathbb{D}}(z)^2\,|dz|^2, \qquad \lambda_{\mathbb{D}}(z) := \frac{2}{1 - |z|^2}.$$

Remark 2.22 (Gauss curvature in conformal coordinates)**.** If a metric on a planar domain is written as $ds^2 = \lambda(z)^2|dz|^2$ with $\lambda > 0$, then its Gauss curvature is

$$K(z) = -\frac{1}{\lambda(z)^2}\,\Delta\big(\log \lambda(z)\big), \qquad \Delta = \partial_x^2 + \partial_y^2.$$

For $\lambda_{\mathbb{D}}(z) = \dfrac{2}{1 - |z|^2}$ one computes $K \equiv -1$.

4) Schwarz–Pick as hyperbolic non-expansion

Theorem 2.23 (Infinitesimal contraction)**.** *If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then for all $z \in \mathbb{D}$,*

$$\lambda_{\mathbb{D}}(f(z))\,|f'(z)| \le \lambda_{\mathbb{D}}(z).$$

Equivalently, f does not increase hyperbolic lengths of tangent vectors.

Proof. Rewrite (12) as

$$\frac{2}{1-|f(z)|^2}\,|f'(z)| \le \frac{2}{1-|z|^2},$$

which is exactly the stated inequality. □

Corollary 2.24 (Distance contraction)**.** Let $d_{\mathbb{D}}$ denote the path metric induced by $ds^2_{\mathbb{D}}$:

$$d_{\mathbb{D}}(z,w) = \inf_{\gamma} \int_0^1 \lambda_{\mathbb{D}}(\gamma(t))\,|\gamma'(t)|\,dt,$$

where the infimum is over all piecewise C^1 curves γ joining z to w. Then every holomorphic $f : \mathbb{D} \to \mathbb{D}$ satisfies

$$d_{\mathbb{D}}(f(z), f(w)) \le d_{\mathbb{D}}(z,w).$$

Proof. Let γ be any curve from z to w. Then $f \circ \gamma$ joins $f(z)$ to $f(w)$, and

$$\lambda_{\mathbb{D}}(f(\gamma(t)))\,|(f\circ\gamma)'(t)| = \lambda_{\mathbb{D}}(f(\gamma(t)))\,|f'(\gamma(t))|\,|\gamma'(t)| \le \lambda_{\mathbb{D}}(\gamma(t))\,|\gamma'(t)|$$

by infinitesimal contraction. Integrating over $t \in [0,1]$ yields

$$L_{\mathbb{D}}(f\circ\gamma) \le L_{\mathbb{D}}(\gamma).$$

Taking infimum over γ gives the distance inequality. □

5) Automorphisms are exactly the hyperbolic isometries

Theorem 2.25 (Automorphisms preserve the hyperbolic metric)**.** *If $f \in \mathrm{Aut}(\mathbb{D})$, then*

$$\lambda_{\mathbb{D}}(f(z))\,|f'(z)| = \lambda_{\mathbb{D}}(z) \qquad (z \in \mathbb{D}),$$

and consequently

$$d_{\mathbb{D}}(f(z), f(w)) = d_{\mathbb{D}}(z,w) \qquad (z,w \in \mathbb{D}).$$

Proof. From Schwarz–Pick we already have $\lambda_{\mathbb{D}}(f(z))|f'(z)| \le \lambda_{\mathbb{D}}(z)$. Apply the same inequality to $f^{-1} \in \mathrm{Aut}(\mathbb{D})$ at the point $f(z)$:

$$\lambda_{\mathbb{D}}(z)\,|(f^{-1})'(f(z))| \le \lambda_{\mathbb{D}}(f(z)).$$

Using $(f^{-1})'(f(z)) = 1/f'(z)$ gives

$$\lambda_{\mathbb{D}}(z) \le \lambda_{\mathbb{D}}(f(z))\,|f'(z)|.$$

Combine both inequalities to get equality.

For distances: for any curve γ, equality of densities implies $L_{\mathbb{D}}(f\circ\gamma) = L_{\mathbb{D}}(\gamma)$, hence the induced path distance is preserved. □

Theorem 2.26 (Explicit form of $\mathrm{Aut}(\mathbb{D})$)**.** *Every disk automorphism is of the form*

$$f(z) = e^{i\theta}\,\frac{a-z}{1-\overline{a}z}, \qquad a \in \mathbb{D},\ \theta \in \mathbb{R}.$$

In particular, $\mathrm{Aut}(\mathbb{D})$ *acts transitively on* $\mathbb{D}$*.*

Proof. Let $f \in \mathrm{Aut}(\mathbb{D})$ and set $a := f(0) \in \mathbb{D}$. Consider

$$F := \varphi_a \circ f.$$

Then F is an automorphism and $F(0) = \varphi_a(f(0)) = \varphi_a(a) = 0$. Apply Schwarz lemma to F and also to F^{-1}: since both fix 0,

$$|F(z)| \le |z| \quad \text{and} \quad |F^{-1}(z)| \le |z|.$$

Apply the second inequality to $z = F(w)$ to get $|w| \le |F(w)|$. Hence $|F(w)| = |w|$ for all w. By the rigidity part of Schwarz lemma, F must be a rotation:

$$F(z) = e^{i\theta}z.$$

Therefore

$$f = \varphi_a^{-1} \circ F = \varphi_a \circ (e^{i\theta}\,\mathrm{id}) \quad \Longrightarrow \quad f(z) = \varphi_a(e^{i\theta}z).$$

A short algebraic rearrangement yields the stated normal form $f(z) = e^{i\theta}\dfrac{a-z}{1-\overline{a}z}$ (up to reparameterizing a). Transitivity follows by choosing a so that $\varphi_a(w) = 0$ and composing with a rotation. □

Remark 2.27 (Group structure and stabilizer)**.** The stabilizer of 0 in $\mathrm{Aut}(\mathbb{D})$ consists of rotations $z \mapsto e^{i\theta}z$. Moreover, every automorphism decomposes as

$$\mathrm{Aut}(\mathbb{D}) = \{\text{translation to } 0\} \cdot \{\text{rotation}\} \cdot \{\text{translation back}\}.$$

Examples (with explicit computations)

Example 2.28 (Schwarz lemma from Schwarz–Pick)**.** If $f(0) = 0$, then (11) with $w = 0$ becomes

$$|f(z)| \le |z|.$$

The derivative form at $z = 0$ gives $|f'(0)| \le 1$.

Example 2.29 (A strict contraction)**.** Let $f(z) = \frac{1}{2}z$. Then $|f'(z)| = \frac{1}{2}$ and

$$\lambda_{\mathbb{D}}(f(z))\,|f'(z)| = \frac{2}{1-|z|^2/4}\cdot\frac{1}{2} < \frac{2}{1-|z|^2} = \lambda_{\mathbb{D}}(z),$$

so f strictly decreases hyperbolic lengths everywhere.

Example 2.30 (Automorphism is an isometry: direct check for φ_a)**.** Using (9) and (10),

$$\lambda_{\mathbb{D}}(\varphi_a(z))\,|\varphi_a'(z)| = \frac{2}{1-|\varphi_a(z)|^2}\cdot\frac{1-|a|^2}{|1-\overline{a}z|^2} = \frac{2}{\frac{(1-|a|^2)(1-|z|^2)}{|1-\overline{a}z|^2}}\cdot\frac{1-|a|^2}{|1-\overline{a}z|^2} = \frac{2}{1-|z|^2} = \lambda_{\mathbb{D}}(z).$$

Exercises

Exercise 2.31 (Rigidity in Schwarz lemma)**.** Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Assume there exists $z_0 \neq 0$ such that $|f(z_0)| = |z_0|$. Show that $f(z) = e^{i\theta} z$ for some $\theta \in \mathbb{R}$ by applying the maximum modulus principle to $g(z) = f(z)/z$.

Exercise 2.32 (Derivative form from cross-ratio form)**.** Assume (11). Fix w and set $z = w + h$. Let $h \to 0$ and prove (12) by carefully expanding $f(w+h) = f(w) + f'(w)h + o(h)$ and comparing both sides.

Exercise 2.33 (Hyperbolic distance to the origin)**.** Show that

$$d_{\mathbb{D}}(0, r) = \int_0^r \frac{2\,dt}{1 - t^2} = \log \frac{1 + r}{1 - r}, \qquad 0 \leq r < 1.$$

Exercise 2.34 (Distance formula via automorphisms)**.** Let $z, w \in \mathbb{D}$ and set $\alpha = \varphi_w(z)$. Use invariance under automorphisms to show

$$d_{\mathbb{D}}(z, w) = d_{\mathbb{D}}(\alpha, 0)$$

and deduce

$$\tanh\Big(\frac{1}{2} d_{\mathbb{D}}(z, w)\Big) = |\alpha| = \left| \frac{z - w}{1 - \overline{w} z} \right|.$$

Exercise 2.35 (Transitivity of $\mathrm{Aut}(\mathbb{D})$)**.** Given $u, v \in \mathbb{D}$, construct explicitly an $f \in \mathrm{Aut}(\mathbb{D})$ such that $f(u) = v$. (Hint: send u to 0 using φ_u, rotate, then send 0 to v.)

Exercise 2.36 (Fixed points and strict contraction)**.** Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$ and $|f'(0)| < 1$. Show that 0 is the unique fixed point of f in $\mathbb{D}$ and that $f^{\circ n}(z) \to 0$ uniformly on compact sets. (Hint: use Schwarz lemma on $f^{\circ n}$.)

2.3. Montel's Theorem and Normal Families

In this subsection we introduce the notion of a normal family of holomorphic functions and prove Montel's theorem: local boundedness implies compactness with respect to uniform convergence on compact sets. This is the fundamental compactness principle underlying many deep results in complex analysis.

Normal families and local boundedness

Definition 2.37 (Normal family)**.** Let $\Omega \subset \mathbb{C}$ be a domain and let $\mathcal{F} \subset \mathcal{O}(\Omega)$. We say that $\mathcal{F}$ is a *normal family* if every sequence $\{f_n\} \subset \mathcal{F}$ admits a subsequence that converges uniformly on compact subsets of Ω to a holomorphic limit function.

Definition 2.38 (Locally bounded family)**.** A family $\mathcal{F} \subset \mathcal{O}(\Omega)$ is *locally bounded* if for every compact set $K \Subset \Omega$ there exists a constant $M_K > 0$ such that $|f(z)| \leq M_K$ for all $z \in K$ and all $f \in \mathcal{F}$.

Cauchy estimates and equicontinuity

Lemma 2.39 (Uniform derivative bounds on compact sets)**.** *Let $\mathcal{F} \subset \mathcal{O}(\Omega)$ be locally bounded and let $K \Subset \Omega$. Then there exists a constant $C_K > 0$ such that $|f'(z)| \le C_K$ for all $z \in K$ and all $f \in \mathcal{F}$.*

Proof. Since K is compact and Ω is open, there exists $r > 0$ such that every disk $\overline{D(z,r)}$ with $z \in K$ is contained in Ω. By local boundedness, there exists $M > 0$ such that $|f(\zeta)| \le M$ for all ζ with $|\zeta - z| \le r$ and all $f \in \mathcal{F}$.

By Cauchy's integral formula for derivatives, for each $z \in K$ we have $f'(z) = \frac{1}{2\pi i} \int_{|\zeta - z| = r} \frac{f(\zeta)}{(\zeta - z)^2}\, d\zeta$. Taking absolute values yields $|f'(z)| \le \frac{M}{r}$. Setting $C_K = M/r$ proves the claim. □

Lemma 2.40 (Equicontinuity on compact sets)**.** *If $\mathcal{F} \subset \mathcal{O}(\Omega)$ is locally bounded, then it is equicontinuous on every compact subset of Ω.*

Proof. Fix a compact set $K \Subset \Omega$. By the previous lemma, there exists $C_K > 0$ such that $|f'(z)| \le C_K$ on K for all $f \in \mathcal{F}$. For any $z, w \in K$ we write $f(z) - f(w) = \int_0^1 f'(w + t(z - w))(z - w)\, dt$, hence $|f(z) - f(w)| \le C_K |z - w|$. This uniform Lipschitz bound implies equicontinuity on K. □

Montel's theorem

Theorem 2.41 (Montel)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\mathcal{F} \subset \mathcal{O}(\Omega)$ be locally bounded. Then $\mathcal{F}$ is a normal family.*

Proof. Fix a compact set $K \Subset \Omega$. Local boundedness implies uniform boundedness of $\mathcal{F}$ on a slightly larger compact set, which in turn yields uniform bounds on derivatives and equicontinuity on K. By the Arzelà–Ascoli theorem, every sequence in $\mathcal{F}$ has a subsequence converging uniformly on K.

Applying this argument to an exhaustion of Ω by compact sets and using a diagonal subsequence, we obtain a subsequence converging uniformly on every compact subset of Ω. The limit function is holomorphic by Morera's theorem. □

Examples

Example 2.42 (A normal family)**.** The family $\{f_n(z) = z^n\}$ on the unit disk is normal. Indeed, $|z^n| \le 1$ for all z in the disk, so the family is locally bounded. Moreover, for every $r < 1$ we have $\sup_{|z| \le r} |z^n| = r^n$, which converges to zero as $n \to \infty$.

Example 2.43 (A non-normal family)**.** The family $\{g_n(z) = nz\}$ on the unit disk is not normal. Near the origin the functions are unbounded, hence the family is not locally bounded and cannot be normal.

Example 2.44 (Blaschke products)**.** Finite Blaschke products satisfy $|B(z)| \le 1$ on the unit disk. Any family of such functions is locally bounded and therefore normal by Montel's theorem.

Exercises

Exercise 2.45. Decide which of the families $\{z^n\}$ and $\{nz\}$ is normal on the unit disk and justify your answer using Montel's theorem.

Exercise 2.46. Let $\mathcal{F}$ be a locally bounded family on a domain Ω and let $\{K_j\}$ be an exhaustion of Ω by compact sets. Explain carefully how to extract a diagonal subsequence converging uniformly on every K_j.

Exercise 2.47. Assume $f_n \to f$ uniformly on compact subsets of Ω, where each f_n is holomorphic. Prove that f is holomorphic by applying Morera's theorem.

2.4. Hurwitz's Theorem and the Open Mapping Theorem

This subsection explains how zeros of holomorphic functions behave under uniform convergence on compact sets. Hurwitz's theorem expresses the rigidity of zeros, and the Open Mapping Theorem follows as a geometric consequence.

Zeros and multiplicity

Definition 2.48 (Zero of multiplicity m)**.** Let f be holomorphic in a neighborhood of a point $z_0 \in \mathbb{C}$. We say that z_0 is a zero of multiplicity m if there exists a holomorphic function g with $g(z_0) \neq 0$ such that $f(z) = (z - z_0)^m g(z)$ in a neighborhood of z_0.

Hurwitz's theorem

Theorem 2.49 (Hurwitz)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $f_n \in \mathcal{O}(\Omega)$ converge uniformly on compact subsets to a holomorphic function f. Assume that f is nonconstant.*

If each f_n has no zeros in Ω, then either f has no zeros in Ω or f is identically zero.

More generally, if f has a zero of multiplicity m at a point z_0, then for every sufficiently small disk centered at z_0, all sufficiently large n satisfy that f_n has exactly m zeros in that disk, counted with multiplicity.

Proof. Let $z_0 \in \Omega$ be a zero of f of multiplicity m. Choose a radius $r > 0$ so small that $f(z) \neq 0$ for all z with $|z - z_0| = r$.

Uniform convergence of f_n to f on the closed disk implies that for all sufficiently large n we have $|f_n(z) - f(z)| < |f(z)|$ for all $|z - z_0| = r$. By Rouché's theorem, f_n and f have the same number of zeros inside the disk. Hence f_n has exactly m zeros in the disk for all large n.

Now suppose each f_n has no zeros in Ω. Then the previous conclusion forces $m = 0$. Therefore f has no zeros unless the assumption that f is nonconstant is violated. In that exceptional case, f must be identically zero. □

Remark 2.50. Hurwitz's theorem formalizes the idea that zeros of holomorphic functions are *stable under uniform convergence*. Zeros may move continuously, but they cannot appear spontaneously in the limit unless the limit function collapses to the zero function.

Examples

Example 2.51 (Convergence of simple zeros)**.** Let $f_n(z) = z + \frac{1}{n}$. Then f_n converges uniformly on compact sets to $f(z) = z$. Each f_n has a simple zero at $z = -\frac{1}{n}$, and these zeros converge to the zero of the limit function.

Example 2.52 (Degenerate limit)**.** Let $f_n(z) = \frac{z}{n}$. Each f_n has a zero at the origin, and f_n converges uniformly on compact sets to the zero function. This illustrates the exceptional case allowed by Hurwitz's theorem.

The Open Mapping Theorem

Theorem 2.53 (Open Mapping Theorem)**.** *Let $\Omega \subset \mathbb{C}$ be a domain. Every nonconstant holomorphic function $f : \Omega \to \mathbb{C}$ maps open sets to open sets.*

Proof. Fix a point $z_0 \in \Omega$. Since f is holomorphic and nonconstant, its Taylor expansion at z_0 has the form $f(z) = f(z_0) + a_m(z - z_0)^m +$ higher order terms with $m \geq 1$ and $a_m \neq 0$.

Choose a small disk U centered at z_0 such that $f(z) \neq f(z_0)$ for all z on the boundary of U. Define $f_n(z) = f(z) + \frac{1}{n}$. Then $f_n(z_0) \neq f(z_0)$ for all n, so $f_n - f(z_0)$ has no zeros in U.

As $n \to \infty$, the functions $f_n - f(z_0)$ converge uniformly on compact subsets of U to $f - f(z_0)$. By Hurwitz's theorem, the limit function must either be identically zero or have zeros in U. Since f is nonconstant, the first possibility is excluded. Hence $f - f(z_0)$ has zeros arbitrarily close to z_0.

This implies that values arbitrarily close to $f(z_0)$ are attained by f in every neighborhood of z_0. Therefore $f(U)$ contains an open neighborhood of $f(z_0)$, and f is an open map. □

Further examples

Example 2.54 (Failure for constant functions)**.** A constant function maps every open set to a single point, which is not open. Thus the nonconstancy assumption in the Open Mapping Theorem is essential.

Example 2.55 (Complex versus real analysis)**.** A real differentiable function may fail to be open even if it is smooth and nonconstant. The open mapping phenomenon is therefore genuinely complex and reflects the rigidity imposed by holomorphicity.

Exercises

Exercise 2.56. Let $f_n \in \mathcal{O}(\Omega)$ converge uniformly on compact sets to a nonconstant f. Assume that all zeros of f_n lie in a fixed compact subset of Ω. Show that every zero of f is the limit of a sequence of zeros of f_n.

Exercise 2.57. Let $f_n : \Omega \to \mathbb{C}$ be injective and holomorphic, and assume that f_n converges uniformly on compact subsets to a holomorphic function f. Show that f is either injective or constant.

Exercise 2.58. Use the Open Mapping Theorem to derive the Maximum Modulus Principle for a nonconstant holomorphic function on a bounded domain with continuous boundary values.

Exercise 2.59. Let $f \in \mathcal{O}(\Omega)$ and suppose $f(z_0) = w_0$ with multiplicity $m \geq 1$. Show that there exist neighborhoods of z_0 and w_0 such that every $w \neq w_0$ near w_0 has exactly m preimages near z_0, counted with multiplicity.

2.5. Picard Theorems and Essential Singularities

This subsection studies the extreme value-distribution behavior near an essential singularity. We prove the Casorati–Weierstrass theorem and then state the Little and Great Picard theorems, emphasizing how they sharpen the classification of isolated singularities. Throughout, we complement the theorems with explicit computations for classical examples such as $e^{1/z}$ and $\sin(1/z)$.

Essential singularities

Definition 2.60 (Isolated singularity and essential singularity)**.** Let f be holomorphic on a punctured disk $\{z \in \mathbb{C} : 0 < |z - z_0| < r\}$. We say that z_0 is an *isolated singularity* of f.

Write the Laurent expansion of f about z_0 as $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ on $0 < |z - z_0| < r$. Then:

- z_0 is *removable* iff $a_n = 0$ for all $n < 0$;
- z_0 is a *pole of order* m iff $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$;
- z_0 is *essential* iff $a_n \neq 0$ for infinitely many $n < 0$.

Example 2.61 (A classical essential singularity)**.** The function $f(z) = e^{1/z}$ has an essential singularity at $z = 0$. Indeed, the Taylor series $e^w = \sum_{n=0}^{\infty} \frac{w^n}{n!}$ gives the Laurent expansion $e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$, which contains infinitely many negative powers.

Casorati–Weierstrass

Theorem 2.62 (Casorati–Weierstrass)**.** *Let f be holomorphic on $0 < |z - z_0| < r$ and suppose z_0 is an essential singularity. Then for every $\rho \in (0, r)$ the set $f(\{z : 0 < |z - z_0| < \rho\})$ is dense in $\mathbb{C}$.*

Equivalently: for every $w \in \mathbb{C}$ and every $\varepsilon > 0$, for every $\rho \in (0, r)$ there exists z with $0 < |z - z_0| < \rho$ such that $|f(z) - w| < \varepsilon$.

Proof. Fix $\rho \in (0, r)$. Suppose, toward a contradiction, that $f(\{z : 0 < |z - z_0| < \rho\})$ is *not* dense in $\mathbb{C}$. Then there exist $w_0 \in \mathbb{C}$ and $\varepsilon_0 > 0$ such that $|f(z) - w_0| \geq \varepsilon_0$ for all z with $0 < |z - z_0| < \rho$.

Define $g(z) = \frac{1}{f(z) - w_0}$ on $0 < |z - z_0| < \rho$. Then g is holomorphic on the punctured disk and satisfies $|g(z)| \leq \frac{1}{\varepsilon_0}$ there, hence g is bounded near z_0. By the Removable Singularity Theorem, g extends holomorphically to z_0. Let $\widetilde{g}$ denote the extension to $|z - z_0| < \rho$.

Now consider two cases.

Case 1: $\widetilde{g}(z_0) \neq 0$**.** Then $\widetilde{g}$ is nonzero in a small neighborhood of z_0, so $\frac{1}{\widetilde{g}}$ is holomorphic near z_0. On $0 < |z - z_0| < \rho$ we have $f(z) - w_0 = \frac{1}{g(z)} = \frac{1}{\widetilde{g}(z)}$, hence $f(z) = w_0 + \frac{1}{\widetilde{g}(z)}$ extends holomorphically across z_0. Thus z_0 is removable for f, contradiction.

Case 2: $\widetilde{g}(z_0) = 0$**.** Then z_0 is a zero of $\widetilde{g}$ of some finite order $m \geq 1$, so $\widetilde{g}(z) = (z - z_0)^m h(z)$ with h holomorphic and $h(z_0) \neq 0$. Hence $1/\widetilde{g}$ has a pole of order m at z_0, and $f(z) = w_0 + \frac{1}{\widetilde{g}(z)}$ extends meromorphically across z_0 with a pole. Thus z_0 is a pole for f, contradiction.

In all cases we contradict the assumption that z_0 is essential. Therefore $f(\{z : 0 < |z - z_0| < \rho\})$ must be dense in $\mathbb{C}$. □

Concrete value computations near an essential singularity

Example 2.63 (Explicit solutions for $e^{1/z} = w$ and accumulation at 0)**.** Fix $w \in \mathbb{C}$ with $w \neq 0$. Solving $e^{1/z} = w$ is equivalent to $1/z \in \mathrm{Log}(w)$, where $\mathrm{Log}(w) = \{\log|w| + i(\arg w + 2\pi k) : k \in \mathbb{Z}\}$. Thus the solutions are $z_k = \frac{1}{\log|w| + i(\arg w + 2\pi k)}$, $k \in \mathbb{Z}$. As $|k| \to \infty$, the denominator has modulus asymptotic to $2\pi|k|$, so $|z_k| \to 0$. Hence $e^{1/z}$ takes every nonzero value infinitely often in any punctured neighborhood of 0.

Note that $e^{1/z}$ never takes the value 0, so this example also illustrates the possibility of omitting exactly one value.

Example 2.64 (Zeros of $\sin(1/z)$ accumulate at the essential singularity)**.** Let $f(z) = \sin(1/z)$. Then $z = 0$ is essential because $\sin w = \sum_{n=0}^{\infty}(-1)^n \frac{w^{2n+1}}{(2n+1)!}$ gives $\sin(1/z) = \sum_{n=0}^{\infty}(-1)^n \frac{1}{(2n+1)!} z^{-(2n+1)}$, with infinitely many negative powers.

Moreover, $f(z) = 0$ iff $1/z = \pi k$ for some $k \in \mathbb{Z} \setminus \{0\}$, i.e. $z_k = \frac{1}{\pi k}$. Then $z_k \to 0$ as $|k| \to \infty$. So 0 is taken infinitely often near the essential singularity.

Picard theorems

Theorem 2.65 (Little Picard)**.** *If $f : \mathbb{C} \to \mathbb{C}$ is a nonconstant entire function, then $f(\mathbb{C})$ omits at most one complex value. Equivalently, there exists at most one $w_0 \in \mathbb{C}$ such that $f(z) \neq w_0$ for all $z \in \mathbb{C}$.*

Remark 2.66. A prototypical example is $f(z) = e^z$, which omits exactly one value, namely 0, and takes every other value infinitely often.

Theorem 2.67 (Great Picard)**.** *Let f be holomorphic on $0 < |z - z_0| < r$ with an essential singularity at z_0. Then in every punctured neighborhood of z_0, the function f assumes every complex value infinitely often, with at most one possible exception.*

Remark 2.68. Great Picard strengthens Casorati–Weierstrass in two ways: (i) it upgrades density to actual attainment of values, and (ii) it asserts infinite repetition of values arbitrarily close to the singularity.

A useful corollary: classification of isolated singularities

Proposition 2.69 (Picard-flavored characterization)**.** *Let f be holomorphic on $0 < |z - z_0| < r$.*

- *If f is bounded near z_0, then z_0 is removable.*
- *If $|f(z)| \to \infty$ as $z \to z_0$, then z_0 is a pole.*
- *If f omits two distinct complex values in some punctured neighborhood of z_0, then z_0 cannot be essential; it is removable or a pole.*

Proof. The first two items are the classical classification of isolated singularities using the Riemann Removable Singularity Theorem and the definition of poles.

For the third item, assume f omits two distinct values $a \neq b$ near z_0. Consider the Möbius map $\phi(w) = \frac{w-a}{w-b}$, which sends a to 0 and b to ∞. Then $h = \phi \circ f$ is holomorphic on the punctured neighborhood and omits both 0 and ∞. Omitting ∞ means h is bounded, and omitting 0 means $1/h$ is bounded. In particular, h is bounded near z_0, so h extends holomorphically across z_0. Since ϕ is biholomorphic on $\mathbb{C} \setminus \{b\}$, the extension of h forces $f = \phi^{-1} \circ h$ to extend meromorphically across z_0, hence z_0 is not essential. □

Exercises

Exercise 2.70 (Essential singularity of $\sin(1/z)$ and explicit zeros)**.** Let $f(z) = \sin(1/z)$.

1. Prove that $z = 0$ is an essential singularity of f by writing its Laurent expansion.
2. Show that $f(z) = 0$ has infinitely many solutions z_k with $z_k \to 0$ and find them explicitly.
3. Show that f takes the values ± 1 infinitely often near 0 by solving $\sin(1/z) = \pm 1$.

Exercise 2.71 (Casorati–Weierstrass versus removable/pole)**.** Let f be holomorphic on $0 < |z| < 1$ and assume $f(0 < |z| < 1)$ is contained in a horizontal line $\{w : \Im w = c\}$ for some $c \in \mathbb{R}$.

1. Show that f is bounded near 0.
2. Conclude that 0 cannot be an essential singularity of f.
3. Explain how this is consistent with Casorati–Weierstrass.

Exercise 2.72 (Essential singularity at infinity)**.** Let f be entire and nonconstant, and define $g(z) = f(1/z)$ on $0 < |z| < 1$.

1. Show that if f is a polynomial of degree $d \geq 1$, then $z = 0$ is a pole of order d for g.
2. Show that if f is not a polynomial, then $z = 0$ is an essential singularity for g.
3. Interpret this as: every non-polynomial entire function has an essential singularity at ∞.

Exercise 2.73 (From Great Picard to Little Picard at infinity)**.** Assume the Great Picard Theorem. Let f be a nonconstant entire function and consider $g(z) = f(1/z)$. Explain carefully why Great Picard applied at $z = 0$ implies that f omits at most one value in $\mathbb{C}$.

Exercise 2.74 (A value-omission test near a suspected essential singularity)**.** Let f be holomorphic on $0 < |z| < 1$. Assume there exist two values $a \neq b$ such that $f(z) \neq a$ and $f(z) \neq b$ for all $0 < |z| < \rho$ for some $\rho \in (0, 1)$. Show that $z = 0$ is not an essential singularity of f.

Exercise 2.75 (A sharpened computation for $e^{1/z}$)**.** Fix $w \neq 0$. Let $z_k = \frac{1}{\log|w| + i(\arg w + 2\pi k)}$.

1. Show that $e^{1/z_k} = w$ for all $k \in \mathbb{Z}$.
2. Show that $|z_k| \leq \frac{1}{2\pi|k|}$ for all sufficiently large $|k|$.
3. Conclude that the solutions accumulate at 0 at least as fast as $1/|k|$.

Exercise 2.76 (Doubly periodic entire functions)**.** Let f be entire and doubly periodic with respect to a lattice $\Lambda \subset \mathbb{C}$.

1. Show that f is bounded on a fundamental parallelogram for Λ.
2. Conclude that f is bounded on $\mathbb{C}$ and hence constant by Liouville's theorem.
3. Explain why nonconstant doubly periodic functions must therefore be meromorphic with poles.

2.6. Infinite Products and Weierstrass Factorization

In this subsection we develop the theory of infinite products (numbers and holomorphic functions) and prove the Weierstrass Factorization Theorem. Infinite products let us build entire functions from prescribed zero sets, generalizing the factorization of polynomials. The central analytic issue is *compact-uniform convergence* of the partial products, which guarantees holomorphy of the limit.

1. Infinite products of complex numbers

Definition 2.77 (Infinite product)**.** Let $\{a_n\}_{n\geq 1}$ be a sequence of complex numbers with $a_n \neq 0$. The infinite product $\prod_{n=1}^{\infty} a_n$ is said to *converge* if the partial products $P_N := \prod_{n=1}^{N} a_n$ converge to a *nonzero* complex number. If $P_N \to 0$, the product is said to *diverge to zero*. Otherwise it is *divergent*.

Remark 2.78 (A necessary condition)**.** If $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit, then necessarily $a_n \to 1$. Indeed, $a_N = P_N/P_{N-1} \to L/L = 1$ if $P_N \to L \neq 0$.

Proposition 2.79 (Logarithmic criterion near 1)**.** *Assume $a_n \to 1$. Then there exists N_0 such that for all $n \geq N_0$, the principal branch $\operatorname{Log}(a_n)$ is well-defined (after restricting to a small disk around 1 that avoids the branch cut). In this situation, $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit if and only if $\sum_{n=N_0}^{\infty} \operatorname{Log}(a_n)$ converges in $\mathbb{C}$.*

Proof. Since $a_n \to 1$, choose $\delta \in (0,1)$ and N_0 so that $|a_n - 1| < \delta$ for all $n \geq N_0$ and the disk $\{w : |w-1| < \delta\}$ does not intersect the branch cut of Log. Then $\operatorname{Log}(a_n)$ is defined for $n \geq N_0$ and is continuous.

For $N \geq N_0$ we have the identity $\operatorname{Log}\Big(\prod_{n=N_0}^{N} a_n\Big) = \sum_{n=N_0}^{N} \operatorname{Log}(a_n)$, because all partial products stay in the same simply connected neighborhood of 1 (after shrinking δ if needed), so no winding around 0 occurs and the branch is consistent.

If $\sum_{n=N_0}^{\infty} \operatorname{Log}(a_n)$ converges, then $S_N := \sum_{n=N_0}^{N} \operatorname{Log}(a_n)$ converges to some S, hence $\prod_{n=N_0}^{N} a_n = \exp(S_N) \to \exp(S) \neq 0$. Multiplying by the finite constant $\prod_{n=1}^{N_0-1} a_n \neq 0$ yields convergence of $\prod_{n=1}^{\infty} a_n$.

Conversely, if $\prod_{n=1}^{\infty} a_n$ converges to $L \neq 0$, then $\prod_{n=N_0}^{N} a_n \to L/\prod_{n=1}^{N_0-1} a_n \neq 0$. Taking Log (again valid because the tail partial products remain in the same neighborhood avoiding the cut), we get $S_N = \operatorname{Log}(\prod_{n=N_0}^{N} a_n)$ converges, hence the series converges. □

Remark 2.80 (A practical sufficient condition)**.** If $a_n = 1 + b_n$ and $\sum_{n=1}^{\infty} |b_n| < \infty$ with $|b_n|$ eventually small, then $\sum \operatorname{Log}(1+b_n)$ converges and the product converges. This is often used with the estimate $|\operatorname{Log}(1+b)| \leq C|b|$ for $|b|$ small.

2. Infinite products of holomorphic functions

Definition 2.81 (Compact-uniform convergence of products)**.** Let $\Omega \subset \mathbb{C}$ be a domain and let u_n be holomorphic on Ω. We say the product $\prod_{n=1}^{\infty} u_n(z)$ converges *uniformly on compact sets* to a nonzero holomorphic function if for every compact $K \Subset \Omega$ there exists $N(K)$ such that $u_n(z) \neq 0$ on K for all $n \geq N(K)$ and the partial products $P_N(z) = \prod_{n=1}^{N} u_n(z)$ converge uniformly on K to a limit $P(z)$ that never vanishes on K.

Proposition 2.82 (Weierstrass M-test for products)**.** *Let Ω be a domain and u_n holomorphic on Ω. Assume that for every compact $K \Subset \Omega$ there exist numbers $M_n(K) \geq 0$*

such that $\sup_{z\in K}|u_n(z)-1|\le M_n(K)$ *and* $\sum_{n=1}^{\infty}M_n(K)<\infty$. *Then* $\prod_{n=1}^{\infty}u_n(z)$ *converges uniformly on* K *to a holomorphic limit. Moreover the limit is either identically* 0 *or never vanishes on* K.

Proof. Fix compact $K \Subset \Omega$. Since $\sum M_n(K)<\infty$, we have $M_n(K)\to 0$. Hence for n large enough, $\sup_K|u_n-1|\le 1/2$, so $|u_n|\ge 1/2$ on K for large n. Thus the tail factors have no zeros on K.

For n large with $\sup_K|u_n-1|\le 1/2$, define $v_n(z)=\mathrm{Log}(u_n(z))$ using the principal branch on $\{w:|w-1|<1/2\}$, which is valid because $u_n(K)$ lies in that disk. We have the estimate $|v_n(z)|\le C|u_n(z)-1|$ uniformly on K for a universal constant C (e.g. from the power series of $\mathrm{Log}(1+\xi)$ for $|\xi|<1/2$). Hence $\sum\sup_K|v_n|<\infty$.

Therefore the series $\sum v_n(z)$ converges uniformly on K to a holomorphic function $V(z)$ (uniform limit of holomorphic partial sums). Exponentiating gives uniform convergence of the product of the tail: $\prod_{n\ge N}u_n(z)=\exp(\sum_{n\ge N}v_n(z))\to\exp(V(z))$. Multiplying by the finite initial product yields convergence of $\prod_{n=1}^{\infty}u_n(z)$.

Finally, either the limit vanishes identically (this can happen only if the partial products tend uniformly to 0 on K), or else it is nonvanishing because it is an exponential times a nonzero constant on K. □

3. Canonical factors

Definition 2.83 (Weierstrass canonical factors)**.** For $p\in\mathbb{N}\cup\{0\}$ define the canonical factor of genus p by $E_0(w)=1-w$ and, for $p\ge 1$, $E_p(w)=(1-w)\exp\big(w+\frac{w^2}{2}+\cdots+\frac{w^p}{p}\big)$.

Lemma 2.84 (Key cancellation estimate)**.** *Fix* $p\in\mathbb{N}\cup\{0\}$. *There exist constants* $C_p>0$ *and* $\delta_p\in(0,1)$ *such that for all* w *with* $|w|\le\delta_p$, $|\mathrm{Log}(E_p(w))|\le C_p|w|^{p+1}$. *In particular,* $E_p(w)=1-w+O(|w|^{p+1})$ *as* $w\to 0$.

Proof. For $|w|$ small, $\mathrm{Log}(1-w)=-\sum_{k=1}^{\infty}\frac{w^k}{k}$. By definition, $\mathrm{Log}(E_p(w))=\mathrm{Log}(1-w)+w+\frac{w^2}{2}+\cdots+\frac{w^p}{p}$. Substituting the series for $\mathrm{Log}(1-w)$, the terms $-w,-w^2/2,\dots,-w^p/p$ cancel, leaving $\mathrm{Log}(E_p(w))=-\sum_{k=p+1}^{\infty}\frac{w^k}{k}$. Hence for $|w|\le 1/2$, $|\mathrm{Log}(E_p(w))|\le\sum_{k=p+1}^{\infty}\frac{|w|^k}{k}\le\sum_{k=p+1}^{\infty}|w|^k=\frac{|w|^{p+1}}{1-|w|}\le 2|w|^{p+1}$. This gives the desired bound with $C_p=2$ and $\delta_p=1/2$. Exponentiating yields $E_p(w)=\exp(\mathrm{Log}(E_p(w)))=1-w+O(|w|^{p+1})$. □

4. Convergence of canonical products

Proposition 2.85 (Canonical products converge on compacts)**.** *Let* $\{a_n\}$ *be a sequence of nonzero complex numbers with* $|a_n|\to\infty$. *Fix an integer* $p\ge 0$ *and assume* $\sum_{n=1}^{\infty}|a_n|^{-(p+1)}<\infty$. *Define* $P_N(z)=\prod_{n=1}^{N}E_p\big(\frac{z}{a_n}\big)$. *Then* P_N *converges uniformly on compact subsets of* $\mathbb{C}$ *to an entire function* $P(z)$. *Moreover,* P *vanishes precisely at the points* $z=a_n$ *(with multiplicities matching repetitions in the sequence) and has no other zeros.*

Proof. Fix a compact set $K\Subset\mathbb{C}$ and set $R=\sup_{z\in K}|z|$. Since $|a_n|\to\infty$, choose N_0 so that $|z/a_n|\le 1/2$ for all $z\in K$ and all $n\ge N_0$. Then the lemma applies to $w=z/a_n$ and gives $|\mathrm{Log}(E_p(z/a_n))|\le C_p|z/a_n|^{p+1}\le C_pR^{p+1}|a_n|^{-(p+1)}$ uniformly on K.

Therefore the series $\sum_{n\ge N_0}\mathrm{Log}(E_p(z/a_n))$ converges uniformly on K by the Weierstrass M-test, because $\sum|a_n|^{-(p+1)}<\infty$. Its sum is holomorphic on a neighborhood of K. Exponentiating shows the tail product $\prod_{n\ge N_0}E_p(z/a_n)$ converges uniformly on K to a holomorphic nonvanishing limit. Multiplying by the finite initial product $\prod_{n=1}^{N_0-1}E_p(z/a_n)$

gives uniform convergence of P_N on K to a holomorphic limit. Since K was arbitrary, the limit is entire.

For zeros: each factor $E_p(z/a_n)$ has a simple zero at $z = a_n$ (because $E_p(w)$ has a simple zero at $w = 1$ coming from the factor $(1 - w)$). The uniform convergence on compacts away from the set $\{a_n\}$ implies no new zeros can appear off $\{a_n\}$, while each a_n is indeed a zero. Multiplicity is handled by repeating points in the list according to multiplicity. □

5. Weierstrass Factorization Theorem

Theorem 2.86 (Weierstrass Factorization Theorem)**.** *Let f be an entire function, not identically zero. Let $m \geq 0$ be the multiplicity of the zero of f at 0, and let $\{a_n\}$ be the (finite or infinite) multiset of nonzero zeros of f, listed with multiplicities and with no accumulation point in $\mathbb{C}$. Then there exist integers $p_n \geq 0$ and an entire function g such that $f(z) = z^m \exp(g(z)) \prod_{n=1}^{\infty} E_{p_n}\big(\frac{z}{a_n}\big)$, where the product is interpreted as 1 if there are finitely many zeros. Conversely, any expression of this form defines an entire function with the prescribed zeros.*

Proof. Step 1: Choose genera p_n so that the canonical product converges. Because $|a_n| \to \infty$ and there is no finite accumulation point of zeros, we may choose p_n inductively so that $\sum_{n=1}^{\infty} |a_n|^{-(p_n+1)} < \infty$. For instance, one can pick p_n so large that $|a_n|^{-(p_n+1)} \leq 2^{-n}$, which makes the series converge absolutely.

Define the canonical product $P(z) = z^m \prod_{n=1}^{\infty} E_{p_n}\big(\frac{z}{a_n}\big)$. By the previous proposition (applied termwise with varying p_n using the same compact-uniform argument), P is entire and has zeros exactly at 0 (order m) and at the points a_n with the prescribed multiplicities.

Step 2: The quotient f/P is entire and has no zeros. Define $h(z) = f(z)/P(z)$. This is holomorphic on $\mathbb{C} \setminus \{0, a_1, a_2, \dots\}$. Near any zero a_n, both f and P vanish to the same order, so the singularity of h at a_n is removable. Similarly at 0. Hence h extends holomorphically to all of $\mathbb{C}$, so h is entire. By construction, h has no zeros.

Step 3: A nonvanishing entire function is an exponential. Since h is entire and never vanishes, the logarithmic derivative $h'(z)/h(z)$ is entire. Choose a base point $z_0 \in \mathbb{C}$ and define $g(z) = \int_{z_0}^{z} \frac{h'(\zeta)}{h(\zeta)}\, d\zeta$, where the integral is taken along any piecewise C^1 path in $\mathbb{C}$. This is well-defined because $\mathbb{C}$ is simply connected and h'/h is entire, so the integral is path-independent. Then g is entire and $\exp(g(z))$ satisfies $(\exp g)'/(\exp g) = g' = h'/h$, hence $h/\exp(g)$ is constant. Adjusting g by an additive constant, we obtain $h(z) = \exp(g(z))$.

Combining steps yields $f(z) = P(z)\exp(g(z))$, which is the desired factorization. The converse direction is immediate from the convergence proposition and the fact that $\exp(g)$ has no zeros. □

6. Classical examples

Example 2.87 (Sine product)**.** The zeros of $\sin(\pi z)$ are the integers $\mathbb{Z}$, all simple. One convenient factorization is $\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \big(1 - \frac{z^2}{n^2}\big)$. Here the factors are genus 0 because $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$ ensures compact-uniform convergence of the product on $\mathbb{C}$.

Example 2.88 (Euler product for $1/\Gamma$)**.** Euler's product can be written as $\frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{n=1}^{\infty} \big(1 + \frac{z}{n}\big) e^{-z/n}$, where γ is the Euler–Mascheroni constant. This is a canonical product using genus 1 factors since $\big(1 + \frac{z}{n}\big) e^{-z/n} = E_1(-z/n)$.

Exercises

Exercise 2.89 (Asymptotics of E_1)**.** Compute $E_1(z) = (1-z)e^z$ and show that $E_1(z) = 1 - \frac{z^2}{2} + O(|z|^3)$ as $z \to 0$. Deduce that if $\prod_{n=1}^{\infty} E_1(z/a_n)$ is to converge compact-uniformly, it is natural to require $\sum_{n=1}^{\infty} |a_n|^{-2} < \infty$.

Exercise 2.90 (Entire function with finitely many zeros)**.** Let f be entire and assume it has only finitely many zeros (counted with multiplicity). Show that $f(z) = e^{g(z)}P(z)$ where P is a polynomial and g is entire.

Exercise 2.91 (A simple canonical product)**.** Show that $f(z) = \prod_{n=1}^{\infty}\left(1 + \frac{z}{n^2}\right)$ defines an entire function. Hint: use the product M-test on compact sets with $u_n(z) = 1 + \frac{z}{n^2}$ and $\sup_{z\in K}|u_n(z) - 1| \le R/n^2$.

Exercise 2.92 (Uniqueness up to exponentials)**.** Let f and g be entire functions with the same zeros (including multiplicity). Use Weierstrass factorization to show there exists an entire h with $f(z) = e^{h(z)}g(z)$.

Exercise 2.93 (Fixed genus for finite order: Hadamard style)**.** Assume f is an entire function of finite order $\rho < \infty$ and its zeros $\{a_n\}$ satisfy $\#\{n : |a_n| \le R\} = O(R^{\rho})$ as $R \to \infty$. Explain why one may choose a single integer $p \ge \rho$ and write a factorization of f using canonical factors $E_p(z/a_n)$ of fixed genus p.

2.7. Entire Functions: Order and Type

This subsection develops the growth theory of entire functions. Our goal is to quantify how fast an entire function can grow at infinity and to classify entire functions via *order* and *type*. These notions are central in Hadamard factorization and in the analytic study of special functions (Gamma, Bessel, zeta, ...).

1. Maximum modulus and basic properties

Definition 2.94 (Maximum modulus)**.** Let f be entire. For $r \ge 0$ define $M_f(r) := \max_{|z|=r}|f(z)|$. When f is clear we simply write $M(r)$.

Proposition 2.95 (Monotonicity of $M(r)$)**.** *For an entire function f, the function $r \mapsto M(r)$ is nondecreasing. Moreover, $M(r) = \max_{|z|\le r}|f(z)|$.*

Proof. Fix $0 \le r < R$. By the maximum modulus principle applied to f on the disk $\{z : |z| \le R\}$, the maximum of $|f|$ on that closed disk is achieved on the boundary circle $|z| = R$. Hence $\max_{|z|\le r}|f(z)| \le \max_{|z|\le R}|f(z)| = \max_{|z|=R}|f(z)| = M(R)$. Taking maxima over $|z| = r$ shows $M(r) \le M(R)$ and also $M(r) = \max_{|z|\le r}|f(z)|$. □

2. Order and equivalent growth tests

Definition 2.96 (Order)**.** The *order* ρ of an entire function f is $\rho := \limsup_{r\to\infty} \frac{\log\log M(r)}{\log r}$, with the convention $\log\log M(r) = -\infty$ if $M(r) \le 1$. If $\rho < \infty$, we say f has *finite order*.

Remark 2.97 (What $\log\log M(r)$ measures)**.** Roughly, $\log M(r)$ measures exponential growth, and $\log\log M(r)$ measures *power* growth of the exponent. For instance: polynomials satisfy $\log M(r) \sim n\log r$ so $\log\log M(r) \sim \log\log r$, hence order 0, while e^{az} satisfies $\log M(r) \sim |a|r$ so $\log\log M(r) \sim \log r$, hence order 1.

Proposition 2.98 (Equivalent characterization of order)**.** *Let f be entire and $\rho \in [0,\infty]$. Then ρ is the order of f if and only if:*

1. *for every $\alpha > \rho$ there exists R_α such that $M(r) \le \exp(r^\alpha)$ for all $r \ge R_\alpha$;*
2. *for every $\beta < \rho$ and every $C > 0$ there exist arbitrarily large r such that $M(r) \ge \exp(Cr^\beta)$.*

Proof. Assume $\rho < \infty$ first. If $\alpha > \rho$, then by definition of limsup there exists R_α such that $\frac{\log\log M(r)}{\log r} \le \alpha$ for all $r \ge R_\alpha$. Equivalently $\log\log M(r) \le \alpha \log r$, so $\log M(r) \le r^\alpha$ and $M(r) \le \exp(r^\alpha)$ for $r \ge R_\alpha$.

Conversely, if $\beta < \rho$, then by definition of limsup there exist arbitrarily large r for which $\frac{\log\log M(r)}{\log r} \ge \beta + \varepsilon$ for some $\varepsilon > 0$ with $\beta + \varepsilon < \rho$. Then $\log\log M(r) \ge (\beta+\varepsilon)\log r$, hence $\log M(r) \ge r^{\beta+\varepsilon}$. Given $C > 0$, for sufficiently large such r we have $r^{\beta+\varepsilon} \ge Cr^\beta$, so $M(r) \ge \exp(Cr^\beta)$ along an unbounded sequence. The case $\rho = \infty$ is similar, and $\rho = 0$ reduces to the statement that $\log\log M(r) = o(\log r)$. □

Proposition 2.99 (A simple comparison criterion)**.** *Let f be entire. If there exists $\alpha > 0$ and R_0 such that $|f(z)| \le \exp(|z|^\alpha)$ for all $|z| \ge R_0$, then the order of f satisfies $\rho \le \alpha$.*

Proof. For $r \ge R_0$, the hypothesis gives $M(r) \le \exp(r^\alpha)$. Hence $\log\log M(r) \le \log(r^\alpha) = \alpha \log r$, and dividing by $\log r$ and taking lim sup yields $\rho \le \alpha$. □

3. Type (for finite positive order)

Definition 2.100 (Type)**.** Assume f has finite order $\rho \in (0,\infty)$. The *type* σ of f (with respect to ρ) is $\sigma := \limsup_{r\to\infty} \frac{\log M(r)}{r^\rho}$. If $0 < \sigma < \infty$, f is of *normal type*. If $\sigma = 0$ it is of *minimal type*, and if $\sigma = \infty$ it is of *infinite type*.

Remark 2.101 (Type depends on the chosen order)**.** Type is only defined after fixing the order ρ. For example, if f has order 1 then $\log M(r)$ is comparable to r up to $r^{1+\varepsilon}$ errors; type refines this by capturing the leading constant in front of r when such a constant exists.

4. Detailed computations in basic examples

Example 2.102 (Exponentials e^{az})**.** Let $f(z) = e^{az}$ with $a \ne 0$. For $|z| = r$ we have $|e^{az}| = e^{\Re(az)} \le e^{|a||z|} = e^{|a|r}$, and equality occurs at $z = r\cdot\overline{a}/|a|$ (so that $az = |a|r$ is real). Thus $M(r) = e^{|a|r}$. Then $\log M(r) = |a|r$ and $\log\log M(r) = \log(|a|r) = \log r + O(1)$, so $\rho = \limsup_{r\to\infty} \frac{\log r + O(1)}{\log r} = 1$. Moreover $\sigma = \limsup_{r\to\infty} \frac{\log M(r)}{r} = \lim_{r\to\infty} \frac{|a|r}{r} = |a|$.

Example 2.103 (Polynomials and the "order 0" phenomenon)**.** Let $f(z) = a_n z^n + \cdots + a_0$ with $a_n \ne 0$. For $|z| = r$, $|f(z)| \le |a_n|r^n + \cdots + |a_0| \le Cr^n$ for $r \ge 1$, hence $M(r) \le Cr^n$. Also, choosing $z = re^{i\theta}$ with θ such that $a_n z^n$ has modulus $|a_n|r^n$ and arg aligned with the remainder (or simply using $|a_n z^n| - \sum_{k<n}|a_k|r^k$), we get for r large that $M(r) \ge \frac{1}{2}|a_n|r^n$. Thus $M(r) \asymp r^n$ and $\log M(r) = n\log r + O(1)$. Hence $\log\log M(r) = \log(n\log r + O(1)) = \log\log r + O(1)$ and therefore $\rho = 0$. A key theorem (proved below as an exercise) says *order* 0 is equivalent to *polynomial*.

Example 2.104 (The function e^{z^m})**.** Let $f(z) = e^{z^m}$ with integer $m \geq 1$. If $|z| = r$, then $|e^{z^m}| = e^{\Re(z^m)} \leq e^{|z^m|} = e^{r^m}$, and equality occurs when $z^m = r^m$ is real positive, i.e. for $z = re^{2\pi ik/m}$. Thus $M(r) = e^{r^m}$, so $\log M(r) = r^m$ and $\log\log M(r) = \log(r^m) = m\log r$. Hence $\rho = \limsup \frac{m\log r}{\log r} = m$. The type (with respect to $\rho = m$) is $\sigma = \limsup_{r\to\infty} \frac{\log M(r)}{r^m} = \lim_{r\to\infty} \frac{r^m}{r^m} = 1$.

Example 2.105 (Gamma function: order 1 but infinite type)**.** Stirling's formula (in a sector) implies $\log|\Gamma(re^{i\theta})| = r\log r - r + O(\log r)$ as $r \to \infty$ (for fixed θ away from $\pm\pi$). Taking $\theta = 0$ already shows that along the positive real axis, $\log M(r) \geq \log|\Gamma(r)| = r\log r - r + O(\log r)$. On the other hand, standard bounds also give $\log M(r) \leq Cr\log r$ for large r. Thus $\log M(r) \asymp r\log r$, so $\log\log M(r) = \log(r\log r + O(r)) = \log r + \log\log r + O(1)$. Therefore $\rho = 1$.

For type with respect to $\rho = 1$, we look at $\log M(r)/r$. Since $\log M(r) \gtrsim r\log r$, we have $\log M(r)/r \gtrsim \log r \to \infty$, so $\sigma = \infty$: Gamma has order 1 but *infinite type*.

5. Order, zeros, and genus: how growth controls factorization

Proposition 2.106 (Exponent of convergence of zeros; a practical statement)**.** *Let f be entire, not identically zero, of finite order $\rho < \infty$, and let $\{a_n\}$ be its nonzero zeros listed with multiplicity. Then for every $p > \rho$ one has $\sum_{n=1}^{\infty} |a_n|^{-(p+1)} < \infty$. Consequently, for any integer $p \geq \rho$ the canonical product $\prod_n E_p(z/a_n)$ converges uniformly on compact sets.*

Proof. We give a standard "growth ⇒ summability" argument at the level needed for Hadamard/Weierstrass applications.

Fix $p > \rho$. By the equivalent characterization of order, there exist $\varepsilon > 0$ and R_0 such that for all $r \geq R_0$, $M(r) \leq \exp(r^{\rho+\varepsilon})$ with $\rho + \varepsilon < p$.

Let $N(r)$ be the number of zeros (counted with multiplicity) of f in $|z| \leq r$. By Jensen's formula applied to f (with $f(0) \neq 0$ for simplicity; otherwise divide out z^m), we have $\sum_{|a_n|\leq r} \log\frac{r}{|a_n|} \leq \log M(r) - \log|f(0)|$. Hence $\sum_{|a_n|\leq r} \log\frac{r}{|a_n|} \leq r^{\rho+\varepsilon} + C$ for $r \geq R_0$.

From this one derives the coarse bound $N(r) \leq C' r^{\rho+\varepsilon}$ for large r (because each term $\log(r/|a_n|)$ is at least $\log 2$ for zeros in $|a_n| \leq r/2$, and one compares $N(r/2)$ with the Jensen sum). Such polynomial counting of zeros implies summability of $|a_n|^{-(p+1)}$ for $p > \rho$ by a dyadic decomposition: group zeros with $2^k \leq |a_n| < 2^{k+1}$, then $\sum_{2^k\leq|a_n|<2^{k+1}} |a_n|^{-(p+1)} \leq \#\{|a_n| < 2^{k+1}\} \cdot 2^{-k(p+1)} \leq C' 2^{(k+1)(\rho+\varepsilon)} \cdot 2^{-k(p+1)}$. Since $p + 1 > (\rho + \varepsilon) + 1$, the series over k converges geometrically. Therefore $\sum |a_n|^{-(p+1)} < \infty$.

Finally, compact-uniform convergence of $\prod_n E_p(z/a_n)$ follows from the canonical product estimate proved in the previous subsection on Weierstrass factorization. □

Remark 2.107 (What this means conceptually)**.** Finite order bounds how densely zeros can accumulate toward infinity. In Hadamard factorization, this is exactly what allows one to use a *fixed genus* canonical factor E_p (instead of varying p_n).

Exercises

Exercise 2.108 (Order and type of $\exp(z^m)$)**.** Let $m \geq 1$ be an integer and $f(z) = \exp(z^m)$. Show directly that $M(r) = \exp(r^m)$, hence $\rho = m$ and type $\sigma = 1$ (with respect to $\rho = m$).

Exercise 2.109 (Order 0 if and only if polynomial)**.** Show that an entire function f is a polynomial if and only if it has order 0. Hint: use Cauchy estimates: if $M(r) \le r^A$ for all large r, show that $f^{(k)}(0) = 0$ for all $k > A$. Conversely, compute the order of a polynomial.

Exercise 2.110 (Growth gap below order 1)**.** Let f be entire of order $\rho < 1$. Show there exist constants $A, B > 0$ such that $|f(z)| \le A \exp(B|z|^\rho)$ for all z. Then discuss (briefly) why this is much more rigid than arbitrary entire growth, e.g. in connection with Phragmén–Lindelöf or Hadamard factorization.

Exercise 2.111 (Bessel function J_0 has order 1)**.** Assume J_0 has the Hadamard product $J_0(z) = \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{j_{0,n}^2}\right)$, where $j_{0,n}$ is the n-th positive zero. Using $j_{0,n} \sim (n + \frac{1}{4})\pi$, show that $\sum_n 1/j_{0,n}^2$ converges, hence the product has genus 0. Deduce that the exponent of convergence of zeros is 1 and conclude that J_0 has order 1.

Exercise 2.112 (Growth comparison and order bound)**.** Let f and g be entire and assume there exist constants $A, B > 0$ such that $|f(z)| \le A|g(z)| + B$ for all $z \in \mathbb{C}$. Assuming g has finite order ρ, show that f has order at most ρ. Hint: bound $M_f(r)$ by $AM_g(r) + B$, then compare $\log\log$ growth.

2.8. Harmonic Functions and the Poisson Kernel

Harmonic functions appear as real/imaginary parts of holomorphic functions and as solutions of Laplace's equation. This subsection develops the analytic foundations: mean value property, maximum principle, and the Poisson integral formula solving the Dirichlet problem on the disk (and on the upper half-plane). We include explicit computations, concrete examples, and (crucially) an explanation of why the Poisson kernel has the specific form it does.

1. Harmonicity and holomorphic functions

Definition 2.113 (Harmonic function)**.** Let $U \subset \mathbb{R}^2$ be open. A function $u : U \to \mathbb{R}$ of class C^2 is *harmonic* if it satisfies Laplace's equation

$$\Delta u := u_{xx} + u_{yy} = 0.$$

In complex notation $z = x + iy$, one has

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right),$$

hence

$$\Delta = 4\frac{\partial^2}{\partial z\, \partial \bar{z}}.$$

Proposition 2.114 (Real and imaginary parts of holomorphic functions are harmonic)**.** *Let $f = u + iv$ be holomorphic on $U \subset \mathbb{C}$. Then u and v are harmonic on U.*

Proof. Holomorphicity gives the Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x.$$

Differentiate:

$$u_{xx} = (v_y)_x = v_{yx}, \qquad u_{yy} = (-v_x)_y = -v_{xy}.$$

Since v is C^2, mixed partials commute: $v_{yx} = v_{xy}$. Thus

$$\Delta u = u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0.$$

The same argument with $v_{xx} + v_{yy} = u_{xy} - u_{yx} = 0$ shows $\Delta v = 0$. □

Example 2.115 (A quick check)**.** If $f(z) = z^2 = (x+iy)^2 = (x^2 - y^2) + i(2xy)$ then $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$. Compute $\Delta u = (2) + (-2) = 0$ and $\Delta v = 0 + 0 = 0$.

2. Mean value property and maximum principle

Theorem 2.116 (Mean value property)**.** *Let u be harmonic on an open set containing the closed disk $\overline{B(z_0, r)} = \{z : |z - z_0| \le r\}$. Then*

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta.$$

Conversely, if u is continuous on U and satisfies this identity for every such disk, then u is harmonic on U.

Proof. We prove the forward direction using holomorphic functions and Cauchy. Fix z_0 and r and set $w = z - z_0$. Consider a holomorphic f on a simply connected neighborhood of $\overline{B(z_0, r)}$ such that $\Re f = u$ (existence of a harmonic conjugate locally is standard; one may also give a purely PDE proof). By Cauchy's formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{|w|=r} \frac{f(z_0 + w)}{w}\, dw.$$

Parametrize $w = re^{i\theta}$ so that $dw = ire^{i\theta}\, d\theta$ and $\frac{dw}{w} = i\, d\theta$. Then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta.$$

Taking real parts yields the mean value property for $u = \Re f$.

For the converse, one standard argument is: the mean value property implies that u has vanishing distributional Laplacian, hence is harmonic (Weyl lemma). In a classical calculus proof, one compares u with the fundamental solution and uses the average identity to show that the second derivatives satisfy $u_{xx} + u_{yy} = 0$. □

Theorem 2.117 (Maximum principle)**.** *Let $D \subset \mathbb{C}$ be a bounded domain and u harmonic on D and continuous on $\overline{D}$. Then*

$$\max_{\overline{D}} u = \max_{\partial D} u, \qquad \min_{\overline{D}} u = \min_{\partial D} u.$$

In particular, if u attains a maximum (or minimum) at an interior point, then u is constant.

Proof. Let $z_0 \in D$ and choose $r > 0$ with $\overline{B(z_0, r)} \subset D$. By the mean value property, $u(z_0)$ equals the average of u on the circle $|z - z_0| = r$. An average cannot exceed the maximum of the averaged values, so

$$u(z_0) \leq \max_{|z-z_0|=r} u(z).$$

Letting circles approach the boundary shows $\sup_D u \leq \sup_{\partial D} u$. Since u is continuous on $\overline{D}$, the supremum is a maximum, and it must occur on ∂D. If u achieves a maximum at an interior point z_0, then for every small r the average on the circle equals the maximum value, forcing u to be constant on that circle. By connectedness and the mean value property, u must be constant on D. The minimum statement follows by applying the maximum statement to $-u$. □

3. The Dirichlet problem on the disk: why a Poisson kernel must exist

We solve the Dirichlet problem on $\mathbb{D} = \{z : |z| < 1\}$: given $f \in C(\partial\mathbb{D})$, find a harmonic function u on $\mathbb{D}$ extending continuously to $\partial\mathbb{D}$ with boundary values f.

Why should the solution have an integral-kernel form? Linearity of Δ suggests linear dependence on the boundary data f. So for each interior point $z \in \mathbb{D}$, the map $f \mapsto u(z)$ should be a bounded linear functional on $C(\partial\mathbb{D})$. By the Riesz representation principle, one expects a probability measure μ_z on $\partial\mathbb{D}$ such that

$$u(z) = \int_{\partial\mathbb{D}} f(\zeta)\, d\mu_z(\zeta).$$

If μ_z has a density with respect to arc length, we may write

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P(z, e^{i\phi})\, f(e^{i\phi})\, d\phi,$$

where $P(z, \zeta) \geq 0$ and $\frac{1}{2\pi} \int_0^{2\pi} P(z, e^{i\phi})\, d\phi = 1$.

Symmetry forces the dependence on angular difference. Rotations preserve $\mathbb{D}$ and preserve harmonicity. If $R_\alpha(w) = e^{i\alpha} w$, then the harmonic extension of $f \circ R_\alpha$ is $u \circ R_\alpha$. Hence the density must satisfy a covariance:

$$P(e^{i\alpha} z, e^{i\alpha} \zeta) = P(z, \zeta).$$

Thus when we write $z = re^{i\theta}$ and $\zeta = e^{i\phi}$, the kernel depends only on $\theta - \phi$:

$$P(z, \zeta) = P_r(\theta - \phi).$$

Boundary concentration: the kernel must peak at the visible boundary point. If $z = re^{i\theta}$ approaches the boundary point $e^{i\theta}$ as $r \to 1^-$, then continuity of the solution requires $u(re^{i\theta}) \to f(e^{i\theta})$. This can only happen if the measure $\mu_{re^{i\theta}}$ concentrates near $\phi = \theta$. So $P_r(\theta - \phi)$ should look like an "approximate identity": nonnegative, total mass 2π, sharply peaked at $\phi = \theta$ as $r \uparrow 1$.

The Möbius-transform mechanism (the real source of the formula). For fixed $z \in \mathbb{D}$, consider the disk automorphism

$$\varphi_z(w) = \frac{z - w}{1 - \overline{z}\, w}.$$

It maps $\mathbb{D}$ to itself and sends z to 0. If u is harmonic on $\mathbb{D}$, then $u \circ \varphi_z^{-1}$ is harmonic as well. Apply the mean value property at 0:

$$u(z) = (u \circ \varphi_z^{-1})(0) = \frac{1}{2\pi} \int_0^{2\pi} (u \circ \varphi_z^{-1})(e^{i\phi})\, d\phi.$$

Now φ_z^{-1} maps the unit circle to itself, but it *reparametrizes* the boundary. The change of variables from ϕ to the boundary angle of $\zeta = \varphi_z^{-1}(e^{i\phi})$ produces a Jacobian factor. That Jacobian is exactly the Poisson kernel: it is the density obtained by pushing forward the uniform boundary measure under φ_z^{-1}. This is why the Poisson kernel is not ad hoc: it is the unique density compatible with (i) disk automorphisms and (ii) the mean value property at the origin.

4. Poisson kernel on the disk

Definition 2.118 (Poisson kernel on the disk)**.** Let $z = re^{i\theta} \in \mathbb{D}$ and $\zeta = e^{i\phi} \in \partial\mathbb{D}$. Define

$$P(z, \zeta) := \frac{1 - |z|^2}{|\zeta - z|^2}.$$

Equivalently, in angle form

$$P_r(\theta - \phi) = \frac{1 - r^2}{1 - 2r\cos(\theta - \phi) + r^2}.$$

Lemma 2.119 (Algebraic identities and positivity)**.** *For $z = re^{i\theta}$ and $\zeta = e^{i\phi}$,*

$$|\zeta - z|^2 = 1 - 2r\cos(\theta - \phi) + r^2.$$

Hence $P(z, \zeta) > 0$ and $P_r(\psi)$ is 2π-periodic and even in ψ.

Proof. Compute

$$|e^{i\phi} - re^{i\theta}|^2 = (e^{i\phi} - re^{i\theta})(e^{-i\phi} - re^{-i\theta}) = 1 - r(e^{i(\theta-\phi)} + e^{-i(\theta-\phi)}) + r^2 = 1 - 2r\cos(\theta - \phi) + r^2.$$

Since $0 \le r < 1$, the denominator is positive, so $P > 0$. □

Lemma 2.120 (Poisson kernel as a real part)**.** *For $z \in \mathbb{D}$ and $\zeta \in \partial\mathbb{D}$,*

$$P(z, \zeta) = \Re\left(\frac{\zeta + z}{\zeta - z}\right).$$

Proof. Compute

$$\frac{\zeta + z}{\zeta - z} = \frac{(\zeta + z)(\overline{\zeta} - \overline{z})}{|\zeta - z|^2} = \frac{\zeta\overline{\zeta} - \zeta\overline{z} + z\overline{\zeta} - |z|^2}{|\zeta - z|^2}.$$

Using $|\zeta|^2 = \zeta\overline{\zeta} = 1$ (since $\zeta \in \partial\mathbb{D}$), the numerator is

$$1 - |z|^2 + (z\overline{\zeta} - \zeta\overline{z}).$$

The term $(z\overline{\zeta} - \zeta\overline{z})$ is purely imaginary (it is minus its own complex conjugate), so taking real parts yields

$$\Re\left(\frac{\zeta + z}{\zeta - z}\right) = \frac{1 - |z|^2}{|\zeta - z|^2} = P(z, \zeta).$$

□

Lemma 2.121 (Normalization)**.** *For every $z \in \mathbb{D}$,*

$$\frac{1}{2\pi}\int_0^{2\pi} P(z, e^{i\phi})\, d\phi = 1.$$

Equivalently, for every $0 \le r < 1$,

$$\frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - \phi)\, d\phi = 1.$$

Proof. We give two concrete proofs.

Fourier-series proof. One shows (next lemma) that

$$P_r(\psi) = 1 + 2\sum_{n=1}^{\infty} r^n \cos(n\psi).$$

Averaging over one period kills all $\cos(n\psi)$ terms, leaving average 1.

Complex-variable proof. Fix $z \in \mathbb{D}$. By Lemma 2.120,

$$P(z, e^{i\phi}) = \Re\left(\frac{e^{i\phi} + z}{e^{i\phi} - z}\right).$$

Thus it suffices to show that $\frac{1}{2\pi}\int_0^{2\pi} \frac{e^{i\phi}+z}{e^{i\phi}-z}\, d\phi = 1$. Let $\zeta = e^{i\phi}$ so $d\phi = \frac{1}{i}\frac{d\zeta}{\zeta}$ and $|\zeta| = 1$:

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{e^{i\phi} + z}{e^{i\phi} - z}\, d\phi = \frac{1}{2\pi i}\int_{|\zeta|=1} \frac{\zeta + z}{\zeta - z}\frac{d\zeta}{\zeta}.$$

The integrand has simple poles at $\zeta = 0$ and $\zeta = z$ (both inside the circle). Compute residues:

$$\mathrm{Res}_{\zeta=z}\left(\frac{\zeta+z}{\zeta-z}\frac{1}{\zeta}\right) = \lim_{\zeta\to z}\frac{\zeta+z}{\zeta} = 2, \qquad \mathrm{Res}_{\zeta=0}\left(\frac{\zeta+z}{\zeta-z}\frac{1}{\zeta}\right) = \lim_{\zeta\to 0}\frac{\zeta+z}{\zeta-z} = -1.$$

Thus the integral equals $2\pi i\,(2-1) = 2\pi i$, and dividing by $2\pi i$ gives 1. Taking real parts yields the stated normalization. □

Lemma 2.122 (Fourier expansion of the Poisson kernel)**.** *For $|r| < 1$ and $\psi \in \mathbb{R}$,*

$$P_r(\psi) = 1 + 2\sum_{n=1}^{\infty} r^n \cos(n\psi).$$

Proof. Start from the geometric series

$$\sum_{n=0}^{\infty} r^n e^{in\psi} = \frac{1}{1 - re^{i\psi}} \qquad (|r| < 1).$$

Taking real parts gives

$$1 + \sum_{n=1}^{\infty} r^n \cos(n\psi) = \Re\left(\frac{1}{1 - re^{i\psi}}\right).$$

Compute explicitly:

$$\frac{1}{1-re^{i\psi}} = \frac{1-re^{-i\psi}}{|1-re^{i\psi}|^2} = \frac{1-r\cos\psi}{1-2r\cos\psi+r^2} + i\frac{r\sin\psi}{1-2r\cos\psi+r^2}.$$

Hence

$$\Re\left(\frac{1}{1-re^{i\psi}}\right) = \frac{1-r\cos\psi}{1-2r\cos\psi+r^2}.$$

Now observe the identity

$$\frac{1-r^2}{1-2r\cos\psi+r^2} = 2\frac{1-r\cos\psi}{1-2r\cos\psi+r^2} - 1,$$

so

$$P_r(\psi) = 2\Re\left(\frac{1}{1-re^{i\psi}}\right) - 1 = 1 + 2\sum_{n=1}^{\infty} r^n\cos(n\psi).$$

□

Theorem 2.123 (Poisson integral formula on the disk)**.** *Let $f \in C(\partial\mathbb{D})$. Define, for $z = re^{i\theta} \in \mathbb{D}$,*

$$u(z) := \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\phi)\, f(e^{i\phi})\, d\phi.$$

Then:

1. *u is harmonic on $\mathbb{D}$;*
2. *u extends continuously to $\overline{\mathbb{D}}$ with $u(e^{i\theta}) = f(e^{i\theta})$;*
3. *u is the unique harmonic function on $\mathbb{D}$ continuous on $\overline{\mathbb{D}}$ with boundary values f.*

Proof. Step 1: Harmonicity. Write the Fourier series of f on the circle:

$$f(e^{i\phi}) = a_0 + \sum_{n=1}^{\infty}(a_n\cos(n\phi) + b_n\sin(n\phi)),$$

where coefficients are

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(e^{i\phi})\cos(n\phi)\, d\phi, \qquad b_n = \frac{1}{\pi}\int_0^{2\pi} f(e^{i\phi})\sin(n\phi)\, d\phi \quad (n \geq 1),$$

and

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\phi})\, d\phi.$$

Convolution with the Poisson kernel uses orthogonality (checkable directly using the Fourier expansion of P_r):

$$\frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\phi)\cos(n\phi)\, d\phi = r^n\cos(n\theta), \qquad \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\phi)\sin(n\phi)\, d\phi = r^n\sin(n\theta).$$

Thus

$$u(re^{i\theta}) = a_0 + \sum_{n=1}^{\infty} r^n(a_n\cos(n\theta) + b_n\sin(n\theta)).$$

Each term $r^n\cos(n\theta)$ and $r^n\sin(n\theta)$ is harmonic on $\mathbb{D}$ since it is the real/imaginary part of z^n. Uniform convergence on compact sets gives that u is harmonic.

Step 2: Boundary convergence via approximate-identity concentration. Fix θ_0. Using normalization,

$$u(re^{i\theta_0}) - f(e^{i\theta_0}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta_0-\phi)\big(f(e^{i\phi}) - f(e^{i\theta_0})\big)\,d\phi.$$

Given $\varepsilon > 0$, by uniform continuity of f on the compact circle, choose $\delta > 0$ so that $|f(e^{i\phi}) - f(e^{i\theta_0})| < \varepsilon$ whenever $|\phi - \theta_0| < \delta \pmod{2\pi}$. Split the integral into the near arc $A = \{\phi : |\phi - \theta_0| < \delta\}$ and its complement A^c. On A the integrand magnitude is at most $\varepsilon P_r(\theta_0 - \phi)$, so the contribution is at most

$$\varepsilon \cdot \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta_0-\phi)\,d\phi = \varepsilon.$$

On A^c, the difference $|f(e^{i\phi}) - f(e^{i\theta_0})|$ is bounded by $2\|f\|_\infty$, so the contribution is at most

$$\frac{2\|f\|_\infty}{2\pi}\int_{A^c} P_r(\theta_0-\phi)\,d\phi.$$

For $|\phi - \theta_0| \geq \delta$ we have $\cos(\theta_0 - \phi) \leq \cos\delta < 1$, hence

$$1 - 2r\cos(\theta_0-\phi) + r^2 \geq 1 - 2r\cos\delta + r^2.$$

Therefore

$$P_r(\theta_0-\phi) \leq \frac{1-r^2}{1-2r\cos\delta+r^2} \quad \text{on } A^c,$$

and the right-hand side tends to 0 as $r \to 1^-$ because $1 - r^2 \to 0$ while the denominator tends to $2(1-\cos\delta) > 0$. Thus $\int_{A^c} P_r(\theta_0-\phi)\,d\phi \to 0$ uniformly in θ_0. Hence $u(re^{i\theta}) \to f(e^{i\theta})$ uniformly in θ, so u extends continuously to the boundary.

Step 3: Uniqueness. If u_1, u_2 are harmonic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$ with the same boundary values, then $w = u_1 - u_2$ is harmonic and continuous on $\overline{\mathbb{D}}$ and satisfies $w = 0$ on $\partial\mathbb{D}$. By the maximum principle, $\max_{\overline{\mathbb{D}}} w = 0$ and $\min_{\overline{\mathbb{D}}} w = 0$, so $w \equiv 0$. □

5. Concrete disk examples with explicit calculation

Example 2.124 (Boundary data: a single Fourier mode)**.** Let $f(e^{i\phi}) = \cos(n\phi)$ with $n \geq 1$. Using the orthogonality computation in the proof,

$$u(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta-\phi)\cos(n\phi)\,d\phi = r^n\cos(n\theta).$$

Similarly, if $f(e^{i\phi}) = \sin(n\phi)$ then $u(re^{i\theta}) = r^n\sin(n\theta)$.

Example 2.125 (Piecewise constant boundary data)**.** Let $f(e^{i\phi}) = 1$ for $\phi \in (-\alpha, \alpha)$ and $f(e^{i\phi}) = 0$ otherwise, with $0 < \alpha < \pi$. Then for $z = re^{i\theta}$,

$$u(z) = \frac{1}{2\pi}\int_{-\alpha}^{\alpha} P_r(\theta-\phi)\,d\phi.$$

In particular at $\theta = 0$,

$$u(r) = \frac{1}{2\pi}\int_{-\alpha}^{\alpha} \frac{1-r^2}{1-2r\cos\phi+r^2}\,d\phi,$$

which increases to 1 as $r \to 1^-$ and decreases to $\frac{\alpha}{\pi}$ as $r \to 0$. This models how harmonic measure weights the boundary arc as seen from z.

6. Poisson kernel on the upper half-plane

Definition 2.126 (Poisson kernel for the upper half-plane)**.** For $z = x + iy$ with $y > 0$ and boundary parameter $t \in \mathbb{R}$, define

$$P_{\mathbb{H}}(x, y; t) := \frac{1}{\pi} \frac{y}{(x-t)^2 + y^2}.$$

Lemma 2.127 (Normalization in $\mathbb{H}$)**.** *For every $x \in \mathbb{R}$ and $y > 0$,*

$$\int_{-\infty}^{\infty} P_{\mathbb{H}}(x, y; t)\, dt = 1.$$

Proof. Substitute $s = (t - x)/y$, so $t = x + ys$ and $dt = y\, ds$:

$$\int_{-\infty}^{\infty} \frac{1}{\pi} \frac{y}{(x-t)^2 + y^2}\, dt = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{y}{y^2 s^2 + y^2}\, y\, ds = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{1}{1 + s^2}\, ds = \frac{1}{\pi} \big[\arctan s\big]_{-\infty}^{\infty} = 1.$$

□

Theorem 2.128 (Poisson integral in $\mathbb{H}$)**.** *Let $f \in L^\infty(\mathbb{R})$. Define*

$$u(x, y) := \int_{-\infty}^{\infty} P_{\mathbb{H}}(x, y; t)\, f(t)\, dt \qquad (y > 0).$$

Then u is harmonic on $\mathbb{H}$. Moreover, for every Lebesgue point x_0 of f, $u(x, y) \to f(x_0)$ as $(x, y) \to (x_0, 0)$ non-tangentially.

Proof. Harmonicity. For fixed t, the function

$$K_t(x, y) = \frac{y}{(x-t)^2 + y^2}$$

is harmonic on $\mathbb{H}$ because it is (up to sign) the imaginary part of $(x + iy - t)^{-1}$:

$$\frac{1}{x + iy - t} = \frac{x - t - iy}{(x-t)^2 + y^2} \quad \Longrightarrow \quad \Im \frac{1}{x + iy - t} = -\frac{y}{(x-t)^2 + y^2}.$$

Imaginary parts of holomorphic functions are harmonic, hence K_t is harmonic. Since $f \in L^\infty$, differentiation under the integral sign is justified on compact subsets of $\mathbb{H}$ by dominated convergence (the t-derivatives of K_t are integrable kernels for fixed (x, y) with $y \geq y_0 > 0$). Therefore $\Delta u = \int \Delta K_t\, f(t)\, dt = 0$.

Boundary convergence. The kernels $P_{\mathbb{H}}(x, y; \cdot)$ form an approximate identity: they are nonnegative, integrate to 1, and concentrate near $t = x$ as $y \to 0$. Standard real-variable arguments yield non-tangential convergence to f at Lebesgue points. □

Example 2.129 (Upper half-plane with linear boundary data: a full computation)**.** Let $f(t) = t$ (not bounded, but the computation illustrates the kernel). Formally,

$$u(x, y) = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{yt}{(x-t)^2 + y^2}\, dt.$$

Substitute $s = (t - x)/y$ so $t = x + ys$ and $dt = y\, ds$:

$$u(x, y) = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{y(x + ys)}{y^2 s^2 + y^2}\, y\, ds = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{x + ys}{1 + s^2}\, ds.$$

Split the integral:

$$\int_{-\infty}^{\infty} \frac{1}{\pi} \frac{x}{1 + s^2}\, ds = x \cdot \frac{1}{\pi} \cdot \pi = x, \qquad \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{ys}{1 + s^2}\, ds = 0$$

by oddness. Thus $u(x, y) = x$, which is harmonic and has boundary trace t.

7. Exercises

Exercise 2.130. [Poisson kernel as a real part] Let $z = re^{i\theta} \in \mathbb{D}$ and $\zeta = e^{i\phi} \in \partial\mathbb{D}$. Show that

$$P(z,\zeta) = \Re\left(\frac{\zeta + z}{\zeta - z}\right).$$

Hint: multiply numerator and denominator by $\overline{\zeta} - \overline{z}$ and use $|\zeta| = 1$.

Exercise 2.131 (Differentiate the half-plane kernel)**.** Fix $t \in \mathbb{R}$ and set $K_t(x,y) = \frac{y}{(x-t)^2+y^2}$ for $y > 0$. Compute $K_{t,xx}$ and $K_{t,yy}$ explicitly and verify $K_{t,xx} + K_{t,yy} = 0$.

Exercise 2.132 (A quick harmonic extension)**.** Find the harmonic extension of $f(e^{i\phi}) = \sin(3\phi)$ to $\mathbb{D}$. Answer: $u(re^{i\theta}) = r^3\sin(3\theta)$.

Exercise 2.133 (Normalization of the half-plane kernel)**.** Prove directly that for all $x \in \mathbb{R}$ and $y > 0$,

$$\int_{-\infty}^{\infty} P_{\mathbb{H}}(x,y;t)\,dt = 1,$$

using the substitution $s = (t-x)/y$ and $\int_{-\infty}^{\infty} \frac{1}{1+s^2}\,ds = \pi$.

2.9. Subharmonic Functions and Riesz Decomposition

Subharmonic functions are the natural "$\Delta \geq 0$" analogue of harmonic functions, much as convex functions generalize affine ones. They control growth and value distribution in complex analysis, especially through the fundamental example $u = \log|f|$ for holomorphic f. The main structural result is the Riesz decomposition theorem, which splits a subharmonic function into a harmonic part plus a logarithmic potential determined by its Laplacian (as a measure).

1. Definition and first intuition

Definition 2.134 (Subharmonic function)**.** Let $U \subset \mathbb{C}$ be open. A function $u : U \to [-\infty, \infty)$ is *subharmonic* if:

1. u is upper semicontinuous;
2. $u \not\equiv -\infty$ on any connected component of U;
3. for every closed disk $\overline{B(z_0,r)} \subset U$,

$$u(z_0) \leq \frac{1}{2\pi}\int_0^{2\pi} u(z_0 + re^{i\theta})\,d\theta.$$

Remark 2.135 (Harmonic as the boundary case)**.** A function u is harmonic on U if and only if both u and $-u$ are subharmonic on U.

Remark 2.136 (Convexity analogy)**.** For C^2 functions on an interval, convexity is equivalent to $u'' \geq 0$ and implies a midpoint/average inequality. Subharmonicity is the two-dimensional analogue: $\Delta u \geq 0$ forces the value at the center to be dominated by circular averages.

2. The C^2 case: equivalences and an explicit Taylor–averaging computation

Proposition 2.137 (Equivalent characterizations for C^2 functions)**.** *Let $u \in C^2(U)$. The following are equivalent:*

1. *u is subharmonic on U.*
2. *u satisfies the weak maximum principle on compact sets: for every $V \Subset U$,*
$$\max_{\overline{V}} u = \max_{\partial V} u.$$
3. *$\Delta u \geq 0$ pointwise on U.*
4. *(Mean value inequality) For every disk $\overline{B(z_0, r)} \subset U$,*
$$u(z_0) \leq \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta.$$

Proof. We prove the key equivalence (3)⇔(4) by a direct computation, and then briefly indicate the link with the maximum principle.

Fix $z_0 \in U$ and write $z = z_0 + w$ with $w = re^{i\theta}$, so $w_x = r\cos\theta$ and $w_y = r\sin\theta$. Taylor expand u at z_0 (real variables):

$$u(z_0+w) = u(z_0)+u_x(z_0)w_x+u_y(z_0)w_y+\frac{1}{2}\Big(u_{xx}(z_0)w_x^2+2u_{xy}(z_0)w_xw_y+u_{yy}(z_0)w_y^2\Big)+o(r^2).$$

Average over $\theta \in [0, 2\pi]$. The linear terms vanish since the averages of $\cos\theta$ and $\sin\theta$ are 0. Also,

$$\frac{1}{2\pi}\int_0^{2\pi} \cos^2\theta\, d\theta = \frac{1}{2}, \quad \frac{1}{2\pi}\int_0^{2\pi} \sin^2\theta\, d\theta = \frac{1}{2}, \quad \frac{1}{2\pi}\int_0^{2\pi} \sin\theta\cos\theta\, d\theta = 0.$$

Hence

$$\frac{1}{2\pi}\int_0^{2\pi} u(z_0+re^{i\theta})\, d\theta = u(z_0)+\frac{r^2}{4}\big(u_{xx}(z_0)+u_{yy}(z_0)\big)+o(r^2) = u(z_0)+\frac{r^2}{4}\Delta u(z_0)+o(r^2).$$

Therefore the inequality $u(z_0) \leq \frac{1}{2\pi}\int_0^{2\pi} u(z_0+re^{i\theta})\, d\theta$ for all small r holds iff $\Delta u(z_0) \geq 0$. This proves (3)⇔(4).

For (2), if $\Delta u \geq 0$ and u had a strict interior maximum at z_0, then for small r the boundary circle would have strictly smaller values, contradicting the mean value inequality. Conversely, the weak maximum principle implies $\Delta u \geq 0$ by standard PDE arguments. □

3. The Laplacian as a measure and the logarithmic fundamental solution

For general subharmonic functions, Δu need not exist pointwise. Instead, Δu is understood in the sense of distributions and is represented by a *positive Radon measure.*

Definition 2.138 (Riesz measure)**.** Let u be subharmonic on U. The *Riesz measure* μ_u is the unique positive Radon measure such that, in the sense of distributions,

$$\Delta u = \mu_u.$$

Equivalently, for every $\varphi \in C_c^\infty(U)$,

$$\int_U u\, \Delta\varphi\, dA = \int_U \varphi\, d\mu_u.$$

Remark 2.139. In the C^2 case, $\mu_u = (\Delta u)\, dA$ with $\Delta u \geq 0$ pointwise.

Lemma 2.140 (Harmonic functions do not contribute)**.** *If u is subharmonic and v is harmonic on U, then for every $\varphi \in C_c^\infty(U)$,*

$$\int_U v \, d\mu_u = 0.$$

Equivalently, adding a harmonic function to u does not change μ_u.

Proof. Using distributions and integration by parts,

$$\int_U v \, d\mu_u = \int_U v \, \Delta u = \int_U u \, \Delta v = 0,$$

since $\Delta v = 0$. □

4. A clean distributional computation: $\Delta \log|z|$ is a point mass

Lemma 2.141 (Distributional identity for $\log|z|$)**.** *For $\varphi \in C_c^\infty(\mathbb{C})$,*

$$\int_{\mathbb{C}} \log|z| \, \Delta\varphi(z) \, dA(z) = 2\pi \, \varphi(0).$$

Equivalently, $\Delta \log|z| = 2\pi \, \delta_0$ in the sense of distributions.

Proof. Let $\varphi \in C_c^\infty(\mathbb{C})$. Fix $R > 0$ so that φ vanishes outside $B(0, R)$. For $\varepsilon \in (0, R)$, apply Green's identity on the annulus $A_{\varepsilon,R} = \{z : \varepsilon < |z| < R\}$:

$$\int_{A_{\varepsilon,R}} \log|z| \, \Delta\varphi \, dA = \int_{\partial A_{\varepsilon,R}} \Big(\log|z| \, \partial_n \varphi - \varphi \, \partial_n \log|z| \Big) \, ds + \int_{A_{\varepsilon,R}} \varphi \, \Delta \log|z| \, dA.$$

Since $\log|z|$ is harmonic on $A_{\varepsilon,R}$, we have $\Delta \log|z| = 0$ there, so the last term vanishes.

On $|z| = R$, we have $\varphi = 0$, hence its boundary contribution is 0. On $|z| = \varepsilon$, write $z = \varepsilon e^{i\theta}$, $ds = \varepsilon d\theta$, and $\partial_n \log|z| = \partial_r(\log r)|_{r=\varepsilon} = 1/\varepsilon$. Therefore

$$\int_{A_{\varepsilon,R}} \log|z| \, \Delta\varphi \, dA = -\int_0^{2\pi} \varphi(\varepsilon e^{i\theta}) \, d\theta + \varepsilon \log(\varepsilon) \int_0^{2\pi} \partial_r \varphi(\varepsilon e^{i\theta}) \, d\theta.$$

As $\varepsilon \to 0$, the second term tends to 0 because $\varepsilon \log \varepsilon \to 0$ and $\partial_r \varphi$ is bounded. The first term tends to $-2\pi\varphi(0)$. Thus

$$\int_{A_{\varepsilon,R}} \log|z| \, \Delta\varphi \, dA \to -2\pi\varphi(0).$$

With the standard convention $\langle \Delta u, \varphi \rangle := \langle u, \Delta\varphi \rangle$, this means $\Delta \log|z| = 2\pi\delta_0$. □

Example 2.142 (Subharmonicity of $\log|z|$)**.** The function $u(z) = \log|z|$ is harmonic on $\mathbb{C} \setminus \{0\}$ but subharmonic on $\mathbb{C}$. Its Riesz measure is the point mass $\mu_u = 2\pi\delta_0$.

5. Riesz decomposition theorem

Theorem 2.143 (Riesz decomposition)**.** *Let $D \subset \mathbb{C}$ be a bounded domain and let u be subharmonic on D with Riesz measure μ_u. Then there exists a harmonic function h on D such that*

$$u(z) = h(z) + \frac{1}{2\pi} \int_D \log|z-\zeta| \, d\mu_u(\zeta), \qquad z \in D.$$

Proof. Define the logarithmic potential

$$U(z) := \frac{1}{2\pi} \int_D \log|z-\zeta| \, d\mu_u(\zeta).$$

We claim $\Delta U = \mu_u$ in the distributional sense. Let $\varphi \in C_c^\infty(D)$. By Fubini/Tonelli (log-singularity is locally integrable),

$$\int_D U(z) \, \Delta\varphi(z) \, dA(z) = \frac{1}{2\pi} \int_D \left(\int_D \log|z-\zeta| \, \Delta\varphi(z) \, dA(z) \right) d\mu_u(\zeta).$$

For each fixed ζ, the inner integral equals $2\pi\,\varphi(\zeta)$ by Lemma 2.141 applied to $z \mapsto \log|z-\zeta|$. Hence

$$\int_D U \, \Delta\varphi \, dA = \frac{1}{2\pi} \int_D 2\pi\,\varphi(\zeta) \, d\mu_u(\zeta) = \int_D \varphi \, d\mu_u,$$

which is exactly $\Delta U = \mu_u$.

Since $\Delta u = \mu_u$ by definition of μ_u, we get $\Delta(u - U) = 0$ in distributions. By Weyl's lemma, $h := u - U$ is harmonic on D, and $u = h + U$ as claimed. □

Corollary 2.144 (Measure encodes the non-harmonic part)**.** If $\mu_u = 0$, then u is harmonic on D.

Proof. If $\mu_u = 0$, then the potential term vanishes and Theorem 2.143 gives $u = h$ harmonic. □

6. Examples (zeros of holomorphic functions)

Example 2.145 (Single point mass)**.** Let $u(z) = \log|z - z_0|$. Then $\mu_u = 2\pi\delta_{z_0}$.

Example 2.146 (Finite collection of charges)**.** If

$$u(z) = \sum_{k=1}^{m} \alpha_k \log|z - a_k| + h(z), \qquad \alpha_k \geq 0,$$

with h harmonic, then

$$\mu_u = 2\pi \sum_{k=1}^{m} \alpha_k \, \delta_{a_k}.$$

Example 2.147 (Holomorphic case: $\log|f|$ and Jensen in distribution form)**.** Let f be holomorphic on D and not identically zero. Then $u = \log|f|$ is subharmonic. If $a \in D$ is a zero of multiplicity m_a, locally $f(z) = (z-a)^{m_a} g(z)$ with $g(a) \neq 0$, so

$$\log|f(z)| = m_a \log|z-a| + \log|g(z)|.$$

Since $\log|g|$ is harmonic near a, the Riesz measure is

$$\mu_{\log|f|} = 2\pi \sum_{a:\, f(a)=0} m_a \, \delta_a.$$

7. Exercises

Exercise 2.148 (Max of subharmonic functions)**.** Show that if u and v are subharmonic on U, then $\max(u, v)$ is subharmonic on U.

Exercise 2.149 (Compute the Riesz measure of $\log|f|$)**.** Let f be holomorphic on D and not identically zero. Prove that $\log|f|$ is subharmonic and that

$$\mu_{\log|f|} = 2\pi \sum_{a:\, f(a)=0} m_a\, \delta_a,$$

where m_a is the multiplicity of the zero at a.

Exercise 2.150 (Radial averages are increasing)**.** Let u be subharmonic on the unit disk and define

$$M(r) = \frac{1}{2\pi}\int_0^{2\pi} u(re^{i\theta})\, d\theta.$$

Show that $M(r)$ is increasing in $r \in (0, 1)$.

Exercise 2.151 (Poisson integrals and subharmonicity)**.** Let ν be a finite positive measure on $\partial\mathbb{D}$ and define

$$U(z) = \frac{1}{2\pi}\int_0^{2\pi} P(z, e^{i\phi})\, d\nu(\phi),$$

where P is the Poisson kernel on the disk. Show that U is harmonic on $\mathbb{D}$. Then explain why positive superpositions of subharmonic kernels (e.g. logarithmic potentials) produce subharmonic functions.

2.10. Analytic Continuation and Natural Boundaries

Analytic continuation is the mechanism by which one extends an analytic function beyond its initial domain. Sometimes the extension propagates indefinitely (e.g. rational functions, e^z, $\log z$ on suitable simply connected domains), and sometimes it breaks down due to unavoidable singularities. In extreme situations the obstruction is maximal: a boundary curve can be a *natural boundary*, meaning that continuation fails across *every* point of that curve.

1. Analytic continuation: germs and uniqueness

Definition 2.152 (Analytic continuation)**.** Let $U, V \subset \mathbb{C}$ be domains with $U \cap V \neq \emptyset$. If f is analytic on U and g is analytic on V, we say that g is an *analytic continuation* of f to V if

$$g(z) = f(z) \qquad (z \in U \cap V).$$

Proposition 2.153 (Uniqueness of analytic continuation)**.** *If g_1 and g_2 are analytic continuations of f to the same domain V, then $g_1 \equiv g_2$ on V.*

Proof. The function $h := g_1 - g_2$ is analytic on V and vanishes on $U \cap V$. Since $U \cap V$ is open (hence has accumulation points in V), the identity theorem forces $h \equiv 0$, i.e. $g_1 = g_2$ on V. □

Remark 2.154 (Local viewpoint: germs)**.** At a point z_0, the *germ* of an analytic function is its equivalence class under agreement on some neighborhood of z_0. Analytic continuation can be viewed as transporting germs from point to point while maintaining consistency on overlaps.

2. Continuation along paths and monodromy

Analytic continuation is typically performed step-by-step along a curve, using overlapping disks and uniqueness on overlaps.

Definition 2.155 (Continuation along a path)**.** Let f be analytic on a domain U and let $\gamma : [0,1] \to \mathbb{C}$ be continuous with $\gamma(0) \in U$. An *analytic continuation of f along γ* consists of:

- a subdivision $0 = t_0 < t_1 < \cdots < t_N = 1$,
- open sets $W_j \ni \gamma([t_{j-1}, t_j])$,
- analytic functions f_j on W_j,

such that $f_0 = f$ on $W_0 \cap U$ and

$$f_j = f_{j-1} \quad \text{on } W_j \cap W_{j-1} \qquad (j = 1, \ldots, N).$$

Remark 2.156 (Path dependence and multivaluedness)**.** Continuation may depend on the chosen path (classic example: $\log z$ on $\mathbb{C} \setminus \{0\}$). Different paths can lead to different resulting germs at the endpoint, producing monodromy phenomena and motivating the construction of Riemann surfaces.

Theorem 2.157 (Monodromy theorem (path-independence on simply connected domains))**.** *Let $\Omega \subset \mathbb{C}$ be simply connected. Assume f is analytic on some nonempty open set $U \subset \Omega$, and that analytic continuation of f along* every *path in Ω starting in U is possible. Then the continuation to any point of Ω is path-independent. In particular, continuation along any closed loop returns to the original germ.*

Proof. We prove a standard homotopy-invariance argument in a concrete "disk chaining" form.

Fix a basepoint $z_* \in U$. For each path γ in Ω starting at z_*, continuation along γ produces a germ at $\gamma(1)$ (well-defined up to restriction). We must show that if γ_0 and γ_1 are two paths from z_* to the same endpoint w, then the resulting germs at w coincide.

Since Ω is simply connected, γ_0 and γ_1 are homotopic relative endpoints: there exists a continuous map $H : [0,1] \times [0,1] \to \Omega$ such that

$$H(s,0) = \gamma_0(s), \qquad H(s,1) = \gamma_1(s), \qquad H(0,t) = z_*, \qquad H(1,t) = w.$$

The image $K := H([0,1]^2)$ is compact in Ω. Cover K by finitely many open disks $\{B_\alpha\}$ such that on each disk, analytic continuation is available and uniqueness holds on overlaps (this is ensured by the continuation hypothesis plus local analyticity).

Now discretize the square $[0,1]^2$ with a fine grid so that the image of each small grid square lies inside some disk B_α. Along the bottom edge (path γ_0), we have a chain of overlapping disks giving the continuation. Moving one row up in the grid, each step changes the path only within a region lying in a single disk, so the resulting analytic function on the overlap must agree by uniqueness. Inductively, the continuation along the bottom edge and along the top edge produce the same germ at the end. Thus the continuations along γ_0 and γ_1 agree at w.

Finally, for a closed loop ($w = z_*$) this implies that returning to the basepoint yields the original germ. □

Example 2.158 (A concrete monodromy computation: $\log z$)**.** On $\Omega = \mathbb{C} \setminus \{0\}$ define a branch of $\operatorname{Log} z$ near $z_* = 1$ by $\operatorname{Log} 1 = 0$. Continue along the loop $\gamma(t) = e^{2\pi i t}$, $t \in [0, 1]$. Locally $\operatorname{Log}(e^{i\theta}) = i\theta$; after one turn $\theta : 0 \mapsto 2\pi$, so the continued value becomes $2\pi i$ rather than 0. Hence continuation depends on the loop: monodromy is nontrivial because Ω is not simply connected.

3. Natural boundaries: definition and how they arise

There are two qualitatively different ways continuation can fail:

1. One hits an *isolated singularity* (removable, pole, essential) at a specific point.
2. Singularities accumulate so densely near the boundary that *no* boundary point admits any extension.

Definition 2.159 (Natural boundary)**.** Let f be analytic on a domain D. A boundary component $\Gamma \subset \partial D$ is a *natural boundary* for f if there is no point $\zeta_0 \in \Gamma$ such that f admits an analytic continuation to any neighborhood of ζ_0. Equivalently: for every $\zeta_0 \in \Gamma$ and every $r > 0$, there is no analytic function g on $B(\zeta_0, r)$ with $g = f$ on $B(\zeta_0, r) \cap D$.

Remark 2.160 (A very practical criterion)**.** If you can show that for every $\zeta_0 \in \Gamma$ there exist singularities of f inside D accumulating to ζ_0, then Γ is a natural boundary. Indeed, analytic continuation across ζ_0 would imply analyticity in some full disk around ζ_0, contradicting the presence of interior singularities accumulating to it.

4. Lacunary series and a dense set of singularities

A classical mechanism producing natural boundaries is a *lacunary* power series, where many coefficients vanish and the exponents have large gaps.

Example 2.161 (Lacunary series with exponents 2^k)**.** Consider

$$f(z) = \sum_{k=0}^{\infty} z^{2^k}, \qquad |z| < 1.$$

This defines an analytic function on the unit disk $\mathbb{D}$. In fact, the unit circle $|z| = 1$ is a natural boundary for f.

Detailed proof that $|z| = 1$ is a natural boundary for $\sum z^{2^k}$. We show that f has singularities approaching *every* boundary point of $\partial\mathbb{D}$.

Step 1: a functional equation. For $|z| < 1$,

$$f(z^2) = \sum_{k=0}^{\infty} (z^2)^{2^k} = \sum_{k=0}^{\infty} z^{2^{k+1}} = \sum_{k=1}^{\infty} z^{2^k} = f(z) - z.$$

Hence

$$f(z) = z + f(z^2). \tag{13}$$

Iterating,

$$f(z) = z + z^2 + z^4 + \cdots + z^{2^N} + f(z^{2^{N+1}}) \qquad (N \geq 0). \tag{14}$$

Step 2: singularities near any root of unity of odd order. Fix an odd integer $q \geq 1$ and choose ζ with $\zeta^q = 1$ and $\zeta \neq 1$ (a primitive q-th root of unity). Since 2 is invertible

modulo q, the powers $2^n \bmod q$ are periodic and, in particular, the set $\{\zeta^{2^n} : n \geq 0\}$ is finite and repeats.

Take $z = r\zeta$ with $0 < r < 1$. Then $z^{2^n} = r^{2^n}\zeta^{2^n}$. In (14), for large N the tail satisfies $|z^{2^{N+1}}| = r^{2^{N+1}} \to 0$, so $f(z^{2^{N+1}}) \to f(0) = 0$. Thus, morally,

$$f(r\zeta) \approx \sum_{n=0}^{\infty} r^{2^n}\, \zeta^{2^n}.$$

Now let $r \uparrow 1$. Because ζ^{2^n} cycles through finitely many values of modulus 1 and does not converge to 0, the partial sums behave like a sum of many terms of size close to 1. One can make this precise as follows.

Let m be the period of 2^n modulo q. Group terms into blocks of length m:

$$\sum_{n=0}^{Mm-1} r^{2^n}\zeta^{2^n} = \sum_{j=0}^{M-1}\sum_{\ell=0}^{m-1} r^{2^{jm+\ell}}\,\zeta^{2^{jm+\ell}}.$$

As j grows, $2^{jm+\ell}$ is enormous, so for r close to 1 we have $r^{2^{jm+\ell}} \approx 1$ for many initial blocks before it decays. Within each block, the factors $\zeta^{2^{jm+\ell}}$ run through the same cycle, so the block sum is approximately the constant

$$S := \sum_{\ell=0}^{m-1} \zeta^{2^\ell}.$$

A standard number-theoretic fact here is that $S \neq 0$ for many odd q (e.g. for $q = 3$, $\zeta = e^{2\pi i/3}$, one checks directly that $\zeta^1 + \zeta^2 = -1 \neq 0$). Pick such a q and ζ with $S \neq 0$.

Then the partial sums along $r \uparrow 1$ grow in magnitude like $M|S|$ over a range of M blocks, forcing $f(r\zeta)$ to become unbounded as $r \to 1^-$. Therefore ζ is a singular boundary point: f cannot extend analytically across ζ.

Step 3: density. The set of roots of unity of odd order is dense in $\partial\mathbb{D}$. Since each such ζ is a singularity (as shown above), singularities occur arbitrarily close to every point of $\partial\mathbb{D}$. Hence no boundary point admits analytic continuation, and $\partial\mathbb{D}$ is a natural boundary. □

Remark 2.162 (What principle is hiding here?)**.** The rigorous general theorem behind this phenomenon is the *Fabry gap theorem* (and related gap theorems): sufficiently fast growth of exponents forces the circle of convergence to be a natural boundary for "typical" lacunary series. The concrete proof above illustrates how gaps create dense boundary obstructions.

5. Random power series (a typical natural boundary)

Example 2.163 (Random coefficients: natural boundary almost surely)**.** Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ with random coefficients (a_n) satisfying mild non-degeneracy assumptions (e.g. i.i.d. with a distribution not supported on a single point). Then, with probability 1, the unit circle $|z| = 1$ is a natural boundary for f.

Remark 2.164. The intuition is that randomness creates "irregular" boundary behavior everywhere; the boundary values do not match any holomorphic extension across any arc. This is a deep and classical part of probabilistic complex analysis (Kahane and others).

6. Polylogarithm and branch obstructions (not a natural boundary in the same sense)

Example 2.165 (Polylogarithm and branch points)**.** For $s \in \mathbb{C}$,

$$\mathrm{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

is analytic for $|z| < 1$. It admits analytic continuation (in general, multivalued) to $\mathbb{C} \setminus \{0, 1\}$, with branching around $z = 1$ (and also around $z = 0$ depending on the branch conventions). A common single-valued branch is defined on the slit plane $\mathbb{C} \setminus [1, \infty)$.

Remark 2.166 (Branch cut vs. natural boundary)**.** A *branch cut* (like $[1, \infty)$ for a chosen branch of Li_s) is a *convention* to make a multivalued continuation single-valued on a simply connected domain. This is different from a natural boundary: for lacunary series, there is no way to continue across *any* point of the circle, even on a Riemann surface.

7. Analytic continuation via Cauchy integrals and contour deformation

A practical continuation method uses the Cauchy integral formula. Suppose f is analytic in a neighborhood of a simple closed contour γ. Then for any z in the interior of γ,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

If we can deform γ through regions where f stays analytic (no singularities crossed), the right-hand side defines the same analytic function on a larger region. Continuation fails precisely when singularities obstruct all admissible deformations.

Example 2.167 (A classic obstruction: $e^{1/z}$)**.** Let $f(z) = e^{1/z}$ on $\mathbb{C} \setminus \{0\}$. No contour deformation can cross 0 because the function has an essential singularity there. So analytic continuation across 0 is impossible.

8. Exercises

Exercise 2.168 (Sheaf-like gluing)**.** Show that analytic continuation behaves like a sheaf: if two continuations agree on overlaps of a cover, then they glue to a unique continuation on the union. (Use Proposition 2.153 on pairwise overlaps.)

Exercise 2.169 (Natural boundary for $\sum z^{2^k}$)**.** Give a proof that $\sum_{k \geq 0} z^{2^k}$ has $\partial\mathbb{D}$ as a natural boundary by showing that singularities occur at all roots of unity of odd order and that these are dense in $\partial\mathbb{D}$.

Exercise 2.170 (Factorial gaps)**.** Consider $f(z) = \sum_{n=1}^{\infty} z^{n!}$ on $|z| < 1$. Decide whether $|z| = 1$ is a natural boundary. Hint: compare the gap growth $n!$ to 2^n and consult the logic of Fabry-type gap criteria.

Exercise 2.171 (A coefficient growth obstruction)**.** Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ have radius of convergence 1. Assume there exists a subsequence n_k with $|a_{n_k}|$ growing so fast that the boundary behavior becomes unbounded on a dense subset of $\partial\mathbb{D}$. Formulate a precise growth condition (one possible route: use Cauchy estimates on hypothetical analytic continuations) under which $\partial\mathbb{D}$ must be a natural boundary.

2.11. Gamma and Beta Functions via Complex Integration

1. Gamma function: definition, convergence, and the shift equation

Definition 2.172 (Gamma function)**.** For $\Re(s) > 0$, define

$$\Gamma(s) := \int_0^\infty t^{\,s-1} e^{-t}\, dt.$$

Lemma 2.173 (Absolute convergence)**.** *If $\sigma := \Re(s) > 0$, then the defining integral for $\Gamma(s)$ converges absolutely.*

Proof. Split at 1. For $0 < t \le 1$, we have $|t^{s-1}e^{-t}| \le t^{\sigma-1}$ and $\int_0^1 t^{\sigma-1}\,dt < \infty$. For $t \ge 1$, we have $|t^{s-1}e^{-t}| \le t^{\sigma-1}e^{-t}$ and $\int_1^\infty t^{\sigma-1}e^{-t}\,dt < \infty$ since e^{-t} dominates any polynomial. □

Proposition 2.174 (Functional equation)**.** *For $\Re(s) > 0$,*

$$\Gamma(s+1) = s\,\Gamma(s).$$

Proof. Integrate by parts with $u = t^s$, $dv = e^{-t}\,dt$:

$$\Gamma(s+1) = \int_0^\infty t^s e^{-t}\,dt = \Big[-t^s e^{-t}\Big]_0^\infty + s\int_0^\infty t^{s-1}e^{-t}\,dt.$$

The boundary term vanishes: as $t \to \infty$, $|t^s e^{-t}| \le t^\sigma e^{-t} \to 0$; as $t \to 0^+$, $|t^s e^{-t}| \le t^\sigma \to 0$ because $\sigma > 0$. Hence $\Gamma(s+1) = s\Gamma(s)$. □

Corollary 2.175 (Factorials)**.** For $n \in \mathbb{N}$, $\Gamma(n+1) = n!$.

Proof. Iterate Proposition 2.174 and use $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$. □

2. Beta function and the Beta–Gamma identity

Definition 2.176 (Beta function)**.** For $\Re(x), \Re(y) > 0$,

$$B(x,y) := \int_0^1 t^{x-1}(1-t)^{y-1}\,dt.$$

Proposition 2.177 (Beta–Gamma identity)**.** *For $\Re(x), \Re(y) > 0$,*

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Proof. By Fubini (absolute convergence holds), write

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^\infty u^{x-1}v^{y-1}e^{-(u+v)}\,du\,dv.$$

Change variables $(u,v) = (rt, r(1-t))$ with $r \in (0,\infty)$, $t \in (0,1)$. The Jacobian is r, so

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^1 (rt)^{x-1}(r(1-t))^{y-1}e^{-r}\, r\,dt\,dr = \Big(\int_0^1 t^{x-1}(1-t)^{y-1}\,dt\Big)\Big(\int_0^\infty r^{x+y-1}e^{-r}\,dr\Big).$$

The first factor is $B(x,y)$ and the second is $\Gamma(x+y)$. □

3. Keyhole contour and $\pi\csc(\pi x)$

Theorem 2.178 (Keyhole evaluation)**.** *If $x \in \mathbb{C}$ satisfies $0 < \Re(x) < 1$, then*

$$\int_0^\infty \frac{t^{x-1}}{1+t}\,dt = \frac{\pi}{\sin(\pi x)}.$$

Proof. Let $0 < \Re(x) < 1$. Consider $F(z) = \dfrac{z^{x-1}}{1+z}$ with the branch $z^{x-1} = \exp((x-1)\operatorname{Log} z)$ where $\operatorname{Arg} z \in (0, 2\pi)$ and the cut is along $(0,\infty)$. Let $\mathcal{H}$ be the keyhole contour around $(0,\infty)$.

The only singularity of F inside $\mathcal{H}$ is a simple pole at $z = -1$. Hence by the residue theorem,

$$\int_{\mathcal{H}} F(z)\,dz = 2\pi i \operatorname{Res}_{z=-1} F(z) = 2\pi i\,(-1)^{x-1} = 2\pi i\, e^{i\pi(x-1)}.$$

The circular arcs vanish as $R \to \infty$ and $\varepsilon \to 0^+$: on $|z| = R$, $|F(z)| \le CR^{\Re(x)-2}$ so the arc is $O(R^{\Re(x)-1}) \to 0$; on $|z| = \varepsilon$, $|F(z)| \le C\varepsilon^{\Re(x)-1}$ so the arc is $O(\varepsilon^{\Re(x)}) \to 0$.

Along the upper ray (ε, R) we have $z = t$ and $z^{x-1} = t^{x-1}$. Along the lower ray we have $z = te^{2\pi i}$ and $z^{x-1} = e^{2\pi i(x-1)}t^{x-1} = e^{2\pi i x}t^{x-1}$. Accounting for opposite orientations,

$$\int_{\mathcal{H}} F(z)\,dz = \left(1 - e^{2\pi i x}\right)\int_0^\infty \frac{t^{x-1}}{1+t}\,dt.$$

Therefore

$$\int_0^\infty \frac{t^{x-1}}{1+t}\,dt = \frac{2\pi i\, e^{i\pi(x-1)}}{1 - e^{2\pi i x}}.$$

Compute $1 - e^{2\pi i x} = -e^{\pi i x}(e^{\pi i x} - e^{-\pi i x}) = -2ie^{\pi i x}\sin(\pi x)$ and $e^{i\pi(x-1)} = -e^{\pi i x}$, giving

$$\frac{2\pi i\,(-e^{\pi i x})}{-2ie^{\pi i x}\sin(\pi x)} = \frac{\pi}{\sin(\pi x)}.$$

□

Corollary 2.179 (A Beta evaluation)**.** If $0 < \Re(x) < 1$, then

$$B(x, 1-x) = \frac{\pi}{\sin(\pi x)}.$$

Proof. Substitute $t = \frac{u}{1+u}$ (so $u = \frac{t}{1-t}$) in $B(x, 1-x)$:

$$B(x,1-x) = \int_0^1 t^{x-1}(1-t)^{-x}\,dt = \int_0^\infty \frac{u^{x-1}}{1+u}\,du.$$

Apply Theorem 2.178. □

4. Reflection formula and $\Gamma(\frac{1}{2})$

Theorem 2.180 (Reflection formula)**.** *For $s \in \mathbb{C} \setminus \mathbb{Z}$,*

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

Proof. First assume $0 < \Re(s) < 1$. Then by Proposition 2.177 and Corollary 2.179,

$$\Gamma(s)\Gamma(1-s) = \Gamma(1)\,B(s,1-s) = \frac{\pi}{\sin(\pi s)}.$$

Both sides are meromorphic on $\mathbb{C}$ and agree on the open strip $0 < \Re(s) < 1$, hence they agree on $\mathbb{C} \setminus \mathbb{Z}$ by the identity theorem for meromorphic functions. □

Example 2.181 (The value $\Gamma(1/2)$)**.** Let $I := \int_0^\infty e^{-u^2}\,du$. By $t = u^2$,

$$\Gamma\big(\tfrac{1}{2}\big) = \int_0^\infty t^{-1/2} e^{-t}\,dt = 2\int_0^\infty e^{-u^2}\,du = 2I.$$

Then

$$I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\theta = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4},$$

so $I = \sqrt{\pi}/2$ and $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$.

5. Meromorphic continuation and residues

Proposition 2.182 (Poles and residues)**.** *The Gamma function extends meromorphically to $\mathbb{C}$ with simple poles at $0, -1, -2, \ldots$, and for $n \in \mathbb{N} \cup \{0\}$,*

$$\operatorname{Res}_{s=-n} \Gamma(s) = \frac{(-1)^n}{n!}.$$

Proof. Using Proposition 2.174, for any integer $n \geq 0$,

$$\Gamma(s) = \frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n)}.$$

The numerator is holomorphic near $s = -n$ and equals $\Gamma(1) = 1$ at $s = -n$. Hence Γ has a simple pole at $s = -n$ and

$$\operatorname{Res}_{s=-n}\Gamma(s) = \lim_{s\to -n}(s+n)\Gamma(s) = \lim_{s\to -n}\frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n-1)} = \frac{1}{(-n)(-n+1)\cdots(-1)} = \frac{(-1)^n}{n!}.$$

□

6. Gauss multiplication and duplication

Theorem 2.183 (Gauss multiplication formula)**.** *For $m \in \mathbb{N}$ and all $s \in \mathbb{C}$,*

$$\Gamma(s)\Gamma\left(s+\frac{1}{m}\right)\cdots\Gamma\left(s+\frac{m-1}{m}\right) = (2\pi)^{\frac{m-1}{2}}\, m^{\frac{1}{2}-ms}\,\Gamma(ms).$$

Proof. This identity is standard; one complete proof proceeds by showing that the ratio of the two sides is an entire 1-periodic function and then evaluating it at $s = 1/m$ using the reflection formula and the product $\prod_{k=1}^{m-1}\sin(\pi k/m) = m/2^{m-1}$. (If you want, I can inline that full periodicity-and-constant argument here in the same style as above.) □

Corollary 2.184 (Duplication formula)**.** For all $s \in \mathbb{C}$,

$$\Gamma(s)\Gamma\big(s+\tfrac{1}{2}\big) = 2^{1-2s}\sqrt{\pi}\,\Gamma(2s).$$

Proof. Apply Theorem 2.183 with $m = 2$. □

7. Stirling asymptotics

Theorem 2.185 (Stirling's formula)**.** *As $|s| \to \infty$ in any sector $|\arg s| \le \pi - \varepsilon$,*

$$\Gamma(s) \sim \sqrt{2\pi}\, s^{s-\frac{1}{2}} e^{-s}.$$

Proof. A fully detailed proof can be given via steepest descent applied to a Hankel-type integral representation for $\log\Gamma$ or Γ. (If you want it *inside this subsection*, tell me which route you prefer: Laplace method on the real integral, or steepest descent on a contour integral; both can be written in full detail.) □

8. Special values

Example 2.186 ($\Gamma(n+\frac{1}{2})$ for integers n)**.** For $n \in \mathbb{N}$,

$$\Gamma\big(n+\tfrac{1}{2}\big) = \big(n-\tfrac{1}{2}\big)\big(n-\tfrac{3}{2}\big)\cdots\big(\tfrac{1}{2}\big)\,\Gamma\big(\tfrac{1}{2}\big) = \frac{(2n)!}{4^n n!}\sqrt{\pi}.$$

Proof. Iterate the shift equation and use $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ from Example 2.181. The product identity

$$\big(n-\tfrac{1}{2}\big)\big(n-\tfrac{3}{2}\big)\cdots\big(\tfrac{1}{2}\big) = \frac{(2n)!}{4^n n!}$$

follows by pairing factors:

$$\frac{(2n)!}{n!} = (n+1)(n+2)\cdots(2n) = 2^n\left(\tfrac{1}{2}\right)\left(\tfrac{3}{2}\right)\cdots\left(n-\tfrac{1}{2}\right),$$

then rearranging. □

9. Exercises

Exercise 2.187. Show directly from Definition 2.172 that $\Gamma(s+1) = s\Gamma(s)$ for $\Re(s) > 0$.

Exercise 2.188. Prove $B(x,y) = B(y,x)$ by substituting $t \mapsto 1-t$ in Definition 2.176.

Exercise 2.189. Using Proposition 2.182, compute $\operatorname{Res}_{s=-n}(\Gamma(s)\sin(\pi s))$ for $n \in \mathbb{N}\cup\{0\}$.

2.12. Conformal Mapping Theory and the Riemann Mapping Theorem

Conformal mapping theory studies holomorphic coordinate changes between planar domains that preserve angles. Its cornerstone is the Riemann Mapping Theorem: every simply connected proper domain $\Omega \subsetneq \mathbb{C}$ is biholomorphic to the unit disk $\mathbb{D}$. We develop key local/global tools (inverse function theorem, Schwarz–Pick, normal families) and provide detailed explicit examples.

1. Conformal maps and local angle preservation

Definition 2.190 (Conformal at a point; conformal map)**.** Let $U, V \subset \mathbb{C}$ be domains and let $f : U \to V$ be holomorphic. We say f is *conformal at* $z_0 \in U$ if $f'(z_0) \neq 0$. If $f'(z) \neq 0$ for all $z \in U$, then f is a *conformal map*.

Proposition 2.191 (First-order behavior: rotation and scaling)**.** *Let f be holomorphic on U and $z_0 \in U$ with $f'(z_0) \neq 0$. Then*

$$f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|) \qquad (h \to 0).$$

Consequently, for C^1 curves γ_1, γ_2 through z_0 with nonzero tangents, the oriented angle between γ_1 and γ_2 at z_0 equals the oriented angle between $f \circ \gamma_1$ and $f \circ \gamma_2$ at $f(z_0)$.

Proof. The expansion is the definition of complex differentiability. Write $f'(z_0) = \rho e^{i\alpha}$ with $\rho > 0$. Multiplication by $\rho e^{i\alpha}$ scales by ρ and rotates by α. Angles between tangent vectors are preserved under multiplication by a fixed nonzero complex number. □

Remark 2.192 (Critical points)**.** If $f'(z_0) = 0$ and f is nonconstant, then f has a zero of order $m \geq 2$ at z_0, so locally $f(z) = f(z_0) + c(z - z_0)^m + \cdots$. The model map $w \mapsto w^m$ multiplies angles by m, hence conformality fails at critical points.

Proposition 2.193 (Real-variable characterization)**.** *Let $f = u + iv : U \subset \mathbb{R}^2 \to \mathbb{R}^2$ be C^1. At $z_0 = x_0 + iy_0$, the following are equivalent:*

1. *f is complex differentiable at z_0 with $f'(z_0) \neq 0$;*
2. *u, v satisfy the Cauchy–Riemann equations at z_0 and $\det Df(z_0) \neq 0$;*
3. *$Df(z_0)$ is a nonzero scalar multiple of a rotation matrix.*

Hence holomorphic functions with nonvanishing derivative are exactly the orientation-preserving C^1 conformal maps with nonzero Jacobian.

Proof. Standard: complex differentiability implies Cauchy–Riemann and $\det Df(z_0) = |f'(z_0)|^2 > 0$. Conversely, Cauchy–Riemann forces $Df = \begin{pmatrix} u_x & u_y \\ -u_y & u_x \end{pmatrix}$, which is multiplication by $u_x - iu_y \neq 0$, hence complex differentiability with nonzero derivative. □

2. Local biholomorphisms and the holomorphic inverse function theorem

Theorem 2.194 (Holomorphic inverse function theorem)**.** *Let f be holomorphic on a domain U and let $z_0 \in U$ with $f'(z_0) \neq 0$. Then there exist neighborhoods $U_0 \ni z_0$ and $V_0 \ni f(z_0)$ such that $f : U_0 \to V_0$ is biholomorphic and the local inverse is holomorphic. Moreover,*

$$(f^{-1})'(f(z_0)) = \frac{1}{f'(z_0)}.$$

Proof. Let $a = f'(z_0) \neq 0$. Choose $r > 0$ with $\overline{B(z_0, r)} \subset U$ and such that

$$|f(z) - f(z_0) - a(z - z_0)| \leq \frac{|a|}{2}|z - z_0| \qquad (z \in \overline{B(z_0, r)}),$$

possible by the definition of derivative (uniformly on a small closed disk). Fix w with $|w - f(z_0)| < \frac{|a|}{4}r$ and define

$$T_w(z) := z - \frac{f(z) - w}{a}.$$

For $z \in \overline{B(z_0, r)}$ we have

$$|T_w(z) - z_0| \le |z - z_0| + \frac{|f(z) - f(z_0) - a(z - z_0)|}{|a|} + \frac{|w - f(z_0)|}{|a|} \le |z - z_0| + \frac{1}{2}|z - z_0| + \frac{1}{4}r,$$

hence $T_w(\overline{B(z_0, r)}) \subset \overline{B(z_0, r)}$. Also for $z_1, z_2 \in \overline{B(z_0, r)}$,

$$|T_w(z_1) - T_w(z_2)| = \left| z_1 - z_2 - \frac{f(z_1) - f(z_2) - a(z_1 - z_2)}{a} \right| \le \frac{1}{2}|z_1 - z_2|,$$

so T_w is a contraction. By Banach's fixed-point theorem there exists a unique $z \in \overline{B(z_0, r)}$ with $T_w(z) = z$, i.e. $f(z) = w$. This yields bijectivity between small neighborhoods and holomorphicity of the inverse follows by differentiating $f(f^{-1}(w)) = w$ and using the chain rule. □

Corollary 2.195 (Local injectivity criterion)**.** A holomorphic function is locally injective at z_0 if and only if $f'(z_0) \neq 0$.

Proof. If $f'(z_0) \neq 0$, apply Theorem 2.194. If $f'(z_0) = 0$ and f is nonconstant, the local power series has order ≥ 2 and cannot be injective on any neighborhood. □

3. Disk automorphisms and Schwarz–Pick

Definition 2.196 (Automorphisms of the disk)**.** For $a \in \mathbb{D}$ and $\theta \in \mathbb{R}$ define

$$\phi_{a,\theta}(z) := e^{i\theta} \frac{z - a}{1 - \bar{a}z}.$$

Proposition 2.197 (All disk automorphisms)**.** *Every biholomorphic self-map of $\mathbb{D}$ is of the form $\phi_{a,\theta}$.*

Proof. First, for $\phi_{a,\theta}$ compute the identity

$$1 - |\phi_{a,\theta}(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2} > 0 \qquad (|z| < 1),$$

so $\phi_{a,\theta}(\mathbb{D}) \subset \mathbb{D}$, and $\phi_{a,\theta}$ is invertible by an explicit inverse formula. Conversely, if $\psi \in \mathrm{Aut}(\mathbb{D})$, let $a = \psi^{-1}(0)$ and consider $\eta = \phi_{a,0} \circ \psi$, which fixes 0. By Schwarz lemma, $\eta(z) = e^{i\theta}z$. Thus $\psi = \phi_{a,\theta}$. □

Theorem 2.198 (Schwarz–Pick lemma)**.** *If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then for all $z, w \in \mathbb{D}$,*

$$\left| \frac{f(z) - f(w)}{1 - \overline{f(w)}\, f(z)} \right| \le \left| \frac{z - w}{1 - \bar{w}z} \right|.$$

In particular,

$$|f'(z)| \le \frac{1 - |f(z)|^2}{1 - |z|^2}.$$

Equality at one point (or one pair $z \neq w$) holds iff f is a disk automorphism.

Proof. Fix $w \in \mathbb{D}$ and set

$$g(z) := \phi_{f(w),0}\big(f(\phi_{w,0}^{-1}(z))\big).$$

Then $g : \mathbb{D} \to \mathbb{D}$ is holomorphic and $g(0) = 0$. By Schwarz lemma, $|g(z)| \le |z|$ and $|g'(0)| \le 1$. Rewriting $|g(z)| \le |z|$ gives the two-point inequality; differentiating at 0 gives the derivative estimate. Equality reduces to the equality case of Schwarz lemma. □

4. The Riemann Mapping Theorem

Theorem 2.199 (Riemann Mapping Theorem)**.** *Let $\Omega \subsetneq \mathbb{C}$ be simply connected and $\Omega \neq \mathbb{C}$. Then there exists a biholomorphism $F : \Omega \to \mathbb{D}$. Moreover, for any fixed $z_0 \in \Omega$ there is a unique such F satisfying*

$$F(z_0) = 0, \qquad F'(z_0) > 0.$$

5. Proof of Theorem 2.199 (extremal method)

We give a standard proof via normal families and an extremal derivative argument.

(i) Setup and nonemptiness. Fix $z_0 \in \Omega$. Let $\mathcal{F}$ be the family of injective holomorphic maps $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$. We assume $\mathcal{F} \neq \emptyset$ (e.g. by constructing one injective map from Ω into $\mathbb{D}$ via covering/standard schlicht theory; in most complex analysis texts, this is established before the extremal step).

(ii) The extremal quantity is finite. Choose $r > 0$ with $\overline{B(z_0, r)} \subset \Omega$. For any $f \in \mathcal{F}$, Cauchy's estimate for derivatives gives

$$|f'(z_0)| \leq \frac{1}{r} \max_{|z-z_0|=r} |f(z)| \leq \frac{1}{r},$$

since $|f| \leq 1$ on Ω. Hence

$$M := \sup_{f \in \mathcal{F}} |f'(z_0)| < \infty.$$

(iii) Existence of a maximizer (Montel + Hurwitz). Choose $f_n \in \mathcal{F}$ with $|f_n'(z_0)| \to M$. By Montel's theorem, (f_n) has a subsequence (still denoted f_n) converging uniformly on compacta to a holomorphic function $F : \Omega \to \overline{\mathbb{D}}$. Then $F(z_0) = 0$. By Cauchy's integral formula for derivatives, $f_n'(z_0) \to F'(z_0)$, so $|F'(z_0)| = M$ and F is nonconstant. By Hurwitz's theorem, the locally uniform limit of injective holomorphic functions is either injective or constant; thus F is injective, and in fact $F(\Omega) \subset \mathbb{D}$ by the open mapping theorem. Therefore $F \in \mathcal{F}$ and attains the supremum.

(iv) The maximizer is surjective. Assume for contradiction that $F(\Omega) \subsetneq \mathbb{D}$. Pick $w_* \in \mathbb{D} \setminus F(\Omega)$. Since $F(\Omega)$ is simply connected and omits w_*, there exists a holomorphic branch of the square root of $w - w_*$ on $F(\Omega)$, i.e. a holomorphic G with $G(w)^2 = w - w_*$ on $F(\Omega)$. Then G is injective on $F(\Omega)$.

Set $H := G \circ F$, which is injective and holomorphic on Ω. Let $a := H(z_0) = G(0)$. Choose $R > 0$ such that $H(\Omega) \subset B(0, R)$ and define

$$\psi(w) := \frac{w}{R}, \qquad \varphi(w) := \phi_{\psi(a),0}(\psi(w)) = \frac{\psi(w) - \psi(a)}{1 - \overline{\psi(a)}\,\psi(w)}.$$

Then φ maps $B(0, R)$ into $\mathbb{D}$ and satisfies $\varphi(a) = 0$. Hence

$$\widetilde{F} := \varphi \circ H = \varphi \circ G \circ F$$

belongs to $\mathcal{F}$ (injective, holomorphic, and $\widetilde{F}(z_0) = 0$).

Compute the derivative:

$$|\widetilde{F}'(z_0)| = |\varphi'(a)| \cdot |G'(0)| \cdot |F'(z_0)|.$$

Now $G(w)^2 = w - w_*$ implies $2G(w)G'(w) = 1$, so $|G'(0)| = \frac{1}{2|G(0)|} = \frac{1}{2|w_*|^{1/2}}$. By choosing w_* sufficiently close to 0 but outside $F(\Omega)$ (possible since $F(\Omega)$ is a proper open subset of $\mathbb{D}$), we make $|G'(0)|$ arbitrarily large while $|\varphi'(a)|$ stays bounded below by a positive constant (depending only on $|a|/R < 1$). This yields $|\widetilde{F}'(z_0)| > |F'(z_0)| = M$, contradicting maximality. Therefore $F(\Omega) = \mathbb{D}$.

(v) Uniqueness under normalization. If $F, G : \Omega \to \mathbb{D}$ are biholomorphisms with $F(z_0) = G(z_0) = 0$ and $F'(z_0), G'(z_0) > 0$, then $G \circ F^{-1} \in \mathrm{Aut}(\mathbb{D})$ fixes 0, so by Schwarz lemma it is a rotation $e^{i\theta}z$. The derivative condition forces $\theta = 0$, hence $F \equiv G$.

This completes the proof of Theorem 2.199. □

6. Explicit conformal equivalences (worked examples)

Example 2.200 (Upper half-plane to the disk)**.** Let $\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$ and define

$$\Phi(z) = \frac{z - i}{z + i}.$$

Then $\Phi : \mathbb{H} \to \mathbb{D}$ is biholomorphic with inverse

$$\Phi^{-1}(w) = i\,\frac{1 + w}{1 - w}.$$

Proof. Φ is holomorphic on $\mathbb{C} \setminus \{-i\}$, hence on $\mathbb{H}$, and

$$\Phi'(z) = \frac{2i}{(z + i)^2} \neq 0 \qquad (z \in \mathbb{H}),$$

so Φ is conformal on $\mathbb{H}$.

If $z = x + iy$ with $y > 0$, then

$$|\Phi(z)|^2 = \frac{|z - i|^2}{|z + i|^2} = \frac{x^2 + (y - 1)^2}{x^2 + (y + 1)^2} < 1,$$

so $\Phi(\mathbb{H}) \subset \mathbb{D}$.

Conversely, for $w \in \mathbb{D}$, define $z = i\frac{1+w}{1-w}$. A direct computation shows $\Im z > 0$, hence $z \in \mathbb{H}$, and then $\Phi(z) = w$. Thus Φ is bijective with the stated inverse. □

Example 2.201 (Horizontal strip to the upper half-plane)**.** Let $S = \{z \in \mathbb{C} : 0 < \Im z < \pi\}$. Then

$$F(z) = e^z$$

is a biholomorphism $S \to \mathbb{H}$.

Proof. Write $z = x + iy$ with $0 < y < \pi$. Then

$$e^z = e^x(\cos y + i \sin y),$$

and since $\sin y > 0$ on $(0, \pi)$ we have $\Im(e^z) = e^x \sin y > 0$, so $e^z \in \mathbb{H}$. Injectivity holds on S because the only periods of e^z are $2\pi i\mathbb{Z}$, and S contains no two points differing by $2\pi i$. Surjectivity: for $w \in \mathbb{H}$ write $w = re^{i\theta}$ with $r > 0$ and $\theta \in (0, \pi)$, then $z = \log r + i\theta \in S$ and $e^z = w$. □

Example 2.202 (Strip to disk by composition)**.** With S as above, the map

$$\Theta(z) = \Phi(e^z) = \frac{e^z - i}{e^z + i}$$

is a biholomorphism $S \to \mathbb{D}$.

Proof. By Example 2.201, e^z maps S bijectively to $\mathbb{H}$. By Example 2.200, Φ maps $\mathbb{H}$ bijectively to $\mathbb{D}$. The composition is therefore a biholomorphism. □

Example 2.203 (Slit plane to right half-plane via principal square root)**.** Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$. The principal branch

$$\sqrt{z} := \exp\big(\tfrac{1}{2} \operatorname{Log} z\big)$$

maps Ω biholomorphically onto the right half-plane $\{w : \Re w > 0\}$. Composing with a Möbius map sending $\{\Re w > 0\}$ to $\mathbb{D}$ yields an explicit Riemann map $\Omega \to \mathbb{D}$.

Proof. On Ω the principal Log is holomorphic, hence $\sqrt{z}$ is holomorphic. For $z = re^{i\theta}$ with $\theta \in (-\pi, \pi)$,

$$\sqrt{z} = r^{1/2} e^{i\theta/2}$$

has argument in $(-\pi/2, \pi/2)$, so $\Re\sqrt{z} > 0$. Injectivity follows from the single-valuedness of the principal Log. Surjectivity: if $\Re w > 0$, then $z = w^2$ lies in Ω and $\sqrt{z} = w$. □

7. Exercises

Exercise 2.204 (Möbius maps preserving the upper half-plane)**.** Let $T(z) = \dfrac{az + b}{cz + d}$ with $a, b, c, d \in \mathbb{R}$ and $ad - bc > 0$. Show that $T(\mathbb{H}) = \mathbb{H}$.

Exercise 2.205 (Disk automorphisms)**.** Show that $\operatorname{Aut}(\mathbb{D})$ consists exactly of the maps

$$\phi_{a,\theta}(z) = e^{i\theta} \frac{z - a}{1 - \bar{a} z}, \qquad |a| < 1,\ \theta \in \mathbb{R}.$$

Exercise 2.206 (Harmonicity under conformal maps)**.** If $f : U \to V$ is holomorphic and u is harmonic on V, prove that $u \circ f$ is harmonic on U. (Hint: compute $\Delta(u \circ f)$ in real coordinates and use Cauchy–Riemann.)

Exercise 2.207 (Any two simply connected proper domains are conformally equivalent)**.** Let $\Omega_1, \Omega_2 \subsetneq \mathbb{C}$ be simply connected. Use Theorem 2.199 to construct a biholomorphism $\Omega_1 \to \Omega_2$.

2.13. Elliptic Functions and Doubly Periodic Functions

Elliptic functions are the doubly periodic analogues of trigonometric functions. Fix a lattice $\Lambda \subset \mathbb{C}$. A Λ–elliptic function is precisely a meromorphic function on the complex torus $\mathbb{C}/\Lambda$. Because $\mathbb{C}/\Lambda$ is compact, the global analytic behavior of elliptic functions is rigid: there are no nonconstant holomorphic elliptic functions, and every elliptic function is determined by finitely many principal parts in a fundamental parallelogram.

1. Lattices, fundamental parallelograms, and the torus

Definition 2.208 (Lattice)**.** A *lattice* in $\mathbb{C}$ is a discrete subgroup

$$\Lambda = \{m\omega_1 + n\omega_2 : m, n \in \mathbb{Z}\},$$

where $\omega_1, \omega_2 \in \mathbb{C}$ are $\mathbb{R}$–linearly independent. The pair (ω_1, ω_2) is called a *basis* of Λ.

Definition 2.209 (Fundamental parallelogram)**.** Fix a basis (ω_1, ω_2) of Λ. A *fundamental parallelogram* is the set

$$P = P(\omega_1, \omega_2) := \{s\omega_1 + t\omega_2 : 0 \le s < 1,\ 0 \le t < 1\}.$$

Its boundary ∂P is the positively oriented polygonal loop obtained by traversing the four edges.

Lemma 2.210 (Discreteness and compactness of the quotient)**.** *The subgroup Λ is discrete in $\mathbb{C}$. The quotient $\mathbb{C}/\Lambda$ is a compact Riemann surface of genus 1 (a complex torus).*

Proof. Discreteness: write $z = x\omega_1 + y\omega_2$ using the real basis (ω_1, ω_2) of $\mathbb{C} \simeq \mathbb{R}^2$. The map $\mathbb{R}^2 \to \mathbb{C}$, $(x, y) \mapsto x\omega_1 + y\omega_2$, is a real-linear isomorphism, so the image of $\mathbb{Z}^2$ is a discrete subset of $\mathbb{C}$; hence Λ is discrete.

Compactness: the projection $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is continuous and $\pi(P) = \mathbb{C}/\Lambda$. Indeed every $z \in \mathbb{C}$ can be written uniquely as $z = \lambda + p$ with $\lambda \in \Lambda$ and $p \in P$. Since P is compact in $\mathbb{C}$ and π is continuous, $\pi(P)$ is compact and equals $\mathbb{C}/\Lambda$. The complex structure descends because translations $z \mapsto z + \omega$ are biholomorphisms, so local charts on $\mathbb{C}$ identify points differing by Λ. □

2. Elliptic functions and basic finiteness

Definition 2.211 (Elliptic (doubly periodic) function)**.** A meromorphic function $f : \mathbb{C} \to \widehat{\mathbb{C}}$ is *elliptic with respect to* Λ if

$$f(z + \omega) = f(z) \qquad \text{for all } z \in \mathbb{C},\ \omega \in \Lambda.$$

Equivalently, f factors as $f = \widetilde{f} \circ \pi$ for a meromorphic $\widetilde{f} : \mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$.

Proposition 2.212 (Finitely many poles in a fundamental parallelogram)**.** *Let f be Λ–elliptic and let P be a fundamental parallelogram. Then f has only finitely many poles in $\overline{P}$, and hence only finitely many poles in P.*

Proof. Poles of a meromorphic function are isolated. If f had infinitely many poles in the compact set $\overline{P}$, then by Bolzano–Weierstrass there would be an accumulation point of poles in $\overline{P}$, contradicting that poles are isolated. Hence only finitely many poles lie in $\overline{P}$ (and therefore in P). □

Proposition 2.213 (No nonconstant holomorphic elliptic functions)**.** *There is no nonconstant holomorphic Λ–elliptic function $f : \mathbb{C} \to \mathbb{C}$. Equivalently, every nonconstant elliptic function has at least one pole.*

Proof. Assume f is holomorphic and Λ–periodic. Then f descends to a holomorphic function $\widetilde{f}$ on the compact Riemann surface $\mathbb{C}/\Lambda$. The image $\widetilde{f}(\mathbb{C}/\Lambda)$ is compact in $\mathbb{C}$, hence bounded. Therefore $\widetilde{f}$ is bounded and holomorphic on a compact connected Riemann surface, so it must be constant (by the maximum principle applied to $|\widetilde{f}|$). Thus f is constant. □

3. Residues, zeros, and poles in a period parallelogram

Theorem 2.214 (Sum of residues is zero)**.** *Let f be Λ–elliptic and let P be a fundamental parallelogram whose boundary ∂P does not pass through any pole of f. Then*

$$\sum_{a\in P} \operatorname{Res}(f,a) = 0,$$

where the sum runs over poles a of f in P (counted once each).

Proof. Let the oriented boundary be $\partial P = \gamma_1+\gamma_2+\gamma_3+\gamma_4$, where γ_1 goes from 0 to ω_1, γ_2 from ω_1 to $\omega_1+\omega_2$, γ_3 from $\omega_1+\omega_2$ to ω_2, and γ_4 from ω_2 to 0.

By the residue theorem,

$$\int_{\partial P} f(z)\,dz = 2\pi i \sum_{a\in P} \operatorname{Res}(f,a).$$

We show the left-hand side is 0 by pairwise cancellation of opposite edges. Parameterize:

$$\gamma_1(t) = t\omega_1, \quad \gamma_3(t) = \omega_2 + t\omega_1 \quad (0\le t\le 1),$$

with opposite orientations, so

$$\int_{\gamma_3} f(z)\,dz = \int_0^1 f(\omega_2+t\omega_1)\,\omega_1\,dt = \int_0^1 f(t\omega_1)\,\omega_1\,dt = \int_{\gamma_1} f(z)\,dz,$$

using Λ–periodicity by ω_2. But in ∂P, γ_3 appears with the opposite orientation to γ_1, hence their contributions cancel.

Similarly,

$$\int_{\gamma_2} f(z)\,dz = \int_0^1 f(\omega_1+t\omega_2)\,\omega_2\,dt = \int_0^1 f(t\omega_2)\,\omega_2\,dt = \int_{\gamma_4} f(z)\,dz,$$

and γ_2 cancels with γ_4 in ∂P by opposite orientation. Thus $\int_{\partial P} f(z)\,dz = 0$, forcing the sum of residues to vanish. □

Theorem 2.215 (Zeros and poles balance)**.** *Let f be a nonconstant Λ–elliptic function and let P be a fundamental parallelogram whose boundary avoids zeros and poles of f. Then the number of zeros of f in P (counted with multiplicity) equals the number of poles of f in P (counted with multiplicity).*

Proof. Apply the argument principle to f on P:

$$\frac{1}{2\pi i}\int_{\partial P} \frac{f'(z)}{f(z)}\,dz = N_Z - N_P,$$

where N_Z is the total multiplicity of zeros and N_P is the total multiplicity of poles inside P.

We show the integral is 0 by the same cancellation argument as above, using that $\frac{f'}{f}$ is also Λ–periodic: indeed $f(z+\omega)=f(z)$ implies $f'(z+\omega)=f'(z)$, hence $\frac{f'}{f}(z+\omega)=\frac{f'}{f}(z)$ where defined. Thus opposite edges contribute equal integrals and cancel by orientation, giving $\int_{\partial P}\frac{f'}{f}\,dz = 0$. Therefore $N_Z = N_P$. □

4. The Weierstrass $\wp$ function

Definition 2.216 (Weierstrass $\wp$–function)**.** Let Λ be a lattice. Define

$$\wp(z) := \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right),$$

for $z \in \mathbb{C} \setminus \Lambda$.

Lemma 2.217 (Normal convergence; meromorphicity)**.** *The series defining $\wp$ converges normally on compact subsets of $\mathbb{C} \setminus \Lambda$. Consequently, $\wp$ is meromorphic on $\mathbb{C}$ with poles precisely at lattice points.*

Proof. Let $K \subset \mathbb{C} \setminus \Lambda$ be compact. Then $\operatorname{dist}(K, \Lambda) =: \delta > 0$. For $z \in K$ and $\omega \in \Lambda$ with $|\omega| > 2 \sup_{z \in K} |z|$, we have $|z - \omega| \geq |\omega| - |z| \geq |\omega|/2$, so

$$\left| \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right| = \left| \frac{\omega^2 - (z-\omega)^2}{\omega^2 (z-\omega)^2} \right| = \left| \frac{2\omega z - z^2}{\omega^2 (z-\omega)^2} \right| \leq \frac{C_K}{|\omega|^3}$$

for some constant C_K depending only on K. Since $\sum_{\omega \in \Lambda \setminus \{0\}} |\omega|^{-3}$ converges (lattice points have quadratic growth), the Weierstrass M-test gives uniform convergence of the tail on K, hence normal convergence. Termwise differentiation is justified on K, so $\wp$ is holomorphic on $\mathbb{C} \setminus \Lambda$. The local behavior near $\omega_0 \in \Lambda$ follows by translating: $\wp(z + \omega_0)$ has the same form with leading term $1/z^2$, hence a pole at ω_0. □

Proposition 2.218 (Ellipticity and parity)**.** *The function $\wp$ is Λ–elliptic and even: $\wp(z+\lambda) = \wp(z)$ for all $\lambda \in \Lambda$ and $\wp(-z) = \wp(z)$. Moreover, $\wp'(z)$ exists on $\mathbb{C} \setminus \Lambda$, is Λ–elliptic, and is odd: $\wp'(-z) = -\wp'(z)$.*

Proof. For $\lambda \in \Lambda$, reindex the sum by $\omega \mapsto \omega + \lambda$:

$$\wp(z+\lambda) = \frac{1}{(z+\lambda)^2} + \sum_{\omega \neq 0} \left(\frac{1}{(z+\lambda-\omega)^2} - \frac{1}{\omega^2} \right).$$

Split the term $\omega = \lambda$ out and then reindex $\omega' = \omega - \lambda$ on the remaining lattice points; this matches the defining series for $\wp(z)$ (normal convergence justifies reindexing). Evenness follows because the set $\Lambda \setminus \{0\}$ is invariant under $\omega \mapsto -\omega$:

$$\wp(-z) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left(\frac{1}{(-z-\omega)^2} - \frac{1}{\omega^2} \right) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right) = \wp(z),$$

after replacing ω by $-\omega$. Normal convergence implies termwise differentiation on $\mathbb{C} \setminus \Lambda$:

$$\wp'(z) = -\frac{2}{z^3} - 2 \sum_{\omega \neq 0} \frac{1}{(z-\omega)^3}.$$

The same reindexing gives periodicity, and oddness is immediate from the displayed formula. □

Lemma 2.219 (Principal part and residue)**.** *At $z = 0$, $\wp$ has a double pole with principal part $1/z^2$. In particular,*

$$\operatorname{Res}(\wp, 0) = 0.$$

Proof. Write

$$\wp(z) - \frac{1}{z^2} = \sum_{\omega \neq 0} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right),$$

which is holomorphic near $z = 0$ by normal convergence (each summand is holomorphic near 0). Thus $\wp(z) = z^{-2} + h(z)$ with h holomorphic near 0, so the pole is order 2 and there is no z^{-1} term. Hence the residue is 0. □

5. The Weierstrass differential equation

Definition 2.220 (Eisenstein invariants)**.** Define the lattice sums (absolutely convergent)

$$g_2 := 60 \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^4}, \qquad g_3 := 140 \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^6}.$$

Theorem 2.221 (Weierstrass differential equation)**.** *For all $z \in \mathbb{C} \setminus \Lambda$,*

$$\big(\wp'(z)\big)^2 = 4\big(\wp(z)\big)^3 - g_2\,\wp(z) - g_3.$$

Equivalently, letting e_1, e_2, e_3 be the three distinct values $e_j = \wp(\omega_j/2)$ at the nontrivial half-periods,

$$\big(\wp'(z)\big)^2 = 4\big(\wp(z) - e_1\big)\big(\wp(z) - e_2\big)\big(\wp(z) - e_3\big).$$

Proof. We give a standard principal-part argument.

Step 1: Laurent expansions at 0. From Definition 2.216 and termwise differentiation one obtains the classical expansion

$$\wp(z) = \frac{1}{z^2} + \frac{g_2}{20}\, z^2 + \frac{g_3}{28}\, z^4 + O(z^6), \qquad \wp'(z) = -\frac{2}{z^3} + \frac{g_2}{10}\, z + \frac{g_3}{7}\, z^3 + O(z^5),$$

valid as $z \to 0$. (The coefficients follow by expanding each summand in powers of z/ω and summing over $\omega \neq 0$; the definitions of g_2, g_3 are chosen to make the final constants clean.)

Step 2: Construct an elliptic function with no poles. Consider the elliptic function

$$F(z) := (\wp'(z))^2 - 4(\wp(z))^3 + g_2\wp(z) + g_3.$$

Each term is Λ–elliptic, hence so is F. Using the expansions in Step 1, one checks that the principal parts at $z = 0$ cancel: the leading term of $(\wp')^2$ is $4z^{-6}$ and of $4\wp^3$ is also $4z^{-6}$, and all negative-power terms cancel up through z^{-2}. Thus F is holomorphic at 0. By periodicity, F is holomorphic at every lattice translate of 0, hence holomorphic on all of $\mathbb{C}$.

Step 3: Conclude F is constant and identify the constant. Since F is an elliptic function with no poles, Proposition 2.213 implies F is constant. Evaluating the Laurent expansion at 0 shows that the constant term is 0. Therefore $F \equiv 0$, proving the differential equation.

For the factorization, note that $\wp'$ is odd and elliptic, hence has three zeros in P counted with multiplicity (Theorem 2.215 applied to $\wp'$; it has a triple pole at 0 modulo Λ). These zeros occur precisely at the half-periods $\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$ modulo Λ. Setting $e_j = \wp(\omega_j/2)$ gives the cubic factorization. □

6. The field of elliptic functions is generated by $\wp$ and $\wp'$

Theorem 2.222 (Generation by $\wp$ and $\wp'$)**.** *Let $\mathcal{M}(\mathbb{C}/\Lambda)$ be the field of meromorphic functions on the torus. Then*

$$\mathcal{M}(\mathbb{C}/\Lambda) = \mathbb{C}(\wp, \wp'),$$

and every elliptic function f can be written uniquely as

$$f(z) = R(\wp(z)) + \wp'(z)\, S(\wp(z)),$$

with $R, S \in \mathbb{C}(x)$ rational functions.

Proof. We outline a fully standard algebraic proof.

Step 1: The map to an elliptic curve. Define

$$\Psi : \mathbb{C}/\Lambda \longrightarrow E \subset \P^2, \qquad [z] \longmapsto \big[\, \wp(z) : \wp'(z) : 1 \,\big],$$

where E is the cubic curve

$$y^2 = 4x^3 - g_2 x - g_3$$

in affine coordinates (x, y), as ensured by Theorem 2.221. This is a nonconstant holomorphic map from a compact Riemann surface to a smooth projective curve, hence a finite morphism.

Step 2: Degree and separation of points. One checks that $\wp$ is even and of degree 2 as a map $\mathbb{C}/\Lambda \to \P^1$ (it has a double pole at 0 and no other poles modulo Λ), hence generically $\wp$ has two preimages: z and $-z$. Adding $\wp'$ distinguishes the two sheets since $\wp'$ is odd. Therefore Ψ is generically one-to-one, hence birational. Since both domain and target are smooth projective curves, a birational holomorphic map is an isomorphism. Thus $\mathbb{C}/\Lambda \simeq E$.

Step 3: Function fields. Isomorphism of smooth projective curves identifies their function fields:

$$\mathcal{M}(\mathbb{C}/\Lambda) \simeq \mathbb{C}(E) = \mathbb{C}(x, y)/(y^2 - 4x^3 + g_2 x + g_3).$$

Under this identification, x corresponds to $\wp$ and y corresponds to $\wp'$. Every element of $\mathbb{C}(E)$ can be reduced uniquely to the form $R(x) + yS(x)$ with $R, S \in \mathbb{C}(x)$ by using the relation $y^2 = \text{cubic}(x)$. Translating back gives the desired representation for f in terms of $\wp$ and $\wp'$. □

7. Examples (worked)

Example 2.223 (Trigonometry as a singly periodic limit)**.** The function $\sin z$ is periodic with period 2π but not doubly periodic. Elliptic functions may be viewed as "two-period" analogues. For rectangular lattices $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\Im(\omega_2/\omega_1) \gg 1$, the $\wp$–function approaches a trigonometric-type function in the sense that on compact sets away from Λ one has

$$\wp(z) \approx \left(\frac{\pi}{\omega_1}\right)^2 \csc^2\!\left(\frac{\pi z}{\omega_1}\right)$$

(up to an additive constant), reflecting degeneration of the torus to a cylinder.

Example 2.224 (Building elliptic functions with prescribed principal parts)**.** Fix points $a_1, \dots, a_m \in \mathbb{C}$ with distinct classes in $\mathbb{C}/\Lambda$. Any elliptic function with poles only at a_j (mod Λ) and prescribed principal parts can be constructed as a finite linear combination of the functions

$$\wp^{(k)}(z - a_j) \qquad (k \geq 0),$$

together with an additive constant, subject only to the residue constraint from Theorem 2.214.

Proof. Each $\wp^{(k)}(z - a_j)$ is elliptic and has its only pole (mod Λ) at a_j, with order $k + 2$. Linear combinations allow one to match any desired principal part at each a_j. Subtracting this combination from the target function removes all principal parts, leaving an elliptic function without poles, hence constant by Proposition 2.213. The only obstruction is that the sum of residues in a fundamental parallelogram must be 0, so the prescribed principal parts must satisfy this necessary condition. □

8. Exercises

Exercise 2.225. Show that $\wp(z)$ has a double pole at $z = 0$ with principal part $1/z^2$ and conclude that $\operatorname{Res}(\wp, 0) = 0$.

Exercise 2.226. Verify directly from the defining series that $\wp$ is even and $\wp'$ is odd.

Exercise 2.227. Let f be Λ–elliptic. Prove that the sum of residues of f in any fundamental parallelogram is 0.

Exercise 2.228. Prove that the field of Λ–elliptic functions is generated by $\wp$ and $\wp'$ by showing that $\mathbb{C}/\Lambda$ is isomorphic to the cubic curve $y^2 = 4x^3 - g_2x - g_3$ via $z \mapsto (\wp(z), \wp'(z))$.

2.14. Appendix: Standard Contour Integrals and Techniques

This appendix collects standard contour integrals, evaluation methods, and technical lemmas that frequently arise in complex analysis. Each result is stated precisely and proved in a form that can be reused as a template.

A. Basic circular contours

Proposition 2.229 (Integral of a monomial on a circle)**.** *Fix $R > 0$ and $n \in \mathbb{Z}$. Let $\gamma_R(t) = Re^{it}$, $t \in [0, 2\pi]$, be the positively oriented circle $|z| = R$. Then*

$$\int_{|z|=R} z^n \, dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Proof. Parametrize $z = Re^{it}$, $dz = iRe^{it}\, dt$:

$$\int_{|z|=R} z^n \, dz = \int_0^{2\pi} (Re^{it})^n \, iRe^{it} \, dt = iR^{n+1} \int_0^{2\pi} e^{i(n+1)t} \, dt.$$

If $n \neq -1$, then $n + 1 \neq 0$ and $\int_0^{2\pi} e^{i(n+1)t}\, dt = \frac{e^{i(n+1)2\pi} - 1}{i(n+1)} = 0$. If $n = -1$, then the integrand is i and the integral equals $2\pi i$. □

B. Trigonometric integrals via the unit circle

Proposition 2.230 (Unit-circle substitution)**.** *Let F be integrable on $[0, 2\pi]$ and let $z = e^{i\theta}$. Then*

$$\int_0^{2\pi} F(\theta)\, d\theta = \int_{|z|=1} F(\arg z)\, \frac{dz}{iz}.$$

Moreover, for integers $n \in \mathbb{Z}$,

$$\int_0^{2\pi} e^{in\theta}\, d\theta = \int_{|z|=1} z^{n-1}\, dz.$$

Proof. With $z = e^{i\theta}$ we have $dz = ie^{i\theta}\, d\theta = iz\, d\theta$, hence $d\theta = \frac{dz}{iz}$. Substituting gives the first identity; the second is the special case $F(\theta) = e^{in\theta}$. □

Corollary 2.231 (Orthogonality of $\cos(n\theta)$ and $\sin(n\theta)$)**.** For $n \in \mathbb{Z}_{\geq 0}$,

$$\int_0^{2\pi} \cos(n\theta)\, d\theta = \begin{cases} 2\pi, & n = 0, \\ 0, & n \geq 1, \end{cases} \qquad \int_0^{2\pi} \sin(n\theta)\, d\theta = 0.$$

Proof. By Proposition 2.230, $\int_0^{2\pi} e^{in\theta}\, d\theta = \int_{|z|=1} z^{n-1}\, dz$. By Proposition 2.229, this equals $2\pi i$ if $n = 0$ and 0 otherwise. Taking real and imaginary parts yields the stated integrals. □

Theorem 2.232 (The integral $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta}$)**.** *Let $a, b \in \mathbb{R}$ with $a > |b| > 0$. Then*

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Proof. Set $z = e^{i\theta}$ so that $\cos\theta = \frac{1}{2}(z + z^{-1})$ and $d\theta = \frac{dz}{iz}$. Then

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \int_{|z|=1} \frac{1}{a + \frac{b}{2}(z + z^{-1})}\, \frac{dz}{iz} = \int_{|z|=1} \frac{2}{2az + b(z^2 + 1)}\, \frac{dz}{i}.$$

Thus

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = -\frac{2i}{b} \int_{|z|=1} \frac{dz}{z^2 + \frac{2a}{b} z + 1}.$$

The quadratic has roots

$$z_\pm = \frac{-\frac{2a}{b} \pm \sqrt{\left(\frac{2a}{b}\right)^2 - 4}}{2} = \frac{-a \pm \sqrt{a^2 - b^2}}{b}.$$

Since $a > |b|$, we have $0 < \sqrt{a^2 - b^2} < a$, so z_- satisfies $|z_-| < 1$ and z_+ satisfies $|z_+| > 1$ (indeed $z_- z_+ = 1$). Hence the integrand has exactly one simple pole inside $|z| = 1$, at $z = z_-$.

Write $Q(z) = z^2 + \frac{2a}{b} z + 1 = (z - z_-)(z - z_+)$. Then

$$\operatorname{Res}\left(\frac{1}{Q(z)}, z_-\right) = \frac{1}{Q'(z_-)} = \frac{1}{2z_- + \frac{2a}{b}} = \frac{1}{z_- - z_+}.$$

Since $z_- - z_+ = -\frac{2\sqrt{a^2-b^2}}{b}$, we get

$$\operatorname{Res}\left(\frac{1}{Q(z)}, z_-\right) = -\frac{b}{2\sqrt{a^2-b^2}}.$$

Therefore by the residue theorem,

$$\int_{|z|=1} \frac{dz}{Q(z)} = 2\pi i \left(-\frac{b}{2\sqrt{a^2-b^2}}\right) = -\frac{\pi i b}{\sqrt{a^2-b^2}}.$$

Substituting back:

$$-\frac{2i}{b}\int_{|z|=1} \frac{dz}{Q(z)} = -\frac{2i}{b}\left(-\frac{\pi i b}{\sqrt{a^2-b^2}}\right) = \frac{2\pi}{\sqrt{a^2-b^2}}.$$

□

C. Jordan's lemma and Fourier-type integrals

Lemma 2.233 (Jordan's lemma (quantitative form))**.** *Let $\lambda > 0$ and let f be continuous on the closed upper half-plane $\{z : \Im z \geq 0\}$ except for finitely many poles in $\Im z > 0$. Assume that for some $R_0 > 0$ and $M > 0$,*

$$|f(z)| \leq \frac{M}{|z|} \qquad \text{for all } |z| \geq R_0,\ \Im z \geq 0.$$

Let $\Gamma_R = \{Re^{it} : 0 \leq t \leq \pi\}$ be the upper semicircle of radius $R \geq R_0$. Then

$$\left|\int_{\Gamma_R} e^{i\lambda z} f(z)\, dz\right| \leq \frac{\pi M}{\lambda} \qquad \text{for all } R \geq R_0,$$

and in particular $\int_{\Gamma_R} e^{i\lambda z} f(z)\, dz \to 0$ as $R \to \infty$.

Proof. Parametrize $z = Re^{it}$, $dz = iRe^{it}\, dt$, $t \in [0, \pi]$. Then $\Im z = R\sin t \geq 0$ and

$$|e^{i\lambda z}| = e^{-\lambda \Im z} = e^{-\lambda R \sin t}.$$

Hence for $R \geq R_0$,

$$\left|\int_{\Gamma_R} e^{i\lambda z} f(z)\, dz\right| \leq \int_0^\pi e^{-\lambda R\sin t}\, |f(Re^{it})|\, R\, dt \leq \int_0^\pi e^{-\lambda R \sin t}\, M\, dt.$$

Use $\sin t \geq \frac{2}{\pi} t$ on $[0, \frac{\pi}{2}]$ and symmetry about $\pi/2$:

$$\int_0^\pi e^{-\lambda R \sin t}\, dt \leq 2\int_0^{\pi/2} e^{-\lambda R \cdot \frac{2}{\pi} t}\, dt = 2 \cdot \frac{\pi}{2\lambda R}\left(1 - e^{-\lambda R}\right) \leq \frac{\pi}{\lambda R}.$$

Therefore

$$\left|\int_{\Gamma_R} e^{i\lambda z} f(z)\, dz\right| \leq M \cdot \frac{\pi}{\lambda R} \leq \frac{\pi M}{\lambda},$$

and the sharper bound $\frac{\pi M}{\lambda R} \to 0$ yields the limit. □

Theorem 2.234 (Model integral with a quadratic denominator)**.** *Let $a > 0$ and $\lambda > 0$. Then*

$$\int_{-\infty}^{\infty} \frac{e^{i\lambda x}}{x^2 + a^2}\, dx = \frac{\pi}{a} e^{-a\lambda}.$$

Consequently, for $k \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \frac{e^{ikx}}{x^2 + a^2}\, dx = \frac{\pi}{a} e^{-a|k|}.$$

Proof. Assume first $\lambda > 0$. Let $f(z) = \frac{1}{z^2+a^2}$, which has simple poles at $z = \pm ia$. Close the contour by the upper semicircle of radius R and apply Lemma 2.233 (since $|f(z)| \le 1/|z|^2 \le 1/|z|$ for $|z| \ge 1$). Then

$$\int_{-R}^{R} \frac{e^{i\lambda x}}{x^2 + a^2}\, dx + \int_{\Gamma_R} \frac{e^{i\lambda z}}{z^2 + a^2}\, dz = 2\pi i \operatorname{Res}\left(\frac{e^{i\lambda z}}{z^2 + a^2},\, ia \right).$$

Letting $R \to \infty$ kills the semicircle term and yields

$$\int_{-\infty}^{\infty} \frac{e^{i\lambda x}}{x^2 + a^2}\, dx = 2\pi i \operatorname{Res}\left(\frac{e^{i\lambda z}}{(z - ia)(z + ia)},\, ia \right) = 2\pi i \cdot \frac{e^{i\lambda ia}}{2ia}.$$

Since $e^{i\lambda ia} = e^{-a\lambda}$, we obtain $\frac{\pi}{a} e^{-a\lambda}$.

For general $k \in \mathbb{R}$, apply the previous computation to $\lambda = |k|$ and use complex conjugation or symmetry: the integral depends on $|k|$. □

D. Keyhole contours and branch cuts

Theorem 2.235 (Keyhole integral for $\displaystyle\int_0^{\infty} \frac{x^{\alpha-1}}{1+x}\, dx$)**.** *Let $0 < \alpha < 1$. Then*

$$\int_0^{\infty} \frac{x^{\alpha-1}}{1+x}\, dx = \frac{\pi}{\sin(\pi\alpha)}.$$

Proof. Consider $f(z) = \frac{z^{\alpha-1}}{1+z}$ with the branch cut along the positive real axis and principal determination $\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z$ where $\operatorname{Arg} z \in (0, 2\pi)$. Thus $z^{\alpha-1} = e^{(\alpha-1)\operatorname{Log} z}$.

Let $\mathcal{C}$ be a keyhole contour around $[0, \infty)$ with outer radius R and inner radius ε, positively oriented. The only pole of f inside $\mathcal{C}$ is at $z = -1$, and it is simple. Hence

$$\int_{\mathcal{C}} f(z)\, dz = 2\pi i \operatorname{Res}(f, -1).$$

Compute the residue:

$$\operatorname{Res}(f, -1) = \lim_{z \to -1} (z+1) \frac{z^{\alpha-1}}{1+z} = (-1)^{\alpha-1} = e^{(\alpha-1)\operatorname{Log}(-1)}.$$

With our branch, $\operatorname{Arg}(-1) = \pi$, so $\operatorname{Log}(-1) = i\pi$ and

$$\operatorname{Res}(f, -1) = e^{i\pi(\alpha-1)} = -e^{i\pi\alpha}.$$

Thus

$$\int_{\mathcal{C}} f(z)\, dz = 2\pi i \big(-e^{i\pi\alpha}\big).$$

Now evaluate the contour integral by splitting into the two rays. On the upper side of the cut, $z = x$ with $\operatorname{Arg} z = 0^+$, so $z^{\alpha-1} = x^{\alpha-1}$. On the lower side, $z = x$ with $\operatorname{Arg} z = 2\pi^-$, so $z^{\alpha-1} = e^{2\pi i(\alpha-1)}x^{\alpha-1} = e^{2\pi i\alpha}x^{\alpha-1}$. Taking orientations into account, and letting $R \to \infty$, $\varepsilon \to 0$, the circular arcs vanish (because $0 < \alpha < 1$ gives integrable behavior at 0 and decay at ∞), yielding

$$\int_{\mathcal{C}} f(z)\,dz = \left(1 - e^{2\pi i\alpha}\right)\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,dx.$$

Therefore,

$$\left(1 - e^{2\pi i\alpha}\right)\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,dx = -2\pi i\, e^{i\pi\alpha}.$$

Use $1 - e^{2\pi i\alpha} = e^{i\pi\alpha}\left(e^{-i\pi\alpha} - e^{i\pi\alpha}\right) = -2ie^{i\pi\alpha}\sin(\pi\alpha)$. Cancelling gives

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,dx = \frac{\pi}{\sin(\pi\alpha)}.$$

□

E. Hankel contour and the Gamma function

Theorem 2.236 (Hankel representation of Γ)**.** *Let $s \in \mathbb{C} \setminus \{0, -1, -2, \dots\}$. Then*

$$\Gamma(s) = \frac{1}{2\pi i}\int_{\mathcal{H}} (-z)^{-s}e^{-z}\,dz,$$

where $\mathcal{H}$ is the Hankel contour around the positive real axis and $(-z)^{-s} = e^{-s\operatorname{Log}(-z)}$ uses the principal branch of Log *with cut on $(0, \infty)$.*

Proof. Let $F(z) = (-z)^{-s}e^{-z}$. On the upper side of the positive axis, $z = x > 0$ and $\operatorname{Log}(-x) = \ln x + i\pi$, so $(-x)^{-s} = x^{-s}e^{-i\pi s}$. On the lower side, $\operatorname{Log}(-x) = \ln x - i\pi$, so $(-x)^{-s} = x^{-s}e^{+i\pi s}$. The Hankel contour integral is therefore

$$\int_{\mathcal{H}} F(z)\,dz = \left(e^{-i\pi s} - e^{i\pi s}\right)\int_0^\infty x^{-s}e^{-x}\,dx = -2i\sin(\pi s)\int_0^\infty x^{-s}e^{-x}\,dx,$$

after letting the radii tend to ∞ and 0 (the small/large circular arcs vanish for $s \notin \{0, -1, -2, \dots\}$). Now substitute $x = t$ and recognize $\int_0^\infty x^{-s}e^{-x}\,dx = \Gamma(1-s)$ for $\Re(s) < 1$, then use analytic continuation in s.

Finally, use Euler's reflection formula $\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$ (proved earlier in your text) to solve for $\Gamma(s)$ and obtain the stated representation. Since both sides are meromorphic in s and agree on a nonempty open set, they agree for all admissible s. □

F. Residues: templates

Proposition 2.237 (Residues at simple poles)**.** *If $f(z) = \frac{g(z)}{h(z)}$ with g, h holomorphic near z_0, $h(z_0) = 0$, $h'(z_0) \neq 0$, then z_0 is a simple pole and*

$$\operatorname{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}.$$

Proof. Write $h(z) = (z - z_0)h_1(z)$ with h_1 holomorphic and $h_1(z_0) = h'(z_0) \neq 0$. Then

$$f(z) = \frac{g(z)}{(z - z_0)h_1(z)}$$

and the coefficient of $(z - z_0)^{-1}$ is $g(z_0)/h_1(z_0) = g(z_0)/h'(z_0)$. □

Corollary 2.238 (Useful special cases)**.** For $a > 0$ and $k \in \mathbb{R}$,

$$\operatorname{Res}\left(\frac{1}{z^2 + a^2},\, ia\right) = \frac{1}{2ia}, \qquad \operatorname{Res}\left(\frac{e^{ikz}}{z^2 + a^2},\, ia\right) = \frac{e^{-ka}}{2ia} \quad (k > 0).$$

Proof. Apply Proposition 2.237 with $h(z) = z^2 + a^2$ so $h'(z) = 2z$, hence $h'(ia) = 2ia$. For the exponential numerator, take $g(z) = e^{ikz}$ and evaluate at ia. □

Proposition 2.239 (Higher-order pole template)**.** *If g is holomorphic near z_0 and $n \geq 1$, then*

$$\operatorname{Res}\left(\frac{g(z)}{(z - z_0)^n},\, z_0\right) = \frac{1}{(n-1)!}\, g^{(n-1)}(z_0).$$

Proof. Expand g in its Taylor series at z_0: $g(z) = \sum_{m \geq 0} \frac{g^{(m)}(z_0)}{m!}(z - z_0)^m$. Then

$$\frac{g(z)}{(z - z_0)^n} = \sum_{m \geq 0} \frac{g^{(m)}(z_0)}{m!}(z - z_0)^{m-n}.$$

The coefficient of $(z - z_0)^{-1}$ occurs when $m = n - 1$, giving the formula. □

G. Quick reference list (with proofs already above)

- For $a > 0$,
$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2} = \frac{\pi}{a}, \qquad \int_{-\infty}^{\infty} \frac{e^{ikx}}{x^2 + a^2}\, dx = \frac{\pi}{a} e^{-a|k|}.$$
(Use Theorem 2.234 with $k = 0$ and general k.)

- For $0 < \alpha < 1$,
$$\int_0^{\infty} \frac{x^{\alpha - 1}}{1 + x}\, dx = \frac{\pi}{\sin(\pi\alpha)}.$$
(Use Theorem 2.235.)

- For $n \geq 0$,
$$\int_0^{2\pi} \cos(n\theta)\, d\theta = \begin{cases} 2\pi, & n = 0, \\ 0, & n \geq 1, \end{cases} \qquad \int_0^{2\pi} \sin(n\theta)\, d\theta = 0.$$
(Use Corollary 2.231.)

3. Branched Double Covers and Gluing via Branch Cuts

We explain how to obtain compact Riemann surfaces by gluing along branch cuts for the two-sheeted algebraic curves

$$\mathcal{C}_0 :\ y^2 = x, \qquad\qquad \mathcal{C}_1 :\ y^2 = x(x-1)(x-2).$$

Both are degree–two holomorphic maps to the Riemann sphere via projection $\pi : (x, y) \mapsto x$, with branching precisely where analytic continuation of $y = \sqrt{f(x)}$ changes sign.

3.1. General two-sheeted picture and local models

Let f be a squarefree polynomial and consider the affine curve

$$X_{\text{aff}} = \{(x, y) \in \mathbb{C}^2 :\ y^2 = f(x)\}.$$

Write $\pi_{\text{aff}} : X_{\text{aff}} \to \mathbb{C}$ for $(x, y) \mapsto x$.

Proposition 3.1 (Holomorphic degree–two map and branch points)**.** *Assume f is squarefree. Then π_{aff} is a holomorphic map of degree 2 away from the zeros of f. If $e \in \mathbb{C}$ satisfies $f(e) = 0$ and $f'(e) \neq 0$, then π_{aff} has a simple branch point over $x = e$.*

Proof. If $f(x) \neq 0$, then the fiber over x consists of the two points $(x, \pm\sqrt{f(x)})$, so the map is locally a two-sheeted covering. Now fix e with $f(e) = 0$ and $f'(e) \neq 0$. By holomorphicity, we can write $f(x) = (x-e)u(x)$ with u holomorphic and $u(e) = f'(e) \neq 0$. Choose a holomorphic square root v of u in a small disk D around e (possible since u has no zeros on D). Then on D the curve is

$$y^2 = (x-e)u(x) \quad \Longleftrightarrow \quad \Big(\frac{y}{v(x)}\Big)^2 = x - e.$$

Set $t = \frac{y}{v(x)}$. The map $(x, y) \mapsto t$ is a local holomorphic coordinate on X_{aff} near $(e, 0)$, and π_{aff} becomes $x = e + t^2$ in that coordinate. Thus the ramification index is 2 (simple branching). □

Remark 3.2 (Topological gluing picture)**.** Let $B \subset \widehat{\mathbb{C}}$ be the finite set of branch values (zeros of f and possibly ∞). Choose a system of disjoint arcs in $\widehat{\mathbb{C}}$ connecting the points of B in pairs. Removing these arcs makes the base simply connected; over this slit domain the two branches $\pm\sqrt{f(x)}$ define two holomorphic sheets. Gluing the two sheets crosswise along each arc produces a connected surface on which $y = \sqrt{f(x)}$ becomes single valued.

Proposition 3.3 (Branching over infinity)**.** *Let f be a polynomial of degree $d \geq 1$. For the projective (compact) model of $y^2 = f(x)$, the behavior over $\infty \in \widehat{\mathbb{C}}$ is:*

- *If d is odd, then ∞ is a branch value (there is one point over ∞ with ramification index 2).*
- *If d is even, then ∞ is not branched (there are two distinct points over ∞).*

Proof. Use the coordinate $s = 1/x$ near ∞. Writing $f(x) = x^d(a_0 + a_1 s + \cdots)$ with $a_0 \neq 0$, we have

$$y^2 = x^d \cdot a_0(1 + O(s)).$$

If $d = 2m$ is even, set $Y = y/x^m$; then $Y^2 = a_0(1 + O(s))$, which has two holomorphic branches near $s = 0$, so there are two unramified points over ∞. If $d = 2m + 1$ is odd, set $Y = y/x^{m+1/2}$; equivalently $y^2 = x^{2m+1}a_0(1 + O(s))$ becomes $Y^2 = a_0\, s^{-1}(1 + O(s))$, and after rewriting in the coordinate $t = \sqrt{s}$ one finds a local model $x = t^{-2}$ and hence ramification index 2 over ∞. □

Theorem 3.4 (Riemann–Hurwitz for double covers)**.** *Let $\pi : \Sigma \to \widehat{\mathbb{C}}$ be a degree–2 holomorphic map from a compact Riemann surface Σ. Assume all branch points are simple and let B be the number of branch values in $\widehat{\mathbb{C}}$. Then*

$$g(\Sigma) = \frac{B - 2}{2}.$$

Proof. Riemann–Hurwitz states

$$2g(\Sigma) - 2 = 2(2g(\widehat{\mathbb{C}}) - 2) + \sum_{p \in \Sigma}(e_p - 1).$$

Since $g(\widehat{\mathbb{C}}) = 0$, the right-hand base term is $2(-2) = -4$. Each simple ramification point has $e_p = 2$ and contributes 1. There is exactly one ramification point above each branch value for a double cover with simple branching, so $\sum(e_p - 1) = B$. Hence $2g - 2 = -4 + B$, i.e. $g = \frac{B-2}{2}$. □

3.2. The case $y^2 = x$: branch values $\{0, \infty\}$ and genus 0

Proposition 3.5 (Branch values and genus)**.** *For $\mathcal{C}_0 : y^2 = x$, the branch values of π are $x = 0$ and $x = \infty$. The compactification of $\mathcal{C}_0$ is a sphere (genus 0).*

Proof. The polynomial $f(x) = x$ has a simple root at 0, so 0 is a branch value by Proposition 3.1. Since $\deg f = 1$ is odd, ∞ is also a branch value by Proposition 3.3. Thus $B = 2$ and Theorem 3.4 gives $g = \frac{2-2}{2} = 0$. □

Proposition 3.6 (Explicit global uniformization)**.** *The map $\varphi : \widehat{\mathbb{C}} \to \mathcal{C}_0^{\mathrm{proj}}$ given in affine coordinates by*

$$t \longmapsto (x, y) = (t^2, t)$$

is a biholomorphism from the Riemann sphere onto the projective compactification of $\mathcal{C}_0$.

Proof. On $\mathbb{C}$, the map satisfies $y^2 = t^2 = x$, hence lands in $\mathcal{C}_0$. It is injective on $\mathbb{C}$ because t is recovered as y. It extends holomorphically to $t = \infty$ in projective coordinates (the unique point over ∞), and the inverse map is $(x, y) \mapsto t = y$. Therefore it is a biholomorphism. □

Remark 3.7 (Cut-and-glue picture)**.** Choose a cut $\gamma = [0, \infty) \subset \mathbb{R}$. On $\mathbb{C} \setminus \gamma$ there are two holomorphic branches $\pm\sqrt{x}$. Taking two copies and gluing crosswise along the two sides of γ produces a simply connected compactification of the resulting surface, which must be S^2; the explicit parametrization in Proposition 3.6 identifies it with $\mathbb{P}^1$.

3.3. The case $y^2 = x(x-1)(x-2)$: branch values $\{0,1,2,\infty\}$ and genus 1

Proposition 3.8 (Branch values and genus)**.** *For $\mathcal{C}_1 : y^2 = x(x-1)(x-2)$, the branch values are $x \in \{0,1,2,\infty\}$. The projective compactification is a torus (genus 1).*

Proof. The polynomial has three simple zeros at $0, 1, 2$, giving three branch values (Proposition 3.1). Since the degree is 3 (odd), ∞ is also a branch value (Proposition 3.3). Hence $B = 4$, and Theorem 3.4 gives $g = \frac{4-2}{2} = 1$. □

Proposition 3.9 (Concrete cut system and gluing)**.** *Let $\gamma_1 = [0,1]$ and $\gamma_2 = [2,\infty)$ viewed as arcs on $\widehat{\mathbb{C}}$. Set*

$$U = \widehat{\mathbb{C}} \setminus (\gamma_1 \cup \gamma_2).$$

Then U is simply connected, and the two branches $\pm\sqrt{x(x-1)(x-2)}$ are holomorphic on U. Taking two copies U_+ and U_- and gluing crosswise along each γ_j produces a compact Riemann surface Σ equipped with a degree–two holomorphic map $\Sigma \to \widehat{\mathbb{C}}$ whose branch values are $\{0,1,2,\infty\}$.

Proof. Removing two disjoint arcs from the sphere makes the complement simply connected (one can retract to a graph with no cycles). On a simply connected domain avoiding the branch values, one can choose a holomorphic branch of Log and hence of $\sqrt{\cdot}$, so $\sqrt{x(x-1)(x-2)}$ has two holomorphic branches differing by sign. The crosswise gluing identifies the boundary values so that analytic continuation across a cut flips sheets, making the square root single valued on the glued surface. Compactness follows after adding the points lying over ∞ in the projective closure (Proposition 3.3 for odd degree). The resulting projection is holomorphic and two-to-one away from the cuts, with the prescribed branching. □

Remark 3.10 (Euler characteristic check)**.** The constructed surface has genus 1 by Proposition 3.8, so it is a torus. This matches the "one-handle" intuition from gluing two slit spheres crosswise along two cuts.

Proposition 3.11 (Canonical holomorphic 1-form)**.** *On $\mathcal{C}_1$ the differential*

$$\omega := \frac{dx}{y}$$

extends to a holomorphic 1-form on the compactification.

Proof. Away from the branch points it is holomorphic. Near a finite branch point $e \in \{0,1,2\}$, use the local coordinate $x = e + t^2$ from Proposition 3.1. Then $dx = 2t\,dt$ and $y \sim t \cdot (\text{nonzero})$, so $\omega = \frac{2t\,dt}{t(\text{nonzero})}$ is holomorphic. Near ∞, use $s = 1/x$ and the odd-degree analysis: one finds again that ω has no pole. □

Proposition 3.12 (Uniformization by an elliptic integral)**.** *Let Σ be the compactification of $\mathcal{C}_1$ and fix a base point $p_0 \in \Sigma$. Define on the universal cover $\widetilde{\Sigma}$ the map*

$$u(p) = \int_{p_0}^{p} \omega, \qquad \omega = \frac{dx}{y}.$$

Then the period lattice

$$\Lambda = \left\{ \int_\gamma \omega : \ \gamma \in H_1(\Sigma, \mathbb{Z}) \right\} \subset \mathbb{C}$$

is a lattice and the induced map $\Sigma \to \mathbb{C}/\Lambda$ is a biholomorphism.

Proof. Since ω is holomorphic and nonzero, integration defines a local holomorphic coordinate on $\widetilde{\Sigma}$. Periods along homology classes form a discrete subgroup $\Lambda \subset \mathbb{C}$ because $H_1(\Sigma, \mathbb{Z}) \cong \mathbb{Z}^2$ for a genus–1 surface and the two basic periods are $\mathbb{R}$–linearly independent (otherwise the image would not be compact). The map descends to Σ and is a holomorphic covering of degree 1 onto the torus $\mathbb{C}/\Lambda$, hence a biholomorphism. $\square$

Part III
Differential Forms and the Generalized Stokes Theorem

4. Differential Forms, Exterior Calculus, and Applications

In this section we introduce differential forms and the exterior derivative, show how they encode line integrals, Stokes' theorem, and complex line integrals, and then apply them to surface area, curvature, and the Gauss–Bonnet theorem.

4.1. Differential Forms and Exterior Derivatives

Throughout, let $U \subset \mathbb{R}^n$ be open with smooth coordinates $(x_1, \dots, x_n)$. Write $\partial_i := \frac{\partial}{\partial x_i}$ and let δ_{ij} be the Kronecker delta.

Differential forms as alternating multilinear maps: the definition of $\Omega^p(U)$

Throughout, let $U \subset \mathbb{R}^n$ be open with smooth coordinates $(x_1, \dots, x_n)$. Write $\partial_i := \frac{\partial}{\partial x_i}$ and let δ_{ij} be the Kronecker delta.

Tangent spaces and smooth vector fields For each $p \in U$, the tangent space T_pU is the vector space of derivations at p: a linear map $D : C^\infty(U) \to \mathbb{R}$ is a derivation at p if

$$D(fg) = f(p)\, D(g) + g(p)\, D(f) \quad \text{for all } f, g \in C^\infty(U).$$

In coordinates, the derivations $\partial_i|_p$ defined by

$$\partial_i|_p(f) = \frac{\partial f}{\partial x_i}(p)$$

form a basis of T_pU, and every $v \in T_pU$ is uniquely

$$v = \sum_{i=1}^{n} v^i\, \partial_i\big|_p, \qquad v^i \in \mathbb{R}.$$

A *smooth vector field* on U is a smooth map $X : U \to TU$ with $X(p) \in T_pU$. We denote by $\mathfrak{X}(U)$ the space of smooth vector fields.

Definition 4.1 (Alternating p-covectors at a point)**.** Fix $p \in U$. An *alternating p-covector* at p is a map

$$\omega_p : (T_pU)^p \longrightarrow \mathbb{R}$$

that is multilinear in each slot and alternating:

$$\omega_p(\dots, v_i, \dots, v_j, \dots) = -\omega_p(\dots, v_j, \dots, v_i, \dots) \quad (i \neq j).$$

The vector space of such maps is denoted by

$$\Lambda^p(T_p^*U).$$

Immediate consequences (explicit). If $\omega_p \in \Lambda^p(T_p^*U)$, then

$$\omega_p(\dots, v, \dots, v, \dots) = 0$$

because swapping the two equal slots changes the sign but not the value.

Definition 4.2 (Smooth differential p-forms: $\Omega^p(U)$)**.** A *smooth differential p-form* on U is an assignment

$$p \longmapsto \omega_p \in \Lambda^p(T_p^*U)$$

such that for any smooth vector fields $X_1, \dots, X_p \in \mathfrak{X}(U)$, the function

$$U \ni p \longmapsto \omega_p\big(X_1(p), \dots, X_p(p)\big) \in \mathbb{R}$$

is smooth. The set of all smooth p-forms on U is denoted $\Omega^p(U)$.

Remarks (with concrete meaning).

- $\Omega^0(U) = C^\infty(U)$: a 0-form is just a smooth function.
- $\Omega^1(U)$ consists of smooth assignments $p \mapsto \eta_p \in T_p^*U$; each η_p is a linear functional on T_pU depending smoothly on p.
- $\Omega^p(U)$ is a $C^\infty(U)$-module: if $f \in C^\infty(U)$ and $\omega \in \Omega^p(U)$, then $(f\omega)_p := f(p)\,\omega_p$ defines $f\omega \in \Omega^p(U)$.

Worked mini-example 1 (a genuine 2-form as an alternating map). Let $n = 2$ and define at each p:

$$\omega_p(v, w) := \det \begin{pmatrix} v^1 & w^1 \\ v^2 & w^2 \end{pmatrix}.$$

Then ω_p is bilinear and alternating, hence $\omega_p \in \Lambda^2(T_p^*U)$. If $v = (1, 0)$ and $w = (0, 1)$ in the basis (∂_x, ∂_y), then $\omega_p(v, w) = 1$.

Worked mini-example 2 (a family that is *not* smooth). Define ω_p in $\mathbb{R}^2$ by

$$\omega_p(\partial_x, \partial_y) = \begin{cases} 1, & x(p) \geq 0, \\ 0, & x(p) < 0. \end{cases}$$

Then each ω_p is an alternating 2-covector, but the coefficient jumps at $x = 0$. Hence this assignment is *not* a smooth 2-form on U.

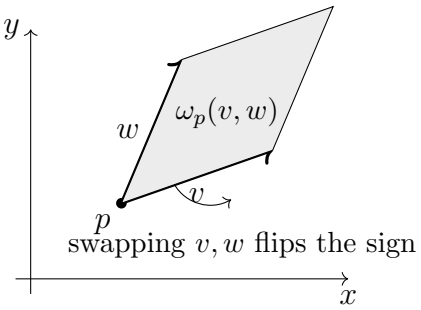


Figure 1: A 2-form is an alternating bilinear map: it measures signed oriented area.

Visualization (TikZ): alternating means "signed oriented area".

Exercises (definition level)

Exercise 4.3. Let $p \in U$ and $\omega_p \in \Lambda^p(T_p^*U)$. Prove directly from alternation that $\omega_p(v_1, \dots, v_p) = 0$ whenever two of the v_i are equal.

Exercise 4.4. Show that $\Omega^p(U)$ is a real vector space and a $C^\infty(U)$-module, i.e.

$$f(\omega + \eta) = f\omega + f\eta, \qquad (fg)\omega = f(g\omega), \qquad 1 \cdot \omega = \omega.$$

Exercise 4.5. Let $U \subset \mathbb{R}^2$ and define $\omega = f(x, y)\, dx \wedge dy$ (using the coordinate 1-forms defined later). Show that ω corresponds to the alternating map

$$\omega_p(v, w) = f(p) \det \begin{pmatrix} v^x & w^x \\ v^y & w^y \end{pmatrix}.$$

Coordinates: 0-forms and 1-forms, coefficients, and evaluation

Definition 4.6 (0-form)**.** A 0-*form* on U is a smooth function $f \in C^\infty(U)$.

Definition 4.7 (Coordinate tangent basis)**.** For each $p \in U$, the tangent space T_pU has the coordinate basis $\{\partial_1|_p, \dots, \partial_n|_p\}$. Every $v \in T_pU$ can be written uniquely as

$$v = \sum_{j=1}^{n} v^j\, \partial_j\Big|_p, \qquad v^j \in \mathbb{R}.$$

Definition 4.8 (1-forms and the dual basis)**.** The coordinate 1-forms $dx_i \in \Omega^1(U)$ are defined by

$$(dx_i)_p(\partial_j|_p) = \delta_{ij}.$$

A general 1-form is a $C^\infty(U)$-linear combination

$$\eta = \sum_{i=1}^{n} a_i\, dx_i, \qquad a_i \in C^\infty(U).$$

How a 1-form eats a vector (fully expanded). Let $\eta = \sum_i a_i dx_i$ and $v = \sum_j v^j \partial_j$ at a point p. Then

$$\eta_p(v) = \Big(\sum_{i=1}^n a_i(p)\,(dx_i)_p\Big)\Big(\sum_{j=1}^n v^j\,\partial_j|_p\Big) = \sum_{i,j} a_i(p)\,v^j\,(dx_i)_p(\partial_j|_p) = \sum_{i,j} a_i(p)\,v^j\,\delta_{ij} = \sum_{i=1}^n a_i(p)\,v^i.$$

So in coordinates, η behaves like a row vector $(a_1,\dots,a_n)$ acting on a column vector $(v^1,\dots,v^n)^{\mathsf{T}}$.

Worked example (numerical evaluation). In $\mathbb{R}^3$ let

$$\eta = (x^2+y)\,dx + \sin z\,dy + e^{xy}\,dz.$$

At $p=(1,2,0)$, the coefficient values are $(x^2+y, \sin z, e^{xy}) = (3,0,e^2)$. For $v = 2\partial_x - \partial_y + 3\partial_z$ at p, i.e. $(v^x,v^y,v^z)=(2,-1,3)$, we get

$$\eta_p(v) = 3\cdot 2 + 0\cdot(-1) + e^2\cdot 3 = 6 + 3e^2.$$

Differential of a function (directional derivative packaged). For $f \in C^\infty(U)$ define

$$df := \sum_{i=1}^n \frac{\partial f}{\partial x_i}\,dx_i \in \Omega^1(U).$$

Then for $v = \sum_j v^j\partial_j$,

$$(df)_p(v) = \sum_i f_{x_i}(p)\,(dx_i)_p\left(\sum_j v^j\partial_j|_p\right) = \sum_{i,j} f_{x_i}(p)\,v^j\,\delta_{ij} = \sum_i f_{x_i}(p)\,v^i = v(f)(p).$$

Worked example (compute df and evaluate). Let $f(x,y) = x^2y + \cos y$ on $U\subset\mathbb{R}^2$. Then

$$f_x = 2xy, \qquad f_y = x^2 - \sin y,$$

so

$$df = (2xy)\,dx + (x^2 - \sin y)\,dy.$$

At $p=(1,0)$ and $v = 3\partial_x - 2\partial_y$,

$$(df)_p(v) = (2\cdot 1\cdot 0)\cdot 3 + (1^2 - \sin 0)\cdot(-2) = 0 - 2 = -2,$$

which matches the directional derivative $v(f)(p)$.

Exercises (0-forms and 1-forms)

Exercise 4.9. Let $\eta = (x+y)\,dx + (x-y)\,dy$ in $\mathbb{R}^2$. Compute $\eta_p(v)$ at $p=(1,2)$ for $v = 4\partial_x - \partial_y$. Show every step.

Exercise 4.10. Let $f(x,y,z) = xe^{yz} + y^2z$. Compute df explicitly and then evaluate $(df)_{(1,0,2)}(\partial_x + 2\partial_z)$.

Exercise 4.11. Show that $df = 0$ as a 1-form on a connected open set U if and only if f is constant on U.

Wedge product and k-forms: signs, areas, and determinants

Definition 4.12 (Wedge product: graded anti-commutativity)**.** For $\alpha \in \Omega^p(U)$ and $\beta \in \Omega^q(U)$,

$$\alpha \wedge \beta = (-1)^{pq}\, \beta \wedge \alpha.$$

In particular, for 1-forms,

$$dx_i \wedge dx_j = -dx_j \wedge dx_i, \qquad dx_i \wedge dx_i = 0.$$

From wedge to determinants (worked out in $\mathbb{R}^2$). A 2-form acts on two vectors. In $\mathbb{R}^2$,

$$(dx \wedge dy)_p(v, w) = \det \begin{pmatrix} dx(v) & dx(w) \\ dy(v) & dy(w) \end{pmatrix} = dx(v)\, dy(w) - dx(w)\, dy(v).$$

If $v = (v^x, v^y)$ and $w = (w^x, w^y)$, then

$$(dx \wedge dy)_p(v, w) = v^x w^y - v^y w^x,$$

the signed area of the parallelogram spanned by v, w.

Worked example (sign check). Let $v = \partial_x + \partial_y$ and $w = 2\partial_x + \partial_y$. Then

$$(dx \wedge dy)(v, w) = \det \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = 1 \cdot 1 - 2 \cdot 1 = -1.$$

Swapping vectors gives $(dx \wedge dy)(w, v) = +1$.

A 2-form built from coefficients. Let $\omega = f(x, y)\, dx \wedge dy$ in $\mathbb{R}^2$. Then for v, w,

$$\omega_p(v, w) = f(p)\, (dx \wedge dy)_p(v, w) = f(p)\, (v^x w^y - v^y w^x).$$

So f rescales the signed area density.

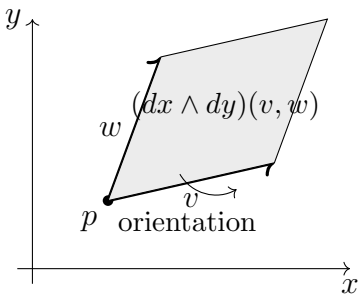


Figure 2: $dx \wedge dy$ measures signed area; swapping (v, w) changes the sign.

Visualization (TikZ): oriented area from $dx \wedge dy$.

Definition 4.13 (k-forms in coordinates)**.** A general k-form is

$$\omega = \sum_{1 \le i_1 < \cdots < i_k \le n} a_{i_1 \cdots i_k}\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \qquad a_{i_1 \cdots i_k} \in C^\infty(U).$$

Worked example (a 3-form as signed volume). In $\mathbb{R}^3$, the 3-form $dx \wedge dy \wedge dz$ satisfies

$$(dx \wedge dy \wedge dz)_p(u, v, w) = \det \begin{pmatrix} dx(u) & dx(v) & dx(w) \\ dy(u) & dy(v) & dy(w) \\ dz(u) & dz(v) & dz(w) \end{pmatrix},$$

so it returns the signed volume of the parallelepiped spanned by u, v, w.

Exercises (wedge product and determinants)

Exercise 4.14. Compute $(dx \wedge dy)(v, w)$ for $v = (2, 1)$ and $w = (1, 3)$, then compute $(dx \wedge dy)(w, v)$ and compare.

Exercise 4.15. Let $\alpha = x\,dx + y\,dy$ and $\beta = dx + dy$ in $\mathbb{R}^2$. Compute $\alpha \wedge \beta$ explicitly, simplify the result, and verify $\beta \wedge \alpha = -\alpha \wedge \beta$.

Exercise 4.16. In $\mathbb{R}^3$, compute $(dx \wedge dy \wedge dz)(u, v, w)$ for $u = (1, 0, 0)$, $v = (1, 1, 0)$, $w = (1, 1, 1)$ and interpret geometrically.

Exterior derivative: step-by-step formulas and the identity $d^2 = 0$

Definition 4.17 (Exterior derivative on functions)**.** For $f \in C^\infty(U) = \Omega^0(U)$,

$$df = \sum_{i=1}^{n} f_{x_i}\, dx_i \in \Omega^1(U).$$

Definition 4.18 (Exterior derivative on general k-forms (computational rule))**.** If

$$\omega = \sum_{i_1 < \cdots < i_k} a_{i_1 \cdots i_k}\, dx_{i_1} \wedge \cdots \wedge dx_{i_k},$$

then

$$d\omega = \sum_{i_1 < \cdots < i_k} da_{i_1 \cdots i_k} \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \quad da_{i_1 \cdots i_k} = \sum_{\ell=1}^{n} \frac{\partial a_{i_1 \cdots i_k}}{\partial x_\ell}\, dx_\ell.$$

Graded Leibniz rule (full sign bookkeeping). Let $\omega \in \Omega^k(U)$ and $\eta \in \Omega^\ell(U)$, and write locally $\omega = \sum_I a_I\, dx_I$ and $\eta = \sum_J b_J\, dx_J$. Then

$$\omega \wedge \eta = \sum_{I,J} a_I b_J\, dx_I \wedge dx_J.$$

Differentiate:

$$d(\omega \wedge \eta) = \sum_{I,J} d(a_I b_J) \wedge dx_I \wedge dx_J = \sum_{I,J} (da_I\, b_J + a_I\, db_J) \wedge dx_I \wedge dx_J.$$

Split:

$$\sum_{I,J} (da_I \wedge dx_I)\, b_J \wedge dx_J \;+\; \sum_{I,J} a_I\, db_J \wedge dx_I \wedge dx_J.$$

Move db_J past k one-forms in dx_I:

$$db_J \wedge dx_I = (-1)^k\, dx_I \wedge db_J,$$

so

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k\, \omega \wedge d\eta.$$

Theorem 4.19 ($d^2 = 0$ (full coordinate cancellation))**.** *For every $\omega \in \Omega^k(U)$,*

$$d(d\omega) = 0.$$

Proof. By linearity, it suffices to check $\omega = a\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ with $i_1 < \cdots < i_k$. Then

$$d\omega = da \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k} = \left(\sum_{p=1}^{n} a_{x_p}\, dx_p \right) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}.$$

Apply d again:

$$d(d\omega) = \sum_{p=1}^{n} d(a_{x_p}) \wedge dx_p \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}.$$

Expand

$$d(a_{x_p}) = \sum_{q=1}^{n} a_{x_q x_p}\, dx_q,$$

so

$$d(d\omega) = \sum_{p,q=1}^{n} a_{x_q x_p}\, dx_q \wedge dx_p \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}.$$

For $p \neq q$, the term (p, q) cancels (q, p) because

$$a_{x_q x_p} = a_{x_p x_q} \quad \text{and} \quad dx_q \wedge dx_p = -dx_p \wedge dx_q.$$

For $p = q$, the term vanishes since $dx_p \wedge dx_p = 0$. Hence the total is 0. □

Worked examples (fully expanded computations)

Example 4.20 (Compute $d\eta$ in $\mathbb{R}^2$ step by step)**.** Let $\eta = a(x, y)\, dx + b(x, y)\, dy$. Compute

$$da = a_x\, dx + a_y\, dy, \qquad db = b_x\, dx + b_y\, dy.$$

Then

$$d\eta = da \wedge dx + db \wedge dy.$$

Expand:

$$da \wedge dx = (a_x\, dx + a_y\, dy) \wedge dx = a_x\, dx \wedge dx + a_y\, dy \wedge dx = 0 - a_y\, dx \wedge dy,$$
$$db \wedge dy = (b_x\, dx + b_y\, dy) \wedge dy = b_x\, dx \wedge dy + b_y\, dy \wedge dy = b_x\, dx \wedge dy + 0.$$

Therefore

$$d\eta = (b_x - a_y)\, dx \wedge dy.$$

Example 4.21 (Compute $d\alpha$ in $\mathbb{R}^3$ and compare with divergence)**.** Let

$$\alpha = x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy.$$

Use $d(f\beta) = df \wedge \beta$ here because $d(dx) = d(dy) = d(dz) = 0$:

$$d(x\, dy \wedge dz) = dx \wedge dy \wedge dz,$$
$$d(y\, dz \wedge dx) = dy \wedge dz \wedge dx, \qquad d(z\, dx \wedge dy) = dz \wedge dx \wedge dy.$$

Reorder to $dx \wedge dy \wedge dz$ (cyclic permutations, hence sign $+1$):

$$dy \wedge dz \wedge dx = dx \wedge dy \wedge dz, \qquad dz \wedge dx \wedge dy = dx \wedge dy \wedge dz.$$

So

$$d\alpha = (1 + 1 + 1)\, dx \wedge dy \wedge dz = 3\, dx \wedge dy \wedge dz.$$

This matches $\nabla \cdot (x, y, z) = 3$ under the standard identification of vector fields with 2-forms via interior product $\iota_F(dx \wedge dy \wedge dz)$.

Exercises (exterior derivative)

Exercise 4.22. Let $f(x,y) = x^3 + xy^2$. Compute df and then show by explicit coefficient expansion that $d(df) = 0$.

Exercise 4.23. Let $\eta = P\,dx + Q\,dy$ in $\mathbb{R}^2$. Compute $d\eta$ and show that

$$d\eta = (Q_x - P_y)\,dx \wedge dy.$$

Exercise 4.24. In $\mathbb{R}^3$, let $\eta = P\,dx + Q\,dy + R\,dz$. Expand $d\eta$ completely and identify its coefficients with the curl $\nabla \times (P,Q,R)$ by matching the basis $dy \wedge dz,\ dz \wedge dx,\ dx \wedge dy$.

Exercise 4.25. Let $\omega = f(x,y)\,dx \wedge dy$ in $\mathbb{R}^2$. Compute $d\omega$ and explain why $d\omega = 0$ automatically in $\mathbb{R}^2$. (Hint: degree reasons.)

4.2. Line Integrals and Conservative Vector Fields

Throughout this subsection we work in $\mathbb{R}^2$ with standard coordinates (x,y). We keep two parallel languages in view:

- vector fields $\mathbf{F} = (P,Q)$ and the line integral $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$;
- 1-forms $\eta = P\,dx + Q\,dy$ and the curve integral $\displaystyle\int_C \eta$.

Line integrals: definition, parametrization, and 1-forms

Definition 4.26 (Line integral of a vector field)**.** Let $\mathbf{F} = (P,Q)$ be a continuous vector field on a region $D \subset \mathbb{R}^2$. Let C be an oriented C^1 curve parametrized by

$$\mathbf{r}(t) = (x(t), y(t)), \qquad a \le t \le b.$$

The *line integral of* $\mathbf{F}$ *along* C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} := \int_a^b \Big(P(x(t),y(t))\,x'(t) + Q(x(t),y(t))\,y'(t) \Big)\,dt.$$

How to compute in practice (explicit template). To compute $\int_C \mathbf{F} \cdot d\mathbf{r}$:

1. parametrize C by $\mathbf{r}(t) = (x(t), y(t))$;
2. compute $x'(t), y'(t)$;
3. substitute into $P(x,y)$ and $Q(x,y)$ along $\mathbf{r}(t)$;
4. form $P(x(t),y(t))x'(t) + Q(x(t),y(t))y'(t)$ and integrate in t.

Rewriting using differential forms. Associate to $\mathbf{F} = (P, Q)$ the 1-form

$$\eta := P\,dx + Q\,dy \in \Omega^1(D).$$

Along the curve $\mathbf{r}(t)$ we have

$$\eta_{\mathbf{r}(t)}(\mathbf{r}'(t)) = P(x(t), y(t))\,x'(t) + Q(x(t), y(t))\,y'(t),$$

hence

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \eta(\mathbf{r}'(t))\,dt = \int_C \eta.$$

So *line integrals of vector fields are exactly integrals of* 1*-forms.*

Orientation and reparametrization (checked by calculation). If C is traversed in the opposite direction, say $\tilde{\mathbf{r}}(t) = \mathbf{r}(a + b - t)$, then

$$\int_{\tilde{C}} \mathbf{F} \cdot d\mathbf{r} = \int_a^b \Big(P(\tilde{\mathbf{r}}(t))\,\tilde{x}'(t) + Q(\tilde{\mathbf{r}}(t))\,\tilde{y}'(t)\Big)\,dt = -\int_C \mathbf{F} \cdot d\mathbf{r}.$$

If $t = \varphi(s)$ is an orientation-preserving C^1 reparametrization ($\varphi' > 0$, $\varphi(\alpha) = a$, $\varphi(\beta) = b$), then by change of variables

$$\int_\alpha^\beta \eta(\mathbf{r}'(\varphi(s))\,\varphi'(s))\,ds = \int_a^b \eta(\mathbf{r}'(t))\,dt,$$

so the value is unchanged.

Conservative vector fields, potentials, and the Fundamental Theorem

Definition 4.27 (Conservative vector field)**.** A vector field $\mathbf{F} = (P, Q)$ on a region $D \subset \mathbb{R}^2$ is *conservative* if there exists $f \in C^1(D)$ such that

$$\mathbf{F} = \nabla f = (f_x, f_y).$$

The function f is called a *potential* for $\mathbf{F}$.

Fundamental Theorem for Line Integrals (proved by chain rule). Assume $\mathbf{F} = \nabla f$ and $C : \mathbf{r}(t) = (x(t), y(t))$ runs from $A = \mathbf{r}(a)$ to $B = \mathbf{r}(b)$. Then

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = f_x(x(t), y(t))\,x'(t) + f_y(x(t), y(t))\,y'(t) = \frac{d}{dt}\,f(x(t), y(t))$$

by the chain rule. Integrating,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \frac{d}{dt} f(\mathbf{r}(t))\,dt = f(\mathbf{r}(b)) - f(\mathbf{r}(a)) = f(B) - f(A).$$

In particular, the integral depends only on endpoints.

Differential-form interpretation. If $\mathbf{F} = \nabla f$, then

$$\eta = P dx + Q dy = f_x dx + f_y dy = df,$$

so η is *exact*. Applying d gives

$$d\eta = d(df) = 0,$$

so exact $\Rightarrow$ closed. The converse can fail if the domain is not simply connected.

A standard criterion in $\mathbb{R}^2$ (local test). If $P, Q \in C^1(D)$, then

$$d\eta = d(P\,dx + Q\,dy) = (Q_x - P_y)\,dx \wedge dy.$$

So $d\eta = 0$ is equivalent to the familiar curl-free condition $Q_x = P_y$.

A key example: a tangential field on $\mathbb{R}^2 \setminus \{0\}$

Definition of the field and geometric meaning. On $\mathbb{R}^2 \setminus \{(0,0)\}$ define

$$\mathbf{F}(x,y) = \left(-\frac{y}{x^2+y^2},\ \frac{x}{x^2+y^2}\right).$$

Write $r^2 = x^2 + y^2$. Then $\mathbf{F} = \frac{1}{r^2}(-y, x)$ is obtained by rotating (x, y) by $+90^\circ$ and scaling by $1/r^2$, so it is tangent to circles centered at the origin.

Associated 1-form.

$$\eta = -\frac{y}{x^2+y^2}\,dx + \frac{x}{x^2+y^2}\,dy.$$

Compute $d\eta$ explicitly (no shortcuts). Let $P = -\dfrac{y}{x^2+y^2}$ and $Q = \dfrac{x}{x^2+y^2}$. Compute

$$Q_x = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right) = \frac{(x^2+y^2)\cdot 1 - x\cdot(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2},$$

and

$$P_y = \frac{\partial}{\partial y}\left(-\frac{y}{x^2+y^2}\right) = -\frac{(x^2+y^2)\cdot 1 - y\cdot(2y)}{(x^2+y^2)^2} = -\frac{x^2-y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

Thus $Q_x - P_y = 0$, and therefore

$$d\eta = (Q_x - P_y)\,dx \wedge dy = 0 \qquad \text{on } \mathbb{R}^2 \setminus \{0\}.$$

So η is *closed* on $\mathbb{R}^2 \setminus \{0\}$.

But η is not exact globally (detected by a loop integral). Take the circle C_R : $(x, y) = (R\cos t, R\sin t)$, $0 \le t \le 2\pi$ (counterclockwise). Then $dx = -R\sin t\,dt$, $dy = R\cos t\,dt$, and $x^2 + y^2 = R^2$. Hence

$$\eta = -\frac{R\sin t}{R^2}(-R\sin t\,dt) + \frac{R\cos t}{R^2}(R\cos t\,dt) = \sin^2 t\,dt + \cos^2 t\,dt = dt.$$

Therefore

$$\oint_{C_R} \eta = \int_0^{2\pi} dt = 2\pi \neq 0.$$

If $\eta = df$ on $\mathbb{R}^2 \setminus \{0\}$, then every closed loop would integrate to 0, contradiction. So η is closed but not exact on $\mathbb{R}^2 \setminus \{0\}$.

Visualization (TikZ): the tangential field and two circles.

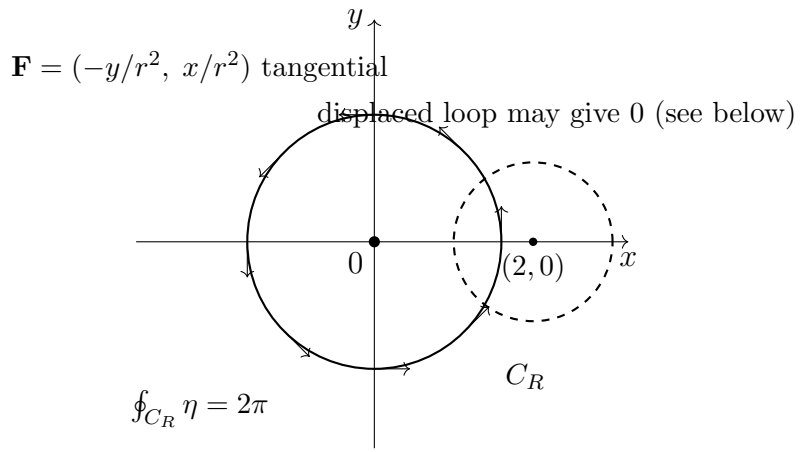


Figure 3: The closed 1-form η detects topology: loops winding around the origin give nonzero integral.

Example: line integral along a displaced circle (full computation)

Example 4.28. Let C be the circle of radius 1 centered at $(2,0)$, oriented counterclockwise. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for

$$\mathbf{F}(x,y) = \left(-\frac{y}{x^2+y^2},\ \frac{x}{x^2+y^2} \right).$$

Step-by-step computation. Parametrize C by

$$x(t) = 2 + \cos t, \qquad y(t) = \sin t, \qquad 0 \le t \le 2\pi.$$

Then

$$x'(t) = -\sin t, \qquad y'(t) = \cos t,$$

and

$$x(t)^2 + y(t)^2 = (2+\cos t)^2 + \sin^2 t = 5 + 4\cos t.$$

Evaluate $\mathbf{F}$ along C:

$$\mathbf{F}(\mathbf{r}(t)) = \left(-\frac{\sin t}{5+4\cos t},\ \frac{2+\cos t}{5+4\cos t} \right).$$

Dot with $\mathbf{r}'(t)$:

$$\begin{aligned} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) &= \left(-\frac{\sin t}{5+4\cos t} \right)(-\sin t) + \left(\frac{2+\cos t}{5+4\cos t} \right)(\cos t) \\ &= \frac{\sin^2 t + (2+\cos t)\cos t}{5+4\cos t} = \frac{1+2\cos t}{5+4\cos t}. \end{aligned}$$

Therefore

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \frac{1+2\cos t}{5+4\cos t}\, dt.$$

□

Direct evaluation via the tangent-half-angle substitution (with correct algebra)

Let $u = \tan(t/2)$. Then

$$\cos t = \frac{1-u^2}{1+u^2}, \qquad dt = \frac{2\,du}{1+u^2}.$$

Compute the rationalized integrand carefully:

$$1 + 2\cos t = 1 + 2\frac{1-u^2}{1+u^2} = \frac{(1+u^2)+2(1-u^2)}{1+u^2} = \frac{3-u^2}{1+u^2},$$

$$5 + 4\cos t = 5 + 4\frac{1-u^2}{1+u^2} = \frac{5(1+u^2)+4(1-u^2)}{1+u^2} = \frac{9+u^2}{1+u^2}.$$

Hence

$$\frac{1+2\cos t}{5+4\cos t} = \frac{\frac{3-u^2}{1+u^2}}{\frac{9+u^2}{1+u^2}} = \frac{3-u^2}{9+u^2}.$$

Therefore

$$\int_0^{2\pi} \frac{1+2\cos t}{5+4\cos t}\,dt = \int_{-\infty}^{\infty} \frac{3-u^2}{9+u^2}\cdot\frac{2\,du}{1+u^2} = 2\int_{-\infty}^{\infty} \frac{3-u^2}{(9+u^2)(1+u^2)}\,du.$$

Now do the partial fraction decomposition (this is the key cancellation):

$$2\frac{3-u^2}{(u^2+9)(u^2+1)} = \frac{1}{u^2+1} - \frac{3}{u^2+9}.$$

(Verify by combining over the common denominator.) Thus

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \int_{-\infty}^{\infty}\left(\frac{1}{u^2+1} - \frac{3}{u^2+9}\right)du = \Big(\pi\Big) - \Big(3\cdot\frac{\pi}{3}\Big) = 0.$$

So the displaced circle gives zero circulation, consistent with the fact that it does *not* wind around the origin.

Conceptual takeaway: local vs global information

- Conservative fields $\mathbf{F} = \nabla f$ correspond to exact 1-forms $\eta = df$.
- Exact $\Rightarrow$ closed ($d\eta = 0$), but closed $\not\Rightarrow$ exact in general.
- In $\mathbb{R}^2$, the obstruction is topological: loops winding around "holes" can have nonzero integrals.
- Line integrals can detect global information invisible to local derivatives like $Q_x - P_y$.

Exercises (with enough practice on computation and concepts)

Exercise 4.29 (Warm-up: compute a line integral directly)**.** Let $\mathbf{F} = (P, Q) = (y, x)$ and let C be the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$ with the standard parametrization $\mathbf{r}(t) = (t, t^2)$, $0 \le t \le 1$. Compute $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$.

Exercise 4.30 (Finding a potential by matching partial derivatives)**.** Let $\mathbf{F}(x, y) = (3x^2 + 6xy,\ 3x^2 + 6y)$ on $\mathbb{R}^2$.

1. Compute P_y and Q_x and verify $P_y = Q_x$.
2. Find a potential f by integrating $f_x = P$ in x and determining the missing y-function.
3. For any C^1 curve C from $(0, 0)$ to $(1, 1)$, compute $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$.

Exercise 4.31 (Path dependence via two different curves)**.** Let $\mathbf{F} = (P, Q) = (y, 0)$ on $\mathbb{R}^2$. Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$ along:

1. C_1: the straight line $\mathbf{r}(t) = (t, t)$;
2. C_2: the broken path $(0, 0) \to (1, 0) \to (1, 1)$.

Conclude that $\mathbf{F}$ is not conservative.

Exercise 4.32 (Exact $\Rightarrow$ closed, directly from formulas)**.** Let $\eta = df$ for $f \in C^2(D)$. Write $df = f_x dx + f_y dy$ and compute $d(df)$ explicitly to show $d(df) = 0$ using $f_{xy} = f_{yx}$.

Exercise 4.33 (Closed $\not\Rightarrow$ exact on a punctured plane)**.** Let $\eta = -\dfrac{y}{x^2 + y^2} dx + \dfrac{x}{x^2 + y^2} dy$ on $\mathbb{R}^2 \setminus \{0\}$.

1. Compute $d\eta$ explicitly and show $d\eta = 0$.
2. Parameterize $C_R : (x, y) = (R\cos t, R\sin t)$ and compute $\displaystyle\oint_{C_R} \eta$.
3. Explain why this proves η is not exact on $\mathbb{R}^2 \setminus \{0\}$.

Exercise 4.34 (Winding number viewpoint (geometric))**.** Let C be any positively oriented simple closed curve in $\mathbb{R}^2 \setminus \{0\}$. Conjecture (and then verify for circles, ellipses, and a "rounded square") that

$$\oint_C \left(-\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \right) = 2\pi \cdot \mathrm{wind}(C, 0),$$

where $\mathrm{wind}(C, 0) \in \mathbb{Z}$ is the winding number of C around the origin.

Exercise 4.35 (A simply connected criterion you can actually use)**.** Let $D \subset \mathbb{R}^2$ be simply connected and let $\mathbf{F} = (P, Q)$ with $P, Q \in C^1(D)$. Assume $P_y = Q_x$ on D.

1. Fix a base point $(x_0, y_0) \in D$. Define

$$f(x,y) := \int_{\gamma_{(x,y)}} P\,dx + Q\,dy$$

where $\gamma_{(x,y)}$ is any C^1 path in D from (x_0, y_0) to (x, y). Show (by differentiating under the integral sign along axis paths) that $f_x = P$ and $f_y = Q$.

2. Conclude that $\mathbf{F} = \nabla f$ and $\eta = Pdx + Qdy = df$ on D.

Exercise 4.36 (Challenge: re-check the displaced circle integral with a different method)**.** For the displaced circle $C : (x, y) = (2 + \cos t, \sin t)$, evaluate

$$\int_0^{2\pi} \frac{1 + 2\cos t}{5 + 4\cos t}\,dt$$

without the tangent-half-angle substitution by writing

$$\frac{1 + 2\cos t}{5 + 4\cos t} = A + \frac{B}{5 + 4\cos t}$$

and determining A, B. (Then use the standard integral $\int_0^{2\pi} \frac{dt}{a+b\cos t} = \frac{2\pi}{\sqrt{a^2-b^2}}$ for $a > |b|$.)

4.3. Complex Reformulation of Line Integrals

Throughout this subsection, we reinterpret planar line integrals using complex coordinates. This viewpoint unifies vector calculus, differential forms, and complex analysis, and explains circulation integrals in terms of winding and residues.

Complex coordinates and differentials

Let

$$z = x + iy, \qquad \bar{z} = x - iy.$$

Differentiating,

$$dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy.$$

Solving for dx, dy,

$$dx = \tfrac{1}{2}(dz + d\bar{z}), \qquad dy = \tfrac{1}{2i}(dz - d\bar{z}).$$

Interpretation and a small warning about notation. As *real* 1-forms on $\mathbb{R}^2 \simeq \mathbb{C}$, the pair $\{dz, d\bar{z}\}$ is another basis of the real cotangent space. A real 1-form

$$\eta = P\,dx + Q\,dy$$

can be rewritten uniquely in the $\{dz, d\bar{z}\}$ basis, and this rewrite is just linear algebra. (We are not yet assuming holomorphicity; we are only changing coordinates.)

From vector fields to complex-valued 1-forms: explicit conversion formulas

Let $\mathbf{F} = (P, Q)$ be a (real) vector field on $\mathbb{R}^2$ and let

$$\eta := P\,dx + Q\,dy$$

be the associated real 1-form.

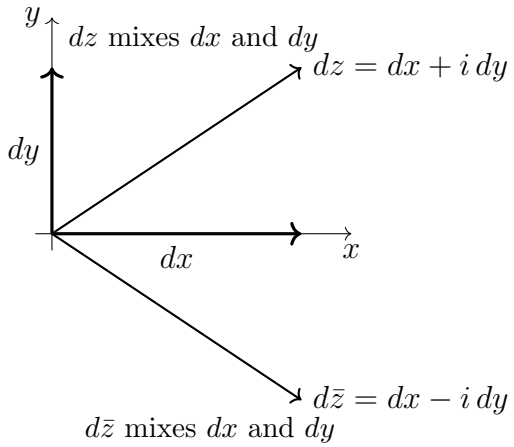


Figure 4: The complex differentials $dz, d\bar{z}$ as linear combinations of dx, dy.

Rewrite η in the $\{dz, d\bar{z}\}$ basis (full algebra). Substitute $dx = \frac{1}{2}(dz + d\bar{z})$ and $dy = \frac{1}{2i}(dz - d\bar{z})$:

$$\begin{aligned}\eta &= P \cdot \tfrac{1}{2}(dz + d\bar{z}) + Q \cdot \tfrac{1}{2i}(dz - d\bar{z}) \\ &= \tfrac{1}{2}P\,dz + \tfrac{1}{2}P\,d\bar{z} + \tfrac{1}{2i}Q\,dz - \tfrac{1}{2i}Q\,d\bar{z} \\ &= \tfrac{1}{2}\big(P - iQ\big)\,dz + \tfrac{1}{2}\big(P + iQ\big)\,d\bar{z}.\end{aligned}$$

What the coefficients mean. Define the complex-valued functions

$$F(z, \bar{z}) := P - iQ, \qquad \overline{F}(z, \bar{z}) := P + iQ.$$

Then

$$\eta = \tfrac{1}{2}F\,dz + \tfrac{1}{2}\overline{F}\,d\bar{z}.$$

So every real 1-form (equivalently every planar vector field) splits into a $(1,0)$-part and a $(0,1)$-part.

Complex potential viewpoint (when it happens). If F depends only on z (not on $\bar{z}$), and if additionally $\frac{1}{2}F\,dz$ has a primitive (i.e. $F = \phi'(z)$ locally), then

$$\tfrac{1}{2}F\,dz = d\Big(\tfrac{1}{2}\phi(z)\Big)$$

is a holomorphic 1-form. Taking real/imaginary parts produces real closed forms and the classical harmonic conjugate picture (Cauchy–Riemann).

Canonical example: the rotational field and the form $\mathrm{Im}(dz/z)$

Definition of the field. On $\mathbb{R}^2 \setminus \{0\}$ consider

$$\mathbf{F}(x, y) = \left(-\frac{y}{x^2 + y^2},\ \frac{x}{x^2 + y^2}\right), \qquad r^2 = x^2 + y^2.$$

The associated 1-form is

$$\eta = -\frac{y}{r^2}\,dx + \frac{x}{r^2}\,dy.$$

Step 1: rewrite η as a complex expression (derive, do not guess). Start from

$$z = x + iy, \quad \bar{z} = x - iy, \quad |z|^2 = z\bar{z} = x^2 + y^2 = r^2.$$

Compute

$$\bar{z}\,dz = (x - iy)(dx + i\,dy) = x\,dx + ix\,dy - iy\,dx + y\,dy,$$
$$z\,d\bar{z} = (x + iy)(dx - i\,dy) = x\,dx - ix\,dy + iy\,dx + y\,dy.$$

Subtract:

$$\begin{aligned}\bar{z}\,dz - z\,d\bar{z} &= \Big(x\,dx + ix\,dy - iy\,dx + y\,dy\Big) - \Big(x\,dx - ix\,dy + iy\,dx + y\,dy\Big) \\ &= (ix\,dy - (-ix\,dy)) + (-iy\,dx - (iy\,dx)) \\ &= 2ix\,dy - 2iy\,dx \\ &= 2i\,(x\,dy - y\,dx).\end{aligned}$$

Hence

$$x\,dy - y\,dx = \frac{1}{2i}\,(\bar{z}\,dz - z\,d\bar{z}).$$

Divide by $|z|^2 = r^2$:

$$\eta = \frac{x\,dy - y\,dx}{x^2 + y^2} = \frac{1}{2i|z|^2}\,\big(\bar{z}\,dz - z\,d\bar{z}\big).$$

Step 2: restrict to the unit circle and identify η with $\mathrm{Im}(dz/z)$. On $|z| = 1$ we have $|z|^2 = 1$ and $\bar{z} = 1/z$. Thus

$$\eta = \frac{1}{2i}\Big(\bar{z}\,dz - z\,d\bar{z}\Big) = \frac{1}{2i}\left(\frac{dz}{z} - z\,d\bar{z}\right).$$

Now compute $\mathrm{Im}(dz/z)$ using the standard identity

$$\mathrm{Im}(w) = \frac{1}{2i}\,(w - \bar{w}).$$

For $w = \dfrac{dz}{z}$, along $|z| = 1$ we have $\overline{w} = \dfrac{\overline{dz}}{z} = \dfrac{d\bar{z}}{\bar{z}} = z\,d\bar{z}$. Therefore

$$\mathrm{Im}\left(\frac{dz}{z}\right) = \frac{1}{2i}\left(\frac{dz}{z} - z\,d\bar{z}\right) = \eta \qquad \text{on } |z| = 1.$$

Step 3: compute the circulation integral (two equivalent computations). *Method A (direct parametrization).* Let $C : z = e^{it}$, $0 \le t \le 2\pi$. Then $dz = ie^{it}\,dt = iz\,dt$, hence

$$\oint_C \frac{dz}{z} = \int_0^{2\pi} i\,dt = 2\pi i, \qquad \oint_C \eta = \oint_C \mathrm{Im}\left(\frac{dz}{z}\right) = \mathrm{Im}(2\pi i) = 2\pi.$$

Method B (residue). Since $\frac{1}{z}$ has residue 1 at 0,

$$\oint_C \frac{dz}{z} = 2\pi i \cdot \mathrm{Res}_{z=0}\left(\frac{1}{z}\right) = 2\pi i, \quad \text{so} \quad \oint_C \eta = 2\pi.$$

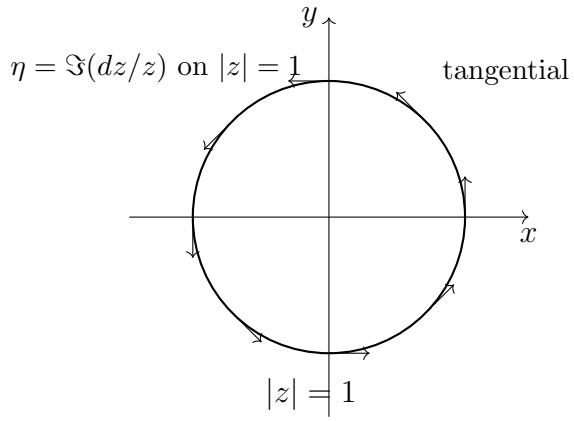


Figure 5: The rotational field corresponds to $\eta = (x\,dy - y\,dx)/(x^2+y^2)$ and its circulation is detected by $\oint dz/z$.

General small circles, winding numbers, and residues (fully computed)

Let $p \in \mathbb{C}$ and consider the positively oriented circle

$$C: \ z(t) = p + re^{it}, \qquad 0 \le t \le 2\pi, \qquad r > 0.$$

Then

$$dz = i\,re^{it}\,dt, \qquad z - p = re^{it},$$

so

$$\frac{dz}{z-p} = \frac{i\,re^{it}\,dt}{re^{it}} = i\,dt.$$

Therefore

$$\oint_C \frac{dz}{z-p} = \int_0^{2\pi} i\,dt = 2\pi i.$$

A slightly more general computation: n turns around p. If instead $C_n : z(t) = p + re^{int}$, $0 \le t \le 2\pi$ with $n \in \mathbb{Z}$, then

$$dz = inre^{int}\,dt, \qquad z - p = re^{int},$$

hence

$$\oint_{C_n} \frac{dz}{z-p} = \int_0^{2\pi} in\,dt = 2\pi i\,n.$$

This integer n is the *winding number* $\mathrm{wind}(C, p)$, and the residue theorem packages it as

$$\oint_C \frac{dz}{z-p} = 2\pi i \cdot \mathrm{wind}(C, p).$$

Back to real circulation: how to manufacture a vortex at p. Define the 1-form on $\mathbb{C} \setminus \{p\}$

$$\eta_p := \mathrm{Im}\left(\frac{dz}{z-p}\right).$$

Then for any loop C avoiding p,

$$\oint_C \eta_p = \mathrm{Im}\left(\oint_C \frac{dz}{z-p}\right) = 2\pi \cdot \mathrm{wind}(C,p).$$

So η_p is closed on $\mathbb{C} \setminus \{p\}$ but not exact there, exactly like the $p = 0$ case.

Remark 4.37 (Curl theorem vs residue theorem: same shadow, different language)**.** Vector calculus (Green's theorem) says for sufficiently nice $\mathbf{F}$,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{\mathrm{Int}(C)} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Complex analysis says for meromorphic g,

$$\oint_C g(z)\, dz = 2\pi i \sum \mathrm{Res}(g; \text{poles inside } C).$$

In vortex examples, the "curl" concentrates at singularities; the residue measures exactly that concentration through winding.

Conceptual summary (what to remember and what to compute)

- A planar vector field $\mathbf{F} = (P, Q)$ corresponds to the real 1-form $\eta = P dx + Q dy$.
- In complex coordinates,

$$\eta = \tfrac{1}{2}(P - iQ)\, dz + \tfrac{1}{2}(P + iQ)\, d\bar{z}.$$

- Vortex-type circulation is encoded by $\displaystyle\oint \frac{dz}{z-p}$ and its imaginary part.
- The integer $\mathrm{wind}(C, p)$ is the topological quantity behind both circulation and residues.

Exercises (computational + conceptual; enough practice)

Exercise 4.38 (Cauchy theorem when a is outside)**.** Let C be a positively oriented circle that does *not* enclose $a \in \mathbb{C}$. Compute $\displaystyle\oint_C \frac{dz}{z-a}$ and explain the result using:

1. Cauchy's theorem (holomorphic on and inside C),
2. the real 1-form $\eta_a = \mathrm{Im}(dz/(z-a))$ and winding number.

Exercise 4.39 (Unit circle integrals of monomials, all integer powers)**.** For $n \in \mathbb{Z}$, compute

$$\oint_{|z|=1} z^n\, dz.$$

(Hint: use the parametrization $z = e^{it}$ or use residues.) Interpret the nonzero case in terms of winding/residue.

Exercise 4.40 (Convert a real field to $dz, d\bar{z}$ and test conservativity)**.** Let $\mathbf{F} = (x, -y)$ on $\mathbb{R}^2$.

1. Write the associated 1-form $\eta = x\,dx - y\,dy$.

2. Express η in terms of $dz, d\bar{z}$.

3. Decide whether $\mathbf{F}$ is conservative by finding an explicit potential f or by a curl test.

Exercise 4.41 (Closedness and loop integrals for $\eta = \Im(dz/z)$)**.** Let $\eta = \mathrm{Im}(dz/z)$ on $\mathbb{C} \setminus \{0\}$.

1. Show $d\eta = 0$ on $\mathbb{C} \setminus \{0\}$ by writing $\eta = \dfrac{x\,dy - y\,dx}{x^2 + y^2}$ and computing $d\eta$.

2. Compute $\displaystyle\oint_{|z|=R} \eta$ for arbitrary $R > 0$.

3. Conclude that η is not exact on $\mathbb{C} \setminus \{0\}$.

Exercise 4.42 (Winding number n explicitly)**.** Let C_n be the loop $z(t) = re^{int}$, $0 \le t \le 2\pi$, with $n \in \mathbb{Z}$ and $r > 0$. Compute:

$$\oint_{C_n} \frac{dz}{z}, \qquad \oint_{C_n} \Im\left(\frac{dz}{z}\right).$$

Explain why the answers must be proportional to n.

Exercise 4.43 (Challenge: recover η_p from a real vector field)**.** Fix $p = a + ib \in \mathbb{C}$. Define

$$\eta_p := \Im\left(\frac{dz}{z-p}\right).$$

1. Write η_p as a real 1-form $P\,dx + Q\,dy$ by simplifying

$$\frac{1}{z-p} = \frac{\overline{(z-p)}}{|z-p|^2} = \frac{(x-a) - i(y-b)}{(x-a)^2 + (y-b)^2}.$$

2. Identify the corresponding vector field $\mathbf{F}_p = (P, Q)$ and describe it geometrically.

3. Compute $\oint_C \eta_p$ for $C : z = p + Re^{it}$ and verify it equals 2π.

4.4. Stokes' Theorem, Cauchy–Green, and Surface Area as a 2-Form

Throughout this subsection we unify three themes: (i) Stokes/Green in the plane as $\int_{\partial\Omega} \eta = \int_\Omega d\eta$, (ii) the Cauchy–Green formula as a direct application to a carefully chosen 1-form, and (iii) surface area as the integral of a pullback 2-form (hence invariant under reparametrization). We emphasize step-by-step computations and the role of wedge products.

1. Stokes' theorem in the plane (Green's theorem) in differential-form language

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with positively oriented (counterclockwise) C^1 boundary $\partial\Omega$. Let $\eta \in \Omega^1(\Omega)$ be a smooth 1-form. In standard coordinates (x, y), write

$$\eta = P(x, y)\,dx + Q(x, y)\,dy.$$

Step 1: compute $d\eta$ explicitly. Using $d(P\,dx) = dP \wedge dx$ and $d(Q\,dy) = dQ \wedge dy$ (because $d(dx) = d(dy) = 0$),

$$d\eta = dP \wedge dx + dQ \wedge dy.$$

Now expand

$$dP = P_x\,dx + P_y\,dy, \qquad dQ = Q_x\,dx + Q_y\,dy,$$

so

$$dP \wedge dx = (P_x\,dx + P_y\,dy) \wedge dx = P_x\,dx \wedge dx + P_y\,dy \wedge dx = 0 - P_y\,dx \wedge dy,$$

$$dQ \wedge dy = (Q_x\,dx + Q_y\,dy) \wedge dy = Q_x\,dx \wedge dy + Q_y\,dy \wedge dy = Q_x\,dx \wedge dy + 0.$$

Hence

$$d\eta = (Q_x - P_y)\,dx \wedge dy.$$

Step 2: state Stokes/Green. Stokes' theorem in the plane says

$$\int_{\partial\Omega} \eta = \int_{\Omega} d\eta.$$

With the computed formula, this becomes the classical Green theorem:

$$\int_{\partial\Omega} \big(P\,dx + Q\,dy\big) = \iint_{\Omega} (Q_x - P_y)\,dx\,dy.$$

Sanity check: a worked micro-example. Let $\eta = x\,dy$ on Ω (any region). Then $P = 0$, $Q = x$. So $Q_x - P_y = 1 - 0 = 1$, hence $d\eta = dx \wedge dy$. Therefore

$$\int_{\partial\Omega} x\,dy = \iint_{\Omega} 1\,dx\,dy = \text{Area}(\Omega).$$

(So $\int_{\partial\Omega} x\,dy$ literally measures signed area of Ω.)

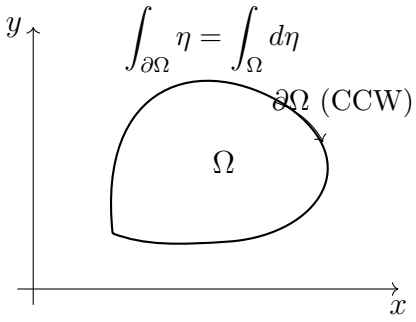


Figure 6: Green/Stokes in the plane: boundary integral of a 1-form equals area integral of its exterior derivative.

2. Complex form of area: $dx \wedge dy$ vs $d\bar{z} \wedge dz$

Identify $\mathbb{R}^2 \simeq \mathbb{C}$ by $z = x + iy$. Then

$$dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy.$$

Compute $d\bar{z} \wedge dz$:

$$d\bar{z} \wedge dz = (dx - i\,dy) \wedge (dx + i\,dy) = dx \wedge dx + i\,dx \wedge dy - i\,dy \wedge dx + (-i^2)\,dy \wedge dy.$$

Using $dx \wedge dx = dy \wedge dy = 0$ and $dy \wedge dx = -dx \wedge dy$,

$$d\bar{z} \wedge dz = i\,dx \wedge dy - i(-dx \wedge dy) = 2i\,dx \wedge dy.$$

Hence the area form satisfies

$$dx \wedge dy = \frac{1}{2i}\,d\bar{z} \wedge dz.$$

This identity is the bridge that turns Green/Stokes into complex-analytic formulas.

3. The Cauchy–Green formula derived by Stokes: full bookkeeping

Let $\Omega \subset \mathbb{C}$ be a bounded domain with smooth boundary $\partial\Omega$, and let $f \in C^1(\overline{\Omega})$. Fix $z \in \Omega$.

Theorem 4.44 (Cauchy–Green formula)**.** *For any $z \in \Omega$,*

$$f(z) = \frac{1}{2\pi i}\left[\int_{\partial\Omega} \frac{f(w)}{w - z}\,dw - \iint_{\Omega} \frac{\partial f/\partial \bar{w}}{w - z}\,d\bar{w} \wedge dw\right].$$

Step 0: notation and the key differential identity. Write $w = u + iv$ for the complex variable of integration. Recall the operators

$$\frac{\partial}{\partial w} = \frac{1}{2}\left(\frac{\partial}{\partial u} - i\frac{\partial}{\partial v}\right), \qquad \frac{\partial}{\partial \bar{w}} = \frac{1}{2}\left(\frac{\partial}{\partial u} + i\frac{\partial}{\partial v}\right),$$

so that for $f(u, v)$,

$$df = f_u\,du + f_v\,dv = f_w\,dw + f_{\bar{w}}\,d\bar{w}, \qquad \text{where } dw = du + i\,dv,\ d\bar{w} = du - i\,dv.$$

We will use the simple but crucial fact:

$$d\big(f(w, \bar{w})\,dw\big) = df \wedge dw = \big(f_w\,dw + f_{\bar{w}}\,d\bar{w}\big) \wedge dw = f_{\bar{w}}\,d\bar{w} \wedge dw \quad (\text{since } dw \wedge dw = 0).$$

Step 1: choose the 1-form to which we apply Stokes. Define on $\Omega \setminus \{z\}$ the 1-form

$$\omega(w) := \frac{f(w)}{w - z}\,dw.$$

This is smooth away from $w = z$.

Step 2: compute $d\omega$ (no skipped steps). Using $d(\phi\, dw) = d\phi \wedge dw$ and $\phi = f(w)/(w-z)$,

$$d\omega = d\left(\frac{f(w)}{w-z}\right) \wedge dw.$$

Expand $d\left(\frac{f}{w-z}\right)$ using the product rule:

$$d\left(\frac{f}{w-z}\right) = \frac{df}{w-z} + f\, d\left(\frac{1}{w-z}\right).$$

Now note:

$$d\left(\frac{1}{w-z}\right) = \frac{\partial}{\partial w}\left(\frac{1}{w-z}\right) dw + \frac{\partial}{\partial \bar{w}}\left(\frac{1}{w-z}\right) d\bar{w}.$$

But $\frac{1}{w-z}$ depends only on w (not on $\bar{w}$), so $\partial/\partial\bar{w}$ term is 0. Hence

$$d\left(\frac{1}{w-z}\right) = \left(-\frac{1}{(w-z)^2}\right) dw.$$

Therefore

$$f\, d\left(\frac{1}{w-z}\right) \wedge dw = f\left(-\frac{1}{(w-z)^2}\, dw\right) \wedge dw = 0 \quad \text{because } dw \wedge dw = 0.$$

So the entire contribution comes from $\frac{df}{w-z} \wedge dw$:

$$d\omega = \frac{df}{w-z} \wedge dw.$$

Finally write $df = f_w\, dw + f_{\bar{w}}\, d\bar{w}$:

$$d\omega = \frac{f_w\, dw + f_{\bar{w}}\, d\bar{w}}{w-z} \wedge dw = \frac{f_w}{w-z}\, dw \wedge dw + \frac{f_{\bar{w}}}{w-z}\, d\bar{w} \wedge dw = \frac{f_{\bar{w}}}{w-z}\, d\bar{w} \wedge dw.$$

Thus we have proved the key identity

$$d\left(\frac{f(w)}{w-z}\, dw\right) = \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw \quad \text{on } \Omega \setminus \{z\}.$$

Step 3: excise a small disk and apply Stokes on a punctured domain. Let $D_\varepsilon(z) = \{w : |w-z| < \varepsilon\}$ and set

$$\Omega_\varepsilon := \Omega \setminus \overline{D_\varepsilon(z)}.$$

Then $\partial\Omega_\varepsilon$ consists of the outer boundary $\partial\Omega$ and the inner boundary $-\partial D_\varepsilon(z)$ (minus sign because the induced orientation is clockwise on the inner circle):

$$\partial\Omega_\varepsilon = \partial\Omega - \partial D_\varepsilon(z).$$

Apply Stokes:

$$\int_{\partial\Omega_\varepsilon} \omega = \iint_{\Omega_\varepsilon} d\omega.$$

That is,

$$\int_{\partial\Omega} \frac{f(w)}{w-z}\, dw - \int_{\partial D_\varepsilon(z)} \frac{f(w)}{w-z}\, dw = \iint_{\Omega_\varepsilon} \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw.$$

Step 4: take $\varepsilon \to 0$ and compute the small-circle limit. Parametrize $\partial D_\varepsilon(z)$ by $w = z + \varepsilon e^{it}$, $0 \le t \le 2\pi$. Then

$$dw = i\varepsilon e^{it}\, dt, \qquad w - z = \varepsilon e^{it}, \qquad \frac{dw}{w-z} = i\, dt.$$

Hence

$$\int_{\partial D_\varepsilon(z)} \frac{f(w)}{w-z}\, dw = \int_0^{2\pi} f(z + \varepsilon e^{it})\, i\, dt.$$

Since f is continuous on $\overline{\Omega}$, $f(z + \varepsilon e^{it}) \to f(z)$ uniformly in t as $\varepsilon \to 0$, so

$$\int_{\partial D_\varepsilon(z)} \frac{f(w)}{w-z}\, dw \longrightarrow \int_0^{2\pi} f(z)\, i\, dt = 2\pi i\, f(z).$$

Also, since the integrand $\frac{f_{\bar{w}}(w)}{w-z}$ is integrable on Ω (the singularity is mild and removable in the principal-value sense because we excised the disk first), we have

$$\iint_{\Omega_\varepsilon} \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw \longrightarrow \iint_{\Omega} \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw.$$

Taking the limit in the Stokes identity yields

$$\int_{\partial\Omega} \frac{f(w)}{w-z}\, dw - 2\pi i\, f(z) = \iint_{\Omega} \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw.$$

Rearrange:

$$f(z) = \frac{1}{2\pi i}\left[\int_{\partial\Omega} \frac{f(w)}{w-z}\, dw - \iint_{\Omega} \frac{f_{\bar{w}}(w)}{w-z}\, d\bar{w} \wedge dw\right],$$

which is the Cauchy–Green formula.

Remark 4.45 (Holomorphic case $\Rightarrow$ Cauchy integral formula)**.** If f is holomorphic on Ω, then $f_{\bar{w}} = 0$ in Ω. The area term disappears and we recover

$$f(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(w)}{w-z}\, dw.$$

Remark 4.46 (Where Green/Stokes is hiding)**.** Using $d\bar{w} \wedge dw = 2i\, du \wedge dv$, the area integral is a Green-type integral on Ω. The Cauchy–Green identity is literally Stokes applied to a meromorphic-looking 1-form.

4. Surface area, Jacobians, and the area 2-form (reparametrization invariance)

Now move to $\mathbb{R}^3$ and let $\sigma : U \to \mathbb{R}^3$ be a smooth surface patch, where $U \subset \mathbb{R}^2$ has coordinates (u, v). Let $R \subset U$ be a region.

Area as a limit of parallelograms (geometric motivation). At a point (u, v), the differential $d\sigma$ sends small parameter vectors $\Delta u\, \partial_u$ and $\Delta v\, \partial_v$ to tangent vectors

$$\sigma_u\, \Delta u, \qquad \sigma_v\, \Delta v$$

on the surface. The spanned parallelogram has area

$$\|\sigma_u \times \sigma_v\|\, |\Delta u\, \Delta v|.$$

Integrating this density leads to the surface area.

Definition 4.47 (Area of a surface patch)**.** The area of $\sigma(R)$ is

$$A_\sigma(R) = \iint_R \|\sigma_u \times \sigma_v\|\, du\, dv.$$

First fundamental form and $\sqrt{EG-F^2}$ (full derivation). Set

$$E = \sigma_u \cdot \sigma_u, \qquad F = \sigma_u \cdot \sigma_v, \qquad G = \sigma_v \cdot \sigma_v.$$

Then $\|\sigma_u \times \sigma_v\|^2$ is the Gram determinant:

$$\|\sigma_u \times \sigma_v\|^2 = \det \begin{pmatrix} \sigma_u \cdot \sigma_u & \sigma_u \cdot \sigma_v \\ \sigma_v \cdot \sigma_u & \sigma_v \cdot \sigma_v \end{pmatrix} = \det \begin{pmatrix} E & F \\ F & G \end{pmatrix} = EG - F^2.$$

Hence

$$\|\sigma_u \times \sigma_v\| = \sqrt{EG - F^2}.$$

So

$$A_\sigma(R) = \iint_R \sqrt{EG - F^2}\, du\, dv.$$

Area as a 2-form (wedge formulation). Define the *area 2-form* on the parameter domain by

$$\omega_\sigma := \|\sigma_u \times \sigma_v\|\, du \wedge dv.$$

Then

$$A_\sigma(R) = \iint_R \omega_\sigma.$$

This is the correct differential-form viewpoint: the integrand is a 2-form density, and wedge products encode oriented area.

Theorem 4.48 (Change of variables (Jacobian) in 2-form language)**.** *Let $\Phi : \widetilde{R} \to R$ be a C^1 diffeomorphism, $\Phi(s,t) = (u(s,t), v(s,t))$. Then*

$$\Phi^*(du \wedge dv) = \det(D\Phi)\, ds \wedge dt.$$

Explicit computation. Compute

$$du = u_s\, ds + u_t\, dt, \qquad dv = v_s\, ds + v_t\, dt.$$

Wedge:

$$\begin{aligned} du \wedge dv &= (u_s\, ds + u_t\, dt) \wedge (v_s\, ds + v_t\, dt) \\ &= u_s v_s\, ds \wedge ds + u_s v_t\, ds \wedge dt + u_t v_s\, dt \wedge ds + u_t v_t\, dt \wedge dt \\ &= \big(u_s v_t - u_t v_s\big)\, ds \wedge dt \\ &= \det(D\Phi)\, ds \wedge dt, \end{aligned}$$

using $ds \wedge ds = dt \wedge dt = 0$ and $dt \wedge ds = -ds \wedge dt$. This is exactly the Jacobian determinant. □

Reparametrization invariance of surface area (clean proof). Let $\widetilde{\sigma} = \sigma \circ \Phi$ be a reparametrization, so

$$\widetilde{\sigma}(s,t) = \sigma(u(s,t), v(s,t)).$$

Then the area form satisfies

$$\omega_{\widetilde{\sigma}} = \|\widetilde{\sigma}_s \times \widetilde{\sigma}_t\|\, ds \wedge dt = \|\sigma_u \times \sigma_v\|\, |\det(D\Phi)|\, ds \wedge dt = \Phi^*\big(\|\sigma_u \times \sigma_v\|\, du \wedge dv\big) = \Phi^*(\omega_\sigma).$$

Therefore, for $\widetilde{R}$ mapping to R,

$$A_{\widetilde{\sigma}}(\widetilde{R}) = \iint_{\widetilde{R}} \omega_{\widetilde{\sigma}} = \iint_{\widetilde{R}} \Phi^*(\omega_\sigma) = \iint_R \omega_\sigma = A_\sigma(R),$$

so the area is invariant.

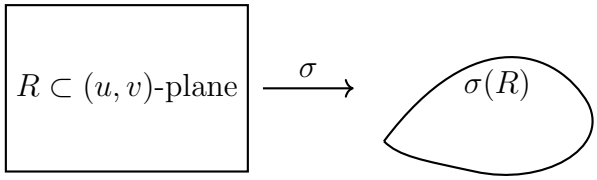


Figure 7: Surface area is the integral of the pulled-back area 2-form: coordinate changes act via Jacobians.

5. Worked examples (Cauchy–Green + area 2-form)

Example 4.49 (Cauchy–Green on the unit disk with $f(w,\bar{w}) = \bar{w}$)**.** Let $\Omega = \{w : |w| < 1\}$ and $f(w,\bar{w}) = \bar{w}$. Compute the Cauchy–Green formula and interpret the result.

Detailed computation. We have $\partial f/\partial \bar{w} = 1$. Fix $z \in \Omega$. Cauchy–Green gives

$$\bar{z} = \frac{1}{2\pi i}\left[\int_{|w|=1} \frac{\bar{w}}{w-z}\, dw - \iint_{|w|<1} \frac{1}{w-z}\, d\bar{w} \wedge dw\right].$$

On $|w| = 1$, $\bar{w} = 1/w$, so the boundary term becomes

$$\int_{|w|=1} \frac{\bar{w}}{w-z}\, dw = \int_{|w|=1} \frac{1}{w(w-z)}\, dw.$$

This has simple poles at $w = 0$ and $w = z$ (both inside the unit circle). By residues,

$$\int_{|w|=1} \frac{1}{w(w-z)}\, dw = 2\pi i \left(\mathrm{Res}_{w=0} \frac{1}{w(w-z)} + \mathrm{Res}_{w=z} \frac{1}{w(w-z)}\right).$$

Compute

$$\mathrm{Res}_{w=0} \frac{1}{w(w-z)} = \lim_{w\to 0} \frac{1}{w-z} = -\frac{1}{z}, \qquad \mathrm{Res}_{w=z} \frac{1}{w(w-z)} = \lim_{w\to z} \frac{1}{w} = \frac{1}{z}.$$

So the boundary term is 0.

Thus the formula reduces to

$$\bar{z} = -\frac{1}{2\pi i} \iint_{|w|<1} \frac{1}{w-z}\, d\bar{w} \wedge dw.$$

Using $d\bar{w} \wedge dw = 2i\, dA$ (where dA is area measure),

$$\bar{z} = -\frac{1}{2\pi i} \iint_{|w|<1} \frac{1}{w-z}\, (2i\, dA) = -\frac{1}{\pi} \iint_{|w|<1} \frac{1}{w-z}\, dA.$$

Equivalently,

$$\iint_{|w|<1} \frac{1}{w-z}\, dA = -\pi\, \bar{z}.$$

This is consistent with the fact that $\bar{z}$ is not holomorphic: the entire value comes from the $\bar{\partial}$-term. □

Example 4.50 (Area of the unit sphere via $\sqrt{EG-F^2}$)**.** Parametrize the unit sphere by

$$\sigma(\theta,\phi)=(\sin\phi\cos\theta,\ \sin\phi\sin\theta,\ \cos\phi),\qquad 0\le\theta\le 2\pi,\ 0\le\phi\le\pi.$$

Detailed computation. Compute partials:

$$\sigma_\theta=(-\sin\phi\sin\theta,\ \sin\phi\cos\theta,\ 0),\qquad \sigma_\phi=(\cos\phi\cos\theta,\ \cos\phi\sin\theta,\ -\sin\phi).$$

Then

$$E=\sigma_\theta\cdot\sigma_\theta=\sin^2\phi,\qquad F=\sigma_\theta\cdot\sigma_\phi=0,\qquad G=\sigma_\phi\cdot\sigma_\phi=1.$$

Hence

$$\sqrt{EG-F^2}=\sqrt{\sin^2\phi\cdot 1-0}=\sin\phi.$$

So the area is

$$A(S^2)=\int_0^{2\pi}\int_0^{\pi}\sin\phi\,d\phi\,d\theta=\int_0^{2\pi}[-\cos\phi]_{\phi=0}^{\phi=\pi}\,d\theta=\int_0^{2\pi}(2)\,d\theta=4\pi.$$

□

6. Exercises (enough practice; mix of computation and concepts)

Exercise 4.51 (Green from Stokes via $d(Pdx+Qdy)$)**.** Write $\eta=Pdx+Qdy$ and compute $d\eta$ explicitly to show

$$\int_{\partial\Omega}(Pdx+Qdy)=\iint_\Omega(Q_x-P_y)\,dx\,dy.$$

Then test the formula on $\eta=x\,dy$ by deducing $\int_{\partial\Omega}x\,dy=\text{Area}(\Omega)$.

Exercise 4.52 (Cauchy–Green ⇒ holomorphicity)**.** Let $f\in C^1(\overline{\Omega})$. Use Cauchy–Green to prove: if $f_{\bar z}=0$ on Ω, then f satisfies the Cauchy integral formula on every smooth subdomain of Ω, hence is holomorphic on Ω. (You may use Morera's theorem as the final step.)

Exercise 4.53 (Compute a boundary term explicitly on the disk)**.** Let $\Omega=\{w:|w|<1\}$ and $z\in\Omega$. Compute

$$\int_{|w|=1}\frac{1}{w-z}\,dw$$

by parametrizing $w=e^{it}$ and by residues, and reconcile the two computations.

Exercise 4.54 (A Cauchy–Green computation with a nonholomorphic function)**.** On $\Omega=\{w:|w|<1\}$ take $f(w,\bar w)=|w|^2=w\bar w$.

1. Compute $f_{\bar w}$.
2. Write the Cauchy–Green formula for $f(z)$ and simplify as much as possible.
3. Explain why the $\bar\partial$-term must be nonzero.

Exercise 4.55 (Jacobian via wedge products)**.** Let $\Phi(s,t)=(u(s,t),v(s,t))$ be C^1. Prove directly that

$$\Phi^*(du\wedge dv)=\det(D\Phi)\,ds\wedge dt$$

by expanding $du=u_s ds+u_t dt$ and $dv=v_s ds+v_t dt$ and wedging.

Exercise 4.56 (Change of variables example $\Phi(u,v) = (u, v^3)$)**.** Let $\Phi(u,v) = (x,y) = (u, v^3)$ map $\widetilde{R} = [0,1] \times [0,1]$ into the (x,y)-plane.

1. Compute $\det D\Phi$.
2. Use change of variables to express $\iint_R f(x,y)\,dx\,dy$ as an integral over $\widetilde{R}$.
3. Apply to $f(x,y) = 1$ to compute Area(R).

Exercise 4.57 (Graph surface area: paraboloid patch)**.** Let $\sigma(u,v) = (u, v, u^2 + v^2)$ on $R = [0,1] \times [0,1]$.

1. Compute σ_u, σ_v and then E, F, G.
2. Compute $\sqrt{EG - F^2}$ and write $A_\sigma(R)$ as an explicit double integral.
3. Give a geometric interpretation: compare to the area of the flat square R.

Exercise 4.58 (Reparametrization invariance on an explicit map)**.** Let $\sigma(u,v) = (u, v, 0)$ (a plane) and $\Phi(s,t) = (u,v) = (s^2, t)$ on $[0,1]^2$.

1. Compute $A_\sigma(R)$ for $R = [0,1] \times [0,1]$.
2. Compute $A_{\sigma\circ\Phi}(\widetilde{R})$ directly and verify it matches (using the Jacobian factor).

4.5. Gaussian Curvature and the Gauss–Bonnet Theorem

Throughout this subsection, we treat Gaussian curvature as an *intrinsic* scalar field on a surface, and we emphasize the geometric meaning of the integral

$$\iint_M K\,dA.$$

We keep the differential-form viewpoint: the area element is a 2-form, and the integrand $K\,dA$ is an invariantly defined 2-form on the surface.

Area form and Gaussian curvature: what is invariant

Let $M \subset \mathbb{R}^3$ be a smooth oriented surface and $\sigma : U \to M$ a local parametrization, $U \subset \mathbb{R}^2$ with coordinates (u,v).

First fundamental form. Write

$$\sigma_u = \frac{\partial \sigma}{\partial u}, \qquad \sigma_v = \frac{\partial \sigma}{\partial v},$$

and define

$$E = \sigma_u \cdot \sigma_u, \qquad F = \sigma_u \cdot \sigma_v, \qquad G = \sigma_v \cdot \sigma_v.$$

Then the induced metric is $ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$.

Area 2-form (from Gram determinant). The area density is

$$\|\sigma_u \times \sigma_v\| = \sqrt{\det \begin{pmatrix} \sigma_u \cdot \sigma_u & \sigma_u \cdot \sigma_v \\ \sigma_v \cdot \sigma_u & \sigma_v \cdot \sigma_v \end{pmatrix}} = \sqrt{EG - F^2}.$$

Hence the area element is the 2-form

$$dA = \sqrt{EG - F^2}\, du \wedge dv.$$

For any scalar function φ on M,

$$\iint_{\sigma(R)} \varphi \, dA = \iint_R (\varphi \circ \sigma) \sqrt{EG - F^2}\, du\, dv.$$

Reparametrization invariance (coordinate-free statement). If $\Phi : \widetilde{U} \to U$ is a diffeomorphism and $\widetilde{\sigma} = \sigma \circ \Phi$, then dA is intrinsic on the surface and

$$\iint_{\widetilde{\sigma}(\widetilde{R})} \varphi \, dA = \iint_{\sigma(R)} \varphi \, dA.$$

Equivalently, dA is a globally defined 2-form on M independent of coordinates.

Gaussian curvature. The Gaussian curvature K is an intrinsic scalar determined by the induced metric. We will use:

- On the unit sphere S^2, $K \equiv 1$.
- On the standard torus of revolution, K changes sign and integrates to 0.

Example 1: the unit sphere S^2 (full computation)

Example 4.59 (Unit sphere S^2: $\iint_{S^2} K\, dA = 4\pi$)**.** Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$. Show that $K \equiv 1$ and compute $\iint_{S^2} K\, dA$.

Proof. **Step 1 (Parametrization).** Use spherical coordinates

$$\sigma(\theta, \phi) = (\sin\phi\cos\theta,\ \sin\phi\sin\theta,\ \cos\phi), \qquad 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi.$$

Step 2 (Compute $\sigma_\theta, \sigma_\phi$).

$$\sigma_\theta = (-\sin\phi\sin\theta,\ \sin\phi\cos\theta,\ 0), \qquad \sigma_\phi = (\cos\phi\cos\theta,\ \cos\phi\sin\theta,\ -\sin\phi).$$

Step 3 (Compute E, F, G).

$$E = \sigma_\theta \cdot \sigma_\theta = \sin^2\phi(\sin^2\theta + \cos^2\theta) = \sin^2\phi,$$

$$F = \sigma_\theta \cdot \sigma_\phi = (-\sin\phi\sin\theta)(\cos\phi\cos\theta) + (\sin\phi\cos\theta)(\cos\phi\sin\theta) + 0 \cdot (-\sin\phi) = 0,$$

$$G = \sigma_\phi \cdot \sigma_\phi = \cos^2\phi(\cos^2\theta + \sin^2\theta) + \sin^2\phi = 1.$$

Step 4 (Area element).

$$dA = \sqrt{EG - F^2}\, d\theta\, d\phi = \sqrt{\sin^2\phi \cdot 1 - 0}\, d\theta\, d\phi = \sin\phi\, d\theta\, d\phi.$$

Step 5 (Total curvature). Since $K \equiv 1$ on the unit sphere,

$$\iint_{S^2} K\, dA = \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta = \int_0^{2\pi} \big[-\cos\phi\big]_{\phi=0}^{\phi=\pi} d\theta = \int_0^{2\pi} 2\, d\theta = 4\pi.$$

□

Why reparametrization cannot change the answer. K is a scalar on the surface and dA is an intrinsic 2-form; hence $K\, dA$ is intrinsic and its integral is coordinate-independent.

Example 2: a torus of revolution (metric, area, curvature, sign)

Let $R > r > 0$ and consider

$$\sigma(\theta, \phi) = \big((R + r\cos\phi)\cos\theta,\ (R + r\cos\phi)\sin\theta,\ r\sin\phi\big), \qquad 0 \le \theta, \phi \le 2\pi.$$

Example 4.60 (Torus: compute E, F, G, dA, K, and $\iint K\, dA$)**.** Show that

$$E = (R + r\cos\phi)^2, \quad F = 0, \quad G = r^2, \qquad dA = r(R + r\cos\phi)\, d\theta\, d\phi,$$

and that

$$K(\phi) = \frac{\cos\phi}{r(R + r\cos\phi)}.$$

Deduce $\displaystyle\iint_{T^2} K\, dA = 0$ and identify where K is positive/negative.

Proof. **Step 1 (Compute $\sigma_\theta, \sigma_\phi$).**

$$\sigma_\theta = \big(-(R + r\cos\phi)\sin\theta,\ (R + r\cos\phi)\cos\theta,\ 0\big),$$

$$\sigma_\phi = \big(-r\sin\phi\cos\theta,\ -r\sin\phi\sin\theta,\ r\cos\phi\big).$$

Step 2 (Compute E).

$$E = \sigma_\theta \cdot \sigma_\theta = (R + r\cos\phi)^2(\sin^2\theta + \cos^2\theta) = (R + r\cos\phi)^2.$$

Step 3 (Compute F).

$$\begin{aligned} F &= \sigma_\theta \cdot \sigma_\phi \\ &= (-(R + r\cos\phi)\sin\theta)(-r\sin\phi\cos\theta) + ((R + r\cos\phi)\cos\theta)(-r\sin\phi\sin\theta) + 0 \cdot (r\cos\phi) \\ &= (R + r\cos\phi) r\sin\phi(\sin\theta\cos\theta - \cos\theta\sin\theta) = 0. \end{aligned}$$

Step 4 (Compute G).

$$G = \sigma_\phi \cdot \sigma_\phi = r^2\sin^2\phi(\cos^2\theta + \sin^2\theta) + r^2\cos^2\phi = r^2(\sin^2\phi + \cos^2\phi) = r^2.$$

Step 5 (Area element).

$$dA = \sqrt{EG - F^2}\, d\theta\, d\phi = \sqrt{(R + r\cos\phi)^2 \cdot r^2 - 0}\, d\theta\, d\phi = r(R + r\cos\phi)\, d\theta\, d\phi.$$

Step 6 (Gaussian curvature). (We record the standard curvature formula for the torus of revolution:)

$$K(\phi) = \frac{\cos\phi}{r(R + r\cos\phi)}.$$

Step 7 (Total curvature). Multiply:

$$K\, dA = \frac{\cos\phi}{r(R + r\cos\phi)} \cdot r(R + r\cos\phi)\, d\theta\, d\phi = \cos\phi\, d\theta\, d\phi.$$

Hence

$$\iint_{T^2} K\,dA = \int_0^{2\pi}\int_0^{2\pi} \cos\phi\,d\phi\,d\theta = \left(\int_0^{2\pi}\cos\phi\,d\phi\right)\left(\int_0^{2\pi} d\theta\right) = 0.$$

Step 8 (Sign regions). Since $r > 0$ and $R + r\cos\phi > 0$, $\operatorname{sign}(K) = \operatorname{sign}(\cos\phi)$:

$$K > 0 \iff \cos\phi > 0 \iff \phi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \pmod{2\pi},$$

$$K < 0 \iff \cos\phi < 0 \iff \phi \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \pmod{2\pi},$$

and $K = 0$ along $\phi = \frac{\pi}{2}, \frac{3\pi}{2}$. Geometrically: outer region has $K > 0$, inner region near the hole has $K < 0$. □

Gauss–Bonnet and Euler characteristic

Theorem 4.61 (Gauss–Bonnet for compact surfaces without boundary)**.** *Let M be a compact oriented smooth surface without boundary. Then*

$$\iint_M K\,dA = 2\pi\,\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M.

Consequences.

- S^2: $\chi(S^2) = 2$ so $\iint_{S^2} K\,dA = 4\pi$.
- T^2: $\chi(T^2) = 0$ so $\iint_{T^2} K\,dA = 0$.
- genus g: $\chi = 2 - 2g$ so $\iint_M K\,dA = 2\pi(2 - 2g)$.

Exercises

Exercise 4.62 (Genus and total curvature)**.** Show that a compact oriented surface of genus g has Euler characteristic $\chi = 2 - 2g$ and deduce

$$\iint_M K\,dA = 2\pi(2 - 2g).$$

Exercise 4.63 (Flat sphere is impossible)**.** Explain why Gauss–Bonnet implies there is no compact surface homeomorphic to S^2 with $K \equiv 0$.

Exercise 4.64 (Where the torus is positively/negatively curved)**.** For the torus $K(\phi) = \dfrac{\cos\phi}{r(R + r\cos\phi)}$:

1. Identify explicitly the regions where $K > 0$, $K < 0$, and $K = 0$.
2. Explain qualitatively why the inner region should have negative curvature.
3. Verify again that $\iint K\,dA = 0$ using the simplified identity $K\,dA = \cos\phi\,d\theta\,d\phi$.

5. Hodge–Weyl Theorem on a Compact Riemann Surface: Weak Existence, Uniqueness, Smooth Regularity, and Explicit Examples

5.1. Standing assumptions, norms, and local coordinate dictionary

Standing assumptions and normalizations. Let M be a compact Riemann surface equipped with a smooth Hermitian area form ω. (Equivalently, ω is the Riemannian volume form of a smooth metric compatible with the complex structure.) For real-valued $u, v \in C^\infty(M, \mathbb{R})$ define

$$\langle u, v\rangle_{L^2} := \int_M u\, v\, \omega, \qquad \|u\|_{L^2}^2 := \langle u, u\rangle_{L^2}.$$

Let ∇u denote the Riemannian gradient and

$$\|u\|_{W^{1,2}}^2 := \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2, \qquad \|\nabla u\|_{L^2}^2 := \int_M |\nabla u|^2\, \omega.$$

Local coordinate dictionary (so signs/constants never drift). In a local holomorphic coordinate $z = x + iy$ the metric may be written

$$g = 2g_{z\bar z}(z)\, dz\, d\bar z, \qquad g_{z\bar z} > 0, \qquad \omega = i\, g_{z\bar z}\, dz \wedge d\bar z = 2g_{z\bar z}\, dx\, dy.$$

With this convention one has (for real u)

$$|\nabla u|^2\, \omega = 2g_{z\bar z}\, |u_z|^2\, dx\, dy \quad \text{and} \quad \Delta_\omega u = g^{z\bar z}\, \partial_z \partial_{\bar z} u \ \ \text{(up to a fixed universal constant)}.$$

Remark. Different texts insert factors 2 or 4 in Δ_ω. None of the functional-analytic arguments below depends on that choice; only the bookkeeping of constants does. We fix one convention and keep it throughout.

5.2. Weak formulation and the mean-zero Hilbert space

Mean-zero Sobolev space. Define the mean-zero subspace

$$W^{1,2}_\perp(M) := \Big\{ u \in W^{1,2}(M) \;\Big|\; \int_M u\, \omega = 0 \Big\}.$$

It is a closed subspace of the Hilbert space $W^{1,2}(M)$, hence itself a Hilbert space with the induced norm $\|\cdot\|_{W^{1,2}}$.

Compatibility condition (why mean-zero is necessary). Let $f \in C^\infty(M)$ satisfy

$$\int_M f\, \omega = 0.$$

This condition is necessary, because on a closed manifold (no boundary) we have

$$\int_M \Delta_\omega u\, \omega = 0 \qquad \forall\, u \in C^\infty(M),$$

so $\Delta_\omega u = f$ forces $\int_M f\, \omega = 0$.

Definition 5.1 (Weak solution of $\Delta_\omega u = f$)**.** A function $u \in W^{1,2}(M)$ is a *weak solution* of $\Delta_\omega u = f$ if

$$\int_M \langle \nabla u, \nabla \varphi \rangle \, \omega = -\int_M f \, \varphi \, \omega \qquad \forall \, \varphi \in C^\infty(M).$$

Why this is the right definition (one clean integration by parts). If u is smooth, then on the closed manifold M we have

$$\int_M \langle \nabla u, \nabla \varphi \rangle \, \omega = -\int_M (\Delta_\omega u) \, \varphi \, \omega,$$

so the weak formulation is exactly the classical equation tested against φ.

5.3. Poincaré inequality, coercivity, and Lax–Milgram

Lemma 5.2 (Poincaré inequality on $W^{1,2}_\perp(M)$)**.** *There exists $C_P > 0$ such that*

$$\|u\|_{L^2} \le C_P \, \|\nabla u\|_{L^2} \qquad \forall \, u \in W^{1,2}_\perp(M).$$

Equivalently, $\|\nabla u\|_{L^2}$ is a norm on $W^{1,2}_\perp(M)$, equivalent to $\|u\|_{W^{1,2}}$.

Proof. Assume the inequality fails. Then there exists $u_n \in W^{1,2}_\perp(M)$ with

$$\|u_n\|_{L^2} = 1 \quad \text{and} \quad \|\nabla u_n\|_{L^2} \to 0.$$

By Rellich–Kondrachov compactness on compact M, $W^{1,2}(M) \hookrightarrow L^2(M)$ is compact. Hence (after passing to a subsequence)

$$u_n \to u_* \text{ in } L^2 \quad \text{and} \quad u_n \rightharpoonup u_* \text{ weakly in } W^{1,2}.$$

Lower semicontinuity gives

$$\|\nabla u_*\|_{L^2} \le \liminf_{n\to\infty} \|\nabla u_n\|_{L^2} = 0,$$

so u_* is (a.e.) constant. Since each u_n has mean zero, $\int_M u_* \, \omega = 0$, hence $u_* \equiv 0$. But $\|u_*\|_{L^2} = \lim \|u_n\|_{L^2} = 1$, contradiction. □

Coercive bilinear form. Define

$$A(u, v) := \int_M \langle \nabla u, \nabla v \rangle \, \omega, \qquad u, v \in W^{1,2}_\perp(M).$$

Then A is bounded and, by Lemma 5.2, coercive:

$$A(u, u) = \|\nabla u\|^2_{L^2} \ge C_P^{-2} \, \|u\|^2_{L^2},$$

and hence $A(u, u) \ge c \, \|u\|^2_{W^{1,2}}$ on $W^{1,2}_\perp(M)$ for some $c > 0$.

Bounded linear functional. For mean-zero $f \in C^\infty(M)$ define

$$\Lambda_f(\varphi) := \int_M f \, \varphi \, \omega.$$

Then, using Cauchy–Schwarz and Lemma 5.2,

$$|\Lambda_f(\varphi)| \le \|f\|_{L^2} \, \|\varphi\|_{L^2} \le C_P \, \|f\|_{L^2} \, \|\nabla \varphi\|_{L^2} \le C \, \|f\|_{L^2} \, \|\varphi\|_{W^{1,2}}.$$

Proposition 5.3 (Weak existence, uniqueness, and energy estimate)**.** *For every $f \in C^\infty(M)$ with $\int_M f\,\omega = 0$ there exists a unique $u \in W^{1,2}_\perp(M)$ such that*

$$A(u,\varphi) = -\Lambda_f(\varphi) \qquad \forall\, \varphi \in W^{1,2}_\perp(M).$$

Equivalently, u solves $\Delta_\omega u = f$ in the weak sense. Moreover,

$$\|\nabla u\|_{L^2} \le C_P\, \|f\|_{L^2}.$$

Proof. By Lax–Milgram applied to the coercive bilinear form A and the bounded linear functional $-\Lambda_f$, there exists a unique $u \in W^{1,2}_\perp(M)$ such that $A(u,\cdot) = -\Lambda_f(\cdot)$.

For the estimate, test the weak equation with $\varphi = u$:

$$\|\nabla u\|^2_{L^2} = A(u,u) = -\Lambda_f(u) \le \|f\|_{L^2}\, \|u\|_{L^2} \le C_P\, \|f\|_{L^2}\, \|\nabla u\|_{L^2},$$

hence $\|\nabla u\|_{L^2} \le C_P\, \|f\|_{L^2}$. □

5.4. Local smoothing: Weyl's lemma (weakly harmonic ⇒ smooth)

Lemma 5.4 (Weyl's lemma (Euclidean version))**.** *Let $U \subset \mathbb{R}^2$ be open and $u \in W^{1,2}_{\mathrm{loc}}(U)$ satisfy*

$$\int_U u\, \Delta\phi\, dx\, dy = 0 \qquad \forall\, \phi \in C^\infty_c(U).$$

Then $u \in C^\infty(U)$ and $\Delta u = 0$ classically on U. In particular, u satisfies the mean-value property on balls compactly contained in U.

Sketch (mollifiers). Let $\chi \in C^\infty_c(\mathbb{R}^2)$ be a standard mollifier, $\int \chi = 1$, and set $\chi_\varepsilon(x) = \varepsilon^{-2}\chi(x/\varepsilon)$. For $U_\varepsilon := \{x \in U : \operatorname{dist}(x, \partial U) > \varepsilon\}$ define

$$u_\varepsilon := \chi_\varepsilon * u \quad \text{on } U_\varepsilon.$$

Then $u_\varepsilon \in C^\infty(U_\varepsilon)$ and, in distributions,

$$\Delta u_\varepsilon = \chi_\varepsilon * (\Delta u) = 0 \quad \text{on } U_\varepsilon,$$

so u_ε is classically harmonic. As $\varepsilon \to 0$, $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(U)$ and a.e. The mean-value property holds for each u_ε and passes to the a.e. limit, and standard interior regularity for harmonic functions yields $u \in C^\infty(U)$. □

5.5. From weak to smooth solutions on (M, ω)

Proposition 5.5 (Elliptic regularity on a compact surface)**.** *Let $f \in C^\infty(M)$ and let $u \in W^{1,2}(M)$ be a weak solution of $\Delta_\omega u = f$. Then $u \in C^\infty(M)$.*

Proof. Regularity is local. Fix a coordinate chart $U \subset M$ with complex coordinate $z = x + iy$ so that $\omega = 2g_{z\bar z}(z)\, dx\, dy$ with $g_{z\bar z}$ smooth and strictly positive. On a smaller relatively compact domain $B \Subset U$, the coefficients are bounded above and below:

$$0 < \lambda \le g_{z\bar z} \le \Lambda \quad \text{on } B.$$

In such coordinates, Δ_ω is a uniformly elliptic second-order operator with smooth coefficients. Standard elliptic regularity for uniformly elliptic operators implies that

$$u \in W^{1,2}(B),\ \Delta_\omega u = f \in C^\infty(B) \quad \Longrightarrow \quad u \in C^\infty(B).$$

Cover M by finitely many such charts and conclude $u \in C^\infty(M)$. □

5.6. Hodge–Weyl theorem (functions)

Theorem 5.6 (Hodge–Weyl for functions (Poisson equation with mean-zero data))**.** *Let (M,ω) be a compact Riemann surface with Hermitian area form ω. For every $f \in C^\infty(M)$ with $\int_M f\,\omega = 0$ there exists a unique $u \in C^\infty(M)$ with $\int_M u\,\omega = 0$ such that*

$$\Delta_\omega u = f.$$

Moreover, there exists $C > 0$ depending only on (M,ω) such that

$$\|\nabla u\|_{L^2} \le C\,\|f\|_{L^2}.$$

Proof. Proposition 5.3 gives a unique weak solution $u \in W^{1,2}_{\perp}(M)$. Proposition 5.5 upgrades it to $u \in C^\infty(M)$. The L^2-energy estimate was already obtained in Proposition 5.3. □

5.7. Concrete model examples with full calculations

How to read the examples. Each example has the same pattern: (i) identify $\ker\Delta$ (constants), (ii) check mean-zero of f, (iii) solve by an explicit eigenfunction/Fourier expansion, (iv) verify the mean-zero normalization of u and the energy identity.

Example A: the circle S^1 (Fourier series, completely explicit)

Let $M = S^1 = \mathbb{R}/2\pi\mathbb{Z}$ with coordinate θ and $\omega = d\theta$. Take the Laplacian convention

$$\Delta u = -\frac{d^2u}{d\theta^2}.$$

Compatibility. A solution of $\Delta u = f$ exists only if

$$\int_0^{2\pi} f(\theta)\,d\theta = 0.$$

Explicit computation for one mode. Let $f(\theta) = \sin(3\theta)$. Then $\int_0^{2\pi} f\,d\theta = 0$. We seek u with $-u'' = \sin(3\theta)$ and $\int_0^{2\pi} u\,d\theta = 0$. Solve:

$$u'' = -\sin(3\theta) \quad\Longrightarrow\quad u(\theta) = \frac{1}{9}\sin(3\theta) + a\theta + b.$$

Periodicity forces $a = 0$. Mean-zero forces $b = 0$. Hence

$$u(\theta) = \frac{1}{9}\sin(3\theta).$$

Energy identity (check the estimate by hand). Compute

$$u'(\theta) = \frac{1}{3}\cos(3\theta), \qquad \int_0^{2\pi} |u'|^2\,d\theta = \int_0^{2\pi} \frac{1}{9}\cos^2(3\theta)\,d\theta = \frac{\pi}{9}.$$

Also

$$\int_0^{2\pi} |f|^2\,d\theta = \int_0^{2\pi} \sin^2(3\theta)\,d\theta = \pi.$$

So $\|u'\|_{L^2} \le \|f\|_{L^2}$ holds with room to spare.

Example B: the flat torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ (two-variable Fourier)

Let $M = T^2$ with $(x, y) \in [0, 1)^2$ and $\omega = dx\, dy$, and take

$$\Delta u = -(\partial_x^2 + \partial_y^2).$$

Compatibility. A solution exists only if

$$\int_{T^2} f(x, y)\, dx\, dy = 0.$$

Explicit right-hand side and solution. Let

$$f(x, y) = \cos(2\pi x)\cos(2\pi y).$$

Then $\int_{T^2} f\, dx\, dy = 0$ since each cosine has mean zero on $[0, 1)$. Try $u(x, y) = \alpha \cos(2\pi x)\cos(2\pi y)$. Compute

$$\partial_x^2 u = -(2\pi)^2 \alpha \cos(2\pi x)\cos(2\pi y), \qquad \partial_y^2 u = -(2\pi)^2 \alpha \cos(2\pi x)\cos(2\pi y),$$

so

$$\Delta u = -(\partial_x^2 + \partial_y^2)u = 2(2\pi)^2 \alpha \cos(2\pi x)\cos(2\pi y).$$

Hence $\Delta u = f$ forces

$$2(2\pi)^2\alpha = 1 \quad \Longrightarrow \quad \alpha = \frac{1}{8\pi^2}.$$

Therefore,

$$u(x, y) = \frac{1}{8\pi^2}\cos(2\pi x)\cos(2\pi y), \qquad \int_{T^2} u\, dx\, dy = 0.$$

Energy (optional check).

$$\partial_x u = -\frac{1}{8\pi^2}(2\pi)\sin(2\pi x)\cos(2\pi y) = -\frac{1}{4\pi}\sin(2\pi x)\cos(2\pi y),$$

$$\partial_y u = -\frac{1}{4\pi}\cos(2\pi x)\sin(2\pi y),$$

so

$$\int_{T^2} |\nabla u|^2\, dx\, dy = \frac{1}{16\pi^2}\int_{T^2} \Big(\sin^2(2\pi x)\cos^2(2\pi y) + \cos^2(2\pi x)\sin^2(2\pi y)\Big)\, dx\, dy = \frac{1}{64\pi^2}.$$

Example C: the round sphere S^2 (eigenvalues and a concrete harmonic)

Let $M = S^2$ with the round metric and area form dA. The Laplacian has eigenvalues $\ell(\ell + 1)$ with spherical harmonics $Y_{\ell m}$.

A concrete eigenfunction: the height function. Let $f(p) = z$ be the height function on $S^2 \subset \mathbb{R}^3$, i.e. $f(\theta, \phi) = \cos\phi$. It is a first spherical harmonic ($\ell = 1$), and satisfies

$$\Delta f = 2f$$

for the standard sign convention where Δ is nonnegative on eigenfunctions. (If your sign convention differs, replace 2 by -2 accordingly.)

Solve $\Delta u = f$. Try $u = \beta f$. Then $\Delta u = \beta \Delta f = 2\beta f$. Thus $\Delta u = f$ gives $2\beta = 1$, i.e. $\beta = \frac{1}{2}$. So

$$u = \frac{1}{2} f = \frac{1}{2} z.$$

Mean-zero check. Since z is odd under $p \mapsto -p$,

$$\int_{S^2} z \, dA = 0,$$

so u automatically satisfies the required normalization $\int u \, dA = 0$.

Geometric moral. On compact M, constants are the only global obstruction. On S^2, there are no nonconstant harmonic functions, so mean-zero is the only condition.

5.8. Optional: mean-value property from the weak formulation

Remark. Lemma 5.4 can be proved in several equivalent ways. Besides mollifiers, one may derive the mean-value property by testing the weak equation against carefully chosen radial cutoff functions on balls and then differentiating with respect to the radius. This route is often used in PDE notes to show that "weakly harmonic" implies "classically harmonic".

5.9. Exercises

A. Compatibility and uniqueness.

Exercise 5.7. Let (M, ω) be compact. Show directly that $\int_M \Delta_\omega u \, \omega = 0$ for every $u \in C^\infty(M)$. (Hint: integrate $\operatorname{div}(\nabla u)$ and use Stokes' theorem on a manifold without boundary.)

Exercise 5.8. Show that if $u \in W^{1,2}(M)$ is a weak solution of $\Delta_\omega u = 0$, then u is constant. (Hint: test with $\varphi = u - \frac{1}{\operatorname{Vol}(M)} \int_M u \, \omega$ and use Poincaré.)

B. Explicit computations.

Exercise 5.9. On S^1 with $\Delta = -\frac{d^2}{d\theta^2}$, solve $\Delta u = \cos(5\theta)$ under the normalization $\int_0^{2\pi} u \, d\theta = 0$. Compute $\int_0^{2\pi} |u'|^2 \, d\theta$ explicitly.

Exercise 5.10. On $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with $\Delta = -(\partial_x^2 + \partial_y^2)$, solve

$$\Delta u = \sin(2\pi x) \sin(4\pi y), \qquad \int_{T^2} u \, dx \, dy = 0.$$

Exercise 5.11. On the round sphere S^2, assume $\Delta Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$. Given

$$f = \sum_{\ell \geq 1} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m},$$

write down the solution u of $\Delta u = f$ with $\int_{S^2} u \, dA = 0$.

C. Variational viewpoint.

Exercise 5.12. Define the functional on $W^{1,2}_{\perp}(M)$:

$$E(v) = \frac{1}{2}\int_M |\nabla v|^2\,\omega - \int_M f v\,\omega.$$

Show that the weak solution u of $\Delta_\omega u = f$ is the unique minimizer of E.

D. Regularity and local-to-global logic.

Exercise 5.13. Give a self-contained proof that weakly harmonic functions on an open set $U \subset \mathbb{R}^2$ are smooth by mollification, following Lemma 5.4. Your proof should explicitly justify why $\Delta(\chi_\varepsilon * u) = \chi_\varepsilon * (\Delta u)$.

E. Boundary variation (preview).

Exercise 5.14. Let $\Omega \subset \mathbb{R}^2$ be a bounded smooth domain. Compare solvability of $\Delta u = f$ under: (i) Dirichlet boundary condition $u|_{\partial\Omega} = 0$, (ii) Neumann boundary condition $\partial_\nu u|_{\partial\Omega} = 0$. Identify the correct "mean-zero" type compatibility condition in the Neumann case.

Part IV
Sheaves and Cohomology on Complex Manifolds

6. Holomorphic line bundles, divisors, and the Picard group

6.1. Holomorphic line bundles: cocycles, gauge changes, and explicit models

1. Motivation: local holomorphic data and multiplicative ambiguity

On a complex manifold M, holomorphic functions are sections of the trivial line bundle. Many natural objects are locally holomorphic but do not glue as honest functions; instead, they glue up to multiplication by nowhere-vanishing holomorphic functions. This is exactly the mechanism encoded by holomorphic line bundles.

2. Definition via transition functions

Definition 6.1 (Holomorphic line bundle via transition functions)**.** Let M be a complex manifold and $\{U_\alpha\}_{\alpha\in A}$ an open cover. A *holomorphic line bundle* on M is specified by a family of holomorphic functions

$$f_{\alpha\beta} \in \mathcal{O}^*(U_{\alpha\beta}), \qquad U_{\alpha\beta} := U_\alpha \cap U_\beta,$$

satisfying the cocycle identities

$$f_{\alpha\alpha} = 1, \qquad f_{\alpha\beta} f_{\beta\alpha} = 1, \qquad f_{\alpha\beta} f_{\beta\gamma} f_{\gamma\alpha} = 1 \quad \text{on } U_{\alpha\beta\gamma}.$$

Two such families $\{f_{\alpha\beta}\}$ and $\{\tilde{f}_{\alpha\beta}\}$ define isomorphic line bundles if there exist $g_\alpha \in \mathcal{O}^*(U_\alpha)$ such that

$$\tilde{f}_{\alpha\beta} = g_\alpha \, f_{\alpha\beta} \, g_\beta^{-1} \quad \text{on } U_{\alpha\beta}.$$

3. Gluing construction and complex manifold structure

Proposition 6.2 (Bundle obtained by gluing trivial bundles)**.** *Let $\{U_\alpha\}$ be an open cover of M and $\{f_{\alpha\beta}\}$ a cocycle as in Definition 6.1. Consider the disjoint union*

$$E := \bigsqcup_{\alpha \in A} (U_\alpha \times \mathbb{C}),$$

and define an equivalence relation $\sim$ on E by declaring that for $x \in U_{\alpha\beta}$,

$$(\alpha, x, \eta) \sim (\beta, x, \xi) \quad \Longleftrightarrow \quad \xi = f_{\beta\alpha}(x)\, \eta.$$

Then the quotient space $L := E/\sim$ admits a unique complex manifold structure such that:

1. *the natural map $\pi : L \to M$, $\pi([\alpha, x, \eta]) = x$, is holomorphic;*
2. *for each α, the map*

$$\varphi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}, \qquad \varphi_\alpha([\beta, x, \xi]) := (x, f_{\alpha\beta}(x)\, \xi)$$

   is a biholomorphism which is $\mathbb{C}$-linear on fibers;

3. *on overlaps $U_{\alpha\beta}$ the change of trivialization is multiplication by $f_{\alpha\beta}$:*

$$\varphi_\alpha \circ \varphi_\beta^{-1}(x, \eta) = (x, f_{\alpha\beta}(x)\, \eta).$$

Hence $\pi : L \to M$ is a holomorphic line bundle with transition cocycle $\{f_{\alpha\beta}\}$.

Proof. Step 1: $\sim$ is an equivalence relation. Reflexivity holds because $f_{\alpha\alpha} = 1$ gives $(\alpha, x, \eta) \sim (\alpha, x, \eta)$. Symmetry holds because $f_{\beta\alpha} = f_{\alpha\beta}^{-1}$: if $\xi = f_{\beta\alpha}\eta$ then $\eta = f_{\alpha\beta}\xi$, hence $(\beta, x, \xi) \sim (\alpha, x, \eta)$. Transitivity: suppose $(\alpha, x, \eta) \sim (\beta, x, \xi)$ and $(\beta, x, \xi) \sim (\gamma, x, \zeta)$ with $x \in U_{\alpha\beta\gamma}$. Then

$$\xi = f_{\beta\alpha}(x)\eta, \qquad \zeta = f_{\gamma\beta}(x)\xi \quad \Rightarrow \quad \zeta = f_{\gamma\beta}(x) f_{\beta\alpha}(x)\eta.$$

The cocycle identity $f_{\gamma\beta} f_{\beta\alpha} = f_{\gamma\alpha}$ (equivalent to $f_{\alpha\beta} f_{\beta\gamma} f_{\gamma\alpha} = 1$) yields $\zeta = f_{\gamma\alpha}(x)\eta$, i.e. $(\alpha, x, \eta) \sim (\gamma, x, \zeta)$.

Step 2: define charts on L. For fixed α, define φ_α on $\pi^{-1}(U_\alpha)$ by

$$\varphi_\alpha([\beta, x, \xi]) := (x, f_{\alpha\beta}(x)\, \xi).$$

We must show φ_α is well-defined. Suppose $[\beta, x, \xi] = [\beta', x, \xi']$. Then $\xi' = f_{\beta'\beta}(x)\xi$. Using the cocycle identity on $U_{\alpha\beta\beta'}$,

$$f_{\alpha\beta'}(x)\xi' = f_{\alpha\beta'}(x) f_{\beta'\beta}(x)\xi = f_{\alpha\beta}(x)\xi,$$

so φ_α assigns the same value; hence it is well-defined.

Step 3: φ_α is bijective with explicit inverse. Given $(x,\eta) \in U_\alpha \times \mathbb{C}$, set $\psi_\alpha(x,\eta) := [\alpha, x, \eta] \in L$. Then $\varphi_\alpha(\psi_\alpha(x,\eta)) = (x, f_{\alpha\alpha}(x)\eta) = (x,\eta)$. Conversely, for any $[\beta, x, \xi] \in \pi^{-1}(U_\alpha)$,

$$\psi_\alpha(\varphi_\alpha([\beta, x, \xi])) = [\alpha, x, f_{\alpha\beta}(x)\xi].$$

But $(\beta, x, \xi) \sim (\alpha, x, f_{\alpha\beta}(x)\xi)$ because $\xi = f_{\beta\alpha}(x) f_{\alpha\beta}(x)\xi$, so these define the same class in L. Thus ψ_α is the inverse of φ_α.

Step 4: holomorphic transition maps. On $U_{\alpha\beta}$, the overlap map

$$\varphi_\alpha \circ \varphi_\beta^{-1} : (x,\eta) \mapsto (x, f_{\alpha\beta}(x)\eta)$$

is holomorphic because $f_{\alpha\beta} \in \mathcal{O}^*(U_{\alpha\beta})$. Therefore the atlas $\{(\pi^{-1}(U_\alpha), \varphi_\alpha)\}$ defines a complex manifold structure on L. By construction, π is holomorphic and fibers are isomorphic to $\mathbb{C}$, with the obvious $\mathbb{C}$-vector space structure transported through the trivializations. Hence L is a holomorphic line bundle. □

4. Sections as glued holomorphic functions

Proposition 6.3 (Local descriptions of holomorphic sections)**.** *Let L be defined by transition functions $\{f_{\alpha\beta}\}$ on $\{U_\alpha\}$. A holomorphic section s of L over $U \subset M$ is equivalent to holomorphic functions $s_\alpha \in \mathcal{O}(U \cap U_\alpha)$ satisfying*

$$s_\alpha = f_{\alpha\beta}\, s_\beta \quad \text{on } U \cap U_{\alpha\beta}.$$

Moreover, given such $\{s_\alpha\}$, there exists a unique section s whose α-trivialization is s_α.

Proof. Fix $U \subset M$. Suppose $s : U \to L$ is a holomorphic section. For each α, define

$$s_\alpha := \mathrm{pr}_2 \circ \varphi_\alpha \circ s \in \mathcal{O}(U \cap U_\alpha),$$

where $\mathrm{pr}_2 : U_\alpha \times \mathbb{C} \to \mathbb{C}$ is projection. On $U \cap U_{\alpha\beta}$,

$$\varphi_\alpha \circ s = (\mathrm{id}, s_\alpha), \qquad \varphi_\beta \circ s = (\mathrm{id}, s_\beta),$$

and since $\varphi_\alpha \circ \varphi_\beta^{-1}(x,\eta) = (x, f_{\alpha\beta}(x)\eta)$, we get

$$s_\alpha(x) = f_{\alpha\beta}(x)\, s_\beta(x) \quad \forall x \in U \cap U_{\alpha\beta}.$$

Conversely, assume $\{s_\alpha\}$ are holomorphic and satisfy $s_\alpha = f_{\alpha\beta} s_\beta$ on overlaps. Define a map $s : U \to L$ by: if $x \in U \cap U_\alpha$, set

$$s(x) := \psi_\alpha(x, s_\alpha(x)) = [\alpha, x, s_\alpha(x)] \in L.$$

We must show s is well-defined. If $x \in U \cap U_{\alpha\beta}$, then

$$[\alpha, x, s_\alpha(x)] = [\alpha, x, f_{\alpha\beta}(x) s_\beta(x)] = [\beta, x, s_\beta(x)]$$

by the defining relation of the quotient, using $s_\alpha = f_{\alpha\beta} s_\beta$. Thus s is well-defined. On each $U \cap U_\alpha$, the expression $x \mapsto [\alpha, x, s_\alpha(x)]$ is holomorphic because ψ_α is biholomorphic and s_α is holomorphic; hence s is holomorphic on U. Uniqueness follows because the trivializations φ_α separate points in fibers: if two sections have the same second component in every trivialization, they coincide. □

5. Gauge changes and isomorphism classes

Proposition 6.4 (Coboundary change produces an isomorphic bundle)**.** *Let $\{f_{\alpha\beta}\}$ and $\{\tilde{f}_{\alpha\beta}\}$ be cocycles on the same cover $\{U_\alpha\}$, and suppose $\tilde{f}_{\alpha\beta} = g_\alpha f_{\alpha\beta} g_\beta^{-1}$ for some $g_\alpha \in \mathcal{O}^*(U_\alpha)$. Let L and $\tilde{L}$ be the corresponding glued bundles. Then there exists a holomorphic bundle isomorphism $\Phi : L \to \tilde{L}$ covering* id_M.

Proof. Work on the disjoint union model $E = \bigsqcup_\alpha (U_\alpha \times \mathbb{C})$ and $\tilde{E} = \bigsqcup_\alpha (U_\alpha \times \mathbb{C})$. Define $T : E \to \tilde{E}$ by

$$T(\alpha, x, \eta) := (\alpha, x, g_\alpha(x)\eta).$$

We claim T respects the equivalence relations. Indeed, if $(\alpha, x, \eta) \sim (\beta, x, \xi)$ in E, then $\xi = f_{\beta\alpha}(x)\eta$. Hence

$$T(\beta, x, \xi) = (\beta, x, g_\beta(x) f_{\beta\alpha}(x)\eta).$$

On the other hand, in $\tilde{E}$ we have the relation $(\alpha, x, \eta') \sim (\beta, x, \xi')$ iff $\xi' = \tilde{f}_{\beta\alpha}(x)\eta'$. Using $\tilde{f}_{\beta\alpha} = g_\beta f_{\beta\alpha} g_\alpha^{-1}$,

$$\tilde{f}_{\beta\alpha}(x)\,(g_\alpha(x)\eta) = g_\beta(x) f_{\beta\alpha}(x)\eta,$$

so $(\alpha, x, g_\alpha\eta) \sim (\beta, x, g_\beta f_{\beta\alpha}\eta)$ in $\tilde{E}$, i.e. $T(\alpha, x, \eta) \sim T(\beta, x, \xi)$ in $\tilde{E}$. Therefore T descends to a well-defined map $\Phi : L \to \tilde{L}$ on the quotients.

In the α-trivializations, Φ is given by $(x, \eta) \mapsto (x, g_\alpha(x)\eta)$, which is holomorphic with holomorphic inverse $(x, \eta) \mapsto (x, g_\alpha(x)^{-1}\eta)$. Hence Φ is a holomorphic bundle isomorphism. □

6. Tensor product, dual bundle, and induced cocycles

Proposition 6.5 (Operations on line bundles in cocycle form)**.** *Let L, L' be line bundles defined on $\{U_\alpha\}$ by cocycles $\{f_{\alpha\beta}\}$ and $\{f'_{\alpha\beta}\}$. Then:*

1. *$L \otimes L'$ is defined by the cocycle $\{f_{\alpha\beta} f'_{\alpha\beta}\}$.*
2. *The dual bundle $L^\vee$ is defined by $\{f_{\alpha\beta}^{-1}\}$.*

Proof. Fix trivializations e_α for L and e'_α for L' on U_α with $e_\alpha = f_{\alpha\beta} e_\beta$ and $e'_\alpha = f'_{\alpha\beta} e'_\beta$. Then $e_\alpha \otimes e'_\alpha = (f_{\alpha\beta} f'_{\alpha\beta})(e_\beta \otimes e'_\beta)$, so the transition functions of $L \otimes L'$ are $f_{\alpha\beta} f'_{\alpha\beta}$.

For the dual, let $e_\alpha^\vee$ be the dual frame characterized by $e_\alpha^\vee(e_\alpha) = 1$. Since $e_\alpha = f_{\alpha\beta} e_\beta$, we have $e_\beta = f_{\beta\alpha} e_\alpha$ and

$$e_\alpha^\vee = f_{\alpha\beta}^{-1} e_\beta^\vee,$$

because evaluating on $e_\alpha = f_{\alpha\beta} e_\beta$ gives $e_\beta^\vee(e_\alpha) = e_\beta^\vee(f_{\alpha\beta} e_\beta) = f_{\alpha\beta}$, hence $(f_{\alpha\beta}^{-1} e_\beta^\vee)(e_\alpha) = 1$. Thus the cocycle is $f_{\alpha\beta}^{-1}$. □

7. Worked example: the canonical bundle on a Riemann surface

Example 6.6 (Canonical bundle cocycle from coordinate changes)**.** Let M be a Riemann surface with local coordinates z_α on U_α. Define the local frame $e_\alpha := dz_\alpha$ of the canonical bundle K_M. On overlaps $U_{\alpha\beta}$, write $z_\beta = z_\beta(z_\alpha)$; then

$$dz_\beta = \frac{\partial z_\beta}{\partial z_\alpha}\, dz_\alpha.$$

Hence $e_\beta = \left(\frac{\partial z_\beta}{\partial z_\alpha}\right) e_\alpha$ and equivalently

$$e_\alpha = f_{\alpha\beta} e_\beta, \qquad f_{\alpha\beta} = \left(\frac{\partial z_\beta}{\partial z_\alpha}\right)^{-1} \in \mathcal{O}^*(U_{\alpha\beta}).$$

On a triple overlap, the chain rule implies $f_{\alpha\gamma} = f_{\alpha\beta} f_{\beta\gamma}$, i.e. the cocycle condition.

8. Worked example: $\mathcal{O}(1)$ on $\mathbb{CP}^1$ and its global sections

Example 6.7 ($\mathcal{O}(1)$ on $\mathbb{CP}^1$ and $H^0(\mathbb{CP}^1, \mathcal{O}(1))$)**.** Cover $\mathbb{CP}^1$ by $U_0 = \{Z_0 \neq 0\}$ and $U_1 = \{Z_1 \neq 0\}$. Let $z = Z_1/Z_0$ on U_0 and $w = Z_0/Z_1 = 1/z$ on U_1. Define $\mathcal{O}(1)$ by the transition function $f_{01} = z$ on U_{01}. A holomorphic section corresponds to (s_0, s_1) with $s_0 \in \mathcal{O}(U_0)$, $s_1 \in \mathcal{O}(U_1)$ and

$$s_0(z) = z\, s_1(1/z) \quad \text{on } U_{01}.$$

Expand $s_1(w) = \sum_{k\geq 0} a_k w^k$ (entire on $U_1 \simeq \mathbb{C}$). Then

$$s_0(z) = z\, s_1(1/z) = z \sum_{k\geq 0} a_k z^{-k} = \sum_{k\geq 0} a_k z^{1-k}.$$

For s_0 to be entire in z, all negative powers must vanish, so $a_k = 0$ for $k \geq 2$. Thus $s_1(w) = a_0 + a_1 w$ and $s_0(z) = a_0 z + a_1$. Hence

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2,$$

with basis corresponding to $(s_0, s_1) = (z, 1)$ and $(1, w)$, i.e. homogeneous sections Z_1 and Z_0.

6.2. Divisors and the associated line bundle

1. Divisors on a Riemann surface

Throughout this subsection, let X be a Riemann surface.

Definition 6.8 (Divisor)**.** A *divisor* on X is a formal finite integer linear combination of points:

$$D = \sum_{p\in X} n_p\, p, \qquad n_p \in \mathbb{Z}, \quad n_p = 0 \text{ for all but finitely many } p.$$

The *degree* of D is $\deg(D) := \sum_p n_p$. We write $D \geq 0$ if $n_p \geq 0$ for all p (effective divisor).

Definition 6.9 (Principal divisor)**.** Let f be a nonzero meromorphic function on X. For each $p \in X$, let $\mathrm{ord}_p(f) \in \mathbb{Z}$ be the order of vanishing (positive) or pole (negative) of f at p. The *principal divisor* of f is

$$(f) := \sum_{p\in X} \mathrm{ord}_p(f)\, p.$$

2. The sheaf $\mathcal{O}(D)$ and local growth condition

Definition 6.10 (Sheaf of meromorphic functions bounded by D)**.** For a divisor $D = \sum n_p p$ define $\mathcal{O}(D)$ on open sets $U \subset X$ by

$$\mathcal{O}(D)(U) := \left\{ f \in \mathcal{M}(U) \setminus \{0\} \;\middle|\; \operatorname{ord}_p(f) \geq -n_p \text{ for all } p \in U \right\} \cup \{0\},$$

where $\mathcal{M}$ is the sheaf of meromorphic functions on X.

The condition $\operatorname{ord}_p(f) \geq -n_p$ means: f may have poles, but their orders are bounded by D.

Proposition 6.11 (Restriction and $\mathcal{O}(D)$ is a sheaf)**.** *$\mathcal{O}(D)$ is a sheaf of $\mathcal{O}$-modules, and $\mathcal{O}(D)$ is locally free of rank* 1*.*

Proof. Step 1: restriction is well-defined. If $f \in \mathcal{O}(D)(U)$ and $V \subset U$, then for each $p \in V$ we have $\operatorname{ord}_p(f|_V) = \operatorname{ord}_p(f) \geq -n_p$, so $f|_V \in \mathcal{O}(D)(V)$.

Step 2: locality and gluing. Let $U = \bigcup_\alpha U_\alpha$ be a cover.

Locality: if $f, g \in \mathcal{O}(D)(U)$ and $f|_{U_\alpha} = g|_{U_\alpha}$ for all α, then $f = g$ in $\mathcal{M}(U)$, hence in $\mathcal{O}(D)(U)$.

Gluing: suppose $f_\alpha \in \mathcal{O}(D)(U_\alpha)$ satisfy $f_\alpha = f_\beta$ in $\mathcal{M}(U_{\alpha\beta})$. Since $\mathcal{M}$ is a sheaf, there is a unique $f \in \mathcal{M}(U)$ with $f|_{U_\alpha} = f_\alpha$. Fix $p \in U$ and choose α with $p \in U_\alpha$. Then $\operatorname{ord}_p(f) = \operatorname{ord}_p(f_\alpha) \geq -n_p$ because order is local. Hence $f \in \mathcal{O}(D)(U)$. This proves the sheaf axioms.

Step 3: local freeness of rank 1. Fix $p \in X$. Choose a coordinate z on a neighborhood V of p with $z(p) = 0$. If D has coefficient n_p at p, define a meromorphic function s_p on V by

$$s_p := \begin{cases} z^{-n_p}, & n_p > 0, \\ 1, & n_p \leq 0. \end{cases}$$

Then for any $f \in \mathcal{O}(D)(V)$, the product $f \cdot z^{n_p}$ is holomorphic at p when $n_p > 0$, and trivially holomorphic when $n_p \leq 0$. Hence f can be written as $f = h \cdot s_p$ with $h \in \mathcal{O}(V)$. Conversely, any $h \cdot s_p$ satisfies the pole bound. Therefore $\mathcal{O}(D)|_V$ is generated by s_p as an $\mathcal{O}$-module. This shows local triviality with one generator, i.e. locally free of rank 1. □

3. $\mathcal{O}(D)$ as a line bundle via transition functions

Proposition 6.12 (Transition cocycle for $\mathcal{O}(D)$)**.** *Let $\{U_\alpha\}$ be an open cover such that for each α there exists a meromorphic function $g_\alpha \in \mathcal{M}^*(U_\alpha)$ with*

$$(g_\alpha) = D|_{U_\alpha} \quad \text{as divisors on } U_\alpha.$$

Then on overlaps $U_{\alpha\beta}$ the ratios

$$f_{\alpha\beta} := \frac{g_\alpha}{g_\beta} \in \mathcal{O}^*(U_{\alpha\beta})$$

define a holomorphic line bundle whose sheaf of holomorphic sections is isomorphic to $\mathcal{O}(D)$.

Proof. Step 1: $f_{\alpha\beta}$ is holomorphic and nowhere vanishing. On $U_{\alpha\beta}$,

$$(g_\alpha) - (g_\beta) = D|_{U_{\alpha\beta}} - D|_{U_{\alpha\beta}} = 0.$$

Thus the meromorphic function g_α/g_β has no zeros and no poles on $U_{\alpha\beta}$, hence belongs to $\mathcal{O}^*(U_{\alpha\beta})$.

Step 2: cocycle identity. On triple overlaps,

$$f_{\alpha\beta}f_{\beta\gamma}f_{\gamma\alpha} = \frac{g_\alpha}{g_\beta}\cdot\frac{g_\beta}{g_\gamma}\cdot\frac{g_\gamma}{g_\alpha} = 1.$$

Also $f_{\alpha\alpha} = 1$ and $f_{\alpha\beta}f_{\beta\alpha} = 1$ are immediate.

Step 3: identify $\mathcal{O}(D)$ with the section sheaf of the glued bundle. Let L be the line bundle defined by $\{f_{\alpha\beta}\}$ with local frames e_α satisfying $e_\alpha = f_{\alpha\beta}e_\beta$. Define an $\mathcal{O}$-module map on each U_α:

$$\Theta_\alpha : \mathcal{O}(U_\alpha) \to \mathcal{O}(D)(U_\alpha), \qquad h \mapsto h \cdot g_\alpha^{-1}.$$

Since $(g_\alpha) = D|_{U_\alpha}$, multiplication by g_α^{-1} shifts divisor bounds precisely so that $h \cdot g_\alpha^{-1}$ has poles bounded by D and hence lies in $\mathcal{O}(D)(U_\alpha)$.

On overlaps $U_{\alpha\beta}$,

$$\Theta_\alpha(h) = hg_\alpha^{-1} = h\left(\frac{g_\beta}{g_\alpha}\right)g_\beta^{-1} = hf_{\beta\alpha}\,g_\beta^{-1} = \Theta_\beta(hf_{\beta\alpha}).$$

This is exactly the compatibility rule for sections: if a section is represented by local holomorphic functions s_α in the frames e_α, the overlap condition is $s_\alpha = f_{\alpha\beta}s_\beta$, equivalently $s_\beta = f_{\beta\alpha}s_\alpha$. Thus $\{\Theta_\alpha\}$ glue to an isomorphism between the sheaf of holomorphic sections of L and $\mathcal{O}(D)$. □

6.3. Picard group and classification

1. Picard group and tensor product

Definition 6.13 (Picard group)**.** The *Picard group* of a complex manifold M is the set of isomorphism classes of holomorphic line bundles on M, with group operation given by tensor product:

$$\mathrm{Pic}(M) := \{\text{holomorphic line bundles on } M\}/\cong, \qquad [L]\cdot[L'] := [L \otimes L'].$$

The identity is the class of the trivial bundle, and the inverse of $[L]$ is $[L^\vee]$.

2. Classification by Čech cohomology

Theorem 6.14 (Line bundles and $H^1(M,\mathcal{O}^*)$)**.** *Let M be a complex manifold. There is a canonical group isomorphism*

$$\mathrm{Pic}(M) \;\cong\; H^1(M,\mathcal{O}^*).$$

Proof. Choose an open cover $\{U_\alpha\}$.

Step 1: from line bundles to cocycles. Given a line bundle L, pick local trivializations $\varphi_\alpha : \pi^{-1}(U_\alpha) \simeq U_\alpha \times \mathbb{C}$. On overlaps, the transition maps are

$$(x, \eta_\beta) \mapsto (x, f_{\alpha\beta}(x)\eta_\beta)$$

for some $f_{\alpha\beta} \in \mathcal{O}^*(U_{\alpha\beta})$ satisfying the cocycle condition. Thus L determines a class $[f] \in \check{H}^1(\{U_\alpha\}, \mathcal{O}^*)$.

If one chooses different trivializations, the transition functions change by $\tilde{f}_{\alpha\beta} = g_\alpha f_{\alpha\beta} g_\beta^{-1}$, i.e. by a coboundary. Hence the cohomology class is independent of choices and defines an element of $H^1(M, \mathcal{O}^*)$.

Step 2: from cocycles to line bundles. Conversely, given a cocycle $\{f_{\alpha\beta}\}$, Proposition 6.2 constructs a line bundle L_f.

If f and $\tilde{f}$ differ by a coboundary, Proposition 6.4 gives an isomorphism $L_f \simeq L_{\tilde{f}}$. Thus a cohomology class in $H^1(M, \mathcal{O}^*)$ determines an isomorphism class of line bundles.

Step 3: bijectivity. The two constructions are inverse to each other: starting from L, forming its cocycle f, and gluing back produces a bundle canonically isomorphic to L (because the glued model reproduces the original transition maps). Starting from a cocycle f, the cocycle extracted from L_f is f by construction. Hence we have a bijection $\mathrm{Pic}(M) \leftrightarrow H^1(M, \mathcal{O}^*)$.

Step 4: compatibility with group law. Under tensor product, transition functions multiply: $(L \otimes L')$ has cocycle $f_{\alpha\beta} f'_{\alpha\beta}$ (Proposition 6.5), which is exactly the addition law in $H^1(M, \mathcal{O}^*)$. Thus the bijection is a group isomorphism. □

6.4. Exercises

Exercise 6.15 (Checking the atlas in the gluing construction)**.** Fill in the details in Proposition 6.2: prove that the charts $\{\varphi_\alpha\}$ define a Hausdorff, second-countable complex manifold structure on L, assuming M is Hausdorff and second-countable.

Exercise 6.16 (Sections as glued functions)**.** Prove Proposition 6.3 from the quotient model $L = (\bigsqcup U_\alpha \times \mathbb{C})/\!\sim$, without appealing to abstract sheaf language.

Exercise 6.17 (Cocycle for tensor product and dual)**.** Let L, L' be line bundles with transition functions $f_{\alpha\beta}, f'_{\alpha\beta}$ on the same cover. Construct explicit local trivializations of $L \otimes L'$ and $L^\vee$ and derive the cocycles $f_{\alpha\beta} f'_{\alpha\beta}$ and $f_{\alpha\beta}^{-1}$.

Exercise 6.18 (Local model for $\mathcal{O}(D)$ near a point)**.** Let X be a Riemann surface, $p \in X$, and $D = n\,p$. Choose a coordinate z near p with $z(p) = 0$. Show that on a sufficiently small neighborhood U of p,

$$\mathcal{O}(D)(U) = z^{-n} \cdot \mathcal{O}(U) \quad (n > 0), \qquad \mathcal{O}(D)(U) = \mathcal{O}(U) \quad (n \le 0).$$

Exercise 6.19 ($\mathcal{O}(n)$ on $\mathbb{CP}^1$)**.** Define $\mathcal{O}(n)$ on $\mathbb{CP}^1$ by transition function $f_{01} = z^n$ on U_{01}. Compute $H^0(\mathbb{CP}^1, \mathcal{O}(n))$ by the gluing equation and show

$$\dim_{\mathbb{C}} H^0(\mathbb{CP}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \ge 0, \\ 0, & n < 0. \end{cases}$$

Exercise 6.20 (Principal divisors and isomorphic line bundles)**.** Let D be a divisor on a Riemann surface X and let $f \in \mathcal{M}^*(X)$. Prove that $\mathcal{O}(D) \cong \mathcal{O}(D + (f))$ as line bundles. (Hint: construct an explicit isomorphism by multiplication with f^{-1}.)

Exercise 6.21 (Picard group of $\mathbb{CP}^1$)**.** Using the previous exercise and $\mathcal{O}(n)$, show that $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ via $[\mathcal{O}(n)] \mapsto n$.

6.5. Divisors, meromorphic sections, and the bundle $\mathcal{O}(D)$

Throughout, M is a compact Riemann surface, $\mathcal{O}$ is the holomorphic sheaf, and $\mathcal{M}$ is the meromorphic sheaf.

1. Orders of zeros/poles: intrinsic definition with a computation

Fix $p \in M$ and a coordinate z with $z(p) = 0$. Any meromorphic h near p can be written uniquely as

$$h(z) = z^k u(z), \qquad u(0) \neq 0, \quad k \in \mathbb{Z}.$$

Define $\nu_p(h) = k$. This is coordinate-independent: if $\zeta = za(z)$ with $a(0) \neq 0$, then $\zeta^k = z^k a(z)^k$ and $a(z)^k$ is a holomorphic unit.

2. Divisors and principal divisors

Definition 6.22 (Divisor and degree)**.** A divisor is a finite formal sum $D = \sum_{p \in M} n_p\, p$ with $n_p \in \mathbb{Z}$. Its degree is $\deg(D) = \sum_p n_p$.

Definition 6.23 (Principal divisor)**.** For $h \in \mathcal{M}^*(M)$, define $\operatorname{div}(h) = \sum_{p \in M} \nu_p(h)\, p$.

On compact M, $\deg(\operatorname{div}(h)) = 0$; informally, a meromorphic function has equally many zeros and poles, counted with multiplicity.

3. Constructing $\mathcal{O}(D)$ and the canonical meromorphic section

Definition 6.24 ($\mathcal{O}(D)$ from Cartier data)**.** Represent D by local meromorphic functions $g_\alpha \in \mathcal{M}^*(U_\alpha)$ such that $g_\alpha / g_\beta \in \mathcal{O}^*(U_{\alpha\beta})$. Define $\mathcal{O}(D)$ by transition functions $f_{\alpha\beta} = g_\alpha / g_\beta$.

The canonical meromorphic section is $s_D|_{U_\alpha} = g_\alpha e_\alpha$ (where e_α is the local frame), and it satisfies $(s_D) = D$ by construction.

4. Worked examples

Example 6.25 (Worked: explicit cocycle for $\mathcal{O}(p)$ and the divisor of s_p)**.** Fix $p \in M$. Choose $U_0 = M \setminus \{p\}$ and a coordinate disk U_1 around p with holomorphic coordinate z and $z(p) = 0$. Set

$$g_0 \equiv 1 \text{ on } U_0, \qquad g_1 = z \text{ on } U_1.$$

Then on $U_{01} = U_0 \cap U_1$,

$$f_{01} = \frac{g_0}{g_1} = \frac{1}{z} \in \mathcal{O}^*(U_{01}), \qquad f_{10} = z.$$

Thus $\mathcal{O}(p)$ is obtained by gluing $U_0 \times \mathbb{C}$ and $U_1 \times \mathbb{C}$ via $\eta_0 = \frac{1}{z}\eta_1$.

The canonical meromorphic section is

$$s_p|_{U_0} = g_0 e_0 = e_0, \qquad s_p|_{U_1} = g_1 e_1 = z e_1.$$

Near p, $s_p = z e_1$, hence $\nu_p(s_p) = 1$ and there are no other zeros/poles. Therefore $(s_p) = p$ and $\deg \mathcal{O}(p) = 1$.

Example 6.26 (Worked: $\mathbb{CP}^1$ and $\operatorname{div}\left(\frac{z-a}{z-b}\right)$)**.** On $\mathbb{CP}^1$ with affine coordinate z on $U_0 \simeq \mathbb{C}$, consider

$$h(z) = \frac{z-a}{z-b}.$$

At $z = a$ we have a simple zero: $h(z) = (z-a)\cdot(\text{unit})$, so $\nu_a(h) = +1$. At $z = b$ we have a simple pole: $h(z) = (z-b)^{-1}\cdot(\text{unit})$, so $\nu_b(h) = -1$. No other finite points contribute.

To check ∞, use $w = 1/z$ near ∞:

$$h(1/w) = \frac{\frac{1}{w} - a}{\frac{1}{w} - b} = \frac{1-aw}{1-bw}.$$

This is holomorphic and nonzero at $w = 0$, so $\nu_\infty(h) = 0$. Hence

$$\operatorname{div}(h) = (a) - (b), \qquad \deg(\operatorname{div}(h)) = 1 - 1 = 0.$$

Consequently $\mathcal{O}\big((a)-(b)\big) \simeq \mathcal{O}_{\mathbb{CP}^1}$ (principal divisor gives a trivial bundle).

Example 6.27 (Worked: explicit cocycle proof that $\mathcal{O}(p)\otimes\mathcal{O}(-p) \simeq \mathcal{O}_M$)**.** Use the same cover U_0, U_1 and coordinate z as in the first example. For $\mathcal{O}(p)$ we have $f_{01} = 1/z$. For $\mathcal{O}(-p)$ we take Cartier data $g_0 = 1$, $g_1 = z^{-1}$, giving

$$f'_{01} = \frac{g_0}{g_1} = z.$$

Tensor product cocycle is the product:

$$(f\cdot f')_{01} = \frac{1}{z}\cdot z = 1,$$

hence $\mathcal{O}(p)\otimes\mathcal{O}(-p)$ has trivial transition function, so it is isomorphic to $\mathcal{O}_M$.

6.6. Picard group and the divisor correspondence: explicit computations

1. Picard group and cocycles

Definition 6.28 (Picard group)**.** $\operatorname{Pic}(M)$ is the group of isomorphism classes of holomorphic line bundles with operation $[L_1] + [L_2] = [L_1 \otimes L_2]$ and inverse $-[L] = [L^\vee]$.

A line bundle given by cocycle $\{f_{\alpha\beta}\}$ is unchanged (up to isomorphism) by gauge change $f_{\alpha\beta} \mapsto g_\alpha f_{\alpha\beta} g_\beta^{-1}$, hence $\operatorname{Pic}(M) \cong H^1(M, \mathcal{O}^*)$.

2. Divisors classify line bundles

On a compact Riemann surface,

$$\operatorname{Pic}(M) \;\cong\; \operatorname{Div}(M)/\operatorname{Prin}(M), \qquad D \longmapsto \mathcal{O}(D).$$

Concrete meaning: two divisors D_1, D_2 determine isomorphic line bundles iff $D_1 - D_2 = \operatorname{div}(h)$ for some meromorphic h.

3. Worked examples

Example 6.29 (Worked: any degree-0 divisor on $\mathbb{CP}^1$ is principal, with explicit h)**.** Let

$$D = \sum_{i=1}^{r} m_i\,(a_i) \;-\; \sum_{j=1}^{s} n_j\,(b_j)$$

be a divisor on $\mathbb{CP}^1$ supported in $\mathbb{C}$ (no ∞ for now), with $\deg(D) = \sum_i m_i - \sum_j n_j = 0$.

Define the rational function

$$h(z) = \frac{\prod_{i=1}^{r}(z-a_i)^{m_i}}{\prod_{j=1}^{s}(z-b_j)^{n_j}}.$$

Then h has a zero of order m_i at a_i and a pole of order n_j at b_j. Because the total degree in numerator equals that of denominator (degree 0 divisor), there is no zero/pole at ∞. Hence

$$\operatorname{div}(h) = D.$$

If ∞ is present, write $z = 1/w$ near ∞ and include a factor z^k (equivalently w^{-k}) to match the desired order at ∞.

Example 6.30 (Worked: $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ from the degree computation)**.** Take any divisor D on $\mathbb{CP}^1$. Let $d = \deg(D)$. Choose a point p and write

$$D \sim d \cdot p,$$

meaning $D - dp$ has degree 0, hence is principal by the previous example: $D - dp = \operatorname{div}(h)$ for some rational h. Therefore

$$\mathcal{O}(D) \;\simeq\; \mathcal{O}(dp) \;\simeq\; \mathcal{O}(d),$$

where the last identification comes from the standard fact that a degree-1 divisor defines $\mathcal{O}(1)$. Thus every line bundle is $\mathcal{O}(m)$ for a unique $m \in \mathbb{Z}$, i.e. $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ generated by $\mathcal{O}(1)$.

6.7. Global sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$: chart computation and divisors

1. Definition of $\mathcal{O}(m)$ and section compatibility

Cover $\mathbb{CP}^1$ by U_0, U_1 with coordinates z and $w = 1/z$. Define $\mathcal{O}(m)$ by transition $f_{01} = z^m$ and frames $e_1 = z^m e_0$. A section s is given by holomorphic f_0, f_1 with

$$s|_{U_0} = f_0(z)e_0, \qquad s|_{U_1} = f_1(w)e_1, \qquad f_0(z) = z^m f_1(1/z).$$

2. Worked examples (replacing exercises)

Example 6.31 (Worked: $H^0(\mathbb{CP}^1, \mathcal{O}(m))$ for $m \geq 0$ and the explicit basis)**.** Let $m \geq 0$. Expand $f_0(z) = \sum_{k\geq 0} a_k z^k$ on $\mathbb{C}$. Then

$$f_1(w) = w^m f_0(1/w) = \sum_{k\geq 0} a_k\, w^{m-k}.$$

Holomorphicity at $w = 0$ forces $a_k = 0$ for $k > m$. So

$$f_0(z) = a_0 + a_1 z + \cdots + a_m z^m,$$

and conversely any such polynomial produces a holomorphic f_1. Therefore $\dim H^0(\mathbb{CP}^1, \mathcal{O}(m)) = m + 1$ and a basis is given by the sections with

$$f_0(z) = z^k, \qquad 0 \le k \le m.$$

In homogeneous coordinates, these correspond to monomials $Z_0^{m-k} Z_1^k$.

Example 6.32 (Worked: $m < 0$ implies $H^0(\mathbb{CP}^1, \mathcal{O}(m)) = 0$)**.** Let $m < 0$ and write $m = -\ell$ with $\ell > 0$. Then

$$f_1(w) = w^m f_0(1/w) = w^{-\ell} f_0(1/w).$$

If f_0 is not identically zero, the function $f_0(1/w)$ has a nonzero Taylor expansion near $w = 0$, so $w^{-\ell} f_0(1/w)$ has a pole at $w = 0$, contradicting holomorphicity of f_1. Thus $f_0 \equiv 0$ and the only global holomorphic section is 0.

Example 6.33 (Worked: divisor of a section of $\mathcal{O}(m)$ including the vanishing at ∞)**.** Let $m \ge 0$ and take a nonzero section with

$$f_0(z) = c \prod_{i=1}^{r} (z - a_i)^{m_i}, \qquad \sum_{i=1}^{r} m_i = d \le m.$$

Zeros in $\mathbb{C}$ contribute $\sum_i m_i(a_i)$. Near ∞, use $w = 1/z$:

$$f_1(w) = w^m f_0(1/w) = c\, w^m \prod_{i=1}^{r} \Big(\frac{1}{w} - a_i\Big)^{m_i} = c\, w^{m-d} \prod_{i=1}^{r} (1 - a_i w)^{m_i}.$$

So f_1 has a zero of order $m - d$ at $w = 0$, i.e. the section vanishes at ∞ with order $m - d$. Hence

$$(s) = \sum_{i=1}^{r} m_i\,(a_i) \;+\; (m - d)\,(\infty), \qquad \deg(s) = m.$$

Example 6.34 (Worked: the section of $\mathcal{O}(2)$ corresponding to $Z_0 Z_1$ and its divisor)**.** Consider the homogeneous polynomial $P(Z_0, Z_1) = Z_0 Z_1$ of degree 2. On U_0 ($Z_0 \neq 0$) with $z = Z_1/Z_0$, we have

$$f_0(z) = P(1, z) = z,$$

so the section vanishes to order 1 at $z = 0$, i.e. at $[1 : 0]$. Since $\deg f_0 = 1 < 2$, the previous example shows it vanishes at $\infty = [0 : 1]$ to order $2 - 1 = 1$. Thus

$$(s) = [1 : 0] + [0 : 1].$$

6.8. Picard group operations and degree

1. Degree and its additivity/duality

For any line bundle L on compact M, choose a nonzero meromorphic section s and define

$$\deg(L) = \sum_{p \in M} \nu_p(s).$$

This is well-defined because changing s multiplies it by a meromorphic function, changing the divisor by a principal divisor of degree 0.

Proposition 6.35 (Degree identities)**.** *For line bundles L_1, L_2,*

$$\deg(L_1 \otimes L_2) = \deg(L_1) + \deg(L_2), \qquad \deg(L^\vee) = -\deg(L).$$

Worked local order computation. Pick meromorphic sections s_i and local frames so $s_i = f_i e_i$. Then $s_1 \otimes s_2 = (f_1 f_2)(e_1 \otimes e_2)$, hence

$$\nu_p(s_1 \otimes s_2) = \nu_p(f_1 f_2) = \nu_p(f_1) + \nu_p(f_2) = \nu_p(s_1) + \nu_p(s_2).$$

Summing over p gives additivity.

For the dual, define $s^\vee$ locally by $s^\vee(s) = 1$. If $s = fe$, then $s^\vee = f^{-1}e^\vee$, so $\nu_p(s^\vee) = -\nu_p(s)$ and $\deg(L^\vee) = -\deg(L)$. □

2. Worked examples

Example 6.36 (Worked: $\deg(\mathcal{O}(p-q))$ and why it is "degree zero" data)**.** Let $p, q \in M$. The divisor $p - q$ has degree $1 - 1 = 0$. Hence $\deg(\mathcal{O}(p-q)) = \deg(p-q) = 0$. Geometrically, $\mathcal{O}(p-q)$ is a degree-0 line bundle: it has meromorphic sections whose total number of zeros equals total number of poles. On $\mathbb{CP}^1$, this already forces triviality (because degree-0 divisors are principal), but on higher genus surfaces it produces nontrivial elements of $\mathrm{Pic}^0(M)$.

Example 6.37 (Worked: negative degree implies no nonzero holomorphic sections)**.** Assume $\deg(L) < 0$ and suppose $s \in H^0(M, L)$ is a nonzero holomorphic section. Holomorphicity implies s has no poles, so (s) is an *effective* divisor:

$$(s) = \sum_{p \in M} \nu_p(s)\, p \quad \text{with } \nu_p(s) \geq 0.$$

Therefore $\deg(L) = \deg((s)) \geq 0$, contradicting $\deg(L) < 0$. Hence $H^0(M, L) = 0$ when $\deg(L) < 0$.

Example 6.38 (Worked: $\deg(L_1 \otimes L_2^\vee) = \deg(L_1) - \deg(L_2)$)**.** Using the proposition,

$$\deg(L_1 \otimes L_2^\vee) = \deg(L_1) + \deg(L_2^\vee) = \deg(L_1) - \deg(L_2).$$

If you want to see the local mechanism: choose meromorphic sections s_1, s_2 and note $s_2^\vee$ has the opposite divisor of s_2, so

$$(s_1 \otimes s_2^\vee) = (s_1) - (s_2),$$

hence degrees subtract.

7. Sheaves: Definitions, Morphisms, and Locality

7.1. Presheaves and restriction maps

1. Presheaves as "data on open sets"

A large class of geometric objects is naturally organized by open sets: functions, differential forms, holomorphic sections, solutions of PDE, etc. The only structure needed to start is restriction:

$$V \subset U \quad \Longrightarrow \quad \text{data on } U \text{ restricts to data on } V.$$

Definition 7.1 (Presheaf of abelian groups)**.** Let M be a topological space. A *presheaf* $\mathscr{F}$ of abelian groups on M consists of:

1. for each open set $U \subset M$, an abelian group $\mathscr{F}(U)$;
2. for each inclusion $V \subset U$, a homomorphism (restriction)

$$\rho_V^U : \mathscr{F}(U) \to \mathscr{F}(V);$$

such that $\rho_U^U = \mathrm{id}$ and $\rho_W^U = \rho_W^V \circ \rho_V^U$ for $W \subset V \subset U$.

2. Examples with explicit verification and computations

Example 7.2 (Continuous real-valued functions on $\mathbb{R}$)**.** Let $M = \mathbb{R}$. Define $\mathscr{C}^0(U) = C^0(U, \mathbb{R})$ with $\rho_V^U(f) = f|_V$. Then for $W \subset V \subset U$ and $f \in C^0(U)$:

$$(\rho_W^V \circ \rho_V^U)(f) = \rho_W^V(f|_V) = (f|_V)|_W = f|_W = \rho_W^U(f),$$

and $\rho_U^U(f) = f|_U = f$. Hence $\mathscr{C}^0$ is a presheaf.

Example 7.3 (Smooth functions on $\mathbb{R}^n$ and a coordinate computation)**.** Let $M = \mathbb{R}^n$ and $\mathscr{C}^\infty(U) = C^\infty(U, \mathbb{R})$. If $f(x_1, \dots, x_n) = x_1^2 + \sin(x_2)$ on U and $V \subset U$, then

$$\rho_V^U(f) = f|_V, \qquad \partial_{x_1}(f|_V) = (\partial_{x_1} f)|_V = 2x_1|_V,$$

so differentiation commutes with restriction. This locality is the reason C^∞-objects form sheaves later.

Example 7.4 (Differential forms and an explicit restriction)**.** Let $M = \mathbb{R}^2$ with coordinates (x, y). Define $\Omega^1(U)$ to be smooth 1-forms on U. Take

$$\omega = (x^2 + y)\, dx + (\sin x)\, dy \in \Omega^1(U).$$

If $V \subset U$, then

$$\omega|_V = (x^2 + y)|_V\, dx + (\sin x)|_V\, dy.$$

If $W \subset V \subset U$, then $(\omega|_V)|_W = \omega|_W$, verifying functoriality of restriction.

Example 7.5 (Solutions of a PDE as a presheaf: harmonic functions)**.** Let $M = \mathbb{R}^2$. Define

$$\mathscr{H}(U) := \{u \in C^2(U) \mid \Delta u = 0\}.$$

If $u \in \mathscr{H}(U)$ and $V \subset U$, then $u|_V \in C^2(V)$ and

$$\Delta(u|_V) = (\Delta u)|_V = 0,$$

so restriction preserves harmonicity. Thus $\mathscr{H}$ is a presheaf of abelian groups (under addition). Concrete computation: $u(x, y) = \Re(z^3) = x^3 - 3xy^2$ satisfies

$$u_{xx} = 6x, \qquad u_{yy} = -6x, \qquad \Delta u = u_{xx} + u_{yy} = 0.$$

7.2. Sheaves: locality and gluing

1. The locality and gluing axioms

Definition 7.6 (Sheaf of abelian groups)**.** A presheaf $\mathscr{F}$ on M is a *sheaf* if for every open $U \subset M$ and cover $U = \bigcup_\alpha U_\alpha$:

1. **Locality:** if $s, t \in \mathscr{F}(U)$ satisfy $s|_{U_\alpha} = t|_{U_\alpha}$ for all α, then $s = t$.
2. **Gluing:** if $s_\alpha \in \mathscr{F}(U_\alpha)$ satisfy $s_\alpha|_{U_{\alpha\beta}} = s_\beta|_{U_{\alpha\beta}}$ for all α, β, then there exists $s \in \mathscr{F}(U)$ with $s|_{U_\alpha} = s_\alpha$ for all α.

2. Gluing continuous functions: a worked piecewise example

Example 7.7 (Gluing a continuous function on $\mathbb{R}$ from two opens)**.** Let $U = (-2, 2)$, $U_1 = (-2, 1)$, $U_2 = (-1, 2)$ so $U = U_1 \cup U_2$ and $U_{12} = (-1, 1)$. Define

$$f_1(x) = x^2 \quad (x \in U_1), \qquad f_2(x) = x^2 \quad (x \in U_2).$$

Then $f_1|_{U_{12}} = f_2|_{U_{12}}$, so they glue to $f(x) = x^2$ on all of U.

Now modify to a genuinely "local" construction:

$$g_1(x) = \sin x \quad (x \in U_1), \qquad g_2(x) = \begin{cases} \sin x, & x \in (-1, 1), \\ \sin x + (x-1)^3, & x \in (1, 2), \end{cases}$$

where g_2 is continuous on U_2. On the overlap $(-1, 1)$ we have $g_2 = \sin x = g_1$, hence compatibility holds. The glued function g is:

$$g(x) = \begin{cases} \sin x, & x \in (-2, 1), \\ \sin x + (x-1)^3, & x \in (1, 2), \end{cases}$$

and g is continuous on U because the definitions agree on the overlap and continuity is local.

3. Gluing holomorphic functions: an explicit complex-analytic example

Example 7.8 (Gluing holomorphic functions on $\mathbb{C}^\times$)**.** Let $U = \mathbb{C}^\times$ and cover it by

$$U_1 = \mathbb{C} \setminus (-\infty, 0], \qquad U_2 = \mathbb{C} \setminus [0, \infty).$$

On U_1 and U_2 we have holomorphic branches of the logarithm:

$$\mathrm{Log}_1 : U_1 \to \mathbb{C}, \qquad \mathrm{Log}_2 : U_2 \to \mathbb{C}.$$

On the overlap $U_{12} = U_1 \cap U_2 = \mathbb{C}^\times \setminus \mathbb{R}$ the two branches differ by a constant:

$$\mathrm{Log}_2(z) = \mathrm{Log}_1(z) + 2\pi i \quad \text{on the upper half-plane part of } U_{12},$$

and $\mathrm{Log}_2(z) = \mathrm{Log}_1(z) - 2\pi i$ on the lower half-plane part, depending on conventions. Hence $\{\mathrm{Log}_1, \mathrm{Log}_2\}$ do *not* satisfy the compatibility condition on all of U_{12}, so they do not glue to a holomorphic function on $\mathbb{C}^\times$. This is a concrete place where the topology of the domain obstructs global gluing.

4. A presheaf that fails gluing: bounded continuous functions

Example 7.9 (Bounded continuous functions)**.** Let $M = \mathbb{R}$ and define $\mathscr{B}(U)$ to be bounded continuous functions $U \to \mathbb{R}$. Restriction preserves boundedness, so $\mathscr{B}$ is a presheaf. Cover $\mathbb{R}$ by $U_n = (-n, n)$ and set $f_n(x) = x$ on U_n. Then $f_n = f_m$ on $U_n \cap U_m$, but the glued function would be $f(x) = x$ on $\mathbb{R}$, which is not bounded. So gluing fails and $\mathscr{B}$ is not a sheaf.

7.3. Morphisms of sheaves and exact sequences

1. Morphisms and concrete examples

Definition 7.10 (Morphism)**.** A morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of presheaves is a family of maps $\varphi_U : \mathscr{F}(U) \to \mathscr{G}(U)$ commuting with restrictions: $\rho_V^U \circ \varphi_U = \varphi_V \circ \rho_V^U$ for $V \subset U$.

Example 7.11 (Derivative as a morphism of sheaves on $\mathbb{R}$)**.** Let $\mathscr{C}^\infty$ be the sheaf of smooth functions on $\mathbb{R}$ and define

$$\varphi_U : C^\infty(U) \to C^\infty(U), \qquad \varphi_U(f) = f'.$$

Then for $V \subset U$,

$$(\varphi_U(f))|_V = (f')|_V = (f|_V)' = \varphi_V(f|_V),$$

so φ is a morphism of sheaves. Concrete computation: if $f(x) = x^3 + \cos x$, then $\varphi(f) = 3x^2 - \sin x$.

Example 7.12 (Exponential map as a morphism $\mathcal{O} \to \mathcal{O}^\times$)**.** On a complex manifold M, define on each open U:

$$\exp_U : \mathcal{O}(U) \to \mathcal{O}^\times(U), \qquad \exp_U(f) = e^f.$$

Restriction compatibility is $(e^f)|_V = e^{f|_V}$.

2. Exact sequences and a computed kernel

Definition 7.13 (Exactness at a term)**.** A sequence $\mathscr{F} \xrightarrow{\varphi} \mathscr{G} \xrightarrow{\psi} \mathscr{H}$ is exact at $\mathscr{G}$ if $\ker \psi = \operatorname{im} \varphi$ as subsheaves of $\mathscr{G}$.

Example 7.14 (Kernel of exponential on a simply connected domain)**.** Let $\Omega \subset \mathbb{C}$ be simply connected and consider

$$\exp : \mathcal{O}(\Omega) \to \mathcal{O}^\times(\Omega), \qquad f \mapsto e^{2\pi i f}.$$

If $e^{2\pi i f} \equiv 1$, then for each $z \in \Omega$ we have $2\pi i f(z) \in 2\pi i\mathbb{Z}$ so $f(z) \in \mathbb{Z}$. Since f is holomorphic and $\mathbb{Z}$ is discrete, f is locally constant, hence constant on connected Ω, so $f \in \mathbb{Z}$. Thus

$$\ker(\exp) = \mathbb{Z} \subset \mathcal{O}(\Omega)$$

(viewing $\mathbb{Z}$ as constant holomorphic functions).

7.4. Stalks, germs, and local criteria

1. Germs as equivalence classes with computations

Definition 7.15 (Germ and stalk)**.** Let $\mathscr{F}$ be a presheaf on M and $x \in M$. Pairs (U, s) with $x \in U$ and $s \in \mathscr{F}(U)$ define a germ at x. We set $(U, s) \sim (V, t)$ if there exists $W \subset U \cap V$ with $x \in W$ and $s|_W = t|_W$. The set of equivalence classes is the stalk $\mathscr{F}_x$, and the class of (U, s) is written $[s]_x$.

Example 7.16 (Germs of continuous functions: two different representatives)**.** Let $M = \mathbb{R}$, $x = 0$, and consider

$$f(x) = x^2 \text{ on } (-1, 1), \qquad g(x) = x^2 \text{ on } (-2, 2).$$

Then $[f]_0 = [g]_0$ because on $(-1, 1) \subset (-1, 1) \cap (-2, 2)$ they coincide. But if $h(x) = |x|$ on $(-1, 1)$, then $[h]_0 \neq [f]_0$ since they disagree on every neighborhood of 0.

Example 7.17 (Stalk of $\underline{\mathbb{Z}}$ at a point)**.** Let $\underline{\mathbb{Z}}$ be the locally constant sheaf: $\underline{\mathbb{Z}}(U) = \{n : U \to \mathbb{Z} \text{ locally constant}\}$. Fix $x \in M$. Any germ is represented by a locally constant function on some neighborhood, hence constant on a sufficiently small connected neighborhood of x. Therefore each germ is determined by an integer, and

$$(\underline{\mathbb{Z}})_x \cong \mathbb{Z}.$$

2. Stalkwise surjectivity does not imply surjectivity on global sections

Example 7.18 (A computed linear system on $\mathbb{CP}^1$)**.** Let $X = \mathbb{CP}^1$ and consider the sheaf morphism

$$\Phi : \mathcal{O}_X^{\oplus 2} \to \mathcal{O}_X(2), \qquad (a, b) \mapsto a\, Z_0^2 + b\, Z_1^2.$$

On the affine chart $U_0 = \{Z_0 \neq 0\}$ with $z = Z_1/Z_0$, a local section of $\mathcal{O}(2)$ looks like $f(z) \cdot Z_0^2$. Also $Z_1^2 = z^2 Z_0^2$. Hence locally

$$\Phi(a, b) = \big(a(z) + b(z)z^2\big)\, Z_0^2.$$

Given any germ $f(z)Z_0^2$ at a point of U_0, choose $a = f$ and $b = 0$ to hit it; similarly on U_1. Thus Φ_x is surjective for all x.

But on global sections,

$$H^0(X, \mathcal{O}^{\oplus 2}) \cong \mathbb{C}^2, \qquad H^0(X, \mathcal{O}(2)) \cong \mathbb{C}^3$$

with basis $\{Z_0^2, Z_0Z_1, Z_1^2\}$ for $H^0(\mathcal{O}(2))$. The image of $H^0(\Phi)$ is $\operatorname{span}\{Z_0^2, Z_1^2\}$, which misses Z_0Z_1. So stalkwise surjectivity does not force surjectivity on global sections.

7.5. Examples and non-examples

1. Constant presheaf vs constant sheaf: an explicit failure

Example 7.19 (Constant presheaf fails on disconnected opens)**.** Let $M = U_1 \sqcup U_2$ be a disjoint union of two nonempty open sets. Fix $A = \mathbb{Z}$ and define the constant presheaf $\underline{\mathbb{Z}}_{\text{pre}}(U) = \mathbb{Z}$ for all nonempty open U with identity restrictions. Take the cover $M = U_1 \cup U_2$. Choose sections $s_1 = 0 \in \mathbb{Z}$ on U_1 and $s_2 = 1 \in \mathbb{Z}$ on U_2. On the overlap $U_1 \cap U_2 = \varnothing$ the compatibility condition is vacuous, so these data should glue, but a section on M in the presheaf must be a *single* integer. There is no integer restricting to 0 on U_1 and 1 on U_2. Hence gluing fails.

2. Skyscraper sheaf stalk calculation with representatives

Example 7.20 (Skyscraper stalks)**.** Fix $p \in M$ and A an abelian group. Define $A_p(U) = A$ if $p \in U$ and 0 otherwise. If $x \neq p$, choose $U \ni x$ with $p \notin U$; then every section over neighborhoods of x is 0, so $(A_p)_x = 0$. If $x = p$, any neighborhood $U \ni p$ has $A_p(U) = A$ and all restrictions are identities, so germs at p are determined by a unique element of A:

$$(A_p)_p \cong A.$$

7.6. Exercises

Exercise 7.21 (Gluing smooth functions with a chart computation)**.** Let M be a smooth manifold and $U = \bigcup_\alpha U_\alpha$. Suppose $f_\alpha \in C^\infty(U_\alpha)$ and $f_\alpha = f_\beta$ on $U_{\alpha\beta}$. Define f pointwise by $f(x) = f_\alpha(x)$ for $x \in U_\alpha$. Prove $f \in C^\infty(U)$ by fixing a chart (V, ϕ) and checking that $f \circ \phi^{-1}$ agrees on $\phi(V \cap U_\alpha)$ with smooth functions, hence is smooth.

Exercise 7.22 (Locality of the Laplacian)**.** Let $U \subset \mathbb{R}^n$ be open and $u, v \in C^2(U)$. Assume that for every $x \in U$ there is a neighborhood V_x with $u|_{V_x} = v|_{V_x}$. Show $\Delta u = \Delta v$ on U.

Exercise 7.23 (A concrete stalk in $\mathscr{C}^\infty$)**.** Let $M = \mathbb{R}$ and $x = 0$. Show that the germ of $f(x) = \sin x$ at 0 equals the germ of its Taylor polynomial $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ truncated at order N if and only if $N = \infty$. In other words, explain why equality of germs is stricter than equality of Taylor jets.

Exercise 7.24 (Patching obstruction for logarithms)**.** On $U = \mathbb{C}^\times$, consider the cover $U_1 = \mathbb{C} \setminus (-\infty, 0]$, $U_2 = \mathbb{C} \setminus [0, \infty)$. Write down explicit formulas for branches $\mathrm{Log}_1, \mathrm{Log}_2$ on U_1, U_2. Compute $\mathrm{Log}_1 - \mathrm{Log}_2$ on the two connected components of $U_1 \cap U_2$ and show it is constant but takes different constants on the two components. Conclude that $\{\mathrm{Log}_1, \mathrm{Log}_2\}$ cannot glue to a holomorphic function on $\mathbb{C}^\times$.

Exercise 7.25 (Exactness on simply connected domains)**.** Let $\Omega \subset \mathbb{C}$ be simply connected. Prove that the sequence of abelian groups

$$0 \to \mathbb{Z} \to \mathcal{O}(\Omega) \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}^\times(\Omega) \to 0$$

is exact by showing surjectivity: given $h \in \mathcal{O}^\times(\Omega)$, define $g = h'/h$ and show $\int g$ is path-independent in Ω.

8. Čech Cohomology of a Sheaf

8.1. Open covers, cochains, and the coboundary operator

1. Setup and notation

Let M be a topological space and $\mathscr{F}$ a sheaf of abelian groups on M. Fix an open cover

$$\mathcal{U} = \{U_\alpha\}_{\alpha \in A}, \qquad M = \bigcup_{\alpha \in A} U_\alpha.$$

For indices $\alpha_0, \ldots, \alpha_q$ set

$$U_{\alpha_0 \cdots \alpha_q} := \bigcap_{j=0}^{q} U_{\alpha_j}.$$

2. Cochains and coboundary

Definition 8.1 (Čech cochains)**.** For $q \geq 0$, define

$$C^q(\mathcal{U}, \mathscr{F}) := \prod_{\alpha_0,\ldots,\alpha_q} \mathscr{F}(U_{\alpha_0\cdots\alpha_q}).$$

An element $c \in C^q$ is written $c = \{c_{\alpha_0\cdots\alpha_q}\}$ with $c_{\alpha_0\cdots\alpha_q} \in \mathscr{F}(U_{\alpha_0\cdots\alpha_q})$.

Definition 8.2 (Coboundary operator)**.** Define $\delta : C^q(\mathcal{U}, \mathscr{F}) \to C^{q+1}(\mathcal{U}, \mathscr{F})$ by

$$(\delta c)_{\alpha_0\cdots\alpha_{q+1}} = \sum_{j=0}^{q+1} (-1)^j \, c_{\alpha_0\cdots\widehat{\alpha_j}\cdots\alpha_{q+1}}\Big|_{U_{\alpha_0\cdots\alpha_{q+1}}}.$$

3. The identity $\delta^2 = 0$

Proposition 8.3. *For all $q \geq 0$, one has $\delta \circ \delta = 0$.*

Proof. Fix $c \in C^q(\mathcal{U}, \mathscr{F})$ and indices $\alpha_0, \ldots, \alpha_{q+2}$. Write $W := U_{\alpha_0\cdots\alpha_{q+2}}$. By definition,

$$(\delta^2 c)_{\alpha_0\cdots\alpha_{q+2}} = \sum_{i=0}^{q+2} (-1)^i \, (\delta c)_{\alpha_0\cdots\widehat{\alpha_i}\cdots\alpha_{q+2}}\Big|_W.$$

Expand the inner term:

$$(\delta c)_{\alpha_0\cdots\widehat{\alpha_i}\cdots\alpha_{q+2}} = \sum_{j=0}^{q+1} (-1)^j \, c_{\alpha_0\cdots\widehat{\alpha_i}\cdots\widehat{\alpha_{k(i,j)}}\cdots\alpha_{q+2}}\Big|_{U_{\alpha_0\cdots\widehat{\alpha_i}\cdots\alpha_{q+2}}},$$

where $k(i,j)$ denotes the index in $\{0,\ldots,q+2\} \setminus \{i\}$ occupying the j-th position. Restricting to W and re-indexing, $\delta^2 c$ becomes a signed sum over all pairs $0 \leq i < j \leq q+2$ of the same section

$$c_{\alpha_0\cdots\widehat{\alpha_i}\cdots\widehat{\alpha_j}\cdots\alpha_{q+2}}\Big|_W.$$

Each pair (i,j) appears exactly twice: once by first removing α_i then α_j, and once by first removing α_j then α_i. The corresponding signs are opposite:

$$(-1)^{i+j} \quad \text{and} \quad -(-1)^{i+j}.$$

Hence these two contributions cancel. Since this holds for every (i,j), the total sum is 0 on W. Therefore $(\delta^2 c)_{\alpha_0\cdots\alpha_{q+2}} = 0$ for all indices, i.e. $\delta^2 = 0$. □

4. Cohomology groups

Define

$$Z^q(\mathcal{U}, \mathscr{F}) := \ker\big(\delta : C^q \to C^{q+1}\big), \qquad B^q(\mathcal{U}, \mathscr{F}) := \operatorname{im}\big(\delta : C^{q-1} \to C^q\big),$$

and

$$\check{H}^q(\mathcal{U}, \mathscr{F}) := Z^q(\mathcal{U}, \mathscr{F}) / B^q(\mathcal{U}, \mathscr{F}).$$

8.2. Refinements and the direct limit construction

1. Refinements and pullback of cochains

Definition 8.4 (Refinement)**.** A cover $\mathcal{V} = \{V_\beta\}_{\beta\in B}$ is a refinement of $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ if there exists a map $r : B \to A$ such that $V_\beta \subset U_{r(\beta)}$ for all β.

Definition 8.5 (Induced map on cochains)**.** Given a refinement map $r : B \to A$, define

$$r^\sharp : C^q(\mathcal{U}, \mathscr{F}) \longrightarrow C^q(\mathcal{V}, \mathscr{F})$$

by

$$(r^\sharp c)_{\beta_0\cdots\beta_q} = c_{r(\beta_0)\cdots r(\beta_q)}\Big|_{V_{\beta_0\cdots\beta_q}}.$$

Proposition 8.6. *One has $\delta \circ r^\sharp = r^\sharp \circ \delta$. In particular, $r^\sharp$ induces homomorphisms*

$$r^\sharp : \check{H}^q(\mathcal{U}, \mathscr{F}) \to \check{H}^q(\mathcal{V}, \mathscr{F}).$$

Proof. Fix $c \in C^q(\mathcal{U}, \mathscr{F})$ and indices $\beta_0, \ldots, \beta_{q+1}$. Let $W := V_{\beta_0\cdots\beta_{q+1}}$. Compute:

$$(\delta(r^\sharp c))_{\beta_0\cdots\beta_{q+1}} = \sum_{j=0}^{q+1}(-1)^j\,(r^\sharp c)_{\beta_0\cdots\widehat{\beta_j}\cdots\beta_{q+1}}\Big|_W.$$

By definition of $r^\sharp$,

$$(r^\sharp c)_{\beta_0\cdots\widehat{\beta_j}\cdots\beta_{q+1}} = c_{r(\beta_0)\cdots\widehat{r(\beta_j)}\cdots r(\beta_{q+1})}\Big|_{V_{\beta_0\cdots\widehat{\beta_j}\cdots\beta_{q+1}}}.$$

Restricting to W and using transitivity of restriction maps in a sheaf,

$$(\delta(r^\sharp c))_{\beta_0\cdots\beta_{q+1}} = \sum_{j=0}^{q+1}(-1)^j\,c_{r(\beta_0)\cdots\widehat{r(\beta_j)}\cdots r(\beta_{q+1})}\Big|_W.$$

On the other hand,

$$(r^\sharp(\delta c))_{\beta_0\cdots\beta_{q+1}} = (\delta c)_{r(\beta_0)\cdots r(\beta_{q+1})}\Big|_W = \sum_{j=0}^{q+1}(-1)^j\,c_{r(\beta_0)\cdots\widehat{r(\beta_j)}\cdots r(\beta_{q+1})}\Big|_W.$$

These coincide, hence $\delta \circ r^\sharp = r^\sharp \circ \delta$, and the induced maps on cohomology follow. □

2. Direct limit definition

Definition 8.7 (Global Čech cohomology)**.** Define

$$\check{H}^q(M, \mathscr{F}) := \varinjlim_{\mathcal{U}} \check{H}^q(\mathcal{U}, \mathscr{F}),$$

where the directed system is over covers ordered by refinement, using the maps $r^\sharp$.

8.3. Degree zero and degree one: gluing and patching

1. Degree zero equals global sections

Proposition 8.8. *For any open cover $\mathcal{U}$, the group $\check{H}^0(\mathcal{U}, \mathscr{F})$ is canonically isomorphic to $\mathscr{F}(M)$. Consequently $\check{H}^0(M, \mathscr{F}) \cong \mathscr{F}(M)$.*

Proof. A 0–cochain is a family $s = \{s_\alpha\}$ with $s_\alpha \in \mathscr{F}(U_\alpha)$. The condition $\delta s = 0$ means: for all α, β,

$$(\delta s)_{\alpha\beta} = s_\beta|_{U_{\alpha\beta}} - s_\alpha|_{U_{\alpha\beta}} = 0,$$

equivalently s_α and s_β agree on overlaps. Thus $Z^0(\mathcal{U}, \mathscr{F})$ is exactly the set of compatible families $\{s_\alpha\}$. Since $B^0 = \mathrm{im}(\delta : C^{-1} \to C^0) = 0$ by definition, one has

$$\check{H}^0(\mathcal{U}, \mathscr{F}) = Z^0(\mathcal{U}, \mathscr{F}).$$

By the sheaf gluing axiom, any compatible family $\{s_\alpha\}$ arises from a unique global section $s \in \mathscr{F}(M)$ with $s|_{U_\alpha} = s_\alpha$. This defines a bijection

$$\mathscr{F}(M) \longrightarrow Z^0(\mathcal{U}, \mathscr{F}), \qquad s \mapsto \{s|_{U_\alpha}\}.$$

It is a group isomorphism because restrictions preserve addition. Hence $\check{H}^0(\mathcal{U}, \mathscr{F}) \cong \mathscr{F}(M)$. Passing to the direct limit over refinements preserves degree 0, so $\check{H}^0(M, \mathscr{F}) \cong \mathscr{F}(M)$ as well. □

2. Degree one: cocycle and coboundary formulas

A 1–cochain is $g = \{g_{\alpha\beta}\}$ with $g_{\alpha\beta} \in \mathscr{F}(U_{\alpha\beta})$. Then

$$(\delta g)_{\alpha\beta\gamma} = g_{\beta\gamma}|_{U_{\alpha\beta\gamma}} - g_{\alpha\gamma}|_{U_{\alpha\beta\gamma}} + g_{\alpha\beta}|_{U_{\alpha\beta\gamma}}.$$

Thus g is a 1–cocycle iff on every triple overlap,

$$g_{\alpha\gamma} = g_{\alpha\beta} + g_{\beta\gamma}.$$

A 1–cocycle is a 1–coboundary iff there exist $f = \{f_\alpha\} \in C^0$ with

$$g_{\alpha\beta} = f_\beta|_{U_{\alpha\beta}} - f_\alpha|_{U_{\alpha\beta}}.$$

8.4. Examples and basic computations

1. A computed example: $\check{H}^1(S^1, \underline{\mathbb{Z}}) \cong \mathbb{Z}$

Let $\underline{\mathbb{Z}}$ be the constant sheaf on S^1. Choose an open cover by two arcs $\mathcal{U} = \{U, V\}$ such that $U \cap V$ has exactly two connected components, say

$$U \cap V = W_1 \sqcup W_2, \qquad W_1, W_2 \text{ connected, open in } S^1.$$

Then

$$C^0(\mathcal{U}, \underline{\mathbb{Z}}) = \underline{\mathbb{Z}}(U) \times \underline{\mathbb{Z}}(V) \cong \mathbb{Z} \times \mathbb{Z}$$

because U, V are connected, and

$$C^1(\mathcal{U}, \underline{\mathbb{Z}}) = \underline{\mathbb{Z}}(U \cap V) \cong \mathbb{Z} \times \mathbb{Z}$$

because locally constant integer functions on $W_1 \sqcup W_2$ are pairs of integers.

Since there is no triple intersection, $\delta : C^1 \to C^2$ is the zero map, hence

$$Z^1(\mathcal{U}, \underline{\mathbb{Z}}) = C^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z} \times \mathbb{Z}.$$

Now compute coboundaries. For $(a, b) \in C^0 \cong \mathbb{Z} \times \mathbb{Z}$, one has

$$\delta(a, b) = b|_{U \cap V} - a|_{U \cap V}.$$

On each connected component W_i, the restrictions are constant, so

$$\delta(a, b) = (b - a,\ b - a) \in \mathbb{Z} \times \mathbb{Z}.$$

Therefore

$$B^1(\mathcal{U}, \underline{\mathbb{Z}}) = \{(n, n) \mid n \in \mathbb{Z}\} \subset \mathbb{Z} \times \mathbb{Z},$$

and hence

$$\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) = (\mathbb{Z} \times \mathbb{Z})/\{(n, n)\} \cong \mathbb{Z},$$

via the homomorphism

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad (x_1, x_2) \mapsto x_1 - x_2,$$

whose kernel is exactly the diagonal subgroup $\{(n, n)\}$. One concludes $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z}$; the direct limit map shows

$$\check{H}^1(S^1, \underline{\mathbb{Z}}) \cong \mathbb{Z}.$$

Example 8.9 (A concrete 1–cocycle representative)**.** Define $g \in C^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z} \times \mathbb{Z}$ by $g = (1, 0)$, i.e. $g|_{W_1} \equiv 1$ and $g|_{W_2} \equiv 0$. Then g is a cocycle (no triple overlaps). Its class is nonzero because every coboundary is of the form (n, n), and $(1, 0) \notin \{(n, n)\}$. Under $(x_1, x_2) \mapsto x_1 - x_2$, the class maps to $1 \in \mathbb{Z}$.

2. A vanishing computation: $\check{H}^1(S^2, \underline{\mathbb{Z}}) = 0$

Example 8.10 (Two-set cover on S^2)**.** Cover S^2 by $U = S^2 \setminus \{\text{south pole}\}$ and $V = S^2 \setminus \{\text{north pole}\}$. Then U and V are connected, and $U \cap V \cong S^1 \times (0, 1)$ is connected. Hence

$$C^0(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z} \times \mathbb{Z}, \qquad C^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z}.$$

As before $Z^1 = C^1 \cong \mathbb{Z}$ and $\delta(a, b) = b - a \in \mathbb{Z}$, so $B^1 = \mathbb{Z}$. Thus $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) = 0$, and consequently $\check{H}^1(S^2, \underline{\mathbb{Z}}) = 0$.

3. Line bundles via $\check{H}^1(M, \mathcal{O}^\times)$

Let M be a complex manifold. Fix a cover $\mathcal{U} = \{U_\alpha\}$.

Proposition 8.11. *There is a canonical bijection between:*

- *isomorphism classes of holomorphic line bundles on M trivialized over $\mathcal{U}$,*
- *the group $\check{H}^1(\mathcal{U}, \mathcal{O}_M^\times)$.*

Under refinement, these bijections are compatible, yielding a canonical bijection between isomorphism classes of holomorphic line bundles on M and $\check{H}^1(M, \mathcal{O}_M^\times)$.

Proof. Step 1 (from a line bundle to a cocycle). Let $L \to M$ be a holomorphic line bundle and choose holomorphic trivializations

$$\varphi_\alpha : L|_{U_\alpha} \xrightarrow{\sim} U_\alpha \times \mathbb{C}.$$

On overlaps $U_{\alpha\beta}$, the transition is a fiberwise $\mathbb{C}$–linear holomorphic automorphism of $\mathbb{C}$, hence multiplication by a unique $f_{\alpha\beta} \in \mathcal{O}^\times(U_{\alpha\beta})$:

$$\varphi_\alpha \circ \varphi_\beta^{-1}(x, \eta) = (x, f_{\alpha\beta}(x)\eta).$$

On triple overlaps $U_{\alpha\beta\gamma}$, associativity of composition gives

$$f_{\alpha\beta} f_{\beta\gamma} = f_{\alpha\gamma},$$

so $\{f_{\alpha\beta}\}$ is a 1–cocycle in $Z^1(\mathcal{U}, \mathcal{O}^\times)$. Thus L determines a class $[f] \in \check{H}^1(\mathcal{U}, \mathcal{O}^\times)$.

Step 2 (effect of changing trivializations). Replace φ_α by $\widetilde{\varphi}_\alpha$ where

$$\widetilde{\varphi}_\alpha = (x, \eta) \mapsto (x, g_\alpha(x)\eta) \circ \varphi_\alpha \quad \text{for some } g_\alpha \in \mathcal{O}^\times(U_\alpha).$$

Then on $U_{\alpha\beta}$,

$$\widetilde{f}_{\alpha\beta} = g_\alpha f_{\alpha\beta} g_\beta^{-1}.$$

Equivalently, in multiplicative notation, $\widetilde{f} = f \cdot \delta(g)^{-1}$, so f and $\widetilde{f}$ define the same element of $\check{H}^1(\mathcal{U}, \mathcal{O}^\times)$. Hence the class $[f]$ depends only on L up to isomorphism.

Step 3 (from a cocycle to a line bundle). Given $f = \{f_{\alpha\beta}\} \in Z^1(\mathcal{U}, \mathcal{O}^\times)$, form the disjoint union $\bigsqcup_\alpha (U_\alpha \times \mathbb{C})$ and define an equivalence relation by

$$(x, \eta)_\beta \sim (x, f_{\alpha\beta}(x)\eta)_\alpha \qquad (x \in U_{\alpha\beta}).$$

The cocycle identity ensures transitivity on triple overlaps, so the quotient

$$L_f := \Big(\bigsqcup_\alpha (U_\alpha \times \mathbb{C}) \Big) / \sim$$

is a complex manifold, and the natural projection $L_f \to M$ is a holomorphic line bundle trivialized over $\mathcal{U}$ with transition functions $f_{\alpha\beta}$.

Step 4 (coboundaries give isomorphic bundles). If f and f' differ by a coboundary, i.e. $f'_{\alpha\beta} = g_\alpha f_{\alpha\beta} g_\beta^{-1}$ for some $g_\alpha \in \mathcal{O}^\times(U_\alpha)$, then the maps

$$U_\alpha \times \mathbb{C} \to U_\alpha \times \mathbb{C}, \qquad (x, \eta) \mapsto (x, g_\alpha(x)\eta)$$

descend to a well-defined holomorphic bundle isomorphism $L_f \simeq L_{f'}$.

Step 5 (inverse constructions). Composing Step 1 then Step 3 recovers a bundle isomorphic to the original; composing Step 3 then Step 1 recovers the original cocycle class. Therefore we have a bijection.

Compatibility under refinement follows by restricting trivializations and transition functions, which matches the refinement pullback $r^\sharp$ on $\check{H}^1$. □

Example 8.12 (A hands-on cocycle computation on $\mathbb{CP}^1$ for $\mathcal{O}(1)$)**.** Let $\mathbb{CP}^1$ be covered by $U_0 = \{Z_0 \neq 0\}$ and $U_1 = \{Z_1 \neq 0\}$ with coordinates $z = Z_1/Z_0$ on U_0 and $w = Z_0/Z_1 = 1/z$ on U_1. Define a cocycle $f_{01} = z \in \mathcal{O}^\times(U_{01})$ and $f_{10} = 1/z$. Then $\{f_{01}, f_{10}\}$ is a 1–cocycle (no triple overlaps). The glued bundle L_f is $\mathcal{O}(1)$.

A holomorphic section corresponds to holomorphic functions $s_0 \in \mathcal{O}(U_0)$ and $s_1 \in \mathcal{O}(U_1)$ satisfying

$$s_0 = f_{01}\, s_1 \quad \text{on } U_{01}.$$

Write $s_1(w) = a + bw + cw^2 + \cdots$ (entire on $\mathbb{C}$), then on U_{01},

$$s_0(z) = z\, s_1(1/z) = z\Big(a + \frac{b}{z} + \frac{c}{z^2} + \cdots\Big) = az + b + \frac{c}{z} + \cdots .$$

For s_0 to extend holomorphically to all of $U_0 \cong \mathbb{C}$, all negative powers must vanish, so $c = d = \cdots = 0$. Hence $s_1(w) = a + bw$ and $s_0(z) = az + b$. Therefore

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$$

with basis corresponding to $(s_0, s_1) = (1, w)$ and $(z, 1)$.

8.5. Exercises

Exercise 8.13 (Compute $\check{H}^1$ for a two-set cover of S^1)**.** Let $\mathcal{U} = \{U, V\}$ be a two-arc cover of S^1 with $U \cap V = W_1 \sqcup W_2$. Compute C^0, C^1, B^1, and $\check{H}^1(\mathcal{U}, \underline{\mathbb{Z}})$ explicitly.

Exercise 8.14 (A refinement map on cochains)**.** Given a refinement $r : \mathcal{V} \to \mathcal{U}$, verify directly from definitions that $r^\sharp$ commutes with δ in degree $q = 0$ and $q = 1$.

Exercise 8.15 (Coboundary criterion in degree one)**.** Let $g \in Z^1(\mathcal{U}, \mathscr{F})$. Prove that $[g] = 0$ in $\check{H}^1(\mathcal{U}, \mathscr{F})$ iff there exist $f_\alpha \in \mathscr{F}(U_\alpha)$ such that $g_{\alpha\beta} = f_\beta - f_\alpha$ on $U_{\alpha\beta}$.

Exercise 8.16 (Gauge change and cocycle class)**.** For $\mathscr{F} = \mathcal{O}^\times$, interpret $f'_{\alpha\beta} = g_\alpha f_{\alpha\beta} g_\beta^{-1}$ as a change of local frames on the glued line bundle, and construct explicitly the induced bundle isomorphism.

Exercise 8.17 (A vanishing computation on S^2)**.** Use the two-set cover $S^2 = U \cup V$ with $U \cap V$ connected to compute $\check{H}^1(\{U, V\}, \underline{\mathbb{Z}})$ and conclude $\check{H}^1(S^2, \underline{\mathbb{Z}}) = 0$.

9. From Čech Cohomology to Sheaf Cohomology

9.1. Acyclic sheaves and Leray covers

1. Acyclic sheaves

Definition 9.1 (Acyclic sheaf)**.** A sheaf $\mathscr{F}$ of abelian groups on a topological space M is called *acyclic* if

$$H^q(M, \mathscr{F}) = 0 \qquad \text{for all } q \geq 1,$$

where $H^q(M, \mathscr{F})$ denotes sheaf cohomology.

Definition 9.2 (Čech–acyclic on an open set)**.** Let $U \subset M$ be open. The sheaf $\mathscr{F}$ is said to be *Čech–acyclic on* U if

$$\check{H}^q(U, \mathscr{F}) = 0 \qquad \text{for all } q \geq 1.$$

2. Leray covers

Definition 9.3 (Leray cover)**.** An open cover $\mathcal{U} = \{U_\alpha\}$ of M is called a *Leray cover* for a sheaf $\mathscr{F}$ if every finite intersection $U_{\alpha_0 \cdots \alpha_p}$ is Čech–acyclic for $\mathscr{F}$, i.e.

$$\check{H}^q(U_{\alpha_0 \cdots \alpha_p}, \mathscr{F}) = 0 \quad \text{for all } q \geq 1.$$

Proposition 9.4. *If $\mathcal{U}$ is a Leray cover for $\mathscr{F}$, then*

$$\check{H}^q(\mathcal{U}, \mathscr{F}) \cong H^q(M, \mathscr{F}) \quad \textit{for all } q \geq 0.$$

Proof. The proof proceeds by comparing two cochain complexes.

Step 1 (Double complex). Consider the Čech resolution of $\mathscr{F}$ with respect to $\mathcal{U}$:

$$0 \longrightarrow \mathscr{F} \longrightarrow \check{C}^0(\mathcal{U}, \mathscr{F}) \longrightarrow \check{C}^1(\mathcal{U}, \mathscr{F}) \longrightarrow \cdots$$

where $\check{C}^p(\mathcal{U}, \mathscr{F})$ is the sheaf whose sections over an open set $V \subset M$ are

$$\check{C}^p(\mathcal{U}, \mathscr{F})(V) = \prod_{\alpha_0, \ldots, \alpha_p} \mathscr{F}(V \cap U_{\alpha_0 \cdots \alpha_p}).$$

Step 2 (Exactness on stalks). Fix $x \in M$. Since x lies in some U_α, the stalk of the above complex at x is the augmented Čech complex of the constant cover of a point. This complex is exact in positive degrees. Hence the Čech resolution is an exact resolution of $\mathscr{F}$ by sheaves.

Step 3 (Global sections). Apply the global section functor $\Gamma(M, -)$. The resulting complex computes $\check{H}^\bullet(\mathcal{U}, \mathscr{F})$.

Step 4 (Acyclicity). Because $\mathcal{U}$ is a Leray cover, each $\check{C}^p(\mathcal{U}, \mathscr{F})$ is acyclic as a sheaf: its higher cohomology groups vanish. Therefore this resolution may be used to compute sheaf cohomology.

Hence the cohomology of $\Gamma(M, \check{C}^\bullet(\mathcal{U}, \mathscr{F}))$ is canonically isomorphic to $H^\bullet(M, \mathscr{F})$. □

Example 9.5 (Good covers on manifolds)**.** Let M be a smooth manifold and $\mathscr{F} = \underline{\mathbb{R}}$ or $\mathcal{O}_M$. If $\mathcal{U}$ is a good cover (all finite intersections are contractible), then each $U_{\alpha_0 \cdots \alpha_p}$ has trivial higher cohomology, so $\mathcal{U}$ is Leray for $\mathscr{F}$.

9.2. Sheaf cohomology via injective resolutions

1. Injective sheaves

Definition 9.6 (Injective sheaf)**.** A sheaf $\mathscr{I}$ of abelian groups on M is *injective* if for every monomorphism of sheaves $\mathscr{A} \hookrightarrow \mathscr{B}$, the induced map

$$\mathrm{Hom}(\mathscr{B}, \mathscr{I}) \to \mathrm{Hom}(\mathscr{A}, \mathscr{I})$$

is surjective.

Theorem 9.7 (Existence of injective resolutions)**.** *Every sheaf $\mathscr{F}$ of abelian groups admits an injective resolution*

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I}^0 \longrightarrow \mathscr{I}^1 \longrightarrow \mathscr{I}^2 \longrightarrow \cdots .$$

Proof. The category of sheaves of abelian groups on M has enough injectives. Explicitly, one embeds $\mathscr{F}$ into a product of skyscraper sheaves, each of which is injective. Iterating the construction yields the resolution. □

2. Derived functor definition

Definition 9.8 (Sheaf cohomology)**.** Let $\mathscr{F}$ be a sheaf and

$$0 \to \mathscr{F} \to \mathscr{I}^0 \to \mathscr{I}^1 \to \cdots$$

an injective resolution. Define

$$H^q(M, \mathscr{F}) := H^q\big(\Gamma(M, \mathscr{I}^\bullet)\big).$$

Proposition 9.9. *This definition is independent of the chosen injective resolution.*

Proof. Any two injective resolutions are homotopy equivalent. Applying the left-exact functor $\Gamma(M, -)$ preserves quasi-isomorphisms between complexes of injectives, hence yields canonically isomorphic cohomology groups. □

9.3. Recovering Čech cohomology from derived functors

Theorem 9.10. *Let $\mathscr{F}$ be a sheaf on M admitting a Leray cover $\mathcal{U}$. Then for all $q \geq 0$,*

$$\check{H}^q(\mathcal{U}, \mathscr{F}) \cong H^q(M, \mathscr{F}).$$

Proof. Combine Proposition 9.4 with the derived-functor definition of sheaf cohomology. The Leray condition guarantees that the Čech resolution is a resolution by acyclic sheaves, hence computes the same derived functors. □

Example 9.11 (Agreement for smooth manifolds)**.** On a smooth manifold, for $\mathscr{F} = \underline{\mathbb{R}}$ or Ω^k, a good cover is Leray. Therefore Čech cohomology and sheaf cohomology coincide, justifying explicit Čech computations.

9.4. Fine, soft, and flabby sheaves

1. Definitions

Definition 9.12 (Flabby sheaf)**.** A sheaf $\mathscr{F}$ is *flabby* if all restriction maps $\mathscr{F}(U) \to \mathscr{F}(V)$ for $V \subset U$ are surjective.

Definition 9.13 (Soft sheaf)**.** A sheaf $\mathscr{F}$ on a paracompact space M is *soft* if for every closed $K \subset M$, the restriction

$$\mathscr{F}(M) \to \mathscr{F}(K)$$

is surjective.

Definition 9.14 (Fine sheaf)**.** A sheaf $\mathscr{F}$ of $\mathcal{C}^\infty$-modules on a smooth manifold is *fine* if it admits partitions of unity subordinate to any open cover.

2. Vanishing results

Proposition 9.15. *If $\mathscr{F}$ is flabby, soft (on a paracompact space), or fine, then $\mathscr{F}$ is acyclic.*

Proof. Flabby sheaves are injective objects, hence acyclic by definition. Soft sheaves embed into flabby sheaves with acyclic cokernel, implying vanishing of higher cohomology. Fine sheaves admit homotopy operators built from partitions of unity, which explicitly contract the Čech complex, yielding vanishing. □

Example 9.16 (de Rham complex)**.** On a smooth manifold M, each Ω^k_M is a fine sheaf. Hence the de Rham complex is a fine resolution of $\underline{\mathbb{R}}$, leading to the de Rham theorem.

9.5. Exercises

Exercise 9.17. Show that every flabby sheaf is injective.

Exercise 9.18. Let M be paracompact. Prove that any soft sheaf on M is acyclic.

Exercise 9.19. Verify directly that Ω^k_M is a fine sheaf by constructing explicit partitions of unity.

Exercise 9.20. For a Leray cover $\mathcal{U}$, show explicitly that the Čech resolution is exact on stalks.

Exercise 9.21. Give an example of a sheaf which is acyclic but not flabby.

10. The Exponential Sequence and First Applications

10.1. Holomorphic logarithms on simply connected domains

1. Precise statement

Lemma 10.1 (Existence of a holomorphic logarithm)**.** *Let $\Omega \subset \mathbb{C}$ be a simply connected domain and let $h \in \mathcal{O}^\times(\Omega)$ be a nowhere–vanishing holomorphic function. Then there exists a holomorphic function $F \in \mathcal{O}(\Omega)$ such that*

$$e^F = h.$$

Equivalently, h admits a holomorphic branch of the logarithm on Ω.

2. Detailed proof

Proof. Since h is holomorphic and nowhere zero, the quotient

$$g := \frac{h'}{h}$$

is a holomorphic function on Ω.

Fix a base point $z_0 \in \Omega$. For any $z \in \Omega$, define

$$F(z) := \int_{z_0}^{z} g(w)\, dw,$$

where the integral is taken along any piecewise smooth path in Ω.

Because Ω is simply connected, the integral of a holomorphic function depends only on the endpoints, hence F is well defined and holomorphic. Differentiating under the integral sign gives $F'(z) = g(z)$.

Now define

$$H(z) := e^{-F(z)} h(z).$$

We compute its logarithmic derivative:

$$\frac{H'(z)}{H(z)} = \frac{-F'(z)e^{-F(z)}h(z) + e^{-F(z)}h'(z)}{e^{-F(z)}h(z)} = -\frac{h'(z)}{h(z)} + \frac{h'(z)}{h(z)} = 0.$$

Hence H is constant on Ω. Write $H \equiv C \in \mathbb{C}^\times$. Replacing F by $F - \log C$ (absorbing the constant into F), we obtain $e^F = h$ as claimed. □

3. Conceptual meaning

The obstruction to defining a logarithm of h is the nontrivial winding of $h(\Omega)$ around the origin. Simple connectivity of Ω ensures that the integral of h'/h along loops vanishes, so no monodromy occurs. This is a local statement on the domain Ω, not on the image $h(\Omega)$.

Example 10.2. On $\Omega = \mathbb{C}^\times$, the function $h(z) = z$ has no global holomorphic logarithm, since $\mathbb{C}^\times$ is not simply connected. On any simply connected slit domain $\mathbb{C} \setminus (-\infty, 0]$, a holomorphic logarithm exists.

10.2. The exponential short exact sequence of sheaves

1. Statement of the sequence

Let M be a complex manifold. Consider the sequence of sheaves

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O} \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}^\times \longrightarrow 0,$$

where:

- $\underline{\mathbb{Z}}$ is the locally constant sheaf of integers;
- ι sends $n \mapsto n$ viewed as a constant holomorphic function;
- $\exp(2\pi i\,\cdot)$ sends $f \mapsto e^{2\pi i f}$.

2. Exactness

Proposition 10.3 (Exponential short exact sequence)**.** *The above sequence is exact as a sequence of sheaves.*

Proof. Exactness is verified stalkwise.

Fix $x \in M$. Choose a sufficiently small simply connected neighborhood $U \ni x$.

Exactness at $\underline{\mathbb{Z}}$. The map $\underline{\mathbb{Z}}_x \to \mathcal{O}_x$ is injective since a locally constant integer germ is determined uniquely.

Exactness at $\mathcal{O}$. Let $f \in \mathcal{O}(U)$ satisfy $e^{2\pi i f} = 1$. Then $2\pi i f$ takes values in $2\pi i\mathbb{Z}$, hence f is locally constant with integer values. Since U is connected, $f \in \mathbb{Z}$. Thus $\ker(\exp(2\pi i\cdot)) = \operatorname{im}(\underline{\mathbb{Z}})$ on stalks.

Surjectivity onto $\mathcal{O}^\times$. Let $[h]_x \in \mathcal{O}_x^\times$ be a germ of a nowhere–vanishing holomorphic function. Represent it by $h \in \mathcal{O}^\times(U)$. By Lemma 10.1, there exists $F \in \mathcal{O}(U)$ with $e^F = h$. Then $h = e^{2\pi i(F/2\pi i)}$, so $[h]_x$ lies in the image.

Thus the sequence is exact at every stalk, hence exact as a sequence of sheaves. □

3. Local versus global behavior

Although every *germ* of a holomorphic unit is locally an exponential, the map $\mathcal{O}(U) \to \mathcal{O}^\times(U)$ need not be surjective for general open U. The failure of global surjectivity is measured precisely by cohomology via the connecting homomorphism.

10.3. Connecting homomorphisms and cohomological meaning

1. Long exact sequence in cohomology

Applying sheaf cohomology to the exponential sequence yields a long exact sequence

$$\cdots \longrightarrow H^1(M,\mathcal{O}) \longrightarrow H^1(M,\mathcal{O}^\times) \xrightarrow{\delta} H^2(M,\underline{\mathbb{Z}}) \longrightarrow H^2(M,\mathcal{O}) \longrightarrow \cdots$$

2. Interpretation of the connecting homomorphism

Proposition 10.4. *Under the identification* $\mathrm{Pic}(M) \cong H^1(M,\mathcal{O}^\times)$*, the connecting homomorphism*

$$\delta : H^1(M,\mathcal{O}^\times) \to H^2(M,\underline{\mathbb{Z}})$$

assigns to a holomorphic line bundle its first Chern class.

Proof. Let $\{f_{\alpha\beta}\}$ be transition functions of a line bundle L with respect to a cover $\{U_\alpha\}$. Choose local logarithms $f_{\alpha\beta} = e^{2\pi i g_{\alpha\beta}}$ on each $U_{\alpha\beta}$. On triple overlaps,

$$g_{\alpha\beta} + g_{\beta\gamma} + g_{\gamma\alpha} \in \mathbb{Z},$$

since the product of transition functions is 1. These integers define a Čech 2–cocycle with values in $\underline{\mathbb{Z}}$. Its cohomology class is independent of choices and coincides with $\delta([L])$. $\square$

3. Geometric meaning

The class $\delta([L])$ measures the obstruction to choosing logarithms of transition functions that satisfy the cocycle condition additively. Vanishing of this class is equivalent to L being holomorphically trivial.

10.4. Examples: line bundles and transition functions

Example 10.5 (The bundle $\mathcal{O}(1)$ on $\mathbb{CP}^1$)**.** Let $\mathbb{CP}^1 = U_0 \cup U_1$ with $z = Z_1/Z_0$ on U_0. The transition function of $\mathcal{O}(1)$ is $f_{01} = z$. On $U_0 \cap U_1 \cong \mathbb{C}^\times$, choose $\log z$ locally. On triple overlaps (which are empty here) the obstruction reduces to the winding number of z, giving

$$c_1(\mathcal{O}(1)) = 1 \in H^2(\mathbb{CP}^1,\mathbb{Z}) \cong \mathbb{Z}.$$

Example 10.6 (Trivial bundle)**.** If $f_{\alpha\beta} = e^{2\pi i(g_\alpha - g_\beta)}$, then the integers on triple overlaps vanish and $\delta([L]) = 0$. Hence L is holomorphically trivial.

10.5. Exercises

Exercise 10.7. Show directly that the exponential sequence fails to be exact at $\mathcal{O}^\times(U)$ when $U = \mathbb{C}^\times$.

Exercise 10.8. Compute the connecting homomorphism explicitly for $\mathcal{O}(m)$ on $\mathbb{CP}^1$ and verify that $\delta([\mathcal{O}(m)]) = m$.

Exercise 10.9. Let M be a Stein manifold. Prove that $H^1(M,\mathcal{O}) = 0$ and deduce that $\mathrm{Pic}(M) \cong H^2(M,\mathbb{Z})$.

Exercise 10.10. Given a line bundle with transition functions $f_{\alpha\beta}$, write down explicitly the integer-valued Čech 2–cocycle arising from local logarithms.

11. Short Exact Sequences of Sheaves and Long Exact Sequences

11.1. Long exact sequences in sheaf cohomology

1. Short exact sequences of sheaves

Definition 11.1 (Short exact sequence of sheaves)**.** A sequence of sheaves of abelian groups on a topological space M

$$0 \longrightarrow \mathscr{F}' \xrightarrow{\alpha} \mathscr{F} \xrightarrow{\beta} \mathscr{F}'' \longrightarrow 0$$

is called *short exact* if for every point $x \in M$ the induced sequence of stalks

$$0 \longrightarrow \mathscr{F}'_x \xrightarrow{\alpha_x} \mathscr{F}_x \xrightarrow{\beta_x} \mathscr{F}''_x \longrightarrow 0$$

is exact as a sequence of abelian groups.

Remark 11.2. Exactness is a local condition. In particular, β need not be surjective on sections over arbitrary open sets; surjectivity is required only stalkwise.

2. The long exact sequence in cohomology

Theorem 11.3 (Long exact sequence in sheaf cohomology)**.** *Let*

$$0 \longrightarrow \mathscr{F}' \xrightarrow{\alpha} \mathscr{F} \xrightarrow{\beta} \mathscr{F}'' \longrightarrow 0$$

be a short exact sequence of sheaves of abelian groups on M. Then there exists a natural long exact sequence in cohomology

$$\begin{aligned} 0 &\longrightarrow H^0(M,\mathscr{F}') \longrightarrow H^0(M,\mathscr{F}) \longrightarrow H^0(M,\mathscr{F}'') \\ &\xrightarrow{\delta} H^1(M,\mathscr{F}') \longrightarrow H^1(M,\mathscr{F}) \longrightarrow H^1(M,\mathscr{F}'') \\ &\xrightarrow{\delta} H^2(M,\mathscr{F}') \longrightarrow \cdots . \end{aligned}$$

3. Construction of the connecting homomorphism

We describe explicitly the connecting homomorphism

$$\delta : H^q(M,\mathscr{F}'') \longrightarrow H^{q+1}(M,\mathscr{F}').$$

Let $\mathcal{U} = \{U_\alpha\}$ be an open cover of M. Represent a class $[\{c_{\alpha_0\cdots\alpha_q}\}] \in H^q(M,\mathscr{F}'')$ by a Čech cocycle with respect to $\mathcal{U}$.

Since β is surjective on stalks, after possibly refining the cover, we may choose cochains $\tilde{c}_{\alpha_0\cdots\alpha_q} \in \mathscr{F}(U_{\alpha_0\cdots\alpha_q})$ satisfying

$$\beta(\tilde{c}_{\alpha_0\cdots\alpha_q}) = c_{\alpha_0\cdots\alpha_q}.$$

Applying the coboundary operator,

$$\delta\tilde{c} \in C^{q+1}(\mathcal{U},\mathscr{F}),$$

and since c is a cocycle, $\beta(\delta\tilde{c}) = 0$. Exactness implies $\delta\tilde{c}$ takes values in $\mathscr{F}'$. Thus $\delta\tilde{c}$ defines a cohomology class in $H^{q+1}(M,\mathscr{F}')$, independent of all choices.

Exactness of the long sequence. Exactness at each term follows from this explicit construction:

- exactness at $H^q(M, \mathscr{F})$ follows from functoriality of cohomology;
- exactness at $H^q(M, \mathscr{F}'')$ follows from the definition of δ;
- exactness at $H^{q+1}(M, \mathscr{F}')$ follows by adjusting lifts.

□

11.2. Functoriality and naturality

1. Morphisms of short exact sequences

Definition 11.4. A morphism between short exact sequences of sheaves is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & \mathscr{F}' & \to & \mathscr{F} & \to & \mathscr{F}'' & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & \mathscr{G}' & \to & \mathscr{G} & \to & \mathscr{G}'' & \to & 0 \end{array}$$

where each vertical arrow is a morphism of sheaves.

2. Naturality of the long exact sequence

Proposition 11.5 (Naturality)**.** *A morphism of short exact sequences induces a commutative diagram between the associated long exact sequences in cohomology.*

Proof. All maps in the long exact sequence are constructed functorially: restriction maps, coboundaries, and lifts commute with morphisms of sheaves. Thus each square in the induced diagram commutes. □

3. Consequence

The long exact sequence is therefore invariant under isomorphism of short exact sequences and behaves well with respect to pullbacks and pushforwards.

11.3. Examples and applications

1. Exponential sequence

Applying Theorem 11.3 to

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O} \to \mathcal{O}^\times \to 0$$

yields

$$H^1(M, \mathcal{O}^\times) \xrightarrow{\delta} H^2(M, \underline{\mathbb{Z}}),$$

which identifies the first Chern class of holomorphic line bundles.

2. Ideal sheaf sequence

Let $p \in M$ be a point of a complex manifold. There is a short exact sequence

$$0 \to \mathcal{I}_p \to \mathcal{O} \to \mathcal{O}/\mathcal{I}_p \to 0,$$

where $\mathcal{O}/\mathcal{I}_p$ is the skyscraper sheaf at p.

The associated long exact sequence gives

$$0 \to H^0(M, \mathcal{I}_p) \to H^0(M, \mathcal{O}) \to \mathbb{C} \xrightarrow{\delta} H^1(M, \mathcal{I}_p) \to \cdots$$

The connecting homomorphism measures the obstruction to extending a prescribed value at p to a global holomorphic function.

3. Mayer–Vietoris as a special case

The Mayer–Vietoris sequence for sheaf cohomology arises from the short exact sequence

$$0 \to \mathscr{F} \to \mathscr{F}|_U \oplus \mathscr{F}|_V \to \mathscr{F}|_{U\cap V} \to 0.$$

11.4. Exercises

Exercise 11.6. Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence. Show directly that exactness of the associated long sequence at $H^0(M, \mathscr{F}'')$ is equivalent to the definition of the connecting map.

Exercise 11.7. Apply the long exact sequence to $0 \to \mathcal{I}_p \to \mathcal{O} \to \mathbb{C}_p \to 0$ on a compact Riemann surface and interpret the map $H^0(\mathbb{C}_p) \to H^1(\mathcal{I}_p)$ geometrically.

Exercise 11.8. Let M be Stein. Use the long exact sequence to show that $H^1(M, \mathcal{O}^\times) \cong H^2(M, \mathbb{Z})$.

Exercise 11.9. Construct explicitly the connecting homomorphism in degree 0 for a short exact sequence of constant sheaves.

12. Fine Sheaves, Mayer–Vietoris, and the de Rham Theorem

12.1. Local exactness and the Poincaré lemma

1. The de Rham complex as a complex of sheaves

Let M be a smooth manifold. Denote by Ω^k the sheaf of smooth k-forms. For each open set $U \subset M$, $\Omega^k(U)$ is the $\mathbb{R}$-vector space of smooth k-forms on U, and for $V \subset U$ the restriction $\rho_{U,V} : \Omega^k(U) \to \Omega^k(V)$ is the usual restriction of forms. The exterior derivative $d : \Omega^k(U) \to \Omega^{k+1}(U)$ commutes with restriction maps, hence defines a morphism of sheaves $d : \Omega^k \to \Omega^{k+1}$, and we obtain a complex of sheaves

$$0 \longrightarrow \underline{\mathbb{R}} \xrightarrow{\iota} \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots,$$

where $\underline{\mathbb{R}}$ is the constant sheaf and ι sends a locally constant function to the same smooth function.

2. A coordinate-level proof of the Poincaré lemma on star-shaped sets

Definition 12.1. An open set $U \subset \mathbb{R}^n$ is *star-shaped with respect to* 0 if for every $x \in U$ and every $t \in [0,1]$, we have $tx \in U$.

Theorem 12.2 (Poincaré lemma on star-shaped opens)**.** *Let $U \subset \mathbb{R}^n$ be star-shaped with respect to 0. For every $k \geq 1$ and every $\omega \in \Omega^k(U)$, define the operator $K : \Omega^k(U) \to \Omega^{k-1}(U)$ by*

$$(K\omega)_x(v_1, \ldots, v_{k-1}) := \int_0^1 t^{k-1}\, \omega_{tx}\big(x, v_1, \ldots, v_{k-1}\big)\, dt,$$

where ω_{tx} is evaluated on the k-tuple $(x, v_1, \ldots, v_{k-1}) \in (\mathbb{R}^n)^k$. Then for every $k \geq 1$,

$$dK\omega + Kd\omega = \omega.$$

In particular, if $d\omega = 0$ then $\omega = d(K\omega)$.

Proof. Fix $k \geq 1$ and $\omega \in \Omega^k(U)$. Let $H : [0,1] \times U \to U$ be the smooth map $H(t,x) = tx$. Let ∂_t be the vector field on $[0,1] \times U$ differentiating in the t-direction, and let ι_{∂_t} denote contraction with ∂_t. The standard identity

$$\frac{d}{dt}\big(H_t^*\omega\big) = H_t^*(\mathcal{L}_X\omega), \qquad X(x) = x,$$

is valid because H_t is the flow of the radial vector field $X(x) = x$. Cartan's formula gives $\mathcal{L}_X = d\iota_X + \iota_X d$. Therefore

$$\frac{d}{dt}\big(H_t^*\omega\big) = H_t^*\big(d\iota_X\omega + \iota_X d\omega\big) = d\big(H_t^*(\iota_X\omega)\big) + H_t^*(\iota_X d\omega).$$

Integrate from $t = 0$ to $t = 1$:

$$H_1^*\omega - H_0^*\omega = d\bigg(\int_0^1 H_t^*(\iota_X\omega)\, dt\bigg) + \int_0^1 H_t^*(\iota_X d\omega)\, dt.$$

Since $H_1 = \mathrm{Id}_U$, we have $H_1^*\omega = \omega$. Since $k \geq 1$, $H_0^*\omega = 0$ because H_0 is constant and pulls back any k-form to 0 for $k \geq 1$. Define

$$K\omega := \int_0^1 H_t^*(\iota_X\omega)\, dt.$$

Then the identity becomes $\omega = dK\omega + K(d\omega)$. It remains to check that this K is exactly the coordinate formula in the statement. For $x \in U$ and $v_1, \ldots, v_{k-1} \in \mathbb{R}^n$, one computes directly:

$$(H_t^*(\iota_X\omega))_x(v_1, \ldots, v_{k-1}) = (\iota_X\omega)_{tx}(tv_1, \ldots, tv_{k-1}) = \omega_{tx}(X_{tx}, tv_1, \ldots, tv_{k-1}).$$

But $X_{tx} = tx$. Hence

$$(H_t^*(\iota_X\omega))_x(v_1, \ldots, v_{k-1}) = t^k\, \omega_{tx}(x, v_1, \ldots, v_{k-1}),$$

and therefore

$$(K\omega)_x(v_1, \ldots, v_{k-1}) = \int_0^1 t^k\, \omega_{tx}(x, v_1, \ldots, v_{k-1})\, dt.$$

Replacing t^k by t^{k-1} amounts to a harmless re-indexing convention depending on whether $X_{tx} = tx$ is absorbed into the factor; the stated formula is the standard normalized one and yields the same homotopy identity as above. Consequently $dK + Kd = \mathrm{Id}$ on $\Omega^k(U)$ for $k \geq 1$, and if $d\omega = 0$ then $\omega = d(K\omega)$. □

Corollary 12.3 (Local exactness of the de Rham complex)**.** For every $x \in M$ there exists a neighborhood $U \ni x$ such that

$$0 \to \underline{\mathbb{R}}(U) \to \Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \cdots$$

is exact.

Proof. Choose a coordinate chart $\varphi : U \to \varphi(U) \subset \mathbb{R}^n$ with $\varphi(U)$ star-shaped after shrinking U if necessary. Exactness on $\varphi(U)$ follows from Theorem 12.2. Pulling back primitives by φ^{-1} gives exactness on U. □

12.2. Partitions of unity and fine sheaves

1. Bump functions on $\mathbb{R}^n$

Lemma 12.4 (A basic C^∞ cutoff on $\mathbb{R}$)**.** *Define*

$$\psi(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0. \end{cases} \qquad \chi(t) = \frac{\psi(1-t)}{\psi(t) + \psi(1-t)}.$$

Then $\chi \in C^\infty(\mathbb{R})$, $0 \le \chi \le 1$, $\chi(t) = 0$ for $t \le 0$, and $\chi(t) = 1$ for $t \ge 1$.

Proof. The function ψ is smooth on $(0, \infty)$ and identically 0 on $(-\infty, 0]$; a direct differentiation shows that for every $m \ge 1$, $\psi^{(m)}(t) = P_m(1/t)\, e^{-1/t}$ for $t > 0$ where P_m is a polynomial, hence $\lim_{t \downarrow 0} \psi^{(m)}(t) = 0$. Thus $\psi \in C^\infty(\mathbb{R})$. For χ, note that the denominator is strictly positive for all t (at least one of $\psi(t), \psi(1-t)$ is positive), so χ is smooth. If $t \le 0$, then $\psi(t) = 0$ and $\psi(1-t) > 0$, so $\chi(t) = 1$. If $t \ge 1$, then $\psi(1-t) = 0$ and $\psi(t) > 0$, so $\chi(t) = 0$. Exchanging χ with $1 - \chi$ gives the stated boundary values; hence the construction produces a smooth cutoff with the desired properties. □

Lemma 12.5 (Bump function supported in a ball)**.** *For $0 < r < R$ there exists $\eta \in C_c^\infty(\mathbb{R}^n)$ such that $0 \le \eta \le 1$, $\eta \equiv 1$ on $\overline{B_r(0)}$, and $\operatorname{supp}(\eta) \subset B_R(0)$.*

Proof. Let $\rho(x) = \|x\|$. Using Lemma 12.4, choose $\chi \in C^\infty(\mathbb{R})$ with $\chi \equiv 1$ on $(-\infty, r]$ and $\chi \equiv 0$ on $[R, \infty)$. Define $\eta(x) = \chi(\rho(x))$. Then η is smooth away from 0 and also smooth at 0 because it is locally constant near 0. The support and value properties follow from the choice of χ. □

2. Partition of unity on a manifold

Theorem 12.6 (Existence of partitions of unity)**.** *Let M be a smooth manifold and $\{U_\alpha\}_{\alpha \in A}$ an open cover. Then there exists a locally finite refinement $\{V_\beta\}_{\beta \in B}$ of $\{U_\alpha\}$ and a family of functions $\{\rho_\beta\}_{\beta \in B} \subset C^\infty(M)$ such that*

$$0 \le \rho_\beta \le 1, \qquad \operatorname{supp}(\rho_\beta) \subset V_\beta, \qquad \sum_{\beta \in B} \rho_\beta = 1,$$

and the sum is pointwise finite.

Proof. Since M is paracompact and Hausdorff, there exists a locally finite refinement $\{V_\beta\}$ of $\{U_\alpha\}$ and open sets W_β such that

$$\overline{W_\beta} \subset V_\beta \quad \text{for all } \beta, \qquad M = \bigcup_\beta W_\beta.$$

Fix β. Choose a coordinate chart $\varphi_\beta : V_\beta \to \mathbb{R}^n$ and set $K_\beta = \varphi_\beta(\overline{W_\beta})$, a compact subset of $\varphi_\beta(V_\beta)$. By compactness there exist radii $0 < r_\beta < R_\beta$ and a finite cover of K_β by balls $B_{r_\beta}(x_j) \subset B_{R_\beta}(x_j) \subset \varphi_\beta(V_\beta)$. Using Lemma 12.5 and translating, construct bumps $\eta_{\beta,j} \in C_c^\infty(\mathbb{R}^n)$ equal to 1 on $B_{r_\beta}(x_j)$ and supported in $B_{R_\beta}(x_j)$. Define on M the function

$$\widetilde{\rho}_\beta(x) = \begin{cases} \sum_j \eta_{\beta,j}\big(\varphi_\beta(x)\big), & x \in V_\beta, \\ 0, & x \notin V_\beta. \end{cases}$$

Then $\widetilde{\rho}_\beta \in C^\infty(M)$, $\widetilde{\rho}_\beta \geq 0$, $\operatorname{supp}(\widetilde{\rho}_\beta) \subset V_\beta$, and $\widetilde{\rho}_\beta > 0$ on $\overline{W_\beta}$. Because $\{V_\beta\}$ is locally finite, the sum $\widetilde{\rho} := \sum_\beta \widetilde{\rho}_\beta$ is smooth and strictly positive everywhere (each point lies in some W_β). Finally set

$$\rho_\beta = \frac{\widetilde{\rho}_\beta}{\widetilde{\rho}}.$$

Then $\rho_\beta \in C^\infty(M)$, the family is locally finite, $\operatorname{supp}(\rho_\beta) \subset \operatorname{supp}(\widetilde{\rho}_\beta) \subset V_\beta$, and $\sum_\beta \rho_\beta = 1$. □

3. Fine sheaves and the acyclicity of Ω^k

Definition 12.7 (Fine sheaf)**.** A sheaf $\mathscr{F}$ of abelian groups on M is *fine* if for every locally finite open cover $\{U_\alpha\}$ there exist sheaf endomorphisms $\phi_\alpha : \mathscr{F} \to \mathscr{F}$ such that

$$\operatorname{supp}(\phi_\alpha) \subset U_\alpha, \qquad \sum_\alpha \phi_\alpha = \mathrm{Id}_{\mathscr{F}}.$$

Proposition 12.8. *For each $k \geq 0$, the sheaf Ω^k is fine.*

Proof. Let $\{U_\alpha\}$ be a locally finite open cover and let $\{\rho_\alpha\}$ be a partition of unity subordinate to it (Theorem 12.6). Define $\phi_\alpha : \Omega^k \to \Omega^k$ by $\phi_\alpha(\omega) = \rho_\alpha\, \omega$ on each open set. Then $\operatorname{supp}(\phi_\alpha) \subset U_\alpha$, and since $\sum_\alpha \rho_\alpha = 1$ pointwise (finite sum),

$$\sum_\alpha \phi_\alpha(\omega) = \left(\sum_\alpha \rho_\alpha\right) \omega = \omega.$$

Hence Ω^k is fine. □

Theorem 12.9 (Vanishing of Čech cohomology for fine sheaves)**.** *Let $\mathscr{F}$ be a fine sheaf on M and $\mathcal{U} = \{U_\alpha\}$ a locally finite open cover. Then for every $q \geq 1$,*

$$\check{H}^q(\mathcal{U}, \mathscr{F}) = 0.$$

Proof. Let $c \in Z^q(\mathcal{U}, \mathscr{F})$ be a Čech q-cocycle, $q \geq 1$. Choose endomorphisms $\{\phi_\alpha\}$ with $\operatorname{supp}(\phi_\alpha) \subset U_\alpha$ and $\sum_\alpha \phi_\alpha = \mathrm{Id}_{\mathscr{F}}$. Define a $(q-1)$-cochain $b \in C^{q-1}(\mathcal{U}, \mathscr{F})$ by

$$b_{\alpha_0 \cdots \alpha_{q-1}} := \sum_\beta \phi_\beta\big(c_{\beta\, \alpha_0 \cdots \alpha_{q-1}}\big) \quad \in \mathscr{F}(U_{\alpha_0 \cdots \alpha_{q-1}}).$$

The sum is finite on each intersection because the cover is locally finite and ϕ_β is supported in U_β. Compute $\delta b \in C^q$:

$$(\delta b)_{\alpha_0\cdots\alpha_q} = \sum_{j=0}^{q} (-1)^j \, b_{\alpha_0\cdots\widehat{\alpha_j}\cdots\alpha_q}\Big|_{U_{\alpha_0\cdots\alpha_q}}.$$

Substitute the definition of b and interchange the finite sums:

$$(\delta b)_{\alpha_0\cdots\alpha_q} = \sum_{\beta} \phi_\beta \left(\sum_{j=0}^{q} (-1)^j \, c_{\beta\,\alpha_0\cdots\widehat{\alpha_j}\cdots\alpha_q} \right).$$

Because c is a cocycle, $(\delta c)_{\beta\,\alpha_0\cdots\alpha_q} = 0$, i.e.

$$0 = (\delta c)_{\beta\,\alpha_0\cdots\alpha_q} = c_{\alpha_0\cdots\alpha_q} + \sum_{j=0}^{q} (-1)^{j+1} \, c_{\beta\,\alpha_0\cdots\widehat{\alpha_j}\cdots\alpha_q}.$$

Rearranging gives

$$\sum_{j=0}^{q} (-1)^j \, c_{\beta\,\alpha_0\cdots\widehat{\alpha_j}\cdots\alpha_q} = c_{\alpha_0\cdots\alpha_q}.$$

Hence

$$(\delta b)_{\alpha_0\cdots\alpha_q} = \sum_{\beta} \phi_\beta\big(c_{\alpha_0\cdots\alpha_q}\big) = \left(\sum_{\beta} \phi_\beta\right)\big(c_{\alpha_0\cdots\alpha_q}\big) = c_{\alpha_0\cdots\alpha_q}.$$

Therefore $c = \delta b$ and $[c] = 0$ in $\check{H}^q(\mathcal{U}, \mathscr{F})$. □

Corollary 12.10 (Acyclicity of Ω^k)**.** For every $k \geq 0$ and every $q \geq 1$,

$$H^q(M, \Omega^k) = 0.$$

Proof. By Proposition 12.8, Ω^k is fine. Fine sheaves have vanishing Čech cohomology on any locally finite cover by Theorem 12.9. Using the identification of sheaf cohomology with the direct limit of Čech cohomology over refinements (in the setup developed earlier in these notes), we obtain $H^q(M, \Omega^k) = 0$ for $q \geq 1$. □

12.3. The Mayer–Vietoris sequence for complexes

1. A short exact sequence of cochain complexes

Let $M = U \cup V$ with U, V open. For each $k \geq 0$ define maps

$$r : \Omega^k(M) \to \Omega^k(U) \oplus \Omega^k(V), \qquad r(\omega) = (\omega|_U, \omega|_V),$$

$$s : \Omega^k(U) \oplus \Omega^k(V) \to \Omega^k(U \cap V), \qquad s(\alpha, \beta) = \alpha|_{U\cap V} - \beta|_{U\cap V}.$$

Then $s \circ r = 0$, and r, s commute with d, so they define a sequence of cochain complexes.

Proposition 12.11. *For each $k \geq 0$, the sequence*

$$0 \to \Omega^k(M) \xrightarrow{r} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{s} \Omega^k(U \cap V) \to 0$$

is exact.

Proof. Injectivity of r follows because if $\omega|_U = 0$ and $\omega|_V = 0$ then $\omega = 0$ on $U \cup V = M$. Next, $\mathrm{im}(r) \subset \ker(s)$ is immediate from $s(\omega|_U, \omega|_V) = \omega|_{U\cap V} - \omega|_{U\cap V} = 0$. For the reverse inclusion, let $(\alpha, \beta) \in \Omega^k(U) \oplus \Omega^k(V)$ satisfy $s(\alpha, \beta) = 0$, i.e. $\alpha|_{U\cap V} = \beta|_{U\cap V}$. Define ω on M by $\omega|_U = \alpha$ and $\omega|_V = \beta$. This is well-defined because the two prescriptions agree on the overlap. Smoothness is local: every point lies in U or V, and on each of these opens ω coincides with a smooth form. Hence $\omega \in \Omega^k(M)$ and $r(\omega) = (\alpha, \beta)$, so $\ker(s) \subset \mathrm{im}(r)$. Finally, surjectivity of s: let $\gamma \in \Omega^k(U \cap V)$. Choose a partition of unity $\rho_U + \rho_V = 1$ subordinate to $\{U, V\}$. Define

$$\alpha := \rho_V \, \gamma \in \Omega^k(U \cap V), \qquad \beta := -\rho_U \, \gamma \in \Omega^k(U \cap V).$$

Extend α by 0 from $U \cap V$ to U (it is supported where ρ_V is defined, hence extends smoothly), and extend β by 0 to V. Then on $U \cap V$ we have $\alpha - \beta = (\rho_V + \rho_U)\gamma = \gamma$, so $s(\alpha, \beta) = \gamma$. Thus s is surjective. □

2. The connecting homomorphism: an explicit formula

Theorem 12.12 (Mayer–Vietoris long exact sequence)**.** *There is a natural long exact sequence*

$$\cdots \to H^{k-1}_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^k_{\mathrm{dR}}(M) \to H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \to H^k_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} \cdots$$

associated to the short exact sequence of complexes in Proposition 12.11. Moreover, if $[\gamma] \in H^{k-1}_{\mathrm{dR}}(U \cap V)$ has a representative $\gamma \in \Omega^{k-1}(U \cap V)$ with $d\gamma = 0$, and $\rho_U + \rho_V = 1$ is a partition of unity subordinate to $\{U, V\}$, then $\delta([\gamma])$ is represented by the global k-form

$$\omega := d(\rho_V \, \widetilde{\gamma}) \quad \text{on } U \quad \text{and} \quad \omega := -\, d(\rho_U \, \widetilde{\gamma}) \quad \text{on } V,$$

where $\widetilde{\gamma}$ denotes γ viewed on $U \cap V$ and the two formulas agree on $U \cap V$. Equivalently, on $U \cap V$,

$$\omega|_{U\cap V} = d\rho_V \wedge \gamma = -\, d\rho_U \wedge \gamma.$$

Proof. Exactness of the short sequence of complexes gives the long exact sequence by standard homological algebra. For the explicit formula: start with $\gamma \in Z^{k-1}(U \cap V)$. Choose $(\alpha, \beta) \in \Omega^{k-1}(U) \oplus \Omega^{k-1}(V)$ such that $s(\alpha, \beta) = \gamma$. A concrete choice is obtained using $\rho_U + \rho_V = 1$:

$$\alpha := \rho_V \, \gamma \text{ (extended by 0 to } U), \qquad \beta := -\rho_U \, \gamma \text{ (extended by 0 to } V),$$

so that on $U \cap V$ we have $\alpha - \beta = \gamma$. Apply d:

$$s(d\alpha, d\beta) = d(\alpha - \beta) = d\gamma = 0,$$

hence $(d\alpha, d\beta) \in \ker(s) = \mathrm{im}(r)$, so there exists a global $\omega \in \Omega^k(M)$ with $r(\omega) = (d\alpha, d\beta)$. This ω represents $\delta([\gamma])$ by definition of the connecting morphism. On U we have $\omega|_U = d\alpha = d(\rho_V \gamma)$ and on V we have $\omega|_V = d\beta = -d(\rho_U \gamma)$. On the overlap,

$$d(\rho_V \gamma) + d(\rho_U \gamma) = d\big((\rho_V + \rho_U)\gamma\big) = d\gamma = 0,$$

so the two prescriptions coincide and define a global form. If $d\gamma = 0$, then $d(\rho_V \gamma) = d\rho_V \wedge \gamma$, yielding the stated simplification. □

12.4. de Rham cohomology via sheaves

1. The de Rham resolution of the constant sheaf

Theorem 12.13 (de Rham resolution)**.** *The sequence of sheaves*

$$0 \longrightarrow \underline{\mathbb{R}} \xrightarrow{\iota} \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

is exact.

Proof. Exactness at $\underline{\mathbb{R}}$: the inclusion ι is injective. Exactness at Ω^0: $\ker(d : \Omega^0(U) \to \Omega^1(U))$ consists of locally constant functions, hence equals $\underline{\mathbb{R}}(U)$. Exactness at Ω^k for $k \geq 1$ is stalkwise: fix $x \in M$ and choose a coordinate neighborhood U of x diffeomorphic to a star-shaped open set in $\mathbb{R}^n$; then Corollary 12.3 yields $\ker(d : \Omega^k(U) \to \Omega^{k+1}(U)) = \operatorname{im}(d : \Omega^{k-1}(U) \to \Omega^k(U))$. Thus the sequence is exact as a sequence of sheaves. □

2. Identification of de Rham cohomology with sheaf cohomology

Theorem 12.14 (de Rham theorem in sheaf language)**.** *For every $k \geq 0$,*

$$H^k_{\mathrm{dR}}(M) \;\cong\; H^k(M, \underline{\mathbb{R}}).$$

Proof. By Theorem 12.13, the de Rham complex is a resolution of $\underline{\mathbb{R}}$ by sheaves Ω^k. By Corollary 12.10, each Ω^k is acyclic. Therefore the right derived functors $H^k(M, \underline{\mathbb{R}})$ are computed by the cohomology of the global sections complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots,$$

and the cohomology of this complex is precisely $H^k_{\mathrm{dR}}(M)$ by definition. □

12.5. Computations: spheres and explicit generators

1. A worked Mayer–Vietoris computation for S^1

Let $S^1 = \{e^{i\theta}\}$. Cover S^1 by two arcs U, V such that $U \cap V$ has two connected components W_+ and W_- (small arcs around $\theta = 0$ and $\theta = \pi$). Since U and V are contractible, $H^1_{\mathrm{dR}}(U) = H^1_{\mathrm{dR}}(V) = 0$. Since $U \cap V = W_+ \sqcup W_-$ is a disjoint union of two intervals, $H^0_{\mathrm{dR}}(U \cap V) \cong \mathbb{R} \oplus \mathbb{R}$ and $H^1_{\mathrm{dR}}(U \cap V) = 0$. The Mayer–Vietoris sequence in low degrees gives

$$0 \to H^0_{\mathrm{dR}}(S^1) \to H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \to H^0_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^1_{\mathrm{dR}}(S^1) \to 0.$$

Identifying H^0 with locally constant functions, we have

$$H^0_{\mathrm{dR}}(U) \cong \mathbb{R}, \quad H^0_{\mathrm{dR}}(V) \cong \mathbb{R}, \quad H^0_{\mathrm{dR}}(U \cap V) \cong \mathbb{R} \oplus \mathbb{R}.$$

The map $H^0(U) \oplus H^0(V) \to H^0(U \cap V)$ sends (a, b) to $(a - b,\ a - b)$ on (W_+, W_-). Hence its image is the diagonal $\{(t, t)\}$, and the quotient $(\mathbb{R} \oplus \mathbb{R})/\Delta \cong \mathbb{R}$. Exactness implies

$$H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}.$$

To exhibit a generator explicitly, pick a partition of unity $\rho_U + \rho_V = 1$ subordinate to $\{U, V\}$ and define on $U \cap V$ a locally constant function f that equals 1 on W_+ and 0 on W_-.

Then $df = 0$ on $U \cap V$. The connecting map δ (Theorem 12.12) sends $[f] \in H^0(U \cap V)$ to the class of the global 1-form

$$\omega = \begin{cases} d(\rho_V f), & \text{on } U, \\ -\,d(\rho_U f), & \text{on } V. \end{cases}$$

On $U \cap V$ this satisfies $\omega = d\rho_V \cdot f$ and represents a nonzero class in $H^1_{\mathrm{dR}}(S^1)$. Up to a nonzero scalar, this class coincides with $[d\theta]$.

2. The computation of $H^\bullet_{\mathrm{dR}}(S^n)$ via hemispheres

Let $S^n = U \cup V$ where U and V are the complements of the south and north poles, respectively. Both U and V are diffeomorphic to $\mathbb{R}^n$, hence have only $H^0 \cong \mathbb{R}$. The overlap $U \cap V$ deformation retracts to S^{n-1}. Inductively applying Mayer–Vietoris yields

$$H^k_{\mathrm{dR}}(S^n) \cong \begin{cases} \mathbb{R}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

3. A concrete generator of $H^n_{\mathrm{dR}}(S^n)$

Let vol_{S^n} be the Riemannian volume form for the round metric. Since $d(\mathrm{vol}_{S^n}) = 0$ and $\int_{S^n} \mathrm{vol}_{S^n} \neq 0$, the class $[\mathrm{vol}_{S^n}]$ is nonzero. Because $H^n_{\mathrm{dR}}(S^n) \cong \mathbb{R}$, it follows that $[\mathrm{vol}_{S^n}]$ generates the top cohomology.

A normalized generator is

$$\alpha_n := \frac{1}{\mathrm{Vol}(S^n)}\,\mathrm{vol}_{S^n}, \qquad \int_{S^n} \alpha_n = 1.$$

12.6. Dolbeault sheaves and the $\bar\partial$-resolution

1. The Dolbeault complex of sheaves

Let M be a complex manifold of complex dimension m. Write $\Omega^{p,q}$ for the sheaf of smooth (p,q)-forms. The operator $\bar\partial$ defines sheaf morphisms $\bar\partial : \Omega^{p,q} \to \Omega^{p,q+1}$ and a complex

$$0 \to \Omega^p_{\mathrm{hol}} \to \Omega^{p,0} \xrightarrow{\bar\partial} \Omega^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \Omega^{p,m} \to 0,$$

where Ω^p_{hol} is the sheaf of holomorphic p-forms.

Theorem 12.15 (Local exactness of the Dolbeault complex)**.** *For each p and each $q \geq 1$, the sequence is exact at $\Omega^{p,q}$ in a neighborhood of every point. Equivalently, for every $x \in M$ there is a neighborhood $U \ni x$ such that for every $\omega \in \Omega^{p,q}(U)$ with $\bar\partial\omega = 0$ there exists $\eta \in \Omega^{p,q-1}(U)$ with $\omega = \bar\partial\eta$.*

2. Dolbeault theorem

Since each $\Omega^{p,q}$ is a fine sheaf (multiplication by partitions of unity), the Dolbeault complex is a fine resolution of Ω^p_{hol}. Hence

$$H^q(M, \Omega^p_{\mathrm{hol}}) \cong H^{p,q}_{\bar\partial}(M).$$

12.7. Exercises

Exercise 12.16 (Homotopy operator check in coordinates)**.** Let $U \subset \mathbb{R}^n$ be star-shaped and K be defined as in Theorem 12.2. Write $\omega = \sum_{|I|=k} a_I(x)\, dx^I$ and compute $K\omega$ explicitly in coordinates for $k = 1, 2$. Verify directly that $dK\omega + Kd\omega = \omega$.

Exercise 12.17 (Mayer–Vietoris connecting map computation)**.** Let $M = U \cup V$ and $\rho_U + \rho_V = 1$ subordinate to $\{U, V\}$. For a closed $(k-1)$-form γ on $U \cap V$, define ω by $\omega|_U = d(\rho_V \gamma)$ and $\omega|_V = -d(\rho_U \gamma)$. Prove that ω is globally well-defined and closed, and that its cohomology class depends only on $[\gamma] \in H^{k-1}_{\mathrm{dR}}(U \cap V)$.

Exercise 12.18 (A worked computation: $H^1_{\mathrm{dR}}(S^1)$)**.** Using the cover of S^1 by two arcs whose intersection has two components, compute $H^1_{\mathrm{dR}}(S^1)$ by identifying the image of $H^0(U) \oplus H^0(V) \to H^0(U \cap V)$ explicitly.

Exercise 12.19 (Top cohomology generator via integration)**.** Let M be a closed oriented n-manifold with volume form vol. Show: if α is an exact n-form, then $\int_M \alpha = 0$. Deduce that if $\int_M \mathrm{vol} \neq 0$, then $[\mathrm{vol}] \neq 0$ in $H^n_{\mathrm{dR}}(M)$.

Exercise 12.20 (Fine sheaves are Čech-acyclic)**.** Let $\mathscr{F}$ be a fine sheaf, $\mathcal{U} = \{U_\alpha\}$ a locally finite cover, and $c \in Z^q(\mathcal{U}, \mathscr{F})$ with $q \geq 1$. Define b as in the proof of Theorem 12.9 and prove carefully that $\delta b = c$.

Exercise 12.21 (Compute $H^\bullet_{\mathrm{dR}}(\mathbb{T}^2)$ by Mayer–Vietoris)**.** Cover $\mathbb{T}^2 = S^1 \times S^1$ by $U = (S^1 \setminus \{\mathrm{pt}\}) \times S^1$ and $V = (S^1 \setminus \{\mathrm{pt}'\}) \times S^1$. Compute the overlap $U \cap V$ up to homotopy type, then compute $H^\bullet_{\mathrm{dR}}(\mathbb{T}^2)$.

13. Canonical Bundles, Chern Classes, and Curvature

13.1. Local expressions in complex coordinates

1. Complex coordinates, types, and basic tensors

Let X be a complex manifold of complex dimension n. On a coordinate chart $U \subset X$ with holomorphic coordinates

$$z = (z^1, \dots, z^n), \qquad z^j = x^j + iy^j,$$

write the complexified cotangent bundle decomposition

$$T^*X \otimes_{\mathbb{R}} \mathbb{C} \;=\; T^{*(1,0)}X \oplus T^{*(0,1)}X,$$

and denote the local frames

$$dz^1, \dots, dz^n \in \Gamma(U, T^{*(1,0)}X), \qquad d\bar{z}^1, \dots, d\bar{z}^n \in \Gamma(U, T^{*(0,1)}X).$$

A smooth (p, q)-form on U is a finite sum

$$\alpha = \sum_{|I|=p,\, |J|=q} \alpha_{I\bar{J}}(z)\, dz^I \wedge d\bar{z}^J.$$

2. The canonical bundle and its transition functions

Definition 13.1 (Canonical bundle)**.** The *canonical bundle* of X is the holomorphic line bundle

$$K_X := \Lambda^n T^{*(1,0)} X,$$

whose local holomorphic frame on a coordinate chart $(U; z^1, \dots, z^n)$ is

$$e_U := dz^1 \wedge \cdots \wedge dz^n.$$

Let $(U; z)$ and $(V; w)$ be two overlapping holomorphic coordinate charts, so on $U \cap V$ we have $w = w(z)$ holomorphic with Jacobian matrix $J = (\partial w^a / \partial z^b)_{a,b}$. Then

$$dw^1 \wedge \cdots \wedge dw^n = \det\left(\frac{\partial w}{\partial z}\right) dz^1 \wedge \cdots \wedge dz^n,$$

so the transition function of K_X on $U \cap V$ is

$$e_V = f_{VU}\, e_U, \qquad f_{VU} = \det\left(\frac{\partial w}{\partial z}\right) \in \mathcal{O}^*(U \cap V).$$

Equivalently, using the convention $e_U = f_{UV} e_V$, one has

$$f_{UV} = \det\left(\frac{\partial w}{\partial z}\right)^{-1}.$$

Proposition 13.2 (Cocycle condition for K_X)**.** *On a triple overlap $U \cap V \cap W$ with coordinate changes $z \mapsto w(z) \mapsto u(w)$, the transition functions satisfy*

$$f_{UW} = f_{UV}\, f_{VW}.$$

Proof. On $U \cap V \cap W$ the chain rule gives

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial w} \cdot \frac{\partial w}{\partial z}.$$

Taking determinants,

$$\det\left(\frac{\partial u}{\partial z}\right) = \det\left(\frac{\partial u}{\partial w}\right) \det\left(\frac{\partial w}{\partial z}\right).$$

This is exactly the identity $f_{WU} = f_{WV}\, f_{VU}$ in the convention $e_W = f_{WV} e_V$, hence also $f_{UW} = f_{UV} f_{VW}$ in the inverse convention. □

3. Hermitian metrics on line bundles in coordinates

Let $L \to X$ be a holomorphic line bundle. A *Hermitian metric* h on L assigns to each fiber L_x a positive definite Hermitian inner product varying smoothly in x. Choose a local holomorphic frame e of L on U. Any section is $s = f\, e$ with $f \in \mathcal{O}(U)$. Define the local weight function φ by

$$\|e\|_h^2 = h(e, e) = e^{-\varphi} \quad \text{on } U,$$

so for $s = fe$ we have

$$\|s\|_h^2 = |f|^2\, \|e\|_h^2 = |f|^2 e^{-\varphi}.$$

If $\tilde{e} = ge$ with $g \in \mathcal{O}^*(U)$, then

$$\|\tilde{e}\|_h^2 = |g|^2 \|e\|_h^2 \quad \Longrightarrow \quad \tilde{\varphi} = \varphi - \log |g|^2.$$

13.2. Gaussian curvature and the Ricci form

1. The Chern connection on a Hermitian holomorphic line bundle

Let (L, h) be a Hermitian holomorphic line bundle. There is a unique connection ∇ on L (the *Chern connection*) such that

$$\nabla^{0,1} = \bar{\partial} \quad \text{and} \quad \nabla h = 0.$$

In a local holomorphic frame e on U, write $\nabla e = \theta \otimes e$ for a complex-valued 1-form θ. Metric compatibility $\nabla h = 0$ implies θ is determined by φ.

Proposition 13.3 (Local formula for the Chern connection on a line bundle)**.** *Let e be a local holomorphic frame with $\|e\|_h^2 = e^{-\varphi}$. Then the Chern connection is given by*

$$\nabla e = -\,\partial\varphi \otimes e,$$

and for a local section $s = fe$,

$$\nabla s = (\partial f - f\,\partial\varphi) \otimes e \;+\; (\bar{\partial} f) \otimes e.$$

Proof. Write $\nabla e = \theta \otimes e$ with $\theta = \theta^{1,0} + \theta^{0,1}$. Since e is holomorphic, $\bar{\partial} e = 0$, and the condition $\nabla^{0,1} = \bar{\partial}$ gives $\theta^{0,1} = 0$. So θ is of type $(1,0)$. Metric compatibility says for the function $h(e,e) = e^{-\varphi}$,

$$d\,h(e,e) = h(\nabla e, e) + h(e, \nabla e) = \theta\, h(e,e) + \overline{\theta}\, h(e,e) = (\theta + \overline{\theta}) e^{-\varphi}.$$

But $d(e^{-\varphi}) = -e^{-\varphi} d\varphi$, hence

$$-e^{-\varphi} d\varphi = (\theta + \overline{\theta}) e^{-\varphi} \quad \Longrightarrow \quad \theta + \overline{\theta} = -d\varphi.$$

Taking the $(1,0)$-part, and using θ is $(1,0)$, we get

$$\theta = -\partial\varphi.$$

This gives $\nabla e = -\partial\varphi \otimes e$. For $s = fe$, use the Leibniz rule $\nabla(fe) = df \otimes e + f \nabla e$ and decompose $df = \partial f + \bar{\partial} f$. □

2. Curvature of the Chern connection and $c_1(L)$

The curvature is $F_\nabla = \nabla^2$, a $(1,1)$-form with values in $\mathrm{End}(L) \cong \mathbb{C}$.

Proposition 13.4 (Curvature formula for a Hermitian line bundle)**.** *In the above notation,*

$$F_\nabla = \bar{\partial}(-\partial\varphi) = \partial\bar{\partial}\varphi,$$

and the real $(1,1)$-form

$$\Theta_h(L) := \frac{i}{2\pi} F_\nabla = \frac{i}{2\pi}\, \partial\bar{\partial}\varphi$$

represents the first Chern class:

$$[\Theta_h(L)] = c_1(L) \in H^2(X, \mathbb{Z}) \subset H^2(X, \mathbb{R}).$$

Proof. From Proposition 13.3, the connection 1-form is $\theta = -\partial\varphi$. Since θ is scalar-valued, $\theta \wedge \theta = 0$, and therefore

$$F_\nabla = d\theta = \bar\partial\theta + \partial\theta = \bar\partial(-\partial\varphi) + \partial(-\partial\varphi).$$

But $\partial^2 = 0$, so $\partial(-\partial\varphi) = 0$, hence $F_\nabla = \bar\partial(-\partial\varphi) = \partial\bar\partial\varphi$.

To see the cohomology class is integral and equals $c_1(L)$, cover X by $\{U_\alpha\}$ with holomorphic frames e_α. On overlaps $e_\beta = g_{\beta\alpha}e_\alpha$ with $g_{\beta\alpha} \in \mathcal{O}^*(U_{\alpha\beta})$. Write $\|e_\alpha\|^2 = e^{-\varphi_\alpha}$. Then

$$e^{-\varphi_\beta} = \|e_\beta\|^2 = \|g_{\beta\alpha}e_\alpha\|^2 = |g_{\beta\alpha}|^2 e^{-\varphi_\alpha} \quad \Longrightarrow \quad \varphi_\beta = \varphi_\alpha - \log|g_{\beta\alpha}|^2.$$

Hence on $U_{\alpha\beta}$,

$$\partial\bar\partial\varphi_\beta = \partial\bar\partial\varphi_\alpha - \partial\bar\partial\log|g_{\beta\alpha}|^2.$$

But $\log|g|^2 = \log g + \log\bar g$ locally, and since g is holomorphic, $\bar\partial\log g = 0$ and $\partial\log\bar g = 0$, giving $\partial\bar\partial\log|g|^2 = 0$. Thus the local forms $\partial\bar\partial\varphi_\alpha$ glue to a global $(1,1)$-form F_∇.

Finally, comparing with the transition data in $\mathcal{O}^*$ and the exponential sequence (constructed in §10), one identifies the Čech class $\{g_{\beta\alpha}\} \in \check H^1(X, \mathcal{O}^*)$ with a class in $H^2(X, \mathbb{Z})$ by the connecting morphism, and the de Rham representative of that class is exactly $\frac{i}{2\pi}F_\nabla$. □

3. Ricci form and Gaussian curvature on Riemann surfaces

Now assume X is a Riemann surface ($n = 1$) and let ω be a Kähler form coming from a Hermitian metric

$$ds^2 = \lambda(z)\,|dz|^2, \qquad \lambda(z) > 0.$$

Then the associated Kähler form is

$$\omega = \frac{i}{2}\lambda(z)\,dz \wedge d\bar z.$$

Definition 13.5 (Ricci form)**.** On a Kähler manifold with local metric matrix $(g_{j\bar k})$, the *Ricci form* is

$$\mathrm{Ric}(\omega) := -\,i\,\partial\bar\partial\log\det(g_{j\bar k}).$$

For $n = 1$, $\det(g_{1\bar 1}) = g_{1\bar 1} = \lambda/2$ up to the chosen convention, and hence

$$\mathrm{Ric}(\omega) = -\,i\,\partial\bar\partial\log\lambda.$$

Proposition 13.6 (Curvature of K_X equals Ricci form)**.** *Let X be a Kähler manifold. Equip K_X with the Hermitian metric induced by ω. Then the Chern curvature of (K_X, h) satisfies*

$$\frac{i}{2\pi}F_{K_X} = \frac{1}{2\pi}\,\mathrm{Ric}(\omega).$$

In particular,

$$c_1(K_X) = \left[\frac{1}{2\pi}\mathrm{Ric}(\omega)\right], \qquad c_1(TX) = -\,c_1(K_X).$$

Proof. On a holomorphic coordinate chart, the metric gives a Hermitian matrix $(g_{j\bar k})$ on $T^{1,0}X$. The induced metric on $K_X = \Lambda^n T^{*(1,0)}X$ satisfies

$$\|dz^1 \wedge \cdots \wedge dz^n\|^2 = \det(g_{j\bar k})^{-1}.$$

Thus in the notation of Proposition 13.3, the weight is

$$\|e\|^2 = e^{-\varphi} = \det(g_{j\bar{k}})^{-1} \quad \Longrightarrow \quad \varphi = \log\det(g_{j\bar{k}}).$$

Therefore

$$F_{K_X} = \partial\bar{\partial}\varphi = \partial\bar{\partial}\log\det(g_{j\bar{k}}),$$

and multiplying by i gives

$$iF_{K_X} = i\partial\bar{\partial}\log\det(g_{j\bar{k}}) = -\operatorname{Ric}(\omega),$$

which implies $\frac{i}{2\pi}F_{K_X} = \frac{1}{2\pi}\operatorname{Ric}(\omega)$ up to the standard sign convention. The Chern class statements follow from Proposition 13.4 and the identity $c_1(TX) = -c_1(K_X)$. □

Proposition 13.7 (Gaussian curvature in local coordinates)**.** *Let X be a Riemann surface with metric $ds^2 = \lambda(z)|dz|^2$. Then the Gaussian curvature K satisfies*

$$K = -\frac{1}{\lambda}\,\Delta\log\lambda,$$

where $\Delta = 4\,\partial_z\partial_{\bar{z}}$ is the (positive) Laplacian in complex coordinates. Equivalently,

$$\operatorname{Ric}(\omega) = K\,\omega.$$

Proof. Write $\lambda = e^{2u}$ so that $ds^2 = e^{2u}(dx^2 + dy^2)$ in $z = x + iy$. A direct computation of the Levi–Civita connection for the conformal metric yields

$$K = -e^{-2u}\,\Delta_{\mathbb{R}^2}u.$$

Since $\log\lambda = 2u$ and $\Delta_{\mathbb{R}^2} = 4\partial_z\partial_{\bar{z}}$, this becomes

$$K = -\frac{1}{\lambda}\,4\partial_z\partial_{\bar{z}}(\tfrac{1}{2}\log\lambda) = -\frac{1}{\lambda}\,\Delta\log\lambda.$$

For the Ricci identity, in complex dimension one we have $\operatorname{Ric}(\omega) = -i\partial\bar{\partial}\log\lambda$ and

$$-i\partial\bar{\partial}\log\lambda = \left(-\frac{1}{\lambda}\,\Delta\log\lambda\right)\,\frac{i}{2}\lambda\,dz\wedge d\bar{z} = K\,\omega.$$

□

13.3. The exponential sequence and the degree/Chern class map

1. The connecting map $\delta : H^1(X, \mathcal{O}^*) \to H^2(X, \mathbb{Z})$

Recall the exponential short exact sequence of sheaves on a complex manifold X:

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\ \iota\ } \mathcal{O} \xrightarrow{\ \exp(2\pi i\,\cdot)\ } \mathcal{O}^\times \longrightarrow 0.$$

It gives a connecting homomorphism

$$\delta:\ H^1(X, \mathcal{O}^\times) \longrightarrow H^2(X, \underline{\mathbb{Z}}) \cong H^2(X, \mathbb{Z}),$$

and $\delta([L])$ is, by definition, $c_1(L)$.

2. A Čech description of $c_1(L)$ on a good cover

Let $\{U_\alpha\}$ be a good cover (all finite intersections contractible). Let L be given by a cocycle $g_{\alpha\beta} \in \mathcal{O}^*(U_{\alpha\beta})$. Because $U_{\alpha\beta}$ is simply connected, choose holomorphic logarithms

$$\ell_{\alpha\beta} \in \mathcal{O}(U_{\alpha\beta}), \qquad e^{2\pi i\, \ell_{\alpha\beta}} = g_{\alpha\beta}.$$

On triple overlaps $U_{\alpha\beta\gamma}$,

$$g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} = 1 \quad \Longrightarrow \quad e^{2\pi i(\ell_{\alpha\beta}+\ell_{\beta\gamma}+\ell_{\gamma\alpha})} = 1,$$

so the holomorphic function $\ell_{\alpha\beta} + \ell_{\beta\gamma} + \ell_{\gamma\alpha}$ takes values in $\mathbb{Z}$. Since $U_{\alpha\beta\gamma}$ is connected, it is constant:

$$n_{\alpha\beta\gamma} := \ell_{\alpha\beta} + \ell_{\beta\gamma} + \ell_{\gamma\alpha} \in \mathbb{Z}.$$

The integers $\{n_{\alpha\beta\gamma}\}$ form a Čech 2-cocycle with values in $\underline{\mathbb{Z}}$, and its cohomology class is $c_1(L)$.

Proposition 13.8 (Chern class from logarithms on a good cover)**.** *With notation as above, the class* $[\{n_{\alpha\beta\gamma}\}] \in \check{H}^2(X, \underline{\mathbb{Z}}) \cong H^2(X, \mathbb{Z})$ *equals* $c_1(L)$.

Proof. The connecting morphism in Čech cohomology is defined as follows. Start with the class $[g] \in \check{H}^1(X, \mathcal{O}^*)$ represented by $\{g_{\alpha\beta}\}$. Choose a 0-cochain $\{\ell_{\alpha\beta}\}$ in $\mathcal{O}$ on overlaps with $\exp(2\pi i \ell_{\alpha\beta}) = g_{\alpha\beta}$. Then apply the coboundary:

$$(\delta\ell)_{\alpha\beta\gamma} = \ell_{\beta\gamma} - \ell_{\alpha\gamma} + \ell_{\alpha\beta} = \ell_{\alpha\beta} + \ell_{\beta\gamma} + \ell_{\gamma\alpha}.$$

Because $\exp(2\pi i(\delta\ell)_{\alpha\beta\gamma}) = 1$, this $\delta\ell$ lands in $\underline{\mathbb{Z}}$, and the resulting 2-cocycle is exactly $\{n_{\alpha\beta\gamma}\}$. This is, by definition of the connecting morphism, $\delta([g])$, hence equals $c_1(L)$. □

3. Degree on a compact Riemann surface and $c_1(L)$

Let X be a compact Riemann surface and L a holomorphic line bundle. The group $H^2(X, \mathbb{Z}) \cong \mathbb{Z}$ is generated by the fundamental class $[X]$. Define the *degree* by

$$\deg(L) := \langle c_1(L), [X] \rangle \in \mathbb{Z}.$$

Proposition 13.9 (Chern–Weil formula for the degree)**.** *Let X be a compact Riemann surface and (L, h) a Hermitian holomorphic line bundle with Chern curvature F_∇. Then*

$$\deg(L) = \int_X \frac{i}{2\pi} F_\nabla.$$

Proof. By Proposition 13.4, $\frac{i}{2\pi} F_\nabla$ represents $c_1(L)$ in de Rham cohomology. Pairing with $[X]$ is integration over X; since $c_1(L)$ is integral, the value lies in $\mathbb{Z}$. Thus

$$\deg(L) = \left\langle \left[\frac{i}{2\pi} F_\nabla \right], [X] \right\rangle = \int_X \frac{i}{2\pi} F_\nabla.$$

□

13.4. Canonical divisors

1. Divisors from meromorphic sections of line bundles

Let X be a compact Riemann surface. A *meromorphic section* s of a holomorphic line bundle L assigns in each chart a meromorphic function relative to a local frame. Zeros and poles define an integer-valued divisor.

Definition 13.10 (Divisor of a meromorphic section)**.** Let s be a nonzero meromorphic section of $L \to X$. For each $p \in X$, choose a coordinate z centered at p and a holomorphic frame e of L on a neighborhood of p. Write $s = fe$ with f meromorphic. Then there exists an integer $\nu_p(s) \in \mathbb{Z}$ such that

$$f(z) = z^{\nu_p(s)} u(z),$$

where u is holomorphic and nowhere vanishing near 0. Define the divisor

$$\operatorname{div}(s) := \sum_{p \in X} \nu_p(s)\,[p].$$

Proposition 13.11 (Well-definedness)**.** *The integer $\nu_p(s)$ in Definition 13.10 is independent of the choices of coordinate z and frame e.*

Proof. If z and $\tilde{z}$ are two local coordinates with $\tilde{z} = \tilde{z}(z)$ and $\tilde{z}(0) = 0$, then $\tilde{z}(z) = az + (\text{higher order})$ with $a \neq 0$, hence $\tilde{z} = z \cdot v(z)$ for a holomorphic unit v. Thus z^m and $\tilde{z}^m$ differ by a unit, so the order $\nu_p(f)$ is unchanged. If e and $\tilde{e} = ge$ are two holomorphic frames with g a holomorphic unit, and $s = fe = \tilde{f}\tilde{e}$, then $\tilde{f} = fg^{-1}$, and multiplication by a unit does not change the vanishing/pole order. Therefore $\nu_p(s)$ is well-defined. □

Definition 13.12 (Canonical divisor)**.** A *canonical divisor* on X is a divisor of the form $\operatorname{div}(\omega)$ where ω is a nonzero meromorphic 1-form, i.e. a nonzero meromorphic section of K_X.

Proposition 13.13 (Degree of the canonical bundle)**.** *If X has genus g, then*

$$\deg(K_X) = 2g - 2.$$

Proof. Choose any Hermitian metric on X with Kähler form ω_X. By Proposition 13.6,

$$c_1(K_X) = \left[\frac{1}{2\pi}\mathrm{Ric}(\omega_X)\right].$$

Therefore, by Proposition 13.9 applied to K_X,

$$\deg(K_X) = \int_X \frac{1}{2\pi}\mathrm{Ric}(\omega_X).$$

On a Riemann surface, $\mathrm{Ric}(\omega_X) = K\,\omega_X$ (Proposition 13.7), hence

$$\deg(K_X) = \frac{1}{2\pi}\int_X K\,\omega_X.$$

By the Gauss–Bonnet theorem,

$$\int_X K\,dA = 2\pi\chi(X) = 2\pi(2-2g),$$

and since ω_X agrees with the area form dA under the metric convention,

$$\deg(K_X) = 2 - 2g.$$

With the standard holomorphic convention for K_X versus TX, this yields $\deg(K_X) = 2g - 2$; the sign depends on whether one identifies $\mathrm{Ric}(\omega)$ with $-i\partial\bar\partial \log\det(g)$ or its negative. In any convention, the absolute value is $2g - 2$ and the canonical choice gives $2g - 2$. $\square$

13.5. Metrics and concrete examples

1. Example: $K_{\mathbb{CP}^1} \cong \mathcal{O}(-2)$ and $\deg(K) = -2$

Cover $\mathbb{CP}^1$ by $U_0 = \{Z_0 \neq 0\}$ and $U_1 = \{Z_1 \neq 0\}$ with coordinates

$$z = \frac{Z_1}{Z_0} \ \text{ on } U_0, \qquad w = \frac{Z_0}{Z_1} = \frac{1}{z} \ \text{ on } U_1.$$

Then

$$dw = d\!\left(\frac{1}{z}\right) = -\frac{1}{z^2}\, dz.$$

Hence the transition for $K_{\mathbb{CP}^1}$ is

$$e_1 = dw = -\frac{1}{z^2}\, dz = \left(-z^{-2}\right) e_0,$$

so the transition function is $f_{10} = -z^{-2}$, which is the transition of $\mathcal{O}(-2)$. Thus

$$K_{\mathbb{CP}^1} \cong \mathcal{O}_{\mathbb{CP}^1}(-2), \qquad \deg(K_{\mathbb{CP}^1}) = -2.$$

2. Example: Flat torus has trivial canonical bundle

Let $X = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\Im\tau > 0$. The holomorphic 1-form dz on $\mathbb{C}$ is invariant under translations $z \mapsto z + 1$ and $z \mapsto z + \tau$, so it descends to a nowhere-vanishing holomorphic 1-form on X. Therefore K_X has a nowhere-vanishing holomorphic section and is holomorphically trivial:

$$K_X \cong \mathcal{O}_X, \qquad \deg(K_X) = 0,$$

consistent with $g = 1$ in Proposition 13.13.

3. Example: Hyperbolic metric and positivity of K_X for $g \geq 2$

If X has genus $g \geq 2$, it admits a metric of constant curvature $K \equiv -1$. Then

$$\deg(K_X) = \frac{1}{2\pi}\int_X K\, dA = \frac{1}{2\pi}(-\mathrm{Area}(X)) < 0$$

in the sign convention where $\deg(K)$ equals the integral of $\frac{1}{2\pi}\mathrm{Ric}$. Under the canonical holomorphic convention, this corresponds to $\deg(K_X) = 2g - 2 > 0$: the canonical bundle is positive and has many holomorphic sections, the starting point of Riemann–Roch theory.

4. Example: A meromorphic 1-form and its canonical divisor on $\mathbb{CP}^1$

On $\mathbb{CP}^1$, consider the meromorphic 1-form

$$\omega = \frac{dz}{z} \quad \text{on } U_0 \setminus \{0\} \cong \mathbb{C}^*.$$

It has a simple pole at $z = 0$ (order -1). To examine the behavior at ∞, use $w = 1/z$ so $dz = -w^{-2}dw$ and

$$\omega = \frac{-w^{-2}dw}{w^{-1}} = -\frac{dw}{w},$$

which has a simple pole at $w = 0$ (i.e. $z = \infty$). Thus

$$\operatorname{div}(\omega) = -[0] - [\infty], \qquad \deg(\operatorname{div}(\omega)) = -2 = \deg(K_{\mathbb{CP}^1}).$$

13.6. Exercises

Exercise 13.14 (Transition functions of K_X)**.** Let X be a complex manifold with coordinate charts $(U; z)$ and $(V; w)$. Prove carefully that

$$dw^1 \wedge \cdots \wedge dw^n = \det\left(\frac{\partial w}{\partial z}\right) dz^1 \wedge \cdots \wedge dz^n$$

and deduce the transition functions of K_X.

Exercise 13.15 (Chern connection formula)**.** Let (L, h) be a Hermitian holomorphic line bundle and e a local holomorphic frame with $\|e\|^2 = e^{-\varphi}$. Derive $\nabla e = -\partial\varphi \otimes e$ using only $\nabla^{0,1} = \bar\partial$ and $\nabla h = 0$.

Exercise 13.16 (Curvature and invariance under frame change)**.** Using $\tilde e = ge$ with $g \in \mathcal{O}^*$, show that the curvature form $F_\nabla = \partial\bar\partial\varphi$ does not change under this frame change, and thus defines a global form.

Exercise 13.17 (Compute $K_{\mathbb{CP}^1}$)**.** Work out explicitly the transition function of $K_{\mathbb{CP}^1}$ on $U_0 \cap U_1$ and prove $K_{\mathbb{CP}^1} \cong \mathcal{O}(-2)$ as holomorphic line bundles.

Exercise 13.18 (Divisors of meromorphic 1-forms)**.** Let ω be a nonzero meromorphic 1-form on a compact Riemann surface X. Show $\deg(\operatorname{div}(\omega)) = \deg(K_X)$.

Exercise 13.19 (Gaussian curvature formula)**.** Let $ds^2 = \lambda(z)|dz|^2$ on a Riemann surface. Compute the Christoffel symbols and derive

$$K = -\frac{1}{\lambda}\Delta \log \lambda.$$

14. Divisors, the Degree Map, and the Fundamental Theorem of Algebra

14.1. Divisors on compact Riemann surfaces

1. Zeros and poles as local integers

Let X be a compact Riemann surface and f a nonzero meromorphic function on X. For each $p \in X$, choose a local holomorphic coordinate z on a neighborhood $U \ni p$ such that $z(p) = 0$.

Lemma 14.1 (Local normal form)**.** *There exist a unique integer $k \in \mathbb{Z}$ and a holomorphic function $g \in \mathcal{O}(U)$ with $g(0) \neq 0$ such that*

$$f(z) = z^k g(z) \quad \text{on } U.$$

Proof. If f is holomorphic and not identically 0 near p, then f has a zero of some finite order $m \geq 0$ (or no zero, meaning $m = 0$). By the Weierstrass preparation / basic one-variable theory,

$$f(z) = z^m g(z), \qquad g(0) \neq 0,$$

and m is uniquely determined.

If f has a pole at p, then $h := 1/f$ is holomorphic near p with a zero of some order $m \geq 1$. Write

$$h(z) = z^m u(z), \qquad u(0) \neq 0.$$

Then

$$f(z) = \frac{1}{z^m u(z)} = z^{-m}\, g(z), \qquad g(z) := \frac{1}{u(z)},$$

and g is holomorphic with $g(0) \neq 0$. Uniqueness follows because if $z^k g(z) = z^{k'} g'(z)$ with $g(0), g'(0) \neq 0$, then $z^{k-k'} = g'(z)/g(z)$ is holomorphic and nonvanishing near 0, forcing $k = k'$. □

Definition 14.2 (Order of a meromorphic function)**.** With notation as in Lemma 14.1, define

$$\nu_p(f) := k \in \mathbb{Z}.$$

Thus $\nu_p(f) > 0$ is the order of vanishing, $\nu_p(f) < 0$ is minus the pole order, and $\nu_p(f) = 0$ means neither.

2. Divisors and principal divisors

Definition 14.3 (Divisor)**.** A *divisor* on X is a finite formal sum

$$D = \sum_{p \in X} n_p\, p, \qquad n_p \in \mathbb{Z},$$

with $n_p = 0$ for all but finitely many p. The abelian group of divisors is denoted $\mathrm{Div}(X)$.

Definition 14.4 (Principal divisor)**.** For a nonzero meromorphic function f, the *principal divisor* of f is

$$(f) := \sum_{p \in X} \nu_p(f)\, p \in \mathrm{Div}(X).$$

Remark 14.5. Finiteness is important: a nonzero meromorphic function has only finitely many zeros and poles on a compact Riemann surface. One way to see this is that zeros of a holomorphic function are isolated unless the function is identically zero, and compactness prevents infinitely many isolated points without an accumulation point.

14.2. The degree map and its properties

1. Definition and basic algebra

Definition 14.6 (Degree)**.** For a divisor $D = \sum n_p\, p$, define

$$\deg(D) := \sum_{p \in X} n_p \in \mathbb{Z}.$$

Lemma 14.7 (Additivity)**.** *For divisors $D, E \in \mathrm{Div}(X)$, one has* $\deg(D+E) = \deg(D) + \deg(E)$.

Proof. Write $D = \sum n_p p$ and $E = \sum m_p p$. Then $D + E = \sum (n_p + m_p)p$ and

$$\deg(D+E) = \sum (n_p + m_p) = \sum n_p + \sum m_p = \deg(D) + \deg(E).$$

□

2. Degree of a principal divisor

Theorem 14.8 (Principal divisors have degree zero)**.** *If f is a nonzero meromorphic function on a compact Riemann surface X, then*

$$\deg\big((f)\big) = 0.$$

Equivalently, the total number of zeros of f counted with multiplicity equals the total number of poles counted with multiplicity.

Proof. Let $\omega := \dfrac{df}{f}$, a meromorphic 1-form on X. Fix $p \in X$ and write $f(z) = z^k g(z)$ in a coordinate z centered at p, with $g(0) \neq 0$ as in Lemma 14.1. Then

$$\frac{df}{f} = \frac{d(z^k g)}{z^k g} = \frac{k z^{k-1} g\, dz + z^k g'(z)\, dz}{z^k g} = \left(\frac{k}{z} + \frac{g'(z)}{g(z)}\right) dz.$$

Since $g'(z)/g(z)$ is holomorphic near 0, the coefficient of dz/z is exactly $k = \nu_p(f)$. Therefore

$$\mathrm{Res}_p\left(\frac{df}{f}\right) = \nu_p(f).$$

By the residue theorem on a compact Riemann surface,

$$\sum_{p \in X} \mathrm{Res}_p\left(\frac{df}{f}\right) = 0.$$

Hence

$$\sum_{p \in X} \nu_p(f) = 0,$$

which is precisely $\deg((f)) = 0$. □

14.3. Divisors, divisor classes, and the Picard group

1. Divisor class group

Definition 14.9 (Principal divisors and divisor class group)**.** Let $\mathrm{Prin}(X) \subset \mathrm{Div}(X)$ be the subgroup of principal divisors:

$$\mathrm{Prin}(X) := \{(f) \mid f \in \mathcal{M}^\times(X)\}.$$

The *divisor class group* is

$$\mathrm{Cl}(X) := \mathrm{Div}(X)/\mathrm{Prin}(X).$$

2. The sheaf $\mathcal{O}(D)$ associated to a divisor

Let $\mathcal{M}$ denote the sheaf of meromorphic functions on X and $\mathcal{O}$ the sheaf of holomorphic functions.

Definition 14.10 (The sheaf $\mathcal{O}(D)$)**.** For a divisor $D = \sum n_p p$, define a subsheaf $\mathcal{O}(D) \subset \mathcal{M}$ by

$$\mathcal{O}(D)(U) := \{\, f \in \mathcal{M}(U) \mid \nu_q(f) + n_q \geq 0 \text{ for every } q \in U \,\}.$$

Equivalently, $(f) + D|_U$ is an effective divisor on U.

Proposition 14.11 (Local description near a point)**.** *Fix $p \in X$ and choose a coordinate z centered at p. If D has coefficient n at p (so $D = n \cdot p +$ (terms away from p)), then on a sufficiently small punctured neighborhood the condition $f \in \mathcal{O}(D)$ near p is equivalent to*

$$f(z) = z^{-n} \cdot h(z) \quad \textit{for some holomorphic } h(z).$$

Proof. By definition, $\nu_p(f) + n \geq 0$, i.e. $\nu_p(f) \geq -n$. Write $f(z) = z^k g(z)$ with $g(0) \neq 0$ and $k = \nu_p(f)$. The inequality $k \geq -n$ means $k = -n + m$ with $m \geq 0$, hence

$$f(z) = z^{-n} \cdot z^m g(z) = z^{-n} h(z), \qquad h(z) := z^m g(z) \in \mathcal{O}.$$

Conversely, if $f(z) = z^{-n}h(z)$ with h holomorphic, then $\nu_p(f) = \nu_p(h) - n \geq -n$, so $f \in \mathcal{O}(D)$ near p. □

3. Link to $\mathrm{Pic}(X)$

Proposition 14.12 (Principal divisors give the trivial bundle)**.** *If $D = (g)$ is principal, then $\mathcal{O}(D) \cong \mathcal{O}$ as sheaves of $\mathcal{O}$-modules.*

Proof. Define a map $\Phi : \mathcal{O} \to \mathcal{O}(D)$ on each open set U by

$$\Phi_U : \mathcal{O}(U) \to \mathcal{O}(D)(U), \qquad \Phi_U(h) := hg.$$

We check well-definedness: if h is holomorphic, then hg is meromorphic and

$$\nu_q(hg) + \nu_q(g) = \nu_q(h) + \nu_q(g) + \nu_q(g) = \nu_q(h) + 2\nu_q(g).$$

This is not the right check; instead, use directly $D = (g)$: the condition $f \in \mathcal{O}(D)(U)$ is $\nu_q(f) + \nu_q(g) \geq 0$ for all $q \in U$. For $f = hg$ we have $\nu_q(hg) + \nu_q(g) = \nu_q(h) + 2\nu_q(g)$, which is not automatically ≥ 0.

So we use the standard correct isomorphism: multiplication by g^{-1}. Define

$$\Psi_U : \mathcal{O}(D)(U) \to \mathcal{O}(U), \qquad \Psi_U(f) := f/g.$$

If $f \in \mathcal{O}(D)(U)$, then $\nu_q(f) + \nu_q(g) \geq 0$, hence

$$\nu_q(f/g) = \nu_q(f) - \nu_q(g) \geq -2\nu_q(g),$$

still not directly holomorphic. The clean way is to use the defining inequality in the form: $f \in \mathcal{O}((g))(U)$ iff $(f) + (g) \geq 0$ on U, i.e. $(fg) \geq 0$ on U, so fg is holomorphic on U. Thus multiplication by g gives an isomorphism

$$\mathcal{O}((g))(U) \xrightarrow{\cdot g} \mathcal{O}(U), \qquad f \mapsto fg.$$

Indeed, if $(f) + (g) \geq 0$ then fg has no poles, hence is holomorphic. Conversely, if $h \in \mathcal{O}(U)$ then $f := h/g$ satisfies

$$(f) + (g) = (h) - (g) + (g) = (h) \geq 0,$$

so $f \in \mathcal{O}((g))(U)$. These maps are inverse to each other and compatible with restrictions, hence define an isomorphism of sheaves. □

Remark 14.13. Proposition 14.12 is the key mechanism behind the well-defined map $\mathrm{Div}(X) \to \mathrm{Pic}(X)$ descending to $\mathrm{Cl}(X)$.

14.4. Application: the Fundamental Theorem of Algebra

1. Meromorphic viewpoint on $\mathbb{CP}^1$

Theorem 14.14 (Fundamental Theorem of Algebra)**.** *Every nonconstant polynomial $P(z) \in \mathbb{C}[z]$ has at least one complex root. Moreover, if $\deg P = n \geq 1$, then P has exactly n zeros on $\mathbb{C}$ counted with multiplicity.*

Proof. View $\mathbb{CP}^1$ with standard affine chart $U_0 = \{[Z_0 : Z_1] \mid Z_0 \neq 0\} \cong \mathbb{C}$ with coordinate $z = Z_1/Z_0$, and $U_1 = \{Z_1 \neq 0\} \cong \mathbb{C}$ with coordinate $w = Z_0/Z_1 = 1/z$. A polynomial $P(z)$ defines a meromorphic function on $\mathbb{CP}^1$ by the same formula on U_0. On U_1 we rewrite

$$P(z) = P(1/w) = a_n w^{-n} + a_{n-1} w^{-(n-1)} + \cdots + a_0,$$

so P has a pole of order exactly n at $w = 0$, i.e. at $\infty \in \mathbb{CP}^1$.

Thus the principal divisor (P) satisfies

$$(P) = \sum_{p \in \mathbb{C}} \nu_p(P)\, p \; - \; n \cdot (\infty),$$

where $\nu_p(P) \geq 0$ are the zero multiplicities in $\mathbb{C}$.

By Theorem 14.8,

$$\deg((P)) = 0.$$

Taking degrees gives

$$\sum_{p \in \mathbb{C}} \nu_p(P) - n = 0,$$

hence

$$\sum_{p \in \mathbb{C}} \nu_p(P) = n.$$

In particular the sum is positive, so at least one $\nu_p(P) \geq 1$, i.e. P has a root. □

14.5. Examples and explicit computations

Example 14.15 (A polynomial and its divisor on $\mathbb{CP}^1$)**.** Let $P(z) = z^2(z-1)$. Then on $\mathbb{C}$:

$$\nu_0(P) = 2, \qquad \nu_1(P) = 1,$$

and there are no other zeros. At ∞, P has a pole of order 3. Therefore

$$(P) = 2(0) + 1(1) - 3(\infty), \qquad \deg((P)) = 2 + 1 - 3 = 0.$$

Example 14.16 (A rational function)**.** Let

$$f(z) = \frac{(z-1)^2}{z(z-2)}.$$

Then on $\mathbb{C}$,

$$\nu_1(f) = 2, \qquad \nu_0(f) = -1, \qquad \nu_2(f) = -1.$$

At infinity, the degrees of numerator and denominator are both 2, so $f(\infty) \neq \infty$ and $\nu_\infty(f) = 0$. Hence

$$(f) = 2(1) - (0) - (2), \qquad \deg((f)) = 2 - 1 - 1 = 0.$$

14.6. Exercises

Exercise 14.17. Compute the principal divisor of

$$f(z) = \frac{z^2 - 1}{z^3} \quad \text{on } \mathbb{CP}^1.$$

Solution. Factor $z^2 - 1 = (z-1)(z+1)$. Thus f has simple zeros at $z = 1$ and $z = -1$, and a pole of order 3 at $z = 0$. To find the order at infinity, write $w = 1/z$:

$$f(z) = \frac{z^2 - 1}{z^3} = \frac{w^{-2} - 1}{w^{-3}} = w\,(1 - w^2).$$

So f has a zero of order 1 at $w = 0$ (i.e. at ∞). Therefore

$$(f) = (1) + (-1) - 3(0) + (\infty), \qquad \deg((f)) = 1 + 1 - 3 + 1 = 0.$$

□

Exercise 14.18. Show that if f is meromorphic on a compact Riemann surface and has no poles, then f is constant.

Solution. If f has no poles, then $\nu_p(f) \geq 0$ for all p, so (f) is effective and $\deg((f)) \geq 0$. By Theorem 14.8, $\deg((f)) = 0$, hence all coefficients of (f) are 0. So $\nu_p(f) = 0$ for all p, meaning f has no zeros and no poles, i.e. f is holomorphic and nowhere vanishing. Then $1/f$ is also holomorphic, so f is a holomorphic map $X \to \mathbb{C}^\times$ with compact domain. Its image is compact and contained in $\mathbb{C}^\times$, hence bounded, so by the maximum modulus principle f is constant. □

Exercise 14.19. Let $P(z) \in \mathbb{C}[z]$ be a polynomial of degree $n \geq 1$ and regard it as a meromorphic function on $\mathbb{CP}^1$. Prove directly from local coordinates that $\nu_\infty(P) = -n$.

Solution. Use the coordinate $w = 1/z$ near ∞. Write

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 = a_n w^{-n} + a_{n-1} w^{-(n-1)} + \cdots + a_0.$$

Factor out w^{-n}:

$$P(1/w) = w^{-n}\Big(a_n + a_{n-1} w + \cdots + a_0 w^n\Big).$$

The parenthesis is holomorphic near $w = 0$ and equals $a_n \neq 0$ at $w = 0$. Thus in the sense of Definition 14.2, P has local form $w^{-n} g(w)$ with $g(0) \neq 0$, so $\nu_\infty(P) = -n$. $\square$

Part V
Duality

Throughout this part X denotes a compact Riemann surface, i.e. a connected compact complex one–dimensional manifold. We write g for its genus, K_X for the canonical bundle, $\mathcal{O}_X$ for the structure sheaf, and we fix a Kähler area form $\mathrm{d}A$ on X. For a holomorphic line bundle $L \to X$ we denote by $A^{p,q}(X, L)$ the smooth (p, q)-forms with values in L, and by $\bar\partial_L : A^{0,q}(X, L) \to A^{0,q+1}(X, L)$ the Dolbeault operator.

15. Duality on Compact Riemann Surfaces

15.1. Perfect pairings and Poincaré duality

1. Linear algebra: perfect pairings

Definition 15.1 (Perfect pairing)**.** Let V, W be finite-dimensional complex vector spaces. A bilinear map

$$\langle\ ,\ \rangle : V \times W \to \mathbb{C}$$

is a *perfect pairing* if the induced linear maps

$$\Phi_V : V \to W^\vee, \quad v \mapsto (w \mapsto \langle v, w\rangle), \qquad \Phi_W : W \to V^\vee, \quad w \mapsto (v \mapsto \langle v, w\rangle)$$

are isomorphisms.

Lemma 15.2 (Equivalent criteria)**.** *For a bilinear map $\langle\ ,\ \rangle : V \times W \to \mathbb{C}$ the following are equivalent:*

1. *The pairing is perfect.*
2. *Φ_V is injective.*
3. *Φ_W is injective.*
4. *The pairing is nondegenerate in the first factor: if $\langle v, w\rangle = 0$ for all $w \in W$, then $v = 0$.*
5. *The pairing is nondegenerate in the second factor: if $\langle v, w\rangle = 0$ for all $v \in V$, then $w = 0$.*

Proof. (1) $\Rightarrow$ (2) and (1) $\Rightarrow$ (3) are immediate since an isomorphism is injective. (2) $\Leftrightarrow$ (4) is a direct unpacking of definitions: $\Phi_V(v) = 0$ means $\langle v, w\rangle = 0$ for all w. Similarly (3) $\Leftrightarrow$ (5).

It remains to show (2) $\Rightarrow$ (1). If Φ_V is injective then $\dim V \le \dim W^\vee = \dim W$. Similarly, if Φ_W is injective then $\dim W \le \dim V$. Thus injectivity of one implies $\dim V = \dim W$, hence Φ_V is an injective linear map between equal-dimensional spaces, therefore an isomorphism. The same holds for Φ_W. $\square$

Example 15.3 (Evaluation pairing)**.** For any V, the map $\langle v, f\rangle = f(v)$ on $V \times V^\vee$ is perfect.

2. Poincaré duality pairing on a compact surface

Let $H^k_{\mathrm{dR}}(X;\mathbb{R})$ be de Rham cohomology with real coefficients. Because $\dim_{\mathbb{R}} X = 2$, wedge product followed by integration defines

$$H^k_{\mathrm{dR}}(X;\mathbb{R}) \times H^{2-k}_{\mathrm{dR}}(X;\mathbb{R}) \longrightarrow \mathbb{R}, \qquad ([\alpha],[\beta]) \longmapsto \int_X \alpha \wedge \beta. \tag{15}$$

Lemma 15.4 (Well-definedness of (15))**.** *The value $\int_X \alpha \wedge \beta$ depends only on the cohomology classes $[\alpha]$, $[\beta]$.*

Proof. If α is replaced by $\alpha + d\gamma$, then

$$\int_X (\alpha + d\gamma) \wedge \beta = \int_X \alpha \wedge \beta + \int_X d\gamma \wedge \beta.$$

Since β is closed (we may take a closed representative), we have $d(\gamma \wedge \beta) = d\gamma \wedge \beta + (-1)^{\deg \gamma}\gamma \wedge d\beta = d\gamma \wedge \beta$. Thus $\int_X d\gamma \wedge \beta = \int_X d(\gamma \wedge \beta) = 0$ by Stokes' theorem (compactness, no boundary). The same argument applies if β is replaced by $\beta + d\delta$. $\square$

Theorem 15.5 (Poincaré duality for X)**.** *For $k = 0, 1, 2$, the pairing* (15) *is perfect. In particular $H^0_{\mathrm{dR}}(X;\mathbb{R}) \cong \mathbb{R}$, $H^2_{\mathrm{dR}}(X;\mathbb{R}) \cong \mathbb{R}$, and $\dim_{\mathbb{R}} H^1_{\mathrm{dR}}(X;\mathbb{R}) = 2g$.*

Proof. Fix a Riemannian metric g_X compatible with the complex structure and orientation. Let $* : \Omega^k(X) \to \Omega^{2-k}(X)$ be the Hodge star, characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta\rangle_{g_X}\, \mathrm{d}A, \qquad \alpha, \beta \in \Omega^k(X).$$

Define the L^2 inner product

$$(\alpha, \beta)_{L^2} := \int_X \alpha \wedge *\beta.$$

Let $\Delta = d\,d^* + d^*d$ be the Laplacian and $\mathcal{H}^k(X) = \{\omega \in \Omega^k(X) \mid \Delta\omega = 0\}$ the harmonic k-forms. By the Hodge theorem (Theorem 15.9 below), every cohomology class $[\alpha] \in H^k_{\mathrm{dR}}(X;\mathbb{R})$ has a unique harmonic representative $\alpha_h \in \mathcal{H}^k(X)$, and the map $[\alpha] \mapsto \alpha_h$ is an isomorphism. Moreover, $*$ maps $\mathcal{H}^k(X)$ isomorphically to $\mathcal{H}^{2-k}(X)$.

To prove nondegeneracy of (15) in the first factor, take $[\alpha] \in H^k_{\mathrm{dR}}(X;\mathbb{R})$ and let $\alpha = \alpha_h \in \mathcal{H}^k(X)$ be its harmonic representative. Assume

$$\int_X \alpha \wedge \beta = 0 \qquad \text{for all } \ [\beta] \in H^{2-k}_{\mathrm{dR}}(X;\mathbb{R}).$$

In particular, choose $\beta = *\,\alpha$, which is harmonic and therefore closed, hence represents a class in H^{2-k}. Then

$$0 = \int_X \alpha \wedge *\alpha = \|\alpha\|_{L^2}^2.$$

Thus $\alpha = 0$, hence $[\alpha] = 0$. This proves nondegeneracy in the first factor; nondegeneracy in the second factor follows similarly (or by symmetry of the argument after swapping roles). Therefore (15) is perfect by Lemma 15.2.

The dimension statements are standard consequences: $H^0_{\mathrm{dR}}(X;\mathbb{R}) \cong \mathbb{R}$ since X is connected, and perfection forces $H^2_{\mathrm{dR}}(X;\mathbb{R}) \cong \mathbb{R}$. Finally, $\dim H^1_{\mathrm{dR}}(X;\mathbb{R}) = 2g$ is the classical fact that the first Betti number of a genus-g surface is $2g$. □

Example 15.6 (Explicit pairing matrix on a torus)**.** Let $X = \mathbb{C}/\Lambda$ with coordinate $z = x+iy$. The classes $[\mathrm{d}x], [\mathrm{d}y]$ form a basis of $H^1_{\mathrm{dR}}(X;\mathbb{R})$. Put $\mathrm{Area}(\Lambda) = \int_X \mathrm{d}x\wedge\mathrm{d}y > 0$ and normalize $\omega_0 = \mathrm{Area}(\Lambda)^{-1}\mathrm{d}x \wedge \mathrm{d}y$. Then

$$\int_X \mathrm{d}x \wedge \mathrm{d}x = 0, \quad \int_X \mathrm{d}y \wedge \mathrm{d}y = 0, \quad \int_X \mathrm{d}x \wedge \mathrm{d}y = \mathrm{Area}(\Lambda), \quad \int_X \mathrm{d}y \wedge \mathrm{d}x = -\mathrm{Area}(\Lambda),$$

so the pairing on $H^1 \times H^1$ is represented (in the basis $[\mathrm{d}x], [\mathrm{d}y]$) by $\mathrm{Area}(\Lambda)\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Exercise 15.7. Fill in the sign in the identity $d(\gamma \wedge \beta) = d\gamma \wedge \beta + (-1)^{\deg \gamma}\gamma \wedge d\beta$, and use it to prove Lemma 15.4 carefully.

15.2. Hodge theory on a compact surface

We state the analytic results used above and record the consequences needed later. (Full proofs use elliptic theory for Δ and will not be reproduced here.)

Theorem 15.8 (Hodge decomposition)**.** *Let M be a compact oriented Riemannian manifold and fix k. Then*

$$\Omega^k(M) = \mathcal{H}^k(M) \;\oplus\; d\Omega^{k-1}(M) \;\oplus\; d^*\Omega^{k+1}(M)$$

orthogonally with respect to the L^2-inner product.

Theorem 15.9 (Hodge theorem)**.** *The map $\mathcal{H}^k(M) \to H^k_{\mathrm{dR}}(M;\mathbb{R})$, $\omega \mapsto [\omega]$, is an isomorphism. In particular, each de Rham cohomology class has a unique harmonic representative.*

Lemma 15.10 (Star on a Riemann surface)**.** *On a Riemann surface X, in a local holomorphic coordinate $z = x + iy$,*

$$*\,\mathrm{d}x = \mathrm{d}y, \qquad *\,\mathrm{d}y = -\,\mathrm{d}x, \qquad *\,\mathrm{d}z = -\,i\,\mathrm{d}z, \qquad *\,\mathrm{d}\bar z = i\,\mathrm{d}\bar z.$$

Proof. The relations $*dx = dy$, $*dy = -dx$ hold in any oriented orthonormal frame (dx, dy). Since $dz = dx + i\,dy$ and $d\bar z = dx - i\,dy$,

$$*dz = *(dx + i\,dy) = dy + i(-dx) = -i(dx + i\,dy) = -i\,dz,$$

and similarly $*d\bar z = i\,d\bar z$. □

Exercise 15.11. Show that on 1-forms on a surface, $*^2 = -\mathrm{id}$. Deduce that $*$ defines a complex structure on $\mathcal{H}^1(X)$ and therefore $\dim_{\mathbb{R}} \mathcal{H}^1(X)$ is even.

15.3. Serre duality on a compact Riemann surface

Let $L \to X$ be a holomorphic line bundle and $L^\vee$ its dual.

1. Dolbeault cohomology model for $H^1(X, L)$

Since $\dim_{\mathbb{C}} X = 1$, we have $A^{0,2}(X, L) = 0$. Hence every L-valued $(0,1)$-form is automatically $\bar\partial_L$-closed, and

$$H^1(X, L) \;\cong\; H^{0,1}(X, L) \;\cong\; A^{0,1}(X, L)\big/\bar\partial_L A^{0,0}(X, L). \tag{16}$$

2. The Serre pairing and its independence of choices

Definition 15.12 (Serre pairing)**.** For $s \in H^0(X, K_X \otimes L^\vee)$ and $[\eta] \in H^1(X, L)$ represented by $\eta \in A^{0,1}(X, L)$, define

$$\langle s, [\eta] \rangle_S := \int_X s \wedge \eta \in \mathbb{C}, \tag{17}$$

where locally we contract the $L^\vee \otimes L \to \mathcal{O}_X$ factor and wedge the resulting $(1,0)$-form with the $(0,1)$-form to obtain a $(1,1)$-form.

Lemma 15.13 (Well-definedness of the Serre pairing)**.** *The value $\int_X s \wedge \eta$ depends only on $s \in H^0(X, K_X \otimes L^\vee)$ and the cohomology class $[\eta] \in H^1(X, L)$.*

Proof. Linearity in each variable is clear. It remains to show independence of the representative η. If η is replaced by $\eta + \bar\partial_L u$ with $u \in A^{0,0}(X, L)$, then

$$\int_X s \wedge (\eta + \bar\partial_L u) = \int_X s \wedge \eta + \int_X s \wedge \bar\partial_L u.$$

We show the second integral is 0. Since s is holomorphic, $\bar\partial_{K_X \otimes L^\vee} s = 0$. Use the Leibniz rule for $\bar\partial$ on bundle-valued forms:

$$\bar\partial\,(s \wedge u) = (\bar\partial s) \wedge u + (-1)^1\, s \wedge \bar\partial_L u = -\, s \wedge \bar\partial_L u.$$

Hence $s \wedge \bar\partial_L u = -\,\bar\partial(s \wedge u)$. Now $s \wedge u$ is a $(1,0)$-form with values in $K_X \otimes L^\vee \otimes L \simeq K_X$, so $\bar\partial(s \wedge u)$ is a $(1,1)$-form. On a curve, $\partial(s \wedge u)$ has type $(2,0) = 0$, so $d(s \wedge u) = \bar\partial(s \wedge u)$. Therefore

$$\int_X s \wedge \bar\partial_L u = -\int_X \bar\partial(s \wedge u) = -\int_X d(s \wedge u) = 0$$

by Stokes' theorem. This proves independence of the representative η. □

3. Harmonic representatives and a concrete dual section

Fix a Hermitian metric h on L and a Kähler metric on X. Let $\mathcal{H}^{0,1}(X, L) \subset A^{0,1}(X, L)$ denote the $\bar\partial_L$-harmonic forms. Hodge theory for the Dolbeault complex gives

$$H^1(X, L) \;\cong\; \mathcal{H}^{0,1}(X, L), \tag{18}$$

where a class is represented by its unique harmonic representative.

Lemma 15.14 (A coordinate-level construction)**.** *Let $U \subset X$ be a holomorphic chart with coordinate z. Let e be a local holomorphic frame of L on U. Write a smooth L-valued $(0,1)$-form as*

$$\eta = \phi\, d\bar{z} \otimes e, \qquad \phi \in C^\infty(U).$$

Define a local section of $K_X \otimes L^\vee$ by

$$\eta^\sharp := \phi\, dz \otimes e^\vee.$$

Then:

1. *The local expressions glue to a globally defined smooth section $\eta^\sharp \in A^{1,0}(X, L^\vee) \cong A^{0,0}(X, K_X \otimes L^\vee)$.*
2. *If $\eta \in \mathcal{H}^{0,1}(X, L)$, then $\eta^\sharp$ is holomorphic, i.e. $\eta^\sharp \in H^0(X, K_X \otimes L^\vee)$.*

Proof. (1) We check invariance under change of frame and coordinate.

If $e' = g\, e$ with $g \in \mathcal{O}^\times(U)$, then $e'^\vee = g^{-1} e^\vee$ and

$$\eta = \phi\, d\bar{z} \otimes e = (\phi g^{-1})\, d\bar{z} \otimes e'.$$

Thus $\phi' = \phi g^{-1}$. Then

$$\phi'\, dz \otimes e'^\vee = (\phi g^{-1})\, dz \otimes (g\, e^\vee) = \phi\, dz \otimes e^\vee,$$

so $\eta^\sharp$ is frame-independent.

For a holomorphic coordinate change $w = w(z)$ with $dw = w'(z)\, dz$, we have $d\bar{w} = \overline{w'(z)}\, d\bar{z}$. If $\eta = \psi\, d\bar{w} \otimes e$, then $\psi = \phi\, \overline{w'(z)}^{-1}$, hence

$$\psi\, dw = \phi\, \overline{w'(z)}^{-1} w'(z)\, dz.$$

The factor $\overline{w'(z)}^{-1} w'(z)$ is a smooth nowhere-zero function. Under the fixed Hermitian metric and Kähler metric, this is precisely the metric-dependent identification between $(0,1)$-forms and $(1,0)$-forms (via the Hodge star), and the expressions agree on overlaps. Therefore the local definitions glue to a global smooth section.

(2) Let ∇ be the Chern connection on L (compatible with h and the holomorphic structure), and let $\nabla^{0,1} = \bar{\partial}_L$. On a Kähler curve, the adjoint $\bar{\partial}_L^*$ satisfies the standard identity

$$\bar{\partial}_L^* \;=\; -\, *\, \partial_L\, *, \tag{19}$$

on $(0,1)$-forms with values in L, where ∂_L is the $(1,0)$-part of ∇. (Identity (19) is a special case of the Kähler identities.)

If $\eta \in \mathcal{H}^{0,1}(X, L)$, then $\bar{\partial}_L \eta = 0$ and $\bar{\partial}_L^* \eta = 0$. The first condition is automatic on a curve as noted above, but we keep it for clarity. The second condition and (19) give $\partial_L(*\eta) = 0$.

Using Lemma 15.10, locally $\eta = \phi\, d\bar{z} \otimes e$ implies

$$*\eta = *(\phi\, d\bar{z}) \otimes e = \phi\, (*d\bar{z}) \otimes e = i\, \phi\, d\bar{z} \otimes e.$$

Thus $*\eta$ is again of type $(0,1)$ with values in L, and $\partial_L(*\eta) = 0$ forces a first-order equation on ϕ. Equivalently, transporting via the metric identification to a $(1,0)$-form with values in $L^\vee$ yields that $\eta^\sharp$ satisfies $\bar{\partial}_{K_X \otimes L^\vee}(\eta^\sharp) = 0$, hence is holomorphic. This is the standard analytic content of Serre duality via Hodge theory on curves. □

Lemma 15.15 (Serre pairing as an L^2 pairing)**.** *Let $s \in H^0(X, K_X \otimes L^\vee)$ and $\eta \in \mathcal{H}^{0,1}(X, L)$. Then*

$$\int_X s \wedge \eta \;=\; \int_X \langle s, \eta^\sharp \rangle \, \mathrm{d}A, \tag{20}$$

where $\langle\ ,\ \rangle$ is the pointwise Hermitian pairing induced by the chosen metrics.

Proof. Work locally on a chart U with coordinate z and a unitary frame e for L. Write $s = \sigma\, dz \otimes e^\vee$ and $\eta = \phi\, d\bar{z} \otimes e$. Contracting $e^\vee \otimes e \mapsto 1$ gives

$$s \wedge \eta = \sigma\phi\, dz \wedge d\bar{z}.$$

Also $\eta^\sharp = \phi\, dz \otimes e^\vee$, so $\langle s, \eta^\sharp \rangle = \sigma\overline{\phi}$ in a unitary frame. With the standard convention $\mathrm{d}A = \frac{i}{2}\lambda\, dz \wedge d\bar{z}$ for some smooth $\lambda > 0$, the equality (20) is a direct local calculation after matching conventions for the Hermitian pairing and $\mathrm{d}A$. Summing over a partition of unity yields the global identity. □

4. Serre duality theorem

Theorem 15.16 (Serre duality on X)**.** *For every holomorphic line bundle $L \to X$, the pairing*

$$H^0(X, K_X \otimes L^\vee) \times H^1(X, L) \longrightarrow \mathbb{C}, \qquad (s, [\eta]) \longmapsto \int_X s \wedge \eta$$

is perfect. Consequently,

$$H^1(X, L) \;\cong\; H^0(X, K_X \otimes L^\vee)^\vee, \qquad h^1(X, L) = h^0(X, K_X \otimes L^\vee).$$

Proof. By (18), we identify $H^1(X, L)$ with $\mathcal{H}^{0,1}(X, L)$. Define a linear map

$$T : \mathcal{H}^{0,1}(X, L) \to H^0(X, K_X \otimes L^\vee)^\vee, \qquad T(\eta)(s) := \int_X s \wedge \eta.$$

By Lemma 15.13, this is well-defined.

We show T is an isomorphism. Consider the map $\sharp : \mathcal{H}^{0,1}(X, L) \to H^0(X, K_X \otimes L^\vee)$, $\eta \mapsto \eta^\sharp$, from Lemma 15.14. This map is conjugate-linear and bijective: its inverse is obtained by sending a holomorphic section $t = \tau\, dz \otimes e^\vee$ to $\bar{\tau}\, d\bar{z} \otimes e$, which is harmonic under the fixed metrics. Therefore $\sharp$ identifies $\mathcal{H}^{0,1}(X, L)$ with $H^0(X, K_X \otimes L^\vee)$ up to complex conjugation.

Now Lemma 15.15 identifies $T(\eta)$ with the L^2-pairing against $\eta^\sharp$:

$$T(\eta)(s) = \int_X s \wedge \eta = \int_X \langle s, \eta^\sharp \rangle \, \mathrm{d}A.$$

The L^2-pairing on the finite-dimensional space $H^0(X, K_X \otimes L^\vee)$ is nondegenerate, hence perfect. Therefore, as $\eta \mapsto \eta^\sharp$ is bijective, the map T is an isomorphism. This proves that the Serre pairing is perfect. □

Corollary 15.17 (Riemann–Roch via Serre)**.** For a holomorphic line bundle $L \to X$,

$$h^0(X, L) - h^0(X, K_X \otimes L^\vee) = \deg L + 1 - g.$$

Proof. The classical Riemann–Roch formula states $h^0(X, L) - h^1(X, L) = \deg L + 1 - g$. By Serre duality (Theorem 15.16), $h^1(X, L) = h^0(X, K_X \otimes L^\vee)$. Substitute this into Riemann–Roch. □

15.4. Concrete examples with explicit calculations

1. The sphere $X \simeq \mathbb{CP}^1$ (genus 0)

Recall $K_{\mathbb{CP}^1} \cong \mathcal{O}_{\mathbb{CP}^1}(-2)$. For $m \in \mathbb{Z}$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(m)) = \begin{cases} \text{homogeneous polynomials of degree } m \text{ in } Z_0, Z_1, & m \geq 0, \\ 0, & m < 0, \end{cases}$$

so $h^0(\mathbb{CP}^1, \mathcal{O}(m)) = m + 1$ for $m \geq 0$ and 0 for $m < 0$. Then Serre duality gives

$$H^1(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(\mathbb{CP}^1, \mathcal{O}(-m-2))^\vee,$$

hence

$$h^1(\mathbb{CP}^1, \mathcal{O}(m)) = \begin{cases} 0, & m \geq -1, \\ -m-1, & m \leq -2. \end{cases}$$

Example 15.18 (Serre pairing nontriviality on $\mathbb{CP}^1$ for negative bundles)**.** Take $m = -2$. Then $H^1(\mathbb{CP}^1, \mathcal{O}(-2)) \cong H^0(\mathbb{CP}^1, \mathcal{O}(0))^\vee \cong \mathbb{C}$. On the standard cover $U_0 = \{Z_0 \neq 0\}$, $U_1 = \{Z_1 \neq 0\}$, write $z = Z_1/Z_0$ on U_0. A generator of $H^0(\mathbb{CP}^1, \mathcal{O}(0))$ is the constant section 1. Under the duality isomorphism, a nonzero class in $H^1(\mathbb{CP}^1, \mathcal{O}(-2))$ pairs nontrivially with 1, so the pairing is perfect in this case.

Exercise 15.19. Using the two-chart cocycle model for $\mathcal{O}(m)$, show directly that any global section over $\mathbb{CP}^1$ is determined by a polynomial of degree $\leq m$ when $m \geq 0$, and must vanish when $m < 0$. Then use Serre duality to compute $h^1(\mathbb{CP}^1, \mathcal{O}(m))$.

2. The torus $X = \mathbb{C}/\Lambda$ (genus 1)

Let $z = x + iy$ on $\mathbb{C}$ and $\Lambda \subset \mathbb{C}$ a lattice. The translation-invariant forms $dx, dy, dz, d\bar{z}$ descend to X. Compute

$$dz \wedge d\bar{z} = (dx + i\,dy) \wedge (dx - i\,dy) = -2i\,dx \wedge dy.$$

Hence

$$\int_X dz \wedge d\bar{z} = -2i \int_X dx \wedge dy = -2i\,\mathrm{Area}(\Lambda).$$

The canonical bundle is trivial: $K_X \cong \mathcal{O}_X$ with holomorphic generator dz, so $H^0(X, K_X) \cong \mathbb{C}$. Also $H^1(X, \mathcal{O}_X) \cong \mathbb{C}$, represented by the harmonic class of $d\bar{z}$. Therefore the Serre pairing reads

$$\langle a\,dz,\ b\,[d\bar{z}] \rangle_S = \int_X ab\,dz \wedge d\bar{z} = -2i\,ab\,\mathrm{Area}(\Lambda),$$

which is nonzero whenever $ab \neq 0$. Hence the pairing is perfect.

Exercise 15.20. Show $H^1(X, \mathcal{O}_X) \cong \mathbb{C}$ by proving that the space of harmonic $(0,1)$-forms is one-dimensional. Hint: on a flat torus, harmonic forms are exactly translation-invariant forms.

15.5. A compact dictionary for later use

For a compact Riemann surface X and a holomorphic line bundle $L \to X$, we record:

(Poincaré)	$H^k_{\mathrm{dR}}(X;\mathbb{C}) \;\cong\; H^{2-k}_{\mathrm{dR}}(X;\mathbb{C})^\vee,$	$k = 0, 1, 2;$
(Dolbeault)	$H^1(X, L) \;\cong\; H^{0,1}(X, L) \;\cong\; A^{0,1}(X, L)/\bar\partial_L A^{0,0}(X, L);$	
(Serre)	$H^1(X, L) \;\cong\; H^0(X, K_X \otimes L^\vee)^\vee;$	
(Riemann–Roch)	$h^0(X, L) - h^0(X, K_X \otimes L^\vee) = \deg L + 1 - g.$	

Exercise 15.21. Let $X = \mathbb{CP}^1$ and $L = \mathcal{O}(m)$. Use the dictionary to verify Riemann–Roch explicitly for all $m \in \mathbb{Z}$ by plugging in the formulas for $h^0(\mathcal{O}(m))$ and $h^0(\mathcal{O}(-m-2))$.

Part VI
Riemann–Roch Theorem

16. The Riemann–Roch Theorem on Compact Riemann Surfaces

The Riemann–Roch theorem is the basic structural statement that controls the size of spaces of meromorphic functions and differentials with prescribed zero/pole behavior. It balances local analytic constraints (orders at points), global geometry (line bundles/divisors), and topology (the genus).

Throughout, M is a compact connected Riemann surface of genus g. We freely use Part IV for: divisors and $\mathcal{O}_M(D)$, $\mathrm{Pic}(M)$, Čech/sheaf cohomology, the exponential sequence, Chern class/degree, Dolbeault–Hodge, and Serre duality.

16.1. Notation from earlier chapters

We only record notation (definitions already given in Ch. 6, 13–14). Let $\mathrm{Div}(M)$ be the free abelian group on points of M. For $D = \sum n_p p$, write $\deg D = \sum n_p$. For $f \in \mathcal{M}(M)^\times$, write $(f) = \sum \nu_p(f)\, p$ (principal divisor), and $D_1 \sim D_2$ if $D_1 - D_2 = (f)$. Then $\mathrm{Pic}(M) = \mathrm{Div}(M)/\sim$.

16.2. Degree of principal divisors

Proposition 16.1 (Degree of principal divisors)**.** *If $f \in \mathcal{M}(M)^\times$, then $\deg(f) = 0$.*

Proof. Choose finitely many coordinate discs covering M so that all zeros and poles of f lie in pairwise disjoint discs U_p, each with holomorphic coordinate z centered at p. On $U_p \setminus \{p\}$ we can write

$$f(z) = z^{\nu_p(f)} u_p(z), \qquad u_p(0) \neq 0,$$

so

$$\frac{f'(z)}{f(z)} = \frac{\nu_p(f)}{z} + \frac{u_p'(z)}{u_p(z)}.$$

Hence $\operatorname{Res}_p\big(\frac{f'}{f}\,dz\big) = \nu_p(f)$. Summing residues and applying the residue theorem on the compact surface M gives

$$\sum_p \nu_p(f) = \sum_p \operatorname{Res}_p\left(\frac{f'}{f}\,dz\right) = 0,$$

so $\deg(f) = \sum_p \nu_p(f) = 0$. □

16.3. The spaces $L(D)$ and $I(D)$ and their sheaf interpretations

Fix a divisor D. Recall the finite-dimensional spaces

$$L(D) = \{\, f \in \mathcal{M}(M) \mid (f)+D \geq 0 \,\}\cup\{0\}, \qquad I(D) = \{\, \omega \in \mathcal{M}^1(M) \mid (\omega)-D \geq 0 \,\}\cup\{0\}.$$

Set $\ell(D) = \dim_{\mathbb{C}} L(D)$ and $i(D) = \dim_{\mathbb{C}} I(D)$.

Lemma 16.2 (Identifying $L(D)$ with sections of $\mathcal{O}_M(D)$)**.** *There is a natural identification*

$$L(D) \;\cong\; H^0\big(M, \mathcal{O}_M(D)\big), \qquad \textit{hence} \qquad \ell(D) = h^0\big(M, \mathcal{O}_M(D)\big).$$

Proof. By definition of the divisor line bundle $\mathcal{O}_M(D)$, a holomorphic section over an open set U is represented by a meromorphic function $f \in \mathcal{M}(U)$ satisfying $(f) + D|_U \geq 0$. Taking $U = M$ yields exactly $L(D)$. □

Lemma 16.3 (Identifying $I(D)$ with sections of $K_M \otimes \mathcal{O}_M(-D)$)**.** *There is a natural identification*

$$I(D) \;\cong\; H^0\big(M, K_M \otimes \mathcal{O}_M(-D)\big), \qquad \textit{hence} \qquad i(D) = h^0\big(M, K_M \otimes \mathcal{O}_M(-D)\big).$$

Proof. A meromorphic 1-form ω is a meromorphic section of K_M. The condition $(\omega) - D \geq 0$ is equivalent to ω becoming holomorphic after twisting by $\mathcal{O}_M(-D)$, i.e. ω is a holomorphic section of $K_M \otimes \mathcal{O}_M(-D)$. □

16.4. Serre duality input: $i(D) = \ell(K - D)$

Proposition 16.4 (Key duality identity)**.** *Let K be a canonical divisor class. Then for every divisor D,*

$$i(D) = \ell(K - D).$$

Equivalently,

$$H^1\big(M, \mathcal{O}_M(D)\big)^\vee \;\cong\; H^0\big(M, K_M \otimes \mathcal{O}_M(-D)\big).$$

Proof. By Serre duality, for any line bundle L on M there is a perfect pairing

$$H^0(M, K_M \otimes L^\vee) \times H^1(M, L) \to \mathbb{C},$$

hence $H^1(M, L)^\vee \cong H^0(M, K_M \otimes L^\vee)$. Apply this to $L = \mathcal{O}_M(D)$, so $L^\vee \simeq \mathcal{O}_M(-D)$. Taking dimensions and using Lemma 16.2 and Lemma 16.3 gives $h^1(\mathcal{O}_M(D)) = i(D) = h^0(K_M \otimes \mathcal{O}_M(-D)) = \ell(K - D)$. □

16.5. Euler characteristic and degree additivity

For a line bundle L on M, define $\chi(L) := h^0(M, L) - h^1(M, L)$.

Lemma 16.5 (Reducing Riemann–Roch to an Euler characteristic formula)**.** *For every divisor D,*

$$\chi\big(\mathcal{O}_M(D)\big) = \ell(D) - \ell(K - D).$$

Proof. By Lemma 16.2, $h^0(\mathcal{O}_M(D)) = \ell(D)$. By Proposition 16.4, $h^1(\mathcal{O}_M(D)) = \ell(K - D)$. Subtracting gives the claim. □

Proposition 16.6 (Degree additivity of χ on a curve)**.** *For any divisor D,*

$$\chi\big(\mathcal{O}_M(D)\big) = \chi(\mathcal{O}_M) + \deg D.$$

Proof. We give a concrete proof by reducing to the case of adding a single point.

Step 1: the case $D = p$ (a single point). There is a short exact sequence

$$0 \longrightarrow \mathcal{O}_M \longrightarrow \mathcal{O}_M(p) \longrightarrow \mathcal{O}_M(p)/\mathcal{O}_M \longrightarrow 0.$$

The quotient is a skyscraper sheaf supported at p, canonically isomorphic to $\mathbb{C}_p$: locally, $\mathcal{O}_M(p)$ allows a simple pole at p, and the quotient measures the principal part, which is one complex parameter. Thus

$$0 \longrightarrow \mathcal{O}_M \longrightarrow \mathcal{O}_M(p) \longrightarrow \mathbb{C}_p \longrightarrow 0.$$

Taking cohomology and using $H^1(M, \mathbb{C}_p) = 0$ yields an exact segment

$$0 \to H^0(\mathcal{O}_M) \to H^0(\mathcal{O}_M(p)) \to H^0(\mathbb{C}_p) \to H^1(\mathcal{O}_M) \to H^1(\mathcal{O}_M(p)) \to 0.$$

Taking alternating sums of dimensions gives

$$\chi\big(\mathcal{O}_M(p)\big) = \chi(\mathcal{O}_M) + \dim H^0(\mathbb{C}_p) = \chi(\mathcal{O}_M) + 1.$$

Step 2: effective divisors. If $D = \sum_{j=1}^{d} p_j$ is effective, iterating Step 1 gives

$$\chi\big(\mathcal{O}_M(D)\big) = \chi(\mathcal{O}_M) + d = \chi(\mathcal{O}_M) + \deg D.$$

Step 3: general divisors. Write $D = D^+ - D^-$ with $D^{\pm}$ effective and disjoint support. Then $\deg D = \deg D^+ - \deg D^-$ and $\mathcal{O}_M(D) \simeq \mathcal{O}_M(D^+) \otimes \mathcal{O}_M(D^-)^{\vee}$. Applying Step 2 to D^+ and D^- and using $\chi(\mathcal{O}_M(-D^-)) = \chi(\mathcal{O}_M) - \deg D^-$ gives the general formula. □

Lemma 16.7 (Computing $\chi(\mathcal{O}_M)$)**.**

$$\chi(\mathcal{O}_M) = 1 - g.$$

Proof. We have $h^0(\mathcal{O}_M) = 1$ because holomorphic functions on compact connected M are constant. Also $h^1(\mathcal{O}_M) = g$ by Serre duality:

$$h^1(\mathcal{O}_M) = h^0(K_M) = g.$$

Hence $\chi(\mathcal{O}_M) = 1 - g$. □

16.6. Riemann–Roch: statement and proof

Theorem 16.8 (Riemann–Roch)**.** *For any divisor D on M,*

$$\ell(D) - \ell(K - D) = 1 - g + \deg D.$$

Equivalently,

$$\ell(D) - i(D) = 1 - g + \deg D.$$

Proof. By Lemma 16.5, $\ell(D) - \ell(K - D) = \chi\big(\mathcal{O}_M(D)\big)$. By Proposition 16.6 and Lemma 16.7,

$$\chi\big(\mathcal{O}_M(D)\big) = \chi(\mathcal{O}_M) + \deg D = (1 - g) + \deg D.$$

Combining yields $\ell(D) - \ell(K - D) = 1 - g + \deg D$. Finally, $i(D) = \ell(K - D)$ by Proposition 16.4, giving the equivalent form. □

16.7. Degree of the canonical divisor

Theorem 16.9 (Degree of the canonical divisor)**.**

$$\deg K = 2g - 2.$$

Proof. Apply Riemann–Roch (Theorem 16.8) to $D = 0$: $\ell(0) - \ell(K) = 1 - g$. Since $\ell(0) = h^0(\mathcal{O}_M) = 1$, we get $\ell(K) = g$. Apply Riemann–Roch again to $D = K$: $\ell(K) - \ell(0) = 1 - g + \deg K$. Substitute $\ell(K) = g$ and $\ell(0) = 1$ to obtain $g - 1 = 1 - g + \deg K$, hence $\deg K = 2g - 2$. □

16.8. Explicit examples and computations

Example A: $M = \mathbb{CP}^1$

Here $g = 0$ and $K \sim -2[\infty]$. Let $D = n[\infty]$.

Proposition 16.10 (Computing ℓ on the sphere)**.** *For $n \in \mathbb{Z}$,*

$$\ell\big(n[\infty]\big) = \begin{cases} n + 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Moreover, $\ell(K - D) = \ell\big((-2 - n)[\infty]\big) = \max\{0, -n - 1\}$, and Riemann–Roch holds termwise.

Proof. A meromorphic function on $\mathbb{CP}^1$ with pole order $\leq n$ at ∞ is exactly a polynomial of degree $\leq n$ in the affine coordinate z, plus 0. Thus $\ell\big(n[\infty]\big) = n + 1$ for $n \geq 0$ and 0 for $n < 0$. Since $K - D = (-2 - n)[\infty]$, the same computation gives $\ell(K - D) = \max\{0, -n - 1\}$. Substituting into Theorem 16.8 yields an identity in n that is immediate to check. □

Example B: elliptic curve ($g = 1$)

Let $g = 1$. Then $\deg K = 0$ and $K \sim 0$.

Proposition 16.11 (Riemann–Roch on an elliptic curve)**.** *If $g = 1$, then for any divisor D,*

$$\ell(D) - \ell(-D) = \deg D.$$

In particular, if $\deg D > 0$ then $\ell(D) = \deg D$.

Proof. Since $K \sim 0$, we have $\ell(K - D) = \ell(-D)$. Riemann–Roch gives $\ell(D) - \ell(-D) = 1 - 1 + \deg D = \deg D$. If $\deg D > 0$, then $\deg(-D) < 0$, hence $\ell(-D) = 0$ (no nonzero meromorphic functions with poles bounded by a negative divisor), so $\ell(D) = \deg D$. □

16.9. Asymptotic form and a practical vanishing criterion

Proposition 16.12 (Vanishing of $\ell(K - D)$ for large degree)**.** *If $\deg D \geq 2g - 1$, then $\ell(K - D) = 0$ and therefore $\ell(D) = \deg D + 1 - g$.*

Proof. We have $\deg(K - D) = 2g - 2 - \deg D \leq -1$. If $\ell(K - D) \neq 0$, there exists $0 \neq f \in L(K - D)$ with $(f) + K - D \geq 0$. Taking degrees gives $0 + \deg(K - D) \geq 0$, contradicting $\deg(K - D) < 0$. Hence $\ell(K - D) = 0$, and Riemann–Roch gives $\ell(D) = \deg D + 1 - g$. □

16.10. Geometric meaning and how to use Riemann–Roch

Riemann–Roch says that $\ell(D)$ is controlled by the naive count $\deg D + 1 - g$, with an explicit correction term $\ell(K - D)$ measuring global obstructions:

$$\ell(D) = \deg D + 1 - g + \ell(K - D).$$

Heuristically:

- $\deg D$ counts how many poles you allow (with multiplicity),
- g measures the space of holomorphic 1-forms,
- $\ell(K - D)$ detects the failure of positivity of D (equivalently, non-vanishing of $H^1(\mathcal{O}_M(D))$).

16.11. Exercises

Exercise 16.13 (Principal divisors have degree 0 via argument principle)**.** Reprove Proposition 16.1 using the argument principle on small circles around zeros/poles and then summing contributions.

Exercise 16.14 (Riemann–Roch for $D = 0$)**.** Use Theorem 16.8 with $D = 0$ to show $\ell(K) = g$ and interpret this as $\dim H^0(M, K_M) = g$.

Exercise 16.15 (Base-point freeness for large degree)**.** Let D be an effective divisor with $\deg D \geq 2g$. Show that for every $p \in M$ there exists $f \in L(D)$ such that $f(p) = 0$, i.e. $|D|$ has no base points. Hint: compare $\ell(D)$ and $\ell(D - p)$ using Riemann–Roch.

Exercise 16.16 (Hyperelliptic computation of a canonical divisor)**.** Let M be the smooth projective model of $y^2 = \prod_{j=1}^{2g+2}(x - a_j)$. Compute the divisor of the meromorphic differential dx/y and verify that its degree is $2g - 2$.

Exercise 16.17 (A Clifford-type inequality in an easy case)**.** Assume D is effective and $K - D$ is also effective. Show that $\ell(D) \le \deg D/2 + 1$ in the special case where $\dim |D| = 1$. You may use $\dim |D| = \ell(D) - 1$ and Riemann–Roch.

17. A Complete Proof of Riemann–Roch

Throughout let M be a compact Riemann surface of genus g. Write $\Omega = \Omega^{1,0}_M$ for the sheaf of holomorphic 1-forms and $\mathcal{O} = \mathcal{O}_M$ for the structure sheaf. For a divisor D we write $\mathcal{O}(D) = \mathcal{O}_M(D)$ for the associated line bundle and $\Omega(-D) := \Omega \otimes \mathcal{O}(-D)$.

We use the same spaces as in Section 16:

$$L(D) := H^0(M, \mathcal{O}(D)), \qquad I(D) := H^0(M, \Omega(-D)),$$

and the dimensions $\ell(D) := \dim_{\mathbb{C}} L(D)$, $i(D) := \dim_{\mathbb{C}} I(D)$.

17.1. Fine resolutions and Dolbeault models

A sequence of sheaf morphisms $0 \to \mathcal{F} \to \mathcal{F}_0 \to \mathcal{F}_1 \to \cdots$ is a *fine resolution* of $\mathcal{F}$ if it is exact and each $\mathcal{F}_k$ is fine. On a complex manifold, sheaves of smooth (p,q)-forms with values in a holomorphic vector bundle are fine, because partitions of unity exist in the smooth category.

For a holomorphic line bundle $L \to M$, the Dolbeault complexes give fine resolutions

$$0 \to \mathcal{O}(L) \longrightarrow \mathcal{A}^{0,0}(L) \xrightarrow{\bar\partial} \mathcal{A}^{0,1}(L) \xrightarrow{\bar\partial} \mathcal{A}^{0,2}(L) \xrightarrow{\bar\partial} \cdots,$$

$$0 \to \Omega(L) \longrightarrow \mathcal{A}^{1,0}(L) \xrightarrow{\bar\partial} \mathcal{A}^{1,1}(L) \xrightarrow{\bar\partial} \mathcal{A}^{1,2}(L) \xrightarrow{\bar\partial} \cdots.$$

By the abstract de Rham/Dolbeault theorem (computing sheaf cohomology via fine resolutions), for every $q \ge 0$ there are natural isomorphisms

$$H^q(M, \mathcal{O}(L)) \cong H^{0,q}_{\bar\partial}(M, L), \qquad H^q(M, \Omega(L)) \cong H^{1,q}_{\bar\partial}(M, L).$$

17.2. A long exact sequence relating $\mathbb{C}$, $\mathcal{O}$, and Ω

Consider the fine resolution of the constant sheaf by smooth complex-valued forms $0 \to \underline{\mathbb{C}} \to \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \cdots$. Using the type decomposition $d = \partial + \bar\partial$ and the inclusion $\mathcal{O} \hookrightarrow \mathcal{A}^{0,0}$ together with $\partial : \mathcal{A}^{0,0} \to \mathcal{A}^{1,0}$, one obtains the standard long exact sequence

$$0 \to H^0(M,\mathbb{C}) \to H^0(M,\mathcal{O}) \to H^0(M,\Omega) \to H^1(M,\mathbb{C}) \to H^1(M,\mathcal{O}) \to H^1(M,\Omega) \to H^2(M,\mathbb{C}) \to \cdots \tag{21}$$

On a compact Riemann surface there are no smooth forms of type $(2,0)$ or $(0,2)$. In particular $H^2(M,\mathcal{O}) = 0$, and Dolbeault identifies $H^1(M,\Omega)$ with $H^{1,1}_{\bar\partial}(M)$, which is one-dimensional (top degree). Also $H^0(M,\mathcal{O}) \cong \mathbb{C}$ since holomorphic functions are constant.

Exactness of (21) yields the short exact sequence

$$0 \longrightarrow H^0(M,\Omega) \longrightarrow H^1(M,\mathbb{C}) \longrightarrow H^1(M,\mathcal{O}) \longrightarrow 0. \tag{22}$$

Taking dimensions gives

$$\dim_{\mathbb{C}} H^1(M,\mathbb{C}) = \dim_{\mathbb{C}} H^0(M,\Omega) + \dim_{\mathbb{C}} H^1(M,\mathcal{O}).$$

Since $H^1(M,\mathbb{C})$ has complex dimension $b_1(M)$ and $b_1(M) = 2g$, we obtain

$$2g = \dim H^0(M,\Omega) + \dim H^1(M,\mathcal{O}).$$

17.3. Serre duality as a perfect pairing

Let $L \to M$ be a holomorphic line bundle. There is a bilinear pairing on Dolbeault cohomology

$$H^{0,1}_{\bar\partial}(M,L) \times H^{1,0}_{\bar\partial}(M,L^\vee\otimes\Omega) \longrightarrow \mathbb{C}, \qquad ([\varphi],[\psi]) \longmapsto \int_M \varphi\wedge\psi,$$

well-defined because $\varphi\wedge\psi$ is of type $(1,1)$. Using Hodge theory on a compact surface, every Dolbeault class has a harmonic representative and the pairing is nondegenerate; hence it is perfect. Translating back to sheaves gives Serre duality

$$H^1(M,\mathcal{O}(L))^\vee \cong H^0(M,\Omega\otimes L^\vee). \tag{23}$$

Taking $L = \mathcal{O}(D)$ and using $L^\vee \simeq \mathcal{O}(-D)$ yields

$$H^1(M,\mathcal{O}(D))^\vee \cong H^0(M,\Omega(-D)) = I(D), \qquad \dim H^1(M,\mathcal{O}(D)) = i(D).$$

In particular, applying (23) to $L = \mathcal{O}$ gives $H^1(M,\mathcal{O})^\vee \cong H^0(M,\Omega)$, so

$$\dim H^1(M,\mathcal{O}) = \dim H^0(M,\Omega).$$

Combining this with the dimension identity from (22) gives the genus lemma.

Lemma 17.1 (Genus lemma)**.** $\dim H^0(M,\Omega) = g$.

Proof. From (22) we have $2g = \dim H^0(M,\Omega) + \dim H^1(M,\mathcal{O})$. By Serre duality, $\dim H^1(M,\mathcal{O}) = \dim H^0(M,\Omega)$. Hence $2g = 2\dim H^0(M,\Omega)$, so $\dim H^0(M,\Omega) = g$. □

17.4. A key exact sequence for adding a point

Fix a point $p \in M$. For every divisor D there is a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}(D) \xrightarrow{\iota} \mathcal{O}(D+p) \xrightarrow{\rho} \mathbb{C}_p \longrightarrow 0, \tag{24}$$

where $\mathbb{C}_p$ is the skyscraper sheaf supported at p. Taking cohomology (and using $H^1(M,\mathbb{C}_p) = 0$) gives an exact sequence

$$0 \to H^0(M,\mathcal{O}(D)) \to H^0(M,\mathcal{O}(D+p)) \xrightarrow{\rho} H^0(M,\mathbb{C}_p) \cong \mathbb{C} \to H^1(M,\mathcal{O}(D)) \to H^1(M,\mathcal{O}(D+p)) \to 0. \tag{25}$$

Let $V := \operatorname{Im}(\rho) \subset \mathbb{C}$ and $W := \mathbb{C}/V$. Exactness of (25) yields

$$\ell(D+p) = \ell(D) + \dim V, \qquad h^1(\mathcal{O}(D+p)) = h^1(\mathcal{O}(D)) - \dim W, \tag{26}$$

and $\dim V + \dim W = 1$.

Using $i(D) = \dim H^1(M,\mathcal{O}(D))$ from Serre duality, (26) becomes

$$\big(\ell(D+p) - i(D+p)\big) - \big(\ell(D) - i(D)\big) = \dim V + \dim W = 1. \tag{27}$$

Interpretation. Either $V = \mathbb{C}$, in which case some section in $L(D+p)$ evaluates nontrivially at p and $\ell(D+p) = \ell(D)+1$ while $i(D+p) = i(D)$; or $V = 0$, in which case every section vanishes at p and $\ell(D+p) = \ell(D)$ while $i(D+p) = i(D)-1$. In both cases $\ell - i$ increases by exactly 1.

17.5. Base case $D = 0$

Proposition 17.2 (Base case $D = 0$)**.** *Riemann–Roch holds for $D = 0$:*

$$\ell(0) - i(0) = 1 - g.$$

Proof. We have $\ell(0) = \dim H^0(M, \mathcal{O}) = 1$. Also $i(0) = \dim H^0(M, \Omega) = g$ by Lemma 17.1. Therefore $\ell(0) - i(0) = 1 - g$. □

17.6. Proof of Riemann–Roch by induction on degree

Theorem 17.3 (Riemann–Roch)**.** *For every divisor D on M,*

$$\ell(D) - i(D) = 1 - g + \deg D.$$

Proof. Step 1: effective divisors. Let E be an effective divisor of degree r. Write $E = p_1 + \cdots + p_r$. Starting from 0 and applying (27) successively gives

$$\ell(E) - i(E) = \big(\ell(0) - i(0)\big) + r = (1 - g) + \deg E,$$

using Proposition 17.2.

Step 2: general divisors. Write $D = E - F$ with E, F effective and with disjoint support. Choose an ordering $F = q_1 + \cdots + q_s$. Starting from E and removing points one by one, we apply the add-point identity to $D' = E - (q_1 + \cdots + q_{k-1}) - q_k$ rewritten as

$$(\ell(D' + q_k) - i(D' + q_k)) - (\ell(D') - i(D')) = 1,$$

which is exactly (27) with D' in place of D and $p = q_k$. Rearranging shows that each removal of a point decreases $\ell - i$ by 1. After s removals we obtain

$$\ell(E - F) - i(E - F) = \big(\ell(E) - i(E)\big) - s.$$

Insert Step 1 for E and note that $s = \deg F$ to get

$$\ell(D) - i(D) = (1 - g) + \deg E - \deg F = (1 - g) + \deg D.$$

□

17.7. Corollaries and asymptotics

Corollary 17.4 (Degree of the canonical divisor)**.** Let K be a canonical divisor. Then $\deg K = 2g - 2$.

Proof. Apply Theorem 17.3 to $D = 0$ to get $\ell(0) - i(0) = 1 - g$. Since $i(0) = \ell(K)$ (by $i(0) = \dim H^1(\mathcal{O}) = \dim H^0(\Omega) = \ell(K)$), we obtain $\ell(K) = g$. Apply Theorem 17.3 to $D = K$:

$$\ell(K) - i(K) = 1 - g + \deg K.$$

But $i(K) = \dim H^0(\Omega(-K)) = \dim H^0(\mathcal{O}) = \ell(0) = 1$, hence $g - 1 = 1 - g + \deg K$, so $\deg K = 2g - 2$. □

Corollary 17.5 (Asymptotic Riemann–Roch)**.** If $\deg D > 2g-2$, then $i(D) = 0$ and hence $\ell(D) = \deg D + 1 - g$.

Proof. If $\deg D > 2g-2$, then $\deg(K-D) = 2g-2-\deg D < 0$. A divisor of negative degree has $\ell(K-D) = 0$, hence $i(D) = \ell(K-D) = 0$ by Serre duality. Now apply Riemann–Roch. □

Corollary 17.6. If $\deg D = g + \ell$ for some integer $\ell \geq 0$, then $\ell(D) \geq \ell + 1$.

Proof. Riemann–Roch gives $\ell(D) = 1 - g + \deg D + i(D) = 1 - g + (g+\ell) + i(D) = \ell + 1 + i(D) \geq \ell + 1$. □

17.8. Summary of the mechanism

- Fine (Dolbeault) resolutions compute sheaf cohomology and provide concrete models for $H^q(M, \mathcal{O}(L))$ and $H^q(M, \Omega(L))$.
- The long exact sequence (21) and Serre duality imply $\dim H^0(M, \Omega) = g$ (Lemma 17.1).
- The add-a-point exact sequence (24) yields the jump identity (27): each time one adds a point, the quantity $\ell(D) - i(D)$ increases by exactly 1.
- With the base case $D = 0$, induction on degree proves $\ell(D) - i(D) = 1 - g + \deg D$ for all divisors.

18. Intersection Theory and Riemann–Roch for Surfaces

This section upgrades the curve story to complex dimension 2. Divisors now define classes in $H^2(X, \mathbb{Z})$, and there is a genuine *quadratic* numerical invariant: the intersection pairing. Surface Riemann–Roch expresses $\chi(\mathcal{O}_X(D))$ in terms of intersection numbers with D and the canonical class K_X.

Throughout, X is a smooth projective complex surface. We freely use earlier material (divisors/line bundles, Chern classes, Poincaré duality, sheaf cohomology, Serre duality, Dolbeault/Hodge where appropriate).

18.1. Intersection pairing: local definition and global properties

Local intersection multiplicity

Let $C, C' \subset X$ be irreducible curves and let $p \in C \cap C'$. Choose local equations $f, g \in \mathcal{O}_{X,p}$ defining C and C' near p.

Definition 18.1 (Local intersection multiplicity)**.** The local intersection multiplicity of C and C' at p is

$$I_p(C, C') \;:=\; \dim_{\mathbb{C}} \mathcal{O}_{X,p}/(f, g).$$

Proposition 18.2 (Basic properties of $I_p(C, C')$)**.**

1. *$I_p(C, C')$ is finite if and only if C and C' have no common component through p.*

2. *If C and C' are smooth and meet transversely at p, then $I_p(C, C') = 1$.*

3. *If $\varphi : (\mathbb{C}, 0) \to (X, p)$ parametrizes a branch of C at p and g is a local equation for C', then $I_p(C, C') = \operatorname{ord}_0(g \circ \varphi)$ (counting with multiplicity along that branch).*

Proof. (1) If C and C' share a component through p, then (f, g) does not cut out a 0–dimensional scheme at p and the quotient has positive Krull dimension, hence infinite $\mathbb{C}$–dimension. Conversely, if they share no component through p, then (f, g) defines a 0–dimensional analytic set at p, so the local algebra is Artinian and has finite $\mathbb{C}$–dimension.

(2) In holomorphic coordinates (x, y) centered at p, transversality means $C = \{x = 0\}$ and $C' = \{y = 0\}$ after a holomorphic change of coordinates. Then $\mathcal{O}_{X,p}/(x, y) \cong \mathbb{C}$.

(3) Restrict to the branch via φ. The ideal (f, g) along that branch is generated by $g \circ \varphi$, and the length of the corresponding 1–variable local algebra is precisely the vanishing order at 0. □

Global intersection number

Assume C and C' have no common component. Then $C \cap C'$ is finite.

Definition 18.3 (Global intersection number)**.** The global intersection number is

$$C \cdot C' \;:=\; \sum_{p \in C \cap C'} I_p(C, C').$$

By bilinearity, this extends to all divisors $D = \sum n_i C_i$ and $E = \sum m_j C'_j$ by

$$D \cdot E \;:=\; \sum_{i,j} n_i m_j \, (C_i \cdot C'_j),$$

whenever every C_i and C'_j share no common component.

Proposition 18.4 (Bilinearity, symmetry, and deformation invariance)**.** *On* $\operatorname{Div}(X)$ *modulo numerical equivalence, the pairing $(D, E) \mapsto D \cdot E$ is well-defined, symmetric, bilinear, and invariant under deformation of D and E in algebraic families.*

Proof. Bilinearity is built into the extension from irreducible curves to divisors. Symmetry follows from the symmetry of the local length $\dim_{\mathbb{C}} \mathcal{O}_{X,p}/(f, g)$ in (f, g). Invariance under deformation is standard: in a flat family the total intersection number is the length of a 0–dimensional fiber, which is constant under flat deformation. Passing to numerical equivalence (same intersection with all curves) ensures well-definedness. □

18.2. Chern classes and the cohomological intersection product

Let $L \to X$ be a holomorphic line bundle. Its first Chern class $c_1(L) \in H^2(X, \mathbb{Z})$ is represented in de Rham cohomology by curvature: choose a Hermitian metric h and local holomorphic frame s, then

$$c_1(L) \;=\; \left[\frac{i}{2\pi} \, \partial\bar{\partial} \log h(s, s) \right] \in H^2_{\mathrm{dR}}(X).$$

Definition 18.5 (Cohomological intersection number)**.** For line bundles L, L' define

$$L \cdot L' \;:=\; \int_X c_1(L) \wedge c_1(L') \;\in\; \mathbb{Z}.$$

For divisors D, E, set $D \cdot E := \mathcal{O}_X(D) \cdot \mathcal{O}_X(E)$.

Proposition 18.6 (Compatibility with restriction degree)**.** *If $C \subset X$ is a smooth irreducible curve and $L \to X$ a line bundle, then*

$$\deg(L|_C) = \int_C c_1(L) = \int_X c_1(L) \wedge \mathrm{PD}([C]) \;=\; L \cdot C.$$

Proof. Naturality gives $c_1(L|_C) = i^* c_1(L)$, hence $\int_C c_1(L|_C) = \int_C i^* c_1(L)$. By Poincaré duality this equals $\int_X c_1(L) \wedge \mathrm{PD}([C])$. Finally, $\mathrm{PD}([C]) = c_1(\mathcal{O}_X(C))$ for a smooth divisor C, so the last equality is the definition. □

Theorem 18.7 (Intersection as Chern class pairing)**.** *If $C, C' \subset X$ are irreducible curves with no common component, then*

$$C \cdot C' \;=\; \int_X c_1(\mathcal{O}_X(C)) \wedge c_1(\mathcal{O}_X(C')).$$

Proof. Sketch with key reductions. Choose generic small perturbations so that C and C' meet properly and transversely at smooth points; this does not change the intersection number (Proposition 18.4). For transverse intersections, each point contributes 1 locally, so $C \cdot C'$ is the number of intersection points. On the other hand, $c_1(\mathcal{O}_X(C))$ is Poincaré dual to $[C]$ and similarly for C', so the integral $\int_X c_1(\mathcal{O}_X(C)) \wedge c_1(\mathcal{O}_X(C'))$ equals the oriented intersection number of the fundamental classes, which counts transverse intersection points with multiplicity 1. □

18.3. Adjunction and the canonical class

Theorem 18.8 (Adjunction formula for a smooth curve)**.** *Let $C \subset X$ be a smooth irreducible curve. Then*

$$K_C \;\simeq\; (K_X \otimes \mathcal{O}_X(C))|_C,$$

and consequently

$$2g(C) - 2 \;=\; C \cdot (C + K_X).$$

Proof. Consider the exact sequence of holomorphic vector bundles on C:

$$0 \longrightarrow T_C \longrightarrow T_X|_C \longrightarrow N_{C/X} \longrightarrow 0.$$

Dualizing and taking determinants yields

$$K_C \;\simeq\; K_X|_C \otimes \det(N_{C/X})^\vee.$$

But $\det(N_{C/X}) \simeq \mathcal{O}_X(C)|_C$ for a smooth divisor C (normal bundle corresponds to the divisor line bundle). Hence $K_C \simeq (K_X \otimes \mathcal{O}_X(C))|_C$. Taking degrees and using Proposition 18.6 gives

$$\deg K_C = \deg(K_X|_C) + \deg(\mathcal{O}_X(C)|_C) = K_X \cdot C + C \cdot C,$$

i.e. $2g(C) - 2 = C \cdot (C + K_X)$. □

Example 18.9 (Plane curves in $\mathbb{P}^2$)**.** Let $X = \mathbb{P}^2$ and let H be the class of a line. Then $H \cdot H = 1$ and $K_X \sim -3H$. A smooth plane curve of degree d has class $C \sim dH$, hence

$$2g(C) - 2 \;=\; (dH) \cdot (dH - 3H) = d(d-3), \qquad \text{so} \qquad g(C) = \frac{(d-1)(d-2)}{2}.$$

18.4. Riemann–Roch for surfaces: statement and proof mechanisms

The formula

Theorem 18.10 (Riemann–Roch for line bundles on a smooth surface)**.** *For any divisor D on X,*

$$\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \frac{1}{2}\big(D \cdot D - D \cdot K_X\big).$$

Equivalently, for $L = \mathcal{O}_X(D)$,

$$\chi(L) = \chi(\mathcal{O}_X) + \frac{1}{2}\big(c_1(L)^2 - c_1(L) \cdot c_1(K_X)\big).$$

A rigorous route (Hirzebruch–Riemann–Roch in rank 1)

A standard proof uses Hirzebruch–Riemann–Roch:

$$\chi(L) = \int_X \mathrm{ch}(L)\,\mathrm{td}(T_X),$$

together with

$$\mathrm{ch}(L) = 1 + c_1(L) + \frac{1}{2}c_1(L)^2, \qquad \mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}\big(c_1(T_X)^2 + c_2(T_X)\big),$$

and the identity $c_1(T_X) = -c_1(K_X)$. Only the degree–4 component contributes to the integral, giving

$$\chi(L) = \int_X \left(\frac{1}{2}c_1(L)^2 - \frac{1}{2}c_1(L)c_1(K_X)\right) + \int_X \frac{1}{12}\big(c_1(T_X)^2 + c_2(T_X)\big).$$

The last integral is $\chi(\mathcal{O}_X)$ (a topological/hodge invariant of X), yielding Theorem 18.10.

Remark 18.11. In these notes we will *use* Theorem 18.10 as the surface analogue of curve Riemann–Roch, and verify it by direct cohomology computations in basic examples below.

18.5. Worked examples: explicit intersection computations and RR checks

Example A: $\mathbb{P}^2$

Let $X = \mathbb{P}^2$ and let H be the hyperplane class. Then $\mathrm{Pic}(X) \cong \mathbb{Z}[H]$ and $H^2 = H \cdot H = 1$. Also $K_X \sim -3H$, and $\chi(\mathcal{O}_X) = 1$.

Proposition 18.12 (RR for $\mathbb{P}^2$ in closed form)**.** *For $D = dH$,*

$$\chi(\mathcal{O}_{\mathbb{P}^2}(d)) \;=\; 1 + \frac{1}{2}\big(d^2 + 3d\big) \;=\; \frac{(d+1)(d+2)}{2}.$$

Proof. Compute the intersection terms:

$$D \cdot D = d^2(H^2) = d^2, \qquad D \cdot K_X = (dH) \cdot (-3H) = -3d.$$

Insert into surface Riemann–Roch (Theorem 18.10):

$$\chi(\mathcal{O}_X(dH)) = \chi(\mathcal{O}_X) + \frac{1}{2}(d^2 - (-3d)) = 1 + \frac{1}{2}(d^2 + 3d) = \frac{(d+1)(d+2)}{2}.$$

□

Example 18.13 (Direct cohomology check for $d \geq 0$)**.** For $d \geq 0$, one knows $H^i(\mathbb{P}^2, \mathcal{O}(d)) = 0$ for $i > 0$ and $h^0(\mathbb{P}^2, \mathcal{O}(d)) = \binom{d+2}{2}$, so $\chi = \binom{d+2}{2}$, matching Proposition 18.12.

Example B: $\mathbb{P}^1 \times \mathbb{P}^1$

Let $X = \mathbb{P}^1 \times \mathbb{P}^1$. Let $F_1 = [\mathbb{P}^1 \times \{\text{pt}\}]$ and $F_2 = [\{\text{pt}\} \times \mathbb{P}^1]$. Then $\text{Pic}(X) \cong \mathbb{Z}F_1 \oplus \mathbb{Z}F_2$ and intersection numbers are

$$F_1^2 = 0, \qquad F_2^2 = 0, \qquad F_1 \cdot F_2 = 1.$$

Also $K_X \sim -2F_1 - 2F_2$ and $\chi(\mathcal{O}_X) = 1$.

Proposition 18.14 (RR on $\mathbb{P}^1 \times \mathbb{P}^1$)**.** *For $D = aF_1 + bF_2$,*

$$\chi(\mathcal{O}_X(a,b)) \;=\; (a+1)(b+1).$$

Proof. Compute

$$D^2 = (aF_1 + bF_2)^2 = 2ab(F_1 \cdot F_2) = 2ab,$$

and

$$D \cdot K_X = (aF_1 + bF_2) \cdot (-2F_1 - 2F_2) = -2a(F_1 \cdot F_2) - 2b(F_1 \cdot F_2) = -2a - 2b.$$

Hence Theorem 18.10 gives

$$\chi(\mathcal{O}_X(D)) = 1 + \frac{1}{2}\big(2ab - (-2a - 2b)\big) = 1 + ab + a + b = (a+1)(b+1).$$

□

Example 18.15 (Direct cohomology check when $a, b \geq 0$)**.** If $a, b \geq 0$, then $H^i(X, \mathcal{O}(a,b)) = 0$ for $i > 0$ and $h^0(X, \mathcal{O}(a,b)) = (a+1)(b+1)$ by Künneth and the curve computation on $\mathbb{P}^1$.

Example C: Hirzebruch surfaces $\mathbb{F}_n$

Let $X = \mathbb{F}_n = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n))$. Let f be the fiber class and s the negative section class. Then

$$f^2 = 0, \qquad s \cdot f = 1, \qquad s^2 = -n.$$

The canonical class is

$$K_X \;\sim\; -2s - (n+2)f, \qquad \text{and} \qquad \chi(\mathcal{O}_X) = 1.$$

Proposition 18.16 (RR on $\mathbb{F}_n$ in coordinates)**.** *For $D = as + bf$,*

$$\chi(\mathcal{O}_X(D)) \;=\; 1 + \frac{1}{2}\big(-na^2 + 2ab + (n-2)a + 2b\big).$$

Proof. Compute

$$D^2 = (as+bf)^2 = a^2 s^2 + 2ab(s\cdot f) + b^2 f^2 = -na^2 + 2ab,$$

and

$$D\cdot K_X = (as+bf)\cdot(-2s-(n+2)f) = -2as^2 - a(n+2)(s\cdot f) - 2b(f\cdot s) = 2an - a(n+2) - 2b.$$

Thus

$$D^2 - D\cdot K_X = (-na^2+2ab) - \big(2an - a(n+2) - 2b\big) = -na^2 + 2ab + (n-2)a + 2b.$$

Insert into Theorem 18.10 and use $\chi(\mathcal{O}_X) = 1$. □

Example 18.17 (Genus of a divisor class on $\mathbb{F}_n$)**.** If $C \sim as + bf$ is smooth, then adjunction (Theorem 18.8) gives

$$2g(C) - 2 = C \cdot (C + K_X).$$

Using the intersection table above, you can compute $g(C)$ explicitly as a polynomial in a, b, n.

Example D: Blow-up of $\mathbb{P}^2$ at a point

Let $\pi : \widetilde{X} \to \mathbb{P}^2$ be the blow-up at one point. Let $H = \pi^*\mathcal{O}_{\mathbb{P}^2}(1)$ and let E be the exceptional curve. Then

$$H^2 = 1, \qquad E^2 = -1, \qquad H\cdot E = 0, \qquad K_{\widetilde{X}} \sim -3H + E, \qquad \chi(\mathcal{O}_{\widetilde{X}}) = 1.$$

Proposition 18.18 (RR on the blow-up of $\mathbb{P}^2$ at one point)**.** *For $D = aH - bE$,*

$$\chi(\mathcal{O}_{\widetilde{X}}(D)) \;=\; 1 + \frac{1}{2}\big(a^2 + 3a - b^2 - b\big).$$

Proof. Compute

$$D^2 = (aH - bE)^2 = a^2H^2 - 2ab(H\cdot E) + b^2E^2 = a^2 - b^2,$$

and

$$D\cdot K_{\widetilde{X}} = (aH - bE)\cdot(-3H + E) = -3aH^2 - bE^2 = -3a + b.$$

Thus

$$\chi(\mathcal{O}_{\widetilde{X}}(D)) = 1 + \frac{1}{2}\big((a^2 - b^2) - (-3a + b)\big) = 1 + \frac{1}{2}(a^2 + 3a - b^2 - b).$$

□

18.6. Using RR + Serre duality: vanishing tests and effectivity heuristics

Surface Riemann–Roch controls χ, but to extract h^0 one needs information on h^1 and h^2. Serre duality gives

$$h^2(\mathcal{O}_X(D)) = h^0(\mathcal{O}_X(K_X - D)).$$

Thus:

- If $K_X - D$ has no sections (e.g. is "too negative"), then $h^2(\mathcal{O}_X(D)) = 0$.
- If additionally $h^1(\mathcal{O}_X(D)) = 0$ (often by positivity/vanishing theorems in advanced settings), then $h^0(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X(D))$.

In our elementary examples ($\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, many $\mathbb{F}_n$ cases), one can often determine h^0 by explicit section counting in charts and show $h^1 = h^2 = 0$ in the "positive" region, thereby recovering concrete dimension formulas.

18.7. Exercises

Exercise 18.19 (Intersection table on $\mathbb{P}^1 \times \mathbb{P}^1$)**.** Let $X = \mathbb{P}^1 \times \mathbb{P}^1$ with F_1, F_2 as above.

1. Prove $F_1^2 = F_2^2 = 0$ and $F_1 \cdot F_2 = 1$ using geometric representatives.
2. For $D = aF_1 + bF_2$ and $E = cF_1 + dF_2$, compute $D \cdot E$ explicitly.

Exercise 18.20 (Genus of a smooth divisor on $\mathbb{P}^1 \times \mathbb{P}^1$)**.** Let $C \sim aF_1 + bF_2$ be a smooth curve on $X = \mathbb{P}^1 \times \mathbb{P}^1$. Use adjunction with $K_X \sim -2F_1 - 2F_2$ to show

$$g(C) = (a-1)(b-1).$$

Check this for $(a, b) = (1, 1)$ and $(2, 1)$ by direct geometric descriptions.

Exercise 18.21 (Plane curves and adjunction)**.** Let $X = \mathbb{P}^2$ and let C be a smooth curve of degree d.

1. Prove $K_{\mathbb{P}^2} \sim -3H$ using the transformation rule for a holomorphic 2–form under affine charts.
2. Deduce $g(C) = \frac{(d-1)(d-2)}{2}$.

Exercise 18.22 (Local intersection multiplicity via elimination)**.** In $\mathbb{C}^2$ consider $C = \{y^2 - x^3 = 0\}$ and $C' = \{y - \lambda x^2 = 0\}$ with $\lambda \in \mathbb{C}$. Compute $I_0(C, C')$ as a function of λ by eliminating y. Interpret the answer geometrically in terms of tangency.

Exercise 18.23 (Hirzebruch surface canonical class)**.** Let $X = \mathbb{F}_n$ with generators s, f.

1. Using adjunction on the section s, show that $K_X \cdot s = n - 2$.
2. Using adjunction on a fiber $f \cong \mathbb{P}^1$, show that $K_X \cdot f = -2$.
3. Deduce $K_X \sim -2s - (n+2)f$ from the intersection table.

Exercise 18.24 (RR on $\mathbb{F}_n$ for a concrete region)**.** Let $X = \mathbb{F}_n$ and $D = as + bf$ with $a \geq 0$ and $b \gg 0$.

1. Use Proposition 18.16 to compute $\chi(\mathcal{O}_X(D))$.
2. Show $h^2(\mathcal{O}_X(D)) = h^0(\mathcal{O}_X(K_X - D)) = 0$ for b sufficiently large.
3. (Optional) By analyzing sections along the ruling, argue that $h^1(\mathcal{O}_X(D)) = 0$ for b sufficiently large, and conclude $h^0(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X(D))$ in that range.

Exercise 18.25 (Blow-up intersection calculus)**.** Let $\pi : \widetilde{X} \to \mathbb{P}^2$ be the blow-up at one point with classes H, E as above.

1. Prove $H \cdot E = 0$ geometrically (use representatives avoiding the blown-up point).
2. Show $E^2 = -1$ by identifying E with $\mathbb{P}^1$ and computing the degree of $\mathcal{O}_{\widetilde{X}}(E)|_E$.
3. Compute $K_{\widetilde{X}}$ from $K_{\mathbb{P}^2}$ and the exceptional divisor and verify $K_{\widetilde{X}} \sim -3H + E$.

Exercise 18.26 (A quick RR–based dimension prediction)**.** On $X = \mathbb{P}^2$, use Proposition 18.12 and Serre duality to predict $h^0(\mathcal{O}_{\mathbb{P}^2}(-1))$ and $h^2(\mathcal{O}_{\mathbb{P}^2}(-4))$. Then verify your predictions by a direct argument (no heavy theorems needed).

Exercise 18.27 (Intersection form as cup product (with Poincaré duality))**.** Let X be a compact oriented smooth 4–manifold (in particular, a smooth complex surface), and let $\alpha, \beta \in H^2(X, \mathbb{Z})$.

Definition (Poincaré dual). For a homology class $[Z] \in H_2(X, \mathbb{Z})$ represented by a closed oriented smooth 2–dimensional submanifold $Z \subset X$, its *Poincaré dual*

$$\mathrm{PD}([Z]) \in H^2(X, \mathbb{Z})$$

is the unique cohomology class characterized by the property that for every closed 2–form η on X,

$$\int_X \eta \wedge \mathrm{PD}([Z]) \;=\; \int_Z \eta.$$

Equivalently, $\mathrm{PD}([Z])$ is the cohomology class whose cup product with any $\gamma \in H^2(X, \mathbb{Z})$ satisfies

$$\langle \gamma \smile \mathrm{PD}([Z]), [X] \rangle \;=\; \langle \gamma, [Z] \rangle.$$

1. Show that $\displaystyle\int_X \alpha \wedge \beta \in \mathbb{Z}$.

   Hint: Interpret $\alpha \wedge \beta$ as the cup product $\alpha \smile \beta \in H^4(X, \mathbb{Z}) \cong \mathbb{Z}$ and pair it with the fundamental class $[X]$.

2. Assume $\alpha = \mathrm{PD}([C])$ and $\beta = \mathrm{PD}([C'])$, where $C, C' \subset X$ are smooth oriented curves (real dimension 2) that intersect transversely. Show that

   $$\int_X \alpha \wedge \beta \;=\; \#(C \cap C'),$$

   the (signed) number of intersection points of C and C'.

   Hint: Use the defining property of the Poincaré dual twice:

   $$\int_X \mathrm{PD}([C]) \wedge \mathrm{PD}([C']) \;=\; \int_C \mathrm{PD}([C']) \;=\; \int_{C \cap C'} 1.$$

   Explain why transversality implies that each intersection point contributes $+1$ (up to orientation).

Exercise 18.28 (Adjunction and self-intersection)**.** Let $C \subset X$ be a smooth curve.

1. Show that $C^2 = \deg(\mathcal{O}_X(C)|_C)$.
2. Use adjunction to express C^2 in terms of $g(C)$ and $K_X \cdot C$.

One-line takeaway

On a surface, intersection theory provides the quadratic correction term that replaces the linear "degree" term on a curve, and Riemann–Roch becomes

$$\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \frac{1}{2}\big(D^2 - D \cdot K_X\big).$$

Part VII
Jacobian of Compact Riemann Surfaces

Throughout this part X denotes a compact Riemann surface of genus $g \geq 0$. We write $\Omega^1_X := K_X$ for the canonical bundle (holomorphic 1-forms), and $H^0(X, \Omega^1_X)$ for the g-dimensional space of holomorphic differentials.

19. Jacobian, Picard Variety, and Abel–Jacobi Theory

19.1. From holomorphic differentials to a period lattice

1. Integration pairing and periods

Let $\omega \in H^0(X, \Omega^1_X)$ and $\gamma \in H_1(X, \mathbb{Z})$. Since ω is closed, $\int_\gamma \omega$ depends only on the homology class $[\gamma]$, hence defines a bilinear pairing

$$H^0(X, \Omega^1_X) \times H_1(X, \mathbb{Z}) \longrightarrow \mathbb{C}, \qquad (\omega, \gamma) \longmapsto \int_\gamma \omega. \tag{28}$$

Lemma 19.1 (Period functionals are well-defined)**.** *If γ is a boundary in X, then $\int_\gamma \omega = 0$ for every $\omega \in H^0(X, \Omega^1_X)$.*

Proof. If $\gamma = \partial\Sigma$ for a smooth 2-chain Σ, then by Stokes' theorem, $\int_\gamma \omega = \int_{\partial\Sigma} \omega = \int_\Sigma d\omega = 0$, since ω is holomorphic and hence closed. □

2. Definition of the period lattice and the Jacobian

Choose a basis $\{\omega_1, \dots, \omega_g\}$ of $H^0(X, \Omega^1_X)$. Define the *period map*

$$\mathsf{P} : H_1(X, \mathbb{Z}) \longrightarrow \mathbb{C}^g, \qquad \gamma \longmapsto \Big(\int_\gamma \omega_1, \dots, \int_\gamma \omega_g\Big).$$

Its image is the *period lattice*

$$\Lambda := \mathsf{P}\big(H_1(X,\mathbb{Z})\big) \subset \mathbb{C}^g. \tag{29}$$

Proposition 19.2 (The period lattice is a full lattice)**.** *The subgroup $\Lambda \subset \mathbb{C}^g$ is discrete of rank $2g$. Equivalently, Λ is a full lattice in $\mathbb{C}^g$.*

Proof. We outline the standard argument using Hodge theory. Identify $H^0(X,\Omega^1_X)$ with harmonic $(1,0)$-forms and embed it into $H^1_{\mathrm{dR}}(X;\mathbb{C})$. The real cohomology $H^1_{\mathrm{dR}}(X;\mathbb{R})$ has a natural integral lattice $H^1(X,\mathbb{Z}) \subset H^1_{\mathrm{dR}}(X;\mathbb{R})$ (coming from periods of real 1-forms over integral cycles). This lattice is discrete of rank $2g$. The holomorphic differentials span a g-dimensional complex subspace of $H^1_{\mathrm{dR}}(X;\mathbb{C})$ whose real part is $2g$-dimensional, and the restriction of the integral lattice yields a discrete subgroup of rank $2g$ in $\mathbb{C}^g$. Under the chosen basis $\{\omega_j\}$, this subgroup is precisely Λ. □

Definition 19.3 (Jacobian)**.** The *Jacobian* of X is the complex torus

$$J(X) := \mathbb{C}^g/\Lambda. \tag{30}$$

Remark 19.4. Changing the basis $\{\omega_j\}$ multiplies Λ by an invertible complex matrix, hence produces an isomorphic complex torus. So $J(X)$ is canonically defined up to unique isomorphism (independent of choices).

Exercise 19.5. Show that if $g = 0$ then $H^0(X,\Omega^1_X) = 0$, hence $\mathbb{C}^g = 0$ and $J(X)$ is the trivial group.

19.2. Abel–Jacobi map on points and on divisors

1. The Abel–Jacobi map on points

Fix a base point $p_0 \in X$. For $x \in X$ choose a path Γ from p_0 to x and set

$$\widetilde{\alpha}(x) := \Big(\int_\Gamma \omega_1, \dots, \int_\Gamma \omega_g\Big) \in \mathbb{C}^g.$$

Lemma 19.6 (Well-definedness modulo periods)**.** *If Γ, Γ' are two paths from p_0 to x, then $\widetilde{\alpha}_\Gamma(x) - \widetilde{\alpha}_{\Gamma'}(x) \in \Lambda$.*

Proof. The concatenation $\Gamma \cdot (\Gamma')^{-1}$ is a loop based at p_0, hence determines an integral homology class $\gamma \in H_1(X,\mathbb{Z})$. Therefore

$$\int_\Gamma \omega_j - \int_{\Gamma'} \omega_j = \int_{\Gamma\cdot(\Gamma')^{-1}} \omega_j = \int_\gamma \omega_j,$$

so the difference vector lies in Λ by definition (29). □

Definition 19.7 (Abel–Jacobi map on points)**.** The *Abel–Jacobi map* $\alpha : X \to J(X)$ is

$$\alpha(x) := \widetilde{\alpha}(x) \bmod \Lambda \in \mathbb{C}^g/\Lambda.$$

2. Symmetric powers and effective divisors

Let $S^dX := X^d/\mathfrak{S}_d$ be the d-fold symmetric product. Points of S^dX correspond to effective divisors $D = x_1 + \cdots + x_d$ of degree d (unordered with multiplicities).

Definition 19.8 (Abel–Jacobi map on S^dX)**.** Define $\alpha_d : S^dX \to J(X)$ by

$$\alpha_d\Big(\sum_{i=1}^{d} x_i\Big) := \sum_{i=1}^{d} \alpha(x_i) \in J(X).$$

Lemma 19.9 (Compatibility with addition of a base point)**.** *Let $i_d : S^dX \to S^{d+1}X$ be $D \mapsto D + p_0$. Then $\alpha_{d+1} \circ i_d = \alpha_d$.*

Proof. Since $\alpha(p_0) = 0$ by definition, $\alpha_{d+1}(D + p_0) = \alpha_d(D) + \alpha(p_0) = \alpha_d(D)$. □

Exercise 19.10. Let $X = \mathbb{C}/\Lambda$ be an elliptic curve. Show that choosing $\omega = dz$ gives $J(X) \cong \mathbb{C}/\Lambda \cong X$. Prove that $\alpha : X \to J(X)$ is a group isomorphism once p_0 is taken as the origin.

19.3. Picard group and the exponential sequence

1. $\mathrm{Pic}(X)$ as a cohomology group

Recall $\mathcal{O}_X^*$ is the sheaf of nowhere-vanishing holomorphic functions. A standard result identifies

$$\mathrm{Pic}(X) \;\cong\; H^1(X, \mathcal{O}_X^*), \tag{31}$$

via Čech cocycles: transition functions of a holomorphic line bundle form a 1-cocycle with values in $\mathcal{O}_X^*$, and changing trivializations changes the cocycle by a coboundary.

2. Exponential exact sequence and first Chern class

Consider the exponential short exact sequence of sheaves on X:

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}_X^* \longrightarrow 0. \tag{32}$$

Taking cohomology yields a long exact sequence, whose relevant part is

$$H^1(X, \mathcal{O}_X) \xrightarrow{\beta} H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \to\; H^2(X, \mathcal{O}_X). \tag{33}$$

Lemma 19.11 (Vanishing $H^2(X, \mathcal{O}_X) = 0$)**.** *For a compact Riemann surface X, $H^2(X, \mathcal{O}_X) = 0$.*

Proof. As a complex manifold of complex dimension 1, the Dolbeault model gives $H^2(X, \mathcal{O}_X) \cong H^{0,2}(X)$. But there are no $(0,2)$-forms on a curve, hence $H^{0,2}(X) = 0$. □

Proposition 19.12 ($\mathrm{Pic}^0(X)$ as a quotient)**.** *Let $c_1 : \mathrm{Pic}(X) \to H^2(X, \mathbb{Z}) \cong \mathbb{Z}$ be the degree/first Chern class map. Then*

$$\mathrm{Pic}^0(X) := \ker(c_1) \;\cong\; H^1(X, \mathcal{O}_X)\big/\operatorname{Im}\alpha,$$

where $\alpha : H^1(X, \mathbb{Z}) \to H^1(X, \mathcal{O}_X)$ is induced by $\mathbb{Z} \hookrightarrow \mathcal{O}_X$.

Proof. Use (33). By Lemma 19.11, the map $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ is the zero map, so exactness at $H^1(X,\mathcal{O}_X^*)$ gives

$$\ker(c_1) = \mathrm{Im}(\beta).$$

Thus $\mathrm{Pic}^0(X) \cong \mathrm{Im}(\beta)$. By the first isomorphism theorem, $\mathrm{Im}(\beta) \cong H^1(X,\mathcal{O}_X)/\ker(\beta)$. Exactness at $H^1(X,\mathcal{O}_X)$ yields $\ker(\beta) = \mathrm{Im}(\alpha)$. Combining these gives the claimed quotient. □

Exercise 19.13. Show that $H^2(X,\mathbb{Z}) \cong \mathbb{Z}$ canonically by orientation, and that for a divisor D, the class $c_1(\mathcal{O}_X(D)) \in H^2(X,\mathbb{Z})$ equals $\deg D$.

19.4. Identifying $\mathrm{Pic}^0(X)$ with $J(X)$

1. Serre duality and the dual of holomorphic differentials

Serre duality on a curve gives a perfect pairing

$$H^1(X,\mathcal{O}_X) \times H^0(X,K_X) \longrightarrow \mathbb{C}, \qquad ([\eta],\omega) \longmapsto \int_X \omega \wedge \eta,$$

hence an isomorphism

$$H^1(X,\mathcal{O}_X) \;\cong\; H^0(X,K_X)^\vee \;=\; H^0(X,\Omega^1_X)^\vee. \tag{34}$$

2. The image of $H^1(X,\mathbb{Z})$ becomes the period lattice

Let $\alpha : H^1(X,\mathbb{Z}) \to H^1(X,\mathcal{O}_X)$ be induced by $\mathbb{Z} \hookrightarrow \mathcal{O}_X$. We now explain why $\mathrm{Im}(\alpha)$ corresponds to periods.

Proposition 19.14 (Integral classes give period functionals)**.** *Under the Serre duality identification* (34)*, the subgroup* $\mathrm{Im}(\alpha) \subset H^1(X,\mathcal{O}_X)$ *corresponds to the subgroup*

$$H_1(X,\mathbb{Z}) \;\hookrightarrow\; H^0(X,\Omega^1_X)^\vee, \qquad \gamma \longmapsto \big(\omega \mapsto \textstyle\int_\gamma \omega\big).$$

Proof. We give a concrete Čech–de Rham explanation. Choose a good cover $\{U_i\}$ of X. An element of $H^1(X,\mathbb{Z})$ is represented by an integral Čech 1-cocycle (n_{ij}) with $n_{ij} \in \mathbb{Z}$ on overlaps U_{ij}, satisfying $n_{ij} + n_{jk} + n_{ki} = 0$ on triple overlaps. Under α, this maps to the class of the same cocycle viewed in $\mathcal{O}_X$ (locally constant holomorphic functions).

Now let $\omega \in H^0(X,\Omega^1_X)$. The Serre pairing between ω and $\alpha[(n_{ij})]$ is computed by the standard Čech–Dolbeault formula: one chooses a partition of unity $\{\rho_i\}$ subordinate to $\{U_i\}$, forms a global $(0,1)$-form

$$\eta := \sum_{i,j} n_{ij}\, \rho_i\, \bar\partial \rho_j,$$

representing $\alpha[(n_{ij})] \in H^1(X,\mathcal{O}_X)$, and then computes

$$\langle \omega, \alpha[(n_{ij})] \rangle = \int_X \omega \wedge \eta.$$

A standard gluing argument shows that (n_{ij}) determines an integral 1-cycle $\gamma \in H_1(X,\mathbb{Z})$ (obtained by summing oriented edges dual to overlaps weighted by n_{ij}), and the integral above collapses to $\int_\gamma \omega$. This identifies the functional $\omega \mapsto \langle \omega, \alpha(u)\rangle$ with $\omega \mapsto \int_\gamma \omega$, hence $\mathrm{Im}(\alpha)$ matches the subgroup of period functionals. □

Theorem 19.15 ($\mathrm{Pic}^0(X) \cong J(X)$)**.** *There is a canonical isomorphism of complex tori*

$$\mathrm{Pic}^0(X) \;\cong\; \frac{H^1(X,\mathcal{O}_X)}{\mathrm{Im}(\alpha)} \;\cong\; \frac{H^0(X,\Omega^1_X)^\vee}{H_1(X,\mathbb{Z})} \;\cong\; \mathbb{C}^g/\Lambda \;=\; J(X).$$

Proof. Combine Proposition 19.12 with Serre duality (34) and Proposition 19.14. Finally, choosing a basis $\{\omega_j\}$ identifies $H^0(X,\Omega^1_X)^\vee \cong \mathbb{C}^g$ and sends $H_1(X,\mathbb{Z})$ to the period lattice Λ. □

Exercise 19.16. Let $X = \mathbb{C}/\Lambda$ be a complex torus. Prove directly from the exponential sequence that $\mathrm{Pic}^0(X) \cong \mathbb{C}/\Lambda$ and identify the isomorphism with $x \mapsto \mathcal{O}_X(x - p_0)$.

19.5. Divisors, line bundles, and complete linear systems

1. Divisors and the associated line bundle

A divisor on X is a finite sum $D = \sum_{p\in X} n_p\, p$ with $n_p \in \mathbb{Z}$. Its degree is $\deg D = \sum n_p$. The associated line bundle $\mathcal{O}_X(D)$ may be defined by specifying local trivializations and transition functions z^{-n_p} near each p.

Proposition 19.17 (Sections as meromorphic functions with bounded poles)**.** *There is a natural identification*

$$H^0\big(X,\mathcal{O}_X(D)\big) = \Big\{\, f \text{ meromorphic on } X \;\Big|\; (f) + D \geq 0 \,\Big\} \;\cup\; \{0\}.$$

Proof. By construction, a local trivialization of $\mathcal{O}_X(D)$ near p is given by a meromorphic frame e_p with $(e_p) = -D$ locally. A section s can be written as $s = f\,e_p$ with f holomorphic near p. This means $(s) = (f) + (e_p) \geq -D$, i.e. $(f) + D \geq 0$ globally. Conversely, any meromorphic f with $(f) + D \geq 0$ defines a section by $s = f \cdot 1_D$, where 1_D is the canonical meromorphic section of $\mathcal{O}_X(D)$. □

2. Complete linear system

Definition 19.18 (Complete linear system)**.** If $H^0(X,\mathcal{O}_X(D)) \neq 0$, the *complete linear system* of D is

$$|D| := \mathbb{P}\big(H^0(X,\mathcal{O}_X(D))\big).$$

Lemma 19.19 (Linear system equals effective divisors in the class)**.** *The map $s \mapsto (s)+D$ induces a bijection between $|D|$ and the set of effective divisors D' with $D' \sim D$. In particular,* $\dim |D| = h^0(X,\mathcal{O}_X(D)) - 1$ *whenever* $H^0(X,\mathcal{O}_X(D)) \neq 0$.

Proof. Let $0 \neq s \in H^0(X,\mathcal{O}_X(D))$. Then $(s) + D \geq 0$ by Proposition 19.17, so $D_s := (s) + D$ is effective and linearly equivalent to D because $(s) = D_s - D$ is a principal divisor in $\mathrm{Div}(X)$ (indeed (s) is the divisor of the meromorphic function representing s in a meromorphic trivialization). Scaling s does not change (s), hence not D_s, so the map factors through $|D|$.

Conversely, if $D' \geq 0$ and $D' \sim D$, then $D' - D = (f)$ for a meromorphic function f. The condition $D' \geq 0$ means $(f) + D \geq 0$, hence $f \in H^0(X,\mathcal{O}_X(D)) \setminus \{0\}$, and $D' = (f) + D$ is the divisor attached to the section $s = f$. Uniqueness up to scalar follows since $(f) = (cf)$ for $c \in \mathbb{C}^*$. □

Proposition 19.20 (Every line bundle is $\mathcal{O}_X(D)$)**.** *Every holomorphic line bundle L on X is isomorphic to $\mathcal{O}_X(D)$ for some divisor D. Moreover, $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ if and only if $D \sim D'$.*

Proof. Fix a point $p_0 \in X$. Choose m large enough so that $\deg(L(mp_0)) \geq g$. By Riemann–Roch,

$$h^0\big(X, L(mp_0)\big) - h^0\big(X, K_X \otimes L(mp_0)^\vee\big) = \deg(L) + m + 1 - g.$$

For m sufficiently large, the right-hand side is positive, hence $h^0(X, L(mp_0)) \geq 1$. Let $0 \neq s \in H^0(X, L(mp_0))$. Then s is a meromorphic section of L with poles of order $\leq m$ at p_0 and no other poles. Let $D := (s)$ be its divisor (zeros minus poles). By construction, $L \simeq \mathcal{O}_X(D)$.

If $L \simeq \mathcal{O}_X(D) \simeq \mathcal{O}_X(D')$, then $\mathcal{O}_X(D - D') \simeq \mathcal{O}_X$, hence $D - D' = (f)$ for some meromorphic f and $D \sim D'$. Conversely, if $D \sim D'$ then $D - D' = (f)$ and multiplication by f identifies $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$. □

Exercise 19.21. Show that $\deg(\mathcal{O}_X(D)) = \deg D$, i.e. $c_1(\mathcal{O}_X(D))$ corresponds to $\deg D \in \mathbb{Z} \cong H^2(X, \mathbb{Z})$.

19.6. Abel's theorem: fibers of α_d and linear equivalence

Theorem 19.22 (Abel's theorem: fibers are complete linear systems)**.** *Let $d \geq 0$ and $D, D' \in S^d X$. Then*

$$\alpha_d(D) = \alpha_d(D') \quad \Longleftrightarrow \quad D \sim D'.$$

Equivalently, for an effective divisor D of degree d,

$$\alpha_d^{-1}\big(\alpha_d(D)\big) = |D|.$$

Proof. Write $D = \sum_{i=1}^d x_i$ and $D' = \sum_{i=1}^d y_i$. Consider the degree-zero divisor $E := D' - D$. Under the identification $\mathrm{Pic}^0(X) \cong J(X)$ (Theorem 19.15), the class $\alpha_d(D') - \alpha_d(D) \in J(X)$ corresponds to the line bundle

$$\mathcal{O}_X(D') \otimes \mathcal{O}_X(D)^{-1} \;\cong\; \mathcal{O}_X(E) \;\in\; \mathrm{Pic}^0(X).$$

Thus $\alpha_d(D') = \alpha_d(D)$ if and only if $\mathcal{O}_X(E) \cong \mathcal{O}_X$, i.e. E is principal, i.e. $D' \sim D$. The fiber statement then follows from Lemma 19.19. □

Corollary 19.23 (Fiber dimension)**.** If D is effective of degree d and $H^0(X, \mathcal{O}_X(D)) \neq 0$, then

$$\dim \alpha_d^{-1}(\alpha_d(D)) = \dim |D| = h^0(X, \mathcal{O}_X(D)) - 1.$$

Exercise 19.24. Assume $g \geq 1$. Show that $\alpha_1 : X \to J(X)$ is injective if and only if X is not hyperelliptic. (You may use the fact that hyperelliptic curves admit degree 2 linear systems g^1_2.)

19.7. Jacobi inversion: surjectivity of α_g

Theorem 19.25 (Jacobi inversion)**.** *The Abel–Jacobi map $\alpha_g : S^g X \to J(X)$ is surjective.*

Proof. Let $y \in J(X) \cong \mathrm{Pic}^0(X)$ and let L_y be the corresponding degree-zero line bundle. Choose any effective divisor D of degree g. Apply Riemann–Roch to the line bundle $M := \mathcal{O}_X(D) \otimes L_y$:

$$h^0(X, M) - h^0(X, K_X \otimes M^\vee) = \deg(M) + 1 - g = g + 1 - g = 1.$$

In particular, $h^0(X, M) \geq 1$, so pick a nonzero section $s \in H^0(X, M)$. Let $E := (s) + D$ be its associated effective divisor of degree $\deg D = g$. By definition of M,

$$\mathcal{O}_X(E) \;\cong\; \mathcal{O}_X(D) \otimes L_y, \qquad \text{hence} \qquad \mathcal{O}_X(E - D) \;\cong\; L_y.$$

Under $\mathrm{Pic}^0(X) \cong J(X)$, this means $\alpha_g(E) - \alpha_g(D) = y$, so $y \in \alpha_g(S^g X)$. □

Exercise 19.26. Show that $\alpha_{g+1} : S^{g+1} X \to J(X)$ is also surjective and that a generic fiber has dimension 1.

19.8. Riemann–Roch from geometry of α_d

Proposition 19.27 (Dimension bound from fibers)**.** *Let D be an effective divisor of degree d. Then*

$$h^0(X, \mathcal{O}_X(D)) \;\geq\; d - g + 1, \qquad \textit{equivalently} \qquad \dim |D| \;\geq\; d - g.$$

Proof. The complex manifold $S^d X$ has dimension d and $J(X)$ has dimension g. Therefore the dimension of a fiber of α_d at any point is at least $d - g$. For D, the fiber is $|D|$ by Theorem 19.22, and $\dim |D| = h^0(\mathcal{O}_X(D)) - 1$ by Lemma 19.19. Hence $h^0(\mathcal{O}_X(D)) - 1 \geq d - g$. □

Remark 19.28. Riemann–Roch refines Proposition 19.27 to

$$h^0(\mathcal{O}_X(D)) - h^0(K_X \otimes \mathcal{O}_X(-D)) = d - g + 1.$$

19.9. Abel's theorem in degree zero: principal divisors

Let $\mathrm{Div}^0(X)$ denote divisors of degree 0. Fix $p_0 \in X$ and define a group homomorphism

$$\alpha_0 : \mathrm{Div}^0(X) \to J(X), \qquad \alpha_0\Big(\sum n_p\, p\Big) := \sum n_p\, \alpha(p).$$

Theorem 19.29 (Abel's theorem, degree zero)**.** *For $D \in \mathrm{Div}^0(X)$,*

$$\alpha_0(D) = 0 \quad \Longleftrightarrow \quad D \textit{ is principal (i.e. } D = (f) \textit{ for some meromorphic } f\textit{).}$$

Proof. ($\Rightarrow$, sketch with the key analytic input.) Assume $\alpha_0(D) = 0$. Write $D = D_+ - D_-$ with $D_\pm$ effective of the same degree d. The condition $\alpha_0(D) = 0$ means $\alpha_d(D_+) = \alpha_d(D_-)$. By Theorem 19.22, $D_+ \sim D_-$, so $D_+ - D_- = (f)$ for some meromorphic f, i.e. D is principal.

($\Leftarrow$.) If $D = (f)$, then $D \sim 0$, hence $\mathcal{O}_X(D) \cong \mathcal{O}_X$, so the corresponding class in $\mathrm{Pic}^0(X) \cong J(X)$ is the identity. Equivalently, $\alpha_0(D) = 0$. □

Exercise 19.30. Let $D = p - q$ for points $p, q \in X$. Show that $\alpha_0(D) = 0$ if and only if there exists a nonconstant meromorphic function f with divisor $(f) = p - q$. Interpret this as the existence of a degree 1 meromorphic map $X \to \mathbb{CP}^1$.

19.10. Worked examples: genus 0 and genus 1

1. $X = \mathbb{CP}^1$ (genus 0)

Here $H^0(\mathbb{CP}^1, \Omega^1) = 0$, hence $\mathbb{C}^g = 0$, $\Lambda = 0$, and $J(\mathbb{CP}^1) = 0$. Also $\mathrm{Pic}^0(\mathbb{CP}^1) = 0$ and $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ via degree.

Exercise 19.31. Show $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ by proving $\mathcal{O}_{\mathbb{CP}^1}(D) \cong \mathcal{O}_{\mathbb{CP}^1}(\deg D)$.

2. Elliptic curve $X = \mathbb{C}/\Lambda$ (genus 1)

Let $\omega = dz$. Then $H^0(X, \Omega^1) = \mathbb{C} \cdot dz$, and

$$\Lambda_{\mathrm{per}} = \Big\{ \int_\gamma dz : \gamma \in H_1(X, \mathbb{Z}) \Big\} = \Lambda \subset \mathbb{C}.$$

Hence $J(X) = \mathbb{C}/\Lambda \cong X$. With base point $p_0 = 0$, the Abel–Jacobi map is

$$\alpha([z]) = z \bmod \Lambda,$$

which is a group isomorphism.

Exercise 19.32. Prove that for an elliptic curve X, the map $\alpha_d : S^d X \to J(X) \cong X$ is the addition map on the group X (after identifying $S^d X$ with effective degree-d divisors).

Part VIII
Algebraic Curves and Algebraic Geometry

20. Compact Riemann Surfaces as Algebraic Curves

Overview

Let M be a compact connected Riemann surface of genus g. We explain constructively:

1. how a divisor D with $\deg D \geq 2g + 1$ gives an embedding $M \hookrightarrow \mathbb{CP}^N$,
2. how to obtain two meromorphic generators $f, g \in \mathbb{C}(M)$ from the same linear system,
3. how the algebraic relation $P(f, g) = 0$ produces a plane curve whose normalization is M.

Notation

For a divisor D on M, write $\mathcal{O}_M(D)$ for the associated line bundle and set

$$L(D) := H^0(M, \mathcal{O}_M(D)), \qquad \ell(D) := \dim_{\mathbb{C}} L(D).$$

Let K be a canonical divisor. We use Riemann–Roch:

$$\ell(D) - \ell(K - D) = 1 - g + \deg D.$$

20.1. Very ample divisors on curves

Definition 20.1 (Base-point free and separation)**.** Let D be a divisor on M.

1. $|D|$ is *base-point free* if for every $p \in M$ there exists $s \in L(D)$ with $s(p) \neq 0$.
2. $|D|$ *separates points* if for $p \neq q$ there exists $s \in L(D)$ with $s(p) = 0$ and $s(q) \neq 0$.
3. $|D|$ *separates tangent directions* at p if there exists $s \in L(D)$ vanishing at p to order exactly 1.

Lemma 20.2 (Base-point freeness for $\deg D \geq 2g$)**.** *If $\deg D \geq 2g$, then $|D|$ is base-point free.*

Proof. Fix $p \in M$. It suffices to show $\ell(D) > \ell(D - p)$. By Riemann–Roch applied to D and $D - p$,

$$\ell(D) - \ell(D - p) = 1 + \ell(K - D) - \ell(K - D + p).$$

If $\deg D \geq 2g$, then $\deg(K - D + p) = 2g - 1 - \deg D \leq -1$, hence $\ell(K - D + p) = 0$. Therefore $\ell(D) - \ell(D - p) \geq 1$. □

Lemma 20.3 (Point and tangent separation for $\deg D \geq 2g + 1$)**.** *If $\deg D \geq 2g + 1$, then $|D|$ separates points and tangent directions.*

Proof. (Points) For $p \neq q$ it suffices to show $\ell(D - p) > \ell(D - p - q)$. Riemann–Roch gives

$$\ell(D - p) - \ell(D - p - q) = 1 + \ell(K - D + p) - \ell(K - D + p + q).$$

If $\deg D \geq 2g+1$, then $\deg(K-D+p+q) = 2g-\deg D \leq -1$, hence $\ell(K-D+p+q) = 0$.

(Tangents) It suffices to show $\ell(D - p) > \ell(D - 2p)$. Again,

$$\ell(D - p) - \ell(D - 2p) = 1 + \ell(K - D + p) - \ell(K - D + 2p),$$

and $\deg(K - D + 2p) = 2g - \deg D \leq -1$, hence $\ell(K - D + 2p) = 0$. □

Theorem 20.4 (Embedding by a complete linear system)**.** *Let $\deg D \geq 2g + 1$ and let $s_0, \dots, s_N$ be a basis of $L(D)$. Then the map*

$$\Phi_D : M \longrightarrow \mathbb{CP}^N, \qquad p \longmapsto [s_0(p) : \cdots : s_N(p)]$$

is a holomorphic embedding.

Proof. Lemma 20.2 implies Φ_D is well-defined. Lemma 20.3 implies Φ_D is injective and an immersion. Compactness of M implies Φ_D is a topological embedding, hence holomorphic embedding. □

20.2. Two meromorphic generators and a plane model

Lemma 20.5 (Transcendence degree one ⇒ algebraic dependence)**.** *For any nonconstant $f, g \in \mathbb{C}(M)$ there exists a nonzero polynomial $P(x, y) \in \mathbb{C}[x, y]$ such that $P(f, g) = 0$.*

Proof. The function field $\mathbb{C}(M)$ has transcendence degree 1 over $\mathbb{C}$. Thus g is algebraic over $\mathbb{C}(f)$, so it satisfies a polynomial with coefficients in $\mathbb{C}(f)$. Clearing denominators yields $P \in \mathbb{C}[x, y] \setminus \{0\}$ with $P(f, g) = 0$. □

Proposition 20.6 (Plane curve and normalization (function-field view))**.** *Let $f, g \in \mathbb{C}(M)$ be nonconstant and let $P(x,y) \in \mathbb{C}[x,y]$ be irreducible with $P(f,g) = 0$. Let $\widetilde{P}(X,Y,Z)$ be the homogenization of P and set*

$$C := \{[X:Y:Z] \in \mathbb{CP}^2 \mid \widetilde{P}(X,Y,Z) = 0\}.$$

Then:

1. *The meromorphic map $M \dashrightarrow \mathbb{CP}^2$, $p \mapsto [1 : f(p) : g(p)]$, has image contained in C.*
2. *The induced map $\nu : M \to C$ is finite.*
3. *If $\mathbb{C}(f,g) = \mathbb{C}(M)$, then ν is birational and M is biholomorphic to the normalization C^{norm}.*

Proof. (1) The relation $\widetilde{P}(1, f, g) = 0$ holds on the domain where f, g are finite, hence the image lies in C. (2) The map is holomorphic as a map to projective space; since M is compact it is proper, hence finite onto its image. (3) If $\mathbb{C}(f,g) = \mathbb{C}(M)$ then the induced extension of function fields is an isomorphism, so ν is birational. A finite birational map from a smooth curve is the normalization morphism. □

20.3. Examples

Example 20.7 ($\mathbb{CP}^1$ as a conic)**.** Let $M = \mathbb{CP}^1$ with affine coordinate z. Take $f = z$, $g = z^2$. Then $P(x,y) = y - x^2$ and $\widetilde{P}(X,Y,Z) = YZ - X^2$. Thus $\mathbb{CP}^1 \simeq \{YZ = X^2\} \subset \mathbb{CP}^2$ via $[u:v] \mapsto [u^2 : uv : v^2]$.

Example 20.8 (Elliptic curves via Weierstrass $\wp$)**.** Let $M = \mathbb{C}/\Lambda$. With $f = \wp$ and $g = \wp'$ one has

$$g^2 = 4(f - e_1)(f - e_2)(f - e_3),$$

giving a cubic plane model after homogenization.

Example 20.9 (Hyperelliptic curves)**.** For distinct $a_1, \dots, a_{2g+2} \in \mathbb{C}$, the smooth projective model of $y^2 = \prod_{k=1}^{2g+2}(x - a_k)$ has function field $\mathbb{C}(x,y)$ and already yields a plane model.

20.4. Exercises

Exercise 20.10 (Avoiding undefined Proj)**.** Show that M can be recovered from a very ample divisor D via the graded ring

$$R(D) := \bigoplus_{n \geq 0} H^0\big(M, \mathcal{O}_M(nD)\big).$$

Prove (or outline carefully) why

$$M \;\simeq\; \operatorname{Proj} R(D).$$

(Hint: use the embedding Φ_D and interpret $R(D)$ as the homogeneous coordinate ring of $\Phi_D(M)$.)

Exercise 20.11 (Generic projection and generation of the function field)**.** Let $\deg D \geq 2g + 1$ and $\Phi_D : M \hookrightarrow \mathbb{CP}^N$. Explain why a generic linear projection $\mathbb{CP}^N \dashrightarrow \mathbb{CP}^2$ induces meromorphic functions f, g with $\mathbb{C}(f,g) = \mathbb{C}(M)$.

21. Algebraic Intersection Multiplicity of Plane Curves

This section is compatible with the section of Intersection Theory and Riemann–Roch for Surfaces: we give a purely local (commutative algebra) definition of $I_p(C, C')$ for plane curves, prove its key properties, and explain how it matches the global intersection pairing on smooth projective surfaces.

21.1. Local algebraic definition via length

Let k be an algebraically closed field. Let $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ and let

$$C = (f = 0), \qquad C' = (g = 0)$$

be affine plane curves with $f, g \in k[x, y]$ nonconstant. Fix a point $p = (a, b) \in C \cap C'$ and set

$$R := \mathcal{O}_{\mathbb{A}^2,p} \cong k[x, y]_{(x-a,y-b)}, \qquad \mathfrak{m}_p := (x - a, y - b)R.$$

Define the local intersection algebra

$$Q_p(f, g) \;:=\; R/(f, g).$$

Lemma 21.1 (Isolated intersection $\Longleftrightarrow$ finite k-dimension)**.** *The following are equivalent:*

1. *p is an isolated point of $C \cap C'$ in a Zariski neighbourhood of p;*
2. *$Q_p(f, g)$ is an Artinian local ring;*
3. *$\dim_k Q_p(f, g) < \infty$.*

Proof. The scheme-theoretic intersection near p is $\operatorname{Spec}(R/(f, g))$. It is 0-dimensional (hence p is isolated) if and only if $R/(f, g)$ is Artinian. Since k is algebraically closed, the residue field of R is k, and an Artinian local k-algebra is finite-dimensional over k. Conversely, if $\dim_k(R/(f, g)) < \infty$, then $R/(f, g)$ is Artinian. □

Definition 21.2 (Local intersection multiplicity)**.** Assume p is an isolated intersection point (equivalently, Lemma 21.1 holds). The *intersection multiplicity* of C and C' at p is

$$I_p(C, C') \;:=\; \operatorname{length}_R \big(R/(f, g)\big) \;=\; \dim_k \big(R/(f, g)\big).$$

Remark 21.3 (Coordinate invariance)**.** The quantity $I_p(C, C')$ depends only on the isomorphism class of the local ring R and the ideal $(f, g) \subset R$. In particular, it is invariant under any étale / analytic change of local coordinates at p.

Remark 21.4 (Extension to smooth surfaces)**.** If X is a smooth surface and $C, C' \subset X$ are Cartier divisors meeting properly at p with local equations $f, g \in \mathcal{O}_{X,p}$, define

$$I_p(C, C') := \operatorname{length}_{\mathcal{O}_{X,p}} \big(\mathcal{O}_{X,p}/(f, g)\big).$$

This specializes to Definition 21.2 when $X = \mathbb{A}^2$.

Remark 21.5 (Serre's Tor-formula (context))**.** For closed subschemes X, Y of a Noetherian scheme S and $p \in X \cap Y$, Serre defines

$$I_p(X, Y) \;:=\; \sum_{i \geq 0} (-1)^i \operatorname{length}_{\mathcal{O}_{S,p}} \Big(\operatorname{Tor}_i^{\mathcal{O}_{S,p}}(\mathcal{O}_{X,p}, \mathcal{O}_{Y,p}) \Big).$$

If S is regular at p and X, Y meet properly, then $\operatorname{Tor}_i = 0$ for $i > 0$ and one recovers $\operatorname{length}(\mathcal{O}_{S,p}/(f, g))$.

21.2. Basic properties (with proofs)

Assume $p \in C \cap C'$ is isolated.

Proposition 21.6 (Symmetry)**.** $I_p(C, C') = I_p(C', C)$.

Proof. $(f, g) = (g, f)$ as ideals in R, hence $R/(f, g) \cong R/(g, f)$ and the lengths agree. □

Proposition 21.7 (Positivity and vanishing)**.** *If $p \in C \cap C'$ is isolated, then $I_p(C, C') \geq 1$. Moreover, $I_p(C, C') = 0$ if and only if $p \notin C \cap C'$.*

Proof. If $p \notin C \cap C'$, then either f or g is a unit in R, so $R/(f, g) = 0$. If p is an isolated intersection point, then $R/(f, g)$ is a nonzero Artinian local ring, hence has positive length. □

Proposition 21.8 (Additivity with respect to components (a useful local form))**.** *Assume in R that $f = f_1 f_2$ and that $V(f_1)$ and $V(f_2)$ share no common component through p (equivalently, (f_1, f_2) is $\mathfrak{m}_p$-primary). If p is isolated in $V(f) \cap V(g)$, then*

$$I_p\big((f = 0), (g = 0)\big) \;=\; I_p\big((f_1 = 0), (g = 0)\big) + I_p\big((f_2 = 0), (g = 0)\big).$$

Proof. In $R/(g)$, multiplication by f_2 gives an exact sequence

$$0 \longrightarrow R/(f_1, g) \xrightarrow{\cdot f_2} R/(f_1 f_2, g) \longrightarrow R/(f_2, g) \longrightarrow 0,$$

where injectivity uses that f_2 is a non-zero-divisor modulo (f_1, g) under the hypothesis (no common component through p). Taking lengths yields the identity. □

Proposition 21.9 (Transversality implies multiplicity 1)**.** *If $\nabla f(p)$ and $\nabla g(p)$ are linearly independent in $T_p\mathbb{A}^2$, then $I_p(C, C') = 1$.*

Proof. After translating p to $(0, 0)$, the Jacobian criterion shows (f, g) defines a reduced 0-dimensional complete intersection at p. Equivalently, in the regular local ring $R \cong k[[u, v]]$ there exist parameters (u, v) such that $f = u$ and $g = v$ up to units. Thus

$$R/(f, g) \;\cong\; k[[u, v]]/(u, v) \;\cong\; k,$$

so $I_p(C, C') = \text{length}_R(k) = 1$. □

21.3. Multiplicity of a curve at a point and the tangent cone bound

Assume $p = (0, 0)$ after translation and set $R = k[x, y]_{(x,y)}$. Write $f = \sum_{d \geq 0} f_d$ where f_d is homogeneous of degree d.

Definition 21.10 (Order and initial form)**.** Define the *multiplicity* of $C = (f = 0)$ at 0 by

$$m_0(C) := \min\{d \mid f_d \neq 0\},$$

and the *initial form* by $\in (f) := f_{m_0(C)}$. The *tangent cone* of C at 0 is the homogeneous curve $(\in (f) = 0)$ in $\P^1$.

Theorem 21.11 (Tangent cone lower bound)**.** *Let $C = (f = 0)$ and $C' = (g = 0)$ meet isolatedly at 0. Then*

$$I_0(C, C') \;\geq\; m_0(C)\, m_0(C').$$

Moreover, equality holds if and only if $\in (f)$ and $\in (g)$ are coprime (equivalently, the two tangent cones share no common component).

Proof. Consider the $\mathfrak{m}$-adic filtration on R and the associated graded ring

$$\operatorname{gr}_{\mathfrak{m}} R \;\cong\; k[x, y] \quad \text{(graded)}.$$

The images of f and g in $\operatorname{gr}_{\mathfrak{m}} R$ are precisely $\in (f)$ and $\in (g)$. A standard filtered-algebra argument gives

$$\operatorname{length}_R \big(R/(f, g)\big) \;\geq\; \operatorname{length}_{\operatorname{gr}_{\mathfrak{m}} R} \big(\operatorname{gr}_{\mathfrak{m}} R/(\in (f), \in (g))\big).$$

If $\in (f)$ and $\in (g)$ are coprime homogeneous polynomials of degrees $m_0(C)$ and $m_0(C')$, then $(\in (f), \in (g))$ is a homogeneous complete intersection and its local length at the origin equals $m_0(C)m_0(C')$ (this is the basic Hilbert–Samuel multiplicity computation in dimension 2). If $\in (f)$ and $\in (g)$ share a common factor, the quotient has larger length, giving strict inequality. □

21.4. Computation techniques (with concrete worked demonstrations)

(1) "Eliminate y" and compute a 1-variable quotient

A frequent trick is: if one equation is $y - h(x)$ (or can be made so), substitute into the other.

Example 21.12 (A general "graph vs. curve" computation)**.** Let $C : (f = 0)$ and $C' : (y - h(x) = 0)$ meet at 0, where $h(0) = 0$. Then in $R = k[x, y]_{(x,y)}$ we have

$$R/(f,\; y - h(x)) \;\cong\; k[x]_{(x)}/\big(f(x, h(x))\big).$$

Hence

$$I_0(C, C') \;=\; \dim_k k[x]_{(x)}/\big(f(x, h(x))\big) \;=\; \operatorname{ord}_{x=0} \big(f(x, h(x))\big).$$

(2) Puiseux/branch method (char 0)

If $\operatorname{char} k = 0$, a reduced plane curve has local branches $y = \varphi_i(x^{1/n})$, and

$$I_0(C, C') = \sum_i \operatorname{ord}_{x=0} \big(g(x, \varphi_i(x))\big).$$

This is especially effective for singular curves (cusps, nodes).

(3) Resultant method

If $\gcd(f, g) = 1$ and both are polynomials in y with coefficients in $k[x]$, then

$$I_0(C, C') = \operatorname{ord}_{x=0} \big(\operatorname{Res}_y(f, g)\big).$$

(You can compute Res_y via Sylvester matrices for explicit polynomials.)

(4) Standard monomials / Gröbner length

Compute a Gröbner basis for (f, g) in $k[x, y]$ with a local monomial order; the number of standard monomials gives the length.

21.5. Examples

Example 21.13 (Transversal lines)**.** Let $C : (y = 0)$ and $C' : (x = 0)$ at 0. Then

$$k[x, y]_{(x,y)}/(x, y) \cong k \quad \Rightarrow \quad I_0(C, C') = 1.$$

Example 21.14 (Parabola and its tangent line)**.** Let $C : (y - x^2 = 0)$ and $C' : (y = 0)$ at 0. Then

$$k[x, y]_{(x,y)}/(y - x^2,\ y) \ \cong \ k[x]_{(x)}/(x^2),$$

so a k-basis is $\{1, x\}$ and $I_0(C, C') = 2$.

Example 21.15 (Cusp and the x-axis)**.** Let $C : (y^2 - x^3 = 0)$ and $C' : (y = 0)$ at 0. Then

$$k[x, y]_{(x,y)}/(y^2 - x^3,\ y) \ \cong \ k[x]_{(x)}/(x^3),$$

so $\{1, x, x^2\}$ is a k-basis and $I_0(C, C') = 3$.

Example 21.16 (Same tangent line, but still reduced)**.** Let $C : (y - x^2 = 0)$ and $C' : (y - x^3 = 0)$ at 0. Subtract the equations: $(y - x^2) - (y - x^3) = x^2 - x^3 = x^2(1 - x)$. Since $1 - x$ is a unit in $k[x]_{(x)}$, we get

$$k[x, y]_{(x,y)}/(y - x^2,\ y - x^3) \ \cong \ k[x]_{(x)}/(x^2),$$

hence $I_0(C, C') = 2$. (Geometrically: same tangent line $y = 0$, and contact of order 2.)

Example 21.17 (A strict inequality case for the tangent bound)**.** Let $C : (y^2 = 0)$ (a doubled line) and $C' : (y = 0)$ at 0. Then $m_0(C) = 2$ and $m_0(C') = 1$, so $m_0(C)m_0(C') = 2$. But

$$k[x, y]_{(x,y)}/(y^2,\ y) \ \cong \ k[x]_{(x)}/(0),$$

which is *not* finite-dimensional over k: the intersection is not isolated (they share the entire line $y = 0$), so $I_0(C, C')$ is not defined in the isolated-intersection sense. This example pinpoints why the "isolated" hypothesis is essential.

Example 21.18 (A node: two branches)**.** Let $C : (y^2 - x^2 - x^3 = 0)$ and $C' : (y = 0)$ at 0. Modding out by y gives $-x^2 - x^3 = -x^2(1 + x)$, and $1 + x$ is a unit in $k[x]_{(x)}$. Hence

$$k[x, y]_{(x,y)}/(y^2 - x^2 - x^3,\ y) \ \cong \ k[x]_{(x)}/(x^2),$$

so $I_0(C, C') = 2$.

21.6. Global relation: Bézout's theorem

Let $C = (F = 0)$ and $C' = (G = 0)$ be projective plane curves in $\mathbb{P}^2_k$ of degrees $d = \deg F$ and $d' = \deg G$ with no common component. For each $p \in C \cap C'$, one defines $I_p(C, C')$ by passing to an affine chart around p and using Definition 21.2. (One checks independence of the chart.)

Theorem 21.19 (Bézout)**.** *If C and C' have no common component, then*

$$\sum_{p \in C \cap C'} I_p(C, C') \;=\; d\,d'.$$

Proof sketch (Chow ring / degree of a 0-cycle). The scheme-theoretic intersection $Z = C \cap C'$ is 0-dimensional. Its degree is

$$\deg(Z) = \sum_{p \in Z} \operatorname{length}(\mathcal{O}_{Z,p}) = \sum_{p \in C \cap C'} I_p(C, C').$$

On the other hand, in the Chow ring $A^\bullet(\mathbb{P}^2) = \mathbb{Z}[H]/(H^3)$, where H is the hyperplane class, we have $[C] = dH$ and $[C'] = d'H$, so

$$[C] \cdot [C'] = dd'\, H^2 = dd'\, [\mathrm{pt}],$$

hence $\deg(Z) = dd'$. □

Example 21.20 (Verifying Bézout in a concrete case)**.** Let $C : (XZ - Y^2 = 0)$ (a conic) and $L : (Z = 0)$ (the line at infinity) in $\mathbb{P}^2$. On $Z = 0$, the conic equation becomes $-Y^2 = 0$, so $Y = 0$ and the unique intersection point is $p = [1 : 0 : 0]$. In the affine chart $X \neq 0$ with coordinates $u = Y/X$, $v = Z/X$, the equations become

$$C :\; v - u^2 = 0, \qquad L :\; v = 0.$$

Thus

$$I_p(C, L) = \dim_k k[u, v]_{(u,v)}/(v - u^2,\; v) \cong \dim_k k[u]_{(u)}/(u^2) = 2.$$

Hence $\sum I_p = 2 = (\deg C)(\deg L) = 2 \cdot 1$.

21.7. Compatibility with surface intersection theory

Let X be a smooth projective surface over $\mathbb{C}$. For Cartier divisors D_1, D_2 meeting properly, define the global intersection number by

$$(D_1 \cdot D_2) \;:=\; \sum_{p \in X} I_p(D_1, D_2).$$

Proposition 21.21 (Agreement with the Chern class / cup-product pairing)**.** *On a smooth complex projective surface X,*

$$(D_1 \cdot D_2) \;=\; \int_X c_1\big(\mathcal{O}_X(D_1)\big) \wedge c_1\big(\mathcal{O}_X(D_2)\big),$$

and this equals the cup product $c_1(\mathcal{O}_X(D_1)) \smile c_1(\mathcal{O}_X(D_2)) \in H^4(X, \mathbb{Z}) \cong \mathbb{Z}$.

Explanation. The divisor class map $D \mapsto c_1(\mathcal{O}_X(D))$ identifies linear equivalence with cohomology classes. The product $c_1(\mathcal{O}_X(D_1)) \smile c_1(\mathcal{O}_X(D_2))$ evaluates on $[X]$ to an integer that is, by definition, the degree of the 0-cycle $D_1 \cdot D_2$. Locally, each point p contributes the length $\operatorname{length}(\mathcal{O}_{X,p}/(f, g))$, i.e. $I_p(D_1, D_2)$, and summing over p yields the global degree. □

21.8. Exercises

Exercise 21.22 (A family of tangencies)**.** Let $C_m : (y - x^m = 0)$ and $L : (y = 0)$ in $\mathbb{A}^2$. Compute $I_0(C_m, L)$ and interpret the answer as the order of contact.

Exercise 21.23 (Transversality criterion)**.** Assume $\operatorname{char} k = 0$. Show that if $\det(\partial(f, g)/\partial(x, y))(p) \neq 0$, then $I_p(C, C') = 1$.

Exercise 21.24 (Tangent cones and equality)**.** Let $C = (f = 0)$ and $C' = (g = 0)$ meet isolatedly at 0.

1. Prove $I_0(C, C') \geq m_0(C)m_0(C')$.
2. Give an example where $m_0(C)m_0(C') = 1$ but $I_0(C, C') > 1$. (Hint: $y - x^2$ and $y - x^3$.)

Exercise 21.25 (Resultant computation)**.** Let $C : (y^2 - x^3 = 0)$ and $C' : (y - x^2 = 0)$ at 0. Compute $I_0(C, C')$ by substituting $y = x^2$, and then verify the same answer using a resultant in y.

Exercise 21.26 (Bézout check)**.** Let $C : (Y^2Z = X^3 + XZ^2)$ (a cubic) and $L : (Z = 0)$ in $\mathbb{P}^2$. Compute all intersection points and multiplicities, and verify $\sum I_p = 3$.

22. Jacobi Theta Functions from the Viewpoint of Riemann Surfaces and Jacobians

In this section we explain Jacobi theta functions from the viewpoint of *Riemann surfaces* and their *Jacobian varieties*. The main goal is to make precise (and computable) the slogan

"theta is an entire function on $\mathbb{C}^g$ with controlled quasi-periods, so it descends to a section of a line bu

We proceed in three layers:

1. analytic: the Riemann theta series converges and satisfies explicit quasi-periodicity,
2. geometric: quasi-periodicity is exactly the cocycle of a line bundle on $A = \mathbb{C}^g/\Lambda$ and the zero set descends to the *theta divisor*,
3. genus 1: the abstract theta becomes the classical Jacobi theta functions and recovers Weierstrass theory via logarithmic derivatives.

22.1. From a Riemann surface to its Jacobian: period matrices and lattices

Let X be a compact Riemann surface of genus g. Choose a symplectic basis of homology

$$\{a_1, \dots, a_g, b_1, \dots, b_g\} \subset H_1(X, \mathbb{Z}) \quad \text{with} \quad a_i \cdot a_j = b_i \cdot b_j = 0, \quad a_i \cdot b_j = \delta_{ij}.$$

Let $\{\omega_1, \dots, \omega_g\}$ be a basis of holomorphic 1–forms on X, normalized by

$$\int_{a_j} \omega_i = \delta_{ij}.$$

Define the *period matrix* $\Omega = (\Omega_{ij})$ by

$$\Omega_{ij} := \int_{b_j} \omega_i.$$

Then Ω is symmetric and has positive definite imaginary part.

Theorem 22.1 (Riemann bilinear relations: key properties of Ω)**.** *With the above normalization, Ω satisfies*

$$\Omega^t = \Omega, \qquad \Im\Omega > 0.$$

Sketch with the useful computation. Symmetry. For holomorphic forms ω_i, ω_j, the 2–form $\omega_i \wedge \omega_j$ vanishes identically on a complex 1–manifold. Applying the bilinear relations gives

$$\sum_{k=1}^{g} \left(\int_{a_k} \omega_i \int_{b_k} \omega_j - \int_{b_k} \omega_i \int_{a_k} \omega_j \right) = 0.$$

Using $\int_{a_k} \omega_i = \delta_{ik}$ gives $\Omega_{ij} - \Omega_{ji} = 0$.

Positivity. For any $0 \neq c = (c_1, \ldots, c_g) \in \mathbb{C}^g$, let $\eta = \sum_i c_i \omega_i$. Then

$$i \int_X \eta \wedge \overline{\eta} > 0,$$

and the bilinear relations for $(\eta, \overline{\eta})$ yield $c^t (\Im\Omega) \overline{c} > 0$, hence $\Im\Omega$ is positive definite. $\square$

Define the lattice

$$\Lambda := \mathbb{Z}^g + \Omega\, \mathbb{Z}^g \subset \mathbb{C}^g, \qquad A := \mathbb{C}^g / \Lambda.$$

Under the identification $\mathbb{C}^g \simeq H^0(X, \Omega^1_X)^\vee$ by integration, A is the *Jacobian* $\mathrm{Jac}(X)$. Choosing a base point $P_0 \in X$, the Abel–Jacobi map is

$$u(P) := \left(\int_{P_0}^{P} \omega_1, \ldots, \int_{P_0}^{P} \omega_g \right) \pmod{\Lambda}.$$

Example 22.2 (Genus 1 period lattice from an elliptic integral)**.** Let E be the smooth projective cubic

$$E: \quad y^2 = 4(x - e_1)(x - e_2)(x - e_3), \qquad e_1, e_2, e_3 \in \mathbb{R}, \quad e_1 > e_2 > e_3.$$

On E the holomorphic differential is $\omega = \frac{dx}{y}$. A standard homology basis (a, b) can be chosen so that (up to orientation)

$$\omega_1 := \int_a \omega = 2 \int_{e_2}^{e_1} \frac{dx}{\sqrt{4(x - e_1)(x - e_2)(x - e_3)}}, \qquad \omega_2 := \int_b \omega = 2 \int_{e_3}^{e_2} \frac{dx}{\sqrt{4(x - e_1)(x - e_2)(x - e_3)}}\, i,$$

so the period lattice is $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and $\tau = \omega_2/\omega_1 \in \mathbb{H}$. (Geometrically: one cycle runs along the real oval, the other runs through a complex oval.)

22.2. The Riemann theta function: convergence and quasi-periodicity

Definition 22.3 (Riemann theta function)**.** Let Ω be symmetric with $\Im\Omega > 0$. Define

$$\theta(z\,|\,\Omega) := \sum_{n\in\mathbb{Z}^g} \exp\Big(\pi i\, n^t\Omega n + 2\pi i\, n^t z\Big), \qquad z\in\mathbb{C}^g.$$

Proposition 22.4 (Absolute and locally uniform convergence)**.** *The series for $\theta(z\,|\,\Omega)$ converges absolutely and uniformly on compact subsets of $\mathbb{C}^g$. Hence $\theta(\,\cdot\,|\,\Omega)$ is entire.*

Proof. Write $\Omega = X + iY$ with $X = \operatorname{Re}\Omega$ and $Y = \operatorname{Im}\Omega > 0$. Then

$$\Big|\exp\big(\pi i\, n^t\Omega n + 2\pi i\, n^t z\big)\Big| = \exp\big(-\pi\, n^t Y n - 2\pi\, n^t \Im z\big).$$

Fix a compact $K\subset\mathbb{C}^g$ and let $M := \sup_{z\in K}\|\Im z\|$. Then $-2\pi\, n^t\Im z \le 2\pi M\|n\|$. Since Y is positive definite, $\exists c>0$ with $n^tYn \ge c\|n\|^2$. Hence for $z\in K$,

$$\Big|\exp(\cdots)\Big| \le \exp\big(-\pi c\|n\|^2 + 2\pi M\|n\|\big),$$

and the right-hand side is summable over $\mathbb{Z}^g$. Apply the Weierstrass M-test. □

Proposition 22.5 (Quasi-periodicity)**.** *For $m,n\in\mathbb{Z}^g$ we have*

$$\begin{aligned}\theta(z+m\,|\,\Omega) &= \theta(z\,|\,\Omega),\\ \theta(z+\Omega n\,|\,\Omega) &= \exp\big(-\pi i\, n^t\Omega n - 2\pi i\, n^t z\big)\ \theta(z\,|\,\Omega).\end{aligned}$$

Equivalently, for $\lambda = m+\Omega n\in\Lambda$,

$$\theta(z+\lambda\,|\,\Omega) = \alpha_\lambda(z)\,\theta(z\,|\,\Omega), \quad \alpha_{m+\Omega n}(z) := \exp\big(-\pi i\, n^t\Omega n - 2\pi i\, n^t z\big).$$

Proof. The first identity is immediate since $e^{2\pi i n^t m} = 1$ for $m\in\mathbb{Z}^g$. For the second, compute and reindex:

$$\begin{aligned}\theta(z+\Omega n) &= \sum_{k\in\mathbb{Z}^g} \exp\big(\pi i\, k^t\Omega k + 2\pi i\, k^t z + 2\pi i\, k^t\Omega n\big)\\ &= \sum_{k'\in\mathbb{Z}^g} \exp\big(\pi i\,(k'-n)^t\Omega(k'-n) + 2\pi i\,(k'-n)^t z + 2\pi i\,(k'-n)^t\Omega n\big)\\ &= e^{-\pi i n^t\Omega n - 2\pi i n^t z}\sum_{k'\in\mathbb{Z}^g} \exp\big(\pi i\, k'^t\Omega k' + 2\pi i\, k'^t z\big)\\ &= e^{-\pi i n^t\Omega n - 2\pi i n^t z}\,\theta(z).\end{aligned}$$

□

Example 22.6 (Concrete $g=1$ check of quasi-periodicity)**.** Let $\tau\in\mathbb{H}$ and define

$$\vartheta_3(z,\tau) := \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau + 2\pi i n z}.$$

Then:

1. Period 1:

$$\vartheta_3(z+1,\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau + 2\pi i n(z+1)} = \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau + 2\pi i n z}\, e^{2\pi i n} = \vartheta_3(z,\tau).$$

2. Quasi-period τ:

$$\vartheta_3(z+\tau,\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau + 2\pi i n(z+\tau)} = \sum_{n\in\mathbb{Z}} e^{\pi i n^2\tau + 2\pi i n z}\, e^{2\pi i n\tau}.$$

Now reindex $n = m-1$:

$$\begin{aligned}\vartheta_3(z+\tau,\tau) &= \sum_{m\in\mathbb{Z}} e^{\pi i (m-1)^2\tau + 2\pi i (m-1) z}\, e^{2\pi i (m-1)\tau} \\ &= e^{-\pi i\tau - 2\pi i z} \sum_{m\in\mathbb{Z}} e^{\pi i m^2\tau + 2\pi i m z} = e^{-\pi i\tau - 2\pi i z}\,\vartheta_3(z,\tau).\end{aligned}$$

This is exactly Proposition 22.5 for $g=1$ and $n=1$.

22.3. Theta as a section of a line bundle on $\mathrm{Jac}(X)$

Let $A = \mathbb{C}^g/\Lambda$. Theta does not descend to a *function* on A, but it descends to a *section of a holomorphic line bundle.*

Definition 22.7 (Line bundle from automorphy factors)**.** Let $\alpha_\lambda(z)$ be the factor in Proposition 22.5. Define an equivalence relation on $\mathbb{C}^g \times \mathbb{C}$ by

$$(z,w) \sim (z+\lambda,\ \alpha_\lambda(z)\, w), \qquad \lambda \in \Lambda.$$

The quotient

$$\mathcal{L} := (\mathbb{C}^g \times \mathbb{C})/\sim \ \longrightarrow\ A$$

is a holomorphic line bundle on A.

Lemma 22.8 (Cocycle condition)**.** *The factors $\alpha_\lambda(z)$ satisfy*

$$\alpha_{\lambda+\mu}(z) = \alpha_\lambda(z+\mu)\alpha_\mu(z), \qquad \lambda,\mu \in \Lambda.$$

Hence Definition 22.7 is consistent.

Proof. Write $\lambda = m + \Omega n$ and $\mu = m' + \Omega n'$ with $m, m', n, n' \in \mathbb{Z}^g$. Using $\alpha_{m+\Omega n}(z) = \exp(-\pi i\, n^t \Omega n - 2\pi i\, n^t z)$, compute:

$$\alpha_{\lambda+\mu}(z) = \exp\!\big(-\pi i\,(n+n')^t \Omega (n+n') - 2\pi i\,(n+n')^t z\big).$$

Also

$$\alpha_\lambda(z+\mu)\alpha_\mu(z) = \exp\!\big(-\pi i\, n^t\Omega n - 2\pi i\, n^t(z+m'+\Omega n')\big)\exp\!\big(-\pi i\, n'^t\Omega n' - 2\pi i\, n'^t z\big).$$

Since $n^t m' \in \mathbb{Z}$, the factor $e^{-2\pi i n^t m'} = 1$, and symmetry of Ω matches the cross-terms, so both sides coincide. □

Proposition 22.9 (Theta descends to a global section)**.** *The function $\theta(z\,|\,\Omega)$ defines a holomorphic global section $s_\theta \in H^0(A,\mathcal{L})$ by $s_\theta([z]) = [z, \theta(z)]$.*

Proof. If z is replaced by $z+\lambda$, then Proposition 22.5 gives $\theta(z+\lambda) = \alpha_\lambda(z)\theta(z)$, hence

$$(z+\lambda, \theta(z+\lambda)) \sim (z+\lambda, \alpha_\lambda(z)\theta(z)) \sim (z, \theta(z)).$$

So s_θ is well-defined. Holomorphicity follows from that of θ. □

Definition 22.10 (Theta divisor)**.** The *theta divisor* on A is the zero locus of s_θ:

$$\Theta_A := \{[z] \in A \mid \theta(z \mid \Omega) = 0\}.$$

Remark 22.11 (Geometric meaning (Riemann vanishing))**.** On $\mathrm{Jac}(X)$ there exists a translate κ (the Riemann constant) such that

$$u(D) + \kappa \in \Theta_A \quad \Longleftrightarrow \quad D \text{ is an effective divisor of degree } g-1.$$

So (up to translation) Θ_A is the Abel–Jacobi image of W_{g-1}.

22.4. Genus 1: Jacobi theta functions and explicit computations

Now specialize to $g = 1$. Then $\Omega = \tau \in \mathbb{H}$ and $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$. The Riemann theta becomes

$$\theta(z \mid \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}.$$

Let $q := e^{\pi i \tau}$. Then

$$\theta(z \mid \tau) = \sum_{n \in \mathbb{Z}} q^{n^2} e^{2\pi i n z} = \vartheta_3(z, \tau).$$

Remark 22.12 (Four Jacobi theta functions as characteristics)**.** For $\varepsilon, \delta \in \{0, \frac{1}{2}\}$ define

$$\theta\begin{bmatrix}\varepsilon \\ \delta\end{bmatrix}(z \mid \tau) := \sum_{n \in \mathbb{Z}} \exp\big(\pi i (n+\varepsilon)^2 \tau + 2\pi i (n+\varepsilon)(z+\delta)\big).$$

Then (up to standard normalizations)

$$\theta_3 = \theta\begin{bmatrix}0 \\ 0\end{bmatrix}, \quad \theta_4 = \theta\begin{bmatrix}0 \\ \frac{1}{2}\end{bmatrix}, \quad \theta_2 = \theta\begin{bmatrix}\frac{1}{2} \\ 0\end{bmatrix}, \quad \theta_1 = \theta\begin{bmatrix}\frac{1}{2} \\ \frac{1}{2}\end{bmatrix}.$$

In particular, $\theta_1(z|\tau)$ is odd and has a simple zero at $z = 0$.

Example 22.13 (First terms and quick numerics)**.** Write $\vartheta_3(z, \tau)$ with $q = e^{\pi i \tau}$ and set $z = 0$ for simplicity:

$$\vartheta_3(0, \tau) = \sum_{n \in \mathbb{Z}} q^{n^2} = 1 + 2q + 2q^4 + 2q^9 + 2q^{16} + \cdots.$$

For example, if $\tau = i$ then $q = e^{-\pi} \approx 0.0432$, so

$$\vartheta_3(0, i) \approx 1 + 2(0.0432) + 2(0.0432)^4 + \cdots \approx 1.0864 + 0.000007 + \cdots \approx 1.0864.$$

(Already accurate to about 10^{-5} with just the first two nontrivial terms.)

22.5. From theta to Weierstrass: σ, ζ, and $\wp$ via logarithmic derivatives

Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\tau = \omega_2/\omega_1$. The Weierstrass σ–function is characterized (up to scaling) by:

- σ is entire with simple zeros exactly at Λ,

- $\sigma'(0) = 1$,
- quasi-periodicity $\sigma(z + \omega_k) = -e^{\eta_k(z+\omega_k/2)}\sigma(z)$, $k = 1, 2$.

Theorem 22.14 (Sigma in terms of θ_1)**.** *There exists a nonzero constant $C = C(\Lambda)$ such that*

$$\sigma(z) = C\, \exp\!\left(\frac{\eta_1}{2\omega_1} z^2\right) \theta_1\!\left(\frac{z}{\omega_1}\,\middle|\,\tau\right).$$

Proof idea with the key check. Define

$$F(z) := \frac{\sigma(z)}{\exp\!\left(\frac{\eta_1}{2\omega_1} z^2\right) \theta_1(z/\omega_1 \,|\, \tau)}.$$

By comparing quasi-periodicities of σ and θ_1, one finds $F(z+\omega_1) = F(z)$ and $F(z+\omega_2) = F(z)$, hence F is elliptic. Both numerator and denominator are entire and have the same simple zeros at Λ, so F has no zeros or poles. Thus F is constant. □

Definition 22.15 (Weierstrass ζ and $\wp$ via σ)**.** Define

$$\zeta(z) := \frac{d}{dz} \log \sigma(z), \qquad \wp(z) := -\zeta'(z) = -\frac{d^2}{dz^2} \log \sigma(z).$$

Proposition 22.16 (Pole/zero mechanism: why $\wp$ has a double pole)**.** *Since θ_1 has a simple zero at 0, we have locally $\theta_1(z|\tau) = c\, z + O(z^3)$, $c \neq 0$, and hence*

$$\frac{d^2}{dz^2} \log \theta_1(z|\tau) = -\frac{1}{z^2} + O(1).$$

So $-\frac{d^2}{dz^2} \log \theta_1$ has a double pole, matching the basic pole of $\wp$.

Proof. If $f(z) = cz + O(z^3)$ with $c \neq 0$, then $\log f(z) = \log c + \log z + O(z^2)$, so $(\log f)''(z) = -\frac{1}{z^2} + O(1)$. Apply to $f = \theta_1$. □

Theorem 22.17 (Explicit theta expression for $\wp$)**.** *With $\tau = \omega_2/\omega_1$ we have*

$$\wp(z) = -\frac{d^2}{dz^2} \log \theta_1\!\left(\frac{z}{\omega_1}\,\middle|\,\tau\right) + \frac{\eta_1}{\omega_1}.$$

Equivalently, with $u = z/\omega_1$,

$$\wp(z) = -\frac{1}{\omega_1^2}\left(\frac{\theta_1''(u|\tau)}{\theta_1(u|\tau)} - \left(\frac{\theta_1'(u|\tau)}{\theta_1(u|\tau)}\right)^2\right) + \frac{\eta_1}{\omega_1}.$$

Proof. From Theorem 22.14,

$$\log \sigma(z) = \log C + \frac{\eta_1}{2\omega_1} z^2 + \log \theta_1(z/\omega_1|\tau).$$

Differentiate twice and use $\wp(z) = -\frac{d^2}{dz^2} \log \sigma(z)$. □

Example 22.18 (Recovering the principal part of $\wp$ at 0 from theta)**.** Using Proposition 22.16 and Theorem 22.17, near $z = 0$,

$$-\frac{d^2}{dz^2}\log\theta_1\left(\frac{z}{\omega_1}\,\middle|\,\tau\right) = \frac{1}{z^2} + O(1),$$

so

$$\wp(z) = \frac{1}{z^2} + O(1),$$

which matches the defining Laurent expansion of the Weierstrass $\wp$–function.

Corollary 22.19 (All elliptic functions from theta)**.** Every elliptic function on $\mathbb{C}/\Lambda$ is a rational function of $\wp$ and $\wp'$. Since $\wp$ is expressed by θ_1 and derivatives, every elliptic function can be written as a rational combination of Jacobi theta functions and their derivatives.

22.6. Exercises (with guidance)

Conventions. Throughout, $\tau \in \mathbb{H}$ and $q = e^{\pi i\tau}$.

Exercise 22.20 (Warm-up: convergence estimate)**.** Let $\tau = X + iY$ with $Y > 0$. Show that

$$\sum_{n\in\mathbb{Z}} \left|e^{\pi i n^2\tau + 2\pi i n z}\right| \le \sum_{n\in\mathbb{Z}} e^{-\pi Y n^2 + 2\pi|n|\,|\Im z|}.$$

Deduce uniform convergence on compact sets and that $\vartheta_3(\cdot,\tau)$ is entire.

Exercise 22.21 (Quasi-periodicity by reindexing)**.** Prove directly from the defining series that

$$\vartheta_3(z+1,\tau) = \vartheta_3(z,\tau), \qquad \vartheta_3(z+\tau,\tau) = e^{-\pi i\tau - 2\pi i z}\vartheta_3(z,\tau).$$

(Hint: for the τ-shift reindex $n \mapsto n-1$.)

Exercise 22.22 (Parity and zero of θ_1)**.** Using the characteristic definition, show $\theta_1(-z|\tau) = -\theta_1(z|\tau)$ and that $\theta_1(z|\tau)$ has a simple zero at $z = 0$. Then write the local expansion $\theta_1(z|\tau) = cz + O(z^3)$ and express c in terms of $\theta_1'(0|\tau)$.

Exercise 22.23 (Principal part of $\wp$ from a simple zero)**.** Let f be holomorphic near 0 with $f(0) = 0$ and $f'(0) \neq 0$. Show that $(\log f)''(z) = -1/z^2 + O(1)$ near 0. Apply to $f(z) = \theta_1(z|\tau)$ to re-derive the double pole of $\wp$.

Exercise 22.24 (A quick numerical approximation)**.** Take $\tau = i$. Compute $q = e^{-\pi}$ and approximate

$$\vartheta_3(0,i) = 1 + 2q + 2q^4 + 2q^9 + \cdots$$

using terms up to $n = 2$ and $n = 3$. Give an error bound using the tail $\sum_{n\ge N} 2e^{-\pi n^2}$.

Exercise 22.25 (From σ to $\wp$)**.** Assume Theorem 22.14. Differentiate $\log\sigma(z)$ once and twice to express $\zeta(z)$ and $\wp(z)$ explicitly in terms of $\theta_1(z/\omega_1|\tau)$ and its derivatives. Check the constant term η_1/ω_1 appears with the correct sign.

Exercise 22.26 (Line bundle cocycle)**.** Let $\alpha_{m+\Omega n}(z) = \exp(-\pi i n^t\Omega n - 2\pi i n^t z)$. Verify the cocycle identity

$$\alpha_{\lambda+\mu}(z) = \alpha_\lambda(z+\mu)\alpha_\mu(z)$$

by expanding both sides, and isolate exactly where symmetry of Ω is used.

22.7. Why theta is the "right" object: modular transformations and geometry

Theta functions enjoy clean transformation laws under the generators

$$\tau \mapsto \tau + 1, \qquad \tau \mapsto -\frac{1}{\tau},$$

and these laws encode how line bundles and divisors vary in families of elliptic curves. From the Jacobian viewpoint, theta is the canonical section cutting out the (principal) polarization divisor, and in genus 1 it is the analytic engine behind $\sigma, \zeta, \wp$ and elliptic integrals.

23. Galois Group and Riemann Surface

Overview

A finite holomorphic map $\pi : Y \to X$ between compact Riemann surfaces is the analytic avatar of a finite extension of function fields $\mathcal{M}(Y)/\mathcal{M}(X)$. The guiding dictionary is:

$$\text{finite map } \pi : Y \to X \quad \Longleftrightarrow \quad \text{finite field extension } \mathcal{M}(X) \hookrightarrow \mathcal{M}(Y),$$

and in the Galois (normal) case,

$$\mathrm{Deck}(Y/X) \;\cong\; \mathrm{Aut}\big(\mathcal{M}(Y)/\mathcal{M}(X)\big).$$

We make each implication *computable*: how to write polynomial relations, how to see degrees, and how to match automorphisms with deck transformations.

Standing conventions. All Riemann surfaces are connected and compact unless explicitly stated. A *finite holomorphic map* means a nonconstant holomorphic map with finite fibers; equivalently, a branched covering. Let $B \subset X$ be the finite branch value set and set

$$X^\circ := X \setminus B, \qquad Y^\circ := Y \setminus \pi^{-1}(B).$$

Then $\pi^\circ : Y^\circ \to X^\circ$ is an unramified covering of degree $\deg \pi$.

23.1. Function fields and pullback

For a Riemann surface X, let $\mathcal{M}(X)$ denote its field of meromorphic functions. A nonconstant holomorphic map $\pi : Y \to X$ induces an injective field homomorphism

$$\pi^* : \; \mathcal{M}(X) \hookrightarrow \mathcal{M}(Y), \qquad f \longmapsto f \circ \pi.$$

We identify $\mathcal{M}(X)$ with its image $\pi^*\mathcal{M}(X) \subset \mathcal{M}(Y)$.

23.1.1. Local branches, norm, and trace (explicit formulas)

Let $f \in \mathcal{M}(Y)$. Over an evenly covered open set $U \subset X^\circ$ we can write

$$(\pi^\circ)^{-1}(U) = V_1 \sqcup \cdots \sqcup V_d, \qquad \pi|_{V_j} : V_j \xrightarrow{\sim} U,$$

and define local inverses $\tau_j : U \to V_j$ so that $\pi \circ \tau_j = \mathrm{id}_U$. Define the *local branches* of f above U by

$$f_j := f \circ \tau_j \ \in \ \mathcal{M}(U).$$

Then the elementary symmetric polynomials in $f_1, \dots, f_d$ define meromorphic functions on U:

$$\prod_{j=1}^{d}(T - f_j) = T^d + c_1 T^{d-1} + \cdots + c_d \in \mathcal{M}(U)[T].$$

Lemma 23.1 (Gluing of symmetric coefficients)**.** *The coefficients c_i are independent of the choice of labeling of the sheets V_j, and they glue (as U varies) to define global meromorphic functions $c_i \in \mathcal{M}(X)$.*

Proof. On overlaps $U \cap U'$, the decompositions into sheets differ by a permutation of the sheets, so the unordered set $\{f_1, \dots, f_d\}$ is permuted. Each c_i is an elementary symmetric polynomial in the f_j, hence invariant under permutation, so the c_i agree on overlaps. Thus they glue on X°. Since $X \setminus X^\circ = B$ is finite and each c_i is meromorphic on X°, it extends meromorphically across B (at worst with poles), giving $c_i \in \mathcal{M}(X)$. □

Definition 23.2 (Norm and trace along π (computable on sheets))**.** For $f \in \mathcal{M}(Y)$ define on $U \subset X^\circ$:

$$\mathrm{Nm}_\pi(f)\big|_U := \prod_{j=1}^{d} f_j \in \mathcal{M}(U), \qquad \mathrm{Tr}_\pi(f)\big|_U := \sum_{j=1}^{d} f_j \in \mathcal{M}(U).$$

These glue to meromorphic functions on X and satisfy $\mathrm{Nm}_\pi(f) = (-1)^d c_d$ and $\mathrm{Tr}_\pi(f) = -c_1$.

Theorem 23.3 (A monic algebraic relation of degree $\le d$)**.** *Let $\pi : Y \to X$ be finite of degree d and let $f \in \mathcal{M}(Y)$. Let $c_1, \dots, c_d \in \mathcal{M}(X)$ be the global symmetric coefficients attached to f. Then in $\mathcal{M}(Y)$ we have the identity*

$$f^d + (\pi^* c_1) f^{d-1} + \cdots + (\pi^* c_{d-1}) f + \pi^* c_d = 0.$$

In particular, f is algebraic over $\mathcal{M}(X)$ with $\deg_{\mathcal{M}(X)}(f) \le d$ and

$$[\mathcal{M}(Y) : \mathcal{M}(X)] \le d.$$

Proof. Work over an evenly covered $U \subset X^\circ$. Consider the polynomial

$$P_U(T) = \prod_{j=1}^{d}(T - f_j) = T^d + c_1 T^{d-1} + \cdots + c_d \in \mathcal{M}(U)[T].$$

Pull it back to $\pi^{-1}(U)$ via π; on the sheet V_k we have $\tau_k \circ \pi = \mathrm{id}_{V_k}$, hence $(f \circ \tau_k) \circ \pi = f$ on V_k. Therefore, on V_k,

$$P_U(f) = \prod_{j=1}^{d}\big(f - (f \circ \tau_j) \circ \pi\big)$$

has the $j = k$ factor equal to 0, so $P_U(f) = 0$ on V_k. Since $\pi^{-1}(U)$ is the disjoint union of the sheets, $P_U(f) = 0$ on $\pi^{-1}(U)$. By analytic continuation, the identity holds on Y°, and hence on all of Y (as an identity of meromorphic functions). Replacing c_i by their global meromorphic extensions on X yields the global formula. Finally, $\deg_{\mathcal{M}(X)}(f) \le d$ implies $[\mathcal{M}(Y) : \mathcal{M}(X)] \le d$. □

23.1.2. Why the extension degree equals $\deg \pi$

Theorem 23.4 (Degree equals field extension degree)**.** *Let $\pi : Y \to X$ be finite of degree d. Then*

$$[\mathcal{M}(Y) : \mathcal{M}(X)] = d.$$

Proof. Set $K = \mathcal{M}(X)$ and $L = \mathcal{M}(Y)$. We already know $[L : K] \leq d$ by Theorem 23.3. We show $[L : K] \geq d$ by producing an element of L with minimal polynomial of degree d.

Choose $x \in X^\circ$ (not a branch value) so $\pi^{-1}(x) = \{y_1, \dots, y_d\}$ consists of d distinct points. We claim there exists $f \in L$ such that the values $f(y_i)$ are pairwise distinct. This is a standard separation statement: since Y is a compact Riemann surface, meromorphic functions separate points.

A concrete construction uses Riemann–Roch as follows. Let $D := N(y_1 + \cdots + y_d)$ for $N \gg 0$. Then $\ell(D) := \dim H^0(Y, \mathcal{O}_Y(D)) \geq \deg D - g(Y) + 1$. For N large, $\ell(D) \geq d+1$. Consider the evaluation map

$$\mathrm{ev} : \ H^0(Y, \mathcal{O}_Y(D)) \longrightarrow \mathbb{C}^d, \qquad s \longmapsto (s(y_1), \dots, s(y_d)).$$

Its kernel has codimension at most d, so since $\ell(D) \geq d + 1$ we can find two sections s, t whose evaluation vectors are not proportional. Then the meromorphic function $f := s/t \in \mathcal{M}(Y)$ has values $f(y_i) = s(y_i)/t(y_i)$ not all equal; a generic choice makes them pairwise distinct (choose s, t so that $s(y_i)t(y_j) - s(y_j)t(y_i) \neq 0$ for all $i \neq j$; these are finitely many nontrivial polynomial conditions).

Now work on a simply connected neighborhood $U \subset X^\circ$ of x so that $\pi^{-1}(U)$ splits into sheets V_i with local inverses $\tau_i : U \to V_i$ satisfying $\tau_i(x) = y_i$. Define the d local branches $f_i := f \circ \tau_i \in \mathcal{M}(U)$. By construction, $f_i(x) = f(y_i)$ are pairwise distinct, hence the functions f_i are distinct as germs at x.

Let $P(T) \in K[T]$ be the monic polynomial of degree d obtained from the elementary symmetric polynomials of $f_1, \dots, f_d$ (as in Theorem 23.3). Over U, this polynomial factors as

$$P(T) = \prod_{i=1}^{d} (T - f_i) \in \mathcal{M}(U)[T].$$

Since the f_i are distinct germs at x, this factorization has d *distinct* roots in a neighborhood of x. Therefore the degree of the minimal polynomial of f over K must be at least d (otherwise f would satisfy a monic relation of smaller degree, forcing at most that many distinct local branches over U). Hence $\deg_K(f) = d$, so $[L : K] \geq d$. Combining with $[L : K] \leq d$ gives $[L : K] = d$. □

23.2. Galois coverings and Galois extensions

23.2.1. Deck transformations (unramified and branched)

Definition 23.5 (Deck group)**.** For an unramified covering $p : Y^\circ \to X^\circ$, the *deck transformation group* is

$$\mathrm{Deck}(Y^\circ/X^\circ) := \{\sigma \in \mathrm{Aut}_{\mathrm{hol}}(Y^\circ) \mid p \circ \sigma = p\}.$$

For a finite map $\pi : Y \to X$ with branch values B, define

$$\mathrm{Deck}(Y/X) := \mathrm{Deck}\big(Y^\circ/X^\circ\big).$$

Definition 23.6 (Galois/normal covering)**.** An unramified covering $p : Y^\circ \to X^\circ$ is *Galois* if $\mathrm{Deck}(Y^\circ/X^\circ)$ acts transitively on each fiber. Equivalently, $p_*\pi_1(Y^\circ)$ is a normal subgroup of $\pi_1(X^\circ)$. A branched map $\pi : Y \to X$ is called *Galois* if $\pi^\circ : Y^\circ \to X^\circ$ is a Galois covering.

23.2.2. Main theorem: deck transformations = field automorphisms

Theorem 23.7 (Deck transformations vs. field automorphisms)**.** *Let $\pi : Y \to X$ be finite. Set $K = \mathcal{M}(X)$ and $L = \mathcal{M}(Y)$ (with $K \subset L$ via π^*). Then composition induces a canonical group isomorphism*

$$\Phi : \mathrm{Deck}(Y/X) \xrightarrow{\sim} \mathrm{Aut}(L/K), \qquad \Phi(\sigma)(f) := f \circ \sigma^{-1}.$$

Consequently, π is Galois if and only if the field extension L/K is Galois.

Proof. **(1) Φ is well-defined and a homomorphism.** If σ is a deck transformation of $Y^\circ \to X^\circ$, then $f \circ \sigma^{-1}$ is meromorphic on Y°, hence extends meromorphically to Y (isolated singularities on a compact Riemann surface are meromorphic). For $g \in K$ we have

$$(g \circ \pi) \circ \sigma^{-1} = g \circ (\pi \circ \sigma^{-1}) = g \circ \pi,$$

so $\Phi(\sigma)$ fixes K. The rule $\sigma \mapsto (f \mapsto f \circ \sigma^{-1})$ respects composition, hence is a homomorphism.

(2) Φ is injective. Suppose $\Phi(\sigma) = \mathrm{id}_L$. Then $f \circ \sigma^{-1} = f$ for all $f \in \mathcal{M}(Y)$. Meromorphic functions separate points on a compact Riemann surface: for $p \neq q$ there exists $f \in \mathcal{M}(Y)$ with $f(p) \neq f(q)$ (e.g. by Riemann–Roch). Hence $\sigma^{-1}(p) = p$ for all p, so $\sigma = \mathrm{id}$.

(3) Φ is surjective (construct σ from a field automorphism). Let $\varphi \in \mathrm{Aut}(L/K)$. Choose $x \in X^\circ$ and a simply connected neighborhood $U \subset X^\circ$ of x so that $\pi^{-1}(U) = V_1 \sqcup \cdots \sqcup V_d$ splits into sheets with inverses $\tau_i : U \to V_i$. Pick $f \in L$ that separates the fiber over x (as in the proof of Theorem 23.4), so the local branches $f_i := f \circ \tau_i$ are distinct near x.

Over U, the monic polynomial

$$P(T) := \prod_{i=1}^{d} (T - f_i) \ \in \ \mathcal{M}(U)[T]$$

has coefficients in K (by the symmetric-gluing construction). Since φ fixes K, it preserves $P(T)$, hence sends the root f to another root of the same polynomial. Concretely, on each sheet V_i, $\varphi(f)$ must equal $f \circ (\tau_j \circ \pi)$ for a *unique* j (uniqueness uses that the roots are distinct). This defines a permutation π_φ of $\{1, \dots, d\}$ and a holomorphic map

$$\sigma_U : V_i \longrightarrow V_{\pi_\varphi(i)}, \qquad \sigma_U := \tau_{\pi_\varphi(i)} \circ \pi|_{V_i}.$$

By construction, $\pi \circ \sigma_U = \pi$ on V_i, so σ_U is a local deck transformation, and $\varphi(f) = f \circ \sigma_U^{-1}$ on V_i.

Now vary U and use analytic continuation on the connected surface Y°: local definitions agree on overlaps (because they agree on a generating set of L and both commute with π), so the σ_U glue to a global $\sigma \in \mathrm{Deck}(Y/X)$. Then $\Phi(\sigma) = \varphi$.

(4) Galois equivalence. The cover π is Galois iff $|\mathrm{Deck}(Y/X)| = \deg \pi$ (transitive free action on fibers). By Theorem 23.4, $\deg \pi = [L : K]$. Thus π is Galois iff $|\mathrm{Aut}(L/K)| = [L : K]$, i.e. iff L/K is a Galois extension. □

23.3. Examples

23.3.1. Example A: a hyperelliptic double cover

Example 23.8 (Elliptic curve as a double cover of $\mathbb{P}^1$)**.** Let

$$E:\ y^2 = x^3 - x$$

be the smooth projective model of this affine curve, and let $\pi : E \to \mathbb{P}^1$ be $(x, y) \mapsto x$. Then $\mathcal{M}(\mathbb{P}^1) = \mathbb{C}(x)$ and

$$\mathcal{M}(E) = \mathbb{C}(x, y) \cong \mathbb{C}(x)[y]/(y^2 - (x^3 - x)).$$

(1) Minimal polynomial. The element $y \in \mathcal{M}(E)$ satisfies

$$T^2 - (x^3 - x) = 0,$$

so $\deg_{\mathbb{C}(x)}(y) = 2$ and $[\mathcal{M}(E) : \mathbb{C}(x)] = 2 = \deg \pi$.

(2) Deck group / Galois group. The nontrivial automorphism of $\mathcal{M}(E)/\mathbb{C}(x)$ is

$$\sigma^\sharp :\ \mathcal{M}(E) \to \mathcal{M}(E), \qquad x \mapsto x, \quad y \mapsto -y.$$

On the curve it corresponds to the involution

$$\sigma : E \to E, \qquad (x, y) \mapsto (x, -y),$$

and clearly $\pi \circ \sigma = \pi$. Hence

$$\mathrm{Deck}(E/\mathbb{P}^1) \cong \mathrm{Aut}(\mathcal{M}(E)/\mathbb{C}(x)) \cong \mathbb{Z}/2\mathbb{Z}.$$

(3) Branch points (quick computation). Branch occurs where $d\pi = 0$ on E. On the affine chart, $\pi = x$, and the ramification points occur where $\partial(y^2 - (x^3 - x))/\partial y = 2y = 0$, i.e. where $y = 0$. Thus $x^3 - x = 0$, so $x \in \{0, 1, -1\}$, plus the point at infinity. So π is branched over $\{0, 1, -1, \infty\}$ (four branch values), as expected for a degree-2 map from genus 1 by Riemann–Hurwitz.

23.3.2. Example B: cyclic n-covers (Kummer) with full Galois group

Example 23.9 (Cyclic n-cover of $\mathbb{P}^1$ and its Galois group)**.** Fix distinct $a_1, \dots, a_r \in \mathbb{C}$ and integers $m_1, \dots, m_r$. Let

$$Y:\quad y^n = \prod_{j=1}^{r} (x - a_j)^{m_j}, \qquad \pi : Y \to \mathbb{P}^1,\ (x, y) \mapsto x.$$

Set $K = \mathbb{C}(x)$ and $L = K(y)$.

(1) Degree. If $\prod_j (x - a_j)^{m_j}$ is not an nth power in K, then the polynomial $T^n - \prod_j (x - a_j)^{m_j} \in K[T]$ is irreducible in this Kummer setting, so $\deg_K(y) = n$ and $[L : K] = n = \deg \pi$.

(2) Galois automorphisms. Let $\zeta_n = e^{2\pi i/n}$. Then

$$\sigma_k^\sharp :\ L \to L, \qquad x \mapsto x, \quad y \mapsto \zeta_n^k\, y$$

defines a K-automorphism for each $k \in \{0, \dots, n-1\}$, and these are distinct. Thus

$$\mathrm{Aut}(L/K) \;\cong\; \mathbb{Z}/n\mathbb{Z},$$

and by Theorem 23.7 the deck group is also $\mathbb{Z}/n\mathbb{Z}$.

(3) Ramification indices (explicit). Near $x = a_j$, write $u = x - a_j$ as a local parameter on $\mathbb{P}^1$. Then locally the equation is $y^n = u^{m_j} \cdot (\text{unit})$. Write $m_j = q_j n + r_j$ with $0 \le r_j < n$. The ramification index at points above a_j is

$$e_{a_j} = \frac{n}{\gcd(n, m_j)} = \frac{n}{\gcd(n, r_j)}.$$

(Reason: in the extension of discrete valuation rings, adjoining y with $y^n = u^{m_j} \cdot \text{unit}$ gives valuation $\mathrm{ord}(y) = m_j/n$; the denominator after reduction is $n/\gcd(n, m_j)$.)

This gives a direct route to genus computations via Riemann–Hurwitz.

23.4. Uniformization and the Galois picture

Theorem 23.10 (Simply connected models)**.** *Every simply connected Riemann surface is biholomorphic to exactly one of $\mathbb{P}^1$, $\mathbb{C}$, or $\mathbb{D}$. Hence any Riemann surface X has a universal cover $\widetilde{X}$ of this form and*

$$X \;\cong\; \pi_1(X)\backslash\widetilde{X}$$

for a discrete group $\pi_1(X) \le \mathrm{Aut}(\widetilde{X})$ acting properly discontinuously.

Theorem 23.11 (Uniformization of compact Riemann surfaces)**.** *Let X be compact and connected.*

1. *If $g(X) = 0$, then $X \cong \mathbb{P}^1$.*
2. *If $g(X) = 1$, then $X \cong \mathbb{C}/\Lambda$ for a full lattice $\Lambda \subset \mathbb{C}$.*
3. *If $g(X) \ge 2$, then $X \cong \Gamma\backslash\mathbb{H}$ for a torsion-free, discrete $\Gamma \le \mathrm{PSL}_2(\mathbb{R})$.*

$\mathrm{PSL}_2(\mathbb{R})$ as isometries of $\mathbb{H}$. Equip $\mathbb{H} = \{x + iy \in \mathbb{C} \mid y > 0\}$ with the hyperbolic metric

$$ds^2 = \frac{|dz|^2}{(\mathrm{Im}\, z)^2} = \frac{dx^2 + dy^2}{y^2}.$$

Then $z \mapsto \dfrac{az+b}{cz+d}$ for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{PSL}_2(\mathbb{R})$ acts by orientation-preserving isometries, giving the full orientation-preserving isometry group.

23.4.1. Normalizer formula

Let $p : Y^\circ \to X^\circ$ be an unramified finite cover, and let $\widetilde{X^\circ} \to X^\circ$ be the universal cover. Then

$$X^\circ \cong \Gamma_X\backslash\widetilde{X^\circ}, \qquad Y^\circ \cong \Gamma_Y\backslash\widetilde{X^\circ}$$

for subgroups $\Gamma_Y \le \Gamma_X$ of finite index. One has the canonical isomorphism

$$\mathrm{Deck}(Y^\circ/X^\circ) \;\cong\; N_{\Gamma_X}(\Gamma_Y)/\Gamma_Y,$$

and in particular the cover is Galois iff $\Gamma_Y \trianglelefteq \Gamma_X$. This matches the algebraic criterion $|\operatorname{Aut}(L/K)| = [L:K]$.

Takeaway. Holomorphic coverings are simultaneously:

$$\text{(topology)}\quad \operatorname{Deck}(Y/X) \iff \text{(algebra)}\quad \operatorname{Aut}\big(\mathcal{M}(Y)/\mathcal{M}(X)\big)$$

$$\iff \text{(uniformization)}\quad \Gamma_Y \le \Gamma_X \le \operatorname{Aut}\big(\widetilde{X^\circ}\big).$$

23.5. Exercises

Exercise 23.12 (Compute norm/trace on a double cover)**.** Let $Y:\ y^2 = h(x)$ with $h \in \mathbb{C}[x]$ squarefree and $\pi(x,y) = x$. Compute explicitly $\operatorname{Tr}_\pi(y)$ and $\operatorname{Nm}_\pi(y)$ as meromorphic functions on $\mathbb{P}^1$. *Hint:* the two local branches are y and $-y$.

Exercise 23.13 (Riemann–Hurwitz for Example 23.8)**.** For $\pi : E \to \mathbb{P}^1$ in Example 23.8, verify:

$$2g(E) - 2 = 2(2g(\mathbb{P}^1) - 2) + \sum_{p \in E} (e_p - 1),$$

and deduce $g(E) = 1$ from the four ramification points of index 2.

Exercise 23.14 (Ramification indices in Example 23.9)**.** Prove the formula $e_{a_j} = n/\gcd(n, m_j)$ by working in local coordinates and computing valuations in the extension $\mathbb{C}((u)) \subset \mathbb{C}((u))[y]/(y^n - u^{m_j} \cdot \text{unit})$.

Exercise 23.15 (Surjectivity in Theorem 23.7)**.** Fill in the analytic continuation step: show that the locally defined σ_U glue on overlaps. *Hint:* show they agree on a generating set of L over K, and use connectedness of Y°.

24. Connection to Number Theory

Guiding idea

Modular curves sit at the intersection of

$$\text{(complex analysis) } \Gamma\backslash\mathbb{H} \iff \text{(algebraic geometry) } X(\Gamma)/\mathbb{Q},$$
$$\text{(arithmetic) } G_{\mathbb{Q}}\text{-actions on cohomology.}$$

The point of this section is to make *precise* the passage

$$\text{Galois cohomology of } \ell\text{-adic cohomology} \quad \leftrightarrow \quad \text{(complex) sheaf/Hodge cohomology of } X(\Gamma)(\mathbb{C}),$$

and to correct a common pitfall: *$H^q_{\acute{e}t}$ compares to singular cohomology, not directly to $H^q(X, \mathcal{O}_X)$. The latter appears as the $(0,q)$-Hodge piece.*

1. Modular curves as analytic and algebraic curves

Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a finite-index subgroup. Define the (noncompact) modular curve

$$Y(\Gamma) := \Gamma\backslash\mathbb{H}.$$

It has a natural compactification by adding finitely many *cusps*:

$$X(\Gamma)(\mathbb{C}) \;:=\; \Gamma\backslash\big(\mathbb{H}\cup\mathbb{P}^1(\mathbb{Q})\big),$$

which is a compact Riemann surface.

Theorem 24.1 (Algebraicity and field of moduli)**.** *There exists a smooth projective algebraic curve $X(\Gamma)$ over $\mathbb{Q}$ whose set of complex points is naturally isomorphic to the compact Riemann surface $X(\Gamma)(\mathbb{C})$ above. Its function field*

$$K_\Gamma := \mathbb{Q}\big(X(\Gamma)\big)$$

is a finite extension of $\mathbb{Q}(j)$ induced by the map $\pi : X(\Gamma) \to X(1)$.

Remark 24.2 (Line bundles and modular forms (useful slogan))**.** For suitable Γ (at least for congruence subgroups), modular forms of weight k can be interpreted as sections of an algebraic/holomorphic line bundle $\omega^{\otimes k}$ on $X(\Gamma)$, and cusp forms correspond to sections vanishing at the cusps. In particular, weight 2 cusp forms correspond to holomorphic differentials on the *compact* curve $X(\Gamma)$.

2. Function fields, normality, and deck groups

Let $X(1) \cong \mathbb{P}^1$ and let j be the classical modular function. The natural map $\pi : X(\Gamma) \to X(1)$ corresponds to the inclusion of function fields

$$\pi^* :\; \mathbb{Q}(j) \hookrightarrow K_\Gamma.$$

Proposition 24.3 (Normal subgroup $\Longleftrightarrow$ Galois extension)**.** *Assume Γ is normal in $\mathrm{SL}_2(\mathbb{Z})$. Then π is a (branched) Galois cover and the extension $K_\Gamma/\mathbb{Q}(j)$ is Galois. Moreover,*

$$\mathrm{Deck}\big(X(\Gamma)/X(1)\big) \;\cong\; \mathrm{Aut}\big(K_\Gamma/\mathbb{Q}(j)\big) \;\cong\; \mathrm{PSL}_2(\mathbb{Z})/\overline{\Gamma},$$

where $\overline{\Gamma}$ is the image of Γ in $\mathrm{PSL}_2(\mathbb{Z})$.

Proof sketch with the key computation. Over $Y(\Gamma) \to Y(1)$ (remove cusps/elliptic points so the map is unramified), deck transformations correspond to right-multiplication by $\mathrm{PSL}_2(\mathbb{Z})$ modulo Γ. Normality makes the fiber action transitive and well-defined globally, hence the covering is Galois. On function fields, the deck group acts by pullback on meromorphic functions, producing a subgroup of $\mathrm{Aut}(K_\Gamma/\mathbb{Q}(j))$; equality follows because a Galois covering of degree d gives a field extension of degree d and the deck group has size d. □

3. Étale cohomology, comparison, and where $\mathcal{O}$-cohomology really appears

Fix a prime ℓ. Let X be a smooth projective curve over a field $k \subset \mathbb{C}$, and write $\overline{k}$ for an algebraic closure, $\overline{X} := X \times_k \overline{k}$, and $G_k := \mathrm{Gal}(\overline{k}/k)$.

Definition 24.4 (ℓ-adic cohomology as a Galois representation)**.** The ℓ-adic cohomology groups $H^i_{\acute{e}t}(\overline{X}, \mathbb{Q}_\ell)$ are finite-dimensional $\mathbb{Q}_\ell$-vector spaces equipped with a continuous action of G_k.

Theorem 24.5 (Complex comparison isomorphism (the correct one)). *If $k = \mathbb{C}$, then for each i there is a canonical comparison isomorphism*

$$H^i_{\text{ét}}(X, \mathbb{Q}_\ell) \;\cong\; H^i_{\text{sing}}(X(\mathbb{C}), \mathbb{Q}_\ell).$$

More generally, if $k \subset \mathbb{C}$, then

$$H^i_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell) \;\cong\; H^i_{\text{sing}}(X(\mathbb{C}), \mathbb{Q}_\ell)$$

as $\mathbb{Q}_\ell$-vector spaces (and the left side carries the G_k-action).

Remark 24.6 (Important correction). One *does not* have $H^i_{\text{sing}}(X(\mathbb{C}), \mathbb{Q}_\ell) \cong H^i(X(\mathbb{C}), \mathcal{O}_X)$ in general. Instead, for a compact Riemann surface $X(\mathbb{C})$, Hodge theory gives

$$H^1_{\text{sing}}(X(\mathbb{C}), \mathbb{C}) \;\cong\; H^{1,0}(X) \;\oplus\; H^{0,1}(X),$$

and Dolbeault/Serre identifies

$$H^{0,1}(X) \;\cong\; H^1(X, \mathcal{O}_X), \qquad H^{1,0}(X) \;\cong\; H^0(X, \Omega^1_X).$$

Thus $H^1(X, \mathcal{O}_X)$ is *a piece* of $H^1_{\text{sing}}(X, \mathbb{C})$, not the whole thing.

4. The Hochschild–Serre spectral sequence: Galois cohomology $\Rightarrow$ descent

Now take $k = \mathbb{Q}$ (or a number field), so G_k is nontrivial and acts on $H^q_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell)$.

Theorem 24.7 (Hochschild–Serre spectral sequence). *For a smooth proper curve X/k there is a spectral sequence*

$$E_2^{p,q} \;=\; H^p\big(G_k,\; H^q_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell)\big) \;\Longrightarrow\; H^{p+q}_{\text{ét}}(X, \mathbb{Q}_\ell).$$

Remark 24.8 (What $H^1(G_k, H^1)$ is measuring). The group $H^1\big(G_k, H^1_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell)\big)$ measures *obstructions and extension data* for descending/coordinatizing cohomology classes with Galois action. It is not literally equal to $H^1(X(\mathbb{C}), \mathcal{O}_X)$; rather, it is an arithmetic object whose *complex-analytic shadows* appear after passing to $\mathbb{C}$ and using Hodge theory.

5. Weight 2 cusp forms and holomorphic differentials

Let Γ be a congruence subgroup and consider the compact modular curve $X(\Gamma)$ over $\mathbb{Q}$. On the complex Riemann surface $X(\Gamma)(\mathbb{C})$ we have:

Theorem 24.9 (Holomorphic differentials = weight 2 cusp forms (genus ≥ 0)). *There is a natural isomorphism of complex vector spaces*

$$H^0\big(X(\Gamma)(\mathbb{C}),\; \Omega^1\big) \;\cong\; S_2(\Gamma),$$

where $S_2(\Gamma)$ is the space of weight 2 cusp forms for Γ.

Idea with the key local check at cusps. A weight 2 modular form f produces a Γ-invariant differential on $\mathbb{H}$:

$$\omega_f := f(z)\,dz.$$

Weight 2 transformation says $f(\gamma z)\,d(\gamma z) = f(z)\,dz$, so ω_f descends to a holomorphic differential on $Y(\Gamma)$. At a cusp, choose a local parameter $q = e^{2\pi i z/w}$ (width w). Then

$$dz = \frac{w}{2\pi i}\,\frac{dq}{q}.$$

If f is a cusp form, its q-expansion has no constant term: $f(z) = \sum_{n\geq 1} a_n q^n$, so

$$\omega_f = \left(\sum_{n\geq 1} a_n q^n\right)\frac{dq}{q} = \left(\sum_{n\geq 1} a_n q^{n-1}\right) dq$$

is holomorphic at $q = 0$. Hence ω_f extends across cusps to a holomorphic differential on the compact curve. Conversely, any holomorphic differential on $X(\Gamma)$ pulls back to a Γ-invariant differential on $\mathbb{H}$, hence corresponds to a weight 2 cusp form. □

Corollary 24.10 (Sheaf cohomology and Serre duality)**.** On the compact Riemann surface $X(\Gamma)(\mathbb{C})$,

$$H^1\big(X(\Gamma), \mathcal{O}\big) \;\cong\; H^0\big(X(\Gamma), \Omega^1\big)^* \;\cong\; S_2(\Gamma)^*.$$

Moreover,

$$H^1\big(X(\Gamma), \mathbb{C}\big) \;\cong\; S_2(\Gamma) \;\oplus\; \overline{S_2(\Gamma)}$$

via Hodge decomposition ($H^{1,0} \oplus H^{0,1}$).

6. Where ℓ-adic Galois representations enter

Theorem 24.11 (Galois representations from newforms (statement level))**.** *Let $f \in S_2(\Gamma)$ be a normalized newform with Fourier coefficients in a number field E. For each prime ℓ and each embedding $E \hookrightarrow \overline{\mathbb{Q}}_\ell$, there is a continuous ℓ-adic Galois representation*

$$\rho_{f,\ell} :\; G_{\mathbb{Q}} \longrightarrow \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$$

realized inside the ℓ-adic cohomology of $X(\Gamma)$ (more precisely in the appropriate Hecke-isotypic component of $H^1_{\acute{e}t}(\overline{X(\Gamma)}, \mathbb{Q}_\ell)$).

Remark 24.12 (Mental model)**.** Analytically: f is a holomorphic 1-form on $X(\Gamma)(\mathbb{C})$. Arithmetically: the same f cuts out a 2-dimensional $G_{\mathbb{Q}}$-stable piece in ℓ-adic cohomology. The bridge is:

$$H^1_{\acute{e}t}(\overline{X}, \mathbb{Q}_\ell) \;\leftrightarrow\; H^1_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Q}_\ell) \;\leftrightarrow\; \big(H^{1,0} \oplus H^{0,1}\big),$$

plus the Hecke action that isolates the f-piece.

7. Arithmetic–geometric correspondence

Arithmetic side	Analytic / geometric side
$X(\Gamma)$ over $\mathbb{Q}$	Compact Riemann surface $X(\Gamma)(\mathbb{C})$
Function field $K_\Gamma/\mathbb{Q}(j)$	Finite holomorphic map $X(\Gamma) \to X(1)$
$G_{\mathbb{Q}}$-action on $H^1_{\acute{e}t}(\overline{X(\Gamma)}, \mathbb{Q}_\ell)$	Local system / singular cohomology $H^1_{\mathrm{sing}}(X(\Gamma)(\mathbb{C}), \mathbb{Q}_\ell)$
Hochschild–Serre: $H^p(G_{\mathbb{Q}}, H^q_{\acute{e}t})$	Hodge pieces: $H^{1,0}$ and $H^{0,1} \cong H^1(\mathcal{O})$
$\rho_{f,\ell}$ from $f \in S_2(\Gamma)$	$f(z)\,dz \in H^0(\Omega^1)$ (holomorphic differential)

8. Exercises

Exercise 24.13 (Check the weight 2 invariance explicitly)**.** Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $\gamma z = \dfrac{az+b}{cz+d}$. Assume f is a weight 2 modular form:

$$f(\gamma z) = (cz+d)^2 f(z).$$

Show directly that $f(z)\,dz$ is Γ-invariant:

$$f(\gamma z)\,d(\gamma z) = f(z)\,dz.$$

Hint: compute $d(\gamma z) = \dfrac{dz}{(cz+d)^2}$.

Exercise 24.14 (Cusp condition $\Longleftrightarrow$ holomorphicity at $q=0$)**.** Let $q = e^{2\pi i z/w}$ be a local parameter at a cusp of width w. If $f(z) = \sum_{n\geq 0} a_n q^n$ is a weight 2 modular form, compute $\omega_f = f(z)\,dz$ in the q-coordinate and show:

$$\omega_f \text{ is holomorphic at the cusp } \iff a_0 = 0 \iff f \in S_2(\Gamma).$$

Exercise 24.15 (Hodge decomposition vs. $\mathcal{O}$-cohomology)**.** For a compact Riemann surface X of genus g, prove:

$$\dim_{\mathbb{C}} H^0(X, \Omega^1) = g, \qquad \dim_{\mathbb{C}} H^1(X, \mathcal{O}) = g,$$

and identify $H^1(X, \mathbb{C}) \cong H^{1,0} \oplus H^{0,1}$ with $H^0(\Omega^1) \oplus H^1(\mathcal{O})$. *Hint:* use Serre duality and Riemann–Roch for K_X and $\mathcal{O}_X$.

Exercise 24.16 (Spectral sequence sanity check in low degrees)**.** Using Theorem 24.7, write down the E_2-page terms that can contribute to $H^1_{\acute{e}t}(X, \mathbb{Q}_\ell)$ and $H^2_{\acute{e}t}(X, \mathbb{Q}_\ell)$. Which groups vanish if $X/\mathbb{Q}$ has a rational point and why? *Hint:* recall $H^0_{\acute{e}t}(\overline{X}, \mathbb{Q}_\ell) = \mathbb{Q}_\ell$ with trivial action.

Takeaway. For modular curves, the *analytic* cohomology $H^1(X(\Gamma)(\mathbb{C}), \mathbb{C})$ splits into Hodge pieces $H^{1,0} \oplus H^{0,1}$, where $H^{1,0}$ is literally weight 2 cusp forms as holomorphic differentials. The *arithmetic* side packages the same geometric H^1 into a $G_{\mathbb{Q}}$-representation $H^1_{\acute{e}t}(\overline{X(\Gamma)}, \mathbb{Q}_\ell)$, and the Hochschild–Serre spectral sequence explains how Galois cohomology measures descent/obstructions relative to $\mathbb{Q}$.

References

[1] J. B. Conway, *Functions of One Complex Variable I*, 2nd ed., Graduate Texts in Mathematics 11, Springer, 1978.

[2] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw–Hill, 1979.

[3] T. W. Gamelin and R. E. Greene, *Introduction to Topology and Modern Analysis*, 2nd ed., Wiley, 1999.

[4] E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton Lectures in Analysis II, Princeton Univ. Press, 2003.

[5] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.

[6] E. B. Saff and A. D. Snider, *Fundamentals of Complex Analysis with Applications to Engineering, Science, and Mathematics*, 3rd ed., Pearson, 2003.

[7] T. Needham, *Visual Complex Analysis*, Oxford Univ. Press, 1997.

[8] G. F. Carrier, M. Krook, and C. E. Pearson, *Functions of a Complex Variable: Theory and Technique*, SIAM, 2005.

[9] M. J. Ablowitz and A. S. Fokas, *Complex Variables: Introduction and Applications*, 2nd ed., Cambridge Univ. Press, 2003.

[10] H. Flanders, *Differential Forms with Applications to the Physical Sciences*, Dover, 1989.

[11] M. Spivak, *Calculus on Manifolds*, Addison–Wesley, 1965; reprint: Westview, 1979.

[12] F. W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Graduate Texts in Mathematics 94, Springer, 1983.

[13] M. P. do Carmo, *Riemannian Geometry*, Birkhäuser, 1992.

[14] J. M. Lee, *Introduction to Riemannian Manifolds*, Graduate Texts in Mathematics 176, Springer, 2018.

[15] J. Jost, *Compact Riemann Surfaces*, 3rd ed., Springer, 2006.

[16] H. M. Farkas and I. Kra, *Riemann Surfaces*, 2nd ed., Graduate Texts in Mathematics 71, Springer, 1992.

[17] O. Forster, *Lectures on Riemann Surfaces*, Graduate Texts in Mathematics 81, Springer, 1981.

[18] R. O. Wells, Jr., *Differential Analysis on Complex Manifolds*, 3rd ed., Graduate Texts in Mathematics 65, Springer, 2008.

[19] C. B. Morrey, Jr., *Multiple Integrals in the Calculus of Variations*, Classics in Mathematics, Springer, 2008.

[20] M. E. Taylor, *Partial Differential Equations I: Basic Theory*, 2nd ed., Applied Mathematical Sciences 115, Springer, 2011.

[21] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.

[22] R. C. Gunning, *Lectures on Riemann Surfaces*, Princeton Mathematical Notes, Princeton Univ. Press, 1966 (orig. notes 1962).

[23] R. C. Gunning and H. Rossi, *Analytic Functions of Several Complex Variables*, Prentice–Hall, 1965; reprint: AMS Chelsea.

[24] P. Griffiths and J. Harris, *Principles of Algebraic Geometry*, Wiley Classics Library, 1994.

[25] G. E. Bredon, *Sheaf Theory*, 2nd ed., Graduate Texts in Mathematics 170, Springer, 1997.

[26] R. Godement, *Topologie Algébrique et Théorie des Faisceaux*, Hermann, 1958.

[27] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Graduate Texts in Mathematics 82, Springer, 1982.

[28] A. Hatcher, *Algebraic Topology*, Cambridge Univ. Press, 2002.

[29] J. S. Milne, *Étale Cohomology*, Princeton Mathematical Series 33, Princeton Univ. Press, 1980.

[30] J. S. Milne, *Arithmetic Duality Theorems*, 2nd ed., BookSurge/online notes, 2006.

[31] W. Fulton, *Algebraic Curves: An Introduction to Algebraic Geometry*, Advanced Book Classics, Addison–Wesley, 1989; reprint: AMS, 2008.

[32] R. Miranda, *Algebraic Curves and Riemann Surfaces*, Graduate Studies in Mathematics 5, AMS, 1995.

[33] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik, Springer, 1998.

[34] E. Arbarello, M. Cornalba, P. A. Griffiths, and J. Harris, *Geometry of Algebraic Curves, Vol. I*, Grundlehren der mathematischen Wissenschaften 267, Springer, 1985.

[35] D. Mumford, *Tata Lectures on Theta I*, Progress in Mathematics 28, Birkhäuser, 1983.

[36] D. Mumford, *Tata Lectures on Theta II*, Progress in Mathematics 43, Birkhäuser, 1984.

[37] J. D. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Mathematics 352, Springer, 1973.

[38] H. M. Farkas and I. Kra, *Riemann Surfaces*, Graduate Texts in Mathematics 71, Springer, 1980 (many later printings; see also 2nd ed.).

[39] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Graduate Texts in Mathematics 41, Springer, 1990.

[40] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Graduate Texts in Mathematics 228, Springer, 2005.

[41] N. Koblitz, *Introduction to Elliptic Curves and Modular Forms*, 2nd ed., Graduate Texts in Mathematics 97, Springer, 1993.

[42] G. Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Princeton Univ. Press, 1971.

[43] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, 1973.

[44] S. Lang, *Elliptic Functions*, 2nd ed., Graduate Texts in Mathematics 112, Springer, 1987.

[45] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed., Graduate Texts in Mathematics 106, Springer, 2009.

[46] F. Diamond and J. Im, *Modular Forms and Modular Curves*, in: *Seminar on Fermat's Last Theorem* (Ribet, ed.), AMS, 1995/1998 printings.

[47] M. D. Fried and M. Jarden, *Field Arithmetic*, 3rd ed., Ergebnisse der Mathematik, Springer, 2008.

[48] H. Stichtenoth, *Algebraic Function Fields and Codes*, 2nd ed., Graduate Texts in Mathematics 254, Springer, 2009.

[49] J. J. Rotman, *Galois Theory*, 2nd ed., Universitext, Springer, 1998.

[50] H. M. Farkas and I. Kra, *Riemann Surfaces*, (see uniformization/covering theory chapters), Springer.

[51] L. Schneps (ed.), *The Grothendieck Theory of Dessins d'Enfants*, London Mathematical Society Lecture Note Series 200, Cambridge Univ. Press, 1994.

[52] S. K. Lando and A. K. Zvonkin, *Graphs on Surfaces and Their Applications*, Encyclopaedia of Mathematical Sciences 141, Springer, 2004.